

Commerce Data Standard

Version 0.2 (Review Draft)

Field	Value
Status	vo.2 Review Draft — for review; not an approved standard
Release	CDS vo.2 (single corpus release, ADR-D5)
Date	2026-09-20 (industry-profile expansion revision of the 2026-08-17 draft; changelog: REVIEW-021)
Requirements	1,320 identified (REVIEW-004 index; 1,034 in the 2026-08-17 draft plus 286 added by the industry-profile expansion)
Companion package	CDS-1200_Reference_Package_vo.2 (vo.2.1; verified 33/33 clean-environment)
Licence	Prose CC-BY 4.0 · machine-readable artifacts Apache-2.0
Supersedes	CDS vo.1 Working Corpus (16 chapter PDFs + package vo.1) — changelog: REVIEW-012; the 2026-08-17 sixteen-chapter vo.2 Review Draft — changelog: REVIEW-021

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Closing matter: A. ADR Register · B. Requirements Index summary · C. Bibliography

Reading conventions: RFC 2119/8174 keywords are normative only in upper case within requirement sentences (CDS000-R001). Every section is marked normative or informative. Requirement IDs are stable (CDSnnn-Rmmm).

Commerce Data Standard (CDS)

CDS-000 — Foundations and Core Principles

Field	Value
Status	vo.2 Review Draft (working source; not an approved standard)
Release	CDS v0.2 (single corpus release per ADR-D5)
Date	2026-08-04
Supersedes	CDS-000 Working Draft v0.1 (formerly mislabelled v1.0); the one-page "CDS Constitution" (retired to <code>_superseded/</code> , its unique principles merged here)
Normative status	§3 (Document Conventions), §4 (Versioning and Release Policy) and §6 (Precedence and Extension) are normative. §1, §2, §5 and §7 are informative.
Findings addressed	000-1..6, 300-9, D23, D26; ADR-D1, ADR-D5

1. Foreword and Mission (*informative*)

The Commerce Data Standard (CDS) is an open specification for the modelling, governance, publication and verification of product information. CDS is PIM-first, channel-agnostic, human-readable and governed-AI-ready. It defines how product information is represented independently of any individual commerce platform.

Mission: products are described once, understood everywhere, published many times, and verified after publication.

CDS exists to solve recurring commerce-data failures: product information duplicated and independently edited across systems; storefront filters that expose hundreds of near-duplicate raw values; tags used as an ungoverned database; internal taxonomies confused with storefront collections and marketplace categories; successful API writes treated as proof of correct downstream representation; free-text enrichment that defeats deterministic mapping and validation; cryptic field names that force operators to consult external notes; and AI automation that invents values without evidence or approval.

The core proposition:

Describe once in the PIM → normalise and govern → project to each channel → observe the downstream result → verify expected versus actual → present actionable data-quality health.

2. Scope and Non-Goals (*informative*)

CDS defines: the canonical product model; classification and taxonomy separation; semantic namespaces and naming; controlled dictionaries and reference data; facet and customer-experience projection; publication, observation and verification; governance, ownership and conformance; AI enrichment under governance; implementation profiles for specific platforms and industries.

CDS does not define: ERP, OMS, WMS or accounting systems; pricing strategy; CMS implementation; order, customer or inventory mastering (these may be *observed* by a CDS implementation but are mastered elsewhere — see the per-fact authority model in CDS-200).

CDS is vendor-neutral. Platform-specific behaviour lives only in versioned, dated profiles (CDS-900). No platform — including any reference implementation — shapes the core standard.

3. Document Conventions (*normative*)

CDS000-R001 The key words MUST, MUST NOT, SHOULD, SHOULD NOT and MAY in CDS documents are to be interpreted as described in RFC 2119 and RFC 8174: they are normative **only when they appear in upper case within the body of a requirement sentence**.

CDS000-R002 The keyword SHALL is not used in CDS. Requirement sentences MUST NOT carry a detached requirement-level prefix label; the in-sentence keyword governs. (*Resolves the prefix-corruption defect catalogued in the vo.2 errata manifest.*)

CDS000-R003 Every normative requirement carries a stable requirement identifier of the form `CDSnnn-Rmmm` (chapter number, requirement number). Identifiers are never reused after retirement.

CDS000-R004 Every section is explicitly marked normative or informative. Examples, worked scenarios and platform observations are informative unless individually marked otherwise.

CDS000-R005 Each defined term has exactly one authoritative definition, in CDS-100. Each normative topic has exactly one authoritative home chapter; other chapters cite it and MUST NOT restate its requirements. The authoritative homes are: terminology — CDS-100; entity and authority model — CDS-200; namespaces and naming — CDS-300; dictionaries and value layers — CDS-400; publication, observation, verification and statuses — CDS-500; facets and customer experience — CDS-600; AI — CDS-700; governance roles and change control — CDS-800; platform profiles — CDS-900; conformance levels, tests and claims — CDS-1000; machine-readable contracts — CDS-1100; the industry profile model, industry profile register and jurisdiction requirement register — CDS-1500 (each industry profile chapter, CDS-1500 through CDS-2000, is the home of the profiles it defines); product relationship records, kits and bundles, and dangerous-goods declarations — CDS-1900.

CDS000-R006 Architecture decisions are recorded as ADRs in a single global register, one ADR per decision. A superseded ADR is marked superseded and cross-referenced; the same decision MUST NOT be recorded in multiple ADRs.

4. Versioning and Release Policy (*normative — from ADR-D5*)

CDSooo-R007 The corpus has one release version shared by every chapter and companion package. The current release is **v0.2 Review Draft**. Chapters MUST NOT carry independent versions.

CDSooo-R008 Release gates: v0.9 requires external/domain review, stable profiles and a complete conformance package; v1.0 requires stable governance, independently tested schemas, current verified profiles and documented adopters. No CDS artifact may be labelled v1.0 before those gates pass.

CDSooo-R009 Schema `$ids` are release-independent URNs (`urn:cds:schema:<domain>:<name>`); each schema carries its own semantic version; instances pin their corpus release via the envelope `cds_version`. (Details: CDS-1100.)

CDSooo-R010 Principle identifiers use the form `CDS-P-nn`; they share no identifier space with chapter numbers, requirement IDs or ADR IDs.

5. Core Principles (*informative — each principle anchors requirements that live in its home chapter*)

ID	Principle	Meaning	Normative home
CDS-P-01	Canonical First	Every product fact has one canonical representation and one declared authoritative owner. Two systems never simultaneously master the same fact.	CDS-200
CDS-P-02	PIM First, Channels Downstream	The PIM is the master product-information layer. Channels consume projections; they never become the master. Channel read-back is observed state and never silently overwrites canonical values.	CDS-200, CDS-500
CDS-P-03	Information Before Technology	Model the product, not the platform. Platform constraints shape mappings in profiles, not canonical identity.	CDS-200, CDS-900
CDS-P-04	Human First	Visible field identifiers and namespaces reveal purpose in ordinary work without a glossary lookup. Semantic prefixes are part of the user interface.	CDS-300
CDS-P-05	AI Under Governance	Semantic clarity and structured metadata make product data highly usable by AI — but AI operates under schemas, dictionaries, provenance, confidence and human governance. Prefixes aid AI understanding; they do not replace schema metadata, and AI never infers authority it was not granted.	CDS-700, CDS-300 §22
CDS-P-06	Publish → Observe → Verify	A transport acknowledgement is not proof. Publication is incomplete until the downstream state is independently observed and compared with the expected state.	CDS-500
CDS-P-07	Controlled Vocabularies	Dictionaries and typed values are preferred over free text wherever consistency, filtering, mapping or AI use matters.	CDS-400
CDS-P-08	Facet UX Over Raw Data	Customers see governed, comprehensible facet families; the PIM retains the detailed canonical and source values beneath them.	CDS-600, CDS-400
CDS-P-09	Explicit Domains	Every visible field identifier belongs to a registered semantic namespace. Internal machine keys may differ where a declared reversible mapping exists.	CDS-300
CDS-P-10	Durable Naming	Names survive technology changes. Identifiers are stable; labels may evolve.	CDS-300
CDS-P-11	Extensible by Design	New channels, attributes and organisational extensions are added through registered extension points without breaking the model or colliding with reserved namespaces. (<i>From the retired Constitution.</i>)	CDS-300 §18–19, CDS-1100
CDS-P-12	Governance Over Convenience	Long-term data quality outranks short-term convenience. Shortcuts that bypass dictionaries, authority or verification are non-conformant even when expedient. (<i>From the retired Constitution.</i>)	CDS-800

6. Precedence and Extension (*normative*)

CDSooo-R011 Profile precedence: more specific profiles (platform → industry → organisation) MAY add constraints; CDS core semantics remain authoritative for shared concepts. A profile MUST NOT weaken a core requirement; deviations go through the exception process (CDS-800).

CDSooo-R012 Conformance levels, their tests and claim rules are defined solely in CDS-1000 (Foundation → Structured → Publisher → Verified → Governed, with overlay profiles such as AI-Assured). Levels are strictly cumulative. (*ADR-D1.*)

7. Relationship to Evidence (*informative*)

CDS's requirements trace to: explicit owner decisions and proven production behaviour (highest weight — the legacy Airtable/Shopify pipeline with per-field round-trip verification and traffic-light health); current official platform documentation, verified and dated in profiles; established PIM literature (Abraham 2014 — which supports the canonical model, taxonomy separation and controlled vocabularies, and whose silence on post-publication verification marks Publish → Observe → Verify as a CDS contribution); and reasoned architecture decisions recorded as ADRs. Working-draft prose is never self-evidencing.

Commerce Data Standard (CDS)

CDS-100 — Terminology

Field	Value
Status	vo.2 Review Draft (working source; not an approved standard)
Release	CDS vo.2
Date	2026-08-04
Supersedes	CDS-100 Working Draft vo.1 (formerly mislabelled v1.0)
Normative status	§2–§6 are normative definitions. §7 (lifecycle diagram and example) is informative. §8 (deprecated terms) is normative.
Findings addressed	100-1..5, Matrix 4 conflicts; ADR-D3, ADR-D4; 800-3 (exception), 1100 R023 (variant), CDS-1500-4 (multicolour)

CDS100-R001 The definitions in this chapter are normative. A term has exactly one CDS definition; other chapters **MUST** use these terms consistently and **MUST NOT** redefine them. Where industry usage conflicts, the CDS definition prevails within CDS documents.

CDS100-R002 Each definition names the chapter that owns the term's detailed requirements ("Home"). Definitions state *what a term means*; the home chapter states *what implementations must do*.

2. Core Entities (*normative*)

Product — A logical commercial item described by a single canonical information record. *Home: CDS-200.*

Variant — A distinct sellable unit of a Product, identified by the values of its variant options. A difference becomes a variant option only when it creates a distinct sellable unit. A Product with no variant options has exactly **one** variant, which **MAY** be implicit in storage but is a variant for counting, publication and verification purposes. A variant is normally identified by a stable SKU; an implementation **MAY** use an explicitly documented alternative sellable-unit key. *Home: CDS-200 §5.*

Variant Option — A named dimension (e.g. size, colour) whose values distinguish the variants of one Product. *Home: CDS-200 §5.*

Attribute — A named characteristic describing a Product or Variant.

Attribute Definition — The first-class schema record for an attribute: its identifier, type, scope (product/variant), validation rules, dictionary binding, requiredness, facet/display/search/channel policies and comparison strategy. Products store values; Attribute Definitions store behaviour. *Home: CDS-200 §7.*

Attribute Value — The stored value of an attribute on a specific Product or Variant, with provenance. *Home: CDS-200 §7.*

Classification — The assignment of a Product to nodes of the internal taxonomy. *Home: CDS-200 §6.*

Internal Taxonomy — The organisation's own classification tree, independent of any channel's taxonomy. A Product has exactly one primary category at the most specific applicable level; additional non-projecting secondary placements **MAY** exist. *Home: CDS-200 §6.*

Product Family / Product Type — The upper and middle tiers of the recommended three-tier internal taxonomy (Family → Type → Category). The three-tier stack is a recommended profile shape; core requirements bind to "an internal taxonomy", not to this exact stack. *Home: CDS-200 §6.*

Category — A node of a taxonomy (internal or external). *Home: CDS-200 §6.*

Collection — A merchandising grouping presented to customers, derived from rules, classification or curation. Collections never redefine product identity or replace classification. *Home: CDS-600 §6 (rule: CDS-200).*

Tag — A governed, flat channel signal generated from canonical data (e.g. `collection_*` projections). Tags are integration outputs, never the primary store for structured product facts. *Home: CDS-400 (governance), CDS-500 (publication merge semantics).*

2.1 Industry Profile Terms (*normative — added in the 2026-09-20 industry-profile expansion*)

Industry Profile — A named, versioned set of product-family and category rules, attribute baselines and dictionary bindings that refines the CDS core for a commerce vertical; listed in the Industry Profile Register. *Home: CDS-1500 §2; the profiles themselves in CDS-1500 through CDS-2000.*

Extension Profile — An Industry Profile that refines a parent (base) profile for a product family and is claimed only together with it (for example footwear under apparel). *Home: CDS-1500 §2.1.*

Jurisdiction Requirement Register — An organisation's governed record of the per-market regulatory obligations that bear on product data, each mapped to the CDS fields that carry the data and evaluated at publication preflight for the target market. *Home: CDS-1500 §2.2.*

Product Relationship — A governed, directed or symmetric, typed link from a Product or Variant to another product, model family, standard, application or external part, carried in the product's `relationships` collection with provenance and lifecycle. *Home: CDS-1900 §5.*

Fitment — A Product Relationship of type `fits` whose target is an application record describing a host (vehicle, machine, appliance model) drawn from a declared vocabulary. *Home: CDS-1900 §6.*

Kit — A Product whose sellable content is composed of other Products through `includes` relationships; stocked kits, virtual bundles and assemblies are its declared kinds. *Home: CDS-1900 §11.*

Dangerous-Goods Declaration — The product-level record stating whether, and under which regime and classification, a product is regulated for transport, with an explicit value for every product. *Home: CDS-1900 §12.*

Market Registration — A per-jurisdiction record of a product's regulatory status (registered, listed, notified, exempt, out of scope) under a named scheme. *Home: CDS-1600 §12.*

3. Value Layers and Dictionaries (*normative — per ADR-D4*)

Source Value — The raw value exactly as received from a supplier, import or channel, preserved verbatim with provenance. (*Replaces the former term "Reference Value" — see §8.*) *Home: CDS-400.*

Alias — A recognised source spelling or synonym of a canonical value.

Alias Mapping Record — The governed record mapping one source value (scoped by dictionary, and optionally supplier or locale) to exactly one canonical value. Every source value that has been mapped is retained as an alias with provenance. *Home: CDS-400.*

Dictionary — A governed, versioned collection of canonical values for one semantic domain (e.g. colour), including their aliases, labels, facet memberships and lifecycle states. *Home: CDS-400.*

Dictionary Value / Canonical Value — A governed entry in a dictionary: a stable `value_id` and `snake_case` canonical code with lifecycle status and provenance. Canonical product data stores canonical `value_ids` — a Variant's colour field holds the canonical value, never a raw source spelling. Promotion of a source value to a new canonical value is an explicit governed act; bulk imports never silently create canonical values. *Home: CDS-400.*

Display Label — A human-facing label linked to a canonical value. Display labels MAY preserve supplier or brand names (e.g. "Moonlit Harbour" displayed while canonical `navy` governs). *Home: CDS-400 §9.*

Facet — A customer-facing refinement dimension used for filtering and navigation. *Home: CDS-600.*

Facet Definition — The governed configuration of a facet: identifier, label, eligible categories, value source, selection logic and count method. *Home: CDS-600.*

Facet Value — A governed, customer-facing grouping value that canonical values map to (e.g. canonical `french_navy` → facet `blue`). Facet values are projections; they never replace canonical values. Multicolour is a facet/status value, never a canonical shade. *Home: CDS-400 (mapping), CDS-600 (presentation).*

Search Synonym — A governed query-side term mapped to a preferred concept for search expansion. *Home: CDS-600 §21.*

Channel Representation — The value form projected to a specific channel for a specific attribute, as declared by the Attribute Definition and channel profile (which layer feeds each channel — canonical, display or facet — is declared per attribute, never assumed). *Home: CDS-400 §11, CDS-500 §6.*

Projection — Any deterministic derivation from canonical data to a downstream form: display, facet, search, tag or channel representation, and the Expected Channel State as a whole. *Home: CDS-500.*

Quarantine — The holding state for unknown or ambiguous source values pending human resolution; quarantined items remain in quality denominators. *Home: CDS-400 §17.*

4. Identity, Naming and Authority (*normative*)

Namespace / Prefix — A registered semantic domain marker (e.g. `MF_`, `CH_`, `QA_`) forming the first segment of visible field identifiers. The single normative registry lives in CDS-300. *Home: CDS-300.*

Semantic Identifier — The visible, namespace-prefixed field identifier (e.g. `MF_colour_family`) used in operator-facing contexts. Distinct from both the Human Label and the Machine Key. *Home: CDS-300 §8.*

Human Label — The natural-language label shown in UIs (e.g. "Colour family").

Machine Key — The internal storage key, which MAY differ from the semantic identifier where a declared reversible mapping exists (e.g. lossy export targets). *Home: CDS-300 §8.*

Extension Namespace — An organisation-registered prefix or extension point that cannot collide with any reserved core prefix. *Home: CDS-300 §18–19.*

Authority — The declared right of exactly one system to master a given product fact. Authority is declared **per fact**, not per system: the PIM masters canonical product content; other facts (e.g. inventory, price) MAY be mastered elsewhere and observed. Delegation is an explicit recorded act. *Home: CDS-200 §13.*

System of Record — The system holding authority for a fact.

Owner — The accountable person or role for a governed artifact (field, dictionary, facet, mapping, workflow). *Home: CDS-800.*

Provenance — The recorded origin and history of a value: source, method (manual, formula, import, AI-proposed), actor, time and evidence. *Home: CDS-200 §16, CDS-700.*

Confidence — A bounded score (0–1) attached to a machine-proposed value; never the sole acceptance criterion. *Home: CDS-700.*

Override — A governed record that redirects the expected channel value for a specific field/channel away from the default projection. A verified match against an active override is a true MATCH with `OVERRIDDEN` provenance. *Home: CDS-500 §22.*

Exception — A governance record granting a time-limited, documented, approved deviation from a requirement, with an expiry and a named owner. Expired exceptions never remain silently active. An exception acknowledges non-conformance; it does not convert failure into conformance. (*Single definition — supersedes the conflicting exception/waiver uses in earlier drafts; "waiver" as a noun is retired, see §8. The field-level verification status `WAIVED` — an accepted difference under an active exception that does not redirect the expected value — remains, defined in CDS-500.*) *Home: CDS-800 §16.*

5. Publication, Observation and Verification (*normative*)

Channel — Any downstream consumer of product information (e.g. Shopify, Google Merchant Center, Meta).

Channel Profile — The versioned, dated definition of a channel's capabilities, constraints, field mappings, write semantics and propagation window. *Home: CDS-900 (content), CDS-500 (role).*

Field Mapping — The declared transformation from a canonical attribute (at its declared layer) to a channel target field, including transformation version, write mode and comparison strategy. *Home: CDS-500 §6, CDS-1100.*

Expected Channel State — The persisted, reproducible projection of what a channel should hold for a product: derived from a specific canonical revision, mapping-set version and dictionary versions. Regenerating it with the same inputs yields the same output. *Home: CDS-500 §6.*

Publication — The controlled delivery of an expected channel state to a channel, recorded immutably.

Publication Record — The immutable record of one publication attempt: payload identity (hash), canonical revision, mapping-set version, transport status and acknowledgement. *Home: CDS-500 §8.*

Acknowledgement — The channel's transport-level response to a publication. An acknowledgement is not proof of correct downstream representation. *Home: CDS-500 §9.*

Observation — The independent retrieval of the state a channel currently holds, separate from any outbound payload. *Home: CDS-500 §14.*

Observation Record / Observed State — The stored result of an observation, kept strictly separate from canonical values, with method, time and per-field coverage. *Home: CDS-500 §15.*

Coverage — The per-field account of what an observation could and could not see (observed, empty, not returned, unsupported, permission denied, stale, pending visibility). Missing and unobservable are never conflated. *Home: CDS-500 §15.*

Verification — The comparison of expected channel state against observed state, field by field, under each field's declared comparison strategy. *Home: CDS-500 §16.*

Verification Status — The outcome classification of a verification, drawn from the single normative enum owned by CDS-500 §17. The eight **core statuses** (machine contract) are MATCH, MISSING, MISMATCH, PENDING, UNOBSERVABLE, NOT_APPLICABLE, OVERRIDDEN and ERROR; detailed statuses roll up to these deterministically. (*Summary here is informative; CDS-500 governs.*)

Comparison Strategy — The declared, type-aware method for comparing one field (exact, case-insensitive, unordered set, numeric tolerance, normalised text), drawn from the normalisation registry with test vectors. *Home: CDS-500 §16.*

Reason Code — The machine-readable code (`CDS_ registry`) carried by every non-MATCH field result, explaining why — the layer beneath the traffic light. *Home: CDS-1100 §21 (registry), CDS-500 (use).**

Traffic Light — The presentation layer over verification statuses: Green = expected equals observed; Amber = attention (missing, pending, unobservable, overridden, waived); Red = confirmed contradiction or error; Amber escalates to Red per declared rules. Colours are presentation, never stored statuses. *Home: CDS-500 §18 (per ADR-D3).*

Drift — Downstream state that no longer conforms to the current expected channel state or ownership contract. Republishing is not always the correct repair. *Home: CDS-500 §20, CDS-1400 §15.*

Repair — A governed action resolving drift (republish, accept via override/exception, or correct canonical data). *Home: CDS-500.*

Channel Health — The aggregate quality indication for a channel, computed from expected-state coverage, publication outcomes, observation coverage and verification results, with explicit denominators. Unobserved fields never count as matched. *Home: CDS-1400 §5–6 (metrics), CDS-500 §23 (aggregation).* (*This supersedes the narrower v0.1 definition "derived from verification results".*)

Propagation Window — The channel-declared time allowance between publication and observable effect; PENDING beyond the window escalates. *Home: CDS-900 (declared), CDS-500 (used).*

6. AI and Automation (*normative*)

AI Proposal — A machine-generated suggested value or action, stored separately from canonical state, carrying evidence, confidence, model/workflow version and review state. Acceptance is an auditable act linking the proposal to the canonical revision it produced. *Home: CDS-700.*

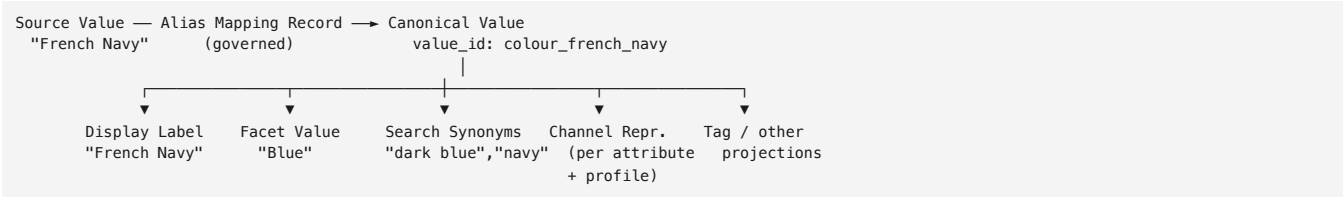
Autonomy Level (Ao–A4) — The declared, per-workflow ceiling on what an AI workflow may do without human action, from suggest-only to bounded auto-acceptance. Never interpreted as unrestricted autonomy. *Home: CDS-700 §5.*

Evidence Class (E1–E4) — The declared strength ranking of the evidence supporting a proposal. *Home: CDS-700 §7.*

Abstention — A first-class AI outcome: declining to propose rather than inventing a value. Absence of evidence never becomes a positive claim. *Home: CDS-700.*

Deterministic Automation — Rule-based processing where identical inputs always produce one correct result; preferred over AI for such tasks, and never obscured by AI. *Home: CDS-700 §20.*

7. Controlled Value Lifecycle (*informative — corrected to the branch model per ADR-D4*)



Projections branch **in parallel** from the canonical value; channels are fed from the layer declared per attribute — never implicitly from the facet.

Worked example. A supplier provides "French Navy". The alias mapping resolves it to canonical `colour_french_navy` (or, in a coarser configuration, aliases it to `colour_navy` — both granularities are conformant). The variant stores the canonical `value_id`. The product page displays the Display Label "French Navy"; the storefront filter presents the Facet Value "Blue"; the channel receives whichever layer the Attribute Definition declares for it. When the mapping changes, republication regenerates every projection deterministically without editing any product.

8. Deprecated and Retired Terms (*normative*)

CDS100-R003 The following terms are retired and MUST NOT be used in vo.2+ documents except when quoting vo.1 drafts or legacy systems:

Retired term	Replacement	Reason
Reference Value	Source Value (the raw input) or Alias Mapping Record (the mapping) or Canonical Value (the governed entry) — whichever layer is meant	Three conflicting published meanings across vo.1 chapters (ADR-D4)
Waiver (noun, governance record)	Exception	Two chapters defined exception/waiver with swapped meanings (finding 800-3). The verification status WAIVED remains (CDS-500)
SHALL	MUST	Single keyword set (CDS000-R002)
Mapping (bare, unqualified)	Field Mapping, Alias Mapping Record, or Taxonomy Mapping — whichever is meant	Ambiguous across three distinct record types
DF_ (prefix, new design)	CH_	ADR-D24; DF_ remains a registered deprecated legacy prefix for migration aliases

9. Design Principles Supported (*informative*)

One concept, one definition (CDS100-R001). Canonical information precedes publication (CDS-P-01/02). Customer experience and internal data are separate concerns (CDS-P-08). Semantic clarity benefits humans, software and governed AI equally (CDS-P-04/05).

Commerce Data Standard (CDS)

CDS-200 — Commerce Information Architecture

Field	Value
Status	vo.2 Review Draft (working source; not an approved standard)
Release	CDS v0.2
Date	2026-08-04
Supersedes	CDS-200 Working Draft v0.1
Normative status	§3 (Canonical Model), §4 (Entity Model), §5 (Product/Variant Boundary), §6 (Classification), §7 (Attribute Architecture), §13 (Ownership and Authority), §14 (State and Audit), §15 (Rules and Validation) are normative. §1–§2, §8–§12, §16–§18 are informative.
Findings addressed	200-1, 200-2, 200-3, 200-4, 200-5, 200-6, 200-7, 200-8, 200-9; SYS-4 (chapter scope); decisions D9/D30 (LC-10); Matrix 5 relocations; ADR-D3, ADR-D4, ADR-D24

This chapter is the authoritative home for the **core entity model, the product/variant boundary, the per-fact authority model, and classification** (CDS000-R005). Terminology is defined in CDS-100 and used here without redefinition.

1. Purpose and Scope (*informative*)

CDS-200 defines the vendor-neutral information architecture that sits above commerce platforms and below business operations: what the master product layer contains, how information is classified and governed, who is authoritative for each fact, and how downstream channels consume projections of it.

The architecture is PIM-first (CDS-P-01/02/03). Sales channels and marketplaces are consumers. They may impose schemas and constraints, but they do not define the canonical identity of a product.

Publication, observation and verification mechanics are defined in CDS-500; naming in CDS-300; dictionaries and value layers in CDS-400; facets in CDS-600; AI participation in CDS-700; platform specifics in CDS-900 profiles.

2. Architectural Objectives (*informative*)

- **Single authority** — every product fact has one clearly defined authoritative source.
- **Separation of concerns** — identity, classification, attributes, merchandising, publication and verification remain distinct.
- **Human legibility** — names and structures communicate purpose without separate notes (CDS-300).
- **Machine determinism** — software can parse, validate, map and compare values without subjective interpretation.
- **AI clarity** — AI agents receive explicit schemas, dictionaries, constraints and confidence rules (CDS-700).
- **Channel durability** — technology-specific changes are absorbed by mappings, not by redesigning the canonical model.
- **Customer usability** — internal richness is simplified into usable facets and navigation (CDS-600).
- **Observability** — publication success is measured by observed downstream state, not merely by API response (CDS-500).

3. The Canonical Commerce Model (*normative*)

The canonical commerce model is the set of product information that is true independently of any channel: identity, classification, attribute values, variant relationships, content, media, compliance and lifecycle state. It excludes channel-specific formatting and channel-owned operational data unless such data is explicitly adopted as canonical.

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Canonical Product Model
|
+-- Identity
+-- Classification
+-- Attribute Values
+-- Variants / Sellable Units
+-- Content
+-- Media
+-- Compliance
+-- Lifecycle
+-- Reference Data Links
|
+-- Publication Projections (per channel; CDS-500)
```

CDS200-R001 A CDS implementation **MUST** maintain a canonical product layer independent of any single sales channel.

CDS200-R002 A channel-specific limitation **MUST NOT** reduce the information fidelity of the canonical model. Channel constraints are absorbed by field mappings and profiles (CDS-500, CDS-900).

CDS200-R003 A downstream read-back MAY inform verification, but MUST NOT silently overwrite canonical values. Observed state is stored separately from canonical state (observation mechanics: CDS-500 §14–§15).

CDS200-R004 Canonical information MUST be stored once and projected many times.

CDS200-R005 A field that exists only because one channel requires it SHOULD be represented as a channel field or mapping (`CH_ scope`, CDS-300), not as canonical product truth.

Informative note: a Google product category is normally a channel taxonomy mapping. Material, colour and dimensions are normally canonical facts.

4. Core Entity Model (*normative*)

CDS defines a logical entity model rather than a flat product table.

CDS200-R006 An implementation MAY store data physically in relational tables, documents, graphs or spreadsheets, but MUST preserve the logical entities below and the rules attached to them.

Entity	Role	Detail home
Product	The logical commercial item and parent information record.	CDS-200 §5
Variant	A sellable unit identified by its variant option values.	CDS-200 §5
Attribute Definition	The first-class schema, behaviour and governance record for an attribute.	CDS-200 §7
Attribute Value	A product- or variant-level value conforming to an Attribute Definition, with provenance.	CDS-200 §7
Dictionary	A governed, versioned set of canonical values, aliases and projections.	CDS-400
Alias Mapping Record	The governed mapping from a source value to exactly one canonical value.	CDS-400
Facet Definition	A customer-facing grouping and filter behaviour.	CDS-600
Category	A node in a taxonomy (internal or external).	CDS-200 §6
Taxonomy Mapping	A relationship between an internal category and an external taxonomy node.	CDS-200 §6
Taxonomy Decision Log	The recorded tie-break decisions for primary-category assignment.	CDS-200 §6
Channel	A downstream consumer of published product information.	CDS-500
Publication Record	The immutable record of one publication attempt.	CDS-500 §8
Observation Record	The state read back from a downstream channel, stored separately.	CDS-500 §15
Verification Result	The comparison of expected versus observed values.	CDS-500 §16–§17
Override Record	A governed redirection of the expected channel value.	CDS-500 §22
Authority Declaration	The per-fact declaration of the mastering system (and any delegation).	CDS-200 §13
Rule	A validation, inheritance, transformation or publication condition.	CDS-200 §15

CDS200-R007 Tags MUST be governed projections generated from canonical data (e.g. `collection_*` signals regenerated from classification rules) and MUST NOT serve as the primary store for structured product facts. Tag governance is defined in CDS-400; tag publication and merge semantics in CDS-500.

5. Product and Variant Boundaries (*normative*)

A Product is a logical record shared by all sellable forms. A Variant is a sellable unit with its own SKU or documented sellable-unit key (CDS-100 §2).

CDS200-R008 A difference MUST create a distinct sellable unit before it is modelled as a variant option. Attributes that do not produce a separately sellable unit MUST NOT be modelled as variant options merely to make them filterable (filterability is a facet concern — CDS-600).

CDS200-R009 A Product with no variant options has exactly one variant, which MAY be implicit in storage but MUST be countable as a variant for publication and verification purposes.

CDS200-R010 Product-level attribute values MUST describe facts shared by all variants of the product.

CDS200-R011 Variant-level attribute values MUST describe facts specific to the sellable unit, and MUST be stored as canonical `value_ids` (never raw source spellings) per ADR-D4.

CDS200-R012 Where a variant-scoped attribute (such as colour on a multi-colour product) is presented at product level for display, faceting or search, the product-level value MUST be a projection derived from the variant values and MUST NOT be an independently authored canonical fact.

```
Product: Linen Shirt
Product attribute values (facts shared by all variants):
  MF_material      = material_linen
  MF_fit           = fit_relaxed
  MF_sleeve_length = sleeve_long
Variants (canonical value_ids per ADR-D4):
  SKU-001: VAR_colour = colour_french_navy, VAR_size = size_s
  SKU-002: VAR_colour = colour_french_navy, VAR_size = size_m
  SKU-003: VAR_colour = colour_white,      VAR_size = size_s
Product-level projections (derived from variants, never authored):
```

```
MF_colour_display = "French Navy / White"    (display projection)
MF_colour_family  = [blue, white]           (facet projection)
```

Informative note: style, occasion and fit usually describe the product. Size and sellable colour commonly define variants. In vo.1 this chapter's worked examples treated colour as both a variant option and an authored product-level value on a multi-colour product; R012 resolves that contradiction — the variant values are canonical, the product-level colour fields are derived projections.

6. Classification Architecture (*normative*)

Classification answers *what the product is*. CDS separates internal classification from external taxonomy mappings and from customer-facing merchandising collections.

CDS200-R013 An implementation **MUST** maintain an internal taxonomy that is independent of any channel taxonomy. Core requirements in this section bind to *an internal taxonomy*, not to any particular tier structure.

CDS200-R014 The three-tier shape **Product Family** → **Product Type** → **Category** (e.g. Apparel → Shirt → Linen Shirts) **SHOULD** be adopted as the recommended profile shape; industry profiles (CDS-1500) refine it. Implementations **MAY** use a different documented internal structure without loss of conformance to this section.

CDS200-R015 A product **MUST** have exactly one primary internal category.

CDS200-R016 The primary category **SHOULD** be the most specific applicable node — a node none of whose child nodes applies to the product.

CDS200-R017 Where more than one category could serve as the primary category, the assignment **MUST** be resolved using an **organisation-declared, documented, ordered set of tie-break rules**, applied in order until one yields a decision and applied consistently across the catalogue. The following ordering is the **recommended default** (*informative*):

1. **Customer mental model** — the category the customer would navigate to or search within when intending to buy the product.
2. **Dominant use** — the function the product is primarily designed to perform.
3. **Material / form** — what the product physically is, where use is ambiguous.
4. **Commercial priority** — where the product's assortment peers sit and where it best serves the range plan.

RESOLVED (was D30 (resolved), owner decision 2026-08-04): the standard requires only *an* organisation-declared documented order (matching the legacy production discipline); the four-rule ordering above is demoted to recommended default. The owner noted this may be revisited after practical use.

CDS200-R018 Every tie-break decision **MUST** be recorded in a taxonomy decision log entry carrying at least: the product type or product concerned, the category chosen, the rule applied, and the date. Subsequent products matching a logged entry **MUST** follow it; changing the decision requires a superseding log entry, not an ad-hoc different assignment.

CDS200-R019 A product **MAY** additionally hold **secondary category placements**. Secondary placements are non-projecting: they **MUST NOT** drive external taxonomy mappings, required-attribute inheritance or collection projection unless a rule explicitly and individually opts a placement in.

CDS200-R020 External taxonomy identifiers (channel or standard taxonomies) **MUST** be stored as Taxonomy Mappings from internal categories (or products) and **MUST NOT** replace the internal taxonomy. Category-level mappings **MAY** be inherited by child categories.

CDS200-R021 A product **MAY** carry a per-channel taxonomy mapping override where the category-level mapping is wrong for that specific product on that specific channel. Such an override **MUST** be recorded as a governed override record (CDS-500 §22), scoped to product + channel.

RESOLVED — accepted as drafted (owner 2026-08-04; was new (resolved)): R021 reuses the CDS-500 §22 override record for taxonomy overrides rather than defining a separate taxonomy-override record; the alternative is a dedicated field on Taxonomy Mapping.

CDS200-R022 A product **MAY** appear in multiple collections without being assigned multiple primary categories. Collections are merchandising groupings and **MUST NOT** redefine classification (presentation rules: CDS-600).

```
Internal category:  Apparel > Tops > Shirts
|
+-- external taxonomy mappings (per channel; inherited or overridden)
+-- inherited attribute family / required-attribute rules
+-- collection projection rules (governed tags per R007)
```

Informative note — taxonomy overlap vs merchandising overlap. If a product genuinely *belongs* in two categories, the category boundaries overlap and should be fixed. If a product merely *appeals* to two audiences, that is a merchandising concern: keep one primary category and reach the second audience via collections, governed tags and cross-links — never via a second primary category.

7. Attribute Architecture (*normative*)

Attributes are governed by Attribute Definitions. The definition is a first-class entity and carries more intelligence than the product record itself.

Definition property	Purpose
identifier	Stable semantic key (registered per CDS-300).
label	Human-readable name.

Definition property	Purpose
description	Plain-language purpose.
scope	Product, variant, category, channel or system.
data type	Text, number, boolean, date, measurement, reference, list or structured object.
cardinality	Single or multiple values.
dictionary	Optional controlled value source (CDS-400).
requiredness	Required, conditionally required or optional.
inheritance	Whether a value may inherit from category, family or parent.
validation	Format, range, units, dependencies and exclusions.
display behaviour	Whether and how the value appears on product pages.
facet behaviour	Whether and how the value participates in filtering (CDS-600).
search behaviour	Indexing, synonyms and weighting.
publication mapping	Destination fields and transformations per channel (CDS-500).
channel source layer	Which value layer (canonical, display, facet) feeds each channel (ADR-D4).
comparison strategy	Exact, normalised text, numeric, set, ordered list or identifier (CDS-500 §16).
authority / delegation	The fact's declared authority and any recorded delegation (§13).
AI behaviour	Extractability, confidence threshold and review requirement (CDS-700).

CDS200-R023 Every governed attribute **MUST** have an Attribute Definition.

CDS200-R024 Products and variants **MUST** store values; Attribute Definitions **MUST** store behaviour.

CDS200-R025 A channel mapping **SHOULD** be defined on the Attribute Definition rather than repeated on every product.

CDS200-R026 The Attribute Definition **MUST** declare which value layer feeds each consuming channel (canonical value, display label or facet value); channels are never fed implicitly from the facet layer (ADR-D4).

8. Dictionaries and Value Layers (*informative*)

Canonical values, alias mappings, display labels, facet values, search synonyms and channel representations follow the branch model **Source Value** → **Alias Mapping Record** → **Canonical Value** → **parallel projections** defined normatively in CDS-400 (model) and CDS-100 §3 (terms), per ADR-D4. This chapter's architecture consumes that model; it does not restate it.

9. Facets and Customer Experience (*informative*)

Facets are governed projections of canonical data, designed for customer usability — never a blind exposure of every distinct internal value. Facet architecture, dictionary design and presentation rules are defined in CDS-600, with facet-mapping governance in CDS-400.

10. Channel and Publication Architecture (*informative*)

A Channel consumes versioned projections of canonical data: canonical data → field mapping → expected channel state → publish → observe → verify. Expected-state calculation, publication records, acknowledgement semantics ("an acknowledgement is not proof") and repair are defined normatively in CDS-500.

11. Verification and Channel Health (*informative*)

Verification compares expected channel state with observed downstream state, field by field, under declared comparison strategies. The single normative verification status enum, reason codes, traffic-light mapping and channel-health aggregation are owned by CDS-500 §16–§18 (per ADR-D3) and CDS-1400. This chapter defines no status vocabulary.

12. Semantic Namespace Layer (*informative*)

Every visible field identifier belongs to a registered semantic namespace (e.g. `STD_`, `CAT_`, `VAR_`, `MF_`, `CH_`, `QA_`). The single normative prefix registry, the identifier grammar and casing rules, registry validation (the testable mechanism replacing v0.1's "prefix must communicate purpose" wording), and the visible-identifier / human-label / machine-key separation are all owned by CDS-300. Namespace-prefixed identifiers appearing in this chapter's examples are illustrative uses of that registry.

13. Data Ownership and Authority (*normative*)

Authority is declared **per fact**, not per system. The PIM is the system of record for canonical product information; specialised systems may own operational facts such as live inventory or order state, which the PIM observes but does not master.

Information	Typical authority
Product identity, classification and descriptive attributes	PIM

Information	Typical authority
Channel mappings and publication configuration	PIM
Product content and media sequencing	PIM unless explicitly delegated
Inventory quantity	Inventory or warehouse authority
Orders and customers	Commerce platform or order-management authority
Observed downstream state	Channel connector read-back (CDS-500)
Verification status	PIM verification engine (CDS-500)

CDS200-R027 Every governed product fact MUST have exactly one declared authority, recorded on the Attribute Definition or in an equivalent authority matrix.

CDS200-R028 Two systems MUST NOT simultaneously claim authoritative ownership of the same product fact.

CDS200-R029 Delegation of authority MUST be an explicit recorded declaration on the Attribute Definition or authority matrix, carrying at least: the delegated fact scope, the delegate system, the approving owner, the effective date and a review date. Undeclared delegation is non-conformant; "the other system just edits it" is not delegation.

CDS200-R030 A channel-specific deviation from the projected canonical value MUST be expressed as a governed override record as defined in CDS-500 §22 (scope, replacement value, reason, owner, approval, time bounds, verification expectation). A downstream edit to a PIM-owned field without an active override record is drift (CDS-500 §11).

CDS200-R031 Read-back (observed) values MUST be stored separately from canonical values (observation record: CDS-500 §15). See also R003.

14. State, Versioning and Audit (*normative*)

Commerce information changes over time; versioning and audit metadata are first-class architecture. Canonical records carry a revision identifier, change timestamps and the actor or process responsible; publication, observation and verification timestamps per channel are defined by the CDS-500 record set.

CDS200-R032 An implementation MUST be able to determine which canonical revision produced a published channel state (record linkage: CDS-500 §8).

CDS200-R033 Mappings and dictionaries SHOULD be versioned, because changes to them alter downstream output without changing the product record (dictionary versioning: CDS-400; mapping-set versioning: CDS-500).

15. Rules and Validation (*normative*)

Rules define requirements, inheritance, transformations and publication conditions, evaluated against the canonical model and channel profiles.

```
IF CAT_product_family = apparel
THEN REQUIRE MF_material, MF_colour, MF_care_instructions

IF CAT_product_type = candle
THEN REQUIRE MF_wax_type, MF_burn_time

IF MF_material = material_leather
THEN REQUIRE MF_country_of_origin
```

CDS200-R034 Conditional requirements MUST be represented as explicit rules rather than undocumented staff knowledge.

CDS200-R035 Validation errors MUST distinguish missing, invalid, conflicting and unmapped values.

CDS200-R036 Rules SHOULD operate on canonical values and declared channel projections, not on uncontrolled display text.

Informative note: channel-imposed identifier requirements (for example Google Merchant Center's GTIN/MPN/brand rules) are platform policy, not core architecture; they are specified, dated and verified in the relevant CDS-900 platform profile and expressed here only as ordinary conditional rules bound to that profile.

16. AI Participation (*informative*)

AI may assist with extraction, classification, enrichment, mapping suggestions and anomaly detection — always as a proposer, never as an authority merely because it generated a value. All normative AI requirements (proposal separation, evidence, confidence, review, autonomy levels, publication restrictions) are owned by CDS-700. AI-proposed values conform to the same Attribute Definitions and dictionaries as human-entered values; that requirement is stated normatively in CDS-700.

17. End-to-End Examples (*informative*)

17.1 Apparel colour (per ADR-D4)

```
Source value (supplier)      = "French Navy"
Alias mapping record         = "French Navy" -> colour_french_navy
Variant stores               : VAR_colour = colour_french_navy (canonical value_id)
Display projection           : MF_colour_display = "French Navy"
```

Facet projection (product)	: MF_colour_family = [blue]	(derived from variants)
Channel representation	: CH_google_colour = "Blue"	(layer declared per R026)
Verification (CDS-500)	: QA_shopify_colour = MATCH	

On a multi-colour product, VAR_colour differs per variant and MF_colour_family derives the full set (e.g. [blue, white]) — the product-level fields are never authored directly (R012).

17.2 Homewares material

Source value	= "Tasmanian Oak"
Canonical value	: MF_material = material_oak
Display projection	: MF_material_display = "Tasmanian Oak"
Facet projection	: MF_material_family = wood
Channel representations	: Shopify "Oak"; Google "Wood" (declared per attribute)
Search synonyms	: timber, oak, tasmanian oak

17.3 Category assignment and projection

Candidate categories	: Homewares > Living Room > Cushions vs Homewares > Textiles > Soft Furnishings
Tie-break (R017)	: rule 1, customer mental model -> Cushions
Decision log entry (R018)	: "Cushions Living Room > Cushions customer mental model 2026-08-04"
Primary category	: Homewares > Living Room > Cushions
Derived outputs	: CAT_product_type = cushion CH_shopify_category = Home & Garden > Decor > Throw Pillows CH_google_product_category = Home & Garden > Decor > Decorative Pillows Collection projection = Cushions (governed tag, R007) Required attributes = colour, material, shape, dimensions, fill

17.4 Round-trip verification

Canonical value	: colour_navy
Expected Shopify value	: "Navy"
Observed Shopify value	: "navy"
Comparison strategy	: case-insensitive canonical text
Result	: MATCH (statuses and traffic light: CDS-500 §17-§18)

18. Conformance (*informative*)

The conformance criteria for this chapter are its requirement identifiers CDS200-R001–R036. Conformance levels, test suites and claim rules are defined solely in CDS-1000; machine-readable expression of this chapter's entities in CDS-1100.

Architecture decisions: the decisions underlying this chapter (PIM-first mastering, value layers, status vocabulary, DF_→CH_) are recorded in the single global ADR register (CDS000-R006) — see ADR-D3, ADR-D4, ADR-D24.

Platform notes: Shopify-specific implementation guidance formerly in Appendix A (taxonomy mapping targets, metafield publication, Search & Discovery facet consumption, read-back population) lives in the CDS-900 Shopify profile. The tags rule formerly there is now normative as R007.

Legacy alignment: the legacy Airtable PIM prefix mapping tables and migration alignment formerly in Appendix B live in the migration annex (CDS-1300).

Commerce Data Standard (CDS)

CDS-300 — Semantic Namespaces and Field Naming

Field	Value
Status	vo.2 Review Draft (working source; not an approved standard)
Release	CDS v0.2 (single corpus release per ADR-D5)
Date	2026-08-04
Supersedes	CDS-300 Working Draft v0.1; the namespace registry tables formerly duplicated in CDS-000, CDS-200 and CDS-900 (this chapter is the single normative registry per CDS000-R005)
Normative status	§3–§23 are normative. §1, §2, §24 and §25 are informative.
Findings addressed	300-1 (ADR-D2), 300-2 (D7), 300-3, 300-4 (ADR-D24), 300-5 (D6), 300-6, 300-7 (D8), 300-8, 300-9; Matrix 1 (all rows); REVIEW-002A LEG-1, LEG-3; CDS-700 §22 registration request

1. Purpose and Scope (*informative*)

CDS-300 defines how commerce fields are named so that their purpose is immediately apparent to humans, deterministic for software and semantically useful to AI. Names are part of the user interface (CDS-P-04): a visible identifier is a compact explanation of why the field exists, which domain owns it and, where necessary, which destination consumes it.

```
MF_material
CH_google_product_category
QA_shopify_material
WF_copy_review
SYS_updated_at
```

This chapter is the **single normative home of the CDS namespace prefix registry** (CDS000-R005). Other chapters cite the registry; they do not restate it. Platform mapping tables live in CDS-900 profiles; they map registered prefixes, they never define them.

The naming system is a visible presentation layer over the information architecture of CDS-200. Implementations need not use identical physical column names, but visible identifiers and exported field names preserve the same semantics (§8).

2. Design Requirements (*informative*)

Requirement	Meaning
Human recognisability	A new operator infers the field purpose within seconds.
Machine safety	Identifiers never require escaping in spreadsheets, CSV, JSON, SQL, APIs or common languages.
AI legibility	Semantic purpose is inferable from the identifier plus its schema metadata.
Stable sorting	Related fields group naturally when sorted alphabetically.
Durability	Names describe business purpose, not temporary delivery technology.
Brevity	Short enough to scan, never cryptic.
Extensibility	Organisations and channels extend the registry without collisions.
Migration clarity	Deprecated identifiers map explicitly to replacements.

Uppercase prefixes create strong visual grouping; the lowercase snake_case remainder is readable and portable. Names optimise for recognition and scanning before typographic elegance. Abbreviations appear only where obvious to the commerce audience or standardised by CDS.

3. Identifier Grammar (*normative*)

The visible CDS field identifier grammar:

```
<NAMESPACE>_<field_name>
CH_<channel>_<field_name>      (channel form)
QA_<channel>_<field_name>      (verification form)
```

CDS300-R001 A visible field identifier **MUST** communicate its semantic role without requiring ordinary users to consult a separate document during routine work.

CDS300-R002 The namespace segment **MUST** be two to eight uppercase ASCII characters, **MUST** begin with a letter, and **MUST** be followed by a single underscore separator.

CDS300-R003 The field-name portion SHOULD use lowercase snake_case. The first field-name segment MUST begin with a lowercase letter; subsequent segments MAY be numeric (e.g. CH_google_custom_label_0).

CDS300-R004 The complete identifier MUST NOT contain spaces, colons, equals signs, dots, slashes, backslashes, commas or Unicode punctuation, and MUST use only ASCII letters, digits and underscores unless a profile explicitly permits another character set.

CDS300-R005 Identifiers MUST match the regular expression:

```
^[A-Z][A-Z0-9]{1,7}_[a-z][a-z0-9]*(?:_[a-z0-9]+)*$
```

This expression is a **syntactic superset**; the registry refines it: an identifier is valid only when its namespace segment is a registered prefix (§22). *(Resolves finding 300-8: the pattern now encodes the 2–8 character namespace bound and the letter-initial first field segment previously stated only in prose.)*

CDS300-R006 A field registry MUST enforce identifier uniqueness case-insensitively.

CDS300-R007 Hyphens MAY appear inside controlled values and tags where CDS-400 permits, but SHOULD NOT be used as a field-name separator.

CDS300-R008 Names MUST NOT use positional codes such as field_1, misc_2 or data_a as permanent identifiers.

Identifier	Assessment
MF_material	Conformant
CH_google_product_category	Conformant
mf_material	Not the visible profile (see §8 lossy targets)
MF.Material, MF:material, MF=material, MF material	Non-conformant
CH_Google_Product_Category	Non-conformant

4. Core Namespace Registry (normative)

This table is the normative CDS prefix registry. It resolves every registry divergence catalogued in the v0.2 review (Matrix 1): the STD_/MF_ boundary (ADR-D2), DF_→CH_ (ADR-D24), the QA_ grammar (D7), PRC_/INV_ membership, AI_ adoption, IMPORT_ and OBS_ registration, and reservation of the full core set.

Prefix	Status	Semantic domain	Example
STD_	Core	Core product field set, defined by the closed enumeration in §5. Platform-neutral: whether a channel stores an STD_ field natively is a profile mapping concern (CDS-900), never part of the definition.	STD_title
CAT_	Core	Internal classification and taxonomy assignments (not external taxonomy mappings — those are CH_).	CAT_product_type
VAR_	Core	Variant and sellable-unit fields.	VAR_size
MF_	Core	Extended structured product attributes in the canonical model, mechanism-neutral (§10, ADR-D2).	MF_material
CH_	Core	Channel-specific mapping, output, override and feed projection (§9, ADR-D24).	CH_google_product_category
SEO_	Core	Metadata whose primary purpose is search presentation, indexing or discoverability. Caution: legacy fields named SEO (DF_Airtable_SEO_*) were data-feed fields, not search metadata (REVIEW-002A LEG-3); they migrate to CH_, never to SEO_.	SEO_title
MED_	Core	Media assets and media-specific metadata.	MED_primary_image
PRC_	Core (conditional)	Pricing and commercial amounts. Registered for use only where the PIM is the declared authority or governed integration surface for pricing (CDS-200 authority model).	PRC_retail_price
INV_	Core (conditional)	Inventory and availability. Same authority condition as PRC_.	INV_available_quantity
WF_	Core	Workflow and operational state.	WF_copy_review
QA_	Core	Field-level channel verification results only, grammar QA_<channel>_<field_name> (§13).	QA_shopify_material
SYS_	Core	System identifiers, technical metadata and computed quality metrics (§12).	SYS_updated_at
AI_	Core	AI proposals, provenance, confidence and workflow metadata (§15; rules in CDS-700).	AI_material_suggestion
IMPORT_	Core	Ingested source records and intake provenance (§16).	IMPORT_supplier_colour
OBS_	Core	Observed channel state captured by read-back (§16; observation model in CDS-500).	OBS_shopify_material
CDS_	Reserved	Standard metadata and conformance artifacts.	CDS_version
TMP_	Reserved	Temporary migration data; prohibited in published schemas.	—
LEG_	Reserved	Explicit legacy-compatibility fields.	LEG_df_google_category
EXT_	Reserved	Generic extension metadata; not a substitute for a meaningful namespace.	—
CMP_	Recommended extension	Compliance and regulatory data.	CMP_country_of_origin

Prefix	Status	Semantic domain	Example
SUP_	Recommended extension	Supplier and sourcing data.	SUP_supplier_sku
DICT_	Recommended extension	Dictionary and reference-data entities. Dictionary value identity itself is governed by CDS-400 (<code>value_id</code>); DICT_ names dictionary <i>entities</i> where an implementation surfaces them as fields.	DICT_colour_canonical
DF_	Deprecated (legacy alias)	Legacy data-feed layer prefix; replaced by CH_ for all new design (ADR-D24). Registered so §22 validation accepts sanctioned migration aliases.	DF_google_product_category → CH_google_product_category
Match_	Deprecated (legacy alias)	Legacy verification traffic lights; replaced by QA_ verification fields backed by CDS-500 verification records.	Match_SEO_title → QA_google_title
Airtable_ (incl. MF_Airtable_*, DF_Airtable_SEO_*, Import_*, Shopify_*)	Deprecated (legacy alias)	Legacy Airtable PIM field families; alias targets per §21.	MF_Airtable_material → MF_material

RESOLVED — accepted as drafted (owner 2026-08-04; was D6 (resolved)): OBS_ is registered here as the observed-channel-state namespace on the strength of the legacy `Shopify_*` mirror-column precedent (REVIEW-002A); prior drafts used "OBS" without defining it. Alternative: leave observed state entirely to CDS-500 record structures with no visible prefix.

CDS300-R009 Every visible field identifier MUST belong to a prefix registered in this registry or in the implementation's extension registry (§20). (*Upholds CDS-P-09.*)

CDS300-R010 An implementation MUST maintain a namespace registry identifying, for every visible prefix: meaning, owner, allowed scope, publication behaviour, lifecycle status (active, deprecated, reserved) and — for deprecated prefixes — replacement and deprecation date.

CDS300-R011 A field MUST use the namespace that describes its semantic responsibility, not its current storage location or delivery mechanism.

CDS300-R012 WF_, QA_, SYS_, AI_, IMPORT_ and OBS_ fields MUST NOT be published to customer-facing channels unless a channel profile explicitly requires them.

5. The STD_ Closed Enumeration (*normative*)

Per ADR-D2, STD_ is defined by a **closed enumeration** published in this registry — never by a judgement about "core-ness". Anything structured and not in the enumeration is MF_.

CDS300-R013 The STD_ namespace MUST contain exactly the fields enumerated in the STD_ registry table. Adding or removing an STD_ field is a registry change under CDS-800 change control.

Initial registry content at v0.2 (the starter enumeration; final list fixed at v0.2 freeze):

Identifier	Meaning
STD_id	Product identity: the primary commercial identifier (product-level SKU or equivalent; variant SKUs are <code>VAR_sku</code>)
STD_title	Product title
STD_description	Product description
STD_brand	Brand
STD_vendor	Vendor / supplier of record
STD_primary_category_ref	Reference to the primary internal category (assignment detail lives in CAT_)
STD_media_set	Reference to the product's governed media set (media items live under MED_)
STD_status	Lifecycle status of the product record

CDS300-R014 A structured product attribute not present in the STD_ enumeration MUST use MF_ (or a more specific registered namespace whose domain it belongs to).

The v0.1 namespace-selection decision tree is deleted (ADR-D2): it was undecidable (`MF_material` matched both "canonical core fact" and "published as metafield"). Namespace assignment is a registry lookup, never a judgement call — which is also how the boundary was successfully maintained in production.

6. Field Name Construction (*normative*)

CDS300-R015 The field-name portion SHOULD state the product fact or operational purpose in plain, specific language, using terminology defined by CDS-100/CDS-400 or by the applicable channel specification.

CDS300-R016 Acronyms such as SKU, GTIN, MPN, URL and SEO MAY be retained where widely recognised; other unexplained abbreviations SHOULD NOT be used.

Preferred	Avoid	Reason
MF_material	MF_mat	Unexplained abbreviation.
CH_google_product_category	CH_google_cat	Preserve the actual channel concept.
QA_shopify_material	QA_mat_status	Identify the verification target.

Preferred	Avoid	Reason
WF_photography_required	WF_photo	State the workflow condition.
SYS_last_verified_at	SYS_lva	Readable outside the originating system.

7. Case and Separators — see §3 (*normative*)

Case, separator and character-set rules are part of the identifier grammar (CDS300-R002–R008). This casing rule is stated normatively only here; CDS-000 carries it as principle CDS-P-09/P-10 without restating it. (*Resolves finding 300-9.*)

8. Semantic Identifier, Human Label and Machine Key (*normative*)

CDS distinguishes three things (definitions: CDS-100 §4): the **Semantic Identifier** (visible, namespace-prefixed), the **Human Label** (natural language) and the **Machine Key** (internal storage/transport key). They MAY differ, provided their relationship is explicit.

```
Semantic identifier: MF_care_instructions
Human label:       Care instructions
Machine key:       product.attributes.care_instructions
Channel mapping:   custom.care_instructions      (profile concern, CDS-900)
```

CDS300-R017 Every machine key exposed to users MUST have a stable semantic identifier or an equally self-describing equivalent.

CDS300-R018 A human label MAY contain spaces and title case but MUST NOT replace the semantic identifier in exports or integrations.

CDS300-R019 The mapping between semantic identifier, machine key and channel destination MUST be discoverable from the Attribute Definition or field registry.

CDS300-R020 Exports and integrations SHOULD preserve identifier case. Where a target is lossy (e.g. a platform that case-folds keys), an export MAY apply case-folding **only where the registry or Attribute Definition declares a reversible mapping** between the semantic identifier and the folded form; the folded form then round-trips deterministically. (*Resolves finding 300-6; upholds CDS-P-09.*)

9. Channel Namespace (CH_) (*normative*)

CH_ identifies information whose semantic responsibility belongs to a specific downstream channel: taxonomy mappings, required outputs, custom labels, channel-only overrides, publication configuration and feed-column projections.

```
CH_shopify_product_category
CH_shopify_title_override
CH_google_product_category
CH_google_custom_label_0
CH_meta_product_category
CH_amazon_browse_node
```

CDS300-R021 The first field-name segment after CH_ MUST identify the channel using its registered lowercase channel key.

CDS300-R022 Channel keys MUST remain stable even when the technical transport changes (file feed, API or otherwise).

CDS300-R023 CH_ MUST NOT be used for a canonical attribute merely because that attribute is sent to a channel; a canonical fact used by many channels remains canonical and SHOULD NOT be duplicated under multiple CH_ fields unless distinct channel representations are required.

CDS300-R024 A field is a feed-layer projection if and only if it is a literal serialized feed column; such projections live under CH_, and the feed artifact is verified end-to-end at the formatted-output layer (verification model: CDS-500). (*ADR-D24, replacing the untestable "retain DF_ where useful".*)

CH_ replaces the legacy DF_ ("Data Feed") prefix for all new design (ADR-D24): "channel" names the durable business destination; "data feed" named one delivery implementation. DF_ remains a registered deprecated prefix for migration aliases (§21).

10. Extended Attribute Namespace (MF_) (*normative*)

MF_ identifies **extended structured product attributes in the canonical model**. The definition is mechanism-neutral: it says nothing about how any platform stores or publishes the attribute.

```
MF_material
MF_pattern
MF_fit
MF_care_instructions
MF_colour_display
MF_colour_family
```

Per ADR-D2 this is an explicit **redefinition, not an inheritance**: in the legacy system MF_ literally meant a platform metafield. The prefix letters are retained for operator familiarity and legacy-mapping continuity; the meaning is now defined solely by this registry. Typical projections of MF_ attributes to metafield-like channel structures are documented informatively in the relevant CDS-900 profile.

CDS300-Ro25 MF_ fields MUST describe product information — never integration status, workflow state or system metadata.

CDS300-Ro26 An MF_ field SHOULD be backed by an Attribute Definition (CDS-200 §7) declaring type, cardinality, dictionary binding, display, facet and channel mappings.

CDS300-Ro27 Where a channel publishes an MF_ attribute as a native or standard field, that is the channel profile's mapping concern; the canonical prefix MUST NOT change.

11. Classification and Variant Namespaces (CAT_, VAR_) (normative)

CAT_product_family	VAR_sku
CAT_product_type	VAR_size
CAT_season	VAR_colour
	VAR_barcode

CDS300-Ro28 CAT_ MUST be used for internal classification and taxonomy assignments, not external taxonomy mappings (those are CH_).

CDS300-Ro29 VAR_ MUST be used for values specific to a sellable unit or variant option (variant boundary rule: CDS-200 §5).

12. Workflow and System Namespaces (WF_, SYS_) (normative)

WF_copy_review	SYS_product_id
WF_photography_status	SYS_created_at
WF_publish_approval	SYS_last_published_at
	SYS_completeness_score

CDS300-Ro30 WF_ fields MUST describe work state, responsibility or approval.

CDS300-Ro31 SYS_ fields MUST represent technical identity, timestamps, versions, process metadata or **computed quality metrics** (e.g. SYS_completeness_score), and SHOULD remain hidden from customer-facing channels.

RESOLVED — accepted as drafted (owner 2026-08-04; was D7 (resolved)): Quality metrics (formerly QA_completeness_score) are homed under SYS_ so that QA_ can carry a single channel-verification grammar. Alternative considered: a formal two-shape QA_ grammar, rejected as re-introducing the parse ambiguity of finding 300-2.

13. Verification Namespace (QA_) (normative)

QA_ is reserved exclusively for field-level channel verification results. It has exactly one grammar:

QA_<channel>_<field_name>
QA_shopify_material
QA_google_gtin
QA_meta_availability

CDS300-Ro32 A QA_ identifier MUST use the form QA_<channel>_<field_name>, where <channel> is a registered channel key. No other QA_ shape is valid. (Resolves finding 300-2: the vo.1 registry's own example QA_completeness_score would have failed the vo.1 §21 validation; it is now SYS_completeness_score.)

CDS300-Ro33 The associated verification record MUST retain expected value, observed value, detailed status, comparison strategy and timestamps even where a UI shows only a traffic-light symbol. Status vocabulary and traffic-light mapping are defined solely in CDS-500.

CDS300-Ro34 Legacy Match_ prefixes MAY be accepted as deprecated aliases (§21) but MUST NOT be used in new design.

14. Search, Media, Pricing and Inventory Namespaces (normative)

SEO_title	MED_primary_image	PRC_retail_price	INV_available_quantity
SEO_description	MED_alt_text	PRC_compare_at_price	INV_backorder_allowed
SEO_search_synonyms	MED_sort_order	PRC_cost_price	INV_expected_restock_at

CDS300-Ro35 SEO_ MUST be used only for metadata whose primary purpose is search presentation, indexing or discoverability. Legacy fields containing the token "SEO" MUST be classified by actual responsibility before migration — production DF_Airtable_SEO_* fields were feed fields and migrate to CH_ (REVIEW-002A LEG-3).

CDS300-Ro36 MED_ MUST be used for media assets and media-specific metadata.

CDS300-Ro37 PRC_ and INV_ MUST be used only when the PIM is the declared authority or governed integration surface for those domains (CDS-200 authority model). Where it is not, price and inventory facts observed from their mastering systems use OBS_ or remain outside the visible registry.

15. AI Namespace (AI_) (normative)

AI_ is formally registered as a core namespace for AI proposals, provenance, confidence and workflow metadata. This satisfies the registration requested by CDS-700 §22, resolves finding 300-5, and follows the legacy Automated_ precedent (machine-generated content state was a production concern, not a novelty).

AI_material_suggestion	AI_review_status	AI_generated_at
AI_material_confidence	AI_model_id	AI_reviewed_by
AI_material_evidence	AI_workflow_version	

CDS300-R038 AI_ fields MUST represent proposals, provenance, confidence or workflow metadata; they MUST NOT replace the canonical field namespace. On acceptance, the value is stored in its semantic canonical field (e.g. MF_material) with AI provenance retained per CDS-700.

All AI governance rules — proposal lifecycle, evidence, confidence, review, autonomy — are defined solely in CDS-700.

16. Ingestion and Observation Namespaces (IMPORT_, OBS_) (normative)

IMPORT_ names ingested source records and intake provenance — the raw layer before normalisation (value layers: CDS-400). It formalises the load-bearing legacy Import_* production namespace (REVIEW-002A) that earlier drafts used (CDS-900 "Layer 1: IMPORT_") without ever registering.

OBS_ names observed channel state captured by read-back — the legacy Shopify_* mirror-column pattern, generalised across channels. Observation semantics, coverage states and the never-overwrite rule are defined solely in CDS-500.

IMPORT_supplier_colour = "French Navy"
OBS_shopify_material = "linen"

CDS300-R039 IMPORT_ fields MUST retain source provenance and MUST NOT be treated as canonical values; promotion to canonical goes through the value-layer model (CDS-400).

CDS300-R040 OBS_ fields MUST hold observed downstream state only and MUST NOT silently overwrite canonical values (CDS-500; CDS-P-02).

17. Dictionary-Backed Field Naming (normative)

Dictionary-backed fields distinguish canonical, display and facet purposes; dictionary value identity (value_id) and layer semantics are governed by CDS-400.

MF_colour_canonical
MF_colour_display
MF_colour_family

CDS300-R041 Where separate fields are materially required, the suffix MUST identify display, canonical or facet purpose explicitly.

CDS300-R042 Implementations SHOULD NOT persist redundant per-product representations that the dictionary can derive deterministically.

18. Localisation (normative)

Localisation is modelled as an **Attribute-Definition dimension** — a locale axis on the attribute's value, declared in the Attribute Definition and represented per the localised-text contract in CDS-1100. It is not encoded in the identifier, and **LOC_ is not a registered prefix**: identifier-encoded locales (LOC_fr_title) scale combinatorially (fields × locales) and had no grammar or locale registry.

CDS300-R043 Locale MUST NOT be encoded in the visible field identifier; localised values are dimensions of one identifier, declared in the Attribute Definition.

RESOLVED — accepted as drafted (owner 2026-08-04; was D8 (resolved)): Localisation as an Attribute-Definition dimension (LOC_ unregistered) is the call made here, resolving finding 300-7. Alternative: identifier-encoded locales with a formal grammar and locale registry.
--

19. Scope, Inheritance and Cardinality (normative)

Field names do not encode every schema property; scope, inheritance and cardinality belong in the Attribute Definition (CDS-200 §7).

MF_material	VAR_size
scope: product	scope: variant
cardinality: multiple	cardinality: single
inheritance: family	inheritance: none

CDS300-R044 The identifier MUST NOT be overloaded with type, requiredness or cardinality suffixes when those properties are represented in the schema; suffixes such as _list, _required or _text SHOULD be avoided unless they carry enduring business meaning.

20. Extension Namespaces and Reservation (normative)

Organisations MAY define additional namespaces for real semantic domains not covered by the core registry.

CDS300-R045 An extension namespace MUST be registered before use, and its registry entry MUST define purpose, owner, examples, publication behaviour and deprecation policy.

CDS300-R046 An extension prefix MUST satisfy the grammar of §3 and MUST NOT collide, case-insensitively, with **any** prefix in the §4 registry — core, conditional, reserved, recommended-extension or deprecated. (*Resolves finding 300-3: the entire core set is reserved, not only CDS_/TMP_/LEG_/EXT_.*)

CDS300-R047 Reserved prefixes MUST NOT be repurposed by an organisation.

CDS300-R048 Forward compatibility: future CDS releases MAY register new core prefixes. Organisation prefixes SHOULD therefore be organisation-distinctive (never generic domain words likely to be standardised). If a later CDS release registers a prefix that collides with an existing organisation extension, the CDS registration prevails and the organisation prefix MUST be migrated using the §21 pattern within the organisation's adoption of that release.

CDS300-R049 Organisation prefixes SHOULD NOT encode temporary project names or staff initials.

21. Deprecation, Aliases and Migration (*normative*)

Field names are integration contracts; silent renaming is prohibited.

```
Legacy:      DF_Airtable_SEO_Product Category
Interim alias: DF_google_product_category      (registry status: deprecated)
Replacement: CH_google_product_category
Migration:    copy -> verify -> switch consumers -> retire
```

CDS300-R050 A deprecated identifier MUST identify its replacement and deprecation date in the registry.

CDS300-R051 Aliases MAY be accepted on import but MUST resolve to exactly one canonical identifier; synonymous fields MUST be consolidated or explicitly linked by alias and deprecation metadata.

CDS300-R052 Every sanctioned alias prefix MUST carry a registry entry with status `deprecated`, so that §22 validation accepts it during migration. (*Resolves finding 300-4, where the v0.1 chapter's own sanctioned alias failed its own validation; per ADR-D24.*)

CDS300-R053 Renaming MUST include impact analysis covering formulas, imports, exports, channel mappings, automation and verification, and a migration SHOULD preserve old and new fields through a verified transition period (copy → verify → switch → retire) wherever data loss is possible.

Legacy alias directions (full legacy Airtable alignment annex: CDS-1300):

Legacy pattern	CDS destination
Import_*	IMPORT_ (source provenance retained; never canonical)
Airtable_*	STD_ / CAT_ / VAR_ / PRC_ / INV_ by semantic responsibility
MF_Airtable_*	MF_ canonical attributes (platform-name segment removed)
MF_Shopify_*, Shopify_*	OBS_ observed publication state; visible verification via QA_
DF_Airtable_SEO_*, DF_*	CH_ channel fields (never SEO_)
Match_*, MF_Match_*	QA_ backed by full CDS-500 verification records
collection_* tags	Governed generated channel projections (CDS-400/CDS-500 tag rules)

22. Parsing and Validation (*normative*)

A CDS validator parses namespace, channel segment and semantic field name deterministically:

```
Input: CH_google_product_category
  namespace = CH
  channel   = google
  field     = product_category
```

CDS300-R054 Validation MUST reject an identifier whose namespace has no registry entry (core, extension or deprecated). Deprecated-prefix identifiers MUST be accepted only as aliases per §21 and SHOULD raise a deprecation warning.

CDS300-R055 Validation MUST reject identifiers failing the §3 grammar, including empty segments, duplicate underscores, leading or trailing underscores and non-ASCII punctuation.

CDS300-R056 Validation MUST verify that the channel key of a CH_, QA_ or OBS_ identifier exists in the channel registry.

CDS300-R057 Validation SHOULD warn on unexplained abbreviations and excessively long identifiers.

CDS300-R058 Identifiers MUST be validated before schema or import changes are accepted, and AI-generated field proposals MUST pass the same registry and naming validation as human proposals.

23. AI Interpretation of Identifiers (*normative*)

Semantic prefixes improve AI understanding; they do not replace schema metadata (CDS-P-05). AI governance — proposal lifecycle, evidence, confidence, autonomy, review — is defined solely in CDS-700; the rules below are naming-specific.

CDS300-R059 AI systems SHOULD receive the semantic identifier together with its human label, description, data type, dictionary binding and examples — never the identifier alone.

CDS300-R060 An AI system MUST NOT infer schema properties from a prefix (e.g. that every MF_ field is free text, that every CH_ field is canonical, or that any prefix implies a storage mechanism).

```
identifier: MF_material
label: Material
description: Canonical materials present in the product
type: list<dictionary_reference>
dictionary: material_canonical
facet_mapping: material_family
channel_mappings: shopify, google, meta
```

24. Worked Examples (*informative*)

Apparel product:

```
STD_title = Linen Relaxed Shirt
STD_brand = Example Brand
CAT_product_family = apparel
CAT_product_type = shirt
STD_primary_category_ref = women/tops/shirts
MF_material = linen
MF_fit = relaxed
MF_colour_display = French Navy
MF_colour_family = blue
VAR_size = m
VAR_colour = french_navy
CH_shopify_product_category = apparel_accessories_clothing_shirts_tops
CH_google_product_category = 212
SEO_title = Linen Relaxed Shirt | Example Brand
QA_shopify_material = MATCH
```

Publication and read-back (statuses per CDS-500):

```
Canonical:      MF_material = linen
Expected projection:  Linen
Observed:      OBS_shopify_material = linen
Comparison strategy: case_insensitive_dictionary_id
Result:        QA_shopify_material = MATCH
```

Platform-specific naming behaviour (metafield projections, taxonomy mapping, tag generation) is documented in the CDS-900 profiles.

25. Architecture Decisions (*informative*)

Decisions live in the global ADR register (CDS000-R006). Affecting this chapter: **ADR-D2** (MF_ retained and redefined mechanism-neutral; STD_/MF_ boundary by closed enumeration; supersedes CDS-ADR-008), **ADR-D24** (DF_→CH_, one consolidated ADR superseding the four parallel vo.1 ADRs; DF_ registered deprecated). Upheld from vo.1: uppercase prefixes with underscore separator (ADR-007); identifier/label/key separation (ADR-010); rejection of colon, equals and dot separators (ADR-011); schema metadata excluded from identifiers (ADR-012).

Commerce Data Standard (CDS)

CDS-400 — Controlled Dictionaries and Reference Data

Field	Value
Status	vo.2 Review Draft (working source; not an approved standard)
Release	CDS vo.2
Date	2026-08-04
Supersedes	CDS-400 Working Draft vo.1
Normative status	§1, §3–§18, §20–§28 and §30 are normative. §2, §19 (baseline note), §29, §31 and all appendices are informative except where individually marked.
Findings addressed	CDS-400-1..10; DICT-1, DICT-2, DICT-3, DICT-5, DICT-9; ADR-D1, ADR-D4; open decisions D28, D29

CDS-400 is the authoritative home for dictionaries and the value layers (CDS000-R005). Terminology is defined in CDS-100 §3; verification statuses and reason-code use live in CDS-500; facet presentation and search behaviour live in CDS-600; AI controls live in CDS-700; conformance levels, tests and claims live in CDS-1000.

1. Purpose (*normative*)

CDS-400 defines how commerce systems govern repeated values such as colours, materials, styles, rooms, finishes, patterns, occasions, conditions, units and channel vocabularies. It separates detailed product truth from the simplified values presented in customer filters, and defines how one canonical value is projected into display labels, facet values, search synonyms and channel-specific outputs without duplicating product data.

CDS400-R001 A CDS implementation **MUST** treat governed values as first-class reference-data entities rather than uncontrolled text wherever consistent reuse, filtering, search, validation or channel mapping is required.

CDS400-R002 The PIM **MUST** remain the authority for canonical dictionary values and their mappings.

CDS400-R003 A customer-facing filter **MUST NOT** be created by exposing every distinct incoming or display value.

2. Problem Statement (*informative*)

Commerce data arrives with rich but inconsistent language. A supplier may describe blue products as Navy, French Navy, Ocean, Denim, Royal, Sky, Petrol or Steel Blue. Publishing those terms directly as filter values produces an unusable panel of near-duplicates. Discarding them and storing only "Blue" destroys product detail and merchandising language.

Source terms: French Navy | Navy | Ocean Blue | Royal Blue | Denim | Sky Blue
Bad storefront facet: the same six near-duplicates exposed to customers
Bad canonical model: all source detail replaced with "Blue"
CDS model: preserve detail -> normalise meaning -> expose a usable facet -> project to channels

The correct response is not less information; it is a layered model that gives each consumer the appropriate representation.

3. Dictionary Principles (*normative*)

Principle	Meaning
Preserve source fidelity	Keep the original value and its provenance even when it maps to a canonical value.
Stable identity	Values use stable identifiers independent of labels, spelling and localisation.
One canonical meaning	Each governed concept has one approved internal semantic representation.
Separate customer facets	Filter groupings are designed for usability, never copied from canonical detail.
Derive channel output	Channel values are generated from dictionary mappings wherever possible.
Govern change	Aliases, deprecations, replacements and mapping changes are versioned and reviewable.
Context matters	The same word can mean different things in different dictionaries or attributes.
Humans, machines and AI	The model is readable by operators, deterministic for software and explicit for AI.

CDS400-R004 A dictionary **MUST** have a declared semantic scope, a named owner and a declared value identity scheme.

CDS400-R005 Labels **MUST NOT** be used as the only permanent identifier for governed values.

4. Controlled Value Lifecycle (*normative — per ADR-D4*)

CDS separates intake, internal meaning and projection. Projections branch in parallel from the canonical value (branch model, CDS-100 §7); the layer that feeds each channel is declared per attribute in the Attribute Definition (CDS-200 §7), never assumed.



Supersession note. The v0.1 term "Reference Value" is retired (ADR-D4; CDS-100 §8). The raw input is a **Source Value**; the governed mapping is an **Alias Mapping Record**; the governed dictionary entry is a **Canonical Value**; everything downstream of canonical is a **projection**.

CDS400-R006 The original source value **MUST** remain available, verbatim with provenance, for audit and reprocessing.

CDS400-R007 An Alias Mapping Record **MUST** resolve to exactly one canonical value within its declared context, unless the mapping is explicitly marked ambiguous and routed for review.

CDS400-R008 Facet values, display labels, search synonyms and channel representations **MUST** be derived from or linked to canonical values rather than maintained as unrelated free text.

Informative: not every implementation exposes every layer as a visible column; the logical distinctions must still exist.

5. Dictionary Entity Model (*normative*)

A Dictionary is the governed container. A Canonical Value is a stable semantic entity. Aliases, facet mappings and channel mappings attach to the value, never repeated on every product. Canonical product data stores the canonical `value_id`; a variant's colour field holds the canonical value, never a raw source spelling (ADR-D4).

Property	Purpose
dictionary_id	Stable identifier for the dictionary.
dictionary_key	Human-readable machine key such as <code>colour</code> or <code>material</code> ; also the context-scoping key for shared codes (§20).
value_id	Stable identifier for one canonical value.
canonical_code	Portable code such as <code>navy</code> or <code>organic_cotton</code> .
canonical_label	Default human-readable label.
status	Value Lifecycle Status (§17.1).
version	Version of this value record; incremented on any governed change (§24).
aliases	Alias Mapping Records: recognised source terms, spellings and abbreviations, each with scope, status, provenance and locale.
display_labels	Optional labels by locale, brand or product (§9).
facet_mappings	Ordered facet memberships; the single authoritative facet mechanism (§10).
search_terms	Synonyms and related query terms (behaviour: CDS-600 §21).
channel_mappings	Values or identifiers required by channels, with mapping_version.
parent_id	Optional broader-value relationship — informative analytics only (§13).
related_values	Optional typed relationships (synonym, similar, substitute, compatible, contrast).
metadata	Swatch, unit, sort order, notes or other dictionary-specific data.
provenance	Who or what introduced and approved the value or mapping.
effective_dates	When a value or mapping becomes valid or invalid.
replacement_id	Successor value when deprecated.

CDS400-R009 Every canonical value **MUST** have a stable `value_id` or equivalent immutable identity.

CDS400-R010 A canonical code **SHOULD** use lowercase ASCII `snake_case` and **MUST** remain stable when a display label changes.

6. Value Identity and Labels (*normative*)

Identity and presentation are separate. A value called Navy in English may be displayed differently by locale, brand or campaign while its identity is unchanged.

value_id:	colour_00017
canonical_code:	navy
canonical_label:	Navy
locale_label_en_AU:	Navy
locale_label_fr_FR:	Bleu marine
facet_mappings:	[blue]
CH_google_colour:	Blue

CDS400-R011 Changing a label **MUST NOT** create a new value identity when the underlying meaning is unchanged.

CDS400-R012 Merging or splitting meanings **MUST** be performed through an explicit governed migration (§24), never a silent label edit.

CDS400-R013 Display labels **MAY** contain spaces, punctuation and brand language; canonical codes **SHOULD** remain portable and stable.

7. Source Values and Alias Mapping Records *(normative)*

A Source Value is the exact content received from a supplier, manufacturer, import, manual entry or extraction process. An Alias Mapping Record recognises that source term and links it to exactly one canonical value within a defined context.

Source value:	"French Navy"		
Source system:	Supplier A		
Source attribute:	Colour		
Normalised match:	french navy		
Alias mapping:	alias_colour_french_navy	(row_type: alias)	
Canonical value:	navy	(row_type: canonical)	

CDS400-R014 Source values MUST be retained exactly as received when traceability is required.

CDS400-R015 An Alias Mapping Record MUST identify the dictionary (via `dictionary_key`) and context in which it is valid.

CDS400-R016 Supplier-specific mappings SHOULD be supported where the same term has different meanings between suppliers.

Informative: alias mappings make supplier onboarding progressively easier — once a term is governed, future imports resolve automatically. Every distinct source value that has been mapped is retained as an alias with provenance (ADR-D4).

8. Canonical Values *(normative)*

A Canonical Value is the approved internal representation used for product truth, rules, analytics and derivation. It expresses meaning at the level of detail the business needs, not merely the broadest customer filter. Both granularities are conformant: an organisation may govern `french_navy` as a canonical shade with facet `blue`, or govern only `navy` and keep "French Navy" as a display label (ADR-D4).

CDS400-R017 Canonical values MUST be semantically distinct within their dictionary.

CDS400-R018 Canonical values SHOULD preserve commercially useful detail that would be lost in a broader facet value.

CDS400-R019 A canonical value MUST NOT be created solely because one supplier uses a unique spelling.

CDS400-R020 A canonical value SHOULD be reused across products and suppliers whenever the meaning is equivalent.

Source values	Canonical value
French Navy, Marine, Deep Navy	-> navy
Sky, Pale Blue, Baby Blue, Light Blue	-> sky_blue
Royal, Cobalt	-> royal_blue
Ocean Blue	-> ocean_blue (only if the business needs the distinction)

Note: the canonical code for the pale-blue shade is `sky_blue`, matching the shipped reference dictionaries and CDS-1500; the vo.1 code `light_blue` is recorded as an alias of `sky_blue` (resolves DICT-3).

9. Display Labels *(normative)*

A Display Label is the wording shown to a customer or operator. It may preserve a supplier or brand name, use a merchandising phrase or provide a localised label. Display labels never define canonical identity.

CDS400-R021 A product MAY use a display label that differs from the value's canonical label.

CDS400-R022 A display label MUST remain linked to a canonical value when the field participates in governed filtering, search or channel publication.

CDS400-R023 A display label SHOULD NOT be used as the only search or facet value.

CDS400-R024 Label precedence MUST be resolved in this order: (1) a product-level display label, valid only when recorded as a governed Display Label link to the canonical value; (2) the value's locale display label for the viewer's locale; (3) the value's canonical label. A product-level free-text label with no Display Label link MUST NOT override the value's labels in governed contexts.

Canonical colour:	navy
Product display:	French Navy (governed Display Label link)
Facet display:	Blue
Search terms:	navy, dark blue, french navy
Google output:	Blue

10. Facet Values and Facet Mappings *(normative)*

A Facet Value is a customer-facing grouping used for filtering and refinement. Facet design is a user-experience activity, not a database distinct-value query. Facet presentation — labels, storefront order, selection logic, counts — is defined in CDS-600; CDS-400 owns the mapping from canonical values to facet values.

CDS400-R025 The Facet Dictionary, referenced through the `facet_mappings` property of canonical values, is the single authoritative mechanism for facet membership. Facet membership MUST NOT be derived from the parent hierarchy (§13), from labels or from any other source. (*Resolves CDS-400-2.*)

CDS400-R026 Facet values MUST be explicitly governed in a Facet Dictionary or equivalent configuration.

CDS400-R027 A facet dictionary SHOULD minimise unnecessary near-duplicate choices while preserving distinctions that materially affect purchase intent. (*Downgraded from MUST: "minimise" is not mechanically testable; the testable mechanism is CDS400-R028.*)

CDS400-R028 A new or materially changed facet dictionary MUST have a recorded usability review, covering the §10.1 checklist, before activation.

CDS400-R029 `facet_mappings` is an ordered list. The first facet value is the primary membership; a canonical value MUST have exactly one primary facet value per facet dimension and MAY have additional secondary memberships where justified.

RESOLVED — accepted as drafted (owner 2026-08-04; was D28 (resolved)): Call made — multi-facet memberships are ordered, first entry is primary. Alternative — unordered set with an explicit `is_primary` flag. The teal exemplar is aligned to the shipped reference data (`[green, blue]`, green-dominant per CDS-1500); the v0.1 Blue-primary example is corrected accordingly.

CDS400-R030 A multi-colour or multi-material product MAY participate in multiple facet values; the storefront MUST treat the product as one result, never a duplicate. Multicolour is a facet or status value, never a canonical shade (CDS-100 §3).

Informative: for colour, Blue is normally a facet; French Navy is a display label or canonical shade. For material, Wood may be a facet while Tasmanian Oak is the canonical material or display label.

10.1 Facet Usability Review (*normative checklist for CDS400-R028*)

The recorded review MUST cover:

- number of values and expected product counts;
- overlap and ambiguity between values;
- labels visible above and below the fold on common devices;
- whether customers search for the broad group or the detailed value;
- whether a value is better represented by search, a swatch, a range or a separate facet;
- behaviour when multiple facet values are selected;
- zero-result and low-count values; and
- localisation and sort order.

11. Channel Representations (*normative*)

A Channel Representation is the value, identifier or label required by a specific downstream channel. It is stored once on the dictionary value or mapping definition and derived for every applicable product. Which layer feeds each channel (canonical, display or facet) is declared per attribute in the Attribute Definition (ADR-D4; CDS-200 §7).

```
Canonical value:      navy
CH_shopify_filter_colour: Blue
CH_google_colour:    Blue
CH_meta_colour:      Blue
Product display label: French Navy
```

CDS400-R031 Channel representations SHOULD be defined at dictionary level rather than repeated on every product.

CDS400-R032 A product-level channel override MAY be used only when the product genuinely requires a different output and the reason is recorded (override semantics: CDS-500 §22).

CDS400-R033 A missing required channel mapping MUST produce a validation or publication error, never silent free-text substitution.

12. Search Terms (*normative*)

Search synonyms attach to canonical values as a projection and extend discovery without expanding the visible facet list. Search behaviour, directionality and query handling are defined in CDS-600 §21.

CDS400-R034 Search synonyms MUST NOT change canonical identity, and a synonym that is ambiguous across dictionaries MUST be scoped to the applicable dictionary or attribute.

13. Hierarchies and Relationships (*normative; parent hierarchy informative*)

A dictionary MAY record a parent hierarchy (e.g. `navy` under `blue`, `oak` under `wood`). In v0.2 the parent hierarchy is an **optional, informative structure for analytics, reporting and operator navigation only**. It carries no authority over facets, channel output or validation; the Facet Dictionary is the only facet-mapping mechanism (CDS400-R025). (*Resolves CDS-400-2.*)

CDS400-R035 Hierarchy and related-value relationships MUST be explicit records and MUST NOT be inferred solely from labels.

CDS400-R036 A value MUST NOT have circular parent relationships.

CDS400-R037 A parent relationship MUST NOT erase, replace or auto-merge the child value.

CDS400-R038 Related-value links SHOULD state their relationship type: synonym, similar, substitute, compatible or contrast.

14. Multi-value and Composite Attributes (*normative*)

A product may have multiple values, and some facts are composite rather than simple lists.

Simple multi-value:	materials = [cotton, linen]
Structured composition:	composition = [{material: cotton, percentage: 80}, {material: polyester, percentage: 20}]
Facet projection:	material facets = [cotton, synthetic]

CDS400-R039 A composition MUST NOT be reduced to an unstructured material label when percentages or component roles are required.

CDS400-R040 Multi-value order MUST be declared significant or insignificant by the Attribute Definition.

CDS400-R041 Facet membership MAY be derived from each component while the structured composition remains canonical.

15. Normalisation and Matching (*normative*)

Normalisation prepares source terms for matching while preserving the original value. The default text pipeline:

1. Preserve original source text
2. Unicode NFC normalisation
3. Trim leading and trailing whitespace
4. Collapse repeated internal whitespace
5. Apply case-insensitive comparison form
6. Apply dictionary-specific punctuation and spelling rules
7. Match exact governed aliases before fuzzy or AI-assisted suggestions
8. Record the rule and mapping used

CDS400-R042 Normalisation MUST NOT overwrite the original source value.

CDS400-R043 Exact governed mappings MUST take precedence over fuzzy or AI-assisted matching.

CDS400-R044 Numeric and measurement values MUST use type-aware comparison rather than text comparison.

CDS400-R045 A dictionary MAY define locale-specific or supplier-specific normalisation rules.

16. Ambiguity and Context (*normative*)

The same term may mean different things in different contexts. Natural may be a colour, a finish, a fibre claim or marketing copy. Large may be a size, a capacity or a package count.

CDS400-R046 Every mapping MUST be scoped at least by dictionary (`dictionary_key`) or attribute.

CDS400-R047 Ambiguous aliases MUST NOT resolve globally without context.

CDS400-R048 Where context is insufficient to resolve a source term, the term MUST enter review or remain unresolved; it MUST NOT be force-mapped.

"Natural" in MF_colour_display	-> colour: natural
"Natural" in MF_finish	-> finish: natural_oil / natural
"Natural" in MF_material_claim	-> no automatic mapping; review required

17. Unknown Values, Resolution and Quarantine (*normative*)

Unknown source values are expected in a living catalogue. They are data-governance events, never a reason to create uncontrolled canonical values automatically.

17.1 Two status enums (*normative — resolves CDS-400-4*)

vo.1 mixed two lifecycles in one vocabulary. vo.2 defines two named enums on two named entities. They share no tokens.

Value Lifecycle Status — attribute `value_lifecycle_status` on the **Canonical Value**:

Status	Meaning
proposed	Candidate canonical value awaiting governance review.
active	Approved for use in canonical data and projections.
deprecated	Still resolvable; identified replacement or declared no-replacement; scheduled for retirement.
retired	No longer assignable; retained for history and audit.

Status	Meaning
rejected	Reviewed and refused; never activated.

Source-Term Resolution Status — attribute `resolution_status` on the **Source Value intake record**:

Status	Meaning
resolved	Mapped to an active canonical value via an Alias Mapping Record.
promotion_proposed	Term proposed as a new canonical value; a linked <code>proposed</code> Canonical Value exists.
quarantined	Excluded from canonical data and projections until resolved (§17.2).
invalid	Not a legitimate value for any target attribute (junk, corruption, wrong field).
ignored	Legitimate content, intentionally not used for the target attribute.

CDS400-Ro49 Implementations **MUST** record the two statuses as separate enums on their respective entities and **MUST NOT** conflate them in schemas, UIs or exports.

CDS400-Ro50 Unknown source values **MUST** be surfaced to a review queue or explicit error process; they **MUST NOT** be silently dropped.

CDS400-Ro51 Bulk imports **MUST NOT** silently create new canonical values. Promotion of a source term to a canonical value is an explicit governed act, performed only by roles authorised under §25 — this applies equally to human, scripted and AI-initiated imports.

CDS400-Ro52 A promotion proposal **MUST** include source evidence, affected products and suggested mappings.

17.2 Publication consequence of quarantine (*normative — resolves CDS-400-5*)

CDS400-Ro53 While a source term for an attribute is `quarantined`, that attribute value **MUST** be omitted from every projection (display, facet, search, channel) for the affected products. The raw source value is never published as a stand-in.

CDS400-Ro54 Verification of an affected field **MUST** carry the reason code `CDS_VALUE_QUARANTINED` (reason-code registry: CDS-1100 §21; use in verification: CDS-500). The field is reported as attention-state, not silently skipped.

CDS400-Ro55 A quarantined value **MUST NOT** block publication of the product as a whole unless the affected attribute is declared requiredness-blocking for that channel in its Attribute Definition or channel profile; in that case the product publication fails validation per CDS400-Ro33.

Informative: quarantined items remain in quality denominators (CDS-1400 §5) — quarantine is visible pressure to resolve, never a way to improve a health score. A dictionary dashboard should show recurring unknown values so one governance action resolves many future imports.

18. Units and Measurements (*normative*)

Units are governed reference data. Measurements store a numeric value and a unit identifier, never units embedded in free text.

```
Measurement: value: 40, unit_id: centimetre
Display:      40 cm      Channel conversion: 15.75 in      Comparison base: 0.4 m
```

CDS400-Ro56 Measurements **MUST** separate numeric value from unit identity.

CDS400-Ro57 Unit conversions **MUST** use declared conversion rules and precision.

CDS400-Ro58 An implementation **MUST** distinguish similarly named units with different definitions, such as short ton and metric tonne.

CDS400-Ro59 Display units **MAY** differ by locale or channel while the canonical measurement remains stable.

19. Colour Dictionaries (*normative; baseline note informative*)

Colour requires multiple representations: product label, canonical shade, broad filter family, search language and channel vocabulary are not always the same.

Layer	Example
Canonical shade	navy
Display label	French Navy
Facet mappings (ordered)	[blue] — or [green, blue] for a green-dominant teal
Search terms	navy blue, dark blue, marine
Swatch metadata	Optional representative colour; not a guarantee of physical appearance
Channel mapping	Blue

CDS400-Ro60 A colour dictionary **SHOULD** distinguish canonical shade from broad colour family.

CDS400-Ro61 Hex, RGB or other digital swatch values **MAY** be stored but **MUST NOT** be treated as an exact representation of physical material colour unless measured under a defined process.

CDS400-Ro62 A product with a branded colour name **SHOULD** retain that name for display while participating in one or more governed colour facets.

CDS400-Ro63 Pattern and colour MUST remain separate attributes even when a supplier combines them in one phrase.

Informative — baseline ownership: the reference colour facet family list (including `natural` as a top-level facet and `multicolour` as a facet-only value) is owned by CDS-1500 Appendix C. CDS-400 appendices cite that baseline and use consistent exemplars (resolves CDS-400-3 / DICT-2).

20. Material and Composition Dictionaries (*normative*)

Material dictionaries support specific materials, broader families, blends and structured composition. A storefront may filter by Cotton while displaying 80% Cotton / 20% Polyester.

CDS400-Ro64 Specific material identity and material family MUST be separate where both are useful.

CDS400-Ro65 Composition percentages SHOULD be stored as structured values and SHOULD sum according to the organisation's declared tolerance rules.

CDS400-Ro66 Claims such as organic, recycled, vegan or certified MUST NOT be inferred solely from the material name unless governed evidence supports the claim.

CDS400-Ro67 There is one materials dictionary per organisation. Where the same canonical code needs different family or facet treatment in different commercial contexts (e.g. cotton in apparel vs homewares), the treatment MUST be expressed as context-scoped family and facet mappings keyed by `dictionary_key` scope (e.g. `material.apparel`, `material.homewares`) on the shared canonical value — never by forking the canonical meaning into duplicate values. (*Resolves DICT-5.*)

RESOLVED — accepted as drafted (owner 2026-08-04; was D29 (resolved)): Call made — one materials dictionary with context scoping via `dictionary_key` scoped families; shared codes keep one meaning with per-context family/facet mappings. Alternative — two independent apparel/homewares dictionaries permitted to diverge on shared codes.

Canonical composition: cotton 80, polyester 20
Facet projection: Cotton, Synthetic
Display: 80% Cotton, 20% Polyester

21. Size and Fit Dictionaries (*normative*)

Size labels are contextual: S, M and L are not universal measurements, and numeric sizes vary by market, gender category, brand and product type.

CDS400-Ro68 A size value MUST be scoped to a declared size system or scale when the label is not globally unambiguous.

CDS400-Ro69 Display size, canonical size code and measurement chart MUST be separable.

CDS400-Ro70 Size conversion tables SHOULD be represented as governed mappings with provenance and effective dates.

CDS400-Ro71 Fit descriptors such as slim, regular, relaxed and oversized MUST be maintained separately from size.

Size system: apparel_womens_au Display size: 10 Canonical code: au_10
Approximate mappings: US 6, EU 38 Product measurements: stored separately Fit: relaxed

22. Localisation (*normative*)

Localisation changes presentation, never canonical identity.

CDS400-Ro72 Canonical codes and value identifiers MUST remain language-neutral or follow a stable declared base-language policy.

CDS400-Ro73 Localised labels MUST identify locale, not merely language, where regional wording differs.

CDS400-Ro74 A missing translation SHOULD fall back according to a declared locale policy and SHOULD be visible as a completeness issue.

CDS400-Ro75 Search synonyms MAY vary by locale and market.

23. Provenance and Audit (*normative*)

Every important mapping answers: who created it, why it exists, which source introduced it and which products it affects.

Field	Purpose
<code>created_by</code>	User, import process or AI agent.
<code>approved_by</code>	Data steward or authorised reviewer.
<code>source_system</code>	Supplier, manufacturer, channel or internal team.
<code>source_value</code>	Exact original value.
<code>evidence</code>	Description, image, document or rule supporting the mapping.
<code>created_at</code> / <code>changed_at</code>	Audit timestamps.
<code>version</code> / <code>mapping_version</code>	Value-record and mapping versions used during a publication or verification run.

Field	Purpose
affected_count	Products or variants currently relying on the mapping.

CDS400-Ro76 A mapping change SHOULD record the previous value, the new value, the actor and the affected product count.

CDS400-Ro77 AI-created suggestions MUST record model or process identity, confidence and evidence where available (full AI provenance requirements: CDS-700).

24. Versioning, Deprecation, Migration and Breaking Changes (*normative*)

Dictionary changes can alter thousands of products without editing those products directly. They therefore require versioning and impact review. Corpus-level version policy is ADR-D5; this section governs value- and mapping-level versioning.

CDS400-Ro78 Every canonical value record MUST carry a `version`, and every alias, facet and channel mapping MUST carry a `mapping_version`, incremented on any governed change.

CDS400-Ro79 A canonical value MUST NOT be deleted while products, aliases, rules or channel mappings reference it.

CDS400-Ro80 Deprecated values MUST identify a replacement or state that no direct replacement exists.

CDS400-Ro81 Mapping changes SHOULD support preview of affected products and expected downstream changes before activation.

24.1 Breaking changes (*normative — resolves CDS-400-7*)

CDS400-Ro82 The following changes are breaking changes to value meaning:

1. **Facet remap** — moving a canonical value's primary facet membership to a different facet value;
2. **Alias retarget** — repointing an existing Alias Mapping Record to a different canonical value;
3. **Meaning narrowing** — restricting what an existing canonical value denotes so that some current usages become incorrect.

A breaking change MUST either (a) create a new value identity (new `value_id`, old value deprecated with `replacement_id`), or (b) be executed through a **governed migration record** that captures the old and new state, the actor and approver, affected product and collection counts, the activation date and the expected verification changes. A breaking change MUST NOT be applied as an in-place edit without one of these two paths.

Status transition: proposed -> active -> deprecated -> retired

Deprecated value:
 canonical_code: petrol_blue
 replacement_id: teal
 migration_note: Consolidated after merchandising review
 active_until: 2026-09-30

25. Dictionary Governance (*normative*)

Governance defines who may propose, approve, change and retire values, and prevents dictionaries from becoming another uncontrolled tag list. Role definitions and change-control machinery are owned by CDS-800; this section states the dictionary-specific obligations.

Role	Responsibility
Dictionary owner	Accountable for scope, quality and policy.
Data steward	Reviews new values, aliases, mappings and deprecations.
Merchandising owner	Approves customer facet labels, grouping and order.
Channel owner	Maintains external channel vocabularies and requirements.
Search owner	Maintains search synonyms and query behaviour.
AI process	May propose; never authoritative by default (CDS-700).

CDS400-Ro83 Every dictionary MUST have a named owner.

CDS400-Ro84 The organisation MUST define which roles may create canonical values and which may only propose them.

CDS400-Ro85 Facet changes SHOULD include merchandising or customer-experience review; channel mapping changes SHOULD include channel validation or test publication.

26. Publication and Verification (*normative*)

Dictionary mappings form part of the expected channel state. Verification compares the value generated by the active mapping version with the value read back from the channel. The verification model, status enum, comparison strategies and reason codes are owned by CDS-500; this section states what dictionaries contribute.

Canonical value: navy Mapping version: colour_map_v12
 Expected Shopify facet: Blue Observed: blue Strategy: case-insensitive Result: MATCH

Expected Google colour: Blue	Observed: blank	Result: MISSING
------------------------------	-----------------	-----------------

CDS400-Ro86 Publication records SHOULD identify the dictionary version and `mapping_version` used to generate channel output, so the expected value is reproducible.

CDS400-Ro87 Verification MUST compare the expected representation (generated from the active mapping version) with the observed representation under the declared comparison strategy (CDS-500 §16).

CDS400-Ro88 A dictionary change that alters expected output SHOULD trigger re-publication or re-verification for affected records.

27. AI Participation (*normative*)

AI may classify source terms, suggest aliases, recommend canonical values, propose facet memberships and detect anomalies — always inside the governed dictionary model. Autonomy levels, evidence classes, proposal storage and review flow are owned by CDS-700; the dictionary-side constraints are:

CDS400-Ro89 AI MUST receive the applicable dictionary, attribute context and allowed values before proposing a mapping where practical.

CDS400-Ro90 AI suggestions outside the dictionary MUST be marked as promotion proposals or unknown and routed under §17 and §25; AI MUST NOT silently create a new active canonical value (CDS400-Ro51 applies; autonomy ceilings: CDS-700 §5).

CDS400-Ro91 AI SHOULD return confidence, evidence and alternative candidates for ambiguous values; abstention is a first-class outcome (CDS-700).

```
Input: "French Navy"    Context: apparel colour
Allowed candidates: navy, royal_blue, ocean_blue, black
AI suggestion: navy     Confidence: 0.97
Evidence: supplier colour name and product image
Decision: auto-accept only if policy threshold and evidence rules pass (CDS-700)
```

28. Performance and Implementation Considerations (*informative except CDS400-Ro92*)

CDS specifies logical behaviour, not a storage engine. Implementations should avoid recalculating the entire catalogue for every dictionary lookup:

- immutable value identifiers and indexed alias lookups;
- cached active dictionary and channel mappings with version identifiers;
- recomputation of affected products when a `mapping_version` changes;
- product-to-value and value-to-product indexes for impact analysis;
- source values and observed channel values kept separate from canonical values;
- bulk review and approval for recurring unknown values;
- dictionary health metrics: unknown rate, deprecated-use count, unmapped-channel count.

CDS400-Ro92 Optimisation MUST NOT remove provenance, `mapping_version` or the ability to reproduce expected output.

29. Worked Examples (*informative*)

29.1 Apparel Colour

```
Source supplier colour: French Navy
Alias Mapping Record:   french navy -> navy
Canonical value stored on variant: value_id of navy
Display label:         French Navy
Facet mappings:        [blue]
Search terms:          navy, dark blue, french navy
Shopify filter value:   Blue           Google colour: Blue
Verification comparison: normalised text (CDS-500)
```

29.2 Teal with Ordered Multi-Facet Membership

```
Canonical value: teal
Facet mappings: [green, blue] (ordered; green primary – green-dominant per CDS-1500 baseline)
Display label:  Deep Teal
Search terms:   teal, blue green, green blue
Policy:         product appears once even when both Green and Blue are selected
```

29.3 Cotton Blend

```
Source description: 80% Cotton, 20% Polyester
Canonical composition: cotton = 80, polyester = 20
Display:            80% Cotton, 20% Polyester
Facets:             Cotton, Synthetic
Google material:    Cotton/Polyester
Claims:             no organic or recycled claim without governed evidence
```

29.4 Tasmanian Oak Homeware

```
Source material:   Tasmanian Oak
Canonical value:  oak   (family via material.homewares scope: wood)
Display label:    Tasmanian Oak
Facet:           Wood
Search terms:     timber, oak, tasmanian oak
Google material:  Wood
```

29.5 Unknown Supplier Colour (Moonlit Harbour)

```
Source value:      Moonlit Harbour
Dictionary match:  none
AI suggestion:     navy (confidence 0.61)
Policy threshold:  0.90
Result:           resolution_status = quarantined
Publication:       colour omitted from projections; verification reason code
                   CDS_VALUE_QUARANTINED; product still publishes (colour not
                   requiredness-blocking on this channel)
Action:           human review
Outcome:          alias "Moonlit Harbour" -> navy; display label remains Moonlit Harbour
```

29.6 Facet Remap as a Governed Migration (breaking change)

Moving `petrol_blue` from facet Blue to facet Green is a **facet remap** — a breaking change under CDS400-Ro82. It cannot be applied as an in-place mapping edit; it proceeds as a governed migration record:

```
Migration record: MIG-2026-014
Change:          petrol_blue facet_mappings [blue] -> [green]
Actor:           merchandising owner   Approver: dictionary owner
Impact:          127 products, 3 Shopify collections, 1 Google supplemental feed
Activation:      2026-09-01
Expected verification changes calculated before activation;
re-publication and re-verification scheduled for affected records (CDS400-R088)
```

30. Conformance (normative)

Conformance levels, test suites and claim rules are owned by CDS-1000 (Foundation -> Structured -> Publisher -> Verified -> Governed; ADR-D1). CDS-400 defines **dictionary capability badges**: cumulative capability declarations inside the Structured evidence set. Each badge requires everything in the badges before it.

Badge	Requires	Minimum capability
Dictionary Core	—	Stable canonical values, aliases, the two lifecycle/resolution enums, quarantine, provenance.
Dictionary Facet	Dictionary Core	Governed Facet Dictionary, ordered facet mappings, recorded usability review.
Dictionary Channel	Dictionary Facet	Channel representations, <code>mapping_versions</code> , channel validation.
Dictionary Round-Trip	Dictionary Channel	Expected-state publication and downstream verification of dictionary-derived output (CDS-500).

Supersession note. The vo.1 badge "Dictionary Verified" is renamed **Dictionary Round-Trip** to avoid colliding with the global Verified level, and the badges are now strictly cumulative (ADR-D1; resolves CDS-400-9).

CDS400-Ro93 A dictionary capability badge **MUST NOT** be claimed unless all lower badges are also satisfied; badge claims are made and evidenced under CDS-1000.

The chapter's core obligations for any conformant implementation are CDS400-Ro01..Ro09 (governed values, layers, identity), Ro25/Ro28/Ro29 (single facet mechanism), Ro49..Ro55 (statuses and quarantine), Ro78..Ro82 (versioning and breaking changes) and Ro86..Ro88 (reproducible expected output).

31. Architecture Decisions (informative)

The vo.1 chapter-local ADR table (CDS-ADR-007..013) is superseded by the single global ADR register (CDS000-Ro06). The decisions binding this chapter are ADR-D1 (conformance ladder and badges), ADR-D4 (value-layer terminology and variant storage) and ADR-D5 (version and identifier policy).

Appendix A — Recommended Dictionary Schema (informative; field-presence rules normative where marked Required)

One row set holds both canonical values and alias mappings; `row_type` disambiguates (ADR-D4; resolves DICT-1, CDS-400-6). Machine-readable schema: CDS-1100.

Field	Type	Requirement	Purpose
dictionary_id	string / UUID	Required	Stable dictionary identity.
dictionary_key	string	Required	Portable semantic key and context scope (§20).
row_type	enum	Required	<code>canonical</code> or <code>alias</code> .
value_id	string / UUID	Required	Stable canonical value identity (target value for alias rows).
canonical_code	string	Required (canonical rows)	Lowercase snake_case portable code.
canonical_label	string	Required (canonical rows)	Default label.
alias_text	string	Required (alias rows)	The recognised source term.
alias_scope	object	Optional	Supplier, locale or attribute scope of the alias.
status	enum	Required	Value Lifecycle Status (canonical rows) or Source-Term Resolution Status (alias rows) — §17.1.
version	integer	Required	Value-record version (§24).
mapping_version	integer	Required (alias/facet/channel mappings)	Mapping version used in publication and verification.
locale	string	Required where localised	Locale of labels and synonyms (e.g. <code>en-AU</code>).
parent_value_id	reference	Optional	Broader value — informative analytics only (§13).
display_labels	localised list	Optional	Merchandising or locale labels.
facet_mappings	ordered list	Optional	Facet memberships; first entry is primary (§10).
search_terms	localised list	Optional	Synonyms and related queries.
channel_mappings	map	Optional	Channel identifiers and labels, each with <code>mapping_version</code> .
metadata	object	Optional	Swatches, sort order, unit data, notes.
provenance	object	Required	Creation, source and approval data (§23).
valid_from / valid_to	datetime	Optional	Effective period.
replacement_value_id	reference	Conditional	Required when deprecated if a direct replacement exists.

Appendix B — Apparel Colour Example *(informative)*

Facet families cite the CDS-1500 Appendix C colour baseline, which owns the reference family list; `natural` is a top-level facet there and is used as such here (resolves CDS-400-3 / DICT-2). Canonical `sky_blue` per DICT-3, with `light blue` as an alias.

Aliases	Canonical	Display	Facet mappings (ordered)	Search	Channel
french navy	navy	French Navy	[blue]	navy; dark blue; marine	Google: Blue; Shopify facet: Blue
royal; cobalt	royal_blue	Royal Blue	[blue]	royal; bright blue	Google: Blue; Shopify facet: Blue
sky; pale blue; baby blue; light blue	sky_blue	Sky Blue	[blue]	sky; pale blue	Google: Blue; Shopify facet: Blue
teal	teal	Teal	[green, blue]	teal; blue green	Google: Green; Shopify facets: Green, Blue
natural; undyed; ecru	natural	Natural	[natural]	natural; undyed; ecru	Google: Beige; Shopify facet: Natural

Appendix C — Homewares Material Example *(informative)*

One materials dictionary; family and facet treatment scoped by `dictionary_key` context (§20, D29 resolved).

Source	Canonical	Scope: material.homewares family	Display	Facet	Search	Channel
Tasmanian Oak	oak	wood	Tasmanian Oak	Wood	timber; oak	Google: Wood
American Walnut	walnut	wood	American Walnut	Wood	timber; walnut	Google: Wood
Stoneware	stoneware	ceramic	Stoneware	Ceramic	ceramic; pottery	Google: Ceramic
Rattan	rattan	natural_fibre	Rattan	Natural Fibre	cane; wicker; rattan	Google: Rattan
80% Cotton / 20% Polyester	structured composition	textile	80% Cotton, 20% Polyester	Cotton; Synthetic	cotton blend	Google: Cotton/Polyester

Appendix D — Legacy Alignment *(informative)*

The legacy production PIM already separated raw supplier input, resolved authoritative values, channel outputs and read-back verification. CDS-400 makes those layers explicit entities: `Import_*` fields correspond to Source Values with provenance; lookup-map normalisation corresponds to Alias Mapping Records; resolved authoritative fields correspond to Canonical Values; legacy `DF_*` feed fields correspond to Channel Representations (`CH_*` in new design, ADR-D24); read-back match fields correspond to verification results (CDS-500). Legacy lookup maps import directly as Alias Mapping Records with provenance; every distinct source value that was mapped is retained as an alias (ADR-D4).

Appendix E — Platform Notes *(informative)*

Platform-specific mappings and constraints live in the versioned, dated profiles of CDS-900. In summary for any storefront platform: platform attribute stores and metafield-equivalents are publication targets, never the dictionary master; storefront filters use governed facet values, not distinct raw values; a product page may display a governed Display Label while the filter shows the facet value; tags are not the primary store for controlled attributes when typed fields exist; read-back is normalised and compared with expected values generated from the active `mapping_version`; manual downstream edits to PIM-owned values are reported as drift unless an approved override exists (CDS-500 §22).

END OF CDS-400 v0.2 REVIEW DRAFT

Commerce Data Standard (CDS)

CDS-500 — Publication, Observation and Verification

Field	Value
Status	vo.2 Review Draft (working source; not an approved standard)
Release	CDS vo.2 (single corpus release per ADR-D5)
Date	2026-08-04
Supersedes	CDS-500 Working Draft vo.1 (3 August 2026)
Normative status	§1, §3–§27 are normative except where individually marked. §2, §28 and the appendices are informative.
Findings addressed	CDS500-1, CDS500-2, CDS500-3, CDS500-4, CDS500-5, CDS500-6, CDS500-7, CDS500-8, CDS500-9, CDS500-10; CDS900-1 (status/colour conflict, resolved here per ADR-D3); LEG-2, LEG-4, LEG-6, LC-1, LC-2; ADR-D3 (implemented in full), ADR-D24 §3; CLEANUP-PASS items 2 (cross-reference side) and 6

Single-home notice. CDS-500 is the sole normative home for: the publication assurance lifecycle, the expected-channel-state model, field ownership contracts, the observation and coverage model, the comparison/normalisation registry obligations, the **verification status enum**, the **traffic-light mapping**, and the rule that **read-back never overwrites canonical values**. Other chapters cite these; they do not restate them (CDS000-R005, Reconciliation Matrix 5). Machine encodings live in CDS-1100; platform specifics live in CDS-900 profiles; terminology lives in CDS-100.

1. Purpose and Scope (*normative*)

CDS-500 defines publication as an assurance lifecycle, not a single API call. It specifies how an implementation converts a canonical product revision into an expected channel representation, validates and dispatches that representation, observes what the channel actually exposes, compares expected and observed values, and reports or repairs discrepancies. It applies to API integrations, file feeds, supplemental feeds, bulk imports, event streams and manual channel exports, independent of channel type.

CDS500-R001 The PIM **MUST** remain the master product-information layer throughout the publication lifecycle (authority model: CDS-200).

CDS500-R002 Publication **MUST NOT** be considered complete solely because a payload was generated, transmitted or acknowledged.

CDS500-R003 Where a channel permits observation, the implementation **MUST** compare observed downstream state with the expected channel state.

CDS500-R004 Where a channel cannot be observed, the implementation **MUST** report that limitation explicitly and **MUST NOT** represent the state as verified.

CDS500-R005 Channel read-back **MAY** inform verification and repair, but observed state **MUST NOT** overwrite canonical values silently. Any adoption of an observed value into the canonical record **MUST** be an explicit, attributed, auditable change. (*This is the single normative home of the read-back rule; CDS-200 and other chapters cite it.*)

2. Publication Principles (*informative*)

Principle	Meaning
Canonical before channel	A channel projection is derived from a versioned canonical revision.
Expected before actual	The expected channel state is calculated before anything is written.
Acknowledgement is not proof	Transport acceptance confirms receipt, not downstream fidelity.
Observe independently	Read-back uses the channel or a trusted downstream surface, never the outbound payload.
Compare semantically	Equivalent values may differ in case, order, formatting, units or identifiers.
Never hide uncertainty	Unsupported, unreadable and stale fields remain explicit states.
Repair by policy	Drift responses are governed, not improvised.
Evidence over assumption	Every status traces to payloads, responses, snapshots and rules.
Human-readable operations	An operator can see why a field is Green, Amber or Red without reading connector code.
Safe automation	Automated repair is bounded by ownership, criticality and confidence.

3. Publication Assurance Lifecycle (*normative*)

The lifecycle runs: canonical revision → expected projection → preflight → dispatch → acknowledgement → independent observation → comparison → drift classification → governed repair → health.

CDS500-R006 Each lifecycle stage **MUST** produce a durable status or evidence record sufficient to explain the current publication state.

CDS500-R007 A later stage **MUST NOT** erase the evidence of an earlier stage.

3.1 Assurance Levels

Level	Name	Meaning
PA-0	MODELLED	Canonical product information exists; no channel projection calculated.
PA-1	PROJECTED	An expected channel state has been calculated and versioned.
PA-2	VALIDATED	The projection passes channel-specific preflight checks.
PA-3	DISPATCHED	A publication attempt has been sent or exported.
PA-4	ACKNOWLEDGED	The transport or channel reports acceptance or successful processing.
PA-5	OBSERVED	A downstream snapshot has been read independently.
PA-6	VERIFIED	Expected and observed values have been compared with declared strategies.
PA-7	HEALTH-MANAGED	Drift, alerts, repair policies and trend reporting operate continuously.

CDS500-R008 An implementation MUST NOT claim a higher publication assurance level than the evidence it maintains. (*Conformance levels, tests and claims: CDS-1000.*)

Informative note: a channel may support only part of the lifecycle; the implementation declares the highest assurance level it can honestly provide in the channel profile.

4. Publication Entity Model (*normative*)

Publication is represented by related, versioned entities rather than mutable flags on a product record.

Entity	Purpose
Channel Profile	Capabilities, constraints, credentials reference, field mappings, propagation window, observation support.
Canonical Revision	Immutable snapshot or revision identifier of master product data.
Publication Projection	Expected channel-specific representation produced from a canonical revision.
Publication Job / Record	One dispatch operation and its immutable attempt records.
Acknowledgement	Transport and processing response from the channel.
Observed Snapshot	Independent read-back of downstream state at a point in time.
Verification Result	Field-level comparison of expected and observed values, with status and reason code.
Health Summary	Aggregated product, batch, channel and organisation view.
Repair Action	Governed remediation attempt and its outcome.
Evidence Artifact	Payload, response, feed row, API result, checksum, screenshot reference or diagnostic report.

CDS500-R009 Publication projections and observed snapshots MUST identify the channel, product, revision and time to which they apply.

CDS500-R010 A verification result MUST reference both the expected projection and the observed snapshot used in the comparison.

Informative note: machine-readable schemas for these entities are defined in CDS-1100; this chapter defines their semantics.

5. Channel Profiles and Capabilities (*normative*)

A Channel Profile declares what a destination can accept, return, transform and verify. Profiles prevent connector assumptions from leaking into the canonical model. Profile *content* for specific platforms lives in CDS-900; this section defines what any profile must declare.

Capability	Description
write_mode	API, file, feed, bulk import, event, manual export or hybrid.
read_mode	API read-back, feed diagnostics, storefront observation, report download or unavailable.
field_support	Writable, readable, required, optional, derived or unsupported fields.
operation_semantics	Create, update, upsert, patch, full replace, append, delete, archive behaviour.
rate_limits	Request, batch, quota and retry constraints.
propagation_window	Declared delay allowance between acknowledgement and observable state (see §13, §17, §18).
normalisation	Known channel casing, truncation, sorting, sanitisation and unit rules.
identity	Product, variant and channel record identifiers.
ownership	Which system controls each field or list partition (§7).
verification_grade	Highest assurance level supported by the connector and channel.
critical_set	Declared fields whose failure blocks publication or escalates presentation (§18, §23).

CDS500-R011 Every channel connector MUST publish a governed, machine-readable or equivalent capability profile.

CDS500-R012 A channel profile MUST declare a propagation window for each write mode it supports. The window bounds PENDING (§17) and drives Amber-to-Red escalation (§18).

CDS500-R013 An organisation claiming verification for a channel MUST declare the channel's critical field set (which MAY be empty). The critical set drives escalation (§18) and the critical-failure cap (§23).

CDS500-R014 A field unsupported by the channel MUST be marked unsupported rather than silently dropped.

6. Canonical Revisions, Projections and Expected Channel State (*normative*)

6.1 Projection identity

A projection is calculated from a specific canonical revision plus a specific channel profile, mapping set, dictionary versions and transformation rules:

```
projection_identity =
  product_id
  + canonical_revision
  + channel_id
  + channel_profile_version
  + mapping_version
  + dictionary_version
  + transformation_version
```

CDS500-R015 The same projection inputs MUST produce the same expected channel state unless a declared non-deterministic process is involved.

CDS500-R016 A projection MUST retain lineage to every canonical field, mapping and rule that contributed to each output field.

CDS500-R017 A new mapping, dictionary or transformation version MUST be able to trigger re-projection even when the canonical product record has not changed.

6.2 Field mappings and transformations

Transformations convert canonical values into channel representations: dictionary lookup, taxonomy mapping, formatting, unit conversion, template rendering, omission and list construction.

CDS500-R018 Every transformation MUST be identified by a versioned rule or function.

CDS500-R019 Transformations SHOULD be deterministic and side-effect free.

CDS500-R020 Lossy transformations MUST be declared so that verification compares against the expected lossy output rather than the richer canonical value.

CDS500-R021 A transformation error MUST be reported separately from a channel rejection or a downstream mismatch (see §17, ERROR family).

6.3 Expected channel state

The expected channel state is the exact semantic representation the implementation expects to observe after publication and channel processing. It is first-class and persisted, not recomputed ad hoc at comparison time.

```
Expected field record (informative example):
  field_key: MF_material
  canonical_value: linen
  expected_channel_value: Linen
  destination: custom.material
  ownership: EXCLUSIVE_PIM
  comparison_strategy: NORMALIZED_TEXT
  criticality: important
  source_revision: product-123@rev-48
```

CDS500-R022 Expected state MUST be calculated before dispatch and MUST be stored, or exactly reproducible from versioned inputs, for later comparison.

CDS500-R023 Expected state MUST reflect declared channel transformations, not merely raw canonical values.

CDS500-R024 Fields intentionally omitted by rule MUST be distinguishable from fields lost during transformation or publication.

7. Field Ownership Contracts (*normative*)

Multi-system commerce fails when ownership is implied rather than declared. Each channel declares field-level ownership modes.

Ownership mode	Meaning
EXCLUSIVE_PIM	The PIM owns the entire field or list. Downstream edits are drift and may be overwritten.
PARTITIONED_PIM	The PIM owns only declared namespaces, prefixes, keys or list members.
MERGED	The connector combines PIM-owned and externally owned values under a deterministic merge rule.
CHANNEL_OWNED	The channel owns the field. The PIM may observe but does not publish it.
OBSERVE_ONLY	The PIM records the field for diagnostics without claiming authority.
UNMANAGED	The field is outside the connector contract and must not influence verification.
OVERRIDE_ALLOWED	The PIM owns the default; a governed channel-specific exception may replace it (§22).

CDS500-R025 Every published field and list MUST have a declared ownership mode.

CDS500-R026 Full-replace operations MUST NOT delete channel-owned or externally owned values unless the ownership contract explicitly permits it.

CDS500-R027 A partitioned or merged full-replace list MUST be rebuilt before dispatch by reading the current downstream list, retaining externally owned members, and regenerating all PIM-owned members from current rules.

CDS500-R028 A manual downstream edit to an EXCLUSIVE_PIM field MUST be classified as drift, not accepted silently into the canonical record.

Informative note: tag lists commonly require PARTITIONED_PIM or MERGED ownership — the PIM may own collection and merchandising namespaces (e.g. collection_*, merch_*, regenerated from rules) while another application owns operational tags. See the worked example in §28.3. Tag governance: CDS-400; the publication merge rule lives here.

8. Preflight Validation, Publication Jobs and Records *(normative)*

8.1 Preflight

Preflight determines whether a projection is fit to publish before external side effects occur. Areas: schema shape; dictionary and taxonomy mappings active; identity (SKU, GTIN, variant, channel record IDs); channel constraints (length, character, image, URL, price, enumeration); ownership (payload changes only authorised fields); business rules (lifecycle, approval, eligibility); dependencies (parents, variants, media); change safety (preview of additions, changes, removals, list-replacement effects); channel readiness (credentials, quota, connector health).

CDS500-R029 Blocking preflight errors MUST prevent dispatch.

CDS500-R030 Preflight warnings MAY permit dispatch but MUST remain visible in the publication record.

CDS500-R031 Bulk publication SHOULD provide a preview of field additions, modifications and removals — including full-replace list effects — before execution.

8.2 Jobs and publication records

A Publication Job is one dispatch operation; a batch groups related jobs. Each attempt produces an immutable Publication Record carrying: job, batch and correlation identifiers; channel, product and projection identity; operation (create, update, upsert, delete, archive, withdraw); requester; timing; attempt count; payload hash; status; evidence references.

CDS500-R032 Every dispatch attempt MUST have a unique, immutable attempt record, even when retries belong to the same logical job.

CDS500-R033 A batch MUST report partial success without converting successful items into failures or hiding failed items inside an overall success status.

9. Transport Acknowledgement *(normative)*

Acknowledgement records what the transport or channel reported after dispatch. It is evidence of processing, not verification of final state.

Acknowledgement	Meaning
ACCEPTED	Request or file accepted for processing.
PARTIAL	Some records or fields accepted, others rejected.
REJECTED	Channel rejected the operation.
QUEUED	Accepted but not yet processed.
PROCESSED	Channel reports processing complete.
UNKNOWN	No reliable acknowledgement is available.
TIMEOUT	Outcome was not received within the declared interval.

CDS500-R034 Acknowledgement status MUST remain separate from verification status.

CDS500-R035 A successful transport response (2xx, successful upload, processed feed) MUST NOT automatically produce MATCH verification results.

10. Idempotency, Delta Publication and Replace Semantics *(normative)*

CDS500-R036 Where the channel supports idempotency keys, the connector SHOULD use them for retrievable write operations.

CDS500-R037 A delta publication MUST be calculated against a known prior expected or observed state, not against assumptions.

CDS500-R038 The connector MUST declare whether an empty value means clear, omit, unknown or no change.

CDS500-R039 Sorting SHOULD be canonicalised for semantically unordered lists to produce stable diffs and checksums.

(Full-replace list safety: CDS500-R026/R027 in §7.)

11. Retries, Errors and Partial Failure *(normative)*

Error classes: TRANSIENT (timeout, rate limit, outage), VALIDATION, AUTHENTICATION, AUTHORISATION, IDENTITY (target record unresolvable), CONFLICT (concurrent change, ownership violation), MAPPING, TRANSFORMATION, PARTIAL, UNKNOWN.

CDS500-R040 Retries MUST use a bounded policy with attempt limits, delay and a terminal state.

CDS500-R041 Deterministic validation, mapping and transformation errors MUST NOT be retried without a relevant data or configuration change.

CDS500-R042 A partial batch failure MUST preserve successful results and separately queue or report failed items.

CDS500-R043 Rate limiting and connector back-pressure MUST be visible as operational states, not misclassified as product-data errors.

12. Rollback, Withdrawal and Republication (*normative*)

Channels differ in whether they support rollback; CDS defines outcomes rather than assuming a universal undo.

CDS500-R044 A rollback MUST identify the exact previously verified projection being restored.

CDS500-R045 Where rollback is unsupported, the connector MAY republish an equivalent prior expected state as a new operation.

CDS500-R046 Withdrawal, deletion, unpublishing and archiving MUST be distinct operations when the channel distinguishes them.

CDS500-R047 Destructive operations MUST require stronger authorisation than ordinary field updates.

13. Scheduling and Verification Cadence (*normative*)

Verification may occur immediately, after propagation delay, on schedule, on demand, on canonical/mapping/override change, on channel events, or by governed sampling.

CDS500-R048 The verification schedule MUST account for the channel's declared propagation window (CDS500-R012) so that in-window latency is reported as PENDING, not drift.

CDS500-R049 Fields in the declared critical set SHOULD be verified more frequently than low-risk descriptive fields.

14. Downstream Observation (*normative*)

Observation independently retrieves the state a channel currently holds or exposes. The observation source and time are part of the evidence.

Observation source	Use
Authoritative channel API	Preferred when the API returns the stored field.
Channel diagnostics or item report	Feed and marketplace processing results.
Storefront or public item endpoint	When the customer-visible result is the verification target.
Search index or catalogue export	When the downstream consumer differs from the write endpoint.
Connector cache	Acceptable only under CDS500-R052.
Manual evidence	Exceptional fields only; not representable as automated verification.

CDS500-R050 Observed snapshots MUST include capture time, source, channel record identity and freshness status.

CDS500-R051 Read-back MUST NOT reuse the outbound payload as the observed state.

CDS500-R052 A connector cache MAY serve as an observation source only when it was populated from channel read operations, and its entries carry capture time and provenance satisfying CDS500-R050. A cache populated from the connector's own writes MUST NOT be used as observed state.

CDS500-R053 Where the write endpoint and the customer-visible endpoint differ, implementations SHOULD verify the surface relevant to the business objective.

(Adoption of observed values into canonical data: CDS500-R005.)

15. Observation Coverage and Unknown State (*normative*)

A missing returned value is not always proof that the channel stored nothing. CDS separates value state from observation capability.

Coverage state	Meaning
OBSERVED_VALUE	The field was returned with a value.
OBSERVED_EMPTY	The field was returned and explicitly empty or null.
NOT_RETURNED	The field was expected in the response but absent.
UNSUPPORTED	The observation interface cannot expose the field.
PERMISSION_DENIED	The connector lacks access to observe the field.
NOT_APPLICABLE	The field does not apply to this item or channel.
STALE	A value was returned but does not meet the freshness requirement.
OBSERVATION_ERROR	The read operation failed.
PENDING_VISIBILITY	The write is acknowledged but the propagation window has not expired.

CDS500-Ro54 An implementation MUST preserve the distinction between absence of data and absence of observability. UNSUPPORTED, PERMISSION_DENIED and NOT_RETURNED MUST NOT be collapsed into the MISSING verification status without the evidence required by the derivation table in §17.4.

16. Normalisation and Comparison (*normative*)

Comparison determines semantic equivalence between expected and observed values. Every field declares a comparison strategy drawn from a documented normalisation registry.

16.1 Strategy catalogue

EXACT, CASE_INSENSITIVE, NORMALIZED_TEXT (unicode/whitespace/punctuation/case), IDENTIFIER, BOOLEAN, NUMERIC, NUMERIC_TOLERANCE (declared absolute or percentage), MONEY (currency + declared rounding), MEASUREMENT (unit conversion then numeric), DATE_TIME (time-zone and precision normalisation), UNORDERED_SET, ORDERED_LIST, MULTISSET, HTML_SEMANTIC (declared sanitisation policy), URL_CANONICAL, IMAGE_IDENTITY, IMAGE_SEQUENCE, CHANNEL_RULE (specialised comparator declared in the channel profile).

16.2 Registry obligations

CDS500-Ro55 Every verifiable field MUST declare a comparison strategy that is an entry in the implementation's normalisation registry.

CDS500-Ro56 Each registry entry MUST declare: a stable comparator identifier, a behaviour version, supported input types, ordered normalisation steps, the equality rule, any tolerance parameters, and test vectors including at least one negative (non-matching) case. A change to normalisation or tolerance behaviour MUST increment the comparator version.

CDS500-Ro57 A comparator's declared behaviour and test vectors are the sole authority for what its normalisation may discard. A comparator MUST NOT be applied outside its declared input types, and an implementation MUST NOT substitute a broader comparator than the one the field declares. (*This replaces the untestable vo.1 "must not hide meaningful data loss" rule: what the comparator may hide is exactly what its registry entry documents and its vectors demonstrate.*)

CDS500-Ro58 Substring containment MUST NOT be used as a text-equality rule. Registries MUST record the substring false-green case of Appendix A.4 as a negative test vector for any text comparator.

CDS500-Ro59 Comparison inputs and normalised forms SHOULD be retained for diagnostics, alongside the comparator identifier and version used.

Informative note: Appendix A seeds the registry with the production-proven legacy tolerance vectors (money rounded to 2 dp, weight to 3 dp, case/whitespace-insensitive text, unordered tag sets) and the documented substring false-green negative case (findings LEG-4/LC-2).

17. Verification Status Model (*normative — the single normative status enum for all of CDS*)

Verification status expresses the relationship between expected state, observation coverage and comparison result. This section implements ADR-D3. All other chapters cite this enum; CDS-1100 encodes it verbatim.

17.1 Core statuses (the machine contract)

CDS500-Ro60 The core verification status enum is exactly: **MATCH**, **MISSING**, **MISMATCH**, **PENDING**, **UNOBSERVABLE**, **NOT_APPLICABLE**, **OVERRIDDEN**, **ERROR**. Implementations MUST NOT add, remove or rename core statuses.

Core status	Meaning
MATCH	Expected and observed values are equivalent under the declared comparator.
MISSING	The channel was observed with sufficient evidence and the expected value is absent.
MISMATCH	Observed state contradicts expected state (differing value, or an unauthorised value present).
PENDING	Resolution is legitimately awaited (propagation latency, or expected/observed revisions not yet aligned).
UNOBSERVABLE	The field cannot currently be read back with sufficient evidence or freshness.
NOT_APPLICABLE	The field is intentionally outside the product-channel contract.
OVERRIDDEN	The result is governed by an active exception rather than by comparison (see §22).
ERROR	The lifecycle failed before a comparison verdict could be produced.

17.2 Detailed statuses and deterministic rollup

CDS500-Ro61 Every field-level verification result MUST carry a core status; it SHOULD additionally carry a detailed status. Where a detailed status is recorded, its core status MUST be derived from the following table with no implementation-defined variation:

Detailed status	Rolls up to	Meaning
MATCH	MATCH	Equivalent under the declared comparator.
MISSING_DOWNSTREAM	MISSING	Observed with evidence; expected value absent.
MISMATCH	MISMATCH	Expected and observed values differ.
UNEXPECTED_DOWNSTREAM	MISMATCH	A value exists downstream where none was expected or authorised.

Detailed status	Rolls up to	Meaning
PENDING	PENDING	Propagation window not yet elapsed.
STALE_EXPECTED	PENDING	Observed state corresponds to a different expected revision; re-projection or re-observation is awaited.
UNOBSERVABLE	UNOBSERVABLE	The observation interface cannot expose the field, or access is denied, or no sufficiently fresh observation exists.
NOT_APPLICABLE	NOT_APPLICABLE	Outside the product-channel contract.
WAIVED	OVERRIDDEN	A governed exception accepts the discrepancy without redirecting the expected value (§22).
TRANSFORMATION_ERROR	ERROR	Expected state could not be calculated.
PUBLICATION_ERROR	ERROR	Dispatch failed or was rejected.
OBSERVATION_ERROR	ERROR	Read-back failed.
COMPARISON_ERROR	ERROR	Values could not be compared under the declared strategy.

RESOLVED — accepted as drafted (owner 2026-08-04; was new (resolved)): WAIVED rolls up to core OVERRIDDEN (both are "governed exception" outcomes, distinguishable by detailed status and reason code). The alternative — a ninth core status — was rejected to keep the machine contract at ADR-D3's eight values.

CDS500-Ro62 SUPERSEDED is a record-retention state, not a live verification status. A verification result belonging to an older projection or publication revision **MUST** be marked superseded when a newer result for the same field exists, **MUST** be retained for audit, and **MUST NOT** appear in current status displays, health denominators or the machine status enum. (*Resolves the STALE_EXPECTED/SUPERSEDED boundary: STALE_EXPECTED is a live condition awaiting realignment; SUPERSEDED is an archival marker on the record.*)

17.3 Reason codes

CDS500-Ro63 Every field-level verification result whose core status is not MATCH **MUST** carry a machine-readable `reason_code` drawn from the CDS_* reason-code registry defined in CDS-1100 §21. A free-text `message` **MAY** accompany the reason code but **MUST NOT** replace it.

CDS500-Ro64 A MATCH produced under a redirected-expected override (§22) **MUST** carry OVERRIDDEN provenance (override identity) visible alongside the status.

Informative note: Appendix B provides a seed reason-code list. The registry itself — identifiers, governance, additions — is owned by CDS-1100 §21. Reason codes are a v0.2 design; they have no legacy precedent.

17.4 Derivation from observation coverage

CDS500-Ro65 Where an expected value exists for a field, the field's detailed verification status **MUST** be derived from the observation coverage state (§15) as follows:

Coverage state	Detailed status	Notes
OBSERVED_VALUE	MATCH, MISMATCH or COMPARISON_ERROR	Per the declared comparator.
OBSERVED_EMPTY	MISSING_DOWNSTREAM	The interface affirmatively returned empty/null.
NOT_RETURNED	UNOBSERVABLE, unless the channel profile declares that the read interface enumerates every stored field, in which case MISSING_DOWNSTREAM	Absence in the response is evidence of absence only when the profile declares complete enumeration.
UNSUPPORTED	UNOBSERVABLE	
PERMISSION_DENIED	UNOBSERVABLE	Reason code distinguishes the cause.
NOT_APPLICABLE	NOT_APPLICABLE	
STALE	UNOBSERVABLE	No sufficiently fresh observation; reason code CDS_STALE_OBSERVATION.
OBSERVATION_ERROR	OBSERVATION_ERROR	The coverage state and detailed status share a name by design; this row is their relation rule.
PENDING_VISIBILITY	PENDING	Only within the declared propagation window (CDS500-Ro66).

(This table promotes the v0.1 §30.2 sketch to normative and replaces the v0.1 §15 reference to a nonexistent "MISSING" status: coverage states derive to MISSING_DOWNSTREAM, never to a bare "MISSING" detailed status.)

RESOLVED — accepted as drafted (owner 2026-08-04; was new (resolved)): NOT_RETURNED derives to UNOBSERVABLE by default, flipping to MISSING_DOWNSTREAM only under a declared complete-enumeration read interface; the alternative (always MISSING) was rejected as unevidenced. STALE derives to UNOBSERVABLE rather than PENDING because staleness is an observation deficiency, not write latency.

17.5 PENDING expiry

CDS500-Ro66 A field result **MUST NOT** remain PENDING beyond the channel's declared propagation window. When the window lapses, the implementation **MUST** re-verify: re-observe and re-derive the status under CDS500-Ro65, recording reason code `CDS_PROPAGATION_WINDOW_EXPIRED` on any non-MATCH outcome. If re-observation is not possible, the status **MUST** become UNOBSERVABLE with that reason code. In all cases the lapsed condition escalates at presentation (CDS500-Ro70).

17.6 Aggregation

CDS500-Ro67 Verification status **MUST** be field-specific before it is aggregated to product, batch or channel level.

CDS500-Ro68 A product-level verified claim **MUST** identify the canonical revision and the verification scope (which fields, which coverage) it covers.

18. Traffic-Light Presentation (*normative*)

A traffic light is a presentation layer over verification statuses. Colours are never stored statuses. This mapping is complete: every core status has exactly one row.

CDS500-Ro69 Where traffic-light presentation is used, the colour mapping **MUST** be:

Colour	Core statuses	Semantics
GREEN	MATCH	Expected equals observed under the declared comparator — with no exceptions. A MATCH under a redirected-expected override is GREEN with its OVERRIDDEN provenance visible (CDS500-Ro64).
AMBER	MISSING, PENDING, UNOBSERVABLE, OVERRIDDEN (including detailed WAIVED)	Attention: absence, latency, blindness or governed exception. Never a confirmed contradiction.
RED	MISMATCH, ERROR	Confirmed contradiction or lifecycle failure.
GREY (optional)	NOT_APPLICABLE	Outside the contract. Implementations that do not use GREY MUST NOT colour NOT_APPLICABLE as GREEN and MUST exclude it from success denominators either way.

CDS500-Ro70 An AMBER indicator **MUST** escalate to RED presentation when either declared condition is met: (a) the channel's declared propagation window for the publication has lapsed without resolution, or (b) the field is a member of the channel's declared critical set (CDS500-Ro13). Escalation changes presentation and alerting; it does not alter the stored status or reason code.

CDS500-Ro71 WAIVED results **MUST** always present as AMBER. A waiver never produces GREEN; only a redirected-expected override that yields a true MATCH does (§22).

CDS500-Ro72 The user interface **MUST** make the detailed status and reason code accessible from every traffic-light indicator. AMBER **MUST NOT** function as a generic category: the indicator **MUST** distinguish — via detailed status and reason code — whether the field is missing, pending, unobservable, overridden or waived.

CDS500-Ro73 Colours **MUST NOT** be stored or exchanged as verification statuses. Machine records carry statuses and reason codes; colour is derived at presentation time.

Informative note: v0.1's BLUE ("in-progress activity") is dropped from the standard per ADR-D3. Implementations remain free to animate or badge in-flight operations as a local presentation choice; such a treatment is not a CDS colour and carries no conformance meaning. GREY and any further presentation variants are optional presentation extensions, never additional statuses.

Informative note: MISSING presents as AMBER, restoring the production-proven legacy semantics (blank on either side = amber; red = confirmed mismatch only — LEG-2) and resolving the v0.1 CDS-500/CDS-900 colour conflict in favour of Amber. The escalation rule (CDS500-Ro70) captures the real risk of long-standing absence without punishing day-one latency.

19. Feed-Output Three-Way Match (*normative — optional layer*)

Channel-formatted projections (the CH_ layer, ADR-D24) are first-class artifacts: a serialized feed column or formatted payload field, distinct from both the canonical value and the channel's stored state. Legacy production practice verified this layer independently (DF_Match_* — LEG-6/LC-1).

CDS500-Ro74 An implementation **MAY** maintain expected/observed/match verification records at the formatted-output layer, in addition to canonical-versus-channel field verification. Where it claims this layer, each formatted-output field **MUST** use the §17 status enum, the §16 registry comparators and the §18 presentation rules, and the feed artifact **MUST** be verified end-to-end (generated output compared against the ingested or reported feed content).

CDS500-Ro75 A formatted-output MATCH **MUST NOT** substitute for downstream observation: it proves the feed said what was intended, not that the channel stored or displays it.

20. Drift Classification (*normative*)

Drift is any downstream state that no longer conforms to the current expected channel state or ownership contract.

Drift type	Description
EXTERNAL_EDIT	A user or application changed a PIM-owned value downstream.
CHANNEL_NORMALISATION	The channel transformed the value in a known but unmatched way.
CHANNEL_TRUNCATION	The channel shortened or removed content.
MAPPING_CHANGE	The expected output changed because a mapping or dictionary changed.
STALE_PUBLICATION	The channel still reflects an older canonical revision.
PARTIAL_PUBLICATION	Only part of a projection was applied.
IDENTITY_MISMATCH	The observed record belongs to the wrong product or variant.
DELETION	The expected record or field was removed.
DUPLICATION	Multiple downstream records represent one expected item.
UNMANAGED_MUTATION	A field outside the ownership contract changed; not PIM drift.
CONNECTOR_DEFECT	The connector transformed, omitted or targeted data incorrectly.

Drift type	Description
UNKNOWN_CAUSE	Evidence is insufficient to assign a cause.

CDS500-Ro76 Drift detection SHOULD assign a cause category separate from the verification status and reason code.

CDS500-Ro77 A known channel normalisation SHOULD be incorporated into expected-state or comparator rules (with a registry entry per §16.2) rather than generating repeated false alarms.

20.1 Drift Attribution (*normative — owner decision A1/D17, 2026-08-04*)

Detection tells you *that* a value changed; attribution helps you find *who or what* changed it — which distinguishes an innocent staff mistake from a connector defect or an unauthorised edit.

CDS500-Ro77a Observation records MUST be retained append-only per field and channel (never overwritten in place), so that for any drifted field the implementation can produce: the **previous observed value**, the **new observed value**, and the **detection interval** (last observation at which the field still matched, first observation at which it differed).

CDS500-Ro77b When drift of type EXTERNAL_EDIT is detected, the implementation SHOULD correlate the detection interval with whatever actor evidence the channel exposes (event logs, audit logs, webhook actor metadata, staff activity records — declared per channel profile, CDS-900) and record the attribution outcome (actor identified, actor class identified, or unattributable) on the drift record.

CDS500-Ro77c Attribution evidence is observed data about the channel, never a verdict about intent. An unattributable drift is still drift; attribution failure MUST NOT delay detection, verification status or escalation.

(Channel-specific attribution mechanisms and their limits are profile content: CDS-900. This restores and extends the legacy production insight that per-field expected/observed comparison — the Match fields — doubles as the change-detection layer.)

(Drift metrics and monitoring: CDS-1400. Republishing is not always the correct repair — see §21.)

21. Reconciliation and Repair Policies (*normative*)

A repair policy determines what happens after drift or failure is detected: REPORT_ONLY, AUTO_REPUBLISH, RETRY_SAME_PROJECTION, QUARANTINE_PRODUCT, SUSPEND_CHANNEL, ACCEPT_EXCEPTION, REMAP_AND_REPROJECT, MANUAL_REPAIR, ROLLBACK, WITHDRAW.

CDS500-Ro78 Automatic repair MUST be limited to fields and operations owned by the PIM under the declared ownership contract.

CDS500-Ro79 Critical or destructive repair actions SHOULD require approval or a pre-authorised policy.

CDS500-Ro80 Every repair action MUST create a new evidence record and MUST trigger re-verification where observation is supported.

22. Overrides and Exceptions (*normative*)

Channel-specific differences are sometimes legitimate. CDS permits explicit, governed exceptions without weakening the canonical model. An override record carries: identifier, scope (product/variant/field/locale/market/channel), replacement value (if any), reason, owner, approver, effective window, review date, verification expectation, and lifecycle status. (*Exception governance: CDS-800.*)

Two forms exist, with different verification outcomes (ADR-D3):

- **Redirected-expected override** — the override supplies an approved channel-specific replacement value. The expected channel state is recalculated to that value; verification compares against it and can yield a **true MATCH**, presented GREEN with OVERRIDDEN provenance (CDS500-Ro64).
- **Waiver (acceptance without redirection)** — a reviewed exception tolerates a known discrepancy without changing the expected value. The result is detailed status **WAIVED** (core OVERRIDDEN), presented AMBER always (CDS500-Ro71).

CDS500-Ro81 An override MUST NOT silently replace canonical product truth; the canonical value is unchanged and the override is a distinct governed record.

CDS500-Ro82 Overrides SHOULD be time-bounded and MUST carry a review date.

CDS500-Ro83 Verification MUST compare against the active redirected expected value when one exists.

23. Channel Health and Aggregation (*normative*)

Health aggregates field-level evidence into product, batch and channel views. A score aids prioritisation; it never replaces reason codes or critical-field rules. Metric definitions, denominators and monitoring cadence are owned by CDS-1400; the aggregation rules below are owned here.

CDS500-Ro84 Health calculations MUST publish their weighting, inclusion and critical-field rules.

CDS500-Ro85 Any field in the declared critical set (CDS500-Ro13) whose presentation is RED — directly or via escalation (CDS500-Ro70) — MUST cap the aggregate product and channel presentation at RED regardless of the numeric score.

CDS500-Ro86 NOT_APPLICABLE and superseded results MUST NOT count as successes in any health denominator; UNOBSERVABLE fields MUST NOT count as matched. (*No 100% match rate for unobserved fields — see CDS-1400.*)

```
Illustrative weighted score (informative):
health = 100 * sum(field_weight * field_pass) / sum(field_weight)
field_pass = 1 for MATCH; configurable partial value for PENDING or
OVERRIDDEN; 0 for MISMATCH or ERROR
```

24. Audit, Lineage and Evidence (*normative*)

Every decision in the publication lifecycle is reconstructable from: the canonical revision used; projection inputs (profile, mappings, dictionaries, rules, overrides); expected payload; dispatch record; acknowledgement; observed snapshot; normalised comparator inputs; verification result with status, reason code and comparator version; repair history; and actor provenance (human, system, schedule or AI agent).

CDS500-R087 Audit records MUST be append-only or otherwise protected against silent alteration.

CDS500-R088 Sensitive payload content and credentials MUST be redacted or securely referenced without destroying operational traceability (hashes, IDs and reason codes retained).

25. Security, Permissions and Secrets (*normative*)

CDS500-R089 Connector credentials MUST be stored in a secret-management mechanism and MUST NOT be embedded in product records, logs or exported evidence.

CDS500-R090 Connectors SHOULD use the least privilege sufficient for their declared operations.

CDS500-R091 Create, update, delete, withdraw, override and repair actions SHOULD support distinct permissions.

26. Bulk Operations and Scale (*normative*)

CDS500-R092 Bulk systems MUST preserve product- and field-level outcomes even when transport is batched.

CDS500-R093 A batch restart SHOULD resume from durable item states rather than replaying the full batch.

CDS500-R094 Checksums MAY be used to avoid unnecessary projection, dispatch or comparison work, provided the checksum inputs are versioned and auditable.

27. AI Participation (*normative*)

AI may assist with anomaly classification, mapping suggestions, repair recommendations, prioritisation and evidence summarisation. It is never the authority for downstream truth. (*AI governance, autonomy levels and proposal lifecycle: CDS-700.*)

CDS500-R095 AI-generated repair actions MUST remain constrained by field ownership, permissions and approval policy (CDS500-R078/R079).

CDS500-R096 AI MAY propose that two values are semantically equivalent, but only a governed comparator or approved mapping MUST determine the verification result.

CDS500-R097 An AI agent MUST NOT convert an UNOBSERVABLE or PENDING field into MATCH based on likelihood or inference.

CDS500-R098 AI repair and triage proposals SHOULD key off reason codes and detailed statuses, not presentation colours.

28. Worked Examples (*informative*)

28.1 Metafield round trip (API-integrated storefront, e.g. Shopify)

Canonical:	MF_material = linen	
Expected:	custom.material = "Linen" (single_line_text_field)	
Acknowledgement:	productUpdate accepted	-> PA-4
Observed:	"Linen" (independent read-back)	-> PA-5
Comparison:	NORMALIZED_TEXT	-> MATCH
Presentation:	GREEN	-> PA-6

28.2 Missing versus unobservable

Field A: expected "Linen"; read-back returns the field as null
coverage = OBSERVED_EMPTY -> MISSING_DOWNSTREAM (core MISSING)
reason_code = CDS_OBSERVED_EMPTY -> AMBER (escalates per CDS500-R070)

Field B: expected "Blue"; read-back interface does not expose the field
coverage = UNSUPPORTED -> UNOBSERVABLE
reason_code = CDS_INTERFACE_UNSUPPORTED -> AMBER

The two fields never share a reason code (§17.4 is the normative derivation).

28.3 Full-replace tag safety

Current channel tags:	collection_Shirts, merch_featured, review_five_star
Ownership:	collection_*, merch_* = PARTITIONED_PIM review_* = external application
New PIM projection:	collection_Shirts, merch_sale
Safe merged payload:	collection_Shirts, merch_sale, review_five_star

Sending only the PIM tags to a full-replace API would delete `review_five_star` — a CDS500-R026/R027 violation. PIM-owned partitions are regenerated from rules on every publication.

28.4 Mapping change without product edit

Canonical revision stays at rev-12; the dictionary mapping changes (`navy` → `Blue`, previously `Navy`). A new projection is required because `dictionary_version` changed (CDS500-R017); verification uses the new expected value.

28.5 Channel health summary

Channel 1 (API round-trip):	PA-7, match 99.6%, coverage 100%, 0 critical failures → GREEN
Channel 2 (feed with diagnostics):	PA-6, match 96.8%, coverage 91%, 4 missing GTINs in the critical set → RED (cap)
Channel 3 (write-only feed mode):	PA-4, no field read-back in this integration mode → AMBER, "acknowledged, not verified"

Channel 3's limitation is a property of the *integration mode in use* (a write-only feed without item-level read-back), not of any named platform; the same platform integrated through an API with read-back could reach PA-6.

29. Conformance and Relationship to Other Chapters (*informative*)

Conformance levels, test suites and claim rules for this chapter's requirements are defined solely in CDS-1000. Machine-readable encodings of the entities, the eight-value core enum, `detailed_status`, `reason_code` and the validation-output contract are defined in CDS-1100 (the enum is encoded verbatim from §17). Platform-specific profile content is defined in CDS-900. Monitoring metrics and drift-age reporting: CDS-1400. The architecture decisions behind this chapter are recorded in the global ADR register (notably ADR-D3, ADR-D24; superseding the vo.1 chapter-local CDS-ADR-020..029, whose substance survives as requirements here).

Appendix A — Normalisation Registry Seed (*informative except A.1*)

A.1 Registry entry shape (*normative*)

CDS500-R099 Each comparator registry entry MUST record: `comparator_id`, `version`, `input_types`, `normalisation_steps` (ordered), `equality_rule`, `tolerance` (where applicable), `diagnostics`, and `test_vectors` including negative cases (per CDS500-R056/R058).

A.2 Seed entries (legacy production tolerance vectors — LEG-4/LC-2)

These are the first entries of the registry, restored from proven production comparison behaviour:

comparator_id	Behaviour	Positive vector	Negative vector
MONEY.round2	Round both sides to 2 dp, then numeric equality (with currency match)	19.999 vs 20.00 → MATCH	19.99 vs 20.00 → MISMATCH
MEASUREMENT.weight_round3	Convert to common unit, round to 3 dp, numeric equality	1.5 kg vs 1500 g → MATCH	1.5005 kg vs 1500 g → MISMATCH
NORMALIZED_TEXT.v1	Case fold, trim, collapse internal whitespace, then equality	" French Navy " vs "french navy" → MATCH	"French Navy" vs "Navy" → MISMATCH
UNORDERED_SET.v1	Same unique members regardless of order	{a,b,c} vs {c,a,b} → MATCH	{a,b} vs {a,b,c} → MISMATCH

A.3 Money and measurement tolerances

Rounding precision (2 dp money, 3 dp weight) is part of the comparator version. Changing precision is a comparator version change (CDS500-R056), never a silent edit.

A.4 Documented negative case: substring false-green

Legacy text comparison used substring containment (spreadsheet `SEARCH()`), which tolerated case and whitespace — and also real prefix differences. Example: expected "Blue", observed "Blue Grey" — substring containment reports a match; the values differ. This case MUST appear as a negative test vector for text comparators (CDS500-R058). Substring containment is not an equality rule in CDS.

Appendix B — Reason-Code Seed List (*informative*)

The reason-code registry (identifiers, governance, additions) is owned by CDS-1100 §21. This seed list illustrates the codes this chapter's rules require:

Reason code	Typical detailed status
CDS_OBSERVED_EMPTY	MISSING_DOWNSTREAM
CDS_FIELD_NOT_RETURNED	MISSING_DOWNSTREAM or UNOBSERVABLE (per §17.4)
CDS_INTERFACE_UNSUPPORTED	UNOBSERVABLE
CDS_PERMISSION_DENIED	UNOBSERVABLE
CDS_STALE_OBSERVATION	UNOBSERVABLE
CDS_PROPAGATION_WINDOW_EXPIRED	any post-lapse non-MATCH (§17.5)
CDS_AWAITING_PROPAGATION	PENDING
CDS_EXPECTED_REVISION_AHEAD	STALE_EXPECTED
CDS_VALUE_DIFFERS	MISMATCH
CDS_UNEXPECTED_VALUE_PRESENT	UNEXPECTED_DOWNSTREAM
CDS_IDENTITY_MISMATCH	MISMATCH / drift IDENTITY_MISMATCH
CDS_CHANNEL_TRUNCATION	MISMATCH / drift CHANNEL_TRUNCATION
CDS_TRANSFORMATION_FAILED	TRANSFORMATION_ERROR
CDS_DISPATCH_REJECTED	PUBLICATION_ERROR
CDS_READBACK_FAILED	OBSERVATION_ERROR
CDS_COMPARATOR_FAILED	COMPARISON_ERROR
CDS_WAIVED_EXCEPTION	WAIVED
CDS_OVERRIDE_ACTIVE	(provenance on overridden results)
CDS_NOT_IN_CONTRACT	NOT_APPLICABLE

Appendix C — Platform, Legacy and Schema Pointers (*informative*)

- **Platform specifics.** The vo.1 Appendices C (Shopify implementation profile) and D (Google Merchant and feed-based channels) are replaced by the versioned, dated implementation profiles in **CDS-900**. Platform behaviour never shapes the requirements of this chapter (Reconciliation Matrix 5).
- **Legacy alignment.** The legacy Airtable/Shopify prefix mapping and migration alignment (vo.1 Appendix E) lives in the consolidated legacy annex (see CDS-1300). The migration mapping of legacy match fields is 1:1: green -> MATCH, amber -> MISSING/PENDING, red -> MISMATCH (ADR-D3); `DF_*` feed fields migrate to `CH_*` per ADR-D24, with the feed-layer verification pattern retained in §19.
- **Schemas.** Recommended entity schemas (the vo.1 Appendix A sketch) are superseded by the machine-readable contracts in **CDS-1100**: verification-result schema carries `status` (8-value core enum), `detailed_status`, `reason_code`, `coverage_ratio` and `comparison_engine_version`.

Appendix D — Operational Checklist (*informative*)

1. Confirm the canonical revision is approved for publication.
2. Generate the projection with current profile, mappings, dictionaries and overrides.
3. Run blocking and warning preflight checks; preview additions, changes, removals and full-replace list effects.
4. Confirm field ownership and destructive-operation permissions.
5. Dispatch with correlation and idempotency identifiers where supported.
6. Record acknowledgement without treating it as verification.
7. Observe downstream state after the declared propagation window.
8. Normalise and compare with declared registry comparators.
9. Derive statuses per §17; present detailed status and reason code beneath any traffic light.
10. Classify drift; execute the governed repair or exception policy; re-verify.
11. Retain payload, response, observation, comparison and actor evidence.

Commerce Data Standard (CDS)

CDS-600 — Customer Experience, Facets and Navigation

Field	Value
Status	vo.2 Review Draft (working source; not an approved standard)
Release	CDS vo.2
Date	2026-08-04
Supersedes	CDS-600 Working Draft vo.1
Normative status	§1, §3, §5–§29 are normative. §2, §4, §30–§32 and all appendices are informative. Examples and notes inside normative sections are informative unless marked otherwise.
Findings addressed	SYS-1 (CLEANUP-PASS §C: all 118 prefix-labelled rules corrected — 76 different-inline-keyword, 42 label-strip-only), SYS-2 (requirement IDs), SYS-4, CDS-600-2, CDS-600-3 (D10), CDS-600-4, CDS-600-5, CDS-600-6 (D11), CDS-600-7, CDS-600-8, CDS-600-9, CDS-600-10, CDS-600-11

CDS-600 is the single normative home for facets and customer-experience projection (CDS000-R005), including zero-result and unavailable facet-value behaviour (§17, per open decision D10). CDS-1500 and other chapters cite this chapter and do not restate its rules.

1. Purpose and Scope (*normative*)

CDS-600 defines how canonical product information is projected into customer-facing discovery experiences: site navigation, category and collection pages, facets, search refinement, sorting, product-card presentation, and the governance that keeps those experiences coherent as a catalogue grows. It does not prescribe a visual design system; it specifies the information, behaviour and quality rules beneath the interface so that different storefronts may implement distinct designs without weakening customer comprehension or data integrity.

CDS600-R001 A CDS implementation **MUST** derive customer-facing discovery structures from governed product information rather than from uncontrolled labels entered independently in the storefront.

CDS600-R002 Customer-facing simplification **MUST NOT** destroy the richer canonical or source information held by the PIM.

CDS600-R003 A storefront **MAY** vary presentation by device, market or audience, provided the underlying meaning remains consistent.

2. Customer Experience Principles (*informative*)

- **Customer language first** — labels use terms customers understand, even when internal or channel taxonomies use different wording.
- **Progressive disclosure** — reveal enough structure to support a decision without presenting the whole data model at once.
- **Controlled simplicity** — rich canonical data is grouped into concise facets and navigation choices (CDS-P-08).
- **Predictable behaviour** — the same control and value behaves consistently across categories unless a documented category profile requires otherwise.
- **No dead ends** — filtering and search minimise preventable zero-result states and provide useful recovery (§17).
- **Availability-aware discovery** — customers are not encouraged toward choices that cannot produce a purchasable result (§16–§17).
- **Accessible by default** — discovery controls remain understandable and operable without relying on colour, pointer input or visual position alone (§24, normative).
- **Measurable quality** — facet, search and navigation performance is evaluated with behavioural and data-quality evidence (§28).

3. Experience Architecture (*normative*)

```
Canonical Product Information
|-- Internal Taxonomy
|-- Controlled Dictionaries
|-- Attribute and Variant Values
`-- Merchandising Rules
    |
    v
Customer Experience Projection
|-- Navigation and Category Pages
|-- Collections and Facets
|-- Search, Sort and Ranking
`-- Product Card Presentation
    |
    v
Experience Analytics and Health
```

CDS600-R004 Customer experience structures **MUST** be treated as projections of canonical information, not as independent product facts.

CDS600-R005 The PIM or governed commerce layer **SHOULD** be able to explain which canonical values and rules produced each customer-facing label, filter and collection membership.

4. Shopping Modes and Customer Intent (*informative*)

Mode	Typical behaviour	Primary support
Directed search	Customer knows a product, brand, model or requirement	Search, synonyms, identifiers, relevance ranking
Category browsing	Customer knows the general kind of product	Navigation, categories, collections, breadcrumbs
Attribute refinement	Customer narrows by colour, size, material, room or specification	Facets and counts
Inspirational discovery	Customer begins with a style, occasion, campaign or need	Editorial collections, merchandising, cross-links
Comparison	Customer evaluates similar products	Consistent attributes, sorting, comparison-ready presentation

A taxonomy answers what a product *is*. Merchandising and facets help customers discover why it may be *relevant* to them.

5. Taxonomy, Navigation, Collections, Facets and Search (*normative*)

Construct	Primary question	Typical source
Taxonomy / category	What is this product?	Internal classification (CDS-200 §6)
Navigation	Where should customers begin browsing?	Selected category and collection nodes
Collection	Which products should be presented together?	Taxonomy, merchandising rules or curation
Facet	Which characteristic should refine the current result set?	Controlled attribute projection (CDS-400)
Search	Which products are relevant to this query?	Index, synonyms, attributes, ranking signals

CDS600-R006 These constructs **MUST** remain logically distinct even when a platform implements more than one of them with the same technical feature.

CDS600-R007 A collection **MUST NOT** be treated as a substitute for a missing controlled attribute when customers need to refine by that attribute.

CDS600-R008 A facet **MUST NOT** be used to redefine a product category.

6. Navigation Architecture (*normative*)

Navigation is a curated entry into the catalogue, not a complete rendering of every internal category. It exposes stable, meaningful destinations and defers fine-grained refinement to facets.

CDS600-R009 Primary navigation **SHOULD** expose no more than the levels required for customers to orient themselves and reach a useful product set.

CDS600-R010 Deep internal taxonomy nodes **MAY** remain available through category pages, search or links without appearing in the main menu.

CDS600-R011 Navigation labels **MUST** have clear, non-overlapping meanings within the same level.

CDS600-R012 A product **MAY** be discoverable from multiple navigation paths while retaining one primary internal category (CDS-200 §6).

Informative note: the one-primary-category rule protects classification integrity. Cross-links and collections solve merchandising overlap without duplicating product identity.

7. Category and Collection Architecture (*normative*)

Category pages represent product types or taxonomy nodes. Collections represent merchandising groupings such as New Arrivals, Winter Edit or Staff Picks. Both may be customer-facing, but they carry different meanings.

CDS600-R013 A category page **SHOULD** inherit its product membership from classification or an equivalent deterministic rule.

CDS600-R014 A merchandising collection **MUST** declare whether membership is computed, manually curated or hybrid.

CDS600-R015 Temporary collections **SHOULD** have an owner, activation window and retirement rule.

CDS600-R016 Parent pages **SHOULD** explain their scope and provide clear paths to child categories or relevant facets.

8. Information Scent and Labels (*normative*)

Information scent is the customer's ability to predict what lies behind a label or control. Weak labels force experimentation; strong labels make the catalogue feel smaller and more understandable.

CDS600-R017 Navigation, collection and facet labels **MUST** be written in customer language and **SHOULD** avoid internal abbreviations.

CDS600-R018 A label **MUST** distinguish neighbouring choices. Terms such as Other, General or Miscellaneous **SHOULD** be avoided unless no clearer grouping exists.

CDS600-R019 Where a term is ambiguous across industries or regions, the label **SHOULD** include qualifying context.

CDS600-R020 Synonyms **MAY** support search, but the interface **SHOULD** present one preferred label per concept.

9. Facet Model and Facet Definitions (*normative*)

A facet is a customer-facing refinement dimension (CDS-100). A Facet Definition governs its label, values, applicability, selection behaviour, ordering, counts, visibility and mapping from canonical attributes.

Property	Requirement
Stable ID	Machine identifier independent of display label
Label	Localisable customer-facing name
Description	Meaning and inclusion boundaries
Source	Canonical attribute or derived rule
Eligibility	Applicable categories, collections or search contexts
Value model	Dictionary, numeric range, boolean or hierarchy
Selection mode	Single-select, multi-select or range
Within-facet operator	OR or AND within the facet
Count unit and method	Products, purchasable variants or another declared unit; count semantics per §16
Ordering	Semantic, numeric, popularity or alphabetic
Disclosure	Expanded, collapsed, hidden or conditional
Unavailable-value behaviour	Per §17
Analytics key	Stable event identifier
Owner	Responsible data or merchandising role

CDS600-R021 Every customer-facing facet **MUST** have a governed Facet Definition.

CDS600-R022 Facet values **MUST** come from controlled canonical or facet dictionaries (CDS-400), not from unrestricted distinct strings.

CDS600-R023 The same logical facet **SHOULD** retain one identifier across channels even where the display label is localised.

CDS600-R024 A Facet Definition **MUST** declare its count unit (products, purchasable variants or another declared unit) and its count method (§16).

CDS600-R025 A Facet Definition **MUST** declare its selection logic; interfaces **MUST NOT** rely on undocumented platform defaults.

CDS600-R026 How facets combine across a result set is a property of the facet set, not of an individual facet. The cross-facet operator **MUST** be declared at the facet-set (category- or context-profile) level, and a Facet Definition **MUST NOT** carry its own cross-facet operator. (*Resolves CDS-600-4; see §15 and Appendix A.*)

10. Facet Eligibility (*normative*)

Not every attribute should become a filter. Eligibility is determined by customer usefulness, coverage, value distribution, category relevance and interface cost:

Criterion	Question
Decision relevance	Does the attribute meaningfully affect product choice?
Coverage	Is the value populated for enough products in the eligible result set?
Discrimination	Does the facet divide the catalogue into useful groups rather than one dominant value and many singletons?
Comprehension	Will customers understand the values without specialist knowledge?
Stability	Are values governed and unlikely to change unpredictably?
Result utility	Will selecting a value usually leave a useful product set?
Maintenance	Can the organisation govern the dictionary and mappings?

CDS600-R027 A field **MUST NOT** become a facet merely because the platform permits filtering on it.

CDS600-R028 Facet eligibility **SHOULD** be reviewed per category profile rather than globally.

CDS600-R029 Low-coverage or highly fragmented facets **SHOULD** remain hidden until data quality and grouping are adequate (thresholds per §28).

11. Value Projection for Facets (*normative*)

The value-layer model (Source Value -> Alias Mapping Record -> Canonical Value -> projections) is owned by CDS-400 and defined per ADR-D4; terms per CDS-100 §3. Facet Values are one projection branch:

```
Canonical Value
  +--> Display Label
  +--> Facet Value
  +--> Search Synonyms
  +--> Channel Representation
```

CDS600-R030 Facet values **MUST** be projections of canonical values and **MUST NOT** replace those canonical values in the PIM.

CDS600-Ro31 Many canonical values MAY map to one broader facet value when that grouping improves customer comprehension.

CDS600-Ro32 A canonical value MAY map to more than one facet group only where the secondary membership is deliberate, documented and non-deceptive.

12. Colour Facets (*normative*)

Colour is a high-risk facet because suppliers and brands use rich marketing names while customers generally need a concise set of recognisable families. CDS separates the sellable colour, its Display Label and its facet family.

CDS600-Ro33 (*Definition — resolves CDS-600-7.*) The **sellable colour** of a Variant is the canonical value of its colour Variant Option (CDS-100 §2–§3). What the customer sees is that canonical value's Display Label (e.g. "French Navy"); what filtering uses is the Facet Value the canonical value maps to (e.g. Blue). Implementations MUST maintain this distinction; the Display Label is never itself the canonical value.

```
Display Label (sellable colour as shown):  French Navy
Canonical value:                          colour_french_navy (shade: navy)
Primary facet family:                      Blue
Optional tone:                            Dark
Search synonyms:                          navy, dark blue, french navy
Swatch:                                   governed visual sample
```

CDS600-Ro34 A colour facet MUST use a controlled colour-family dictionary rather than exposing every display colour as a separate filter value.

CDS600-Ro35 The storefront MAY display the sellable colour's Display Label on the product card or product page while filtering by the broader family.

CDS600-Ro36 Colour swatches MUST NOT be the sole means of communicating a value; a text label MUST remain available (§24).

CDS600-Ro37 Multi-colour, transparent, metallic and natural finishes SHOULD be modelled explicitly rather than forced into an inaccurate basic hue. Multicolour is a facet/status value, never a canonical shade (CDS-100 §3).

CDS600-Ro38 Where a product contains several meaningful colours, the system SHOULD distinguish primary colour, secondary colour and multicolour status.

Informative note (*resolves CDS-600-6 per D11*): the exemplar colour-family baseline — including the core family set and the boundary rules for neighbours such as Cream/Beige/Natural/White and Silver/Grey — is owned by CDS-1500 Appendix C. This chapter deliberately publishes no family list of its own; the exact dictionary remains profile-specific and is adopted from that baseline.

13. Size, Fit and Dimension Facets (*normative*)

Size information is frequently variant-level and may use category-specific scales. Fit describes intended garment shape and normally remains product-level. Dimensions may require units and ranges rather than labels.

CDS600-Ro39 Size facets MUST preserve the distinction between the displayed size label and the normalised size order or scale.

CDS600-Ro40 A category profile MUST define which size system applies (e.g. alpha, numeric apparel, footwear, waist, bedding, dimensional).

CDS600-Ro41 Size values MUST sort semantically rather than alphabetically.

CDS600-Ro42 Fit MUST NOT be inferred from size alone.

CDS600-Ro43 Numeric dimension facets SHOULD use normalised units and MAY present customer-local units.

Informative note: a size label of 10 may represent different measurements in different markets. CDS therefore requires profile and unit context rather than treating the visible label as a universal canonical value.

14. Material, Style, Pattern, Occasion and Room Facets (*normative*)

Facet	Canonical detail	Typical facet projection
Material	linen; cotton; oak; stoneware	Linen; Cotton; Wood; Ceramic
Composition	80% cotton / 20% polyester	Cotton or Cotton Blend, per declared policy
Pattern	fine white pinstripe	Striped
Style	relaxed coastal resort shirt	Coastal and/or Relaxed, per profile
Occasion	beach holiday	Beach / Holiday
Room	living room cushion	Living Room
Finish	natural Tasmanian oak	Natural / Wood

CDS600-Ro44 Material composition MUST remain separately available from the simplified material facet where percentages or blends matter.

CDS600-Ro45 Style and occasion facets SHOULD use a small governed vocabulary and MUST NOT become an uncontrolled list of marketing phrases.

CDS600-Ro46 Homewares facets SHOULD be category-aware; a Room facet may be useful for decor but irrelevant for cookware components. (*Industry baselines: CDS-1500.*)

15. Selection Logic (*normative*)

Selection logic determines how multiple chosen values combine. The intended logic must be explicit because platforms and customers may otherwise interpret the same control differently. Cross-facet combination is declared per facet set (CDS600-R026).

CDS600-R047 Different facets SHOULD combine with AND logic (Blue AND Linen), as the declared facet-set default.

CDS600-R048 Multiple values within the same categorical facet SHOULD combine with OR logic (Blue OR Green).

CDS600-R049 An AND-within-facet mode MAY be used where a product must possess all selected values, but the interface MUST make that behaviour clear.

CDS600-R050 Single-select facets MUST prevent contradictory simultaneous choices.

CDS600-R051 A range facet MUST define inclusive boundaries, units and rounding behaviour.

```
Selected:
  Colour: Blue OR Green
  Material: Linen
  Size: M OR L
Result logic:
  (Blue OR Green) AND Linen AND (M OR L)
```

16. Counts and Variant Availability (*normative*)

Facet counts can represent products or sellable variants. Availability-aware variant counting ensures that selecting Blue and Size M returns products with a purchasable Blue/M combination, not products that merely have Blue somewhere and M somewhere.

CDS600-R052 Facet counts MUST declare their counting unit and availability scope (CDS600-R024).

CDS600-R053 Variant-defined facets SHOULD evaluate valid variant combinations rather than independent product-level unions.

CDS600-R054 Where the storefront platform supports combination-level availability filtering, a product MUST NOT be counted as satisfying a selected combination unless at least one eligible sellable unit satisfies the complete selected combination.

CDS600-R055 Where the storefront platform does not support combination-level availability filtering, the implementation MUST disclose that limitation in its platform profile (CDS-900) and SHOULD approximate combination behaviour as closely as the platform allows. (*Resolves CDS-600-2.*)

RESOLVED (was CDS-600-2 (resolved); verified 2026-08-04, REVIEW-006A): the conditional structure of R054/R055 stands (it is the correct platform-neutral form), and the pending question is answered — **native Shopify Search & Discovery evaluates variant-option and availability filters at variant scope** (`filter.v.option.*`, `filter.v.availability`; different filters AND at variant level), so the Shopify profile declares the capability present (CDS-900 §4.7). Per-value count accuracy under combined selections remains a vertical-slice empirical check.

Informative note (*preserved*): R054 prevents the classic false-positive result where a shirt has Blue only in Small and Medium only in White, yet appears after Blue + Medium is selected.

CDS600-R056 (*Disjunctive count semantics — resolves CDS-600-5.*) Within a multi-select OR facet, the count displayed for an unselected value MUST be computed against the result set produced by all *other* facet selections, excluding that facet's own current selection, so that the count predicts the effect of adding the value rather than reporting the already-filtered set. The count method in force MUST be declared in the Facet Definition (CDS600-R024).

CDS600-R057 The interface SHOULD update counts or availability after each selection so customers can anticipate the effect of another filter.

17. Unavailable Values and Zero-Result Recovery (*normative — single normative home per D10*)

This section is the corpus-wide normative home for zero-result and unavailable facet-value behaviour. CDS-1500 §19 and any other chapter cite this section and MUST NOT restate or vary these rules. The stricter CDS-1500 wording is adopted here as the governing MUST. (*Resolves CDS-600-3.*)

CDS600-R058 A facet value whose selection would produce no truthful available result in the current context MUST be disabled or hidden.

CDS600-R059 As a governed exception to R058's hiding branch, out-of-stock values MAY remain visible for discovery only where the interface presents a clear unavailable state; unavailable values MUST be distinguished from purchasable options and MUST NOT be presented as normally selectable.

CDS600-R060 When no results remain, the interface MUST preserve the selected filters and provide a clear recovery action.

CDS600-R061 Recovery SHOULD identify the narrowest selected constraint or offer removal of one filter at a time.

CDS600-R062 A zero-result message MUST NOT imply that the store has no relevant products when the result is caused by an over-restrictive combination.

18. Ordering, Defaults and Disclosure (*normative*)

Facet order and value order are part of the merchandising experience. Alphabetical sorting is appropriate only when it matches how customers think about the values.

CDS600-R063 High-value decision facets SHOULD appear before secondary descriptive facets.

CDS600-Ro64 Size, numeric ranges and lifecycle stages MUST use semantic order rather than alphabetical order.

CDS600-Ro65 Colour families MAY use a stable merchandising order or a visual palette order, provided labels remain accessible.

CDS600-Ro66 Long value lists SHOULD use search, grouping or progressive disclosure rather than expanding the page indefinitely.

CDS600-Ro67 A facet SHOULD be expanded by default only when it is broadly useful in the current context.

19. Mobile and Responsive Filtering (*normative*)

CDS600-Ro68 Mobile filtering MUST expose the number of active filters and provide a persistent way to clear or edit them.

CDS600-Ro69 An apply action MAY defer result updates inside a mobile filter panel; where it does, the apply control MUST display a preview of the result count that will apply, and selected counts and availability SHOULD remain understandable while the panel is open. (*Resolves CDS-600-9.*)

CDS600-Ro70 Filter controls MUST remain operable with touch targets, keyboard navigation and assistive technology (§24).

CDS600-Ro71 A long mobile facet list SHOULD support internal search or grouping rather than forcing excessive scrolling.

CDS600-Ro72 Closing the filter interface MUST preserve current selections unless the user explicitly cancels them.

20. Search, Synonyms and Query Understanding (*normative*)

Search and facets share dictionaries but serve different interactions: search interprets free-form intent; facets expose governed choices.

CDS600-Ro73 Search indexes SHOULD include canonical values, display labels, approved aliases and relevant facet families.

CDS600-Ro74 A search synonym MUST map to a preferred concept or query expansion and MUST NOT create an uncontrolled duplicate value.

CDS600-Ro75 Brand, SKU, MPN and GTIN identifiers SHOULD be indexed according to access and relevance requirements.

CDS600-Ro76 Search query interpretation MAY infer a facet selection, but the resulting refinement SHOULD be visible to the customer.

CDS600-Ro77 Misspelling tolerance and semantic expansion MUST NOT override exact product identifiers or create materially misleading results.

```
Query: "navy linen shirt"
Interpreted intent:
  Product type = Shirt
  Colour facet = Blue / canonical navy
  Material facet = Linen
Visible refinements:
  Shirts | Blue | Linen
```

21. Sorting and Ranking (*normative*)

CDS600-Ro78 Every sort option MUST declare the data and tie-breakers used.

CDS600-Ro79 Price sorting MUST use the same customer-visible price basis used in the current market and tax context.

CDS600-Ro80 Newness SHOULD use a governed publication or arrival date, not an arbitrary last-edit timestamp.

CDS600-Ro81 Relevance ranking MAY use behavioural signals, but product-data completeness and availability SHOULD remain quality inputs.

CDS600-Ro82 Sponsored or commercially influenced ranking MUST be identifiable where applicable.

Informative note: Best Selling, Featured and Recommended are not self-defining technical fields. Each requires a documented calculation or merchandising owner.

22. Product Cards and Variant Presentation (*normative*)

CDS600-Ro83 A product card SHOULD display the attributes necessary to distinguish neighbouring products without reproducing the entire product page.

CDS600-Ro84 Variant swatches or size indicators MUST reflect actual eligible variants and availability.

CDS600-Ro85 A colour swatch SHOULD use the governed Display Label and visual sample, while its filter behaviour uses the facet family.

CDS600-Ro86 Product-card badges such as New, Sale, Clearance or Exclusive MUST originate from governed lifecycle or merchandising rules.

CDS600-Ro87 A product card MUST NOT show an unavailable price or variant as though it were the active purchasable choice.

23. Breadcrumbs, Cross-Links and Merchandising Overlap (*normative*)

CDS600-Ro88 Breadcrumbs SHOULD reflect the chosen customer navigation context while remaining consistent with the primary category model.

- CDS600-R089** Cross-links MAY surface a product to another audience without assigning a second primary category.
- CDS600-R090** Editorial and campaign pages SHOULD link back to stable category or collection destinations where customers can continue browsing.
- CDS600-R091** A product appearing in several collections MUST retain one canonical identity and one primary classification.

24. Accessibility *(normative)*

- CDS600-R092** Facet groups and controls MUST have programmatically determinable labels.
- CDS600-R093** Selected state, disabled state and result counts MUST be exposed to assistive technology.
- CDS600-R094** Colour swatches and colour indicators MUST include text alternatives, and colour MUST NOT be the sole signal for any value, state or distinction.
- CDS600-R095** Keyboard focus MUST follow a logical order and remain visible.
- CDS600-R096** Changes to result counts MUST be announced to assistive technology without causing disruptive focus movement.
- CDS600-R097** Clear-all and remove-filter controls MUST have unambiguous accessible names.
- CDS600-R098** Range controls MUST expose current values, minimum, maximum and units.

25. Performance and Resilience *(normative)*

- CDS600-R099** An implementation MUST declare a response-time target for facet and filter interactions and MUST monitor performance against that declared target. *(Resolves CDS-600-8; replaces the untestable "quickly enough" wording.)*
- Informative note: the intent of the target is that the interface preserves a direct, perceptible relationship between a selection and its response.
- CDS600-R100** The storefront MUST fail safely when a facet service is unavailable; product browsing SHOULD remain possible.
- CDS600-R101** Cached counts MUST identify their freshness and MUST NOT be presented as live availability when they are materially stale relative to the declared target and propagation windows.
- CDS600-R102** Large dictionaries SHOULD support indexed lookup rather than rendering all values unconditionally.
- CDS600-R103** Performance optimisation MUST NOT change selection semantics or counts without explicit disclosure.

26. Faceted Navigation and SEO *(normative)*

- Faceted combinations can create an unbounded number of URLs. CDS does not prescribe one search-engine policy, but requires explicit governance so that customer refinement does not accidentally generate a duplicate crawl space.
- CDS600-R104** An implementation MUST define which facet combinations are indexable, canonicalised, blocked or represented without unique URLs.
- CDS600-R105** Indexable landing pages SHOULD have stable intent, sufficient product depth and unique explanatory content.
- CDS600-R106** Transient or extremely narrow combinations SHOULD NOT become indexable pages by default.
- CDS600-R107** The canonical category and collection structure MUST remain discoverable independently of facet-generated URLs.
- CDS600-R108** SEO policy MUST NOT weaken customer filter behaviour merely to simplify crawling.

27. Personalisation and AI *(normative)*

- AI proposal, evidence, approval and autonomy rules are owned by CDS-700; dictionary authority by CDS-400. This section carries only the customer-experience rules.
- CDS600-R109** Personalisation MAY reorder facets, values or products but MUST NOT change the semantic meaning of a facet.
- CDS600-R110** A personalised interface MUST retain a predictable way to access the full applicable filter set.
- CDS600-R111** Sensitive personal inference MUST NOT be required for ordinary product discovery.
- Informative note: AI may suggest facet mappings, synonyms, labels and collection memberships only as proposals under CDS-700 governance; governed dictionaries and approval rules remain authoritative (CDS-400, CDS-700).

28. Measurement and Analytics *(normative)*

Measure	Examples <i>(informative)</i>
Facet usage	Selections, removals, clear-all events and sequence

Measure	Examples (<i>informative</i>)
Result quality	Zero-result rate, low-result rate and recovery
Coverage	Percentage of eligible products populated per facet
Fragmentation	Number of values, singleton values and long-tail share
Discovery	Search-to-filter transitions and navigation depth
Conversion support	Outcome after facet or search interaction
Performance	Response latency against the declared target (R099)
Accessibility	Keyboard and assistive-technology defects
Data drift	Unexpected values or missing mappings

CDS600-R112 Analytics identifiers **MUST** remain stable when display labels change.

CDS600-R113 Facet analytics **SHOULD** distinguish exposure, interaction and resulting product-set size.

CDS600-R114 The organisation **MUST** declare its thresholds for facet usage and facet coverage. A facet whose usage exceeds the declared usage threshold while its coverage falls below the declared coverage threshold **MUST** trigger data-quality review rather than automatic removal. (*Resolves CDS-600-11; replaces the untestable "high-use facet with poor coverage".*)

29. Governance and Change Management (*normative*)

Roles, decision rights and change control in general are owned by CDS-800; dictionary and mapping workflows by CDS-400. This section carries the facet-specific rules.

CDS600-R115 Every facet **MUST** have an owner responsible for definition, dictionary, coverage and retirement.

CDS600-R116 New facet values **MUST** enter through the governed dictionary or mapping workflows defined in CDS-400.

CDS600-R117 Renaming a visible label **SHOULD** preserve the stable facet and value identifiers.

CDS600-R118 Merging or splitting values **MUST** include migration, publication, search and analytics impact analysis.

CDS600-R119 Facet changes **SHOULD** be tested against representative categories, devices and product combinations before release.

CDS600-R120 Deprecated values **MUST** retain aliases or redirects long enough to protect imports, saved URLs and historical analytics where applicable.

30. Worked Examples (*informative*)

30.1 Apparel: colour and size

```
Product: Relaxed Linen Shirt
Variants:
  French Navy / S / in stock
  French Navy / M / out of stock
  White / M / in stock
Canonical:
  colour: colour_french_navy -> facet Blue (Display Label "French Navy")
  colour: colour_white      -> facet White
  size scale: Alpha Apparel
Customer selects Blue + M:
  Product excluded – no purchasable Blue/M variant exists (R054).
Customer selects Blue:
  Product included; product card shows "French Navy" (R035).
```

30.2 Homewares: material and room

```
Product: Tasmanian Oak Side Table
Canonical material: oak      Display material: Tasmanian Oak
Material family facet: Wood  Room facets: Living Room, Bedroom
Style facet: Contemporary
Filters on Side Tables collection: Material, Colour, Style, Shape, Width, Price
Room may remain available on a broader Furniture collection (R046).
```

30.3 Pattern simplification

```
Display pattern: Fine White Pinstripe
Canonical pattern: pinstripe
Facet value: Striped
Search aliases: pinstripe, stripe, striped
```

The customer sees "Striped" in filters and "Fine White Pinstripe" on the product page.

31. Conformance *(informative)*

Conformance levels, test suites and claim rules are defined solely in CDS-1000 (CDS000-R012). CDS-600 conformance is assessed against the requirement IDs in this chapter; the UX-oriented test suite (T-UX) covers them per CDS-1000. No conformance content is defined here.

32. Design Decision Summary *(informative)*

Chapter-level ADRs from v0.1 are queued for the single global ADR register (CDS000-R006). Their substance, as carried forward:

vo.1 ADR	Decision as carried into vo.2
CDS-ADR-601	Facet values are projections, separate from canonical values (§11)
CDS-ADR-602	Taxonomy and collections remain distinct (§5, §7)
CDS-ADR-603	Colour filters use controlled families; the exemplar baseline is owned by CDS-1500 App C (§12, D11)
CDS-ADR-604	Within-facet OR and cross-facet AND are the defaults; cross-facet combination is a facet-set property (§15, CDS-600-4)
CDS-ADR-605	Variant combinations determine availability-aware results, conditional on platform capability with mandatory disclosure (§16, CDS-600-2)
CDS-ADR-606	Facet Definitions are first-class governed entities (§9)
CDS-ADR-607	Faceted SEO policy is implementation-specific but mandatory (§26)

Appendix A. Recommended Facet Definition and Facet-Set Schemas *(informative)*

Per-facet definition (cross_facet_operator removed per CDS600-R026):

```
facet_id
label
localised_labels
source_attribute_ids
eligible_category_ids
value_dictionary_id
selection_mode
within_facet_operator
count_unit
count_method          # incl. disjunctive-count semantics per R056
availability_policy    # unavailable-value behaviour per section 17
value_sort_strategy
facet_sort_priority
default_disclosure
mobile_presentation
accessible_label
analytics_id
owner
status
version
created_at
updated_at
```

Facet-set / category-context profile (holds what was per-facet in v0.1):

```
facet_set_id
context                # category, collection or search context
member_facet_ids       # ordered
cross_facet_operator   # normally AND (R047)
response_time_target   # declared per R099
owner
version
```

Appendix B. Industry Facet Baselines *(informative)*

The apparel and homewares facet profiles and the exemplar colour-family baseline formerly in this chapter's Appendices B and C are owned by CDS-1500 (industry dictionaries and profiles; colour families in CDS-1500 Appendix C). *(Resolves CDS-600-10.)*

Appendix C. Platform Implementation Profiles *(informative)*

Platform-specific guidance (including the former Shopify implementation notes) lives in versioned, dated CDS-900 profiles. Platform declarations relevant to this chapter: combination-level availability filtering capability (R054/R055) and any facet value/group limits.

Appendix D. Operational Review Checklist *(informative)*

- Are primary categories and collections clearly distinguished?
- Does every visible facet have a definition, owner and controlled value source?

- Are colour values grouped into a manageable customer-facing dictionary (per the CDS-1500 baseline)?
- Are size values ordered by the correct category and market scale?
- Do variant combinations produce truthful availability-aware results, or is the platform limitation disclosed?
- Are low-coverage and highly fragmented facets hidden or remediated against declared thresholds?
- Can customers recover easily from zero results?
- Are mobile controls accessible, and do deferred-apply controls preview the result count?
- Are search synonyms governed rather than invented per product?
- Are facet URLs and indexing rules documented?
- Are facet analytics stable across display-label changes?
- Are customer-facing values verified after publication (CDS-500)?
- Is the declared response-time target being monitored?

Commerce Data Standard (CDS)

CDS-700 — AI Enrichment, Automation and Human Oversight

Field	Value
Status	vo.2 Review Draft (working source; not an approved standard)
Release	CDS vo.2
Date	2026-08-04
Supersedes	CDS-700 Working Draft vo.1; CDS-800 vo.1 §29 (AI governance rules consolidated here per the single-home rule)
Normative status	§1, §3, §5–§33 are normative. §2, §4, §34 and Appendices A–D are informative.
Findings addressed	SYS-1 (errata manifest §C: all 105 prefix-labelled rules corrected), SYS-2/700-2 (requirement IDs), 700-3, 700-4, 700-5, 700-6, 700-7, 700-8, 700-9; Matrix 2 (D14, open), Matrix 5 (AI single-home); provides the autonomy-scaled monitoring hook for 1400-11

CDS-700 is the single authoritative home for AI and probabilistic-automation requirements in CDS (CDS000-R005). Other chapters cite this chapter and do not restate its rules. This chapter in turn cites: statuses and verification — CDS-500; namespace registry — CDS-300; dictionaries and value layers — CDS-400; facets and customer experience — CDS-600; governance roles and change control — CDS-800; platform profiles — CDS-900; conformance levels and the T-AI test suite — CDS-1000; the machine-readable AI-proposal contract — CDS-1100.

1. Purpose and Scope (*normative*)

CDS-700 defines how artificial intelligence and other probabilistic automation may participate in a PIM-first commerce data environment: extraction, classification, normalisation, dictionary mapping, content generation, translation, media analysis, channel assistance, human review, evaluation and operational governance. It does not select a model vendor, prompting framework or orchestration technology; it defines the information contracts and assurance boundaries that hold regardless of which model, agent or platform performs the work.

CDS700-R001 AI MUST operate as a producer of proposals, evidence or bounded actions; it MUST NOT become the implicit source of truth for product information.

CDS700-R002 AI-generated information MUST conform to the same schemas, dictionaries, validation rules and publication controls as information entered by a human or imported from a supplier.

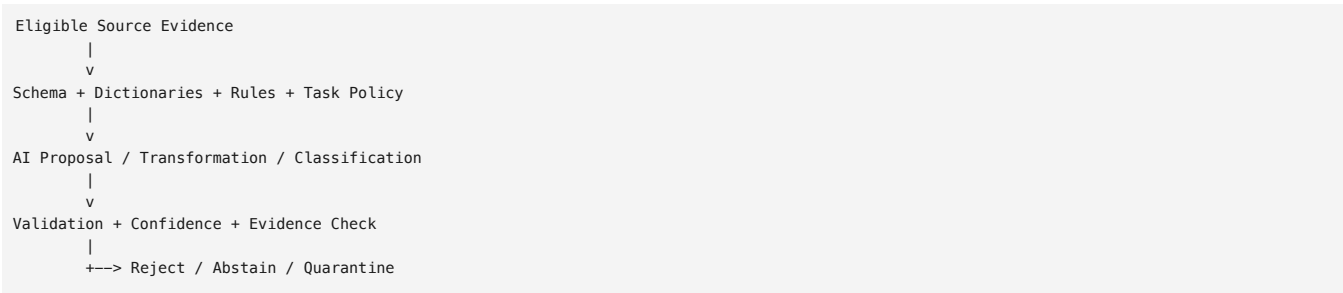
CDS700-R003 The PIM MUST remain the master product-information layer when AI is used (CDS-200; CDS-P-02).

CDS700-R004 An implementation MUST make AI participation visible and auditable: authorised operators MUST be able to enumerate the active AI workflows (via the registry of §33) and inspect the work objects (§6) behind any AI-originated value.

2. AI Principles (*informative*)

- **Canonical authority remains human-governed.** AI may propose or transform; acceptance into canonical state follows declared policy.
- **Evidence before assertion.** Product facts are supported by eligible evidence (§7).
- **Schema before prose.** AI fills governed fields and controlled values before producing unconstrained text.
- **Abstention is valid.** Returning unknown or requesting review is a first-class outcome, safer than a confident invention.
- **Risk determines autonomy.** Low-risk, reversible tasks may be automated more aggressively than pricing, compliance, safety or product claims.
- **Deterministic rules stay deterministic.** Validation, arithmetic, unit conversion and exact mappings are not delegated to probabilistic reasoning when reliable rules exist.
- **Humans remain legible participants.** AI workflows and field names are understandable without specialist prompt-engineering knowledge (CDS-P-04).
- **Evaluation precedes scale.** A workflow is tested on representative products before it touches a whole catalogue.
- **Publication remains assured.** AI does not bypass CDS-500 expected-state, observation and verification controls.
- **Provenance survives acceptance.** Accepted values retain their source, evidence and AI-generation history without polluting the canonical field name.

3. AI Participation Architecture (*normative*)





CDS700-R005 Every AI workflow MUST identify its inputs, expected output schema, validation rules, approval policy and permitted destination state.

CDS700-R006 An AI workflow MUST NOT write directly into a canonical field unless its autonomy level and the field's risk class explicitly permit that action.

Informative note: the AI system is one participant in the information lifecycle. The schema, dictionaries, rule engine and verification engine remain separate authorities.

4. Roles of AI in Commerce Information (*informative, except R007*)

Role	Typical tasks	Default risk posture
Extractor	Identify material, colour, dimensions, features or model identifiers from eligible sources	Low to medium
Classifier	Suggest product family, type, category or taxonomy mapping	Medium
Normalizer	Map source wording to canonical dictionary values	Low to medium
Writer	Draft titles, descriptions, bullets, care text or summaries	Medium
Translator	Translate and localise governed content	Medium
Media analyst	Describe images, detect visible attributes, propose alt text or sequence	Medium
Channel assistant	Suggest channel categories, labels, titles or policy fixes	Medium to high
Quality analyst	Detect anomalies, missing values, conflicts and likely drift	Low to medium
Merchandising assistant	Suggest collections, facets, synonyms, related products or campaign groupings	Medium
Workflow coordinator	Route work, request evidence and invoke approved tools	Depends on permissions

CDS700-R007 (*normative*) An implementation MUST declare the role an AI workflow performs rather than describing all AI activity as a single generic enrichment step.

5. Autonomy Levels (*normative*)

Level	Name	Permitted behaviour
A0	Disabled	No AI processing for the task or field.
A1	Assistive	AI proposes values or content; a human must accept each result.
A2	Governed draft	AI may populate a non-canonical draft or review queue after validation.
A3	Bounded acceptance	AI may accept eligible values into canonical state only under the bounded-A3 controls of R011.
A4	Bounded execution	AI may trigger approved downstream actions for specified low-risk fields or workflows, subject to publication and verification controls (§30) and the enumeration rule R012.

CDS700-R008 An organisation MUST declare the autonomy level for each AI workflow.

CDS700-R009 A4 MUST NOT be interpreted as unrestricted autonomy.

CDS700-R010 High-risk fields — price, tax, regulated claims, hazardous-use instructions, legal warranties and safety information — MUST NOT use A3 or A4 without an explicit specialist-approved policy.

CDS700-R011 Bounded A3 is A3 operated under all four of the following controls. A workflow claiming A3 MUST implement all four; R2-risk tasks (Appendix B) MAY use A3 only in this bounded form: 1. a **per-field policy** declaring which fields and value classes are eligible for automatic acceptance; 2. a **calibrated confidence threshold** derived from representative reviewed examples for that task and field (§17); 3. **mandatory sampled human review** of accepted results at a declared sampling rate; 4. **automatic suspension** of automatic acceptance when the reviewer correction rate breaches a declared threshold.

CDS700-R012 An A4 workflow MUST enumerate its permitted actions in its workflow definition; any action not enumerated is prohibited for that workflow.

Informative note: a single catalogue may use A3 for mapping supplier colour names while requiring A1 for compliance claims and A2 for marketing descriptions.

6. AI Work Objects and Review States (*normative*)

AI work is represented as durable objects, not transient chat responses. A work object records the task, source evidence, schema, output, validation, review and final disposition. The machine-readable contract for this record is the CDS-1100 ai-proposal schema; Appendix A lists the recommended semantic fields.

Property	Purpose
work_id	Stable identifier for the AI task

Property	Purpose
product_id / variant_id	Target information record
task_type	Extraction, mapping, classification, writing, translation or analysis
workflow_version	Approved workflow or agent definition
model_id	Model and version used
input_evidence	References to eligible source material
output_schema	Expected structured result
proposal	Generated values or content
confidence	Field- or task-level confidence
validation_results	Dictionary, schema and business-rule outcomes
review_state	See R015
reviewer	Human or policy that authorised acceptance
timestamps	Created, completed, reviewed and accepted times
cost_and_usage	Optional operational usage data

CDS700-R013 AI proposals MUST be stored separately from canonical values until accepted by the declared approval policy.

CDS700-R014 Rejecting or superseding a proposal MUST NOT erase its audit record.

CDS700-R015 The review state of an AI proposal MUST use the shared enum: `proposed`, `review_required`, `accepted`, `rejected`, `superseded`, `expired`. This enum is identical in CDS-700 and the CDS-1100 ai-proposal schema — they describe the same record. Execution failure (timeout, malformed output, tool error) is a work-object disposition (§28), not a review state.

RESOLVED — accepted as drafted (owner 2026-08-04; was D14 (resolved)): Adopted the merged enum `proposed/review_required/accepted/rejected/superseded/expired` ("accepted" replaces v0.1 "approved"; "expired" adopted from CDS-1100; "review_required" retained from CDS-700; v0.1 "failed" moved to work-object disposition). Alternative: keep two vocabularies with a mapping table — rejected because Matrix 2 shows they describe one record.

7. Source Evidence and Input Eligibility (*normative*)

AI quality is constrained by the information it receives. CDS distinguishes eligible evidence from contextual material that may inspire wording but cannot establish a product fact.

Class	Eligibility criteria	Examples	Permitted use
E1 — Authoritative	Publisher is the manufacturer, the contracted supplier or the organisation itself, with an identifiable authority chain to the product (specification, signed record, approved internal measurement)	Manufacturer specification, signed supplier record, approved internal measurement	May establish canonical facts subject to validation
E2 — Strong secondary	Publisher identity is known and the content originates from a brand-controlled or certified source: an official product page (a page whose origin is controlled by the brand owner or authorised distributor), packaging produced by the brand, or a certified feed (a data feed whose certifier is named in the evidence record)	Official product page, packaging image, certified data feed	May support facts with provenance and review policy
E3 — Observational	Direct observation of the product or its current presentation; publisher authority not required, but the observation itself is verifiable	Product photographs, video, existing storefront copy	May support visible or descriptive attributes; ambiguity must remain explicit
E4 — Contextual	No qualifying publisher identity or authority chain	Competitor listings, general web content, related-product text	May inform terminology or review; must not establish unverified product facts

CDS700-R016 An evidence record MUST identify the publisher and the basis for its class assignment; evidence that cannot meet the E1–E3 criteria MUST be classed E4. For E2 "certified feed" evidence the certifier MUST be named.

CDS700-R017 Every factual AI proposal MUST identify the evidence used or explicitly declare that no eligible evidence was available.

CDS700-R018 AI MUST NOT convert absence of evidence into a positive product claim.

CDS700-R019 Conflicting eligible sources MUST be surfaced as a conflict rather than silently resolved by model preference.

8. Schema-Constrained Generation (*normative*)

AI outputs are constrained by Attribute Definitions, controlled dictionaries, units, cardinality and conditional rules (CDS-200 §7, CDS-400). Free-form generation is reserved for content fields that are intentionally unstructured.

Task: Extract apparel attributes
 Allowed output:
 MF_material: dictionary(materials), multi-value
 MF_pattern: dictionary(patterns), single-value
 MF_fit: dictionary(fits), single-value

```
MF_sleeve_length: dictionary(sleeve_lengths), single-value
AI_evidence: source references
AI_confidence: 0.00-1.00
Not allowed:
  new dictionary values
  price claims
  sustainability claims without evidence
```

CDS700-R020 An AI workflow **MUST** receive an explicit output schema.

CDS700-R021 Out-of-schema fields **MUST** be rejected or quarantined.

CDS700-R022 When an output requires a dictionary value, the AI **MUST** select from the allowed identifiers or return an unknown-value proposal (CDS-400 §17).

9. Extraction and Normalisation (*normative*)

Extraction identifies candidate facts from source material. Normalisation converts those candidates into canonical units, identifiers and dictionary values. The two stages remain distinguishable for audit and correction.

```
Observed source text: "80% cotton / 20% recycled poly"
Extracted components:
  cotton = 80 percent
  recycled polyester = 20 percent
Canonical composition:
  material.cotton = 80
  material.recycled_polyester = 20
Facet projection:
  Cotton
  Recycled material
```

CDS700-R023 The original extracted text **MUST** be preserved when the normalised result may be disputed or reprocessed.

CDS700-R024 Unit conversion and arithmetic **SHOULD** be performed by deterministic code after AI extraction, not by unchecked model calculation.

CDS700-R025 A normalisation step **MUST** identify whether the result was exact, alias-based, transformed or inferred.

10. Classification and Taxonomy Assistance (*normative*)

AI may suggest internal categories, product types and external taxonomy mappings. Classification remains governed because it controls inherited attributes, collections, navigation and channel requirements (CDS-200 §6).

CDS700-R026 AI classification **MUST** select from the current approved taxonomy identifiers.

CDS700-R027 The classifier **SHOULD** return ranked candidates, confidence and the evidence that distinguished the selected category from nearby alternatives.

CDS700-R028 A category suggestion **MUST NOT** automatically create a new category.

CDS700-R029 External taxonomy suggestions **MUST** remain channel mappings and **MUST NOT** replace the internal classification.

Informative note: low-confidence classification routes to the taxonomy decision log rather than creating silent inconsistency.

11. Dictionary Mapping and Unknown Values (*normative*)

AI's role in mapping supplier values (colour names, materials, fits) to controlled dictionaries is to propose a mapping, not to hide ambiguity. Dictionary governance, aliases and unknown-value quarantine are defined in CDS-400.

Outcome	Required behaviour
Exact canonical or alias match	Use deterministic mapping; AI is unnecessary
High-confidence contextual match	AI may propose an existing canonical value with evidence
Ambiguous match	Return ranked candidates and require review
No suitable value	Create an unknown-value record or dictionary-extension request
Potential synonym	Propose an alias; do not add it silently

CDS700-R030 AI **MUST NOT** create a canonical dictionary value merely to avoid returning unknown.

CDS700-R031 A proposed alias **MUST** retain the source wording, language, supplier and approval status.

12. Product Content Generation (*normative*)

AI may draft titles, descriptions, feature bullets, care summaries and other commerce content. Generated content remains grounded in canonical facts and eligible evidence.

CDS700-R032 Generated content MUST NOT introduce product capabilities, compatibility, certifications, scarcity, sustainability or performance claims that are absent from approved information.

CDS700-R033 The workflow SHOULD distinguish factual fields, marketing interpretation and stylistic wording.

CDS700-R034 A generated title MUST preserve the identifiers and distinguishing attributes required by the organisation's declared product-title policy (R035).

CDS700-R035 An organisation using AI title generation MUST declare a **product-title policy**: a per-catalogue or per-category-profile statement of which identifiers (e.g. brand, model number) and which distinguishing attributes (e.g. colour, size, capacity) a title must preserve, and any ordering or length constraints. The policy is the testable reference for R034.

RESOLVED — accepted as drafted (owner 2026-08-04; was new (resolved)): The product-title policy had no home chapter in vo.1 (referenced only here), so it is defined here as a declared-policy mechanism. Alternative: relocate to CDS-600 (customer experience) or CDS-200 (canonical model) if a later revision gives titles a fuller treatment.

CDS700-R036 Generated content SHOULD cite or internally link the canonical fields used, so reviewers can trace each factual statement.

Informative note: a strong writing workflow generates from governed facts. It does not treat an old product description as unquestionable truth.

13. Translation and Localisation (*normative*)

Translation changes language. Localisation adapts terminology, units, spelling, regulatory wording and market conventions. These are modelled separately.

CDS700-R037 The canonical source language and approved translation MUST remain linked by version.

CDS700-R038 Dictionary values SHOULD use locale-specific display labels (CDS-400) rather than being translated independently in every product record.

CDS700-R039 AI translation MUST preserve product identifiers, measurements, brand names and regulated wording according to field policy.

CDS700-R040 A translation workflow MUST report omitted, ambiguous or culturally adapted content.

14. Image, Video and Document Analysis (*normative*)

AI may analyse product photographs, packaging, diagrams, manuals and video to propose attributes, alt text, media roles or quality findings.

CDS700-R041 Visual analysis MUST distinguish visible observation from inferred product fact.

CDS700-R042 AI-generated alt text SHOULD describe relevant visible content concisely and MUST NOT repeat unsupported marketing claims.

CDS700-R043 Image-derived colour SHOULD be treated as observational (E3) evidence: lighting, editing, display profiles and material reflectance can all shift apparent colour.

CDS700-R044 Sensitive or private information found in media MUST be handled under the organisation's security policy and MUST NOT be copied into product content by default.

15. Channel Optimisation and Mapping Assistance (*normative*)

Channel optimisation is a projection task; it must not alter canonical facts to satisfy one platform (CDS-P-03).

CDS700-R045 AI channel optimisation MUST operate on expected channel state (CDS-500), not rewrite canonical information to match a channel limitation.

CDS700-R046 Channel-specific generated text MUST be stored as a CH_ override or draft when it differs from canonical content (CDS-300).

CDS700-R047 Policy-sensitive marketplace attributes MUST be validated against current channel rules before publication.

CDS700-R048 An AI suggestion MUST NOT be treated as evidence that a channel will accept or display the value.

16. Search, Facet and Merchandising Assistance (*normative*)

AI may assist with synonyms, query interpretation, related-product suggestions, collection candidates and facet-gap analysis. These outputs influence customer discovery and therefore require measurement and governance; facet and search governance is defined in CDS-600.

CDS700-R049 AI MAY propose search synonyms; dictionary and search owners MUST approve terms that materially broaden or redirect meaning.

CDS700-R050 AI MUST NOT expose every generated concept as a customer facet.

CDS700-R051 Merchandising suggestions MUST remain distinguishable from taxonomy classification.

CDS700-R052 Personalised or generated collection membership SHOULD be explainable by declared signals and MUST respect availability and exclusion rules.

17. Confidence, Uncertainty and Abstention (*normative*)

Confidence is a decision aid, not proof. Model scores may be uncalibrated, task-specific or incomparable across versions. Confidence is interpreted together with evidence and validation:

Signal	Meaning
Model confidence	The system's estimate for its own proposal
Evidence strength	Authority and directness of source material (§7)
Dictionary certainty	Exact, alias, contextual, ambiguous or unknown mapping
Validation outcome	Whether schema and business rules pass
Risk class	Consequence if the result is wrong (Appendix B)
Review requirement	Human, policy or no approval required

CDS700-R053 A confidence score MUST NOT be used as the sole acceptance criterion.

CDS700-R054 The workflow MUST support an explicit abstain or insufficient-evidence outcome.

CDS700-R055 Thresholds SHOULD be calibrated on representative reviewed examples for the specific task and field; for bounded A3 this calibration is mandatory (R011).

18. Evidence, Provenance and Explainability (*normative*)

Explainability in CDS is practical lineage, not a demand that a model reveal internal reasoning.

CDS700-R056 An AI proposal MUST record the source references, model, workflow version, timestamp and validation results.

CDS700-R057 Where feasible, extracted facts SHOULD include a source snippet, document region, image region or field reference.

CDS700-R058 The implementation MUST NOT require or expose private model chain-of-thought as an audit mechanism.

CDS700-R059 A human-readable decision summary SHOULD explain the evidence, mapping and policy outcome without revealing sensitive internal reasoning.

19. Human Review and Approval (*normative*)

Human review is a governed task, not a decorative approve button. Reviewers need the proposal, current value, evidence, validation results, impact and available actions in one place.

Review action	Meaning
Accept	Write the proposal to the permitted target state
Edit and accept	Correct the proposal while preserving original AI output
Reject	Do not apply; record reason
Request evidence	Return to the workflow for additional eligible sources
Create dictionary request	Escalate an unknown or missing controlled value
Defer	Keep pending without acceptance
Bulk accept	Apply only when preview, scope and rollback are available

CDS700-R060 A reviewer MUST be able to see which fields will change before acceptance.

CDS700-R061 Bulk approval MUST provide a summary of affected products, values, risks and validation exceptions.

CDS700-R062 Human corrections SHOULD be captured as evaluation data, subject to privacy and governance policy.

20. Deterministic and Probabilistic Boundaries (*normative*)

A robust system combines AI with deterministic software. AI handles ambiguity and interpretation; deterministic components handle exact rules and repeatable transformations.

Prefer deterministic logic	Prefer AI assistance
Arithmetic and tax calculation	Extracting a price from unstructured text
Unit conversion after unit identification	Identifying a unit in a diagram
Exact dictionary aliases	Suggesting a mapping for a new supplier term
Required-field validation	Inferring which attributes are described in prose
Identifier format validation	Classifying a product from mixed evidence
Known channel transformation	Drafting a channel-specific summary

CDS700-R063 An implementation SHOULD use deterministic logic whenever the same inputs should always produce one objectively correct result.

CDS700-R064 For each task the workflow registry (§33) marks as deterministic-eligible, the registry entry MUST either name the deterministic implementation used or record why no reliable deterministic rule exists. (*This replaces the untestable vo.1 rule "AI MUST NOT be used to obscure a missing deterministic rule"*)

with an auditable mechanism.)

21. Validation and Guardrails *(normative)*

Guardrail layers: schema validation (types, cardinality, required fields, structure); dictionary validation (approved identifiers, aliases, unknown-value handling); business validation (conditional requirements, incompatibilities, range limits); evidence validation (eligible sources, conflicts); content validation (prohibited claims, duplicate wording, field length); channel validation (allowed values, required fields, destination limits); publication validation (authority, workflow state, expected-state generation).

CDS700-Ro65 Guardrails MUST be evaluated after AI output and before canonical acceptance or publication.

CDS700-Ro66 A failed guardrail MUST produce a specific reason code rather than a generic AI error.

CDS700-Ro67 A model instruction alone MUST NOT be treated as a sufficient guardrail for critical rules that can be enforced deterministically.

22. AI Semantic Namespace *(normative)*

AI_ is a registered core semantic namespace in the CDS-300 v0.2 namespace registry, reserved for AI work metadata and proposals. The prefix identifies the role of a field without changing the semantic identity of an accepted canonical value. Naming patterns are listed in Appendix C; the registry itself is owned by CDS-300.

CDS700-Ro68 AI_ fields MUST represent proposals, provenance, confidence or workflow metadata; they MUST NOT replace the canonical field namespace.

CDS700-Ro69 After acceptance, the value MUST be stored in its semantic canonical field (e.g. MF_material) while AI provenance remains linked in the audit record.

Informative note: MF_material means material. AI_material_suggestion means an AI proposal about material. The prefix communicates role immediately to humans, software and AI agents.

23. Prompt, Model and Configuration Versioning *(normative)*

AI behaviour can change when the model, system instruction, prompt template, tool set, retrieval source, dictionary or validation policy changes. Reproducibility depends on a versioned workflow, not the model name alone.

CDS700-Ro70 Every production AI result MUST identify the model and workflow version used.

CDS700-Ro71 Prompt templates, tool permissions, retrieval sources, output schemas and thresholds SHOULD be versioned together as one deployable configuration.

CDS700-Ro72 A workflow change that may alter accepted outputs MUST pass evaluation (§24) before production rollout.

CDS700-Ro73 Historical proposals MUST remain associated with the configuration that generated them.

24. Evaluation and Benchmark Sets *(normative)*

Evaluation uses representative, reviewed commerce examples, including ordinary products and difficult boundary cases.

Metric	Example application
Precision	How often accepted material mappings are correct
Recall	How often available attributes are successfully found
Exact-match rate	Category or dictionary identifier selection
Schema-valid rate	Structured output validity
Unsupported-claim rate	Generated content grounding
Abstention quality	Whether uncertain cases are correctly deferred
Reviewer correction rate	Human effort required after generation
Downstream verification rate	Whether accepted AI output publishes faithfully

CDS700-Ro74 An AI workflow MUST be evaluated before bulk production use.

CDS700-Ro75 The benchmark set SHOULD include diverse suppliers, categories, languages, missing data, conflicting evidence and unknown dictionary values.

CDS700-Ro76 Evaluation results MUST be associated with the exact workflow version.

25. Drift, Regression and Change Detection *(normative)*

Model providers, prompts, dictionaries and product distributions change. A workflow that performed well previously may degrade without an obvious software error.

CDS700-Ro77 Production workflows SHOULD be re-evaluated after material model, prompt, schema, dictionary or source changes.

CDS700-Ro78 The implementation SHOULD monitor changes in acceptance rate, correction rate, unknown-value rate, content violations and downstream mismatch rate.

CDS700-Ro79 A significant regression MUST be able to pause or roll back the affected workflow.

CDS700-Ro80 Previously accepted canonical values MUST NOT be silently regenerated merely because a model version changed.

CDS700-Ro81 Monitoring obligations MUST scale with the declared autonomy level of a workflow: segmented monitoring that distinguishes AI-originated from human-originated values MUST apply to A3 and A4 workflows, SHOULD apply to A2 workflows, and MAY be limited to volume and sampled quality checks for A1. Operational monitoring mechanics are defined in CDS-1400, which cites this rule for its AI segmentation requirement.

26. Security, Privacy and Data Handling (normative)

AI workflows may process supplier records, internal prices, unpublished products, contracts, customer material or credentials. The organisation controls what data leaves its boundary and which tools an agent may invoke.

CDS700-Ro82 Each AI workflow definition MUST declare the data classes it may receive as input; supplying data outside the declared classes is non-conformant. Declared inputs SHOULD be the minimum necessary for the task.

CDS700-Ro83 Secrets and credentials MUST NOT be included in model input. This prohibition admits no policy exception.

CDS700-Ro84 Private customer data MUST NOT be included in model input unless a declared data-handling policy explicitly permits it and states the safeguards applied (such as minimisation, masking or pseudonymisation, retention limits and access controls).

CDS700-Ro85 Tool permissions MUST be scoped to the workflow; read access MUST NOT imply write or publication access.

CDS700-Ro86 Logs and evidence records MUST follow retention and access-control policy.

27. Rights, Claims and Restricted Content (normative)

Commerce content may include copyrighted supplier text, trademarks, certifications, environmental claims and regulated statements. AI does not remove the need to establish rights and factual authority.

CDS700-Ro87 An AI workflow MUST NOT infer regulated or rights-bearing claims — such as certifications, origin claims, sustainability claims, medical claims or legal warranties — from style, context or similar products. *Informative note:* which claim categories are regulated, and how, is jurisdiction-dependent; the categories named here are examples, and organisations MUST rely on their declared claim policies for the binding list in each market.

CDS700-Ro88 Generated content MUST NOT reproduce supplier or third-party content beyond the rights the organisation holds to use it.

CDS700-Ro89 Generated content SHOULD avoid close stylistic imitation of identifiable third-party content even where reproduction rights are not directly at issue.

CDS700-Ro90 Brand and trademark wording MUST follow approved source and brand policy.

CDS700-Ro91 Restricted or regulated categories SHOULD use specialist review profiles and narrower autonomy levels.

28. Failure Modes and Recovery (normative)

Failure	Required response
Malformed output	Reject and retry only under bounded policy
Unsupported dictionary value	Quarantine or dictionary review (CDS-400 §17)
Conflicting evidence	Human review; preserve both sources
Tool or retrieval failure	Mark incomplete; do not fabricate missing input
Model timeout or refusal	Record status and preserve work object
Hallucinated claim	Reject, record reason and include in evaluation set
Bulk anomaly	Pause batch and provide rollback or staged recovery
Publication mismatch	Use CDS-500 reconciliation; do not assume AI output was accepted

CDS700-Ro92 Retries MUST be bounded and MUST NOT conceal repeated failure.

CDS700-Ro93 A fallback model or workflow MUST meet the same schema, evidence and evaluation requirements as the primary.

29. Agent and Workflow Orchestration (normative)

Agentic systems may coordinate extraction, validation, research, review routing and publication tools. An agent is not exempt from field authority, permissions or evidence rules.

Orchestrator

|— Evidence Reader (read only)

```
|-- Attribute Extractor (proposal only)
|-- Dictionary Mapper (proposal only)
|-- Validator (deterministic)
|-- Human Review Queue
`-- Publisher (explicitly authorised, CDS-500 controlled)
```

CDS700-R094 Each agent or tool **MUST** have a declared role and least-privilege permission set.

CDS700-R095 The orchestrator **MUST** preserve task lineage across delegated steps.

CDS700-R096 An agent **MUST NOT** grant itself broader authority based on its own confidence or task interpretation.

CDS700-R097 A human-readable workflow name and state **SHOULD** be visible to operators.

30. Publication Boundaries (*normative*)

AI acceptance and channel publication are separate events. A value may be approved as canonical yet still fail channel mapping, preflight validation or downstream publication.

CDS700-R098 AI workflows **MUST NOT** bypass CDS-500 expected channel state and publication preflight.

CDS700-R099 AI-generated channel overrides **MUST** be visibly channel-specific and independently reviewable.

CDS700-R100 Direct AI publication **MAY** be used only under A4 for an explicitly approved, low-risk workflow with rollback and downstream verification.

CDS700-R101 Price, inventory, legal status and safety-critical changes **SHOULD** remain outside direct AI publication unless governed by a specialist standard and deterministic controls.

31. Verification and Reconciliation (*normative*)

AI output is verified at two boundaries: against the canonical schema and evidence at acceptance, and against the observed downstream state after publication (CDS-500).

```
AI proposal
-> Canonical validation and approval
-> Expected channel representation
-> Publication
-> Channel observation
-> Semantic comparison
-> Health and reconciliation
```

CDS700-R102 An accepted AI value **MUST** be included in ordinary CDS-500 field-level verification when published.

CDS700-R103 Before a downstream mismatch involving AI-originated content is dispositioned as an AI-content defect, its triage record **MUST** evidence that mapping, transformation and channel behaviour were checked and excluded.

CDS700-R104 AI **MAY** assist with anomaly triage; repair actions remain governed by field ownership and CDS-500 policy.

32. Audit, Observability and Cost (*normative*)

Recommended operational telemetry: work volume and completion status; acceptance, rejection and correction rates; unknown and abstention rates; validation and guardrail failures by reason code; model, workflow and dictionary versions; latency, token or compute usage and cost; human review time; downstream publication and verification outcomes.

CDS700-R105 An implementation **MUST** retain sufficient audit data to reconstruct how an AI proposal reached canonical or published state.

CDS700-R106 Operational metrics **SHOULD** distinguish quality, throughput, cost and human effort.

CDS700-R107 A cost-optimisation change **MUST NOT** alter a workflow's declared evidence-class requirements, validation configuration or required review rate except through the workflow change process (§23, §33). (*This replaces the untestable v0.1 rule "cost optimisation MUST NOT silently reduce quality" with a reviewable mechanism.*)

33. Governance, Ownership and the Workflow Registry (*normative*)

AI governance belongs inside normal data governance (CDS-800), not in a separate experimental layer. Attribute owners, dictionary owners, channel owners and risk owners remain accountable for their domains. This section is the single home for AI governance rules; CDS-800 cites it.

CDS700-R108 Every production AI workflow **MUST** have an accountable owner, a declared autonomy level, a risk class, evaluation evidence and approved destination permissions.

CDS700-R109 Organisations operating production AI workflows **MUST** maintain a registry of approved workflows recording, per workflow: owner, role (§4), autonomy level, risk class, permitted fields and destinations, input data classes (R082), deterministic-eligibility disposition (R064) and current version. The registry is the mechanism behind R004's visibility requirement.

CDS700-R110 Workflow changes MUST follow versioning, evaluation, approval and rollback policy (§23–§25); model or prompt changes that can alter output semantics MUST pass change and regression controls before rollout.

CDS700-R111 AI-generated canonical changes MUST retain provenance and applicable human-review evidence (§18–§19).

CDS700-R112 Deprecated workflows MUST be disabled for new work while historical audit records remain readable.

Informative note: AI exceptions and waivers follow the CDS-800 exception process (expiry and review included); conformance levels, the AI-Assured overlay profile and the T-AI test suite are defined in CDS-1000, which traces its tests to the CDS700-Rmmm identifiers in this chapter.

34. Worked Examples (*informative*)

34.1 Apparel colour mapping

```
Source evidence: supplier value "French Navy" (E1 supplier record)
Task: dictionary mapping
AI proposal: canonical colour = navy
Dictionary status: existing canonical value
Facet projection: blue
Autonomy: A3 (bounded, R011 controls active) for approved supplier
Validation: pass
Canonical write: MF_colour = navy
Provenance retained: AI work object + source value
Publication: channel displays "French Navy", filter "Blue"
Verification: observed values match expected
```

34.2 Unknown homewares finish

```
Source evidence: "smoked eucalypt finish"
AI proposal candidates: smoked_wood (0.54), dark_wood (0.49)
Dictionary status: no exact or approved alias
Outcome: ABSTAIN
Action: create dictionary review request
Canonical value: unchanged
Customer facet: not published until approved
```

34.3 Product description generation

```
Inputs: canonical title, material, dimensions, care, included items;
        approved brand voice; prohibited-claim rules
AI output: draft short description and bullets
Validation: no unsupported claims; measurements match canonical fields
Autonomy: A2 governed draft
Human action: edit and accept
Publication: ordinary channel workflow
```

34.4 Channel category assistance

```
Internal category: Homewares > Living Room > Cushions
AI suggestion: candidate mappings for two external channel taxonomies
Result: channel mapping proposals only; no change to internal category
Review: channel specialist approves mappings
Publication: expected channel state generated and verified
```

Appendix A — Recommended AI Work Object Fields (*informative*)

The normative machine-readable contract is the CDS-1100 ai-proposal schema. Recommended semantic fields:

```
AI_work_id, AI_task_type,
SYS_product_id, SYS_variant_id,
AI_workflow_version, AI_model_id,
AI_input_evidence[], AI_output_schema_version,
AI_proposal{}, AI_confidence{},
AI_validation_results[],
AI_review_status, AI_reviewed_by, AI_reviewed_at,
AI_disposition_reason, AI_generated_at,
SYS_created_at, SYS_updated_at
```

The exact storage model may differ; the semantic information and lineage remain available.

Appendix B — Risk and Review Matrix (*informative*)

Category examples in this matrix are informative; the binding list of regulated categories is jurisdiction-dependent (Ro87).

Risk class	Examples	Maximum default autonomy	Review expectation
R1 — Low	Aliases, internal suggestions, alt-text drafts, anomaly flags	A3	Sampled or exception review
R2 — Moderate	Material mapping, classification, search synonyms, ordinary content	A2, or A3 only in bounded form (Ro11)	Human review or calibrated policy with sampling
R3 — High	Compatibility, origin, sustainability claims, channel policy attributes	A1 or A2	Qualified human approval
R4 — Critical	Price, tax, safety, regulated claims, hazardous-use instructions, legal warranty	A1	Specialist approval; deterministic controls required

Appendix C — Recommended AI Field Naming (*informative*)

The AI_ namespace registration lives in the CDS-300 registry; these patterns illustrate its use.

Pattern	Example	Meaning
AI__suggestion	AI_material_suggestion	Proposed value for a governed field
AI__confidence	AI_material_confidence	Task-specific confidence
AI__evidence	AI_material_evidence	Evidence references
AI__status	AI_classification_status	Workflow state
AI_model_id	AI_model_id	Model/version identifier
AI_workflow_version	AI_workflow_version	Versioned instruction, tools and policy
AI_reviewed_by	AI_reviewed_by	Human or policy that authorised disposition

Human-facing identifiers SHOULD remain descriptive even when the underlying implementation uses structured work-object properties.

Appendix D — Platform Profiles (*informative*)

Platform-specific guidance for AI-assisted publication (including the Shopify profile that appeared as vo.1 Appendix D) lives in CDS-900. The invariants are platform-independent: AI suggestions are created and reviewed in the PIM; accepted values publish through ordinary channel mappings; channel-only AI text uses CH_ overrides; read-back verifies AI-originated values exactly as human-originated ones; manual downstream edits to PIM-owned fields are drift unless an authorised override exists.

Appendix E — Operational Review Checklist (*informative*)

- Is the task role and autonomy level declared?
- Are the source materials eligible and referenced, with evidence classes assigned?
- Is the output schema explicit?
- Are dictionary values and unknown handling defined?
- Are deterministic validations applied, and deterministic-eligibility recorded in the registry?
- Can the workflow abstain?
- Is risk-appropriate review configured (including bounded-A3 controls where claimed)?
- Are model and workflow versions recorded?
- Can the change be previewed and rolled back?
- Will accepted values pass through normal publication and read-back verification?
- Are quality, correction rate and downstream mismatch monitored at the level the autonomy class requires?
- Can a new operator understand the AI fields and statuses without separate notes?

Commerce Data Standard (CDS)

CDS-800 — Governance, Ownership and Change Control

RESOLVED (was new (resolved), owner decision 2026-08-04): retitle accepted — "Governance, Ownership and Change Control". File renamed to match.

Field	Value
Status	vo.2 Review Draft (working source; not an approved standard)
Release	CDS vo.2
Date	2026-08-04
Supersedes	CDS-800 Working Draft vo.1, including its §9 conformance ladder (superseded by CDS-1000 per ADR-D1) and its exception/waiver dual vocabulary (superseded by the single Exception definition in CDS-100 §4)
Normative status	§1, §3–§8, §9–§30 are normative except where marked. §2 (principles), §31 (scenarios), Annexes A–C, E (templates/checklists) and Annex F (organisation example) are informative. Annex D is a normative pointer.
Findings addressed	800-1 (via ADR-D1), 800-2, 800-3, 800-5, 800-6, 800-7, 800-8, 800-9, 800-10; SYS-1 (CLEANUP-PASS §C: all 86 prefix-labelled rules corrected, including the one ambiguous case), SYS-2 (requirement IDs), SYS-4 (non-testable MUSTs); ADR-D1, ADR-D24; REVIEW-003 Matrix 5 dedup

1. Purpose and Scope (*normative*)

CDS-800 defines how a CDS implementation is governed over time: authority, ownership, decision rights, change control, quality oversight, audit evidence, security responsibility and the governance of conformance claims. Conformance **levels, tests and claim schemas** are owned by CDS-1000; this chapter governs who may claim, how claims stay honest, and how the organisation operates the standard day to day.

A technically capable PIM still fails if nobody owns the taxonomy, uncontrolled dictionary values are accepted, channel mappings change without review, or editors work around the master data layer. Governance exists to keep those failures from becoming normal operating practice.

CDS800-R001 Every CDS implementation MUST define accountable ownership for the canonical product model, each governed data domain and each publication channel.

Informative note: Governance does not require a large committee. A small organisation may assign several roles to one person (see §18 and ADR-800-008), but responsibilities and decision rights remain explicit. Governance is judged by operating behaviour and its evidence trail (§16), not by the existence of a policy document.

2. Governance Principles (*informative*)

- **Accountability precedes automation.** A system may automate work only after authority, ownership and exception handling are defined.
- **One owner, many contributors.** Many people and systems may contribute to a field; one authority resolves conflict (CDS-200 §13).
- **Enter data where it originates.** Information is captured and corrected in its system of authority, not patched downstream. Downstream edits to upstream-owned facts are drift, not maintenance (CDS-500 §11).
- **The standard is stable; implementations evolve.** Local tools and schemas change without weakening CDS principles (CDS-P-12, Governance Over Convenience).
- **Change is deliberate.** New fields, values, namespaces and mappings follow a visible proposal and review path.
- **Exceptions are explicit.** A deviation is recorded, time-bounded and reviewed — never an undocumented permanent rule.
- **Evidence survives decisions.** Approvals, migrations, mappings and verification results remain auditable.
- **Quality is measured.** Completeness, validity, consistency, mapping coverage and channel fidelity are observable (CDS-1400).
- **Access follows responsibility.** Users receive the minimum permission needed for their role.
- **Customer impact matters.** Governance decisions consider storefront discovery, accessibility, truthfulness and usability.
- **Conformance is testable.** Requirements are capable of objective assessment wherever practical (CDS-1000).

3. Governance Operating Model (*normative*)

```
CDS Standard and Profiles
-> Organisation Policies and Ownership
-> Schemas + Taxonomies + Dictionaries + Rules
-> Product Data Operations and Publication
-> Verification + Metrics + Audit Evidence
-> Review, Change and Continuous Improvement
```

CDS800-R002 The governance model MUST cover definition, operation, monitoring and change, and MUST NOT stop after initial PIM configuration.

CDS800-R003 An organisation SHOULD maintain one identifiable forum or accountable role capable of resolving cross-domain conflicts.

Informative note: The forum may be called a Data Council, Product Information Council, PIM Steering Group or any locally meaningful name. CDS governs function, not branding.

4. Decision Rights and Authority (*normative*)

Decision	Required authority	Typical consultation
Create or retire a product family	PIM Product Owner	Schema, taxonomy, channel and operations owners
Add or rename a category	Taxonomy Owner	Merchandising, SEO, channel and analytics owners
Add a canonical dictionary value	Dictionary Owner	Merchandising and channel owners
Change a facet grouping	Facet and Search Owner	Dictionary, search and merchandising owners
Add or change a canonical attribute	Schema Owner	Domain steward, integration and channel owners
Introduce a semantic namespace	CDS Governance Owner	Architecture, integration and operations owners
Change a channel mapping	Channel Owner	Schema, dictionary and verification owners
Approve an AI workflow for canonical writes	AI Governance Owner and Product Data Owner	Security, product and operations owners
Declare CDS conformance	Conformance Declaration Executive	CDS Governance Owner and assurance functions

CDS800-R004 A decision right **MUST** identify who may approve, who must be consulted and who must be informed.

CDS800-R005 The person requesting a material change **SHOULD NOT** be the only person approving it when the change can affect many products or channels.

CDS800-R006 The organisation **MUST** declare, in its implementation statement (§9), the thresholds that make a change *material* (for example: product count affected, number of channels affected, breaking-change class per §11).

RESOLVED — accepted as drafted (owner 2026-08-04; was new (resolved)): R006 converts the otherwise untestable "material change" qualifier (SYS-4) into a declared-threshold mechanism. Alternative: fix corpus-wide thresholds in CDS-1000 test conditions.

5. Roles and Responsibilities (*normative*)

This table is the **role register** for the CDS corpus. Other chapters and local artefacts use these role names; local titles **MAY** differ where the mapping is documented (§9).

Role	Accountability
CDS Governance Owner	Local interpretation of CDS, policy hierarchy, namespace introduction and unresolved cross-domain decisions.
Conformance Declaration Executive	The executive or accountable system owner who approves and signs conformance declarations (§29).
PIM Product Owner	System capability, roadmap, operational fitness and the master product layer.
Product Data Owner	A business data domain and its fitness for use.
Data Steward	Definitions, values, quality, issue resolution and daily governance within a domain (may specialise: Taxonomy Steward, Dictionary Steward).
Schema Owner	Product families, attribute definitions, requiredness and validation rules.
Taxonomy Owner	Internal category structure, boundaries, mappings and decision log.
Dictionary Owner	Canonical values, aliases, facets, channel representations and value lifecycle.
Facet and Search Owner	Customer-facing filter behaviour, search synonyms, ranking and usability outcomes.
Channel Owner	Destination capability, mappings, policy requirements, publication behaviour and channel health.
Integration Owner	Connector behaviour, transformations, retries, technical monitoring and data lineage.
AI Governance Owner	AI use-case approval, risk classification, evaluation, evidence and human-review policy (controls: CDS-700).
Security and Compliance Owner	Access, data handling, regulated claims, retention and incident obligations.
Auditor / Assurance Reviewer	Independent assessment of conformance, evidence and control operation.

CDS800-R007 One person **MAY** hold multiple roles, but each responsibility **MUST** remain explicitly assigned.

CDS800-R008 Role assignment **MUST** be reviewed when personnel, systems or operating models change.

6. Domain Ownership Contracts (*normative*)

A domain ownership contract states what a domain owns, consumes, enforces and commits to downstream.

Domain: colour Owner: Dictionary Owner Canonical facts: canonical shade, colour family, aliases Consumes: supplier colour text, image evidence Publishes: display label, search terms, channel colour values Controls: allowed values, mappings, deprecation, facet membership Quality measures: unmapped rate, ambiguous rate, channel mismatch rate

CDS800-R009 Each governed domain **SHOULD** have a concise ownership contract accessible to operators and automated agents.

CDS800-R010 A domain contract **MUST** distinguish canonical facts from derived display, facet and channel representations.

7. Policy and Document Hierarchy (*normative*)

Level	Artefact	Function
1	CDS specification	Vendor-neutral normative requirements and terminology
2	CDS profile	Industry, platform or overlay implementation constraints
3	Organisation policy	Local authority, risk, approval and operating rules
4	Data model and registries	Schemas, taxonomies, dictionaries, mappings and namespaces
5	Procedures and runbooks	How operators execute, verify and recover work
6	Records and evidence	Approvals, releases, exceptions, audits and verification results

CDS800-R011 Lower-level artefacts MUST NOT silently contradict higher-level normative requirements.

CDS800-R012 Where an implementation cannot satisfy a requirement, the deviation MUST be represented through the exception process (§14) rather than hidden in procedure.

8. Conformance Levels and Profiles (*normative*)

The single CDS conformance ladder is defined in CDS-1000: **Foundation** → **Structured** → **Publisher** → **Verified** → **Governed**, strictly cumulative, with **overlay profiles** (for example AI-Assured, CX, multi-tenant) claimable alongside a level. Level definitions, test suites, evidence packages and claim schemas are owned by CDS-1000 and are not restated here.

Supersession note (ADR-D1): the v0.1 CDS-800 §9 ladder (Foundation → Governed → Publisher → Verified → AI-Assured) is withdrawn. "CDS Governed" now names only the capstone level of the CDS-1000 ladder; "AI-Assured" is an overlay profile, not a level.

CDS800-R013 A profile MUST identify the CDS release it depends on.

CDS800-R014 A profile MAY add required attributes, dictionaries, rules or mappings, but MUST NOT weaken any base CDS requirement or the cumulativity of the conformance levels. (*The v0.1 exception-based escape clause is deleted — finding 800-2. Implementation-level deviations use the exception process (§14); a profile itself is never a vehicle for weakening.*)

CDS800-R015 Organisation-specific extensions MUST use governed identifiers registered per CDS-300 and MUST NOT collide with reserved CDS namespaces.

Informative note: Profiles make the standard practical without turning platform-specific behaviour into universal product truth.

9. Organisation-Specific Implementation Statement (*normative*)

Local choices — role-name mappings, approval and materiality thresholds (R006), product families, custom attributes and dictionaries, channel enablement, risk classes and human-review thresholds, quality targets, retention and access controls, approved extension namespaces — are recorded in one controlled place.

CDS800-R016 Local implementation choices MUST be documented in an implementation statement or equivalent controlled record.

CDS800-R017 Local naming MAY differ from CDS examples, provided semantic intent and the mapping to CDS terms remain unambiguous.

10. Change Proposal Lifecycle (*normative*)

```
Propose -> Triage -> Impact Analysis -> Consult Owners
-> Approve / Reject / Return -> Implement in Test
-> Validate and Migrate -> Release -> Observe and Review
```

Required proposal field	Purpose
Change identifier	Stable reference for discussion and audit
Problem statement	Explains the need, not only the desired solution
Affected domains	Schemas, products, dictionaries, channels and users
Proposed change	The exact model or rule alteration
Compatibility impact	Breaking and non-breaking effects (§11)
Migration plan	Data, connector and channel transition
Validation plan	How correctness will be tested
Owner and approver	Accountability
Target release	Connection to deployment control (§28)
Rollback or recovery	Response to adverse outcomes

CDS800-R018 Material changes (per the declared thresholds, R006) MUST be represented by a durable change proposal before production implementation.

CDS800-R019 Emergency changes MAY use an accelerated path, but MUST be documented and reviewed retrospectively.

11. Versioning and Release Policy for Governed Artefacts (*normative*)

Corpus and document versioning is defined in CDS-000 §4 (ADR-D5). This section governs **local** artefacts: schemas, dictionaries, taxonomies, mappings and profiles. Semantic-versioning principles apply, adapted to the impact on consuming systems and stored data.

Change class	Typical examples	Version effect
Major / breaking	Remove a field, change meaning, replace identifiers, incompatible cardinality	Major version
Minor / compatible	Add optional field, add dictionary value, add supported mapping	Minor version
Patch / corrective	Correct documentation, label or non-semantic metadata	Patch version

CDS800-Ro20 Version identifiers **MUST** change when a governed artefact changes semantically.

CDS800-Ro21 A release record **MUST** identify included changes, migrations, tests, approvals and effective date.

12. Compatibility and Migration (*normative*)

Migration execution (baselines, cutover, rollback mechanics) is defined in CDS-1300; this section governs the change-control obligations.

CDS800-Ro22 A breaking change **MUST** include a migration plan and a defined compatibility period, unless immediate removal is required for safety, law or severe data-integrity risk.

CDS800-Ro23 Migration tooling **SHOULD** provide a preview of affected products and values before bulk changes are applied.

CDS800-Ro24 Channel mappings and verification logic **MUST** be reviewed when canonical identifiers or value semantics change.

Informative note: Renaming a display label may be a patch. Replacing a stable canonical identifier is a breaking change.

13. Deprecation, Aliases and Removal (*normative*)

Active
-> Deprecated (replacement announced)
-> Migration in progress
-> Read-only alias / compatibility period
-> Removed from new use
-> Retired after evidence confirms no active dependency

CDS800-Ro25 Deprecated fields and values **MUST** identify a replacement or explicitly state that no replacement exists.

CDS800-Ro26 Aliases **MAY** support ingestion and migration, but **MUST** resolve to one canonical identity.

CDS800-Ro27 A deprecated identifier **MUST NOT** be reused for a different meaning.

CDS800-Ro28 Removal **MUST** be blocked while verified active dependencies remain, unless a documented risk decision authorises otherwise.

14. Exceptions (*normative*)

An **Exception** is defined once, in CDS-100 §4: a time-limited, documented, approved deviation from a requirement, with an expiry and a named owner. An exception acknowledges non-conformance; it does not convert failure into conformance. The vo.1 "waiver" noun is retired (CDS-100 §8, finding 800-3); the field-level verification status **WAIVED** remains and is defined in CDS-500.

Required field	Description
Identifier	Stable exception reference
Requirement	The CDS or local control being deviated from (requirement ID where applicable)
Scope	Products, fields, channels, teams or dates affected
Reason	Why conformance is not currently practical
Risk	Customer, operational, legal, security and data-quality effects
Compensating control	What reduces the risk during the exception
Owner and approver	Who accepts accountability
Expiry or review date	Prevents permanent undocumented drift
Resolution plan	How the organisation returns to conformance
Retest result	Evidence at closure that conformance was re-established

CDS800-Ro29 Every exception **MUST** have an owner, scope, rationale, compensating-control assessment and a review or expiry date.

CDS800-Ro30 Expired exceptions **MUST NOT** remain silently active.

CDS800-Ro31 Exception closure **MUST** record a retest result demonstrating that conformance was re-established, or a new approved exception continuing the deviation. (*New in vo.2 — finding 800-6.*)

15. Data Quality Governance (*normative*)

Data quality is assessed against fitness for use, not merely field population. The quality dimensions below are the governance vocabulary; canonical metric formulas, denominators and monitoring live in CDS-1400 (which owns Channel Health and the no-100%-for-unobserved-fields rule).

Dimension	Meaning (<i>informative</i>)
Completeness	Required information is present
Validity	Values conform to definitions, types, units and rules
Consistency	Equivalent facts agree across products and representations
Uniqueness	Identifiers and records do not create unintended duplicates
Timeliness	Information is updated within an appropriate operating window
Provenance	Source and decision history are known
Mapping coverage	Canonical values have required facet and channel mappings
Publication fidelity	Observed channel state matches expected state (CDS-500)
Usability	Customer-facing values support clear discovery and decisions

CDS800-Ro32 Quality rules MUST identify the applicable scope, severity, owner and remediation path.

CDS800-Ro33 A product SHOULD NOT be published while blocking quality rules fail.

CDS800-Ro34 Each governance metric (for example exception age, change failure rate) MUST be defined with an explicit formula and denominator such that repeated calculation produces comparable results.

CDS800-Ro35 Quality targets SHOULD be set by risk and business use rather than one universal threshold.

Informative note: A 100% score can still mislead if the wrong fields are required. Governance reviews the model as well as the metric.

16. Audit Evidence and Traceability (*normative*)

Evidence classes retained: current and historical schema definitions; category, dictionary, mapping and namespace versions; change requests and approvals; publication and observation records; verification outcomes and channel-health calculations; exceptions; AI evidence and workflow versions where applicable; access changes and privileged actions; migration results and rollback evidence; conformance assessments and declared scope.

CDS800-Ro36 Audit evidence MUST be sufficient to reconstruct who changed what, when, under which authority and with what downstream effect.

CDS800-Ro37 Retention periods MUST be declared per record class in the implementation statement and reflect legal, operational and assurance needs.

17. Security and Access Governance (*normative*)

CDS800-Ro38 Access MUST follow least privilege and be granted according to role responsibilities.

CDS800-Ro39 Privileged actions such as schema change, bulk publication, mapping replacement and canonical-value deletion SHOULD require stronger authentication and enhanced audit logging.

CDS800-Ro40 Supplier credentials, channel tokens and other secrets MUST NOT be exposed through human-readable product fields or AI model inputs. (*Absolute — aligned with the CDS-700 §26 credential rule, finding 700-3; the v0.1 "approved need" escape applied only to non-secret confidential source data, which remains policy-gated under CDS-700.*)

CDS800-Ro41 Production and test environments SHOULD be separated.

18. Segregation of Duties (*normative*)

Activity	Recommended separation (<i>informative</i>)
Create schema change	Requester separate from final approver for material changes
Create channel credentials	Credential administrator separate from ordinary product editors
Bulk publish	Operator separate from policy owner for high-impact releases
Approve high-risk AI canonical writes	Reviewer separate from workflow developer (CDS-700)
Declare conformance	Conformance Declaration Executive supported by independent assessment
Close serious incident	Incident owner supported by verification or assurance review

CDS800-Ro42 Material high-risk changes SHOULD involve at least two distinct accountable participants.

Informative note (small teams — ADR-800-008): Where full organisational separation is impractical, small teams MAY substitute system controls, delayed approval, peer review or periodic independent review. The CDS-1300/CDS-1400 small-team profile cross-references this accommodation; roles are hats, responsibilities are not erased.

19. Multi-Tenant and Organisation Isolation (*normative*)

CDS800-Ro43 Organisation-specific products, dictionaries, mappings, credentials, exceptions and audit records MUST be isolated from other organisations.

CDS800-Ro44 A shared CDS baseline MAY provide definitions and templates, but MUST NOT expose or merge tenant-specific values without explicit authorisation.

CDS800-Ro45 Imports and bulk actions MUST execute within a declared organisation context.

CDS800-Ro46 Cross-tenant administrative access MUST be privileged, logged and reviewed.

Test-coverage note (finding 800-7): conformance tests for Ro43–Ro46 are bound to the **multi-tenant overlay profile** defined in CDS-1000; single-tenant implementations declare the profile not applicable.

20. Supplier and Import Governance (*normative*)

CDS800-Ro47 Every supplier source MUST have an identified owner, acquisition method, field mapping, refresh policy and trust classification.

CDS800-Ro48 Raw source values SHOULD be preserved separately from resolved canonical values (value model: CDS-400).

CDS800-Ro49 Bulk imports MUST provide validation results and a preview of high-impact changes before commit.

Unknown-value quarantine and the prohibition on imports silently creating canonical values are defined in CDS-400 §17 and are not restated here.

21. Taxonomy Governance (*normative*)

Governed content per taxonomy: category name, definition and inclusion boundary; parent-child relationships and depth; primary-category assignment rules; tie-breaking decision log; inherited family or attribute rules; external taxonomy mappings; collection and navigation projections.

CDS800-Ro50 Every category MUST have a recorded scope definition that distinguishes what belongs from what does not.

CDS800-Ro51 A taxonomy change MUST analyse product reassignment, collection, navigation, facet, SEO and channel impacts before approval.

CDS800-Ro52 Ambiguous product-type decisions SHOULD be recorded in a taxonomy decision log.

22. Dictionary and Reference-Data Governance (*normative*)

The value model — stable canonical identifiers, alias resolution, promotion of source values, quarantine — is owned by CDS-400. This section governs the change-control obligations around it.

CDS800-Ro53 New canonical values MUST be checked for existing synonyms, aliases and facet membership before approval.

CDS800-Ro54 Facet-grouping changes MUST consider customer usability, product counts, search behaviour and channel effects (facet requirements: CDS-600).

CDS800-Ro55 Dictionary owners MUST monitor unknown, ambiguous and unmapped value rates.

CDS800-Ro56 Channel representations MUST be reviewed when destination vocabularies change.

23. Attribute and Schema Governance (*normative*)

Change question (<i>informative</i>)	Required consideration
Does the attribute describe a stable product fact?	If not, it may belong to workflow, channel or merchandising state
Is an existing attribute semantically equivalent?	Avoid duplicates and near-duplicates
What is the scope and cardinality?	Product, variant, category or channel; single or multi-value
Is a dictionary required?	Prefer controlled values for filterable and mapped information
Who owns the value?	Assign authority and stewardship
How is it validated?	Type, range, unit, dependency and source evidence
How is it displayed and filtered?	Separate canonical, display and facet behaviour
How is it published and verified?	Define channel mappings and comparison strategy

CDS800-Ro57 A new attribute MUST have an Attribute Definition (CDS-200 §7) before product values are created at scale.

CDS800-Ro58 Attribute removal or semantic change MUST follow the compatibility and migration controls of §12.

24. Namespace Governance (*normative*)

The single normative prefix registry, the naming tests and the reserved-prefix rules are owned by CDS-300.

CDS800-Ro59 Introduction, deprecation or retirement of a semantic namespace MUST follow the change proposal lifecycle (§10) and be recorded in the CDS-300 registry.

CDS800-Ro60 Legacy prefix aliases MAY remain during migration, but new fields SHOULD use the current approved namespace.

The DF_ → CH_ transition is recorded once, in ADR-D24 (see CDS-300); it is not restated here.

25. Channel Governance (*normative*)

Channel governance object	Minimum content
Capability profile	Supported fields, limits, types, observability, update semantics (content: CDS-900)
Mapping registry	Canonical source, transformation, destination and version
Ownership contract	Channel owner, integration owner and escalation path
Publication policy	Eligibility, scheduling, retry and approval rules (CDS-500)
Verification policy	Read-back coverage, comparison and health thresholds (CDS-500)
Override policy	Allowed local differences and expiry/review rules (CDS-500 §22)
Change watch	How platform schema and policy changes are detected and assessed

CDS800-Ro61 Each active channel **MUST** have an owner and a maintained capability profile.

CDS800-Ro62 Channel changes **MUST** be assessed against mappings, publication, observation and verification before production rollout.

26. AI Governance (*normative*)

All AI controls — autonomy levels, risk classes, evidence, evaluation, provenance, human review, exception handling for AI outcomes — are owned by CDS-700 and are not restated here (*finding 800-8; the vo.1 §29 duplicates are deleted*). This chapter contributes only the role-assignment rule:

CDS800-Ro63 Every production AI workflow **MUST** have a named owner recorded in the workflow registry (CDS-700).

27. Incident, Drift and Recovery Governance (*normative*)

Incident management process, severity model and monitoring are owned by CDS-1400; drift classification and repair by CDS-500. This section governs the accountability obligations.

Incident class (<i>informative</i>)	Expected response
Data integrity (wrong canonical values, duplicate identities)	Contain, assess scope, correct and verify
Publication (incorrect bulk update, partial channel writes)	Pause, reconcile, retry or roll back
Drift (downstream edits, platform transformations)	Classify ownership, restore or approve override (CDS-500)
Security (credential exposure, cross-tenant leakage)	Revoke, contain, investigate, notify as required
AI quality (systematic hallucination, regression)	Disable workflow, review accepted outputs (CDS-700)
Governance (expired exception, bypassed approval)	Record non-conformance, restore control, review cause

CDS800-Ro64 A material incident **MUST** have an owner, severity, affected scope, containment action and closure evidence.

CDS800-Ro65 Incident closure **MUST** include verification that canonical and downstream states are correct.

CDS800-Ro66 Repeated incidents **SHOULD** trigger root-cause review of schema, process, tooling or authority design.

28. Release and Deployment Governance (*normative*)

```
Approved Change Set -> Build / Configuration -> Test Data Validation
-> Migration Preview -> Approval Gate -> Production Release
-> Publication Observation -> Verification and Health Review
-> Close or Roll Back
```

CDS800-Ro67 A production release **MUST** identify the included change records and the versions of affected governed artefacts.

CDS800-Ro68 High-impact releases **SHOULD** use staged rollout, limited product cohorts or channel canaries.

CDS800-Ro69 A release **MUST NOT** be considered complete until required verification and health checks pass.

29. Conformance Claims Governance (*normative*)

Level definitions, test suites, evidence packages and the claim/manifest schema are owned by CDS-1000 (claim schema: CDS-1000 §23). This section governs claim integrity and authority.

CDS800-Ro70 A conformance claim **MUST** describe the implemented scope and level honestly; an organisation **MUST NOT** claim conformance to capabilities it has not implemented.

CDS800-Ro71 An implementation **MUST** pass all **MUST** and **MUST NOT** requirements applicable to its declared scope.

CDS800-Ro72 SHOULD requirements MAY be unmet only where a documented rationale exists. (*Keyword corrected per the CLEANUP-PASS §C ambiguous-case ruling: the governing modal is MAY; "SHOULD requirements" is a noun reference to the requirement class.*)

CDS800-Ro73 A conformance claim MUST be accompanied by a dated conformance statement conforming to CDS-1000 §23, approved by the Conformance Declaration Executive.

CDS800-Ro74 An organisation MAY declare different conformance levels or overlay profiles for different channels or business units, provided each claim's scope is explicit.

CDS800-Ro75 Assessment method MUST be stated using the CDS-1000 §20 vocabulary: **self-attestation**, **customer assessment** or **independent assessment** (*finding 800-5; replaces vo.1 "self-assessed / partner-reviewed / independent"*).

CDS800-Ro76 A certification mark or badge MUST NOT imply independent assessment when only self-attestation occurred.

CDS800-Ro77 Material scope or control changes SHOULD trigger reassessment before a previous claim is reused.

Informative example claim (CDS-1000 naming): CDS vo.2 · scope: apparel catalogue, AU organisation, Shopify and Google channels · level: **CDS Verified** · overlay profiles: none (AI-Assured overlay not claimed) · assessment: self-attestation, 2026-08-04 · known exceptions: EXC-014, expires 2026-10-01. Full field set: CDS-1000 §23.

30. Review Cadences and Forums (normative)

Cadence (informative)	Typical review
Daily / event-driven	Import failures, publication errors, blocking quality issues, security alerts
Weekly	Unknown values, workflow queues, channel drift, ageing incidents
Monthly	Quality scorecards, mapping coverage, AI evaluation, change backlog
Quarterly	Taxonomy, dictionary, namespace, exception and access review
Per release	Schema, mapping, migration and conformance impact
Annually	Governance model, ownership, profiles, conformance claim, roadmap

Minimum recurring-review requirements are owned by CDS-1400 §24; the table above is a recommended shape, not a mandate.

CDS800-Ro78 Review cadence SHOULD reflect the rate and risk of change in the governed domain.

CDS800-Ro79 Review outcomes SHOULD create assigned actions rather than remain informational only.

31. Worked Governance Scenarios (informative)

31.1 Supplier introduces a new colour name. Supplier value "Midnight Ocean" enters the source layer; dictionary matching finds no approved alias; the value is quarantined (CDS-400 §17). The Dictionary Steward reviews imagery and supplier evidence and aliases it to canonical navy, retaining "Midnight Ocean" as the display label; facet membership is blue; channel values inherit existing navy mappings. The decision and evidence are recorded; affected products are revalidated, published, and read-back verifies the expected result (CDS-500).

31.2 Proposed category rename. "Throw Pillows" → "Cushions": the internal category identifier remains stable; display label, navigation text and storefront collection title change; URLs require redirect review; external taxonomy mappings are unchanged; analytics dimensions need an alias/continuity plan; legacy collection_* tag compatibility is assessed. Approved as a minor compatible release.

31.3 Manual downstream edit creates drift. Expected channel title "Linen Relaxed Shirt"; observed "Summer Linen Shirt"; verification result MISMATCH. The title is PIM-owned with no active override, so this is unauthorised drift: restore the expected title on the next controlled publication, notify the editor of the ownership rule ("enter data where it originates"), retain the observed value and reconciliation evidence.

31.4 High-risk AI workflow change. A request to let AI write compliance claims directly into canonical fields is rejected (claims may require evidence and legal approval — CDS-700). Approved alternative: AI drafts a proposal with mandatory evidence attachment; the compliance owner reviews; the accepted value enters the canonical record with provenance; publication remains subject to channel verification.

32. Architecture Decision Records (normative record; rationale informative)

ADR	Decision	Status	Rationale
CDS-ADR-800-001	Governance is a conformance requirement	Accepted (scope updated)	A sound data model degrades without ownership, controlled change and measurable operation. Under ADR-D1, full governance is the Governed capstone level ; ownership basics (Ro01) bind from Foundation.
CDS-ADR-800-002	Conformance is scoped and levelled	Superseded by ADR-D1	The single ladder and overlay-profile model now live in CDS-1000.
CDS-ADR-800-003	One authority per product fact	Accepted	Multiple simultaneous masters create unresolved drift and accountability gaps (CDS-200 §13).
CDS-ADR-800-004	Exceptions expire or are reviewed	Accepted	Unbounded exceptions become hidden permanent architecture.
CDS-ADR-800-005	Stable identifiers are governed separately from labels	Accepted	Display language evolves without breaking identity and mappings.

ADR	Decision	Status	Rationale
CDS-ADR-800-006	Release completion includes verification	Accepted	Deployment success does not prove correct downstream state.
CDS-ADR-800-007	Certification claims distinguish assessment type	Accepted	Self-attestation, customer assessment and independent assessment are not equivalent (vocabulary: CDS-1000 §20).
CDS-ADR-800-008	Small teams may combine roles but not erase responsibilities	Accepted	Governance must remain practical while preserving accountability. Cross-referenced by the CDS-1300/CDS-1400 small-team profile.

Annex A — Recommended RACI Matrix (*informative*)

Role names are those of the §5 register.

Governance activity	Accountable	Responsible	Consulted	Informed
Canonical product model	PIM Product Owner	Schema Owner	Product Data and Channel Owners	Product teams
Taxonomy change	Taxonomy Owner	Data Steward (taxonomy)	Merchandising, SEO, Channel Owners	Editors and analysts
Dictionary value approval	Dictionary Owner	Data Steward (dictionary)	Merchandising and Channel Owners	Import and content teams
Channel mapping release	Channel Owner	Integration Owner	Schema Owner, verification owners	Commerce operations
AI workflow approval	AI Governance Owner	AI workflow owner (per CDS-700 registry)	Product Data, Security and Compliance Owners	Reviewers and operators
Conformance statement	Conformance Declaration Executive	CDS Governance Owner	Assurance Reviewer and domain owners	Stakeholders

Annex B — Change Proposal Record (*informative template*)

Change ID · Title · Requester · Problem statement · Proposed change · Affected artefacts · Affected product count · Affected channels · Compatibility class · Risk and customer impact · Migration plan · Validation plan · Rollback plan · Required approvals · Target release · Outcome and evidence.

Annex C — Exception Record (*informative template*)

Exception ID · Requirement (ID) · Scope · Reason · Risk · Compensating control · Owner · Approver · Start date · Expiry/review date · Resolution plan · **Retest result** (evidence at closure, Ro31) · Status.

(The vo.1 "Exception and Waiver Record" title is retired with the waiver noun — CDS-100 §8.)

Annex D — Conformance Statement (*normative pointer*)

The conformance statement and claim/manifest schema are defined in CDS-1000 §23 and encoded in CDS-1100. The vo.1 Appendix D template is withdrawn (*finding 800-6*); CDS-800 adds only the governance obligations of §29 (executive approval, honest scope, dated statement, assessment vocabulary).

Annex E — Governance Readiness Checklist (*informative*)

- ☐ PIM and canonical authority are explicitly declared (CDS-200 §13).
- ☐ Every governed domain has an accountable owner.
- ☐ Taxonomy, dictionary, schema and namespace registries exist.
- ☐ Channel owners and capability profiles exist.
- ☐ Change, release, migration and rollback processes are documented.
- ☐ Exceptions have owners, review dates and retest evidence at closure.
- ☐ Quality metrics and review cadences are defined.
- ☐ Publication observation and verification are operational for claimed channels.
- ☐ Access and tenant isolation are reviewed.
- ☐ AI workflows have named owners and CDS-700 controls.
- ☐ Conformance statement scope is accurate and supportable with evidence (CDS-1000).

Annex F — Organisation Example: Legacy Documentation Alignment (*informative — organisation-specific example; not part of the vendor-neutral standard*)

This annex records how one adopting organisation's pre-CDS artefacts map to this chapter (*finding 800-10: all organisation-specific material consolidated here*). It illustrates that CDS governance can be reached bottom-up from working practice; no artefact named here is a CDS requirement.

Existing artefact or pattern	CDS-800 interpretation
Legacy product-data workflow	Operational evidence of source separation, canonical resolution, channel output and verification responsibility
Legacy tag-governance workflow	A governed registry, namespace discipline, deprecation model and PIM authority over channel tags

Existing artefact or pattern	CDS-800 interpretation
Legacy implementation sequencing guide	A controlled implementation sequence in which taxonomy and governance precede product-scale publication
Airtable Import_* / Airtable_* / Shopify_* / Match_* layers	Clear separation of supplier intake, authority, observed state and verification evidence — a working example of §20's source/canonical separation
Traffic-light formulas	Human-facing control status backed by detailed reconciliation logic (now: CDS-500 §18)
Taxonomy decision log	A practical record for ambiguous classification decisions (R052)
Profile H / Profile S choice	An organisation-level implementation policy that must remain consistent or be migrated deliberately (§9)

RESOLVED — accepted as drafted (owner 2026-08-04; was new (resolved)): Annex placement. REVIEW-003 Matrix 5 proposes one corpus-wide legacy-alignment annex (candidate homes: CDS-1300 or standalone). This chapter's organisation-specific material is consolidated here for now; if a corpus-wide annex is created at v0.2 freeze, this annex merges into it and becomes a pointer.

END OF CDS-800 v0.2 REVIEW DRAFT

Commerce Data Standard (CDS)

CDS-900 — Implementation Profiles and Platform Mappings

Field	Value
Status	vo.2 Review Draft (working source; not an approved standard)
Release	CDS vo.2
Date	2026-08-17
Supersedes	CDS-900 Working Draft vo.1; the platform-specific material of CDS-500 vo.1 Appendices C and D (merged here per REVIEW-006 systemic finding 2)
Normative status	§1–§3 (profile model, precedence, mapping contract), the profile rules in §4–§11 and §13 (conformance) are normative. All platform-behaviour statements, mapping tables marked informative, §12 (migration), the appendices and every IMPLEMENTATION NOTE are informative.
Findings addressed	SYS-1 (errata manifest §C: all 64 prefix-labelled rules), CDS900-1, CDS900-3, CDS900-4, CDS900-5, CDS900-6, CDS900-7, CDS900-8, CDS900-9, CDS900-10, 200-7, CDS500-6 (corollary), Matrix 1 (STD_ row), Matrix 5; ADR-D2, ADR-D3, ADR-D24; REVIEW-006

Profile releases defined by this chapter: CDS-SHOPIFY-0.2 (§4), CDS-GMC-0.2 (§5), CDS-META-0.2 (§6). Platform references were retrieved and spot-verified on **2026-08-03**, with Shopify Search & Discovery limits refreshed on **2026-08-17** (REVIEW-006); Appendix D carries per-reference dates.

IMPLEMENTATION NOTE (*informative*): Platform behaviour changes over time. Platform-specific requirements in this chapter are **versioned profile rules, not permanent changes** to the canonical CDS model. Each platform profile carries a release identifier and dated references so that a claim of conformance is always a claim against a stated platform snapshot.

1. Purpose and Profile Model (*normative*)

CDS defines a canonical commerce information model. A profile explains how that model is implemented within a platform, channel, industry or organisation without allowing the implementation target to become the master product layer.

CDS900-R001 Every implementation profile **MUST** identify its scope, mapping rules, platform constraints, publication behaviour, observation coverage and conformance tests.

CDS900-R002 Every platform profile **MUST** carry a profile release identifier and **MUST** date every external platform reference it relies on (retrieved-on date).

CDS900-R003 A platform profile **MUST NOT** redefine canonical product truth merely because the target platform lacks a corresponding field.

CDS900-R004 An industry profile **MAY** require additional attributes and dictionaries, but **MUST** preserve the CDS core semantics.

CDS900-R005 An organisation profile **MAY** extend a platform or industry profile, but **MUST** document every divergence.



2. Profile Hierarchy and Precedence (*normative*)

Profiles are applied in layers.

Layer	Purpose	Example
CDS Core	Vendor-neutral semantics and architecture	Canonical colour, facet colour, channel mapping
Industry Profile	Domain-specific fields and rules (register: CDS-1500 §2.1)	Apparel fit, garment size; homewares room and finish; cosmetic ingredient lists; food allergens; electronics specifications; parts fitment; toy age warnings
Platform Profile	Platform storage and delivery mapping	Shopify product category, Google product_type
Organisation Profile	Store-specific vocabularies and policies	Brand colour names, collection hierarchy
Product Record	Actual values and permitted overrides	French Navy shirt in size M

CDS900-R006 Precedence rule: more specific profiles (industry → platform → organisation) MAY add constraints; **CDS core semantics remain authoritative for every shared concept**. Where two profiles conflict, the implementation MUST preserve the core CDS meaning and MUST record the narrower profile decision explicitly. (This restates CDS-000 §6 CDS000-R011 in profile terms; a profile never weakens a core requirement.)

CDS900-R007 A platform limitation MAY create an exception in the publication projection, but MUST NOT silently alter the canonical record.

3. Common Platform Mapping Contract (*normative*)

Every channel mapping is described by a common contract. This enables consistent publication and verification even when platforms use different APIs, files or terminology.

Contract Property	Meaning
Canonical source	The PIM field or derived canonical value.
Target path	The channel field, metafield, attribute or feed column.
Transformation	The deterministic conversion from canonical to expected channel value.
Requiredness	Always, conditional, optional or unsupported.
Cardinality	Single, list, set or variant-specific.
Write semantics	Replace, merge, append, upsert or derived.
Read-back path	How the downstream value is observed.
Comparison rule	Exact, normalized, numeric, set, ordered list or identifier.
Ownership	PIM, channel, shared by explicit contract, or observed only.
Failure policy	Block, warn, omit, quarantine or retry.

CDS900-R008 Where a transformation is defined, verification MUST compare the expected channel representation with the observed channel representation, not raw PIM text. (Lifecycle, statuses and comparison semantics: CDS-500.)

CDS900-R009 A mapping MUST declare whether the channel value is observable. Missing and unobservable MUST remain different verification states (CDS-500 §17).

CDS900-R010 Where an MF_ attribute is published to a channel's native or standard field rather than a metafield-like structure, that is a mapping concern of the platform profile; the canonical prefix does not change (ADR-D2). A profile MUST document MF_→metafield, MF_→native-field and MF_→feed-column projections per mapping.

4. Shopify Implementation Profile — CDS-SHOPIFY-0.2 (*normative rules; platform statements informative, references retrieved 2026-08-03*)

The Shopify profile maps CDS product information into Shopify standard fields, product categories, category metafields, custom metafields, variants, collections, tags, media and Search & Discovery filters. Shopify is a publication and transaction platform; the PIM remains authoritative for PIM-owned product information (CDS-200).

4.1 Shopify Object Boundaries (*normative*)

CDS Concept	Shopify Target	Authority
Canonical product	Product resource and related custom data	PIM
Sellable variant	Product variant	PIM for descriptive identity; inventory authority may differ
Internal category	No direct equivalent; mapped to product category and store collections	PIM
Shopify standard category	Product category	PIM mapping
Product type	Product type field	PIM
Canonical attribute	Category metafield or custom metafield	PIM
Facet value	Search & Discovery filter source or controlled projection	PIM
Merchandising collection	Automated or manual collection	PIM or governed merchandiser
Workflow state	Normally retained in PIM; optional internal tag/metafield	PIM
Observed state	Read-back through connector/API	Channel observation only

CDS900-R011 PIM-owned product fields edited directly in Shopify SHOULD be considered downstream drift unless an approved override workflow exists (CDS-500 ownership modes).

Edit protection and drift attribution on Shopify (*informative platform facts, verified 2026-08-04 — REVIEW-006A; implements CDS-500 §20.1 for this profile*)

What permissions can and cannot do. Shopify staff permissions for products are: View, View cost, Create and edit, Edit cost, Edit price, Export, Delete — **there is no field-level or per-product edit restriction** on any plan (Plus adds user groups/custom org roles but the product permission set is the same). The one real permission lever: staff who do not need to edit products can have "Create and edit" withheld entirely. Note the metafield implication: anyone with product

edit rights can also add, edit and **delete metafield definitions and values** — merchant-owned metafield definitions cannot be protected from staff; the only edit-proof mechanism is app-owned reserved namespaces (with the D17 visibility trade-off, declared per this profile).

Attribution mechanisms and their limits (for CDS500-R077b correlation): | Mechanism | What it gives | Limits | |---|---|---| | GraphQL `events` query / Events resource | Timestamped events with `author` string, app/admin attribution booleans | Product verbs are create/destroy/published/unpublished — **routine field edits may generate no event**; author is a string, not a typed staff ID; 1-year retention | | Admin Store activity log | Recent admin actions with actor name (person/app/channel) | View-only, max 250 results, no export; bulk edits/background jobs/app syncs can show "Shopify" instead of the actor | | `products/update` webhook | Near-real-time change signal with timestamps | **Payload contains no actor field** — tells you *when*, never *who* | | User-management activity log (CSV export, 6 months) | User/role changes | Does not cover product edits | | PIM's own publication records | Complete provenance for API writes made by the PIM | Covers only the PIM's own writes |

Consequence: on Shopify, drift attribution is **best-effort correlation**: the CDS-500 §20.1 detection interval (previous vs new observed value, bounded by observation timestamps) narrowed against the Store activity log's recent window and webhook `X-Shopify-Triggered-At` timestamps. The implementation SHOULD record the attribution outcome class honestly (actor identified / actor class identified / unattributable) and MUST NOT promise field-level audit trails this platform does not provide.

CDS900-R012 The Shopify connector MUST know, per target field, whether an update replaces or merges the downstream value, and MUST declare this in the mapping's write semantics.

CDS900-R013 The connector MUST use the correct current mutation for each object class and MUST NOT assume one product mutation covers variants.

Platform note (*informative, verified 2026-08-03*): in the current Shopify GraphQL Admin API, `productUpdate` **no longer updates variant data**; variant changes require `productVariantsBulkUpdate` or `productSet`. Connectors written against older API behaviour will silently stop updating variants. Source: [S6].

4.2 Shopify Namespace Mapping (*normative table; CDS-300 is the registry of record*)

The namespace registry lives solely in CDS-300; this table states only how registered namespaces **project onto Shopify targets**. It does not define any namespace.

CDS Namespace (per CDS-300)	Shopify Projection	Example
STD_ (canonical core field, closed enumeration per ADR-D2)	Mapped to the corresponding native Shopify field	STD_title -> Product title
CAT_	Internal classification; mapped product category and product type projections	CAT_shopify_category_id
VAR_	Option and variant data	VAR_size, VAR_colour
MF_ (extended structured attribute, mechanism-neutral per ADR-D2)	Typically a category or custom metafield; MAY project to a native field where one exists	MF_material -> category metafield
CH_shopify_	Shopify-specific output or override	CH_shopify_title_override
SEO_	Search engine listing fields	SEO_title -> search engine listing title
MED_	Product media and alt text	MED_primary_image
OBS_shopify_	Observed Shopify state (read-back), never canonical	OBS_shopify_material
QA_shopify_	Comparison result and verification status	QA_shopify_material
WF_	PIM workflow; not customer-facing, published only by exception	WF_ready_to_publish

IMPLEMENTATION NOTE (*informative*): MF_ identifies the human-recognisable attribute domain. The internal technical key may still use a durable namespace and key such as `cds.material` or `custom.material` (CDS-300 visible-identifier vs internal-key separation).

4.3 Classification and Taxonomy (*normative*)

Platform note (*informative, verified 2026-08-03*): each Shopify product has **one** product category (from Shopify's Standard Product Taxonomy) and **one** merchant-defined product type; assigning a standard category unlocks the category metafields associated with that category. Source: [S1].

CDS900-R014 The PIM internal category MUST remain separate from the Shopify product category and the Shopify product type.

CDS900-R015 The PIM SHOULD map each final internal category to one Shopify Standard Product Taxonomy category identifier or breadcrumb.

CDS900-R016 The mapping SHOULD be stored on the category entity and inherited by products unless a documented product-level exception is required.

CDS900-R017 The Shopify product type SHOULD contain a controlled merchant vocabulary and MUST NOT be a concatenation of descriptive attributes.

```
Internal category:
  Apparel > Women > Tops > Shirts
Shopify projections:
  Product category = Apparel & Accessories > Clothing > Clothing Tops > Shirts
  Product type    = Shirt
  Collections     = Women, Tops, Shirts
  Category metafields = fabric, colour, neckline, sleeve length, target gender...
```

IMPLEMENTATION NOTE (*informative, not spot-verified this pass*): Shopify may update or replace taxonomy nodes. The PIM should version category mappings and treat taxonomy migration as a controlled profile change, not a manual per-product repair.

4.4 Attributes, Metafields and Metaobjects (normative)

Platform note (*informative; unlock mechanism verified 2026-08-03, metaobject-entry detail partially verified*): Shopify category metafields are standardized product attributes associated with a product category; their entries use Shopify metaobjects and standardized base values. Custom metafields remain available for information not represented by the category model. Sources: [S1] [S3].

CDS900-Ro18 When an appropriate Shopify category metafield exists, the Shopify profile SHOULD map the corresponding canonical attribute to it.

CDS900-Ro19 A custom metafield SHOULD be used when the canonical attribute is not available as a category metafield, is store-specific, or requires a different type or lifecycle.

CDS900-Ro20 The choice of category metafield versus custom metafield MUST be recorded in the Attribute Definition mapping (CDS-200 §7), not decided independently for each product.

CDS900-Ro21 A metafield used as a storefront filter MUST have an appropriate metafield definition and a supported data type.

Canonical Attribute	Preferred Shopify Target	Fallback
Material / fabric	Category metafield when available	Custom product metafield
Colour family	Category metafield or dedicated filter metafield	Custom product metafield
Display colour	Variant option / linked category entry	Custom product or variant metafield
Fit	Category metafield when available	Custom product metafield
Care instructions	Custom product metafield	Rich text content block
Supplier identifier	Internal custom metafield only if required downstream	Keep in PIM
Workflow review status	Keep in PIM	Internal-only metafield or tag by exception

IMPLEMENTATION NOTE (*informative*): Shopify allows category entries to be customised for brand presentation, but CDS still distinguishes display label, canonical value and facet value (CDS-400). Renaming a Shopify entry does not by itself redefine the PIM canonical dictionary.

4.5 Variants and Options (normative)

Platform note (*informative, verified 2026-08-03*): current Shopify documentation permits up to **three options and 2,048 variants** per product, while noting that some themes, apps and sales channels may have lower effective support (historically ≤ 100). Source: [S4].

CDS900-Ro22 Only SKU-producing differences MAY become Shopify variant options under the CDS profile (variant boundary rule: CDS-200 §5).

CDS900-Ro23 The connector MUST preflight the projected variant count and option count against the profile's declared platform limits before publication.

CDS900-Ro24 A canonical product model requiring more options than Shopify supports MUST use a documented decomposition strategy rather than dropping information.

CDS900-Ro25 Variant option values SHOULD be connected to reusable category metafield entries where that improves consistency and the target theme or channel supports them.

4.5.1 Variant Decomposition Strategies (informative)

Condition	Preferred Strategy
More than three independent options	Split into logical products or use a configured product mechanism.
Very high variant count	Use combined listings or split by a major customer choice where supported. Caveat: combined listings have historically been restricted by Shopify plan (Plus); availability on the store's plan must be confirmed before this strategy is adopted (not verified this pass — REVIEW-006).
Personalisation that does not create inventory	Use line-item/customer input rather than variants.
Colour-specific product imagery and titles	Retain canonical parent relation; publish separate products only when merchandising requires it.
Size is not inventory-bearing	Do not create variants merely for filtering.

4.6 Collections, Tags and Navigation (normative)

CDS separates classification from merchandising. Shopify collections are customer-facing groupings. Tags are governed integration signals and may drive automated collections, but they are not the canonical source for structured attributes (CDS-400 tag governance).

CDS900-Ro26 Taxonomy collections SHOULD be generated from the PIM category tree or explicit category-to-collection projection rules.

CDS900-Ro27 Campaign and editorial collections MAY use merchandising rules or manual curation, but MUST have a declared owner and lifecycle.

CDS900-Ro28 Structured values such as material, colour family, dimensions and fit SHOULD be stored in metafields or standard attributes rather than duplicated as tags solely for filtering.

CDS900-Ro29 Every published tag MUST be generated, registered or explicitly approved. Uncontrolled tag creation MUST be rejected or quarantined.

CDS900-R030 Tag publication MUST be set-based: the connector MUST regenerate and write the complete governed tag set, or use an explicitly additive operation, and MUST declare which in the mapping's write semantics.

Platform note (*informative, verified 2026-08-03*): in the Shopify GraphQL Admin API, updating the `tags` field via `productUpdate` is a **full replacement** — "Updating tags overwrites any existing tags"; `tagsAdd` exists for additive updates. A connector that writes a partial tag list silently deletes every other tag on the product. Source: [S7]. (*Absorbed from CDS-500 v0.1 Appendix C; the general set-comparison and merge-semantics rules remain in CDS-500.*)

4.6.1 Tag Namespace Profile (*normative table*)

Tag Prefix	Purpose	Generation
collection_	Automated collection or navigation projection	Computed from category/collection rules
merch_	Merchandising signal	Governed manual or rule-based
season_	Seasonal relevance where a binary signal is sufficient	Controlled value / rule-based
workflow_	Internal process signal	PIM only by default; publish by exception
supplier_	Traceability	PIM only by default; publish by exception

IMPLEMENTATION NOTE (*informative*): the legacy `collection_*` model remains valid as a Shopify projection regenerated from governed rules. It is not the canonical category itself. A future connector may use explicit collection membership instead without changing the PIM taxonomy.

4.7 Search, Discovery and Facets (*normative*)

Platform note (*informative, verified 2026-08-17*): Shopify Search & Discovery supports filters based on category, product and variant metafield definitions, including text, numeric, boolean and metaobject reference types, and permits grouping filter values into a single displayed value. Documented limits: **25 filters per store; 100 values displayed per storefront filter; 1,000 values visible per filter in the app; 200 unique values per filter group; 1,000 filter groups store-wide**; filters do not display on collections over **5,000 products** or searches over **100,000 results**; and the **Category filter cannot use value grouping**. Source: [S2].

CDS900-R031 Customer filters MUST use the CDS facet layer (CDS-600) rather than exposing every distinct source or display value.

CDS900-R032 The Shopify profile MUST define one filter source per approved facet and MUST document its data source.

CDS900-R033 A colour filter SHOULD expose broad colour families such as Blue rather than dozens of commercial shade names.

CDS900-R034 Filter value grouping MAY be used as a channel-level convenience, but the authoritative grouping MUST remain in the PIM facet dictionary.

CDS900-R035 Facet projections MUST be preflighted against the declared platform limits (filter count, storefront and app value limits, values per group, store-wide group count, collection/search result limits and non-groupable filters) recorded in the profile's channel capability declaration.

Product display shade:	French Navy
Canonical shade:	Navy
Facet family:	Blue
Shopify product/variant:	French Navy
Shopify filter value:	Blue
Search synonyms:	french navy, navy, dark blue

RESOLVED (was new (resolved); verified 2026-08-04, REVIEW-006A): native Search & Discovery **does evaluate variant-scope filters at variant level**. Variant-option filters use variant-scope parameters (`filter.v.option.color`, `filter.v.option.size`), availability is a variant-specific filter (`filter.v.availability=1`), different filters combine with AND ("selecting Red from Color and 8 from Size returns products that are both red and size 8"), and the product object's `featured_media/url` update to the first variant matching the current filters — variant-level evaluation, not product-level unions. Sources: help.shopify.com Search & Discovery filters; shopify.dev storefront-filtering (both accessed 2026-08-04). The Shopify profile therefore declares the combination-availability capability **present**; empirical confirmation of per-value *count* accuracy under combined selections remains a vertical-slice test item.

4.8 Publication and Verification (*normative*)

The Shopify profile implements the CDS-500 lifecycle: calculate expected state, publish, observe, normalize, compare and report. Shopify success responses are transport acknowledgements, not field-level proof (CDS-P-06).

CDS900-R036 The connector MUST retain Shopify product and variant identifiers after creation.

CDS900-R037 The connector MUST read back the PIM-owned fields required by the verification profile.

CDS900-R038 Tags and list fields MUST be compared as sets unless order has declared meaning.

CDS900-R039 Rich text and HTML MUST be compared using a declared canonicalisation method (normalisation registry: CDS-500 §16).

CDS900-R040 Category and metafield references SHOULD be compared by stable identifiers or base values rather than display text alone.

CDS900-R041 Verification statuses and their traffic-light presentation are defined solely in **CDS-500 §17–§18**; this profile MUST use that enum and mapping and MUST NOT define its own.

Migration note (*informative, per ADR-D3*): the v0.1 CDS-900 status table is deleted. Legacy `WRITE_ERROR` maps to the detailed status `PUBLICATION_ERROR` (core status `ERROR`); `WARNING` is retired — use the appropriate core status plus a `reason_code`; `ACCEPTED_WITH_LIMITED_OBSERVATION` is not a status and is expressed as per-field `MATCH` / `UNOBSERVABLE` results with a coverage ratio (see Appendix A.1).

5. Google Merchant Center Profile — CDS-GMC-0.2 (*normative rules; mapping table informative, references retrieved 2026-08-03*)

The Google profile maps canonical product information to the Merchant Center product data specification. Google uses submitted product data to match products with relevant queries and requires accurate formatting to avoid disapprovals or display problems (*verified 2026-08-03, [G1]*).

5.1 Classification (*normative*)

Platform note (*informative, verified 2026-08-03*): `google_product_category` uses Google's predefined taxonomy and accepts **either** a category ID **or** a full breadcrumb path, not both; Google describes it as an override of automatic categorisation, honoured only in applicable cases (category-specific requirements, Ads targeting, alcohol compliance). `product_type` carries the merchant's own classification path. Sources: [G2] [G3].

CDS900-R042 `CH_google_product_category` MUST contain a valid Google taxonomy identifier or an accepted breadcrumb (one form, not both) when supplied.

CDS900-R043 `CH_google_product_type` SHOULD be derived from the PIM internal category path from broad to specific.

CDS900-R044 Google taxonomy and PIM taxonomy MUST remain separate mappings even where their text appears similar.

```
CAT_path           = Apparel > Women > Tops > Shirts
CH_google_product_type = Apparel > Women > Tops > Shirts
CH_google_product_category = Apparel & Accessories > Clothing > Shirts & Tops
```

5.2 Core Field Mapping (*informative — snapshot of the Google product data specification as retrieved 2026-08-03; the current dated specification governs at publication time*)

CDS Source	Google Attribute	Requiredness (as at 2026-08-03)	Profile Rule
STD_id / SKU	id	Required	Stable unique offer identifier.
STD_title or approved override	title	Required	Channel title may be transformed but canonical title is preserved.
Content description	description	Required	Remove unsupported markup and channel-inappropriate content.
Canonical product URL	link	Required	Must resolve to the correct product or variant landing page.
MED_primary_image	image_link	Required	Use the primary compliant image. Minimum 500x500 px enforced from 31 January 2027 (verified 2026-08-03).
Availability authority	availability	Required	Map controlled stock state to Google vocabulary.
Price authority	price	Required	Include currency and current sell price.
Brand dictionary	brand	Required for new products other than media/books (verified 2026-08-03)	Use canonical brand name.
GTIN / MPN	gtin / mpn	Conditional; omitting an existing GTIN limits visibility/performance (verified 2026-08-03)	Publish valid identifiers when available; identifier policy is a dated profile rule, not a core CDS rule (finding 200-7).
Canonical colour / channel value	color	Conditional (apparel)	Landing-page colour value (Display Label / sellable colour) — see the resolved note below.
Canonical material	material	Conditional	Use controlled channel representation.
Variant relationship	item_group_id	Conditional (variants)	Group related variants consistently.
CAT mapping	google_product_category	Optional (override)	Mapped Google taxonomy (ID or breadcrumb).
CAT path	product_type	Optional	Merchant-defined category breadcrumb.

RESOLVED (was new (resolved); verified 2026-08-04 against support.google.com/merchants/answer/6324487, REVIEW-006A): **the facet-family projection is NOT compliant and is removed as an option**. Google's minimum requirements state the submitted `color` must match the landing page ("if the landing page uses 'Toasted Walnut', submit 'Toasted Walnut' — don't submit 'Brown'"); non-compliance risks disapproval. Normative guidance for this profile: - **color = the landing-page colour value** (the Display Label / sellable colour, e.g. "French Navy"), never the facet family. - The **broad searchable colour name belongs in title** where discoverability is wanted (Google's own best practice for unique colour names). - **Multi-colour**: up to 3 slash-separated values, one primary + up to two secondary ("Navy/White"); never commas or merged strings. - As of May 2026 Google recommends also sending `variant_option` when colour is variant-identifying. The GMC field mapping's `source_layer` for `color` is therefore **display**, and the CDS-400 facet family remains a storefront-filter concern only.

CDS900-R045 The Google profile MUST preserve variant identity and item grouping consistently across updates.

CDS900-R046 The connector MUST validate required and conditional attributes against the current, dated Google product data specification before publication.

CDS900-R047 Disapprovals and diagnostics SHOULD be imported as observed channel quality events, not treated as canonical product values.

5.3 Observation and Verification (*normative*)

CDS900-R048 The Google connector SHOULD observe item status, diagnostics and accepted attribute values where the integration provides access.

CDS900-R049 A product accepted into Merchant Center MUST NOT automatically be treated as fully verified if field-level observation is incomplete; unobserved fields carry UNOBSERVABLE, and coverage is reported per CDS-500.

CDS900-R050 Google-normalised or automatically inferred values MUST be stored as observed or inferred channel state and MUST NOT be silently copied into canonical fields.

6. Meta Catalogue Profile — CDS-META-0.2 (normative rules; platform statements informative, references retrieved 2026-08-03)

Platform note (*informative, verified 2026-08-03*): Meta catalogues can receive product information through partner integrations, manual entry, APIs, scheduled data feeds **and pixel-based ingestion**. Meta publishes product data specifications for catalogue ingestion and supports data-source management in Commerce Manager. Sources: [M1] [M2] [M3].

CDS900-R051 The Meta profile MUST identify the exact ingestion mechanism in use, because write, update, read-back and diagnostic behaviour differ by source.

CDS900-R052 Meta category, gender, age group and other channel values MUST be mapped from canonical dictionaries or channel mappings.

CDS900-R053 The PIM MUST remain authoritative even when Shopify or another partner platform forwards products to Meta.

6.1 Direct versus Indirect Meta Publication (normative)

Mode	Benefits	CDS Requirement
Direct PIM to Meta feed/API	Maximum mapping control and independent verification	PIM owns expected state and source identifiers.
PIM to Shopify to Meta channel	Simpler operations; platform-managed transfer	Treat Meta as a second-hop channel and document reduced observation coverage.
Supplier/partner source to Meta	Fast onboarding	Must not bypass canonical governance for fields the organisation claims to control.

CDS900-R054 Multi-hop publication MUST record the immediate publication target and SHOULD record the final destination channel.

CDS900-R055 Where final Meta values cannot be read back, the verification status MUST be UNOBSERVABLE rather than MATCH.

IMPLEMENTATION NOTE (*informative, corollary of finding CDS500-6*): "Meta read-back is unavailable" is **not a platform fact** — the Graph/Catalog API supports item reads. Read-back availability depends on the **integration mode**: a direct API integration can observe items; a Shopify-forwarded catalogue typically cannot be observed from the PIM. Declare observation coverage per ingestion mechanism, honestly. A second-hop value that cannot be observed is reported as an observation gap, never assumed correct.

7. Supplier Import Profile (normative)

The supplier import profile standardises raw acquisition while preserving supplier-specific source data and provenance. It formalises the proven three-layer pattern from the legacy Airtable PIM:

Layer 1: IMPORT_ raw supplier source record (Source Values per CDS-400; legacy Import_* fields alias to IMPORT_)
Layer 2: canonical PIM value
Layer 3: channel expected and observed values
Supplier value -> normalize -> map dictionary -> validate -> propose/accept canonical value

RESOLVED (was new (resolved)): CDS-300 v0.2 registers IMPORT_ as a core namespace for ingested source records and intake provenance (CDS-300 §4/ §16), formalising the load-bearing legacy Import_* production namespace. Layer 1 uses IMPORT_; legacy Import_* fields are migration aliases to it.

CDS900-R056 Raw supplier values MUST be preserved separately from canonical values.

CDS900-R057 Every supplier mapping MUST record source, field, transformation version and last acquisition time.

CDS900-R058 Unknown dictionary values MUST enter quarantine or review and MUST NOT silently create new canonical values (CDS-400 §17).

CDS900-R059 Supplier categories MUST be treated as source classifications and mapped to the internal taxonomy, not copied directly as canonical categories.

Supplier Input	Canonical Target	Example Transformation
Brand spelling	Brand dictionary	Blackmagic -> Blackmagic Design
Stock text	Availability/quantity authority	IN STOCK -> governed availability state
Colour name	Colour reference dictionary	French Navy -> Navy -> Blue facet
Category text	Internal taxonomy mapping	Cameras - Cinema -> Cameras > Cinema Cameras
Price text	Money value	\$1,476.00 -> 1476.00 AUD
Barcode text	GTIN string	Remove spaces; retain leading zeros
Composition text	Material composition structure	80% Cotton / 20% Polyester -> component list

8. Apparel Industry Profile (*normative*)

The apparel profile adds domain-specific requirements for garment classification, variants, colour, size, fit, material composition, care and customer facets. (Detailed reference dictionaries: CDS-1500.)

8.1 Required Attribute Families (*normative table*)

Group	Common Attributes	Scope
Identity	Brand, style number, season, range	Product
Classification	Department, garment type, target gender, age group	Product
Colour	Display shade, canonical shade, colour family, swatch	Product/variant
Size	Size system, label, normalized size, measurements	Variant
Fit	Fit type, silhouette, rise, leg style	Product
Construction	Material composition, weave/knit, lining, closure	Product
Design	Pattern, neckline, sleeve length, garment length	Product
Care	Wash, dry, iron, bleach, dry-clean instructions	Product
Merchandising	Style, occasion, seasonality	Product
Compliance	Country of origin and required claims	Product/variant

CDS900-Ro60 Apparel colour MUST support at least display shade, canonical shade and customer facet family (value layers: CDS-400).

CDS900-Ro61 Apparel size MUST record its size system and MUST NOT rely on an unqualified label such as M or 10 alone when multiple systems are possible.

CDS900-Ro62 Material composition MUST preserve percentages and component order where supplied.

CDS900-Ro63 Fit and style MUST remain separate attributes; style is not a substitute for garment fit.

8.2 Apparel Facet Profile (*informative*)

Facet	Recommended Customer Values
Colour	Black, White, Grey, Blue, Green, Red, Pink, Purple, Brown, Beige/Cream, Yellow, Orange, Metallic, Multi
Size	Store-specific normalized size groups with visible system labels
Material	Cotton, Linen, Wool, Silk, Denim, Leather, Synthetic, Blended
Fit	Slim, Regular, Relaxed, Oversized, Tailored, Loose
Pattern	Plain, Striped, Checked, Floral, Animal, Abstract, Geometric, Textured
Occasion	Everyday, Work, Formal, Party, Wedding, Travel, Beach, Active
Style	Classic, Minimal, Coastal, Boho, Streetwear, Resort, Athleisure and governed store values

IMPLEMENTATION NOTE (*informative*): the recommended values are informative seeds. Each organisation must validate its customer vocabulary and avoid facet lists that are too long to scan (CDS-600).

9. Homewares Industry Profile (*normative*)

The homewares profile adds room, product form, material, finish, dimensions, shape, indoor/outdoor suitability, care and safety considerations.

Group	Common Attributes	Scope
Classification	Homeware type, room, use	Product
Material	Primary material, secondary materials, composition	Product
Finish	Painted, glazed, polished, brushed, matte, natural	Product/variant
Colour	Display colour, canonical colour, colour family	Product/variant
Dimensions	Height, width, depth, diameter, capacity, weight	Product/variant
Shape	Round, square, rectangular, oval, irregular	Product
Use	Indoor/outdoor, food safe, dishwasher safe, microwave safe	Product
Care	Cleaning and maintenance instructions	Product
Room	Living, bedroom, dining, kitchen, bathroom, office, outdoor	Product
Style	Modern, coastal, Scandinavian, rustic, industrial, classic	Product

CDS900-Ro64 Measurements MUST use structured numeric values and units; display strings alone are insufficient.

CDS900-Ro65 A material family facet MAY group detailed values, for example Tasmanian Oak -> Oak -> Wood.

CDS900-Ro66 Room is a merchandising/facet attribute and MUST NOT replace the product's primary category.

CDS900-R067 Safety and suitability claims MUST be backed by an admissible evidence class (CDS-700) and MUST NOT be generated solely from aesthetic inference.

9A. Other Industry Profiles (*informative pointer*)

The industry profiles added in the 2026-09-20 expansion are defined in their own chapters and registered in CDS-1500 §2.1: beauty, health and personal care (CDS-1600), food, beverage and grocery (CDS-1700), consumer electronics, appliances and technical goods (CDS-1800), parts, automotive, industrial and fitment (CDS-1900), and sports and outdoor, toys and games, and pet supplies (CDS-2000). Each carries its own channel projection guidance for the platform profiles in this chapter and a per-jurisdiction requirement register seed (CDS-1500 §2.2). Platform-specific facts those chapters rely on (for example which countries accept `energy_efficiency_class`, or that the seeded feed channels define no fitment attributes) are dated in the chapters that state them and are candidates for the next revision of the CDS-GMC and CDS-META profiles in §5–§6.

10. Cross-Channel Mapping Matrix (*informative*)

Canonical Concept	Shopify	Google	Meta	PIM Authority
Internal category	Mapped to product category + collections	product_type + taxonomy mapping	category mapping	Yes
Product title	Native title	title	title	Yes
Display colour	Variant option / display metafield	color representation	color representation	Yes
Colour family facet	Filter metafield/category value	Never published to color (non-compliant — §5.2 resolved note); storefront filter concern only	Storefront filters	Yes
Material	Category/custom metafield	material	material where supported	Yes
Variant grouping	Product + variants	item_group_id	item_group_id / retailer group	Yes
Google taxonomy	Optional Shopify feed field/mapping	google_product_category	May inherit/map	Yes — as mapping
Merchandising collection	Collection	custom label by choice	product set / label by choice	Yes
Verification result	PIM QA only	PIM QA only	PIM QA only	Yes

11. Conformance and Testing (*normative*)

A CDS-900 conformant platform implementation passes both structural and behavioural tests. (Levels, claims and test-suite governance: CDS-1000.)

11.1 Common Profile Tests

- Every mapped field declares canonical source, target, transformation, authority and comparison strategy (R008, §3).
- Unsupported canonical values remain preserved in the PIM (R003, R007).
- Channel overrides are distinguishable from canonical values.
- Expected state can be regenerated from the same canonical revision and mapping version (CDS-500).
- Observed values are stored separately from expected values.
- Missing and unobservable are distinct verification states (R009).
- Dictionary and taxonomy mappings are versioned.
- Unknown supplier values enter review rather than silently extending dictionaries (R058).
- The profile release identifier and dated references are present (R002).

11.2 Shopify Profile Tests

- A product category mapping publishes a valid Shopify taxonomy value (R015).
- Product type remains a controlled merchant value distinct from category (R014, R017).
- Category/custom metafield selection is defined at schema level (R020).
- Variant counts and option counts are preflighted (R023).
- Facet projections respect declared Search & Discovery limits (R035).
- Structured facets use metafields or standard attributes rather than uncontrolled tags (R028).
- Collection tags, if used, are regenerated from governed rules; tag writes are set-complete or explicitly additive (R029, R030).
- Variant updates use a variant-capable mutation (R013).
- Read-back verification detects direct Shopify edits to PIM-owned fields (R011, R037).

11.3 Google and Meta Profile Tests

- Internal category and external taxonomy values remain separate (R044).
- Variant grouping identifiers are stable (R045).
- Channel-required values are validated against the current dated specification (R046).
- Diagnostics are imported as observed quality events (R047).
- The ingestion mechanism and its observation coverage are declared (R051; §6.1).

- Multi-hop publication records reduced observation coverage honestly (R054, R055).

12. Migration from the Legacy Airtable PIM (*informative — organisation example*)

This section documents one organisation's legacy conventions and their CDS mapping. It is retained as a worked migration example, not as normative content. The legacy conventions align strongly with CDS; migration preserves their human readability and round-trip verification pattern while replacing formula sprawl with governed schema entities and mappings.

Legacy Field Pattern	CDS-900 Interpretation	Migration Action
Import_*	Supplier import profile raw source field	Retain source data and provenance (§7).
Airtable_*	Canonical or computed standard field	Map to STD_, CAT_, VAR_ or other canonical domain.
MF_Airtable_*	Canonical value intended for metafield publication	Map to MF_ definition and target mappings (ADR-D2).
DF_Airtable_SEO_*	Google feed projection	Map to CH_google_*; feed-layer fields live under CH_ with end-to-end feed verification (ADR-D24).
Shopify_/MF_Shopify_	Observed Shopify state	Map to OBS_shopify_* observation fields.
Match_*	Verification result	Map to QA_shopify_* status and reason code (ADR-D3).
collection_*	Computed Shopify collection signal	Retain as governed Shopify projection if used (§4.6).
Traffic light formulas	Human-facing verification status	Retain UI; detailed machine status underneath per CDS-500 §17–§18.

CDS900-R068 Migration MUST preserve historical source and observed values where they are required for audit or reconciliation.

CDS900-R069 Legacy DF_ names MAY remain as registered deprecated aliases during migration, but new channel mappings SHOULD use CH_. The DF_→CH_ transition is governed by the single consolidated **ADR-D24** (which supersedes CDS-ADR-900-002/003); DF_ is a deprecated legacy prefix in the CDS-300 registry.

CDS900-R070 The migration MUST NOT collapse canonical, expected and observed values into one field.

13. Architecture Decision Records (*informative — global ADR register is authoritative*)

ADR	Decision	Status
CDS-ADR-900-001	Profiles extend CDS core without redefining it	Accepted
CDS-ADR-900-002	CH_ is the default namespace for channel-specific data	Superseded by ADR-D24
CDS-ADR-900-003	DF_ may remain for actual feed artefacts and legacy aliases	Superseded by ADR-D24
CDS-ADR-900-004	Shopify category, product type and collections remain distinct	Accepted
CDS-ADR-900-005	Metafield mappings are schema decisions	Accepted
CDS-ADR-900-006	Facet simplification is canonical PIM logic	Accepted
CDS-ADR-900-007	Multi-hop channels declare observation gaps	Accepted
CDS-ADR-900-008	Industry profiles add requirements without creating new product truth	Accepted

Appendix A — Example Product Records (*informative*)

A.1 Apparel: Linen Relaxed Shirt

STD_product_id	=	P-1001
STD_title	=	Linen Relaxed Shirt
CAT_internal_path	=	Apparel > Women > Tops > Shirts
CAT_product_type	=	Shirt
CAT_shopify_category	=	Apparel & Accessories > Clothing > Clothing Tops > Shirts
VAR_colour_display	=	French Navy
VAR_size	=	M
MF_colour_canonical	=	Navy
MF_colour_facet	=	Blue
MF_material	=	Linen
MF_fit	=	Relaxed
CH_google_product_type	=	Apparel > Women > Tops > Shirts
CH_google_product_category	=	Apparel & Accessories > Clothing > Shirts & Tops
CH_google_color	=	French Navy (landing-page display shade – §5.2; the facet family is never published to color)
CH_google_material	=	Linen
QA_shopify_material	=	MATCH
QA_google_title	=	MATCH
QA_google_color	=	UNOBSERVABLE
QA_google_coverage_ratio	=	0.62 (per-field results + coverage replace the retired aggregate ACCEPTED_WITH_LIMITED_OBSERVATION)

A.2 Homewares: Tasmanian Oak Side Table

```

STD_product_id      = P-2001
STD_title           = Tasmanian Oak Side Table
CAT_internal_path   = Homewares > Living Room > Tables > Side Tables
CAT_product_type    = Side Table
MF_material_display = Tasmanian Oak
MF_material_canonical = Oak
MF_material_facet   = Wood
MF_room            = Living Room
MF_height_mm        = 520
MF_width_mm         = 450
MF_depth_mm         = 450
CH_shopify_filter_material = Wood
CH_google_material  = Wood
QA_shopify_dimensions = MATCH

```

Appendix B — Namespace Registry Excerpt (*informative*)

The registry of record is CDS-300. This excerpt lists only the namespaces this chapter's profiles project, with their publication behaviour; it registers nothing and is not authoritative. Consult CDS-300 for definitions, the STD_ enumeration (ADR-D2), lifecycle states and reserved prefixes.

Namespace (CDS-300)	Role in this chapter	Publication behaviour
STD_	Canonical core field (closed enumeration)	Mapped to native platform fields where they exist
CAT_	Internal classification and taxonomy mappings	May project to category/product-type fields
VAR_	Sellable variant and option value	Projects to platform variant model
MF_	Extended structured attribute (mechanism-neutral)	Projects per schema mapping: metafield, native field or feed column
CH_	Channel-specific mapping, expected output or override	Publishes only to the named channel
SEO_	Search engine listing fields	Projects to listing/SEO fields
MED_	Media and alt text	Projects to platform media model
DF_	Deprecated legacy feed prefix (ADR-D24)	Sanctioned migration aliases only; not for new design
OBS_	Observed downstream state	Never customer-facing; never canonical
QA_	Verification result and quality state	Never customer-facing
WF_	Workflow state	PIM-only by default
AI_	AI proposal, confidence, provenance	PIM-only unless explicitly approved (CDS-700)

Appendix C — Profile Change Checklist (*informative*)

1. Identify the official platform change, its effective date, and the retrieved-on date of the source.
2. Classify whether the change affects vocabulary, requiredness, capability, API transport or observation.
3. Update the channel capability profile (including declared limits and propagation windows).
4. Update mappings and transformation versions.
5. Run reference products through preflight, publication and read-back tests.
6. Assess migration and rollback requirements.
7. Update affected organisation and industry profiles.
8. Publish a versioned profile release (e.g. CDS-SHOPIFY-o.3) and retain the previous mapping for audit.

Appendix D — References (*informative; verification per REVIEW-006, dated*)

Ref	Source	Location	Retrieved / verified
[S1]	Shopify Help Center, Shopify's Standard Product Taxonomy (product category)	help.shopify.com/en/manual/products/details/product-category	Verified 2026-08-03
[S2]	Shopify Help Center, Adding filters with Shopify Search & Discovery	help.shopify.com/manual/online-store/search-and-discovery/filters	Verified 2026-08-17 (25 filters; 100 storefront values; 1,000 app values; 200/group; 1,000 groups; 5,000 collection and 100,000 search-result limits)
[S3]	Shopify Help Center, Metafields	help.shopify.com/en/manual/custom-data/metafields	Retrieved 2026-08-03 (partially verified)
[S4]	Shopify Help Center, Adding variants	help.shopify.com/en/manual/products/variants/add-variants	Verified 2026-08-03
[S5]	Shopify Help Center, Adding color swatches using category metafields	help.shopify.com/en/manual/custom-data/metafields/category-metafields/using-category-metafields	Retrieved 2026-08-03 (not spot-verified)
[S6]	Shopify GraphQL Admin API, productUpdate mutation	shopify.dev/docs/api/admin-graphql/latest/mutations/productUpdate	Verified 2026-08-03 (variants no longer updated)
[S7]	Shopify GraphQL Admin API, ProductUpdateInput (tags)	shopify.dev/docs/api/admin-graphql/latest/input-objects/ProductUpdateInput	Verified 2026-08-03 (tags full replacement)
[G1]	Google Merchant Center Help, Product data specification	support.google.com/merchants/answer/7052112	Verified 2026-08-03
[G2]	Google Merchant Center Help, Google product category	support.google.com/merchants/answer/6324436	Verified 2026-08-03

Ref	Source	Location	Retrieved / verified
[G3]	Google Merchant Center Help, Product type	support.google.com/merchants/answer/6324406	Retrieved 2026-08-03
[G4]	Google Merchant Center Help, color attribute	support.google.com/merchants/answer/6324487	Verified 2026-08-04 (REVIEW-006A) — governs §5.2
[M1]	Meta Business Help Center, Product data specifications for catalogues in Commerce Manager	facebook.com/business/help/120325381656392	Retrieved 2026-08-03
[M2]	Meta Business Help Center, Ways to add products in your catalog	facebook.com/business/help/384041892421495	Verified 2026-08-03 (corroborated via /365831587397584 and /125074381480892)
[M3]	Meta Business Help Center, About catalog data sources (incl. pixel)	facebook.com/business/help/125074381480892	Retrieved 2026-08-03

END OF CDS-900 v0.2 REVIEW DRAFT

Commerce Data Standard (CDS)

CDS-1000 — Conformance, Testing and Assurance

Field	Value
Status	vo.2 Review Draft (working source; not an approved standard)
Release	CDS vo.2 (single corpus release per ADR-D5)
Date	2026-08-04
Supersedes	CDS-1000 Working Draft vo.1; CDS-800 §9 conformance ladder and CDS-800 Appendix D claim template (both replaced by references to this chapter per ADR-D1)
Normative status	§§2–5, 7–19, 21–27 and Appendix B are normative. §1, §6, §20, §28 and Appendices A and C are informative.
Findings addressed	1000-1, 1000-2, 1000-3, 1000-4, 1000-5, 1000-6; 800-1, 800-2, 800-5, 800-6, 800-7 (this chapter's side); CDS1100-6 (corollary); SYS-2; ADR-D1, ADR-D3, ADR-D5
Dependencies	CDS-000 through CDS-900 (all published in the CDS vo.2 corpus release); machine-readable claim and manifest schemas in CDS-1100
Audience	Implementers, PIM vendors, auditors, platform integrators, data owners, standards maintainers

1. Purpose and Scope (*informative*)

CDS-1000 makes the Commerce Data Standard testable. The other chapters state what a conformant implementation does; this chapter defines how an implementation proves it: objective conformance levels, overlay profiles, test suites, evidence requirements, assurance methods and the rules for honest public claims.

This chapter is the **single normative home** for conformance levels, test suites, conformance claims and assurance (CDS-000 Ro05, ADR-D1). Other chapters cite it and do not restate it. In particular, the divergent five-level ladder formerly in CDS-800 §9 and the claim template formerly in CDS-800 Appendix D are superseded by this chapter.

CDS-1000 does not create a certification authority. It defines the framework that a self-assessor, customer, auditor or future certification body may use.

2. Conformance Philosophy (*normative*)

- **Objective.** A test yields the same conclusion when repeated against the same implementation state.
- **Version-specific.** A claim applies to a named CDS release and does not automatically transfer to a later release.
- **Profile-aware.** Platform, industry and overlay profiles add requirements; they never weaken the CDS core.
- **Evidence-based.** Every pass is supported by a retained artefact, query result, export, log or reproducible observation.
- **No compensation.** Strong performance in one area does not cancel failure of a mandatory requirement in another.
- **Honest scope.** A component, connector or workflow may claim limited conformance without implying that the entire organisation is conformant.
- **Repeatable.** The test suite is designed for automation where possible and controlled human review where necessary.

CDS1000-Ro01 A conformance claim MUST identify the CDS version, test-suite version, conformance level, applicable profiles, implementation scope, assessment method and assessment date.

CDS1000-Ro02 An implementation MUST NOT claim general CDS conformance when only a subset of requirements has been tested.

CDS1000-Ro03 Tests MUST evaluate observable system behaviour and evidence, not marketing descriptions or intended future capability.

CDS1000-Ro04 Mandatory tests MUST be pass/fail. A score or maturity percentage MUST NOT convert a failed mandatory test into a conformant result.

CDS1000-Ro05 Informative quality metrics MAY accompany a conformance result but MUST be clearly separated from normative pass/fail status.

3. Conformance Claim Model (*normative*)

A complete claim is a structured statement with these parts: specification version, test-suite version, conformance level, implementation scope, profiles and assessment method.

```
cds_claim:
  specification_version: "0.2"
  test_suite_version: "0.2"
  level: verified
  scope: Product information service and storefront connector
  profiles:
    - apparel-0.2
    - <platform-profile>-0.2
  assessment_method: independent_assessment
  assessed_at: 2026-08-04
  valid_until: 2027-08-04
```

CDS1000-R006 A claim **MUST** name the product, service, component or organisational process being assessed.

CDS1000-R007 A claim **MUST** list every profile — platform, industry and overlay — included in the assessment.

CDS1000-R008 A claim **MUST** state its assessment method as exactly one of the three canonical methods defined in §20: self-attestation, customer assessment or independent assessment.

CDS1000-R045 A claim and its machine-readable manifest (§26) **MUST** record the `test_suite_version` and `assessment_method` used, so that the result can be reproduced against the same test definitions.

4. Conformance Levels (*normative*)

CDS defines exactly five conformance levels, strictly cumulative:

Foundation → Structured → Publisher → Verified → Governed

Level	Primary capability	Minimum proof
CDS Foundation	Canonical authority and semantic clarity	Canonical source, ownership, terminology and namespace evidence
CDS Structured	Governed classification, attributes and dictionaries	Schema, data types, dictionary controls, validation and facet separation
CDS Publisher	Deterministic channel projection and publication	Expected channel state, mappings, preflight and publication records
CDS Verified	Downstream observation and field-level reconciliation	Read-back, comparison strategies, verification results and channel health
CDS Governed	Sustained ownership, change control and assurance	Owners, policies, audit, incident handling, versioning and recurring review

CDS1000-R009 A higher-level claim **MUST** satisfy all mandatory tests of every preceding level. No profile — platform, industry, overlay or organisational — may exempt an implementation from this rule or weaken any core requirement. There is no exception process for level cumulativity.

CDS1000-R010 An implementation **MAY** claim different levels for different components if the scope of each claim is explicit.

Informative notes: - A PIM implementation might be CDS Structured while its channel connector is CDS Verified. This is preferable to an inflated claim covering both. - Governance is deliberately the capstone: a team may prove Publisher and Verified round-trip value before formalising full change-control and audit governance (ADR-D1). - Superseded drafts of CDS-800 defined a different ladder in which "Governed" was level 2 and permitted profiles to relax cumulativity. That ladder and its escape clause are deleted; any prior claim using it must be restated against this section.

5. Overlay Profiles (*normative*)

An overlay profile binds an additional test suite to a claim without being a level. Overlays are claimed alongside a level and appear in the claim's `profiles` list.

Overlay profile	Binds test suite	Additional bound controls	Minimum level
AI-Assured	T-AI (§17)	CDS-700 normative controls for proposal status, provenance, approval and publication boundary	CDS Structured
CX	T-UX (§16)	CDS-600 facet and navigation requirements within the declared storefront scope	CDS Foundation
Multi-Tenant	T-TEN (§18)	CDS-800 §22 tenant-isolation requirements	CDS Foundation

CDS1000-R039 An overlay profile claim **MUST** name the overlay, its version and the level alongside which it is claimed, and **MUST** pass every mandatory test in the bound suite in addition to all mandatory tests of the claimed level.

CDS1000-R040 The AI-Assured overlay **MUST NOT** be claimed below CDS Structured, because T-AI tests presuppose governed attribute definitions and dictionaries for AI output to conform to.

CDS1000-R041 A claim that includes customer-facing facet or navigation scope **MUST** bind the CX overlay for that scope; T-UX tests are mandatory only under this overlay.

CDS1000-R042 An implementation serving more than one organisation from shared infrastructure **MUST** bind the Multi-Tenant overlay before claiming any level for that shared deployment.

Informative note: AI-Assured is not a sixth level (ADR-D1). Implementations without AI enrichment simply do not claim the overlay; their level claims are unaffected.

RESOLVED — accepted as drafted (owner 2026-08-04; was new (resolved)): The T-TEN suite (§18) is newly designed here as a minimal one-test-per-requirement mapping of CDS-800 §22; ADR-D1 assigned the overlay pattern but not the test list. Alternative: a broader penetration-style suite, deferred to a future test-suite version.

6. Adoption Maturity Mapping (*informative*)

CDS-1300's adoption maturity model describes organisational progress; it is not a conformance construct. It maps onto the conformance ladder as follows (ADR-D1):

CDS-1300 maturity level	Conformance ladder equivalent
Level 0 — Fragmented	Pre-Foundation (no claim available)
Level 1 — Foundation (formerly "Canonical")	CDS Foundation

CDS-1300 maturity level	Conformance ladder equivalent
Level 2 — Structured	CDS Structured
Level 3 — Publisher	CDS Publisher
Level 4 — Verified	CDS Verified
Level 5 — Governed	CDS Governed

CDS-400's cumulative dictionary capability badges — Dictionary Core, Dictionary Facet, Dictionary Channel, Dictionary Round-Trip (renamed from "Dictionary Verified") — are not levels or overlays. They are evidence descriptors carried inside the CDS Structured evidence set, defined in CDS-400.

7. Requirement and Test Identification (*normative*)

Every normative requirement and test case has a stable identifier. Requirement identifiers (CDSnnn-Rmmm) describe obligations; test identifiers describe a reproducible method for evaluating them.

Requirement: CDS500-R014
Test: CDS1000-T-VRF-014
Profile test: CDS1000-T-<profile>-023
Evidence: EVID-2026-000184

Prefix	Suite	Bound to
T-FND	Foundation tests	CDS Foundation
T-STR	Structured data tests	CDS Structured
T-PUB	Publisher tests	CDS Publisher
T-VRF	Verified tests	CDS Verified
T-GOV	Governed tests	CDS Governed
T-DICT	Controlled dictionary tests	CDS Structured
T-UX	Customer experience tests	CX overlay
T-AI	AI governance tests	AI-Assured overlay
T-TEN	Tenant isolation tests	Multi-Tenant overlay

Each platform or industry profile in CDS-900 registers its own test prefix in its versioned profile test matrix (Appendix A).

CDS1000-R011 Test identifiers MUST remain stable after publication. Substantive changes to test meaning require a new identifier or a new test-suite version.

CDS1000-R012 Every mandatory requirement in the CDS corpus MUST map to at least one test case or a documented inspection procedure.

Informative note: the corpus-wide retrofit of CDSnnn-Rmmm requirement identifiers is complete in v0.2 (finding SYS-2 resolved); every chapter now carries stable IDs, so this traceability mandate is implementable across the whole standard.

8. Test Categories (*normative*)

Category	Method	Typical evidence
Static schema	Inspect definitions and configuration	Schema export, namespace registry, dictionary definitions
Data sample	Evaluate representative records	Product export, variant records, mapping outputs
Behavioural	Trigger a workflow and observe the result	API logs, before/after states, job records
Round-trip	Publish, read back and compare	Expected state, observed state, comparison result
Negative	Provide invalid or unknown data	Validation error, quarantine record, rejected publication
Security and governance	Inspect permissions and approvals	Role matrix, audit history, change record
Human factors	Review visible labels and operational clarity	UI capture, field list, task observation
Performance	Measure defined operational limits	Timing log, batch result, retry behaviour

CDS1000-R013 A test suite SHOULD include positive, negative and boundary cases where the requirement can fail in materially different ways.

9. Test Artefacts and Evidence Package (*normative*)

The evidence package is the durable record supporting a claim. It must allow another qualified reviewer to reproduce or reasonably verify the assessment conclusion. Contents:

- assessment manifest and scope statement
- implementation, test-suite and profile versions
- test-run identifier and timestamps
- test environment description
- input fixtures and expected outcomes
- actual outputs and comparison results

- screenshots or exports for human-inspection tests
- logs for publication, observation and verification
- list of failed, not-tested and not-applicable tests
- exceptions and corrective-action records
- assessor identity or responsible organisation
- cryptographic hashes or checksums for retained files

CDS1000-R014 Evidence **MUST** be retained for every mandatory test result included in a claim.

CDS1000-R015 Evidence **MUST** identify the implementation version and configuration that was tested.

CDS1000-R016 Sensitive evidence **MAY** be redacted, but the redaction **MUST NOT** remove information necessary to validate the result.

10. Foundation Test Suite (T-FND) (*normative*)

Test ID	Objective	Pass condition
T-FND-001	Canonical source exists	Product facts have a declared master and channel read-back is stored separately
T-FND-002	Authority is explicit	Each tested field has exactly one declared authority
T-FND-003	Channel independence	Canonical values remain valid and retrievable with any single channel disabled
T-FND-004	Self-describing fields	Every sampled visible field identifier resolves to a registered namespace and Attribute Definition (CDS-300/CDS-200) with no out-of-band notes required; sample size and selection rule are declared in the assessment scope
T-FND-005	Semantic namespaces	Field prefixes match the approved namespace registry
T-FND-006	Product/variant boundary	SKU-producing differences are variants; descriptive facts are not misused as options
T-FND-007	Classification separation	Internal category, collection and external taxonomy are distinct concepts
T-FND-008	Audit identity	A change can be attributed to an actor or process

CDS1000-R017 CDS Foundation conformance requires every mandatory Foundation test to pass.

11. Structured Data Test Suite (T-STR) (*normative*)

Test ID	Objective	Pass condition
T-STR-001	Attribute definitions	Every governed attribute has type, scope, cardinality and validation
T-STR-002	Controlled values	Governed enumerations use a dictionary or equivalent reference-data control
T-STR-003	Canonical identity	Dictionary values use stable identifiers independent of display labels
T-STR-004	Source fidelity	Original supplier values are retained when required for provenance
T-STR-005	Facet separation	Facet values are stored distinctly from canonical values for every attribute with a declared facet layer
T-STR-006	Conditional rules	Category- or value-dependent requirements are machine-expressible
T-STR-007	Unknown-value handling	Unmapped values are rejected or quarantined rather than silently accepted
T-STR-008	Units and measurements	Numeric quantities retain declared units and normalisation rules
T-STR-009	Inheritance	Inherited values preserve provenance and can be distinguished from direct values
T-STR-010	Dictionary lifecycle	Aliases, deprecation and replacement are governed

CDS1000-R018 CDS Structured conformance requires Foundation conformance plus all mandatory Structured tests, including the Controlled Dictionary Test Suite (§15).

12. Publisher Test Suite (T-PUB) (*normative*)

Test ID	Objective	Pass condition
T-PUB-001	Expected state generation	The system calculates an expected channel representation before publishing
T-PUB-002	Mapping determinism	The same canonical input and mapping version produce the same expected output
T-PUB-003	Preflight validation	Invalid or incomplete output is blocked before transport
T-PUB-004	Publication record	Each attempt records product, channel, revision, status and timestamps
T-PUB-005	Replace semantics safety	Full-replace operations preserve governed values and remove stale generated values
T-PUB-006	Partial failure handling	Field or object failures are visible and not reported as complete success
T-PUB-007	Retry safety	Retries do not create duplicate or contradictory downstream records
T-PUB-008	Override governance	Channel overrides are explicit, scoped and auditable

Test ID	Objective	Pass condition
T-PUB-009	Mapping version trace	Published output identifies the transformation or mapping version
T-PUB-010	Disabled publication	A disabled channel or product does not publish unintentionally

CDS1000-R019 CDS Publisher conformance requires Structured conformance plus all mandatory Publisher tests.

13. Verified Test Suite (T-VRF) (*normative*)

Test ID	Objective	Pass condition
T-VRF-001	Downstream observation	Published values are read back or otherwise observed
T-VRF-002	Expected vs observed separation	Expected and observed values are stored separately
T-VRF-003	Comparison strategy	Each verified field uses a declared type-aware comparison
T-VRF-004	Status distinction	Missing, unobservable, pending and not-applicable verification states are distinguishable per the CDS-500 status enum
T-VRF-005	Mismatch evidence	A mismatch records expected value, observed value and a machine-readable reason code (CDS-500, CDS-1100)
T-VRF-006	Normalised equivalence	Equivalent case, numeric and unordered-list representations can pass correctly
T-VRF-007	Drift detection	Manual or external downstream changes are detected
T-VRF-008	Channel health	Health reporting is derived from detailed verification results
T-VRF-009	Timestamp integrity	Last publish, last observe and last verify times are separately recorded
T-VRF-010	Repair policy	Automatic repair, alert or accepted override is governed and auditable

CDS1000-R020 CDS Verified conformance requires Publisher conformance plus all mandatory Verified tests.

14. Governed Test Suite (T-GOV) (*normative*)

Test ID	Objective	Pass condition
T-GOV-001	Domain ownership	Every governed domain has an accountable owner
T-GOV-002	Change control	Schema, dictionary, namespace and mapping changes follow an approved process
T-GOV-003	Versioning	Breaking, additive and corrective changes are distinguishable
T-GOV-004	Deprecation	Retired values and fields have aliases, replacement guidance and dates
T-GOV-005	Access control	Roles follow least privilege and separate approval where required
T-GOV-006	Incident response	Data-quality and publication incidents have detection, ownership and recovery procedures
T-GOV-007	Review cadence	Governance reviews occur at a declared interval
T-GOV-008	Exception expiry	Exceptions are scoped, justified, approved and time-limited (CDS-800 §16)
T-GOV-009	Supplier governance	Supplier mappings and imports are controlled and reviewable
T-GOV-010	Claim governance	Public CDS claims match current assessment scope and status

CDS1000-R021 CDS Governed conformance requires Verified conformance plus all mandatory Governed tests.

15. Controlled Dictionary Test Suite (T-DICT) (*normative*)

Dictionary tests verify both data integrity and customer-useful projection. This suite is bound to CDS Structured: it is mandatory for every claim at Structured or above.

Test ID	Fixture	Expected result
T-DICT-001	French Navy	Reference preserved; canonical Navy; facet Blue
T-DICT-002	navy / NAVY / Navy	Aliases resolve to one canonical identifier
T-DICT-003	Blue-Green	Ambiguous value is mapped by rule or sent to review
T-DICT-004	80% cotton, 20% polyester	Composition retained; Cotton and Synthetic facet rules applied explicitly
T-DICT-005	Tasmanian Oak	Canonical Oak; facet Wood; display label may remain Tasmanian Oak
T-DICT-006	Unknown supplier colour	Quarantine or review; no uncontrolled canonical creation
T-DICT-007	Deprecated canonical value	Alias resolves to replacement while history remains auditable
T-DICT-008	Channel vocabulary mismatch	Channel-specific representation is generated without changing canonical identity

CDS1000-R022 A dictionary test MUST verify identifiers and relationships, not only visible labels.

CDS1000-R043 The Controlled Dictionary Test Suite is mandatory at CDS Structured and every higher level; a Structured claim MUST NOT omit it.

16. Customer Experience Test Suite (T-UX) (*normative — mandatory under the CX overlay*)

Test ID	Objective	Pass condition
T-UX-001	Facet size control	Each assessed facet exposes no more values than the maximum declared for its facet family in the assessment scope, and every exposed value is a governed facet value rather than a raw source or canonical shade
T-UX-002	Colour family projection	Detailed shades group into usable colour families
T-UX-003	Truthful availability	Facet counts and product cards reflect available product or variant combinations
T-UX-004	No false zero results	For the filter-combination set and zero-result threshold declared in the assessment scope, misleading empty states do not exceed the threshold. If no combination set and threshold are declared, this test is a WARNING-class advisory: it MUST NOT report PASS and its result MUST NOT support the CX overlay claim
T-UX-005	Progressive disclosure	Long lists collapse or search without hiding essential context
T-UX-006	Accessible labels	Facet names and values remain understandable without colour alone
T-UX-007	Search synonym alignment	Search terms map to the same governed concepts as facets
T-UX-008	Navigation distinction	Navigation categories and filter facets do not duplicate each other unnecessarily

CDS1000-Ro23 A customer-facing facet test MUST evaluate the published experience, not only the PIM configuration.

RESOLVED — accepted as drafted (owner 2026-08-04; was new (resolved)): T-UX-001 and T-UX-004 use declared-threshold criteria (thresholds set in the assessment scope) rather than fixed universal numbers; the alternative — standard-wide numeric limits — was rejected as arbitrary across industries. Finding 1000-2.

17. AI Governance Test Suite (T-AI) (*normative — mandatory under the AI-Assured overlay*)

Test ID	Objective	Pass condition
T-AI-001	Proposal status	AI output is identifiable as a proposal until accepted
T-AI-002	Schema constraint	AI values conform to attribute definitions and dictionaries
T-AI-003	Evidence and provenance	The source evidence, model/workflow version and confidence are retained
T-AI-004	Abstention	Low-confidence or unsupported extraction returns review-required rather than invented data
T-AI-005	Human approval	Required review gates cannot be bypassed accidentally
T-AI-006	Publication boundary	AI does not publish to production outside declared authority
T-AI-007	Regression testing	Model or prompt changes are tested against a benchmark set
T-AI-008	Rights and privacy	Restricted source material and personal data controls are enforced

CDS1000-Ro24 AI tests MUST assess the complete workflow, including validation and approval, rather than only model output quality. The controls under test are defined in CDS-700.

18. Tenant Isolation Test Suite (T-TEN) (*normative — mandatory under the Multi-Tenant overlay*)

Each test maps to one of the four tenant-isolation requirements of CDS-800 §22 (finding 800-7 resolved).

Test ID	Objective	Pass condition
T-TEN-001	Tenant data isolation	A cross-tenant read attempt against organisation-specific products, dictionaries, mappings, credentials, exceptions or audit records is denied and the denial is evidenced
T-TEN-002	Shared baseline boundary	Shared baseline definitions and templates are readable, but no tenant-specific value is exposed to or merged into another tenant without recorded explicit authorisation
T-TEN-003	Organisation-scoped operations	An import or bulk action submitted without a declared organisation context is rejected; the same operation with a declared context affects only that organisation's records
T-TEN-004	Cross-tenant administration	Cross-tenant administrative access requires a privileged role and produces a reviewable log entry for each access

CDS1000-Ro42 (defined in §5) applies here: shared deployments must pass all mandatory T-TEN tests before claiming any level.

19. Platform and Industry Profile Test Suites (*normative*)

A platform or industry profile adds profile-specific tests to the applicable core level. It may constrain mappings, capability declarations and observation methods; it cannot weaken core requirements (CDS1000-Ro09, CDS000-Ro11).

Profile test content — including test matrices, expected platform outputs and observation methods — lives with the versioned, dated profile definitions in CDS-900 (platform) and CDS-1500 through CDS-2000 (industry; register in CDS-1500 §2.1), not in this chapter (see Appendix A).

CDS1000-R025 A profile conformance claim **MUST** pass all mandatory core tests for the claimed level and all mandatory tests in that profile's version-pinned test matrix.

Informative note — typical profile test areas: commerce-platform profiles cover taxonomy mapping, variant constraints, structured metadata definitions, tag/collection safety under full-replace semantics, storefront facet exposure, API read-back and drift handling; feed-destination profiles cover taxonomy and identifier mapping, field validation, disapproval handling and observed item-state reconciliation; supplier-import profiles cover source identification, delta/full behaviour, normalisation, duplicate detection, unknown-value quarantine and provenance; apparel and homewares profiles cover colour, size, fit, composition, materials, dimensions, units and facet truthfulness.

20. Reference Test Vectors (*informative*)

Reference vectors are small, portable records with known expected outcomes, usable in automated tests and demonstrations. Core vectors state **canonical and facet expectations only**; platform-specific expected outputs (e.g. a channel's filter value or feed colour) live in the version-pinned profile test matrices (finding 1000-4), because they change with platform API versions while canonical expectations do not.

20.1 Apparel product

```
input:
  product_type: shirt
  supplier_colour: French Navy
  material_text: 100% Linen
  variants:
    - { size: S, colour: French Navy }
    - { size: M, colour: French Navy }
expected:
  canonical_colour: navy
  facet_colour: blue
  canonical_material: linen
  verification_case_rule: case_insensitive
# Channel-specific expectations for this vector are defined in each
# profile's version-pinned test matrix (CDS-900 / CDS-1500).
```

20.2 Homewares product

```
input:
  product_type: cushion
  supplier_material: Tasmanian Oak buttons and linen cover
  dimensions: 50 x 50 cm
expected:
  canonical_materials: [linen, oak]
  material_facets: [textile, wood]
  dimension_width: 50 cm
  dimension_height: 50 cm
  room_facets: determined_by_governed_rule_or_review
```

20.3 Verification equality

```
expected: [Cotton, Linen]
observed: [linen, cotton]
comparison: unordered_list + case_insensitive + dictionary_identity
result: MATCH
```

21. Test Outcomes (*normative*)

A test result takes exactly one of six outcomes. Spellings are the canonical underscore forms (ADR-D3 alignment). Test outcomes are distinct from field-level verification statuses, whose single normative home is CDS-500.

Outcome	Meaning	Effect on claim
PASS	Mandatory condition is satisfied with evidence	Supports conformance
FAIL	Mandatory condition is not satisfied	Blocks the applicable level/profile claim
WARNING	Non-mandatory guidance or quality concern	Does not block the claim but MUST be reported
NOT_APPLICABLE	Requirement genuinely does not apply to the declared scope	Allowed only with documented rationale
NOT_TESTED	No valid test result exists	Blocks the claim when the test is mandatory
INCONCLUSIVE	Evidence is insufficient or contradictory	Treated as not passed until resolved

CDS1000-R026 A conformance claim **MUST NOT** be made for a level or profile while any of its mandatory tests has a result of FAIL, NOT_TESTED or INCONCLUSIVE.

CDS1000-R027 NOT_APPLICABLE requires a written scope-based justification and reviewer approval.

22. Nonconformity, Exceptions and Corrective Action (*normative*)

A nonconformity is a failed mandatory requirement. An exception (defined in CDS-100; governed by CDS-800 §16) is a time-limited, documented, approved acknowledgement that the implementation does not currently satisfy a requirement. An exception acknowledges non-conformance; it never converts failure into conformance.

An exception record contains:

- identifier and affected requirement
- description and evidence
- risk and affected records/channels
- responsible owner
- corrective action and target completion date
- temporary control
- approval and expiry date
- retest result

CDS1000-R028 An exception MUST NOT be used to claim conformance to a level or profile whose mandatory capability is absent. This rule is absolute; no CDS release, profile or exception class relaxes it.

CDS1000-R029 Corrected nonconformities MUST be retested using the same or a stronger test method.

23. Assurance Methods: Self-Attestation and Independent Assurance (*normative*)

The three assurance methods below are the canonical vocabulary for the entire standard; CDS-800's claims-governance requirements cite this section (finding 800-5 resolved).

Method	Description	Claim wording
Self-attestation	The implementing organisation performs and records its own assessment	"Self-attested CDS Verified conformance"
Customer assessment	A customer or contracting party evaluates the implementation	"Customer-assessed against CDS Publisher"
Independent assessment	A qualified party independent of implementation delivery performs the assessment	"Independently assessed CDS Governed conformance"

CDS1000-R030 The assurance method MUST be disclosed wherever a conformance claim is presented.

CDS1000-R031 Independent assessment MUST include evidence sampling, assessor independence and conflict-of-interest disclosure.

24. Certification Framework (*normative*)

CDS-1000 provides a future-ready certification structure but does not designate a certification body.

CDS1000-R044 Any organisation offering CDS certification MUST publish its assessor-competence, impartiality, surveillance, complaint and certificate-withdrawal processes.

CDS1000-R032 A certificate MUST identify its issuer, scope, CDS version, level, profiles, issue date, expiry date and public verification reference.

CDS1000-R033 A certification mark MUST NOT imply endorsement of product quality beyond the assessed CDS conformance scope.

25. Versioning and Reassessment (*normative*)

Conformance can be invalidated by changes to the system, schema, mappings, dictionaries, channel API behaviour, governance or the CDS specification itself.

Change	Minimum response
Patch implementation release	Risk review and targeted regression tests
New attribute or dictionary values	Structured-data tests for affected domains
Mapping change	Publisher and Verified tests for affected fields/channels
New platform API version	Platform profile regression and observation review
Major architecture change	Full reassessment of affected level and profiles
New CDS minor version	Gap analysis and tests for new/changed requirements
New CDS major version	Formal reassessment before using the new-version claim

CDS1000-R034 A claim MUST be reviewed after any change that could alter a mandatory test outcome.

CDS1000-R035 A claim SHOULD include an expiry or reassessment date.

26. Machine-Readable Conformance Manifest (*normative*)

A machine-readable manifest enables automated reporting, procurement checks and public verification. This chapter owns the manifest's semantics; its JSON Schema is published in CDS-1100 (finding 800-6: the former CDS-800 Appendix D template is a pointer to this section). JSON or YAML may be used, provided the required semantics are preserved.

```
cds_conformance:
  specification_version: "0.2"
  test_suite_version: "0.2"
  level: verified
  scope_id: product-information-service-connector
  profiles:
    - <platform-profile>-0.2
    - apparel-0.2
    - ai-assured-0.2
  assessment_method: independent_assessment
  assessed_at: 2026-08-04T09:00:00+10:00
  valid_until: 2027-08-04
  results:
    passed: 86
    failed: 0
    warnings: 4
    not_applicable: 7
    not_tested: 0
    inconclusive: 0
  evidence_package_hash: sha256:...
  public_report: https://example.invalid/cds/assessment-184
```

CDS1000-R036 The manifest MUST NOT report a conformant level or profile when any mandatory test for that level or profile has a result of FAIL, NOT_TESTED or INCONCLUSIVE.

(Required fields `test_suite_version` and `assessment_method`: CDS1000-R045.)

27. Reporting and Public Claims (normative)

Reports should be understandable to both technical and non-technical readers. Detailed evidence may remain private, while the public summary states scope, result, limitations and assurance method.

```
Conformant claim:
"Product Information Service 4.2 is independently assessed as conformant with
Commerce Data Standard v0.2 at the CDS Verified level for the Apparel v0.2
profile and the AI-Assured v0.2 overlay. Scope excludes inventory and order
management. Assessment date: 4 August 2026."

Nonconformant claim:
"CDS certified" # invalid: no version, scope, level, profile, issuer or method
```

CDS1000-R037 Public claims MUST be precise enough that a reader can determine what was assessed and what was excluded.

CDS1000-R038 Expired, withdrawn or superseded claims MUST be removed or visibly marked.

28. Decision Records (informative)

Architecture decisions are held in the single global ADR register (CDS000-R006). Decisions governing this chapter: **ADR-D1** (one cumulative ladder owned by CDS-1000; governance as capstone; AI-Assured, CX and Multi-Tenant as overlay profiles; T-DICT bound to Structured; assurance vocabulary and claim/manifest ownership here), **ADR-D3** (status-enum spellings; verification statuses homed in CDS-500), **ADR-D5** (single corpus release version; casing of enum values). The chapter-local records CDS-ADR-1000-001..007 from v0.1 (cumulative levels; pass/fail mandatory tests; version- and scope-specific claims; self-attestation with disclosure; warnings separated from conformance; machine-readable manifest; no in-spec certification body) are subsumed into the global register unchanged in substance.

Appendix A — Profile Test Matrices (informative)

Version-pinned profile test matrices — including all platform-specific expected outputs formerly embedded in this chapter's reference vectors and Shopify appendix — are published alongside each profile definition (CDS-900 for platforms, CDS-1500 through CDS-2000 for industries). Each industry chapter added in the 2026-09-20 expansion ends with a "Reference Validation Cases" section listing the negative cases its profile contributes to the cross-industry validation set (REVIEW-020). Each matrix pins: the profile version, the platform API version and date verified, the registered test prefix, the mandatory test list, and the profile-specific expected outputs for the shared core reference vectors (§20). A representative matrix covers: product identity mapping, taxonomy mapping, structured metadata definitions, variant boundaries, tag/collection safety, storefront filter exposure, observation method, verification comparison and drift handling.

Appendix B — Apparel and Homewares Reference Dataset (normative)

Implementations must demonstrate more than a single happy path.

CDS1000-R046 A test dataset used to support a conformance assessment MUST include negative and non-happy-path cases, covering at minimum: an unknown supplier value requiring quarantine, a deprecated alias resolving to a current canonical value, and a deliberate downstream manual edit that must produce detected drift.

The reference dataset SHOULD additionally contain:

- an apparel product with multiple size and colour variants
- colour values French Navy, Sky Blue, Royal Blue and Teal with controlled facet grouping
- an unordered multi-material composition
- a homewares product with dimensions and explicit units
- a homewares product with multiple materials and one dominant display material
- a product whose internal category maps differently to two external taxonomies
- a product with a deliberate channel title override

Implementations claiming an industry profile from CDS-1600 through CDS-2000 SHOULD extend the dataset with that chapter's reference validation cases (for example a cosmetic with a free-from claim contradicted by its ingredient list, a food with an allergen declared only in text, a lithium-battery product without a dangerous-goods declaration, a part with fitment held only in a tag, a toy whose safety age warning is derived from its retail age band) and, for every market claimed, a product that fails a `block` entry of the jurisdiction requirement register (CDS1500-R057).

Appendix C — Example Evidence Package (*informative*)

```
assessment-184/
  manifest.yaml
  scope.md
  environment.yaml
  results/
    T-FND-001.json
    T-STR-001.json
    T-PUB-001.json
    T-VRF-001.json
    ...
  fixtures/
    apparel-shirt.json
    homewares-cushion.json
  exports/
    attribute-definitions.json
    dictionaries.json
    namespace-registry.json
  logs/
    publication.ndjson
    observation.ndjson
    verification.ndjson
  screenshots/
    storefront-filters.png
    channel-health.png
  exceptions/
    none.md
  report.pdf
  checksums.sha256
```

End of CDS-1000 v0.2 Review Draft.

Commerce Data Standard (CDS)

CDS-1100 — Reference Schemas and Machine-Readable Contracts

Field	Value
Status	vo.2 Review Draft (working source; not an approved standard)
Release	CDS vo.2 (single corpus release per ADR-D5)
Date	2026-08-04
Supersedes	CDS-1100 Working Draft vo.1
Normative status	§§1, 3–23 and Appendix A are normative. §2, §§24–26 and Appendices B–D are informative.
Findings addressed	CDS1100-1 through CDS1100-24; ADR-D3, ADR-D4, ADR-D5; open decisions D14 (via CDS-700 §6) and D21
Dependencies	CDS-000 through CDS-1000; reference schema package CDS-1200 vo.2.1
Audience	PIM developers, commerce platform engineers, connector authors, data architects, test-tool developers, AI workflow designers

Single-home notice. CDS-1100 is the sole normative home for: the machine-readable document contracts, the common document envelope, the schema identifier and schema-versioning scheme, the extension-key grammar for document envelopes, the validation-output contract and the **CDS_* reason-code registry** (§21). This chapter *encodes* — and never redefines — semantics owned elsewhere: verification statuses, detailed statuses and the traffic-light mapping are owned by CDS-500 §17–18; conformance levels, test suites and claims by CDS-1000; AI review rules by CDS-700; dictionaries and value layers by CDS-400; visible field-identifier namespaces by CDS-300; terminology by CDS-100 (CDS000-R005).

Relationship to the shipped package. Every schema this chapter describes exists in the CDS-1200 Reference Package vo.2.1 (`schemas/` , 21 schemas at schema version 0.2.0). Where this chapter states what a schema requires, the statement has been checked against the shipped schema text. The published specification governs where package and prose conflict (CDS-1200 README), but as of vo.2 no such conflict is known.

1. Purpose and Scope (*normative*)

CDS-1100 defines the machine-readable contracts that represent the entities and lifecycle records specified by the Commerce Data Standard. It translates the architectural language of products, attributes, dictionaries, channel projections, observations and verification into interoperable documents that software can validate and exchange.

This chapter defines logical contracts and their reference JSON Schema expression. It does not prescribe a database engine, API style, message broker, programming language or storage topology. A relational, document, graph, event-stream or spreadsheet-backed implementation MAY conform if its exported and consumed documents satisfy these contracts.

CDS1100-R001 A CDS machine-readable document MUST declare its document type, schema identifier, corpus release version, document identifier and revision (the envelope, §5).

CDS1100-R002 An implementation MUST validate externally exchanged CDS documents against the applicable schema and profile constraints before accepting them as authoritative input.

CDS1100-R003 A serialization or API transport MUST NOT alter the semantic meaning, type or precision of a CDS value.

2. Contract Design Principles (*informative*)

- **Explicit over inferred.** Types, identifiers, scopes and authorities are declared, never guessed from labels or field order.
- **Stable identity.** Business identity is separate from labels, array positions and channel identifiers.
- **Composable contracts.** Shared definitions (envelope, localised text, measurement, money, entity reference) are referenced, not copied.
- **Human-readable payloads.** Keys and enumerations remain understandable to operators and reviewers.
- **Strict core, governed extension.** Core property sets are complete and closed; extension points are namespaced and governed (§20).
- **Canonical and observed separation.** Authoritative values never share storage locations with downstream read-back values (CDS-200, CDS-500).
- **Validation before mutation.** Invalid input is rejected or quarantined before it changes canonical state (CDS-400 §17).
- **Evidence by design.** Publication and verification contracts retain revisions, versions, hashes and comparison evidence.
- **Round-trip safety.** Serializing and parsing preserves identifiers, values, units, significant order and null semantics.
- **AI containment.** AI outputs are proposals, distinct from accepted canonical values (CDS-700).

CDS1100-R004 A contract MUST distinguish absent, explicitly null, empty string, empty list and zero wherever those states have different business meaning.

CDS1100-R005 A list whose order is semantically significant MUST declare that significance in its Attribute Definition or contract documentation.

3. Serialization and Schema Dialect (*normative*)

JSON is the normative interchange syntax for CDS reference contracts. The reference validation language is JSON Schema Draft 2020-12. Implementations MAY provide YAML 1.2-compatible views for human editing (Appendix C) and MAY expose equivalent Protocol Buffers, Avro, XML or database schemas, provided a

lossless mapping to the normative JSON contract is documented and tested.

Concern	CDS baseline
Character encoding	UTF-8
JSON Schema dialect	Draft 2020-12, declared via <code>\$schema</code>
Media type	<code>application/json</code> or a profile-specific type; APIs SHOULD identify the CDS document type through media type or envelope
Date-time	RFC 3339-compatible timestamp
Duration	ISO 8601 duration string where needed
Language tag	BCP 47-compatible string
Country	ISO 3166-1 alpha-2 where a country code is required; display names are separate from codes
Currency	ISO 4217 alphabetic code
Units	Stable CDS or recognised unit code; value and unit are distinct fields

CDS1100-R006 Exchanged JSON MUST be encoded as UTF-8, and reference schemas MUST declare the Draft 2020-12 dialect via `$schema`.

CDS1100-R007 Timestamps MUST include a timezone (UTC Z or an explicit offset). *The shipped schemas enforce this doubly: every date-time property asserts both `format: date-time` and a pattern requiring the offset, so environments that skip format assertion still reject offset-less timestamps.*

CDS1100-R008 JSON numbers MUST NOT be used for identifiers such as GTIN, SKU, MPN or postal codes; leading zeros and exact textual form are significant, so identifiers are strings.

CDS1100-R009 Monetary values MUST be represented in a decimal-safe form that guarantees declared precision; binary floating-point rounding MUST NOT silently alter a published amount. The reference money contract is defined in §7.

CDS1100-R010 Implicit YAML typing MUST NOT change CDS scalar values; YAML representations MUST be converted to the canonical JSON data model before schema validation (Appendix C).

4. Schema Package Architecture and Identifiers *(normative — implements ADR-D5)*

The reference package is a registry of independently versioned schemas. Shared primitives live in `common/` and are referenced from entity schemas. The shipped vo.2 layout (21 schemas):

```
schemas/  
  common/    envelope, entity-reference, localised-text, measurement, money  
  core/      product, variant, attribute-definition  
  reference/  dictionary, dictionary-value, category, taxonomy-mapping, facet-definition  
  channel/   channel-profile, field-mapping, publication-record,  
            observation-record, verification-result  
  automation/ ai-proposal  
  assurance/  conformance-manifest, validation-output
```

Platform and industry profiles are not schema directories in vo.2: platform-specific constraints live in CDS-900 profiles and industry constraints in CDS-1500 through CDS-2000 (register: CDS-1500 §2.1), applied by composition (§20). The product relationship record (CDS-1900 §5) is carried in the canonical product contract's `relationships` collection (§8); its typed record shape is a candidate component schema for the next package release. The vo.1 `profiles/` directory sketch is withdrawn.

Three identity layers, per ADR-D5:

- Schema identity.** Every schema `$id` is a release-independent URN: `urn:cds:schema:<domain>:<name>` (for example `urn:cds:schema:core:product`). The specification release is *not* part of the URN, so identifiers are stable across releases. The vo.1 `https://schemas.cds.example/v1/...` example URIs are retired.
- Schema version.** Every schema carries its own semantic version in a `version` keyword (all shipped schemas: `0.2.0`), bumped by the compatibility rules of §22, independently of chapter prose revisions.
- Instance release pinning.** The envelope's `cds_version` is a per-release constant (`{"const": "0.2"}` in the vo.2 package). An instance therefore pins the corpus release it was written against, and a validator can refuse mismatched generations.

CDS1100-R011 Every published schema MUST have a stable, release-independent `$id` of the form `urn:cds:schema:<domain>:<name>`, independent of any file-system location.

CDS1100-R012 Every published schema MUST declare its own semantic version in a `version` keyword, distinct from both the corpus release and any document revision.

CDS1100-R013 A schema package release MUST publish a manifest listing schema identifiers, versions, per-file hashes, and a package-level hash. *(The shipped `package-manifest.json` carries a `schema_registry` inventory and a `package_hash`; its self-hash convention is documented in the CDS-1200 README.)*

CDS1100-R014 A profile schema MAY add constraints through composition but MUST NOT redefine the meaning of a core property (§20).

Informative note: schemas are resolved offline from the registry — the reference validator loads all schemas from the package and resolves `$ref` URNs against that in-memory registry. URNs carry no resolution promise; nothing dereferences them over a network. Registered `https` URIs were deferred until a domain and governance body exist (ADR-D5, revisit at vo.9).

5. Common Document Envelope (*normative*)

Every top-level CDS document uses a common envelope (`urn:cds:schema:common:envelope`). The envelope supports routing, validation, audit, multi-tenant labelling and version-specific interpretation without requiring consumers to inspect business fields first.

Property	Type	Required	Purpose
<code>cds_schema</code>	string (URN)	Yes	<code>\$id</code> of the schema this document conforms to
<code>cds_version</code>	const "0.2"	Yes	Corpus release the document targets
<code>document_type</code>	string	Yes	Logical entity or lifecycle record type
<code>document_id</code>	string	Yes	Stable globally unique document identifier
<code>revision</code>	integer ≥ 1 or string	Yes	Monotonic entity revision or immutable revision identifier
<code>status</code>	enum draft, active, deprecated, archived	Yes	Document lifecycle state
<code>tenant_id</code>	string	Optional	Organisation boundary label for multi-tenant implementations (§6)
<code>created_at</code>	date-time	Yes	Creation timestamp of the logical record
<code>updated_at</code>	date-time	Yes	Timestamp of the current revision
<code>source_system</code>	string	Yes	System that produced the document
<code>correlation_id</code>	string	Optional	Links related import, publish, observe and verify operations
<code>extensions</code>	object	Optional	Governed extensions; key grammar per §20

```
{
  "cds_schema": "urn:cds:schema:core:product",
  "cds_version": "0.2",
  "document_type": "product",
  "document_id": "prd_shirt_100",
  "revision": 42,
  "status": "active",
  "tenant_id": "org_reference",
  "created_at": "2026-07-18T03:12:10Z",
  "updated_at": "2026-08-03T00:08:17Z",
  "source_system": "cds-reference-package",
  "correlation_id": "job_01K1PUB9"
}
```

Composition idiom (preserved from vo.1, now safe). Entity schemas compose the envelope with `allOf: [{ $ref: envelope }, { entity properties }]` and close the document with `unevaluatedProperties: false`. In vo.1 this closure was a trap: it forbade properties the prose promised (lifecycle, provenance, reviewer fields). In vo.2 the property sets are complete — every property a contract names is present in its schema — so closure now does what it should: any unnamed property is a defect, not a casualty. (CDS1100-5, CDS1100-23 resolved.)

The envelope `status` enum is the *document* lifecycle. It is deliberately distinct from the dictionary-value lifecycle owned by CDS-400 and from verification statuses owned by CDS-500; the three vocabularies never mix.

CDS1100-R015 Every top-level CDS document MUST carry the envelope, with all envelope-required properties present.

CDS1100-R016 `document_id` MUST remain stable across revisions of the same logical record.

CDS1100-R017 `revision` MUST change whenever any property affecting canonical meaning or downstream projection changes.

CDS1100-R018 A consumer MUST reject or quarantine a document whose `cds_schema` is unknown or unsupported, unless an explicitly configured forward-compatibility policy applies.

CDS1100-R019 `cds_schema` MUST equal the `$id` of the schema against which the document is validated. (*The reference validator cross-checks this on every fixture; a mismatch is a validation failure.*)

6. Identity, References and Tenancy (*normative*)

CDS distinguishes canonical identifiers, human business keys and downstream identifiers. A product may hold an internal CDS identifier, one or more SKUs, supplier identifiers and channel platform identifiers. These are related but never interchangeable.

Identifier class	Example	Authority
CDS entity ID	<code>prd_shirt_100</code>	PIM or CDS entity service
Business key	<code>SHIRT-100</code>	Organisation
Variant SKU	<code>SHIRT-100-NVY-M</code>	Organisation or inventory authority
Supplier SKU	<code>SUP-829103</code>	Supplier
GTIN	<code>09338716007824</code>	Assigned standards authority / brand
Channel ID	<code>gid://shopify/Product/123</code>	Downstream channel
Taxonomy node ID	<code>aa-1-2-3</code>	Taxonomy publisher

Cross-document references use the shared entity-reference contract (`urn:cds:schema:common:entity-reference`): `document_type` + `document_id` required, optional `role` and `tenant_id`.

```
{ "document_type": "dictionary_value", "document_id": "dictval_colour_navy", "role": "canonical_colour" }
```

CDS1100-R020 A channel identifier **MUST** be stored inside a channel identity or mapping structure and **MUST NOT** replace the canonical CDS entity ID.

CDS1100-R021 Cross-document references **MUST** use stable IDs and **MUST** include the referenced document type. (*vo.1's SHOULD is upgraded: the entity-reference schema requires `document_type`, so the requirement is mechanically enforced.*)

6.1 Tenancy — what the schema does and does not guarantee

`tenant_id` is an optional envelope property: it *labels* an organisation boundary, and single-tenant deployments legitimately omit it. Schema validation of one document **cannot** enforce a boundary between documents; tenant isolation is a property of the consuming implementation, not of any JSON Schema.

CDS1100-R022 A multi-tenant implementation **MUST** prevent cross-document references from resolving across tenant boundaries unless a governed shared-reference mechanism is explicitly defined. This is an **implementation obligation**: it is not, and cannot be, a schema-level guarantee. Its test coverage is the mandatory T-TEN suite in CDS-1000 §18; the reference runner's SEM-004 check (all related fixture documents share one `tenant_id`) is an illustration of the join, not the enforcement. (*CDS1100-13 resolved by honesty rather than by pretending the envelope enforces it.*)

7. Type System and Standard Formats (*normative*)

Attribute Definitions declare the applicable type and constraints; values **MUST** conform before becoming canonical. The reference value types (the attribute-definition `value_type` enum, shipped):

CDS type	JSON representation	Key constraints
text	string	Length, pattern, whitespace policy
boolean	boolean	No string substitutes such as yes/no
integer	integer	Minimum, maximum, unit where applicable
decimal	string	Declared scale and rounding mode (decimal-safe)
date	string	Calendar date, no time component
date_time	string	Timezone required (R007)
identifier	string	Pattern and authority required
enum_reference / enum_reference_list	object reference(s)	Must resolve to an active dictionary value; <code>dictionary_id</code> required (§10)
measurement	object	<code>urn:cds:schema:common:measurement</code> — string decimal <code>value</code> + <code>unit</code>
money	object	<code>urn:cds:schema:common:money</code> — string decimal <code>amount</code> + ISO 4217 <code>currency</code>
localised_text	array	Items per <code>urn:cds:schema:common:localised-text</code>
composition_list	array	Dictionary references with percentages
object	object	Nested schema required

Rich text is not a vo.2 reference type. Markup-bearing content is exchanged as `text` under the sanitisation obligations of §23; a structured rich-text contract is deferred.

Single localised-text model. All localised values corpus-wide use one shape: an array of `{ "language": "<BCP 47 tag>", "text": "..." }` objects. The vo.1 mixture of language-keyed maps (`{ "en-AU": "Navy" }`), arrays and bare strings is retired; dictionary-value `canonical_label` and facet `label` are now arrays of localised-text like product content. (CDS1100-15 resolved.) The attribute-definition `label` is an operator-facing plain string in the reference schema; customer-facing labels are the localised ones.

Measurement. `{ "value": "1.250", "unit": "kg" }` — `value` is a string decimal, `unit` a code. The measurement schema is wired into the package: product typed values reference it (measurement branch, §7.1), and `measurement` is an attribute-definition value type. It is no longer dead weight (CDS1100-10 resolved).

Money. `{ "amount": "129.95", "currency": "AUD" }` — `amount` is a string decimal in **major units**; `currency` is an ISO 4217 alphabetic code. The shipped schema documents the design call in its `$comment`.

RESOLVED — accepted as drafted (owner 2026-08-04; was new (resolved)): Money is a string decimal in major units, matching the measurement convention already in the package. Alternative: integer minor units (cents), which avoids scale ambiguity but diverges from every other numeric-as-string-decimal in the corpus. Revisit if a payments-adjacent profile needs exact minor-unit semantics.

CDS1100-R023 A measurement **MUST** store the numeric value separately from its unit.

CDS1100-R024 A localised value **MUST** identify the language of each translation using the localised-text contract.

7.1 Typed attribute values

Canonical product attribute values are constrained to a minimal **typed-value** shape (product schema `$defs/typed_value`): a value is exactly one of

```
{dictionary_value_id, percentage?} | {text: [localised-text]} | {number, unit?} | {measurement} | {money} | {boolean} | {identifier} | {date_time}
```

or a non-empty array of such values. Bare scalars are rejected: `"MF_material": 42` no longer validates (CDS1100-14).

Deliberate limits. The typed-value shape is intentionally minimal. It guarantees that every canonical value has a declared representation; it does **not** encode the binding between a specific attribute and its declared `value_type`, cardinality or dictionary — JSON Schema cannot look up an Attribute Definition while validating a product. Enforcing that a value keyed `MF_material` actually is a `composition_list` against `dict_material` remains an implementation obligation (§21.4). The same limit applies to variant `variant_attributes`, which the shipped schema constrains only to an object.

CDS1100-R025 A canonical attribute value MUST use the typed-value shape (or an array of it), and an implementation MUST additionally validate each value against its Attribute Definition's declared type, cardinality and dictionary before accepting it as canonical.

8. Canonical Product Contract (*normative*)

`urn:cds:schema:core:product`. The Product contract represents the logical commercial item: product-level truth, classification, content and references to variants. Channel projections and observations are separate documents (§15–§17).

Required beyond the envelope: `product_id`, `business_key`, `classification` (`product_family_id`, `product_type_id`, `category_id` — all IDs), `content` (localised title required; `short_description`, `long_description` optional), `attributes`, `variant_ids`. Optional: `brand_id`, `manufacturer_part_number`, `media` and `relationships` (entity-reference arrays), and the `lifecycle` (`introduction_date`, `end_of_sale_date`, `approval_state`) and `provenance` (`source_type`, `source_reference`, `acquired_at`) groups restored in vo.2 (CDS1100-23).

Attribute keys are validated by the strict semantic-name pattern `^[A-Z][A-Z0-9]*_[a-z0-9]+(?:_[a-z0-9]+)*$` (prefix grammar owned by CDS-300); attribute values use §7.1 typed values.

```
{
  "product_id": "prd_shirt_100",
  "business_key": "SHIRT-100",
  "classification": {
    "product_family_id": "family_apparel",
    "product_type_id": "type_shirt",
    "category_id": "cat_womens_shirts"
  },
  "content": { "title": [{ "language": "en-AU", "text": "Relaxed Linen Shirt" }] },
  "attributes": {
    "MF_material": [{ "dictionary_value_id": "material_linen", "percentage": "100.00" }],
    "MF_fit": { "dictionary_value_id": "fit_relaxed" }
  },
  "variant_ids": ["var_shirt_100_navy_s", "var_shirt_100_navy_m"]
}
```

CDS1100-R026 Product attributes MUST be keyed by stable Attribute Definition identifiers conforming to the CDS-300 namespace grammar, never by mutable display labels.

CDS1100-R027 A product contract MUST NOT embed observed channel values in its canonical attributes object.

CDS1100-R028 Variant references MUST resolve to sellable-unit records whose parent `product_id` matches the product. (*Executed as reference semantic test SEM-001; see §21.4.*)

9. Variant and Sellable Unit Contract (*normative*)

`urn:cds:schema:core:variant`. The Variant represents a separately sellable, inventory-addressable unit. Required beyond the envelope: `variant_id`, `product_id`, `sku`, `option_values` (each {`option`, `value_id`, `display`?}), `sellability` (`sellable`, `preorder`, `unavailable`, `retired`). Optional: `barcode_identifiers`, `variant_attributes`.

Barcode identifiers are typed and length-checked per scheme:

Scheme	Length	Note
gtin_8	exactly 8 digits	
gtin_12, upc	exactly 12 digits	
gtin_13, ean	exactly 13 digits	
gtin_14	exactly 14 digits	

Values are strings, so leading zeros are preserved — the positive reference fixture deliberately uses the zero-leading GTIN-14 `09338716007824` (CDS1100-21).

```
{
  "variant_id": "var_shirt_100_navy_m",
  "product_id": "prd_shirt_100",
  "sku": "SHIRT-100-NVY-M",
  "option_values": [
    { "option": "colour", "value_id": "colour_navy", "display": "French Navy" },
    { "option": "size", "value_id": "size_m", "display": "M" }
  ],
  "barcode_identifiers": [{ "scheme": "gtin_13", "value": "9338716007824" }],
  "sellability": "sellable"
}
```

CDS1100-R029 A variant MUST carry a stable SKU or an explicitly documented alternative sellable-unit key.

CDS1100-R030 Option values MUST reference controlled values (`value_id`) when a dictionary exists for that option; display strings are presentation only.

CDS1100-R031 A barcode identifier MUST declare its scheme, MUST be represented as a string preserving leading zeros, and MUST satisfy the exact digit length of its scheme.

CDS1100-R032 Inventory quantity SHOULD remain owned by the declared inventory authority and MAY be referenced rather than duplicated into the canonical record (per-fact authority: CDS-200).

10. Attribute Definition Contract (*normative*)

`urn:cds:schema:core:attribute-definition`. Attribute Definitions carry the schema intelligence of CDS: what a value means, where it applies, how it is validated, displayed, faceted, compared and proposed by AI. Required beyond the envelope: `attribute_id` (semantic-name pattern), `label`, `scope` (`product`, `variant`, `category`, `channel`, `system`), `value_type` (§7 table), `cardinality` (`{minimum, maximum|null}`), `behaviour` (`filterable`, `searchable`, `displayable`, `comparison_strategy` — all required). Optional: `description`, `dictionary_id`, `channel_mappings` (entity references).

```
{
  "attribute_id": "MF_material",
  "label": "Material",
  "scope": "product",
  "value_type": "composition_list",
  "dictionary_id": "dict_material",
  "cardinality": { "minimum": 1, "maximum": 10 },
  "behaviour": {
    "filterable": true, "searchable": true, "displayable": true,
    "comparison_strategy": "unordered_composition"
  }
}
```

CDS1100-R033 An Attribute Definition MUST declare scope, value type, cardinality and behaviour (including a comparison strategy for every attribute that participates in downstream verification).

CDS1100-R034 An Attribute Definition whose `value_type` is `enum_reference` or `enum_reference_list` MUST declare `dictionary_id`. (*Mechanically enforced by if/then in the shipped schema; CDS1100-16.*)

CDS1100-R035 Attribute labels MAY change and be localised at the display layer without changing `attribute_id`.

11. Dictionary Definition and Value Contracts (*normative*)

`urn:cds:schema:reference:dictionary` and `urn:cds:schema:reference:dictionary-value`. CDS separates the Dictionary Definition (name, purpose, `value_type` flat/hierarchical, `default_locale`, `governance_owner`, `dictionary_version` — all required) from its Values. Value semantics, lifecycle and governance are owned by CDS-400; this section defines only the machine shape.

Dictionary Value — required beyond the envelope: `dictionary_value_id`, `dictionary_id`, `canonical_code` (lower snake case pattern), `canonical_label` (localised-text array), `aliases` (unique strings), `facet_memberships` (`{facet_id, value_id, primary}`), `channel_representations` (map of channel key to scalar). Optional: `parent_value_id` for hierarchical dictionaries.

```
{
  "dictionary_value_id": "colour_navy",
  "dictionary_id": "dict_colour",
  "canonical_code": "navy",
  "canonical_label": [{ "language": "en-AU", "text": "Navy" }],
  "aliases": ["navy blue", "french navy", "dark navy"],
  "facet_memberships": [{ "facet_id": "facet_colour_family", "value_id": "facet_blue", "primary": true }],
  "channel_representations": { "google_merchant": "Blue", "shopify_filter": "Blue" }
}
```

CDS1100-R036 Dictionary Value identity MUST NOT depend on the displayed label or array position.

CDS1100-R037 Aliases MUST NOT be treated as additional canonical values (intake and quarantine rules: CDS-400 §17).

CDS1100-R038 Facet memberships MUST reference governed facet values, never arbitrary strings. (*Executed as reference semantic test SEM-002; see §21.4.*)

CDS1100-R039 Channel representations SHOULD be stored at dictionary or mapping level when they apply consistently across products.

12. Classification and Taxonomy Mapping Contracts (*normative*)

`urn:cds:schema:reference:category` and `urn:cds:schema:reference:taxonomy-mapping`. Internal category identity remains stable when external taxonomies change.

Category — required beyond the envelope: `category_id`, `parent_id` (nullable at the root), `name`, `path`, `product_family_id`, `required_attribute_ids`. **path is an ordered array of stable ancestor category_ids ending with the category's own id** — identifiers, not display labels, so renaming a category never rewrites

descendants' paths (CDS1100-15, stable-identity principle).

Taxonomy Mapping — required beyond the envelope: `mapping_id`, `internal_category_id`, `channel`, `taxonomy_version`, `external_node_id`, `mapping_status` (proposed, approved, deprecated, rejected), `inherited`.

```
{
  "mapping_id": "map_cat_womens_shirts_shopify",
  "internal_category_id": "cat_womens_shirts",
  "channel": "shopify",
  "taxonomy_version": "2026-03",
  "external_node_id": "aa-1-2-3",
  "mapping_status": "approved",
  "inherited": false
}
```

CDS1100-Ro40 An external taxonomy mapping MUST identify the external taxonomy version where the publisher provides one.

CDS1100-Ro41 A category path MUST consist of stable category identifiers, and a category parent reference MUST NOT create a cycle. (*Acyclicity is graph-level and remains an implementation obligation — §21.4.*)

CDS1100-Ro42 A taxonomy remap MUST create a new mapping revision and MUST NOT rewrite historical publication evidence.

13. Facet Definition Contract (*normative*)

`urn:cds:schema:reference:facet-definition`. A facet is a customer-experience projection, not merely an attribute flag; facet semantics, UX behaviour, SEO and accessibility rules are owned by CDS-600. Required beyond the envelope: `facet_id`, `label` (localised-text array), `source_attribute_id`, `selection_mode` (`single`, `multi_or`, `multi_and`, `range`), `sort_mode` (`configured`, `alphabetical`, `count`, `numeric`), `value_ids` (unique).

```
{
  "facet_id": "facet_colour_family",
  "label": [{ "language": "en-AU", "text": "Colour" }],
  "source_attribute_id": "VAR_colour",
  "selection_mode": "multi_or",
  "sort_mode": "configured",
  "value_ids": ["facet_black", "facet_white", "facet_blue", "facet_green", "facet_red"]
}
```

CDS1100-Ro43 A customer facet MUST reference a controlled value set (or, for `range` mode, a declared numeric/date range model per CDS-600).

CDS1100-Ro44 Facet value order and selection logic MUST be explicit; raw supplier values MUST NOT be exposed as facet values without dictionary governance (CDS-400) and UX review (CDS-600).

14. Channel Profile and Field Mapping Contracts (*normative*)

`urn:cds:schema:channel:channel-profile` and `urn:cds:schema:channel:field-mapping`. Channel Profile semantics — capabilities, ownership modes, limits, write semantics — are owned by CDS-500 §5 and §7; platform-specific profile content by CDS-900.

Field Mapping is a standalone contract in vo.2 (resolving open decision D21 in favour of standalone): `urn:cds:schema:channel:field-mapping`, with all properties required — `source_field`, `target_field`, `source_layer` (`canonical`, `display`, `facet`, `search` — the ADR-D4 value layers), `transformation`, `transformation_version`, `write_mode` (`replace`, `merge`, `append`), `read_back_path` (nullable when the target is not readable), `comparison_strategy`.

A Channel Profile (required: `channel_profile_id`, `channel`, `profile_version`, `capabilities`, `field_mappings`) MAY inline field mappings; each inlined item conforms to the standalone field-mapping contract by `$ref`. **Authority:** the mapping set pinned by a Publication Record's `mapping_set_version` (§15) is authoritative for interpreting that publication; a channel profile's inlined `field_mappings` express the profile's *current* mapping set and are superseded, for historical interpretation, by whatever version the publication record pinned.

RESOLVED — accepted as drafted (owner 2026-08-04; was D21 (resolved)): Field mapping ships as a standalone envelope-free component schema referenced by channel-profile — standalone won, and the composition keeps one authoritative shape. A top-level *enveloped* "mapping set" document type (independently publishable, revision-controlled) is deferred; today `mapping_set_version` is a version string, not a document reference. Alternative: promote mapping sets to full documents in vo.3.

The shipped capability block is a deliberately minimal core set of four booleans (`supports_readback`, `supports_partial_update`, `supports_variant_attributes`, `supports_structured_metafields`); richer capability declarations (limits, normalisation behaviour, ownership defaults) are profile content per CDS-500 §5 and MAY be carried under `extensions` until standardised.

CDS1100-Ro45 A Field Mapping MUST identify source field, source value layer, target field, transformation and transformation version, write mode, read-back path (or its explicit absence) and comparison strategy.

CDS1100-Ro46 A mapping target path MUST be interpreted within the named Channel Profile, not as a global CDS path.

CDS1100-Ro47 A channel capability declaration MUST distinguish unsupported fields from supported-but-currently-absent values (observation coverage model: CDS-500 §15; encoded in §16).

15. Publication Record Contract (*normative*)

urn:cds:schema:channel:publication-record. The Publication Record is immutable evidence of what the PIM intended to publish: the expected channel state generated from a specific canonical revision under a specific mapping set. Semantics: CDS-500 §6 and §8–9.

Required beyond the envelope: `publication_id`, `product_id`, `channel_profile_id`, `canonical_revision`, `expected_state`, `publication_status` (queued, preflight_failed, published, partially_published, failed), `published_at` (nullable until transport), `transport_reference` (nullable), `mapping_set_version`, `payload_hash` (sha256: + 64 hex), `preflight_result` (passed, failed, not_run), `transport_status` (queued, sent, acknowledged, rejected, failed).

The vo.1 conflation of preflight and transport into one status is resolved: **preflight outcome, transport status and overall publication status are three separate properties**, so "valid but rejected by the channel" and "never dispatched because preflight failed" are distinguishable machine states (CDS1100-3).

```
{
  "publication_id": "pub_shopify_shirt_100_r42",
  "product_id": "prd_shirt_100",
  "channel_profile_id": "chp_shopify_main",
  "canonical_revision": 42,
  "expected_state": { "title": "Relaxed Linen Shirt", "material": ["Linen"] },
  "publication_status": "published",
  "published_at": "2026-08-03T00:10:00Z",
  "transport_reference": "gid://shopify/Product/123456",
  "mapping_set_version": "shopify-1.4.0",
  "payload_hash": "sha256:2af3...",
  "preflight_result": "passed",
  "transport_status": "acknowledged"
}
```

CDS1100-R048 A Publication Record MUST identify the exact canonical revision and mapping-set version used to produce the expected state. (*Revision pinning is executed as reference semantic test SEM-003; see §21.4.*)

CDS1100-R049 Transport acknowledgement MUST remain distinct from preflight outcome and from field-level verification status ("acknowledgement is not proof": CDS-500 §9).

CDS1100-R050 The expected payload's immutable hash MUST be recorded; the full payload MAY be stored externally by reference provided the evidence location is retrievable under CDS-500 §24.

16. Observation Record Contract (*normative*)

urn:cds:schema:channel:observation-record. The Observation Record stores downstream state read back from a channel. It is evidence, not canonical truth. Observation semantics and the coverage model are owned by CDS-500 §14–15.

Required beyond the envelope: `observation_id`, `publication_id`, `product_id`, `channel_profile_id`, `observed_at`, `coverage`, `observed_state`, `channel_revision` (nullable), `observation_method` (api_read, feed_diagnostic, scrape, manual). Optional: `channel_entity_id` (nullable channel-native identity).

coverage is **per-field**: two required lists of JSON Pointer paths, `supported` and `unobservable`. This replaces the vo.1 bare description and makes the missing-versus-unobservable distinction machine-checkable — a field absent from `observed_state` but listed in `supported` is *missing*; a field listed in `unobservable` can never be missing (CDS1100-2).

```
{
  "observation_id": "obs_shopify_shirt_100_001",
  "publication_id": "pub_shopify_shirt_100_r42",
  "product_id": "prd_shirt_100",
  "channel_profile_id": "chp_shopify_main",
  "observed_at": "2026-08-03T00:11:42Z",
  "observation_method": "api_read",
  "coverage": { "supported": ["/title", "/material", "/tags"], "unobservable": ["/search_index_tokenisation"] },
  "observed_state": { "title": "Relaxed Linen Shirt", "material": ["Linen"] },
  "channel_revision": null,
  "channel_entity_id": "gid://shopify/Product/123456"
}
```

CDS1100-R051 Observation data MUST be stored separately from canonical state and from expected channel state.

CDS1100-R052 An Observation Record MUST declare its observation method and timestamp.

CDS1100-R053 The coverage declaration MUST distinguish, per field, a missing value from a field that is unobservable through the selected method.

17. Verification Result Contract (*normative — encodes CDS-500 §17, never redefines it*)

urn:cds:schema:channel:verification-result. Verification Results compare expected with observed field representations under declared comparison strategies.

All status semantics are owned by CDS-500: the core enum and detailed statuses by §17, the traffic-light mapping by §18, aggregation by §17.6. This schema encodes them verbatim per ADR-D3.

Required beyond the envelope: `verification_id`, `publication_id`, `observation_id`, `verified_at`, `overall_status`, `field_results`. Optional: `coverage_ratio` (0..1 — observable fraction of contracted fields, CDS-500 §17.6), `comparison_engine_version` (reproducibility), `presentation_status` (green, amber, red, grey — informative presentation derived per CDS-500 §18, never a stored status).

Each field result carries:

Property	Requirement
<code>field_path</code>	Required. JSON Pointer (RFC 6901) into expected/observed state — replaces the vo.1 bare field name
<code>expected</code> , <code>observed</code>	Required (any type, may be null)
<code>comparison_strategy</code>	Required; from the CDS-500 §16 registry
<code>status</code>	Required; the 8-value core enum: <code>MATCH</code> , <code>MISSING</code> , <code>MISMATCH</code> , <code>PENDING</code> , <code>UNOBSERVABLE</code> , <code>NOT_APPLICABLE</code> , <code>OVERRIDDEN</code> , <code>ERROR</code>
<code>detailed_status</code>	Optional; the CDS-500 §17.2 detailed set with deterministic rollout to the core status
<code>reason_code</code>	Required whenever status is not <code>MATCH</code> (schema-enforced if/then); <code>^CDS_[A-Z_]+\$</code> , from the §21 registry
<code>repair_action</code>	Optional
<code>message</code>	Optional free text; never a substitute for <code>reason_code</code> (CDS500-R063)

```
{
  "field_path": "/material",
  "expected": ["Linen"],
  "observed": [],
  "comparison_strategy": "unordered_text_set",
  "status": "MISSING",
  "detailed_status": "MISSING_DOWNSTREAM",
  "reason_code": "CDS_OBSERVED_EMPTY"
}
```

CDS1100-R054 The verification-result contract MUST encode the CDS-500 §17 status enum verbatim and MUST NOT introduce, remove or re-map statuses.

CDS1100-R055 Every field-level result MUST identify its field by JSON Pointer, its comparison strategy and its core status.

CDS1100-R056 Every field-level result whose core status is not `MATCH` MUST carry a `reason_code` from the §21 registry (encodes CDS500-R063; mechanically enforced by the shipped schema and covered by a negative fixture).

CDS1100-R057 Traffic-light colour MUST NOT be stored as a verification status; it is presentation derived per CDS-500 §18. (*The package's "uppercase GREEN rejected as a status" negative fixture expresses this.*)

18. AI Proposal and Provenance Contract (*normative — encodes CDS-700, never redefines it*)

`urn:cds:schema:automation:ai-proposal`. AI outputs are proposals with evidence and confidence, not canonical values. All AI policy — autonomy levels, review rules, evidence classes, abstention — is owned by CDS-700; this schema encodes the record.

Required beyond the envelope: `proposal_id`, `target_document_id`, `target_field`, `proposed_value` (any type), `confidence` (number 0..1), `evidence` (non-empty array of {`source_type`, `source_reference`, `excerpt`|`null`}), `model_id`, `workflow_version`, `review_state`, `task_type` (extraction, mapping, classification, writing, translation, analysis), `validation_results` (array of §21 validation-output items). Optional: `reviewer` (nullable until reviewed), `accepted_revision` (nullable until accepted).

`review_state` uses the **shared enum** `proposed`, `review_required`, `accepted`, `rejected`, `superseded`, `expired` — identical to CDS-700 §6 (CDS700-R015); CDS-700 and this schema describe the same record, resolving the vo.1 vocabulary split (D14).

```
{
  "proposal_id": "aip_01K1MAT9",
  "target_document_id": "prd_shirt_100",
  "target_field": "/attributes/MF_material",
  "proposed_value": { "dictionary_value_id": "material_linen", "percentage": "100.00" },
  "confidence": 0.97,
  "evidence": [{ "source_type": "supplier_description", "source_reference": "doc_sup_829103", "excerpt": "100% pure linen" }],
  "model_id": "example-extractor-2",
  "workflow_version": "extract-material-1.3.0",
  "review_state": "proposed",
  "task_type": "extraction",
  "validation_results": []
}
```

CDS1100-R058 An AI Proposal MUST remain distinguishable from accepted canonical data until an authorised workflow accepts it.

CDS1100-R059 A proposal MUST identify its evidence, calibrated confidence, model identifier and workflow version, and MUST carry its validation results in the §21 validation-output contract.

CDS1100-R060 Acceptance MUST create an auditable link from the proposal to the canonical revision it affected (`accepted_revision`, plus the reviewing identity in `reviewer`).

19. Conformance Manifest Contract (*normative — encodes CDS-1000 §23, never redefines it*)

urn:cds:schema:assurance:conformance-manifest. Conformance levels, test suites, claim rules and manifest semantics are owned by CDS-1000; this schema is their machine expression.

Required beyond the envelope: `manifest_id`, `implementation_name`, `implementation_version`, `claimed_level` (Foundation, Structured, Publisher, Verified, Governed — the ADR-D1 ladder), `profiles` (unique strings), `test_summary` (passed, failed, not_tested, inconclusive — all four counts required), `evidence_uri`, `test_suite_version`, `assessment_method` (self-attestation, customer-assessment, independent-assessment).

CDS1100-Ro61 A Conformance Manifest MUST identify the implementation scope and MUST NOT imply conformance beyond that scope (claim rules: CDS-1000 §24).

CDS1100-Ro62 The manifest MUST identify the exact test-suite version, the applicable profiles and the assessment method.

CDS1100-Ro63 Every executed test case MUST be counted in exactly one `test_summary` disposition; failed or not-executed mandatory tests MUST NOT be omitted. *(The vo.1 fixture that claimed "Verified" with a count contradicting its own evidence report is retired; the shipped fixture counts match the shipped evidence — CDS1100-6.)*

20. Extension and Profile Rules (*normative*)

CDS supports platform, industry and organisation extensions without uncontrolled schema fragmentation. Envelope `extensions` keys are constrained by the shipped schema's `propertyNames` pattern to **dotted reverse-namespace form** — `^[a-z][a-z0-9_]*(\.[a-z][a-z0-9_]+)$`, e.g. `com.example.photography_status`, `profile.apparel.garment_measurement_set_id` — so namespace governance is mechanically enforced at validation time, not merely urged (CDS1100-12). Visible *field-identifier* prefixes (`MF_`, `VAR_`, ...) are a separate registry owned by CDS-300; the two grammars do not overlap.

```
"extensions": {
  "com.example.warehouse_zone": "A3",
  "profile.apparel.garment_measurement_set_id": "gms_womens_tops_au"
}
```

CDS1100-Ro64 An extension key MUST use the dotted-namespace grammar, and its namespace MUST be organisationally controlled with an identified owner.

CDS1100-Ro65 An extension MUST NOT reuse a core property name with a different meaning, and a profile MAY require additional properties, restrict enumerations or tighten validation but MUST NOT make a core mandatory property optional.

CDS1100-Ro66 Unknown governed extensions SHOULD be preserved during read-modify-write operations unless an explicit policy authorises removal.

21. Validation Output and the Reason-Code Registry (*normative — registry home*)

21.1 Validation output contract

urn:cds:schema:assurance:validation-output — shipped in vo.2 (it was promised and absent in vo.1; CDS1100-1). A validation finding is machine-actionable and human-readable:

Property	Requirement
<code>document_id</code>	Required — the affected document
<code>field_path</code>	Required — JSON Pointer (RFC 6901) to the instance location
<code>reason_code</code>	Required — <code>^CDS_[A-Z_]+\$</code> from the §21.3 registry
<code>severity</code>	Required — <code>error</code> , <code>warning</code> , <code>info</code>
<code>message</code>	Optional free text (nullable)

Validation-output items are embedded by the ai-proposal contract (`validation_results`) and are the recommended shape for import, preflight and API validation responses.

CDS1100-Ro67 A validation finding MUST identify the affected document, the instance location as a JSON Pointer, a registered reason code and a severity; severity MUST distinguish errors, warnings and informational findings.

CDS1100-Ro68 A rejected import SHOULD retain the original value and source provenance in quarantine rather than discarding it (quarantine model: CDS-400 §17).

21.2 Registry governance

CDS1100-Ro69 Machine-readable reason codes MUST match `^CDS_[A-Z_]+$` and MUST be drawn from the `CDS_*` reason-code registry defined here. Additions, deprecations and semantics changes follow CDS-800 change control; a code, once registered, is never reused with a different meaning. Organisation-specific codes are not permitted in the `CDS_` namespace; organisations extend via their own prefix under the CDS-300 extension registry.

21.3 Seed registry (vo.2)

Reason codes are a vo.2 design with no legacy precedent (ADR-D3). The seed list below is the initial registry content; CDS-500 Appendix B mirrors the verification subset informatively.

Reason code	Class	Typical use (detailed status where applicable)
CDS_OBSERVED_EMPTY	verification	MISSING_DOWNSTREAM
CDS_FIELD_NOT_RETURNED	verification	MISSING_DOWNSTREAM or UNOBSERVABLE (CDS-500 §17.4)
CDS_INTERFACE_UNSUPPORTED	verification	UNOBSERVABLE
CDS_PERMISSION_DENIED	verification	UNOBSERVABLE
CDS_STALE_OBSERVATION	verification	UNOBSERVABLE
CDS_AWAITING_PROPAGATION	verification	PENDING
CDS_PROPAGATION_WINDOW_EXPIRED	verification	post-lapse non-MATCH (CDS-500 §17.5)
CDS_EXPECTED_REVISION_AHEAD	verification	STALE_EXPECTED
CDS_VALUE_DIFFERS	verification	MISMATCH
CDS_UNEXPECTED_VALUE_PRESENT	verification	UNEXPECTED_DOWNSTREAM
CDS_IDENTITY_MISMATCH	verification	MISMATCH / drift IDENTITY_MISMATCH
CDS_CHANNEL_TRUNCATION	verification	MISMATCH / drift CHANNEL_TRUNCATION
CDS_TRANSFORMATION_FAILED	verification	TRANSFORMATION_ERROR
CDS_DISPATCH_REJECTED	verification	PUBLICATION_ERROR
CDS_READBACK_FAILED	verification	OBSERVATION_ERROR
CDS_COMPARATOR_FAILED	verification	COMPARISON_ERROR
CDS_WAIVED_EXCEPTION	verification	WAIVED
CDS_OVERRIDE_ACTIVE	verification	provenance on OVERRIDDEN results
CDS_NOT_IN_CONTRACT	verification	NOT_APPLICABLE
CDS_SCHEMA_VALIDATION_FAILED	validation	document fails its declared schema
CDS_SCHEMA_UNKNOWN	validation	<code>cds_schema</code> unknown or unsupported (Ro18)
CDS_VALUE_NOT_IN_DICTIONARY	validation	value does not resolve to an active dictionary value
CDS_VALUE_QUARANTINED	validation	value held in quarantine (CDS400-Ro54)
CDS_REFERENCE_UNRESOLVED	validation	cross-document reference does not resolve
CDS_TENANT_BOUNDARY_VIOLATION	validation	reference crosses a tenant boundary (Ro22)

21.4 Semantic and referential integrity

JSON Schema validates one document at a time. Integrity that spans documents is specified as semantic tests. In vo.2 the reference runner **executes** the semantic catalogue (vo.1 shipped it dead; CDS1100-8):

Test	Integrity class	Encoding requirement
SEM-001	Variant references resolve; parent <code>product_id</code> matches	Ro28
SEM-002	Facet memberships resolve to configured facet values	Ro38
SEM-003	Publication <code>canonical_revision</code> equals the product revision projected	Ro48
SEM-004	Related documents share one tenant boundary	Ro22 (illustration)

The following integrity classes remain **implementation obligations** with no reference-runner check in vo.2: dictionary-reference resolution to *active* values (lifecycle is in the dictionary-value document, not the referencing one); attribute-value conformance to the owning Attribute Definition's type, cardinality and dictionary (§7.1); category-graph acyclicity (Ro41); taxonomy-mapping resolution against the published external taxonomy; and tenant isolation at the storage and API layer (Ro22, tested via CDS-1000 §18 T-TEN).

CDS1100-Ro70 An implementation MUST enforce the cross-document integrity classes above before accepting a document as authoritative, whether or not a reference-runner check exists for them.

22. Versioning and Compatibility (*normative — implements ADR-D5*)

Schema versioning follows the CDS governance model (change control: CDS-800). Per schema: breaking changes require a new major version; additive compatible changes a minor version; corrections that do not change accepted or produced instance meaning a patch version. The `$id` never changes with the version (§4).

Change	Compatibility expectation
Add optional property	Usually backward compatible (minor)
Add required property without default/migration	Breaking
Remove or rename property	Breaking (rename: unless an alias/migration profile is provided)
Tighten enumeration	Potentially breaking
Add enumeration value	May break closed consumers; profile policy required
Change numeric precision or unit semantics	Breaking
Clarify description only	Patch if validation meaning is unchanged
Change mapping transformation	Mapping-set version change; no core schema change implied

CDS1100-R071 A schema MUST declare its semantic version independently of the CDS corpus release and of any document revision.

CDS1100-R072 A breaking contract change MUST publish migration guidance and a new major schema version; the schema `$id` MUST NOT change.

CDS1100-R073 Historical publication and verification records MUST retain the schema versions, `cds_version` and mapping-set versions under which they were created; a corpus release MUST NOT rewrite stored evidence.

Migration note (informative): v0.1 URNs of the form `urn:cds:schema:v0.1:*` map one-to-one to the release-independent URNs of §4 for anything already persisted (ADR-D5).

23. Security, Privacy and Data Minimisation (*normative*)

Product information is usually less sensitive than customer data, but contracts may carry supplier costs, unpublished products, workflow notes, provenance and user identities.

CDS1100-R074 Secrets, API credentials and access tokens MUST NOT be embedded in CDS documents.

CDS1100-R075 A producer MUST publish only the fields required by the destination profile and approved organisation policy; cost and margin fields SHOULD be excluded from channel payloads unless the destination is explicitly authorised.

CDS1100-R076 Inbound documents MUST be treated as untrusted input and validated for size, recursion depth, markup, URLs and extension content; markup-bearing text MUST be sanitised before render. Schema validation MUST NOT be treated as sufficient security validation.

CDS1100-R077 Personal reviewer identities SHOULD be represented by governed user IDs and exposed only as policy permits; logs and evidence packages MUST follow retention and access policies (CDS-500 §24, CDS-800).

Cross-tenant reference rejection: R022. External references SHOULD use allow-listed schemes and hosts where automatic retrieval is supported.

24. Worked Apparel Contract (*informative*)

Separation of supplier value, canonical value, customer facet, channel representation and verification evidence (four-layer value model: CDS-400; worked end-to-end in the reference fixtures):

Supplier input:	<code>colour = "French Navy"; composition = "100% Linen"</code>
Canonical:	<code>VAR_colour -> dictionary_value_id colour_navy (reference form "French Navy" retained per CDS-400 intake) MF_material = [{dictionary_value_id: material_linen, percentage: "100.00"}] facet_colour_family = facet_blue; facet_material_family = linen</code>
Facet projection:	
Shopify expected:	<code>option colour = "French Navy"; metafield colour_family = "Blue"; metafield material = "100% Linen"</code>
Google expected:	<code>color = "Blue"; material = "Linen"</code>
Observed Shopify:	<code>option colour = "French Navy"; colour_family = "blue"; material = "100% linen"</code>
Verification:	<code>/options/colour -> MATCH (exact_text) /colour_family -> MATCH (case_insensitive_enum) /material -> MATCH (normalised_composition)</code>

Display, canonical, facet and channel values may all differ while remaining correctly mapped and verifiable — that is the point of declared comparison strategies.

25. Worked Homewares Contract (*informative*)

A detailed display material coexisting with a broader customer-facing family:

Supplier input:	<code>material = "Tasmanian Oak"; finish = "Natural Oil"; dimensions = "45 x 45 x 50 cm"</code>
Canonical:	<code>MF_material = material_oak; MF_material_origin_label (display text) MF_finish = finish_natural_oil MF_width = {measurement: {value:"45", unit:"cm"}} MF_depth = {measurement: {value:"45", unit:"cm"}} MF_height = {measurement: {value:"50", unit:"cm"}} material_family = wood; finish_family = natural; room = living_room</code>
Facets:	
Channels:	<code>Shopify material = "Oak"; Google material = "Wood"; PDP display = "Tasmanian Oak"</code>
Verification:	<code>dimensions -> numeric-with-unit comparison material -> canonical identifier comparison display -> normalised text comparison</code>

Structured measurements are exchanged as separate value-and-unit fields, not as the unparsed display string, whenever the destination supports structured fields (R023).

26. Conformance, Package and Decision Pointers (*informative*)

- **Conformance.** Levels, test suites, evidence and claim rules for this chapter's requirements are defined solely in CDS-1000; the machine-readable claim is §19's manifest.
- **Reference package.** The CDS-1200 Reference Package v0.2.1 ships the 21 schemas of Appendix A, positive and negative fixtures, the executed structural and semantic test catalogues, the validation runner and the regenerable manifest. Package composition, runner behaviour and evidence conventions are

CDS-1200's subject matter.

- **Decisions.** The vo.1 chapter-local ADR table (CDS-ADR-1101..1110) is superseded by the global ADR register (ADR-D3 statuses and reason codes; ADR-D4 value layers; ADR-D5 identifiers and versioning). Its substance survives as requirements here: JSON + Draft 2020-12 (§3), YAML as a view not a model (§3, Appendix C), the common envelope (§5), canonical/expected/observed/verification separation (§8, §15–17), stable IDs independent of labels (§6, §11), AI-as-proposal (§18), namespaced extensions (§20), payload-by-immutable-reference (§15), stable JSON Pointer error paths (§21).

Appendix A — Reference Schema Registry (*normative*)

The registry of shipped vo.2 schemas. \$ids are release-independent (§4); all shipped schema versions are 0.2.0 in this release. This table is regenerated from the package manifest at each release.

Schema \$id	Version	Purpose
urn:cds:schema:common:envelope	0.2.0	Common envelope and audit metadata
urn:cds:schema:common:entity-reference	0.2.0	Typed cross-document reference
urn:cds:schema:common:localised-text	0.2.0	Language tag plus text
urn:cds:schema:common:measurement	0.2.0	String-decimal value and unit
urn:cds:schema:common:money	0.2.0	String-decimal amount and ISO 4217 currency
urn:cds:schema:core:product	0.2.0	Canonical logical product
urn:cds:schema:core:variant	0.2.0	Sellable unit
urn:cds:schema:core:attribute-definition	0.2.0	Attribute schema and behaviour
urn:cds:schema:reference:dictionary	0.2.0	Dictionary definition
urn:cds:schema:reference:dictionary-value	0.2.0	Controlled canonical value
urn:cds:schema:reference:category	0.2.0	Internal taxonomy node
urn:cds:schema:reference:taxonomy-mapping	0.2.0	External taxonomy projection
urn:cds:schema:reference:facet-definition	0.2.0	Customer filter definition
urn:cds:schema:channel:channel-profile	0.2.0	Channel capabilities and mapping carrier
urn:cds:schema:channel:field-mapping	0.2.0	Canonical-to-channel field transformation
urn:cds:schema:channel:publication-record	0.2.0	Expected state and transport evidence
urn:cds:schema:channel:observation-record	0.2.0	Read-back evidence with per-field coverage
urn:cds:schema:channel:verification-result	0.2.0	Field comparison, statuses and reason codes
urn:cds:schema:automation:ai-proposal	0.2.0	AI candidate, evidence and review
urn:cds:schema:assurance:conformance-manifest	0.2.0	Machine-readable conformance claim
urn:cds:schema:assurance:validation-output	0.2.0	Validation findings with CDS_ * reason codes

The vo.1 registry keys (cds/common/envelope style), the https://schemas.cds.example/... example URIs, and the unshipped vo.1 sketch entries (identifiers, language, provenance as separate schemas; a profiles/ schema directory) are retired. Identifier and provenance shapes live inside the entity schemas; language lives in localised-text.

Appendix B — Illustrative JSON Schema Fragments (*informative*)

Shipped fragments, quoted from the vo.2 package:

```
{
  "$schema": "https://json-schema.org/draft/2020-12/schema",
  "$id": "urn:cds:schema:common:measurement",
  "version": "0.2.0",
  "title": "CDS Measurement",
  "type": "object",
  "properties": {
    "value": { "type": "string", "pattern": "^-?\\d+(\\.\\d+)?$" },
    "unit": { "type": "string", "pattern": "^[a-zA-Z]{a-zA-Z0-9_./-}* $" }
  },
  "required": ["value", "unit"],
  "additionalProperties": false
}
```

```
{
  "$schema": "https://json-schema.org/draft/2020-12/schema",
  "$id": "urn:cds:schema:common:entity-reference",
  "version": "0.2.0",
  "title": "CDS Entity Reference",
  "type": "object",
  "properties": {
    "document_type": { "type": "string", "minLength": 1 },
    "document_id": { "type": "string", "minLength": 3 },
    "role": { "type": "string", "minLength": 1 },
    "tenant_id": { "type": "string", "minLength": 1 }
  }
}
```

```

},
"required": ["document_type", "document_id"],
"additionalProperties": false
}

```

Composition idiom (entity schemas):

```

{
  "$id": "urn:cds:schema:core:variant",
  "allOf": [
    { "$ref": "urn:cds:schema:common:envelope" },
    { "type": "object", "properties": { "...entity properties..." : {} }, "required": ["..."] }
  ],
  "unevaluatedProperties": false
}

```

Appendix C — YAML Representation Rules (*informative*)

- Quote identifier values that may look numeric: GTINs, SKUs with leading zeros, postal codes.
- Quote date-like values when the intended CDS type is plain text rather than date.
- Do not use YAML tags or implementation-specific object constructors in interoperable CDS files.
- Do not rely on anchors and aliases to create business identity; stable document IDs remain authoritative.
- Convert YAML to the canonical JSON data model before schema validation (R010).
- Preserve mapping keys and list order exactly where order is declared significant.

```

document_type: variant
variant_id: var_shirt_100_navy_m
sku: SHIRT-100-NVY-M
barcode_identifiers:
  - scheme: gtin_14
    value: "09338716007824"

```

Appendix D — Standards References (*informative*)

Reference	Location
JSON Schema Draft 2020-12	https://json-schema.org/draft/2020-12
YAML 1.2.2 specification	https://yaml.org/spec/1.2.2/
RFC 3339 — Date and Time on the Internet	https://www.rfc-editor.org/rfc/rfc3339
RFC 6901 — JSON Pointer	https://www.rfc-editor.org/rfc/rfc6901
RFC 8259 — JSON	https://www.rfc-editor.org/rfc/rfc8259
BCP 47 language tags	https://www.rfc-editor.org/info/bcp47
RFC 8141 — URN syntax	https://www.rfc-editor.org/rfc/rfc8141

END OF CDS-1100 vo.2 REVIEW DRAFT

Commerce Data Standard (CDS)

CDS-1200 — Reference Implementation, Schema Package and Test Fixtures

Field	Value
Status	vo.2 Review Draft (working source; not an approved standard)
Release	CDS vo.2
Date	2026-08-04
Supersedes	CDS-1200 Working Draft vo.1 (and Reference Package vo.1.0)
Companion package	CDS-1200 Reference Package vo.2.1 (semver, independent of chapter prose — ADR-D5; schemas remain 0.2.0)
Normative status	§1–§2, §4–§16 and §18 are normative except where marked otherwise. §3, §17, §19–§21, Annex A and Appendices A–B are informative.
Findings addressed	CDS1200-1..13, PKG-1, PKG-2 (REVIEW-010); ADR-D5

A reproducible reference package that turns the CDS machine-readable contracts into executable schemas, fixtures, an executable structural and semantic test suite, and environment-annotated validation evidence.

1. Purpose and Scope (*normative*)

CDS-1200 defines the executable reference implementation that accompanies the machine-readable contracts in CDS-1100. Its purpose is to make the standard reproducible: implementers obtain the same schemas, run the same fixture suite, observe the same expected failures and produce comparable validation evidence.

The reference implementation is deliberately small enough to read in full, but complete enough to demonstrate the core CDS lifecycle from canonical product records through channel publication, observation and verification (CDS-500). It is not a production PIM, commerce connector or certification service, and it confers no conformance claim by itself (claims: CDS-1000).

CDS1200-R001 A CDS reference package **MUST** contain executable schemas, positive fixtures, negative fixtures, a machine-readable expected-results catalogue, a machine-readable semantic test catalogue and a repeatable validation procedure.

CDS1200-R002 A reference fixture **MUST NOT** depend on undocumented database state, a network service, the mutable current time or an external API response.

CDS1200-R003 Where the reference package and the published specification conflict on **normative content** — contract meaning, requirement wording, enumerations, identifier and versioning rules — the specification governs and the package carries the defect. This precedence rule does **NOT** extend to execution evidence: test reports, result counts and environment records are informative snapshots of a recorded run (§16), never specification prose, and the specification cannot make a non-reproducing number authoritative over an actual run. (*Re-scoped per finding CDS1200-8.*)

2. Relationship to CDS-1100 (*normative*)

CDS-1100 specifies the logical contracts and their serialization rules; CDS-1200 packages those contracts into concrete schema files, reusable examples and automated tests. The chapters are complementary, not duplicative. Contract semantics, type rules and extension rules are homed in CDS-1100 and are not restated here.

CDS-1100 responsibility	CDS-1200 responsibility
Define the logical document contracts	Publish executable schema files
Define identifier, type and extension rules	Provide conforming and non-conforming examples
Define validation and compatibility expectations	Run validation and produce evidence
Define the reference schema registry	Distribute the registry as a versioned package
Define document semantics	Demonstrate cross-document and lifecycle behaviour

CDS1200-R004 The reference package **MUST** identify the CDS release its contracts implement: the envelope pins `cds_version` as a per-release constant ("0.2" for this release), and every schema carries its own semantic version (ADR-D5; details CDS-1100).

CDS1200-R005 A package release **MUST NOT** silently change the meaning of a schema without a corresponding specification change or schema version change.

3. Reference Implementation Principles (*informative*)

- **Executable:** every claimed schema rule is represented in a validation artefact where the schema language can express it.
- **Reproducible:** a clean environment, with the declared pinned dependencies installed, runs the suite and obtains the same expected outcomes. An environment that cannot enforce a declared assertion fails loudly instead of passing weakly (§5).
- **Deterministic:** fixture IDs, timestamps, values and expected reports are fixed.
- **Inspectable:** schemas and fixtures are readable without proprietary tooling.
- **Layered:** structural schema validation is separated from semantic cross-document tests — and both layers are executed (§12).

- **Positive and negative:** a rule is demonstrated by valid examples and by intentionally invalid counterexamples that fail for the intended reason (§11).
- **Channel-aware:** the package includes an expected publish, observed state and verification chain (§13).
- **Safe:** fixtures contain synthetic data — no real credentials, customer records or confidential supplier data.
- **Portable:** the logical test catalogues are independent of the reference Python runner.
- **Evidence-producing:** a run emits a structured, environment-annotated report rather than only console text.

4. Distribution Package Layout (*normative*)

```
CDS-1200_Reference_Package_v0.2/
  README.md
  CHANGELOG.md
  LICENSE.txt
  requirements.txt
  package-manifest.json
  schemas/
    common/  core/  reference/  channel/  automation/  assurance/
  fixtures/
    positive/
    negative/
  tests/
    expected-validation-results.json
    semantic-tests.json
  tools/
    validate.py
    make_manifest.py
  evidence/
    test-report.json          (generated)
```

CDS1200-R006 A reference distribution MUST keep schemas, fixtures, test catalogues, tools and generated evidence logically separate.

CDS1200-R007 A consumer MUST be able to identify which files are source artefacts and which are generated evidence. In the reference layout, everything under `evidence/` is generated; everything else is source.

5. Dependency Declaration and Environment Integrity (*normative*)

The v0.1 package shipped without a dependency declaration, and its headline evidence claim did not reproduce in a clean environment: JSON Schema Draft 2020-12 treats `format` as annotation-only by default, and the Python format checker silently skips the `date-time` assertion unless the optional `rfc3339-validator` package is installed. A clean run therefore accepted an invalid timestamp fixture (22/23 against a claimed 23/23; REVIEW-010). The v0.2 package closes both the declaration gap and the silent-weakening gap.

CDS1200-R008 A reference package MUST declare its complete runtime dependencies with pinned versions in a machine-readable file distributed with the package.

CDS1200-R009 The reference runner MUST verify at startup that every validation capability the suite depends on (including the `date-time` format assertion) is actually enforced in the current environment, and MUST exit non-zero with an actionable message when one is not — it MUST NOT continue with silently weakened validation.

Informative note. `requirements.txt` pins `jsonschema==4.26.0`, `referencing==0.37.0` and `rfc3339-validator==0.1.4`. `tools/validate.py` probes a deliberately invalid date-time against a `format: date-time` schema before loading anything; if the probe validates, the runner exits with `FATAL: the 'date-time' format assertion is not enforced in this environment` and the install command. This was proven the same way the original defect was found: in a third clean `virtualenv` with no dependencies installed, the v0.2 runner refuses to run at all — where the v0.1 runner reported a quietly wrong pass — and with the pinned dependencies installed the full suite reproduces (§17).

6. Package Manifest and Integrity (*normative*)

`package-manifest.json` inventories the distributed files with a SHA-256 digest and byte length per file, plus the schema registry inventory (§7). This supports reproducible downloads, release comparison and evidence that a run used the intended artefacts.

The manifest convention (*resolves finding CDS1200-6*): the manifest lists every package file **except itself and `evidence/`**. The manifest cannot contain its own hash without circularity, and evidence is regenerated per run and instead embeds the manifest's hash (`manifest_sha256` in the report). The manifest's top-level `package_hash` is the SHA-256 over the sorted `"path sha256"` lines, giving a single package-level integrity value that covers all listed files. The manifest is regenerated with `tools/make_manifest.py`, run *before* the validator so the report embeds the current manifest hash.

CDS1200-R010 A released package MUST include a machine-readable file manifest with a cryptographic hash for each distributed file, and MUST state its self-exclusion convention.

CDS1200-R011 The manifest MUST include a single package-level hash derived deterministically from the per-file hashes.

CDS1200-R012 Generated evidence intended for assurance MUST embed the hash of the package manifest (or an equivalent package release identifier) so the evidence identifies the exact artefacts evaluated. (*Closes finding CDS1200-10: the v0.2 report records `manifest_sha256`.*)

CDS1200-R013 A validator or assurance process SHOULD verify package hashes before treating results as release evidence.

7. Executable Schema Registry (*normative*)

The vo.2 package ships **21 schemas** implementing the registry needed to demonstrate the canonical and publication lifecycle. Each schema declares JSON Schema Draft 2020-12, a release-independent URN `$id` and its own semantic version.

Domain	Schemas (21)
common (5)	envelope, entity-reference, localised-text, measurement, money
core (3)	product, variant, attribute-definition
reference (5)	dictionary, dictionary-value, category, taxonomy-mapping, facet-definition
channel (5)	channel-profile, field-mapping, publication-record, observation-record, verification-result
automation (1)	ai-proposal
assurance (2)	conformance-manifest, validation-output

Identifier policy (ADR-D5, homed in CDS-000/CDS-1100; restated here only as it binds the package): `$ids` take the form `urn:cds:schema:<domain>:<name>` with **no release version in the URN**, so identifiers are stable across releases; each schema carries a `version` keyword with its own semver (all `0.2.0` at this release); instances pin the corpus release via the envelope `cds_version` constant. (*Resolves finding CDS1200-13: vo.1's "stable \$id" wording contradicted its version-embedded `urn:cds:schema:v0.1:*` URNs.*)

CDS1200-R014 Every reference schema MUST declare the supported schema dialect, a release-independent `$id` and a per-schema semantic version, per the identifier policy in CDS-1100.

CDS1200-R015 The package MUST include all referenced schemas required to validate the distributed fixtures without network access, and the manifest MUST inventory the registry (id, version, path).

8. Shared Definitions and Composition (*normative*)

Common types are defined once and reused by `$ref`. Entity schemas compose the common document envelope with entity-specific properties, so identifiers, revisions, timestamps and source-system fields behave consistently.

```
{
  "$id": "urn:cds:schema:core:product",
  "version": "0.2.0",
  "allOf": [
    {"$ref": "urn:cds:schema:common:envelope"},
    {"type": "object", "properties": {"document_type": {"const": "product"}}}
  ],
  "unevaluatedProperties": false
}
```

The envelope requires `cds_schema`, `cds_version`, `document_type`, `document_id`, `revision`, `status`, `created_at`, `updated_at` and `source_system`; `cds_version` is `{"const": "0.2"}` (*resolves finding CDS1200-7 – vo.1 fixtures declared "1.0" against an unpinning pattern*); extension keys are constrained to dotted reverse-namespace form (extension rules: CDS-1100, CDS-300).

CDS1200-R016 A shared definition MUST have exactly one authoritative schema file within a package release.

CDS1200-R017 Entity schemas SHOULD close unevaluated top-level properties unless a governed extension point is explicitly provided.

CDS1200-R018 The reference runner MUST cross-check each instance's declared `cds_schema` against the `$id` of the schema actually applied, and treat a mismatch as a validation error.

9. Fixture Design Model (*normative*)

A fixture is a stable document designed to demonstrate one or more contract rules, identified by path, declared schema and expected outcome. Positive fixtures model realistic commerce records; negative fixtures isolate a specific invalid condition.

Fixture property	Rule
Synthetic	No personal, confidential or credential data
Minimal but meaningful	Enough business context to explain the rule
Stable	No current timestamps, random IDs or live taxonomy lookups
Single purpose	A negative case fails for one declared principal reason
Self-identifying	The envelope declares document type and schema
Traceable	The expected-results catalogue names the schema, expected outcome and, for negatives, the intended error

CDS1200-R019 Every fixture MUST have a declared expected validation outcome in the machine-readable catalogue.

CDS1200-R020 A negative fixture SHOULD isolate one principal violation so that a failed test remains diagnosable.

CDS1200-R021 Reference fixtures MUST be synthetic and MUST NOT contain live credentials, personal customer information or confidential supplier data.

10. Positive Fixture Suite (*normative*)

The vo.2 suite contains **16 positive structural fixtures**. Together they demonstrate the canonical product, controlled dictionary, classification, channel profile and publication assurance chain.

Fixture group	Demonstrated behaviour
Attribute Definitions (2)	Semantic prefix syntax, scope, value type, cardinality, comparison behaviour, dictionary binding
Dictionaries (2)	Stable dictionary identity, canonical labels (localised-text form), aliases, facet memberships, channel representations
Classification (2)	Internal category path (stable-ID array) and Shopify taxonomy mapping
Product and variants (3)	Canonical product truth, typed attribute values, variant options, SKU and zero-preserving GTIN handling
Channel profile (1)	Capabilities, field mappings (source layer, write mode, comparison strategy), transformations
Publication chain (3)	Expected channel state, observed state, field-level verification (§13)
Facets (1)	Facet family definition and configured values
AI proposal (1)	Proposed value, evidence, confidence, review state, model and workflow version (rules: CDS-700)
Conformance manifest (1)	Declared level, profiles, assessment method, test summary (semantics: CDS-1000 §26)

CDS1200-R022 All positive fixtures **MUST** validate under their declared schemas, and the package **MUST** include at least one complete canonical product-to-verification lifecycle fixture chain.

11. Negative Fixture Suite (*normative*)

The package includes **13 negative fixtures**. Each fails under its declared schema, and — new in vo.2 — each catalogue entry declares the *intended* error via `expected_error_substrings`. Negative tests prove that a validator rejects invalid input for the right reason, not merely that it produced some error.

The mechanism (*resolves findings CDS1200-3 / PKG-2*): because entity schemas compose the envelope via `allOf` and close with `unevaluatedProperties: false`, any subschema failure in Draft 2020-12 drops property annotations and triggers a blanket "Unevaluated properties are not allowed" cascade error alongside the real one. In vo.1 the runner asserted only "any error", so the suite could not detect loss of the constraint actually under test. The vo.2 runner requires at least one declared intended substring among the reported errors and **excludes the `unevaluatedProperties` cascade line from counting as evidence** — a negative case whose only surviving error is the cascade is a suite failure.

Negative fixture	Intended principal failure
product-missing-product-id.json	Missing required <code>product_id</code>
envelope-missing-revision.json	Missing required envelope <code>revision</code>
envelope-bad-status.json	Envelope status not in the controlled enumeration
variant-invalid-gtin.json	Barcode contains a non-digit
attribute-invalid-semantic-name.json	Field identifier violates CDS semantic prefix syntax (CDS-300)
dictionary-value-invalid-code.json	Canonical code is not lowercase snake_case (CDS-400)
publication-invalid-status.json	Publication status not in the controlled enumeration (CDS-500)
observation-invalid-timestamp.json	Timestamp is not RFC 3339 date-time
timestamp-missing-offset.json	Date-time lacks a timezone offset
product-attribute-wrong-type.json	Attribute value violates the typed-value shape
verification-invalid-status.json	Presentational colour label used where a normative verification status is required (CDS-500)
verification-missing-reason-code.json	Non-MATCH field result lacks the required <code>reason_code</code> (CDS-500)
ai-confidence-out-of-range.json	Confidence exceeds 1.0 (CDS-700)

CDS1200-R023 A package **MUST** contain negative fixtures for required fields, controlled enumerations, semantic identifier syntax, numerical ranges and standard formats where those constraints are normative.

CDS1200-R024 A validator run **MUST** be considered failed if a negative fixture unexpectedly validates.

CDS1200-R025 Each negative catalogue entry **MUST** declare the intended error (as one or more expected error substrings or an equivalent machine-checkable assertion), and the runner **MUST** require the intended error to be present, excluding composition-cascade errors from counting as evidence.

12. Semantic and Cross-Document Tests (*normative*)

JSON Schema validates a document in isolation. CDS also requires relationships that span documents, revisions and tenant boundaries. These live in a separate machine-readable semantic catalogue (`tests/semantic-tests.json`) and — new in vo.2 — are **executed by the reference runner**, not merely documented. (*Resolves finding CDS1200-4: in vo.1 the semantic catalogue was aspirational, SEM-001 would have failed if run, and the CI narrative listed steps the runner never performed.*)

Test	Cross-document rule
SEM-001	Product variant references resolve to variants whose <code>product_id</code> matches the product
SEM-002	Dictionary value facet membership resolves to a configured facet value

Test	Cross-document rule
SEM-003	Publication <code>canonical_revision</code> equals the product revision used to calculate expected state
SEM-004	All related documents remain within the same tenant boundary

Scope honesty: SEM-001..004 demonstrate the *mechanism* — a distinct, executed semantic layer over already-validated documents. They are four checks over a small fixture set, not a complete semantic integrity suite. Production-scale referential integrity, authority enforcement and tenancy isolation remain the implementation's obligation, tested under CDS-1000 (for example T-TEN-001 for tenant isolation).

CDS1200-R026 An implementation MUST NOT represent cross-document or semantic integrity as established merely because each individual document passes structural schema validation. (*Applies equally to this chapter's own evidence: §17 reports structural and semantic counts separately.*)

CDS1200-R027 A semantic test MUST identify all documents or services required to evaluate the rule.

CDS1200-R028 The reference runner MUST execute the packaged semantic catalogue, and a run MUST be considered failed when any semantic outcome differs from its declared expectation.

13. Publication Assurance Test Chain (*normative*)

The reference apparel fixtures demonstrate the CDS-500 assurance lifecycle using four linked documents, separating canonical revision, expected state, downstream observation and verification evidence. Verification statuses, detailed statuses and traffic-light presentation are homed in CDS-500 and not restated here.

```
prd_shirt_100 revision 42
-> pub_shopify_shirt_100_r42    expected title, category, material, tags
-> obs_shopify_shirt_100_001    observed downstream representation
-> ver_shopify_shirt_100_001    field comparison + overall MATCH
```

The observation returns the same tags in a different order; the declared comparison strategy treats tags as an unordered set, so the verified result remains MATCH.

CDS1200-R029 A publication assurance fixture MUST preserve the identifiers linking canonical revision, publication, observation and verification result, and MUST keep canonical, observed and verification values in distinct documents.

CDS1200-R030 A verification fixture MUST declare the comparison strategy used for each compared field.

14. Determinism and Reproducibility (*normative*)

A reference suite is useful only when independent implementers can reproduce its results. The package therefore fixes timestamps, revisions, IDs, source systems, channel states and expected reports:

- No fixture calls a live channel or taxonomy service.
- No fixture obtains the current time during validation.
- No test depends on random values or array iteration order.
- All schema references resolve from the local package registry.
- Generated reports use stable case ordering.
- Package and manifest hashes let evidence identify the exact artefacts used.

CDS1200-R031 A reference fixture MUST be deterministic under the documented comparison and validation rules.

CDS1200-R032 Where a test requires randomness or time simulation, the seed and clock MUST be explicitly controlled and recorded.

CDS1200-R033 The packaged validation suite MUST run without network access.

15. Reference Validation Runner and CLI Contract (*normative*)

The package contains a small Python runner as an **informative** executable implementation (the standard is language-neutral; §18). A clean run:

1. Verifies that declared validation capabilities are enforced (§5) — else exits FATAL.
2. Loads every schema file and registers it by `$id` in a local registry.
3. Validates each positive fixture and requires no errors; validates each negative fixture, requires errors and requires the intended error substring (§11).
4. Cross-checks `cds_schema` against the applied schema (§8).
5. Executes the semantic catalogue SEM-001..004 (§12).
6. Writes the evidence report, embedding the manifest hash and environment (§16).
7. Exits 0 only when every structural and semantic case matches its expectation.

CLI behaviour	Contract
Default run	Validate the packaged structural and semantic catalogues
Report output	Print a structured JSON summary; optionally write the full report to a path
Exit 0	All actual outcomes equal expected outcomes
Non-zero exit	Any outcome differs, package loading fails, or a required validation capability is unavailable

CLI behaviour	Contract
Offline	All schema references resolve locally
Diagnostics	Include fixture, schema, expected, actual, outcome and validator errors

```
python3 -m venv .venv
.venv/bin/pip install -r requirements.txt
.venv/bin/python tools/validate.py --write-report evidence/test-report.json
```

CDS1200-R034 A reference runner MUST return a non-zero process status when any actual outcome differs from the expected outcome or the package cannot be loaded and verified.

CDS1200-R035 An alternative implementation MAY replace the reference runner if it evaluates the same catalogues and produces equivalent outcomes.

16. Validation Evidence and Reporting (*normative*)

Validation evidence is a machine-readable record of the package version, environment, test cases, expected and actual outcomes, and error messages. Evidence is **informative**: it is a snapshot of a recorded run in a recorded environment, never governed prose (§1, CDS1200-R003). Evidence feeding a conformance claim is governed by CDS-1000's evidence-package rules.

Evidence field	Purpose
package_version	Identifies the schema and fixture release
generated_at	When the evidence was generated
environment	Python version, platform, and pinned dependency versions actually resolved
manifest_sha256	Hash of the package manifest — ties evidence to exact artefacts (CDS1200-R012)
summary	Structural and semantic case counts, passed/failed, reported separately and combined
fixture / schema_id	Identifies the tested document and governing contract
expected / actual / outcome	PASS only when actual equals expected — an intentionally invalid negative fixture passes its test by failing validation
detail / errors	Intended-error assertion result and preserved validator diagnostics

CDS1200-R036 A validation report MUST distinguish instance validity from test-case outcome.

CDS1200-R037 Generated evidence MUST identify fixture, schema, expected result, actual result, test outcome and the execution environment (language runtime and dependency versions).

CDS1200-R038 Structural and semantic results MUST be separately identifiable in the report and its summary.

17. Execution Snapshot (*informative*)

The following is the recorded outcome of executing Reference Package v0.2.1 in a clean environment with the pinned dependencies installed. It is an evidence snapshot, not a normative claim (§1); reproducing it requires the recorded environment or a compatible one.

Metric	Result
Schemas	21
Positive fixtures	16
Negative fixtures	13
Structural cases	29 / 29 as expected
Semantic cases (SEM-001..004)	4 / 4 as expected
Total cases	33 / 33, 0 unexpected outcomes
Manifest-listed files	60 (manifest itself and evidence/ excluded by convention, §6)
Evidence ties	manifest_sha256 embedded in report; manifest_package_hash present

Recorded environment: Python 3.14.7; jsonschema 4.26.0, referencing 0.37.0, rfc3339-validator 0.1.4 (per evidence/test-report.json, generated 2026-08-17). In a clean environment *without* the pinned dependencies, the runner exits FATAL at startup rather than reporting a weakened pass (§5) — the failure mode that produced v0.1's non-reproducing 23/23 (REVIEW-010) can no longer occur silently.

18. Release Packaging, Versioning and Language Neutrality (*normative*)

The reference package uses semantic versioning independently of chapter prose (ADR-D5). A patch release corrects a fixture or tool without changing schema meaning; a minor release adds backward-compatible schemas or fixtures; a major release introduces incompatible contract changes.

Change	Package version impact
Documentation typo only	Patch
Additional positive or negative fixture	Minor, unless it changes a prior expected outcome
Backward-compatible optional schema property	Minor

Change	Package version impact
New required property	Major
Enumeration removal or meaning change	Major
Validator bug fix restoring specified behaviour	Patch, with documented evidence change

CDS1200-R039 Every package release MUST publish a changelog, a manifest and a test report. (*The vo.2.1 release ships all three; vo.2.0 first closed finding CDS1200-5, under which the vo.1 release failed its own rule.*)

CDS1200-R040 A fixture outcome change between releases MUST be explained as a standard correction, a test correction or an intentional compatibility change.

CDS1200-R041 Conformance MUST be determined by documented outcomes against the packaged catalogues, not by use of a particular programming language or library. Python is used only for the initial informative runner; any language may host an equivalent implementation (CDS1200-R035).

RESOLVED (was D27 (resolved), owner decision 2026-08-04, research REVIEW-006A): dual licence adopted — **Apache-2.0** for machine-readable artifacts (schemas, fixtures, tests, tools, manifests; express patent grant) and **CC-BY 4.0** for specification prose, matching prevailing open-standard practice (OpenAPI, AsyncAPI, CloudEvents use Apache-2.0; W3C separates document and software licences). `LICENSE.txt` declares the split; `LICENSE-CODE.md` and `LICENSE-SPEC.md` contain the full terms; tools carry SPDX headers. The distribution blocker recorded in the original vo.2 draft was closed on 2026-08-17.

19. Implementation Guidance: CI, Release Discipline and Code Generation (*informative*)

The vo.1 draft stated several MUSTs regulating adopters' internal CI pipelines, release processes and generated model classes — untestable against this package and out of its scope (*finding CDS1200-9*). They are retained here as guidance with their normative homes cross-referenced:

- **Run the suite on change.** A maintained implementation should run the reference fixture suite whenever schemas, mappings, serializers or validators change, and should not release a schema change that causes an unexplained change to a golden fixture outcome. (Release evidence obligations for the *package itself* remain normative in §18; conformance testing obligations: CDS-1000.)
- **Reference CI shape.** checkout package → verify manifest → validate schema metaschemas → run structural suite (positive + negative) → run semantic tests → compare evidence summary → publish report and artefact hashes. Unlike vo.1, every step listed here is one the reference runner actually performs, except metaschema validation and hash verification, which `make_manifest.py` and standard tooling cover.
- **Multi-tenant testing.** Production assurance suites should verify that cross-tenant references cannot resolve without an explicit governed sharing mechanism; the normative tenant-isolation test is CDS-1000 T-TEN-001, and SEM-004 demonstrates only the single-boundary fixture case.
- **Code generation.** Generated model classes should preserve unknown governed extensions and CDS scalar semantics when round-tripping documents; the governing serialization and extension rules are CDS-1100's, and round-trip assertions belong to the future suite in §20.
- **Organisation-specific fixtures.** Organisations should add fixtures for known incidents, migrations and channel failure modes; these live alongside, not inside, the reference package.

20. Future Work: Round-Trip, Security and Scale Suites (*informative*)

These suites are deliberately out of scope for vo.2 and are stated as future work rather than implied capability.

Round-trip serialization. Structural validity does not prove an implementation preserves values when reading and writing. A complete harness should parse, serialize, reparse and compare representative documents with type-aware semantics: GTIN leading zeros and exact textual form; decimal/money scale and value; measurement value/unit separation; ordered arrays vs unordered sets; null vs absent; UTF-8 stability under the declared normalization policy; language-tag association. An implementation claiming safe interchange should pass round-trip tests for every supported CDS type (interchange rules: CDS-1100).

Security and multi-tenant fixtures. The public package is synthetic and secret-free (CDS1200-R021). Production assurance additionally needs: cross-tenant reference rejection; quarantine of disallowed script markup; unknown extension namespaces rejected or isolated per policy; fixtures containing credentials blocked from evidence publication; oversized payloads rejected under profile limits; AI proposals unable to mutate canonical state without the required approval transition (CDS-700).

Performance and scale. The vo.2 package prioritises correctness. Future releases should add *generated* scale fixtures while preserving a small hand-reviewable core: bounded validation time and memory across small (100 products) to large (1M products, 8M variants) catalogues, wide products, deep taxonomies and large publication batches. Illustrative sizes are not normative performance targets; service-level expectations belong to implementation profiles (CDS-900).

21. Adoption by a PIM Project (*informative*)

A PIM project can adopt CDS-1200 incrementally. The package does not force internal database schemas to mirror the JSON documents; it defines stable interchange and assurance boundaries.

1. Map existing product and variant records to the CDS product and variant contracts.
2. Export a small canonical fixture set and validate it.
3. Model existing controlled lists as CDS dictionaries and values.
4. Generate expected channel state for one channel product.
5. Pull the product back into an observation record.
6. Produce a field-level verification result.
7. Add organisation-specific negative fixtures for historical failure modes.

8. Run the suite in CI before connector or schema releases.

(Steps renumbered from the vo.1 draft's 8–15 — finding CDS1200-11.)

22. Architecture Decision Records (*informative*)

Decisions are held in the single global ADR register (CDS000-R006). Governing this chapter: **ADR-D5** (single corpus release; release-independent \$id URNs; per-schema semver; envelope cds_version const; changelog, pinned dependencies and informative environment-annotated evidence required of the vo.2 package). The chapter-local records CDS-ADR-1200-001..007 from vo.1 are subsumed into the global register unchanged in substance: ship executable artefacts with the prose standard; use JSON Schema Draft 2020-12; separate structural from semantic tests; require positive and negative fixtures; use deterministic synthetic fixtures; treat the Python runner as informative; record package hashes.

Annex A — Organisation Example: Legacy Hosted-Table PIM (*informative*)

An organisation migrating from a hosted-table PIM (for example *Airtable*) can express its legacy estate directly in package terms: legacy canonical tables and merchandising-fact fields become canonical fixtures; per-channel export tables become observation fixtures; existing match/compare reports become verification fixtures. This turns a migration inventory into an executable regression suite before cutover. The governed migration pathway — including the pre-cutover downstream baseline — is defined in CDS-1300. (Moved from the vo.1 adoption section per finding CDS1200-12.)

Appendix A — Schema and Fixture IDs (*informative*)

Schema \$id	Representative fixture
urn:cds:schema:core:product	relaxed-linen-shirt.product.json
urn:cds:schema:core:variant	relaxed-linen-shirt-navy-s.variant.json, relaxed-linen-shirt-navy-m.variant.json
urn:cds:schema:core:attribute-definition	MF_colour.json, MF_material.json
urn:cds:schema:reference:dictionary	colour.dictionary.json
urn:cds:schema:reference:dictionary-value	colour_navy.value.json
urn:cds:schema:reference:category	womens-shirts.category.json
urn:cds:schema:reference:taxonomy-mapping	womens-shirts.shopify-mapping.json
urn:cds:schema:reference:facet-definition	colour-family.json
urn:cds:schema:channel:channel-profile	shopify.channel-profile.json
urn:cds:schema:channel:publication-record	shopify-shirt.publication.json
urn:cds:schema:channel:observation-record	shopify-shirt.observation.json
urn:cds:schema:channel:verification-result	shopify-shirt.verification.json
urn:cds:schema:automation:ai-proposal	colour-mapping.ai-proposal.json
urn:cds:schema:assurance:conformance-manifest	reference.conformance-manifest.json

common/* (envelope, entity-reference, localised-text, measurement, money), channel/field-mapping and assurance/validation-output are composed by reference and exercised through every fixture rather than by a dedicated fixture file.

Appendix B — Reference Commands (*informative*)

```
# Unpack the companion package
unzip CDS-1200_Reference_Package_v0.2.zip
cd CDS-1200_Reference_Package_v0.2

# Install pinned dependencies (required; the runner exits FATAL without them)
python3 -m venv .venv
.venv/bin/pip install -r requirements.txt

# Run the structural + semantic suite and write evidence
.venv/bin/python tools/validate.py --write-report evidence/test-report.json

# Regenerate the manifest (run BEFORE the validator so evidence embeds the current hash)
.venv/bin/python tools/make_manifest.py
```

End of CDS-1200 vo.2 Review Draft.

Commerce Data Standard (CDS)

CDS-1300 — Migration, Adoption and Operational Rollout

Field	Value
Status	vo.2 Review Draft (working source; not an approved standard)
Release	CDS vo.2
Date	2026-08-04
Supersedes	CDS-1300 Working Draft vo.1
Normative status	§1–§2, §4–§23 are normative except where marked otherwise. §3, §24, Appendices A–C and all worked scenarios are informative.
Findings addressed	SYS-3 (D18), 1300-1..14; ADR-D1 (maturity mapping), ADR-D2, ADR-D4, ADR-D24; Matrix 5 (legacy annex, DF_→CH_ consolidation)

A governed pathway for moving existing commerce data, teams and channels into a CDS-conformant operating model without losing authority, traceability or business continuity.

1. Purpose and Scope (*normative*)

CDS-1300 defines how an organisation adopts CDS while continuing to operate existing products, channels and business processes. It covers migration from spreadsheets, hosted-table PIMs, legacy enterprise PIMs, channel-first catalogues and mixed estates into a PIM-first CDS architecture.

The chapter governs both technical migration and operating-model transition. A structurally correct database is not a successful migration if staff continue editing authoritative fields in downstream channels, dictionaries remain uncontrolled, verification is absent, or operational ownership is unclear.

CDS1300-R001 A migration **MUST** include an operational transition plan, not only a data-load procedure.

CDS1300-R002 Business continuity, rollback and verification **MUST** be designed and approved before production cutover.

Completeness of the migration itself is governed by the inventory-disposition rule (CDS1300-R009): nothing leaves the migration without a recorded disposition. *(This replaces the vo.1 "preserve or explicitly retire every authoritative fact" rule, which was untestable as stated.)*

Informative note: CDS does not require a big-bang replacement. Incremental adoption is preferred where it reduces risk and maintains traceability.

2. Operating Scale and the Small-Team Profile (*normative*)

CDS-1300 names many functions. They are **hats, not headcount**. A sole trader and a hundred-person programme can both conform; what scales is evidence and separation of duties, not the org chart.

CDS1300-R003 Every function named in this chapter **MUST** have a named owner. One person **MAY** hold any number of functions simultaneously. A conformance assessment **MUST NOT** require distinct individuals per function.

CDS1300-R004 Where staffing permits, the person who builds a migration artifact (mapping, transformation, load package) **SHOULD NOT** be the sole acceptor of that artifact. Where one person must do both, the acceptance record **MUST** note the dual role and the compensating control applied (for example: checklist-driven self-review against frozen criteria, evidence archived for later independent review). Segregation-of-duties expectations and compensating controls are defined in CDS-800 (decision-rights and segregation table); this chapter applies them, it does not redefine them.

CDS1300-R005 An organisation **MAY** adopt the **small-team profile** by declaring it in its conformance claim (claim rules: CDS-1000). Under that profile the requirements listed in Table 2-1 apply at the downgraded level shown; **all other requirements apply unchanged**.

Table 2-1 — Small-team profile downgrades

Requirement	Base level	Small-team profile level
CDS1300-R053 (separate read, write and approval migration credentials)	MUST	SHOULD; a single operator MAY hold combined credentials provided every bulk write is attributable, audit-logged, and the log is reviewed at wave exit
CDS1300-R057 (operational readiness demonstrated without project-team intervention)	MUST	The intervention-free clause is waived where the operators and the project team are the same people; the readiness demonstration MUST still be performed and recorded

Requirements that reference "named owners" or "operational owners" are satisfied under either profile by the hats rule (R003); acceptance and evidence records are never waived, only the requirement for the people involved to be different individuals.

RESOLVED (was D18 (resolved), owner decision 2026-08-04): accepted — self-declared eligibility, no headcount threshold.

3. Adoption Principles (*informative*)

- **Authority before movement** — determine who owns each field before copying values.

- **Profile before transformation** — understand the source data before designing target mappings.
- **Preserve provenance** — retain where a value came from and how it was transformed.
- **Canonical before channel** — resolve canonical values before publishing channel projections.
- **Dictionaries before bulk normalisation** — do not mass-convert values until controlled vocabularies and ambiguity rules exist.
- **Verification before trust** — do not declare migration complete until downstream state is observed and compared.
- **Small reversible releases** — prefer bounded migrations that can be measured and rolled back.
- **Human comprehension** — field names, status displays and review queues must remain understandable to operators.
- **No silent loss** — unsupported fields are quarantined, mapped or explicitly retired, never silently dropped.
- **Evidence over confidence** — completion claims require reproducible evidence.

4. Migration Pathways and Starting States (*normative*)

Organisations begin from different levels of structure. CDS defines migration pathways rather than assuming one starting point.

Starting state	Primary migration emphasis
Channel-first store	A commerce platform currently holds the most complete product data. Extract and profile the channel state first, then establish PIM authority.
Spreadsheet or hosted-table PIM	A structured but implementation-specific master exists. Preserve formulas, mappings and validation behaviour while replacing brittle field coupling.
Legacy enterprise PIM	A mature PIM lacks CDS dictionaries, channel verification or human-readable namespaces. Adoption may be layered rather than replacing the platform.
ERP-led catalogue	The ERP owns identity and commercial data; descriptive enrichment is scattered. CDS introduces a canonical enrichment layer without displacing legitimate ERP authority.
Multi-channel fragmented estate	Channels contain competing product versions. Migration requires explicit reconciliation and an authority decision per field.
Greenfield	No production catalogue exists. Sequence taxonomy, schemas, dictionaries, channel profiles and test fixtures before product entry.

CDS1300-R006 The migration plan **MUST** identify the declared starting state and **MUST NOT** assume that a system is authoritative merely because it currently contains the most data.

5. Readiness Assessment (*normative*)

Before migration begins, the organisation assesses readiness. **Minimum evidence for every area below: documented findings, a named owner and a risk rating.**

Area	Assessment scope
Data	Record counts, duplicate rate, missing identifiers, uncontrolled values, rich-text quality, media completeness, historical anomalies.
Architecture	Current product/variant model, taxonomies, field definitions, formulas, scripts, APIs, manual workarounds, hidden dependencies.
Channels	Published fields, channel overrides, collection logic, feed requirements, read-back capability, rate limits.
People	Data owners, editors, approvers, knowledge concentration, training needs, resistance risks.
Governance	Existing naming standards, change control, incident handling, deprecation rules, audit evidence.
Operations	Release windows, peak trading periods, support coverage, rollback capacity, acceptable freeze periods.
Security	Credentials, least privilege, tenant separation, personal data, supplier confidentiality.

CDS1300-R007 Production migration **MUST NOT** begin until every critical readiness risk has an owner, a treatment and an acceptance decision.

6. Source-System and Data Inventory (*normative*)

The source inventory records every system, file, integration and manual process that creates, alters, enriches, publishes or verifies product information.

Source Inventory Record	
- source_id	- identifiers
- system_name	- record_count
- owner	- known_quality_issues
- data_domains	- retention_requirement
- extraction_method	- migration_disposition
- update_frequency	- evidence_location

CDS1300-R008 Every source that can change product information **MUST** appear in the source inventory, including manual spreadsheets and channel-admin edit practices.

CDS1300-R009 Every inventoried source, field, mapping and rule **MUST** carry a `migration_disposition` — one of `migrate`, `transform`, `merge`, `quarantine`, `retire`, `out_of_scope` — with an evidence location. Nothing exits the migration without a recorded disposition.

CDS1300-R010 The inventory **MUST** distinguish authoritative source data, reference data, derived data, channel projections and observed downstream data.

CDS1300-R011 Unknown fields **MUST** be classified (and receive a disposition) before they are transformed or discarded.

7. Authority and Ownership Baseline (*normative*)

Migration is the opportunity to remove competing masters. An authority matrix records the current owner, target owner, transition date and permitted read-back behaviour for each domain or field. (Authority is declared per fact — CDS-200 §13.)

Domain	Common current owner	Target authority	Migration rule
Product identity	ERP or legacy PIM	PIM or ERP by declared contract	Identifiers remain stable.
Descriptive attributes	Files, supplier feeds, channel admin	PIM	Channel edits become drift unless approved overrides exist (CDS-500 §22).
Taxonomy	Channel collections, tags, legacy categories	PIM	External taxonomies remain mappings.
Price	ERP, PIM or channel	Declared commercial authority	Do not move authority without finance approval.
Inventory	Warehouse, ERP or channel	Inventory authority	PIM may consume but not master live stock.
Channel observed state	Channel APIs	Observation store	Never merged into canonical values (CDS-500 §15).
Verification result	Legacy formulas or none	Verification engine	Reason codes beneath traffic lights (CDS-500 §17–18).

CDS1300-R012 Every migrated field **MUST** have exactly one declared target authority.

CDS1300-R013 Read-back and verification fields **MUST** remain separate from authoritative values throughout migration.

CDS1300-R014 Each authority cutover **MUST** be evidenced by at least one of: **(a)** a permission or access change that prevents non-authoritative writes to the field; **(b)** drift detection armed on the field (observation plus comparison per CDS-500); or **(c)** a documented acceptance that enforcement is detection-only, naming the residual risk owner.

Informative note: some platform standard fields cannot be write-protected at all — any admin user can edit them regardless of PIM authority. For such fields option (a) is unavailable by construction; enforcement is necessarily detection-only (b or c), and the training rule CDS1300-R051 is the compensating human control.

8. Target-State Design (*normative*)

The target state is designed before bulk transformation: entity boundaries (CDS-200), semantic namespaces (CDS-300), Attribute Definitions, dictionaries (CDS-400), internal taxonomy, channel profiles (CDS-900), publication and observation records (CDS-500), and operating procedures.

Canonical PIM	External Authorities
+-- Products and Variants	+-- ERP / Pricing
+-- Attribute Definitions	+-- Inventory / Warehouse
+-- Dictionaries and Facets	+-- Orders / Customers
+-- Internal Taxonomy	
+-- Media and Content	
+-- Channel Profiles	
+-- Publication Records	
+-- Observations	
+-- Verification Results	
+-- Governance and Audit	

CDS1300-R015 The target model **MUST** be validated with representative products before full mapping begins.

CDS1300-R016 Target fields **MUST NOT** be created solely to preserve accidental source-system implementation details; legitimate source provenance and business meaning **MUST** be preserved even when field names and storage mechanisms change.

9. Taxonomy and Classification Migration (*normative*)

Taxonomy migration separates what a product **is** from where it is **merchandised**. Steps (informative): extract all category-like values (tags, product types, collection rules, supplier categories); identify synonyms and overlaps; design a non-overlapping internal tree; assign one primary category per product; map external taxonomies at category level where inheritance is valid; define collection and navigation projections separately; keep a taxonomy decision log for ambiguous types; validate counts, orphans and duplication.

CDS1300-R017 A channel's collections, tags or product types **MUST NOT** be assumed to be canonical categories.

CDS1300-R018 After migration each product **MUST** have one primary internal category; collections **MAY** remain many-to-many merchandising projections (rules: CDS-200 §6).

CDS1300-R019 External taxonomy mappings **MUST** retain the source taxonomy version and the mapping rationale.

10. Attribute and Dictionary Migration (*normative*)

Legacy fields mix source values, canonical values, display labels, facet values and channel output. Migration separates these into the value layers defined by ADR-D4 (definitions: CDS-100 §3; model: CDS-400).

Legacy field:	Colour = "French Navy"
Migrated representation (value layers per ADR-D4):	
Source value	= "French Navy" (preserved verbatim, with provenance)
Alias mapping record	= "French Navy" -> colour_french_navy
Canonical value	= colour_french_navy (stored on the variant)

Display label	= "French Navy"	
Facet value	= blue	
CH_google_colour	= "Blue"	(channel representation)

CDS1300-Ro20 Distinct source values MUST be profiled before dictionary normalisation begins.

CDS1300-Ro21 Ambiguous mappings MUST enter review; they MUST NOT be resolved by first-match or other arbitrary logic.

CDS1300-Ro22 Every source value mapped during migration MUST be retained as an alias (an Alias Mapping Record, CDS-400) with provenance. (*vo.1's "where useful" qualifier is deleted: aliases are what future imports and read-back comparison match against — discarding them breaks verification later.*)

CDS1300-Ro23 Facet values MUST be designed for customer usability (CDS-600), not inferred solely from current distinct-value counts.

11. Product and Variant Migration (*normative*)

Product and variant migration resolves parent–child relationships, SKU identity, shared attributes and sellable differences (variant boundary rule: CDS-200 §5).

Control	Migration requirement
Identity continuity	Preserve SKU, GTIN, MPN, source IDs and channel IDs where continuity is required.
Parent grouping	Confirm variants genuinely represent sellable forms of one logical product.
Option definition	Only values that create distinct sellable units become options.
Shared data lifting	Move identical values from variants to the parent product.
Variant truth	Do not create unavailable combinations by Cartesian expansion.
Duplicate detection	Detect same SKU, same GTIN, same MPN/brand and near-identical records.
Lifecycle state	Preserve active, draft, archived, discontinued and pre-order meaning.

CDS1300-Ro24 A migration MUST NOT create new SKU identities merely to fit a target platform unless a governed re-identification plan exists.

CDS1300-Ro25 Every variant consolidation or split MUST be evidenced by an approved old-to-new cross-reference table. The cross-reference table is the **primary** control; mechanical reversibility of a restructure is rarely available on production channels and MUST NOT be assumed or claimed without test evidence.

CDS1300-Ro26 The wave design MUST check migrated products against the declared structural limits of each target channel profile (variants per product, option count, value lengths — CDS-900) **at design time**, before mapping is finalised, not at load time.

12. Content, Media and Digital Assets (*normative*)

Checklist (informative): preserve original source content separately from enriched content where provenance matters; normalise rich text without silently removing meaningful structure; retain asset identifiers, source URLs, checksums, rights and usage restrictions; define primary image, sequence, variant association and alt-text rules; quarantine broken, duplicate, low-resolution or unsupported assets; verify downstream media ordering and variant-image association after publication.

CDS1300-Ro27 Asset migration MUST preserve available rights and provenance metadata.

CDS1300-Ro28 Asset transfer success MUST NOT be treated as proof that the channel displays the correct asset in the correct order; media ordering and variant association are in verification scope (CDS-500).

13. Channel Mapping and Publication Migration (*normative*)

Channel migration converts legacy output fields and direct-admin practices into governed channel profiles and publication records. The canonical model is not redesigned to mirror channels.

Canonical attribute	-> channel mapping	-> expected representation
	-> publication	-> observation -> verification

CDS1300-Ro29 Each migrated channel field MUST identify its canonical source, transformation, destination and comparison strategy.

CDS1300-Ro30 Channel overrides MUST be explicit, attributable and bounded (override records: CDS-500 §22).

CDS1300-Ro31 New channel fields MUST use the CH_ namespace (ADR-D24; registry: CDS-300). Legacy DF_ names MAY persist during transition only as registered deprecated aliases with provenance. Per ADR-D24 §3, a field is a feed-layer projection **iff it is a literal serialized feed column**; such projections live under CH_ and the feed artifact is verified end to end at the formatted-output layer.

CDS1300-Ro32 Generated tags, collections and feed outputs MUST be reproducible from governed source data or rules; they MUST NOT be imported as independent canonical truth.

(*vo.1's "retain DF_ only where useful" and "metafield-equivalent" qualifiers are resolved by ADR-D24 and ADR-D2 respectively; the MF_/STD_ boundary is the CDS-300 enumeration, never a judgement call.*)

14. SEO and URL Continuity (*normative — new in v0.2*)

After data loss, the largest business risk in a commerce migration is losing accumulated search equity. URL and search continuity is migration scope, not an afterthought.

CDS1300-R033 A migration that changes customer-facing URLs, handles, slugs or channel identifiers **MUST**, for every existing customer-facing URL, either preserve the URL or serve a permanent redirect to its successor.

CDS1300-R034 SEO-critical fields — page titles, meta descriptions, canonical URLs, and any structured-data fields declared in the channel profile — **MUST** be included in the pre-cutover baseline (§15) and in post-cutover verification scope.

CDS1300-R035 Redirect coverage **SHOULD** be verified after cutover by sampling pre-cutover URLs, and index-status and organic-traffic signals **SHOULD** be monitored through the wave's proving period.

Informative note: redirect behaviour is a channel-profile capability check. Some platforms create redirects automatically when a handle changes; others require explicit redirect records; some feed-based channels key on identifiers where "redirect" means identifier continuity. Record the mechanism per channel in the CDS-900 profile before cutover.

15. Verification Baseline and Downstream Read-Back (*normative*)

Before changing production data, capture what downstream channels currently contain. This separates pre-existing drift from migration-induced drift — the strongest available defence against blaming the migration for old defects, or missing new ones.

```
Baseline: extract current channel state -> store observed values
         -> compare against current expected state -> record pre-existing mismatches
         -> freeze baseline evidence -> begin migration
```

CDS1300-R036 A downstream baseline **MUST** be captured and frozen before production cutover wherever read-back is technically available, and the frozen baseline **MUST** be retained at least until the wave passes its exit gates (it is also the rollback republication source — R042).

CDS1300-R037 Pre-existing mismatches **MUST** be recorded and distinguished from mismatches introduced by migration.

CDS1300-R038 Fields that cannot be observed **MUST** be recorded with the `UNOBSERVABLE` status from the CDS-500 enum; they **MUST NOT** be reported as verified. Verification statuses, reason codes and traffic-light presentation are owned by CDS-500 §17–§18 and are not restated here.

16. Cutover Strategies (*normative*)

Strategy	Use and risk
Big bang	Everything switches together. Highest coordination risk; only for small, well-tested, easily reversible estates.
Wave by category	Bounded category groups; useful for dictionary and taxonomy validation.
Wave by brand or supplier	Supports source-specific mappings and operational ownership.
Wave by channel	Establish PIM authority, then migrate one downstream channel at a time.
Shadow publication	Generate expected output without writing production; compare against current channel state.
Dual run	Old and new pipelines run simultaneously with controlled reconciliation — see R040.
Pilot cohort	A small representative product set migrated end to end before scale-out.

CDS1300-R039 The chosen strategy **MUST** define scope, entry criteria, exit criteria, rollback point and evidence package.

CDS1300-R040 At any moment, exactly **one** pipeline **MUST** be the declared writer for a given channel field-set. On a channel with a single write target, a "dual run" therefore **MUST** operate as shadow publication on the non-authoritative side: the non-authoritative pipeline generates expected output and compares, and **MUST NOT** write. Reconciliation happens on evidence, never by both pipelines writing.

CDS1300-R041 Each wave plan **MUST** declare its proving period (the post-cutover observation window that must elapse defect-free before exit) before cutover.

Informative note: typical proving periods run one to four weeks and should cover at least one full business cycle for the affected products (including any weekly ordering, feed or campaign rhythm).

17. Rollback, Recovery and Business Continuity (*normative*)

Rollback is a designed capability, not an emergency improvisation.

CDS1300-R042 Every production migration wave **MUST** have a tested rollback or compensating recovery procedure. **Channel-side rollback is defined as republication of the frozen pre-cutover baseline (R036)** through the governed publication path, followed by observation and verification — not as restoration of channel-internal state, which most channels cannot provide.

CDS1300-R043 Rollback **MUST NOT** overwrite the evidence required to determine what changed and why (publication records, observations, verification results, decision log).

Rollback design checklist (informative): freeze pre-cutover source and channel state; version transformation rules, mappings, dictionaries and payloads; maintain old-to-new identifier cross-references; define stop conditions and escalation authority; test restoration of data, credentials and scheduled jobs; reconcile publication attempts made before failure; preserve evidence from failed runs.

Informative — irreversibility classes. Republication restores content state but cannot reverse everything. Classes to enumerate per wave: (1) customer interactions bound to channel identifiers (orders, reviews, wishlists) — never reversible; (2) search-engine indexing and ranking signals accumulated against changed URLs (mitigated by §14, not reversed); (3) feeds, emails and ads already consumed downstream; (4) channel-side identifiers destroyed by deletion (recreated objects get new identities); (5) analytics history continuity; (6) third-party caches and syndication copies. A rollback plan states, per class, whether the exposure exists and what compensating action applies.

Informative — commerce activity during rollback: orders, payments and fulfilment are mastered outside CDS scope (CDS-000 §2) and proceed on their own systems. The migration team's obligation is product-data remediation: identify products sold while incorrect data was live, correct the data forward, and hand any customer-impact assessment to the responsible business owner.

18. Operational Rollout and Run-State Transition (*normative*)

Migration ends only when the new operating model is stable and owned.

Run-state area	Required operational definition
Publishing	Schedules, queue monitoring, retries, rate limits, incident ownership.
Verification	Read-back cadence, mismatch triage, health thresholds, escalation.
Dictionaries	New-value review, alias creation, facet governance, deprecation.
Taxonomy	Category-change procedure, mapping review, decision log.
Imports	Supplier onboarding, schema-change detection, quarantine, delta-sync monitoring.
AI	Proposal review, thresholds, benchmarks, model-change handling (CDS-700).
Support	Operator runbooks, dashboards, alert routing, service expectations (CDS-1400).
Release management	Versioning, change windows, approvals, rollback ownership.

CDS1300-Ro44 Project completion **MUST** include recorded acceptance by the named run-state owners (who **MAY** be the same people as the project team — §2).

CDS1300-Ro45 Temporary migration scripts and credentials **MUST** be retired, governed, or incorporated into supported operations.

19. Data Quality Remediation (*normative*)

Migration surfaces defects that may have existed for years. CDS separates blocking defects, remediable defects and accepted historical imperfections.

Class	Treatment
Blocking	Missing product identity, duplicate SKU, invalid variant relationship, unsafe price, critical compliance data.
High priority	Unmapped category, dictionary ambiguity, missing required channel field, broken media.
Improvement	Weak descriptions, missing optional attributes, poor search synonyms, non-critical legacy formatting.
Accepted debt	Known issue with documented reason, owner, review date and impact assessment.

CDS1300-Ro46 Quality thresholds **MUST** be defined before acceptance testing begins.

CDS1300-Ro47 Quality and completeness denominators **MUST** include failed and quarantined records. A record **MAY** leave a denominator only through a disclosed `migration_disposition` (R009). Reporting completeness over a silently reduced population is non-conformant. (*This is the corpus's strongest anti-gaming control; see also CDS-1400 §6.*)

20. AI-Assisted Migration (*normative*)

AI may accelerate classification, attribute extraction, dictionary suggestions, duplicate review, content cleanup and anomaly detection. It remains a **proposal mechanism** governed entirely by CDS-700 (proposals separate from canonical state, evidence classes, confidence, abstention, autonomy levels, review). This section adds only the migration-specific rules.

CDS1300-Ro48 AI-generated migration values **MUST** carry the provenance CDS-700 requires (model/workflow version, evidence, confidence where available) and **MUST** pass the same schema and dictionary validation as any other input.

CDS1300-Ro49 AI **MUST NOT** silently resolve identifier conflicts, legal claims, safety attributes or other high-risk discrepancies; these **MUST** enter human review.

CDS1300-Ro50 Human review samples **SHOULD** include both accepted and rejected proposals, to detect systematic error in either direction.

21. People, Functions, Training and Change Management (*normative*)

A CDS rollout changes responsibilities and habits. Teams must understand not only new screens but why canonical authority, dictionaries and verification exist.

The migration **functions** (hats — one person **MAY** hold several, per R003):

Function	Primary accountability
Executive sponsor	Scope, risk and organisational priority.
Programme lead	Migration plan, dependencies, acceptance.
Data architect	Target model, mappings, technical integrity.
Data steward	Dictionaries, taxonomy quality, exceptions.
Channel owner	Platform mappings and operational behaviour.
Migration engineer	Extraction, transformation, load and evidence automation.
Verifier / QA	Expected-versus-observed validation and test evidence.
Business editor	Usability, product meaning, workflow fit.
Operations owner	Runbooks, monitoring, incidents, support obligations.

CDS1300-R051 Training MUST cover authority, namespaces, dictionaries, publication states, verification and exception handling — and specifically **which channel-admin edits will be overwritten or treated as drift**.

22. Security, Privacy and Tenant Isolation (*normative*)

Migration often requires broad temporary access and bulk extraction; controls stay proportionate and time-bounded.

CDS1300-R052 Migration tooling MUST preserve tenant and organisation boundaries (no cross-organisation mixing of dictionary, publication or observation records).

CDS1300-R053 Migration credentials MUST follow least privilege, and read, write and approval credentials MUST be separated. (*Small-team profile: downgraded per Table 2-1 — combined credentials permitted with full attributable audit logging reviewed at wave exit.*)

CDS1300-R054 Temporary extracts MUST have a retention and destruction decision; temporary credentials MUST be rotated or revoked after each phase.

Checklist (informative): encrypt exports; restrict supplier-pricing and confidential data; exclude unnecessary customer or order data from product migration datasets; record who executed bulk writes and which package revision was used; validate archive retention against policy.

23. Metrics, Acceptance Criteria and Exit Gates (*normative*)

Acceptance criteria are defined at product, wave, channel and programme levels.

Metric	Acceptance evidence
Record reconciliation	Source, transformed, loaded, quarantined and retired counts reconcile (dispositions per R009).
Identifier integrity	No unapproved duplicate or changed SKU/GTIN/MPN identities.
Schema validity	Required records pass structural and semantic validation.
Dictionary coverage	Distinct values mapped, quarantined or explicitly excluded — with aliases retained (R022).
Taxonomy coverage	Valid primary categories and expected external mappings.
Publication success	Expected payloads accepted without unhandled failure.
Observation coverage	Expected fields read back or explicitly UNOBSERVABLE (CDS-500).
Verification health	Mismatch and missing rates below approved thresholds.
SEO continuity	URL/redirect coverage verified per §14.
Operational stability	Queues, schedules, retries and alerts operate through the declared proving period (R041).
User acceptance	Readiness demonstration per R057.

CDS1300-R055 Acceptance thresholds MUST be approved before the migration wave executes.

CDS1300-R056 A wave MUST NOT exit merely because all jobs completed; quality, verification, SEO-continuity and operational gates also apply.

CDS1300-R057 Operational readiness MUST be demonstrated by named operators completing representative workflows without project-team intervention, and the demonstration recorded. (*Small-team profile: intervention-free clause waived per Table 2-1.*)

24. Conformance (*normative*)

Conformance levels, test suites and claim rules are owned by CDS-1000. A CDS-1300 conformance claim is a claim against the requirement set CDS1300-R001–R057 (with Table 2-1 substitutions where the small-team profile is declared), evidenced per CDS-1000. Recommended practices — pilot cohorts, automated evidence generation, versioned mapping packages, a migration issue/decision register, use of the CDS-1200 fixtures for deterministic validation, and a post-cutover retrospective with debt handover — are SHOULD-level and MAY be cited as supporting evidence in a claim.

25. Worked Migration Scenarios (*informative*)

25.1 Legacy hosted-table PIM (Airtable) to CDS

Legacy layers	CDS role
Import_*	-> source ingestion records
Airtable_*	-> canonical core values
MF_Airtable_*	-> canonical structured attributes
DF_Airtable_SE0_*	-> channel/feed projections (Data Feed layer – ADR-D24)
Shopify_* / MF_Shopify_*	-> downstream observations
Match_* / MF_Match_*	-> verification results

Migration outcome: preserve formula intent as explicit transformations; replace implicit behaviour with Attribute Definitions and rules; retain read-back and traffic-light concepts (CDS-500); map general downstream fields into CH_*; register DF_* as deprecated aliases (feed-column projections continue under CH_ per R031).

CDS1300-R058 Legacy formulas **MUST** be catalogued and tested before they are replaced by code or rules — hidden business rules must not be lost even where implementation details change.

Informative note: Airtable formula *logic* is not extractable through the data API — the API returns computed results, not formula text. Catalogue formulas from the schema/UI, exported documentation, or maintained records, and budget for this as a manual step.

25.2 Channel-First Clothing Store

Current state: platform products are master; collection_* tags drive navigation; colours contain 80+ distinct values; some attributes live in description text.

Migration: 1. extract products, variants, tags, collections and structured fields 2. profile current colour values 3. build canonical colour and facet dictionaries 4. establish internal taxonomy 5. create PIM records and expected channel state 6. shadow compare before write (R040) 7. pilot one category 8. publish, read back and verify 9. expand by category wave

25.3 Multi-Supplier Homewares Catalogue

Current state	Target
Supplier A: "Tasmanian Oak"	Source values preserved; aliases retained (R022)
Supplier B: "Oak Timber"	Canonical material = oak or wood, per evidence
Supplier C: "Natural Wood"	Facet material = wood
	Display label remains source-appropriate
	Channel values follow each channel profile

Ambiguous "Natural Wood" **enters review** rather than being auto-mapped to oak (R021): the term may mean oak, pine, or an unspecified species; first-match mapping would fabricate a material claim.

26. Architecture Decisions (*informative*)

Per CDS000-R006, ADRs live in the single global register. Decisions originating in this chapter, now recorded there: staged reversible adoption preferred; operational handover is part of migration; pre-cutover downstream baseline required; legacy formula intent preserved before replacement; quality denominators include quarantined records; AI assists but does not silently arbitrate conflicts. The v0.1 record CDS-ADR-1300-005 (DF_→CH_) is **superseded by ADR-D24**.

Appendix A — Migration Phase Checklist (*informative*)

Each phase exits on evidence accepted by a named accountable owner (one person MAY own several or all phases — §2; builder/acceptor separation per R004).

Phase	Exit evidence
A. Mobilise	Scope, sponsor, programme lead, risk register, release constraints, communication plan approved.
B. Discover	Sources, fields, formulas, integrations, channels, users and hidden processes inventoried.
C. Profile	Counts, duplicates, nulls, distinct values, taxonomies, variants and media quality measured.
D. Design	Target model, ownership, namespaces, dictionaries, taxonomy, mappings and rules approved.
E. Build	Extraction, transformation, load, publication, observation and verification workflows implemented.
F. Test	Schema, semantic, negative, round-trip, performance, security and rollback tests passed.
G. Pilot	Representative cohort migrated and operated through a proving period.
H. Cut over	Approved wave executed with monitoring, reconciliation and rollback readiness.
I. Stabilise	Mismatches triaged, operational metrics stable, documentation and training completed.
J. Close	Operational acceptance, debt register, retrospective, evidence archive, temporary-access retirement completed.

Appendix B — Legacy Airtable PIM Mapping (*informative — organisation example*)

This table records one organisation's legacy naming as a worked example. It is not normative reference data; the normative prefix registry is CDS-300.

Legacy pattern	CDS role	Migration treatment
Import_*	Source ingestion	Retain as source records; never canonical.
Airtable_*	Resolved core data	Map to canonical product/variant fields.

Legacy pattern	CDS role	Migration treatment
MF_Airtable_*	Resolved structured data	Map to governed attributes and values; MF_ per the ADR-D2 enumeration boundary.
DF_Airtable_SEO_*	Data Feed output (ADR-D24)	Map to CH_google_* or feed-column projections under CH_.
Shopify_*	Observed standard fields	Map to observation records (CDS-500 §15).
MF_Shopify_*	Observed metafields	Map to observation fields.
Match_* / MF_Match_*	Comparison formulas	Map to verification results with reason codes and comparison strategies (CDS-500).
collection_* tags	Computed collection signals	Regenerate from taxonomy/collection rules; never import as independent canonical truth.
Traffic lights	Operator QA display	Retain as presentation over detailed statuses (CDS-500 §18).

Appendix C — Adoption Maturity Mapping (*informative*)

The single conformance ladder is owned by CDS-1000 (ADR-D1): Foundation → Structured → Publisher → Verified → Governed. The vo.1 maturity levels of this chapter map as follows and are retained only as adoption-narrative labels:

vo.1 maturity level	CDS-1000 ladder
Level 0 — Fragmented	pre-Foundation
Level 1 — Canonical	Foundation
Level 2 — Structured	Structured
Level 3 — Publisher	Publisher
Level 4 — Verified	Verified
Level 5 — Governed	Governed

END OF CDS-1300 vo.2 REVIEW DRAFT

Commerce Data Standard (CDS)

CDS-1400 — Monitoring, Incident Management and Continuous Improvement

Field	Value
Status	v0.2 Review Draft (working source; not an approved standard)
Release	CDS v0.2
Date	2026-08-04
Supersedes	CDS-1400 Working Draft v0.1
Normative status	§1–§2, §4–§27 are normative except where marked otherwise. §3, §28, §29 and Appendices A–D are informative. Tables explicitly marked (<i>informative</i>) inside normative sections carry no requirements.
Findings addressed	SYS-3 (D18), SYS-4, 1400-1.14; ADR-D3 (Appendix E deleted — statuses and traffic lights single-homed in CDS-500 §17–§18); Matrix 5 (drift taxonomy → CDS-500 §20; AI rules → CDS-700; conformance claims → CDS-1000; test-outcome spellings → CDS-1000 §21)

An operational assurance framework for detecting product-information defects, controlling incidents, measuring downstream health and converting evidence into sustained improvement.

1. Purpose and Scope (*normative*)

CDS-1400 defines how a CDS-conformant commerce information environment is monitored after adoption and how operational defects are detected, assessed, contained, repaired and prevented from recurring. It covers the full information path from source ingestion and canonical enrichment through channel publication, downstream observation, verification and customer-facing use.

The chapter treats product information as an operational service. Availability alone is insufficient: a PIM can be online while products are missing attributes, filters expose unusable values, channel mappings drift, or downstream listings differ from the expected state.

CDS1400-R001 A CDS operating model **MUST** monitor correctness, completeness, timeliness and downstream fidelity in addition to technical availability.

CDS1400-R002 An organisation **MUST** define how product-data incidents are declared, owned, escalated, resolved and reviewed.

CDS1400-R003 Operational monitoring **MUST** use declared denominators and **MUST NOT** hide quarantined, failed or unobserved records (denominator rules: §6).

Informative note: CDS-1400 does not prescribe a particular monitoring product. The controls may be implemented with dashboards, logs, metrics, queues, scheduled reports or a combination of systems.

2. Operating Scale and the Small-Team Profile (*normative*)

CDS-1400 names many roles, views and reviews. They are **hats, not headcount** — the same rule that governs migration functions in CDS-1300 §2. A sole trader and a hundred-person operations organisation can both conform; what scales is evidence and separation of duties, not the org chart.

CDS1400-R004 Every role, review and monitoring domain named in this chapter **MUST** have a named owner. One person **MAY** hold any number of roles simultaneously, including all incident-command roles. A conformance assessment **MUST NOT** require distinct individuals per role.

CDS1400-R005 Where staffing permits, the person coordinating an incident **SHOULD NOT** be the person performing the most complex repair, and the person who built a monitoring control or automation **SHOULD NOT** be its sole acceptor. Where one person must hold both hats, the incident or acceptance record **MUST** note the dual role and the compensating control applied (for example: checklist-driven self-review against the runbook, evidence archived for later independent review). Segregation-of-duties expectations and compensating controls are defined in CDS-800 §18; this chapter applies them, it does not redefine them. The small-team accommodation mirrors CDS-1300 §2.

CDS1400-R006 An organisation **MAY** adopt the **small-team profile** by declaring it in its conformance claim (claim rules: CDS-1000). Under that profile the requirements listed in Table 2-1 apply at the downgraded level shown; **all other requirements apply unchanged**. Acceptance and evidence records are never waived — only the requirement for the people involved to be different individuals.

Table 2-1 — Small-team profile downgrades

Requirement	Base level	Small-team profile level
CDS1400-R053 (SEV-1 post-incident review includes a participant who was not the primary responder)	MUST	Waived where no second qualified person exists; the review record MUST note the sole-operator condition, and the evidence package MUST be archived so a later independent review remains possible

After the base-level corrections in this revision — the hats rule (R004), the informative cadence table (§26), **SHOULD**-level service objectives (R016), the next-business-day coverage allowance (R033) and combined dashboard views (§10) — R053 is the only remaining requirement that presupposes a second person. Migration-phase downgrades continue to be governed by CDS-1300 Table 2-1.

CDS1400-R007 An organisation MAY declare the **enterprise operations profile** (an overlay profile under CDS-1000 §16). Under that profile: CDS1400-R016 (per-context service objectives) upgrades from SHOULD to MUST, and the minimum-roles column of Appendix B becomes applicable staffing guidance. Outside that profile the minimum-roles column is informative illustration only.

RESOLVED (was D18 (resolved), owner decision 2026-08-04): accepted — self-declared eligibility, no headcount threshold, mirroring CDS-1300.

RESOLVED — accepted as drafted (owner 2026-08-04; was new (resolved)): The enterprise operations profile is modelled as a CDS-1000 overlay profile so that its upgraded MUSTs are testable within the existing claim machinery; the alternative — leaving SLO enforcement to per-organisation declared thresholds only — was rejected because it gives large multi-team operations no claimable assurance tier.

3. Operational Assurance Principles (*informative*)

- **Observe the whole information path** — source, canonical, publication, downstream and customer-experience states.
- **Measure facts, not confidence** — health claims derive from reproducible metrics and evidence.
- **Separate expected from observed** — retain what should exist downstream and what was actually read back.
- **Alert on actionability** — an alert has an owner, a condition and a defined response.
- **Preserve detail beneath summaries** — colour views remain backed by precise statuses, reason codes and affected records.
- **Prefer early containment** — stop incorrect propagation before optimising full repair.
- **Make recovery reversible** — repair, replay and rollback preserve lineage and can be audited.
- **Learn without blame** — post-incident review improves systems and controls, not scapegoats.
- **Track recurrence** — repeated symptoms are a problem-management issue, not isolated tickets.
- **Continuously reduce uncertainty** — monitoring gaps and unobservable fields are explicit, owned risks.

4. Observability Model (*normative*)

CDS observability is the ability to determine the health and history of product information from recorded signals. The minimum model contains the linked states:

Source State

- > Canonical State
- > Expected Channel State
- > Publication Attempt
- > Observed Channel State
- > Verification Result
- > Customer Experience Signal

CDS1400-R008 Monitoring MUST distinguish data state from processing state. A completed job does not prove that the resulting data is correct.

CDS1400-R009 Each material signal SHOULD be traceable to product, variant, field, channel, canonical revision and the relevant mapping or dictionary version.

CDS1400-R010 Where a downstream field cannot be observed, its verification state MUST be recorded as UNOBSERVABLE from the CDS-500 §17 enum. It MUST NOT be recorded or reported as MATCH. (Test executions that were never run use the CDS-1000 §21 outcome NOT_TESTED ; verification statuses and test outcomes are distinct vocabularies.)

5. Monitoring Domains (*normative*)

Representative signals (*informative*):

Domain	Representative signals
Source ingestion	Fetch success, freshness, record counts, duplicate identifiers, schema changes, parsing errors, supplier anomalies.
Canonical product layer	Completeness, validity, conflicts, unresolved overrides, stale values, product/variant integrity.
Taxonomies and dictionaries	Unmapped values, ambiguous aliases, deprecated entries, version changes, unexpected cardinality growth.
Workflow and enrichment	Queue age, blocked approvals, review backlog, failed rules, missing ownership, time-to-ready.
Publication	Preflight failures, attempted writes, acknowledgements, rejections, rate limiting, retries, publish lag.
Observation	Read-back coverage, retrieval failures, stale observations, unsupported fields, API visibility gaps.
Verification	Match rate, missing rate, mismatch rate, comparison errors, unresolved age, drift recurrence.
Customer experience (<i>conditionally observable — §24</i>)	Facet cardinality, zero-result combinations, navigation dead ends, search failures, incorrect counts.
AI and automation	Proposal volume, confidence calibration, rejection and override rates, out-of-dictionary output, model or prompt drift.
Security and tenancy	Credential failures, unauthorised changes, cross-tenant leakage, unusual export volume, evidence tampering.

CDS1400-R011 Every CDS operating profile MUST declare which monitoring domains are in scope and who owns each domain (one person MAY own all — §2).

6. Metrics, Measures and Denominators (*normative*)

A metric is useful only when its population, time window and exclusions are explicit. (*Measure kinds — count, rate, latency, age, distribution, trend, cardinality, coverage — are informative vocabulary; Appendix A gives the reference catalogue.*)

Worked example — why observation coverage and match rate are separate measures:

```
eligible_expected_fields    = 10,000
observable_expected_fields = 9,200  (800 are UNOBSERVABLE on this channel)
observed_fields            = 9,050
matched_fields             = 8,960
mismatched_fields         = 60
missing_fields             = 30

observation_coverage        = 9,050 / 9,200  (how much we can see)
verification_match_rate     = 8,960 / 9,050  (how much of what we see is right)
```

Reporting 8,960 / 10,000 as a "match rate" would silently blend blindness into fidelity; reporting 100% for the 800 unobservable fields would fabricate assurance. Both numbers must travel together.

CDS1400-R012 A rate MUST name its numerator, denominator, eligibility rules, exclusions and measurement window.

CDS1400-R013 Quarantined or failed records MUST remain visible in relevant operational denominators unless the metric explicitly measures a different population. A record MAY leave a denominator only through a disclosed disposition. This applies the corpus's strongest anti-gaming control (CDS1300-R047; quarantine model: CDS-400 §17).

CDS1400-R014 An organisation MUST NOT report a 100 percent match rate — or any match rate — over fields that were not observed. Observation coverage MUST be reported separately from verification match rate and MUST NOT be folded into it.

7. Service Indicators, Objectives and Verification-Health Thresholds (*normative*)

Reference indicators (informative): publication success (accepted / attempted eligible records); verified fidelity (MATCH results / observed expected fields); observation coverage (observed / observable expected fields); freshness (records within permitted age / eligible records); workflow timeliness; facet usability (collection views within approved cardinality and zero-result thresholds); recovery (incidents restored within target / incidents in severity class).

CDS1400-R015 For every verified channel, the organisation MUST declare and monitor verification-health thresholds — at minimum: acceptable mismatch rate, acceptable missing rate, minimum observation coverage and maximum unresolved-mismatch age. These declared thresholds are the **base assurance mechanism** of this chapter. (Channel health aggregation: CDS-500 §23; migration exit thresholds: CDS-1300 §23.)

CDS1400-R016 Per-context service objectives (SLOs) with defined measurement windows SHOULD be defined over the reference indicators. Under the enterprise operations profile (R007) this requirement upgrades to MUST. Verification-health thresholds (R015) already deliver the core assurance; full SLO and error-budget apparatus is proportionate to multi-team operations, not a universal floor.

CDS1400-R017 A breach of a declared threshold or a missed declared objective MUST trigger a review proportional to customer, commercial or compliance impact.

Informative note: an error budget is the permitted amount of objective failure within a period; organisations may use it to balance change velocity against reliability work. CDS does not prescribe universal numeric targets because catalogue size, channel behaviour and business criticality differ (v0.1 ADR CDS-ADR-1400-008, retained).

8. Product, Catalogue and Channel Health (*normative*)

CDS distinguishes health at different aggregation levels — field, product, category, catalogue, channel, customer experience and operations. A single score may be useful for scanning, but it must not replace the underlying dimensions. Health computation and aggregation rules are owned by CDS-500 §23; this section applies them operationally.

CDS1400-R018 A composite health score MUST expose its component measures, weights and unavailable inputs.

CDS1400-R019 A green catalogue or channel score MUST NOT conceal a red condition on any field in the channel's **declared critical set**. The mechanism is CDS-500's: the organisation declares the critical field set per channel (CDS500-R013), and any critical-set field presenting RED caps the aggregate presentation at RED regardless of the numeric score (CDS500-R085). This chapter adds the operational obligation: the declared critical set MUST be reviewed when incidents reveal high-impact fields outside it.

9. Telemetry and Event Model (*normative*)

Operational signals SHOULD be emitted as structured events or equivalent records supporting correlation, replay and audit. *Reference event fields (informative):* event_id, event_type, occurred_at, recorded_at, organisation_id, product_id/variant_id, field_id, channel_id, canonical_revision, mapping_version, dictionary_version, correlation_id, severity_hint, reason_code, evidence_reference.

CDS1400-R020 Telemetry MUST preserve tenant boundaries and MUST NOT expose sensitive supplier or customer data to unauthorised monitoring users (isolation rules: CDS-800 §19).

CDS1400-R021 Event types and reason codes SHOULD be versioned and governed as controlled reference data. Verification reason codes are drawn from the CDS_* registry (CDS-1100 §21; seed list CDS-500 Appendix B), not invented per dashboard.

10. Dashboards and Operational Views (*normative*)

Dashboards are role-specific views over evidence. *The audience catalogue below is informative: it describes focus areas, not mandatory separate dashboards. Views MAY be combined; a small team MAY operate a single view covering all audiences.*

Audience (informative)	Typical focus
Executive	Catalogue and channel health, objective attainment, major incidents, unresolved critical debt.
PIM operations	Ingestion freshness, canonical quality, workflow backlog, quarantines, change effects.
Channel operations	Publication, observation, verification, rate limits, retries, channel-specific failures.
Merchandising / UX	Facet cardinality, zero-result combinations, navigation defects, collection coverage.
Data stewardship	Unknown values, ambiguous mappings, dictionary changes, deprecations, ownership queues.
Engineering	Job health, APIs, transformations, event correlation, deployment regressions.
AI governance	Proposal quality, acceptance, corrections, calibration, prohibited-output incidents.

CDS1400-Ro22 Every summary dashboard MUST support drill-down to affected populations or evidence references.

CDS1400-Ro23 Dashboard freshness and last-successful-update time MUST be visible on the dashboard itself.

CDS1400-Ro24 Colour MUST NOT be the only signal conveying state in any operational view; a text label, status or reason code MUST accompany colour, for accessibility and for diagnosis. Traffic-light colour semantics and the detailed-status accessibility rule for verification indicators are owned by CDS-500 §18 (CDS500-Ro69, CDS500-Ro72–Ro73); this requirement extends the same principle to every operational view, not only verification indicators.

11. Alerting and Noise Control (normative)

Alerts exist to cause timely human or automated action. Reports and informational trends should not be disguised as urgent alerts. *Full alert anatomy (informative): condition, scope, owner, urgency, evidence, runbook link, suppression logic, clear condition — a useful design checklist, not eight mandatory artifacts.*

CDS1400-Ro25 Every production alert MUST have an accountable response owner and a documented first action. This is the normative floor for alert design.

CDS1400-Ro26 Alerts SHOULD be deduplicated by condition, scope and time window to prevent notification storms.

CDS1400-Ro27 An alert that repeatedly produces no action MUST be reviewed and SHOULD be redesigned, downgraded or retired. Alerting is self-limiting by design: an ignored alert is operational debt, not coverage.

Informative note: useful patterns include threshold alerts, absence-of-expected-event alerts, rate-of-change alerts and anomaly alerts. Anomaly alerts require extra caution because they are hard to explain and tune.

12. Incident Classification and Severity (normative)

An incident is an unplanned condition that materially threatens or degrades canonical integrity, publication fidelity, customer experience, compliance, security or operational continuity.

Incident class	Description
Canonical integrity	Incorrect or lost authoritative product facts, identifiers or relationships.
Publication failure	Expected records cannot be published or are rejected.
Downstream drift	Observed channel state differs from expected state (causes: CDS-500 §20).
Taxonomy or dictionary defect	Incorrect classification, mapping, alias or facet grouping affects products.
Customer-experience defect	Filters, search, navigation or product details mislead or block customers.
Security or tenant isolation	Unauthorised access, leakage, tampering or cross-organisation exposure.
AI or automation defect	Automated enrichment or execution produces material incorrect output.
Observability failure	Monitoring or read-back cannot detect or measure a critical state.
Operational capacity	Backlog, rate limit, resource exhaustion or staffing prevents timely processing.

Severity	Impact interpretation
SEV-1 Critical	Widespread or high-risk incorrect data, security/tenant breach, material legal exposure, or loss of control over authoritative data. Immediate coordinated response.
SEV-2 High	Material customer or channel impact, significant publication failure, or major drift affecting a defined business area. Urgent response.
SEV-3 Moderate	Limited impact with available workaround; bounded degradation. Scheduled but prompt response.
SEV-4 Low	Minor defect, cosmetic issue, isolated warning or improvement opportunity. Normal work queue.

CDS1400-Ro28 Severity MUST be based on actual or credible impact, not only the number of affected records.

CDS1400-Ro29 Security, safety, legal or tenant-isolation concerns MUST be allowed to raise severity even when the affected population is small.

CDS1400-Ro30 Severity MAY change as evidence improves; changes MUST be recorded with rationale.

13. Incident Roles and Command (normative)

Incident roles (informative — these are hats per §2, holdable by one person): incident commander; technical or data lead; operations lead; communications lead; scribe / evidence owner; security or compliance lead; business owner; reviewer.

CDS1400-R031 Every SEV-1 and SEV-2 incident **MUST** have a named incident commander. The commander **MAY** simultaneously hold any or all other incident roles (R004).

CDS1400-R032 Where staffing permits, the person coordinating the incident **SHOULD** be distinct from the person performing the most complex repair; the sole-operator case is governed by R005 (dual role noted, compensating control recorded).

CDS1400-R033 Decision authority, escalation contacts and coverage hours **MUST** be declared per severity and incident class before an incident occurs. Declared after-hours coverage **MAY** be next-business-day for any incident class **except security and tenant-isolation classes, which MUST have a declared immediate escalation path at all times.**

14. Detection, Triage and Declaration (*normative*)

Triage determines whether a signal is valid, its scope, likely impact and the next safe action. Early uncertainty is expected; hidden uncertainty is not. *Triage checklist (informative):* confirm the signal is current, not a stale dashboard or duplicate event; identify affected organisation, products, variants, fields, categories and channels; determine whether canonical state, publication state, observed state or customer presentation is wrong; check recent code, mapping, dictionary, taxonomy, supplier or bulk-edit changes; estimate impact and declare provisional severity; assign roles and open a durable incident record; decide whether to pause ingestion, enrichment, publication, automated repair or AI workflows; set the next update time and stakeholder audience.

CDS1400-R034 An incident record **MUST** contain a declared start time, current severity, affected scope, owner and known evidence.

CDS1400-R035 Triage **MUST** distinguish symptom, impact and suspected cause.

CDS1400-R036 If continued automation can increase impact, the relevant automation **MUST** be paused or constrained (using the R040 mechanisms) until safe operation is demonstrated.

15. Containment, Remediation and Automation Safeguards (*normative*)

Containment limits further impact; remediation restores correct operation. *Containment options (informative):* pause publication; disable one mapping or rule; freeze automation and PIM-side edits on a bounded product set; revert a dictionary or taxonomy version; remove or hide affected listings; switch AI or automation to manual approval; preserve an evidence snapshot before repair changes state.

CDS1400-R037 Containment **MUST** prioritise prevention of additional incorrect propagation over speed of full repair.

CDS1400-R038 Repair actions **MUST** identify whether they change canonical values, expected channel state, downstream channel state or verification records.

CDS1400-R039 A bulk repair **MUST** support preview, bounded scope, evidence capture and post-repair verification.

CDS1400-R040 For each automated pipeline stage operated in production — ingestion, enrichment, publication, automated repair and AI workflows — a mechanism to pause or constrain that stage **MUST** exist and be operable **before** the automation runs in production. Containment capability is a system requirement, not an improvisation during the first incident. (*Promotes the capability that vo.1 assumed implicitly in triage and containment steps.*)

16. Drift Diagnosis and Repair Selection (*normative*)

Drift is a difference between the governed expected state and the observed state. The drift cause taxonomy (EXTERNAL_EDIT, CHANNEL_NORMALISATION, MAPPING_CHANGE, STALE_PUBLICATION, CONNECTOR_DEFECT, UNKNOWN_CAUSE and the rest) is owned by CDS-500 §20; repair policies by CDS-500 §21. This section governs their operational use.

CDS1400-R041 Repair policy **MUST** be selected from the diagnosed drift cause. **Republishing MUST NOT be assumed to be the correct repair:** republishing over an unauthorised external edit destroys the evidence of who changed what; republishing "fixes" a channel normalisation only until the next publish; republishing cannot repair a canonical defect, an observation defect or a transformation defect at all.

Diagnosis-to-repair guide (informative, causes per CDS-500 §20):

Diagnosed cause	Typical correct repair family
EXTERNAL_EDIT	Preserve evidence, then republish or escalate per ownership contract; address the editing practice (§28.2).
CHANNEL_NORMALISATION	Register the normalisation (CDS-500 §16.2) and adjust comparator or expected state — not repeated republish.
MAPPING_CHANGE / transformation defect	Fix the mapping or transformation, re-project, then republish.
STALE_PUBLICATION / PARTIAL_PUBLICATION	Republish through the governed path; verify.
Observation defect	Fix read-back; the channel may be correct already.
Canonical defect	Correct the canonical value under governance; then re-project and publish.
Monitoring gap	Record UNOBSERVABLE; treat as owned risk, not as health.

CDS1400-R042 Automatic repair **MAY** be used only for explicitly authorised, deterministic and bounded drift conditions, within PIM-owned fields (CDS500-R078).

CDS1400-R043 A repair **MUST** be followed by observation and verification before the incident or defect is closed (CDS500-R080).

17. Rollback, Replay and Recovery (*normative*)

Recovery restores a known-good operating state and confirms the data path is reliable. *Recovery sequence (informative)*: contain; capture evidence and scope; select rollback, forward-fix or replay; validate on a bounded cohort; execute with traceable batch identity; observe; verify; restore paused automation gradually; monitor for recurrence.

CDS1400-Ro44 Critical mappings, dictionaries, taxonomies and transformations SHOULD have a known rollback mechanism. Channel-side rollback is republication of known-good state through the governed publication path, not restoration of channel-internal state (definition and irreversibility classes: CDS-1300 §17).

CDS1400-Ro45 Replay MUST be idempotent or otherwise protected against duplicate or destructive effects.

CDS1400-Ro46 An incident MUST NOT be declared recovered solely because processing resumed; affected data MUST meet the defined verification exit criteria (declared thresholds per Ro15).

18. Communication and Stakeholder Management (*normative*)

Incident communication states what is known, what is not known, what is being done and when the next update will occur. *Minimum content by stage (informative)*: initial declaration — severity, start time, known impact, scope, containment, next update time; progress — new evidence, changed scope or severity, actions completed, remaining risks; recovery — restored state, verification coverage, residual risk, monitoring period; closure — confirmed outcome, remediation, review and action ownership.

CDS1400-Ro47 Communication MUST distinguish confirmed facts from hypotheses.

CDS1400-Ro48 External notification decisions MUST follow legal, contractual, security and channel obligations.

CDS1400-Ro49 Sensitive evidence MUST NOT be copied into broad communication channels when a controlled reference is sufficient.

19. Post-Incident Review (*normative*)

A post-incident review (PIR) converts an operational failure into durable learning. It reconstructs conditions and decisions without rewriting history after the outcome is known.

CDS1400-Ro50 Every SEV-1 incident MUST receive a documented post-incident review (template: Appendix D.1). Every SEV-2 incident SHOULD receive at least a lightweight review (template: Appendix D.2). (*vo.1 mandated full PIRs for all SEV-2 incidents; under this chapter's SEV-2 definition that produces review-writing as a standing tax at small scale and reviews nobody reads at large scale. The SHOULD, with a template that takes minutes rather than hours, keeps the learning loop without the tax.*)

CDS1400-Ro51 The review MUST identify contributing conditions and control gaps, not merely the final triggering action.

CDS1400-Ro52 Corrective actions arising from a review MUST have named owners and tracked completion evidence.

CDS1400-Ro53 A SEV-1 review MUST include at least one participant who was not the primary incident responder. (*Small-team profile: waived per Table 2-1 with sole-operator note and archived evidence.*)

Informative note: a blameless review does not remove accountability. It separates learning about system conditions from disciplinary or conduct processes, which are handled separately when required.

20. Problem Management and Recurrence Prevention (*normative*)

Problem management addresses underlying causes or repeated patterns not fully resolved by individual incident repair. *Recurrence patterns worth a problem record (informative)*: repeated mismatch in the same field or channel; recurring supplier schema change; growing dictionary quarantine or alias debt; repeated manual channel edits to PIM-owned fields; frequent alert suppression or false positives; repeated facet-cardinality regressions; recurring AI rejection patterns; multiple incidents sharing an undocumented dependency; thresholds missed across consecutive periods.

CDS1400-Ro54 Recurring incidents SHOULD be linked to a problem record with an accountable owner and a long-term remediation plan.

CDS1400-Ro55 A temporary workaround MUST have an expiry date, review date or retirement condition recorded at the time the workaround is adopted.

CDS1400-Ro56 Known operational debt MUST remain visible in operational reporting until it is accepted, remediated or explicitly retired.

21. Change and Regression Monitoring (*normative*)

Most product-data incidents follow a change: supplier import, dictionary edit, taxonomy move, mapping release, bulk update, API version change, theme change, AI model change or staff-process change. *Change controls (informative)*: pre-change baseline; change identifier linking the change to telemetry; canary or bounded cohort; regression tests (CDS-1000 suites, CDS-1200 fixtures); post-change comparison against baseline; observation window covering asynchronous publication and indexing; defined rollback trigger.

CDS1400-Ro57 Material changes MUST be identifiable in operational telemetry (a change identifier linkable to affected events and records).

CDS1400-Ro58 Dictionary, taxonomy, mapping and AI changes SHOULD receive the same regression discipline as code changes.

CDS1400-R059 A change **MUST NOT** be considered successful until its downstream effects have been observed and compared against the pre-change baseline within the declared observation window, with results inside the declared thresholds (R015).

22. Continuous Improvement System (*normative*)

Continuous improvement turns operational evidence into prioritised change: prevent defects, detect earlier, reduce impact, recover faster, remove toil, improve usability, improve assurance. (*The improvement loop and class examples of vo.1 §21 are retained as informative practice.*)

Operational evidence -> trend and incident review -> identify control gap
-> prioritise by impact and recurrence -> implement bounded change
-> test and observe -> confirm improved outcome
-> update standard, profile, runbook or training

CDS1400-R060 Improvement work **SHOULD** be prioritised using impact, recurrence, risk and effort rather than anecdote alone.

CDS1400-R061 Completed improvements **SHOULD** be measured against the outcome they were intended to change.

23. AI and Automation Monitoring (*normative*)

AI and automation require both output-quality monitoring and operational monitoring; high throughput does not imply high quality. AI governance — proposals, evidence classes, confidence, abstention, autonomy levels A0–A4 — is owned by CDS-700; this section defines only the operational measurement obligations. *Reference measures (informative)*: proposal acceptance rate; human correction rate; out-of-dictionary rate; abstention rate; confidence calibration; evidence coverage; override and rollback rate; model or prompt drift; automation blast radius.

CDS1400-R062 AI monitoring segmentation **MUST** scale with the declared autonomy level of each workflow, exactly as defined by CDS700-R081: segmented monitoring distinguishing AI-originated from human-originated values is **MUST** for A3/A4 workflows, **SHOULD** for A2, and **MAY** be limited to volume and sampled quality checks for A1. This chapter supplies the mechanics; CDS-700 owns the scaling rule.

CDS1400-R063 AI quality **MUST** be measured against reviewed outcomes, not the model confidence value alone.

CDS1400-R064 A material decline in quality or an increase in prohibited output **MUST** trigger containment proportional to the workflow's autonomy level (CDS-700), using the pause/constrain mechanisms required by R040.

24. Customer-Experience Signals (*normative*)

Customer-experience health is the hardest domain to observe: most storefronts and marketplaces do not expose interaction telemetry to the PIM side. CDS therefore distinguishes signals computable from governed data from signals requiring channel telemetry.

CDS1400-R065 Facet cardinality and zero-result **construction** (facet values whose product set is empty by construction) **MUST** be monitored. These are the minimum customer-experience signals: they are computable PIM-side from governed dictionaries, facet projections and product data, on every channel, with no storefront telemetry required (facet rules: CDS-600).

CDS1400-R066 Interaction-derived signals — search failure rates, zero-result filter *interactions*, navigation abandonment — are **conditionally observable**. Each channel profile **MUST** declare whether interaction telemetry is available; where it is not, customer-experience health **MUST** be reported as a declared monitoring gap for those signals, never assumed healthy.

CDS1400-R067 Where interaction telemetry is available, it **SHOULD** be monitored. Interaction-metric monitoring is otherwise profile-optional and **MAY** be bound to the customer-experience overlay profile (CDS-1000).

25. Security, Privacy and Evidence Retention (*normative*)

Monitoring and incident records may contain supplier pricing, unpublished product information, credentials, customer-impact evidence or internal system details. *Practice checklist (informative)*: least-privilege access to dashboards, logs and incident records; tenant-scoped telemetry and evidence; redaction or tokenisation of secrets and unnecessary personal data; controlled export and sharing; audit-integrity and timestamp preservation; retention matched to contractual, legal and operational needs; legal hold where required; secure deletion after expiry; monitoring tools reviewed as part of the security boundary.

CDS1400-R068 Monitoring data **MUST** follow the same tenant-isolation and access-control principles as product data (CDS-800 §19).

CDS1400-R069 Secrets, authentication tokens and sensitive personal data **MUST NOT** be intentionally recorded in general operational telemetry.

CDS1400-R070 Incident evidence retention **MUST** be governed and documented.

26. Operating Cadence and Review (*normative*)

Cadence catalogue (informative — a menu, not a mandate):

Cadence	Typical review
Continuous / event-driven	Critical alerts, publication failures, security events, threshold breaches.
Daily	Failed jobs, stale sources, oldest quarantines, unresolved high-impact mismatches.

Cadence	Typical review
Weekly	Quality trends, workflow backlog, mapping debt, recurring alerts, action progress.
Monthly	Objectives/thresholds, incident trends, customer-experience metrics, AI quality, monitoring gaps.
Quarterly	Dictionary and taxonomy health, conformance posture, access review, recovery readiness.
After material change	Regression review and post-change comparison (§21).
After qualifying incident	Post-incident review and action tracking (§19).

CDS1400-R071 The operating model MUST include, at minimum: **(a)** one recurring operational review, at a declared cadence, with recorded outcomes and tracked actions; and **(b)** a defined post-incident review path (§19). *(vo.1 mandated five governed review forums; that is enterprise practice, not a conformance floor. One honest recurring review that produces decisions beats five that produce minutes.)*

CDS1400-R072 A recurring review that repeatedly produces no decisions, ownership or follow-up SHOULD be redesigned or retired — the same self-limiting rule that governs alerts (R027). Reviews are for decisions, not attendance.

27. Conformance (normative)

Conformance levels, test suites, outcomes and claim rules are owned by CDS-1000. A CDS-1400 conformance claim is a claim against the requirement set CDS1400-R001–R072 (with the Table 2-1 substitution where the small-team profile is declared, and the R007 upgrades where the enterprise operations profile is declared), evidenced per CDS-1000.

The recommended practices of this chapter — service objectives and error budgets (§7), separated dashboard audiences (§10), telemetry correlation and versioned reason codes (§9), canary releases (§21), the continuous-improvement loop (§22), AI calibration monitoring (§23), interaction-metric monitoring (§24) and tested runbooks (Appendix C) — are SHOULD-level and MAY be cited as supporting evidence in a claim. *(vo.1's conformance list restated several of these SHOULDs as MUSTs; this section now matches the strengths of the requirements it summarises — finding 1400-8.)* The testable operational-debt controls remain MUST: workaround expiry (R055) and visible known debt (R056).

28. Worked Incident Scenarios (informative)

28.1 Colour Facet Explosion

Signal. Colour facet cardinality rises from 12 to 87 after a supplier import (R065). **Triage.** Canonical colour dictionary remains valid; the import bypassed alias mapping; raw supplier shades were published as facet values. **Containment.** Pause affected collection facet publication; revert the facet projection to the last known-good dictionary version. **Repair.** Quarantine unknown shades (they stay in the denominator — R013); map approved aliases; republish facet values; verify counts and collection filtering. **Prevention.** Facet-cardinality regression test; alert on raw-to-facet ratio change.

28.2 Channel Metafield Drift (SEV-2)

Signal. Verification of the material field on a Shopify channel reports 430 MISMATCH results. Expected Linen, observed Cotton; recent change: a manual bulk edit in the channel admin. Diagnosed drift cause: EXTERNAL_EDIT (CDS-500 §20).

Response. - Declare SEV-2: customer-visible incorrect material claims. - Preserve the observation snapshot **before** repairing — republishing first would destroy the evidence of what was edited and when (R041). - Confirm PIM canonical values are correct (rule out a canonical defect masquerading as drift). - Republish a bounded cohort through the governed path; observe and verify; then repair the remainder in batches (R039, R043).

Follow-up — with real mechanisms. "Freeze affected channel edits" is not an operation most platforms provide; the platform cannot lock a merchant-editable metafield against admin users. The real controls are:

- **Staff instruction and training** — operators are told, specifically, which channel-admin edits will be overwritten or treated as drift (this is the CDS1300-R051 training obligation applied in run-state).
- **Permission narrowing** — restrict which staff accounts hold channel-admin roles that can edit products/metafields at all; this is coarse (role-level, not field-level) but is an enforceable platform mechanism.
- **App-owned namespaces** — move PIM-owned metafields into an app-owned namespace, which admin users cannot edit. **Trade-off:** app-owned metafields are also not visible to merchants in the admin UI, so staff lose the ability to *see* the governed values in their daily tool, which can itself cause errors and shadow spreadsheets.
- **Drift detection stays armed regardless** — enforcement may be detection-only, and that is a legitimate declared posture (CDS1300-R014(c)).

RESOLVED — accepted as drafted (owner 2026-08-04; was D17 (resolved)): For PIM-owned channel metafields there is an unresolved tension between **app-owned namespaces** (edit-proof but invisible to merchant staff) and **merchant-visible metafields** (visible and editable, so protected only by permissions, training and drift detection). This chapter permits either posture provided it is declared per channel profile with its residual risk; the alternative — mandating app-owned namespaces for all PIM-owned fields — was rejected because operator visibility loss is itself an error source.

28.3 Supplier Feed Staleness

Signal. No successful stock-source event for 28 hours; expected cadence is daily (absence-of-expected-event alert). **Triage.** Supplier portal changed HTML format; canonical descriptions and prices unaffected; stock values now stale. **Containment.** Flag source freshness as stale; prevent stale stock from being represented as freshly verified; notify purchasing and channel operations. **Repair.** Update the parser; run a bounded extraction test; replay the source load (idempotent — R045); verify quantity and publication effects. **Prevention.** Keep the absence alert; add a parser fixture from a supplier sample.

28.4 AI Material Misclassification

Signal. Human correction rate for AI material extraction rises from 4% to 21% after a model change (segmented monitoring per Ro62 localises it). **Triage.** Decline limited to homewares image-only extraction; the model confuses rattan-look resin with natural rattan. **Containment.** Reduce autonomy to proposal-only for the affected task (containment proportional to autonomy — Ro64, CDS-700); require text evidence or human review. **Repair.** Revert the model version or update evidence rules; re-evaluate unreviewed proposals. **Prevention.** Benchmark fixtures for imitation materials; quality gate by product family and evidence class.

29. Architecture Decisions (*informative*)

Per CDS000-R006, ADRs live in the single global register. Decisions originating in this chapter, recorded there: monitor data correctness as an operational service; keep observation coverage separate from match rate; base severity on impact, not record count; require verification after repair; treat reference-data and AI changes like code changes; make continuous improvement evidence-driven; prescribe no universal numeric objectives (profiles may). The vo.1 record CDS-ADR-1400-003 (detailed statuses beneath traffic lights) is **subsumed by ADR-D3**, whose normative expression is CDS-500 §17–§18; vo.1 Appendix E (five-colour palette including BLUE) is superseded — see the note after Appendix D.

Appendix A — Reference Metric Catalogue (*informative*)

Metric	Definition	Dimensions	Type
source_freshness_age	Age since last successful authoritative source update	source, organisation	duration
ingestion_failure_rate	Failed eligible source records / attempted eligible source records	source, batch	rate
canonical_required_completeness	Populated required fields / required eligible fields	family, category	rate
dictionary_unmapped_rate	Unmapped reference values / distinct eligible reference values	dictionary, source	rate
quarantine_backlog	Records currently awaiting governed resolution	queue, owner	count and age
publication_lag	Canonical approval to channel acknowledgement	channel, category	duration distribution
publication_failure_rate	Failed publication records / attempted eligible records	channel, reason	rate
observation_coverage	Observed expected fields / observable expected fields	channel, field	rate
verification_match_rate	MATCH results / observed expected fields	channel, field	rate
unresolved_mismatch_age	Age of unresolved MISMATCH	channel, severity	duration distribution
facet_cardinality	Distinct facet values exposed	facet, collection	count
zero_result_combination_rate	Zero-result filter interactions / filter interactions (<i>conditionally observable — §24</i>)	facet, collection	rate
ai_correction_rate	Materially corrected accepted proposals / accepted proposals	task, model	rate
incident_mtta	Declaration time minus detection time	severity, class	duration
incident_mtrr	Verified recovery time minus declaration time	severity, class	duration
incident_recurrence_rate	Recurring incidents / closed incidents	class, period	rate

Appendix B — Severity and Response Matrix (*informative*)

The **minimum-roles** column describes the enterprise operations profile (Ro07); it is informative illustration for all other adopters — every role is a hat holdable by one person (§2).

Severity	Response expectation	Minimum roles (<i>enterprise profile — informative</i>)	Communication	Exit condition
SEV-1	Immediate (coverage per Ro33)	Incident commander; technical/data lead; communications; business owner; security/compliance as relevant	Continuous coordination until contained; frequent updates	Verified containment and business approval to downgrade
SEV-2	Urgent	Incident commander; owning technical/data team; channel/business owner	Regular updates during active response	Material impact stopped; verified recovery plan operating
SEV-3	Prompt business-hours response unless risk increases	Owning team and data/channel owner	Updates at agreed milestones	Scope corrected or controlled workaround accepted (with expiry — Ro55)
SEV-4	Normal queue	Named owner	Status through work management	Fix, acceptance or explicit retirement

Informative note: profiles or organisations may define numeric acknowledgement and restoration targets. Targets should be realistic, measured and reviewed, not copied without operational capacity.

Appendix C — Runbook Template (*informative*)

Section	Content
Runbook identity	Name, owner, version, last tested date, applicable services.
Trigger	Alert conditions, reason codes, scope.
Safety checks	Actions not to be taken without approval; data or channel risks.
Triage	Queries, dashboards, record samples, recent-change checks.

Section	Content
Containment	Safe ways to stop propagation, including the Ro40 pause mechanisms for each stage.
Repair	Deterministic repair, rollback or replay procedure.
Verification	Expected observations, comparison strategy, exit criteria.
Escalation	Contacts, severity triggers, external obligations.
Evidence	Required snapshots, batch IDs, logs, decision records.
Recovery	How paused automation is restored and monitored.
Known limitations	Unobservable fields, manual dependencies, unsupported scenarios.

Appendix D — Post-Incident Review Templates (*informative*)

D.1 Full review (SEV-1 — Ro50)

Incident identifier, title, severity and dates; executive summary; impact and affected populations; detection and declaration; timeline; expected controls and actual behaviour; contributing factors; containment and recovery; verification evidence; what went well; what made response difficult; corrective and preventive actions; owners, due dates and completion evidence; recurrence indicators and review date; changes to CDS profiles, runbooks, tests or training.

D.2 Lightweight review (SEV-2 — Ro50)

One page or less:

```
Incident:      <id, title, SEV-2, dates>
What happened: <two or three sentences>
Impact:       <affected products/fields/channels/customers>
Why:          <contributing conditions and the control gap, not just the trigger>
Fixed by:     <repair + verification evidence reference>
Prevent next: <0..3 actions, each with an owner and a date>
Recurrence of: <link to problem record, or "first occurrence">
```

Appendix E of vo.1 is deleted. Traffic-light presentation is single-homed in **CDS-500 §18** (implementing ADR-D3): GREEN/AMBER/RED plus optional GREY, whose denominator semantics are preserved there — NOT_APPLICABLE is never a success state and is excluded from success denominators either way (CDS500-Ro69). vo.1's **BLUE / INFORMATIONAL tier is dropped** from the standard; in-flight badging remains a local presentation choice with no conformance meaning (CDS-500 §18 note). The accessibility rule survives strengthened as CDS1400-Ro24 (colour never the only signal, in any operational view) alongside CDS500-Ro72.

END OF CDS-1400 vo.2 REVIEW DRAFT

Commerce Data Standard (CDS)

CDS-1500 — Apparel and Homewares Industry Profiles and Reference Dictionaries

Field	Value
Status	vo.2 Review Draft (working source; not an approved standard)
Release	CDS vo.2 (single corpus release per ADR-D5)
Date	2026-09-20 (revision of the 2026-08-04 draft: industry profile register, jurisdiction requirement register and extension profiles added in the industry-profile expansion)
Supersedes	CDS-1500 Working Draft vo.1 and the CDS-1500 Starter Reference Dictionaries vo.1 package structure (data content carried forward, restructured per Appendix E)
Normative status	§1, §2 (including §2.1 and §2.2), §4, §5, §7–§13, §14A, §14B, §15–§18, §23, §24 and Appendices A–B (including A.1, A.2, B.1, B.2) are normative. §3, §6, §14, §19–§22, §25, §26 and Appendices C–G are informative; Appendix G seeds the jurisdiction requirement register whose record structure and obligations are normative in §2.2. Every table is individually marked.
Primary audience	Merchandisers, data stewards, ecommerce operators, PIM architects, catalogue managers, UX teams, developers and AI enrichment designers
Depends on	CDS-000 through CDS-1400, especially CDS-200, CDS-300, CDS-400, CDS-500, CDS-600, CDS-900 and CDS-1100. Cited by CDS-1600 through CDS-2000, which apply the profile model (§2), requirement levels (§4), industry profile register (§2.1) and jurisdiction requirement register (§2.2) homed here
Companion package	CDS Reference Dictionary (unified informative starter data; column contract in Appendix E): Chapters 1–9 and 15–16 for the profiles in this chapter; Chapter 19 (Jurisdictions & Regulatory Schemes) for §2.2
Profile identifiers	Base profiles <code>cds.profile.apparel.v0_2</code> and <code>cds.profile.homewares.v0_2</code> ; extension profiles <code>cds.profile.footwear.v0_2</code> , <code>cds.profile.jewellery.v0_2</code> (§12), <code>cds.profile.furniture.v0_2</code> , <code>cds.profile.garden.v0_2</code> (§14A–§14B). All profile identifiers in the corpus are registered in §2.1
Findings addressed	CDS-1500-1.8; CDS-400-3 and DICT-2 (colour-baseline ownership, D11); CDS-600-3 (zero-result rule re-homed); DICT-1.17; Matrix 2 and Matrix 5 deduplication; ADR-D3, ADR-D4, ADR-D24; D12 (partial — see OPEN flag), D28 (as resolved in CDS-400)

Terminology in this chapter follows CDS-100. In particular, the vo.1 term "reference value" is retired (CDS100-R003); this chapter uses **Source Value** → **Alias Mapping Record** → **Canonical Value** → **Projections** per ADR-D4.

1. Purpose and Scope (*normative*)

CDS-1500 defines two interoperable industry profiles for a commerce catalogue whose primary range is apparel and accessories and whose secondary range includes lifestyle homewares. It translates the vendor-neutral CDS architecture into concrete product-family schemas, dictionary bindings, facet baselines and publishing expectations. **CDS-1500 is the owning chapter for the colour facet baseline** (Appendix C); CDS-400 and CDS-600 cite it and MUST NOT restate divergent copies (*resolves CDS-400-3/DICT-2 in this chapter's favour, decision D11*).

CDS1500-R001 An implementation claiming the Apparel Profile MUST distinguish product identity, sellable variants, canonical attributes, display labels, customer facets and channel representations, per the value-layer model of CDS-400 and CDS-100 §3.

CDS1500-R002 An implementation claiming the Homewares Profile MUST store dimensions, capacity and other measurable specifications as typed values with declared units rather than uncontrolled text.

CDS1500-R003 An industry profile MUST NOT force every possible attribute onto every product. Requiredness MUST be defined per product family and category profile.

CDS1500-R004 A retailer MAY extend the baseline dictionaries, but MUST preserve stable identifiers, aliases, mappings and deprecation history (change control per CDS-400 and CDS-800).

CDS-1500 is also the owning chapter (CDS000-R005) for two mechanisms shared by every industry profile in the corpus: the **Industry Profile Register** (§2.1), which lists every profile identifier, its home chapter and its parent profile, and the **Jurisdiction Requirement Register** (§2.2), which defines how an organisation records the per-country regulatory obligations that bear on product data. The industry profile chapters CDS-1600 (beauty, health and personal care), CDS-1700 (food, beverage and grocery), CDS-1800 (consumer electronics, appliances and technical goods), CDS-1900 (parts, automotive, industrial and fitment) and CDS-2000 (sports and outdoor, toys and games, pet supplies) apply both without restatement; each carries its own seed of the register in its Appendix D, and this chapter's seed for apparel, footwear, jewellery, furniture and garden goods is Appendix G. The selection of profiles follows the global commerce vertical research recorded in REVIEW-020.

2. Industry Profile Model (*normative*)

```
CDS Core
-> Industry Profile
    -> Product Family
        -> Category Profile
            -> Attribute Definitions
            -> Requiredness Rules
            -> Dictionary Assignments
            -> Facet Projection Rules
            -> Channel Mappings
```

CDS1500-R005 Every profile MUST identify which attributes are inherited common attributes and which are category-specific.

CDS1500-R006 A category profile MUST declare a requirement level (§4) for each attribute it governs.

CDS1500-R007 Attributes that create distinct sellable units MUST be modelled at variant scope; descriptive attributes MUST remain at product scope unless values genuinely differ by variant (variant boundary rule: CDS-200 §5).

2.1 Industry Profile Register (normative)

The register is the single list of industry profiles defined by the corpus. A **base profile** stands alone. An **extension profile** refines a base profile for a product family (footwear refines apparel) and is claimed only together with its parent. A conformance claim (CDS-1000) names profile identifiers from this register.

Profile identifier	Profile	Home chapter	Kind	Requires	Dictionary chapters bound
cds.profile.apparel.v0_2	Apparel	CDS-1500 §5–§11	Base	—	1, 2, 3, 4, 5, 6
cds.profile.footwear.v0_2	Footwear	CDS-1500 §12.1	Extension	cds.profile.apparel.v0_2	2, 5, 8
cds.profile.jewellery.v0_2	Jewellery and accessories	CDS-1500 §12.2	Extension	cds.profile.apparel.v0_2	1, 9
cds.profile.homewares.v0_2	Homewares	CDS-1500 §13–§17	Base	—	1, 2, 3, 4, 6 (styles), 7
cds.profile.furniture.v0_2	Furniture	CDS-1500 §14A	Extension	cds.profile.homewares.v0_2	2, 3, 5 (bedding sizes), 7, 15
cds.profile.garden.v0_2	Garden and outdoor living	CDS-1500 §14B	Extension	cds.profile.homewares.v0_2	2, 7, 16
cds.profile.beauty_health.v0_2	Beauty, health and personal care	CDS-1600	Base	—	10, 17 (ingestibles), 18, 19
cds.profile.food_beverage.v0_2	Food, beverage and grocery	CDS-1700	Base	—	17, 18, 19
cds.profile.electronics.v0_2	Consumer electronics, appliances and technical goods	CDS-1800	Base	—	11, 18, 19
cds.profile.parts_fitment.v0_2	Parts, automotive, industrial and fitment	CDS-1900	Base	—	18, 19 (11 where relevant)
cds.profile.sports_outdoor.v0_2	Sports and outdoor	CDS-2000 §5	Base	—	12, 5, 19
cds.profile.toys_games.v0_2	Toys and games	CDS-2000 §6	Base	—	13, 19
cds.profile.pets.v0_2	Pet supplies	CDS-2000 §7	Base	—	14, 17, 19

Cross-industry mechanisms homed in one chapter and cited by the others: the product relationship record, kits and bundles, and the dangerous-goods declaration (CDS-1900 §5, §11, §12); market registration and responsible-person records (CDS-1600 §12); ingredient, allergen, nutrition and net-quantity patterns (CDS-1700 §5–§9); typed technical specifications and per-market compliance records (CDS-1800 §6, §12); the jurisdiction requirement register (§2.2 below).

CDS1500-R054 Every industry profile defined by the corpus MUST appear in the register above with a stable identifier following the CDS-1100 §20 grammar, a home chapter, a kind (base or extension) and, for an extension, the profile it requires; a conformance claim MUST name profile identifiers from this register.

CDS1500-R055 A claim of an extension profile MUST include a claim of the profile it requires; the extension's requirements add to, and MUST NOT weaken, the parent's.

2.2 Jurisdiction Requirement Register (normative; the seed tables in each chapter's Appendix D and this chapter's Appendix G are informative)

Much of what a product record must contain is decided by law, market by market: what must appear on the label, what must be shown online before purchase, what must be registered before sale, what must be declared for transport and what may not be sold at all. The corpus does not embed those rules in the requirement levels of §4, because they change by jurisdiction and over time. Instead every organisation maintains a **jurisdiction requirement register**: a governed record of the obligations that bear on its product data for each market it publishes to, mapped to the CDS fields that carry the data. The industry profile chapters seed the register with the obligations known at the release date; the organisation verifies, extends and reviews it.

Register entry property	Meaning	Requirement
entry_id	Stable identifier	Required
jurisdiction	Governed value from the <code>jurisdiction</code> dictionary (package Chapter 19; ISO 3166 country codes with sub-national codes where a state or province regulates separately, for example US-CA)	Required
regulatory_scheme	Governed value from the <code>regulatory_scheme</code> dictionary (package Chapter 19) or a governed organisation extension, naming the instrument	Required
regulator	The authority administering the scheme	Required
instrument	Citation of the instrument and provision (act, regulation, standard, article or section)	Required
product_scope	Product families, categories or attribute conditions the entry applies to (a governed expression over CAT_ and MF_ attributes)	Required
obligation_type	One or more of: <code>pre_market</code> (registration, notification, listing, approval or certification before sale), <code>labelling_element</code> (a data element that must appear on the product or packaging), <code>listing_element</code> (a data element or artefact that must appear in the online offer before purchase), <code>warning_statement</code> (a mandated warning), <code>rating_label</code> (an energy, water, efficiency or similar rating label or class), <code>restricted_content</code> (a restriction on what may be claimed, shown or sold, including age and channel restrictions), <code>transport</code> (a dangerous-goods or carriage obligation), <code>documentation</code> (a document that must exist or be available on request), <code>post_market</code> (a producer-responsibility, recall, take-back or repair obligation)	Required
data_elements	The data elements the obligation requires, each mapped to the CDS field (namespace and attribute, or record) that carries it, or to a declared organisation extension where no CDS field exists	Required

Register entry property	Meaning	Requirement
trigger	The condition under which the obligation applies (category, attribute value, quantity, sale channel, customer type)	Required
effective_from / effective_to	Dates the obligation applies, including transition periods	Required where dated
publication_consequence	What preflight does for the target market when the obligation is unmet: <code>block</code> , <code>warn</code> or <code>record</code>	Required
verification_status / verified_on / verified_against	Whether the entry was verified against the primary instrument or a regulator publication, when, and against what	Required
owner / review_due	Accountable owner (CDS-800) and next review date	Required
notes	Interpretation notes, exemptions relied on	Optional

CDS1500-R056 An implementation claiming any industry profile **MUST** maintain a jurisdiction requirement register conforming to the record structure above for every market it publishes products to, with each entry carrying jurisdiction, scheme, instrument, product scope, obligation types, data elements mapped to CDS fields, trigger, dates, publication consequence, verification status and owner.

CDS1500-R057 Publication preflight for a target market (CDS-500 §8) **MUST** evaluate every register entry whose jurisdiction and product scope match the product and channel, and **MUST** apply the entry's publication consequence: a `block` entry with unmet data elements blocks publication to that market, a `warn` entry records a warning, and a `record` entry records the evaluation; the evaluation result **MUST** be retained as an observation (CDS-500) with the entry identifier and version.

CDS1500-R058 Every register entry **MUST** carry a verification status and date; an entry that is unverified, past its review date or seeded but not yet adopted **MUST** be visible as such in the register and in preflight results, and an implementation **MUST NOT** present an unverified entry as a verified legal obligation.

CDS1500-R059 Where an obligation requires a data element for which the corpus defines no field, the organisation **MUST** declare an extension attribute or record (CDS-300 §21 grammar; CDS-1100 §20) and map the entry to it; the obligation **MUST NOT** be recorded as satisfied by free text in a description.

CDS1500-R060 The seed tables in the industry profile chapters are informative starter content, dated and marked with their verification status: an organisation **MAY** adopt them as the initial content of its register, at which point they become governed records under that organisation's control and change through CDS-800 governance; an organisation **MUST NOT** rely on a seed entry marked "retrieved", "unverified" or "not seeded" without verifying it against the primary instrument.

CDS1500-R061 Register entries whose obligation type is `listing_element`, `warning_statement` or `rating_label` **MUST** have their required elements projected in the channel representation for the target market and verified under CDS-500 where read-back exists; where the channel exposes no read-back for the element, the verification status **MUST** be UNOBSERVABLE, not MATCH.

3. Shared Product Information Layers (*informative*)

Layer	Purpose	Examples	Normative home
Identity	Stable commercial identity	Product ID, SKU, brand, MPN, GTIN	CDS-200
Classification	What the product is	Family, type, internal category, external taxonomy mappings	CDS-200 §6
Canonical attributes	What the product is like	Material, colour, pattern, fit, room, finish	CDS-200 §7, CDS-400
Variant attributes	What creates a sellable unit	Size, sellable colour, pack count	CDS-200 §5
Content	How it is explained	Title, description, features, care, dimensions	CDS-200
Facet projection	How customers filter	Blue, Linen, Relaxed, Living Room, Wood	CDS-600
Channel projection	How a destination receives it	Metafield-like structures, <code>CH_google_color</code> , marketplace category	CDS-500, CDS-900
Verification	Whether the destination matches expectation	Statuses and traffic lights per the single normative enum in CDS-500 §17	CDS-500

CDS1500-R008 A single product fact **SHOULD** be entered canonically once and reused across display, search, facets, collections and channel mappings.

4. Attribute Requirement Levels (*normative*)

Code	Meaning	Publication consequence
R — Required	Every active product in the profile must contain a valid value.	Publication is blocked when absent or invalid (preflight per CDS-500).
C — Conditional	Required when a declared condition is true.	Publication is blocked only when the condition applies.
REC — Recommended	Expected for strong customer experience and channel quality.	Warning or quality deduction when absent; never blocking.
O — Optional	Useful but not expected for every product.	No consequence when absent.
N/A	Not applicable to the product family.	Field is hidden or ignored.

The *vo.1* code "P — Recommended" is replaced by REC; codes now read unambiguously alongside RFC-2119 keywords (resolves CDS-1500-8's companion code finding).

CDS1500-R009 Requiredness **MUST** be evaluated against the product family and category profile, not against a universal flat field list.

CDS1500-R010 A value of unknown or not-supplied **MUST NOT** be fabricated to satisfy requiredness (quarantine per CDS-400 §17; abstention per CDS-700).

CDS1500-R011 Where a requirement level applies to a dictionary-bound field, the requirement binds to a **governed dictionary conforming to CDS-400** (stable value_ids, lifecycle, provenance). The seed baselines in Appendices C–D and the companion CSV package are **informative starter data**: an organisation

MAY adopt them verbatim as the initial content of its governed dictionaries, at which point they become governed data under that organisation's control. *(Resolves CDS-1500-3; normative requirement levels no longer bind to informative value sets.)*

5. Apparel Product Model *(normative)*

The Apparel Profile treats the Product as the shared style or design and the Variant as the sellable combination of option values. A shirt offered in four sizes and three colours is normally one Product with twelve Variants, provided the descriptive design and construction remain substantially shared.

Sellable colour means the colour-bearing variant option value that identifies a distinct sellable unit — a Variant Option in CDS-100 terms, stored as a canonical value_id.

CDS1500-R012 Size and sellable colour SHOULD be variant attributes when they identify distinct sellable units.

CDS1500-R013 Where colour is variant-defining, the variant field MUST store the canonical value_id (e.g. VAR_colour = colour_french_navy); the product-scope display label and facet value are projections derived from the variants' canonical values, never independently authored parallel facts. *(Per ADR-D4; resolves CDS-1500-6.)*

CDS1500-R014 Display colour, canonical colour and facet colour MUST remain related but distinct values where the richer display name differs from the broad customer filter (layer model: CDS-400; retained from v0.1).

CDS1500-R015 A product SHOULD NOT be split into separate products solely to obtain separate product pages. An organisational profile MAY declare a variants-as-products merchandising model, in which case it MUST preserve family relationships between the split products. *(Informative note: variants-as-products was deliberate legacy production architecture — a supported declared pattern, not an anti-pattern; see REVIEW-002A LEG-8.)*

```
Product: Relaxed Linen Shirt          (informative example)
Product scope:
  CAT_product_type = shirt
  MF_material_primary = linen
  MF_fit = relaxed
  MF_sleeve_length = long
  MF_neckline = collar
Variant scope:
  VAR_colour = colour_french_navy    # canonical value_id
  VAR_size = au_10
  STD_sku = SHIRT-FN-10
```

6. Apparel Product Families and Category Profiles *(informative baseline; the inheritance rule is normative)*

Product family / category	Required	Recommended	Typical variant options
Tops and shirts	material composition; sellable colour; size	fit; sleeve length; neckline/collar; pattern; care	size, colour
Dresses and jumpsuits	material composition; sellable colour; size	dress length; sleeve length; neckline; fit; occasion; lining	size, colour
Knitwear	material composition; sellable colour; size	knit type; weight; fit; neckline; care	size, colour
Jackets and coats	material composition; sellable colour; size	outer material; lining; insulation; closure; weather properties	size, colour
Pants and jeans	material composition; sellable colour; size	rise; leg shape; fit; inseam; stretch; closure	waist/size, length, colour
Skirts and shorts	material composition; sellable colour; size	length; rise; fit; closure; lining	size, colour
Activewear	material composition; sellable colour; size	activity; support; compression; moisture properties	size, colour
Swimwear	material composition; sellable colour; size	coverage; support; lining; chlorine resistance	size, colour
Sleep and lounge	material composition; sellable colour; size	fit; set contents; warmth; care	size, colour
Scarves, hats and soft accessories	material composition; sellable colour	dimensions; construction; season; pattern	colour, size where sellable

CDS1500-R016 Profile attributes MAY be inherited from a parent family, but category-specific requirements MUST remain explicit in the category profile.

7. Apparel Colour Architecture *(normative; the worked chain is informative)*

Colour uses the four-layer value model (ADR-D4): the source value is preserved with provenance, an alias mapping resolves it to a canonical shade, and display, facet, search and channel forms are parallel projections of the canonical value.

```
Source Value:      "French Navy"      (supplier text, preserved verbatim with provenance)
Alias Mapping Record: "French Navy" -> colour_french_navy (dictionary_key: colour)
Canonical Value:    colour_french_navy (code french_navy, status active)
Projections:
  Display Label:    French Navy
  Facet Value:      blue                (single facet; ordered list where multiple)
  Search Synonyms:  french navy, dark navy
  Channel:          CH_google_color = "Blue" (feeding layer declared per attribute)
```

An organisation preferring coarser canonicals MAY instead alias "French Navy" to canonical colour_navy and keep "French Navy" as display label only. Both granularities are conformant (ADR-D4).

CDS1500-R017 The customer colour facet MUST use a deliberately limited colour-family dictionary governed under CDS-400, not the full set of source or display values. (Seed baseline: Appendix C.)

CDS1500-R018 The canonical shade SHOULD remain more specific than the facet, because content, search, swatches and channel mappings need the specificity the facet deliberately discards.

CDS1500-R019 A canonical shade MAY belong to more than one colour facet only where its governed dictionary entry declares an **ordered facet list, primary first** (e.g. teal → green|blue), per the facet-order rule as resolved in CDS-400 (D28). Facet membership is declared in the dictionary, never decided per product.

CDS1500-R020 Facet membership MUST be deterministic at dictionary level. Shade names whose real-world appearance spans facet families MUST be split into distinct canonical shades, one per facet, with intake resolved through alias mappings (which MAY be supplier- or locale-scoped per ADR-D4). The seed baseline applies this split: **charcoal_grey** (facet grey) vs **charcoal_black** (facet black); **rust_orange** (facet orange) vs **rust_brown** (facet brown); **tan_beige** (facet beige) vs **tan_brown** (facet brown). Bare source values ("Charcoal", "Rust", "Tan") map by default alias to charcoal_grey, rust_orange and tan_beige respectively; a supplier whose shade reads otherwise gets a supplier-scoped alias to the sibling canonical. (*Resolves CDS-1500-1: the v0.1 "may map to Grey unless visually near-black" judgement notes are deleted.*)

CDS1500-R021 A product-level facet override (assigning a product a facet other than its canonical shade's dictionary mapping) MAY exist only as a governed override record with provenance, owner and reason, per the override model of CDS-500 §22. It never silently edits the dictionary mapping.

RESOLVED (was D12 (resolved), owner decision 2026-08-04): product-level facet overrides are **kept**, exceptional, provenance-carrying. Schema home: the **CDS-500 §22 override record** (one override mechanism corpus-wide — same record family as field and taxonomy overrides); no separate CDS-400 mapping-exception record is introduced.

CDS1500-R022 A multi-colour product MUST NOT be assigned to every colour present in a complex print. The canonical layer records the dominant and/or component shades; **multicolour is a facet/status value only, never a canonical shade** (ADR-D4; resolves CDS-1500-4). An organisational profile MAY define a dominant-colour extraction rule for prints.

8. Apparel Fibre, Fabric and Composition (*normative; seed table informative*)

Apparel material data is separated into fibre composition, fabric or construction, and product components. This prevents common category errors such as treating Denim, Jersey or Satin as fibre types. (*Preserved from v0.1 — ADR-1500-003.*)

Field	Meaning	Example
MF_fibre_composition	Percentage composition of fibres	55% Linen; 45% Cotton
MF_material_primary	Primary canonical fibre/material	Linen
MF_fabric_type	Fabric construction or named textile	Denim, Jersey, Twill, Satin
MF_surface_treatment	Finish or treatment	Garment-dyed, Brushed, Washed
MF_lining_material	Lining composition when present	100% Viscose
MF_fill_material	Insulation or fill	Recycled Polyester Fill
MF_stretch	Stretch behaviour	None, Slight, Moderate, High

CDS1500-R023 Composition percentages MUST be numeric values associated with canonical material identifiers and SHOULD total 100 percent per component.

CDS1500-R024 Fabric type MUST NOT replace fibre composition.

CDS1500-R025 Marketing claims such as organic, recycled or certified MUST be stored separately from the underlying material identity (fields: §11) and MUST require supporting evidence. Consequences for facets (*resolves CDS-1500-2*): - A **"Recycled" customer facet is claim-driven**: it derives from MF_recycled_content_percent (and its evidence), never from a material family. recycled_polyester remains a canonical material where supply-chain identity warrants it, but its material family is synthetic and its material facet is Polyester. - **Elastane is family synthetic with no material facet**; stretch is a garment property expressed through MF_stretch, not a material family. A "Stretch" filter, where offered, is a property facet fed by MF_stretch. - The v0.1 facet label "Other Natural" is deleted (CDS-600 forbids Other/General/Misc facet labels); hemp gets its own facet, hemp.

Seed material table (*informative — governed adoption per CDS1500-R011; material facet ids reference the material_facets seed, Appendix D*):

Canonical material	Alias	Family	Material facet
cotton	—	plant_fibre	cotton
organic_cotton	—	plant_fibre	cotton (<i>organic claim still requires MF_organic_claim evidence</i>)
linen	—	plant_fibre	linen
hemp	—	plant_fibre	hemp
wool	—	animal_fibre	wool
merino_wool	—	animal_fibre	wool
cashmere	—	animal_fibre	cashmere
silk	—	animal_fibre	silk
viscose	rayon	regenerated_cellulose	viscose
lyocell	—	regenerated_cellulose	lyocell
modal	—	regenerated_cellulose	modal

Canonical material	Alias	Family	Material facet
polyester	—	synthetic	polyester
recycled_polyester	—	synthetic	polyester (<i>Recycled facet claim-driven, see R025</i>)
nylon	polyamide	synthetic	nylon
acrylic	—	synthetic	synthetic
elastane	spandex	synthetic	— (<i>property: MF_stretch</i>)
leather	—	animal_material	leather
suede	—	animal_material	leather
faux_leather	vegan leather	synthetic	faux_leather

9. Apparel Size, Fit and Garment Measurements (*normative; seed table informative*)

Size labels are not globally interchangeable. CDS stores the supplied sellable label, the applicable size system, a stable sort key and optional body or garment measurements.

Field	Scope	Example
VAR_size_label	Variant	10, M, 32, EU 40
VAR_size_system	Variant or product	AU_WOMENS_NUMERIC, ALPHA, EU_FOOTWEAR
VAR_size_sort_key	Variant	030, 040, 050
MF_size_chart_id	Product	sizechart_brand_womens_tops_2026
MF_fit	Product	Relaxed
MF_garment_length_mm	Product or variant	720
MF_inseam_mm	Product or variant	780
MF_model_wears_size	Content metadata	AU 8 / Small

CDS1500-R026 A size conversion MUST NOT be presented as exact unless a brand- or product-specific size chart supports it. (*Preserved — ADR-1500-007: no universal exact size conversion.*)

CDS1500-R027 Size labels MUST be sortable independently of alphabetical order (sort keys or an equivalent declared mechanism).

CDS1500-R028 Fit describes intended silhouette relative to the size specification; it MUST NOT be used as a substitute for size.

Seed fit dictionary (*informative*): skinny, slim, regular, relaxed, oversized, tailored, straight, tapered, wide_leg, bootcut, cropped, longline, fitted, loose.

10. Apparel Pattern, Style, Occasion and Season (*normative; seed tables informative*)

Pattern, style and occasion are separate controlled concepts. A floral garment may be romantic, resort or casual; the pattern does not define the style by itself.

Dictionary	Purpose	Examples
Pattern	Visible repeat or motif	Plain, Striped, Checked, Floral
Style	Aesthetic or merchandising language	Classic, Minimal, Bohemian, Resort
Occasion	Customer use context	Work, Casual, Formal, Travel, Beach
Season relevance	Merchandising relevance, not lifecycle	Summer, Winter, Transitional, All-year
Weather suitability	Functional environmental use	Warm weather, Cold weather, Rain

CDS1500-R029 Style and occasion MAY be multi-value attributes but SHOULD enforce a declared maximum value count per product to prevent indiscriminate tagging.

CDS1500-R030 Season relevance MUST NOT be conflated with a collection campaign or product lifecycle status.

Seed pattern dictionary (*informative*): plain, striped, checked, plaid (alias: tartan), floral, animal, geometric, abstract, paisley, polka_dot, graphic (alias: placement print), colour_block, motif. *Note (DICT-17): the vo.1 entries textured and marbled are surface textures, not print patterns; they are re-homed as finish/surface values (§17 finishes for homewares, MF_surface_treatment for apparel). plain doubles as "no pattern"; organisations wanting the distinction may add an explicit none value.*

Seed style dictionary (*informative*): classic, minimal, modern, contemporary, casual, formal, romantic, bohemian, coastal, resort, streetwear, athleisure, workwear, vintage_inspired, scandinavian, industrial, rustic, country, luxury.

Seed occasion dictionary (*informative*): everyday, work, casual, formal, party, wedding_guest, travel, holiday, beach, active, outdoor, lounge.

11. Apparel Care, Origin and Claims (*normative*)

Attribute	Requirement guidance	Control
MF_care_instructions	REC for all apparel; R where special care applies	Structured symbols plus human-readable text where possible

Attribute	Requirement guidance	Control
MF_country_of_origin	C on legal, channel or organisational requirements	Governed country reference
MF_certifications	O unless a claim is made	Reference evidence and expiry or scope
MF_recycled_content_percent	R when a quantified recycled claim is published	Numeric, component-aware
MF_organic_claim	R when organic is claimed	Evidence source and applicable component
MF_animal_material	C for leather, wool, silk, down and similar	Canonical material plus care/compliance controls

CDS1500-R031 Claims MUST remain distinct from descriptive attributes and MUST NOT be inferred from supplier marketing language without evidence (evidence classes: CDS-700 §7).

CDS1500-R032 The profile MUST support withdrawal or expiry of a claim without changing the canonical material identity.

12. Footwear and Accessories Extension Profiles (*informative baseline; R033 and §12.1–§12.2 normative*)

Category	Required	Recommended	Variant options
Footwear	size; size system; sellable colour; upper material	sole material; lining; closure; heel height; width; toe shape	size, colour, width
Bags	sellable colour; primary material	dimensions; strap; closure; compartments; capacity	colour, size where applicable
Belts	size; sellable colour; primary material	width; buckle material; closure	size, colour
Jewellery	primary material; colour/finish	dimensions; stone; plating; closure; care	finish, size where applicable
Hats	size or adjustable status; sellable colour; material	brim, crown, fit, care	size, colour

CDS1500-R033 Footwear size systems MUST be identified explicitly and MUST NOT be merged into a single unqualified number.

12.1 Footwear Extension Profile (*normative; cds.profile.footwear.v0_2, requires cds.profile.apparel.v0_2*)

The footwear profile binds the Chapter 8 vocabularies (styles, heel heights, closures, toe shapes, width fittings) and the footwear size systems of Chapter 5, and adds the attributes that distinguish footwear from garments: upper, lining, sole and insole materials as component materials; width fitting as a variant option where offered; and measured fit (internal length) as the basis for size guidance.

CDS1500-R062 Footwear MUST model size and, where offered, width fitting as variant options with an explicit size system (R033) and MUST store the size system's sort order (R027); internal length in millimetres SHOULD be stored per size where the brand publishes it, and a size conversion MUST follow R026.

CDS1500-R063 Upper, lining, sole and insole materials MUST be component materials (the R035 pattern: canonical material per component role) rather than a single material field; the primary material facet derives from the upper unless the category profile declares otherwise.

CDS1500-R064 Footwear style, heel height, closure, toe shape and width fitting MUST bind to governed dictionaries conforming to CDS-400 where those facets are exposed (R011); heel height MUST be stored as a typed measurement in millimetres with the governed band derived, never authored.

12.2 Jewellery and Accessories Extension Profile (*normative; cds.profile.jewellery.v0_2, requires cds.profile.apparel.v0_2*)

The jewellery and accessories profile binds the Chapter 9 vocabularies (metals, stones, closures, piece styles) and covers jewellery, watches, bags, belts, hats, eyewear and small leather goods. Its distinctive facts are material composition with purity and plating, stone identity with treatment and origin claims, and sizes measured in ring, chain-length and circumference systems.

CDS1500-R065 Metal, purity or fineness, plating and stone MUST be separate governed attributes: the metal (a `jewellery_metal` value) with its fineness as a typed value (karat or parts per thousand) and the plating or coating as a distinct value; a plated item MUST NOT be described by the plating metal alone.

CDS1500-R066 Stone identity MUST be a governed `jewellery_stone` value; natural, laboratory-grown, treated and simulated status MUST be an explicit governed attribute per stone, and origin, carat weight and grading are claims or typed values with evidence (R031), never inferred.

CDS1500-R067 Ring, chain, bracelet and hat sizes MUST identify their size system (`size_system` values such as `au_ring`, `us_ring`, `eu_ring`, `head_circumference_cm`) and store the measured basis (inside circumference or diameter, length, head circumference) as typed measurements where the brand publishes it.

CDS1500-R068 Precious-metal marking, nickel-release and lead or cadmium content obligations that a market imposes (Appendix G) MUST be evaluated at preflight for that market through the jurisdiction requirement register (§2.2).

13. Homewares Product Model (*normative*)

The Homewares Profile covers lifestyle products used in rooms and domestic settings. It emphasises typed measurements, material and finish separation, product composition, pack contents and use context.

```
Product: Stoneware Table Lamp           (informative example)
  CAT_product_type = table_lamp
  MF_material_primary = stoneware
  MF_finish = glazed
  MF_colour_canonical = colour_oatmeal
  MF_colour_display = Oatmeal
```

```
MF_colour_facet = beige
MF_room = [living_room, bedroom]
MF_height_mm = 520
MF_width_mm = 280
VAR_plug_type = AU
VAR_voltage = 230 V
```

CDS1500-R034 Dimensions, weight, capacity, power and other measures MUST be typed values with declared units.

CDS1500-R035 Material and finish MUST be separate attributes wherever the same material may appear in multiple finishes.

CDS1500-R036 Room and style are merchandising attributes and MUST NOT replace functional product classification (CDS-200 §6).

Electrical compatibility fields use VAR_plug_type (the v0.1 variant spelling VAR_plug_profile is retired — resolves CDS-1500-8 / CLEANUP-PASS B.7).

14. Homewares Product Families and Category Profiles (*informative*)

Product family / category	Required	Recommended	Typical variants
Cushions and decorative pillows	dimensions; cover material; colour	fill material; closure; removable cover; pattern; room	colour, size, insert option
Throws and blankets	dimensions; material composition; colour	weight/warmth; weave; pattern; care	colour, size
Bedding	bed size; material composition; colour; set contents	thread count; weave; closure; care	size, colour
Towels and bath textiles	dimensions; material composition; colour	weight/GSM; absorbency; set contents; care	size, colour, pack
Rugs and mats	dimensions; primary material; colour	pile height; construction; backing; indoor/outdoor; care	size, colour
Candles and home fragrance	product type; net weight/volume; fragrance; burn/diffusion metric	wax type; vessel material; notes; safety; refillability	fragrance, size
Vases and decor	dimensions; primary material; colour	finish; shape; room; style; handmade status	colour, size
Tableware and serveware	product type; material; dimensions/capacity; set count	finish; microwave/dishwasher suitability; food contact	colour, size, set count
Drinkware	capacity; material; set count	thermal properties; dishwasher suitability; lid/straw	colour, capacity, pack
Storage and baskets	dimensions/capacity; material	lid; handle; stackability; room; load guidance	colour, size
Lighting	type; dimensions; electrical compatibility	material; finish; bulb base; max wattage; dimmable; cable length	colour/finish, plug type
Outdoor lifestyle	dimensions; material; outdoor suitability	UV/water resistance; storage; care; assembly	colour, size

14A. Furniture Extension Profile (*normative; cds.profile.furniture.v0_2, requires cds.profile.homewares.v0_2*)

Furniture (seating, tables, storage, beds, mattresses, desks, outdoor furniture) is the highest-value and most logistics-intensive homewares family. The profile binds Chapter 15 (assembly types) and the bedding size systems of Chapter 5, reuses the Chapter 6 style vocabulary and Chapter 2–3 materials and fabrics for upholstery, and adds the attributes that furniture retail and regulation demand: assembled and packaged dimensions, weight and load ratings, assembly type and time, seat and bed dimensions, mattress firmness and construction, and stability, flammability and toppling obligations that differ by market.

Field	Meaning	Type
MF_assembly_type	Governed value (assembly_type: fully assembled, partial, flat-pack, professional installation)	Product
MF_assembly_time_min / MF_assembly_person_count	Assembly guidance	Integers
MF_dimensions_assembled / MF_dimensions_packaged (per package)	Typed dimensions with unit (R039)	Product / Variant
MF_weight_kg / MF_packaged_weight_kg / MF_package_count	Weights and package count	Typed measurements; integer
MF_load_rating_kg / MF_seat_height_mm / MF_seat_depth_mm / MF_arm_height_mm	Ergonomic and load measurements	Typed measurements
MF_bed_size_system + VAR_bed_size	Bedding size (size_system values au_bedding, us_bedding, uk_bedding, eu_bedding) with the mattress dimensions the size denotes	Variant
MF_mattress_firmness / MF_mattress_construction / MF_mattress_depth_mm	Governed firmness (mattress_firmness), construction as governed organisation value, depth	Product / Variant
MF_upholstery_material / MF_upholstery_fabric / MF_frame_material / MF_fill_material	Component materials per R035	Product
MF_material_components	Component list with roles	Product
MF_delivery_class	Governed logistics class (parcel, oversized, two-person, white-glove) derived under governed rules	Product / Variant
MF_warning_statements / CMP_compliance_declarations	Stability, toppling, flammability and general-safety obligations per market (§2.2; Appendix G)	Per market
Relationship records	Modular and sectional configurations, matching pieces, replacement parts (CDS-1900 §5)	Product

CDS1500-R069 Furniture MUST store assembled dimensions and, per package, packaged dimensions and weight as typed measurements (R039), with package count; a product shipped in several packages MUST NOT report a single packaged dimension.

CDS1500-R070 Assembly type MUST be a governed `assembly_type` value; delivery class MUST derive from packaged measurements and category rules under governed logic, not be authored per product.

CDS1500-R071 Bed and mattress sizes MUST identify their size system and MUST store the mattress dimensions the size denotes in that system; a size label such as "Queen" MUST NOT be published without its system where the organisation sells into more than one market.

CDS1500-R072 Mattress firmness MUST bind to a governed `mattress_firmness` dictionary where the facet is exposed; a brand's numeric firmness scale MUST be mapped to the governed value through an alias mapping with its basis recorded, and the mapping MUST NOT be presented as a measured property.

CDS1500-R073 Upholstery fabric, frame and fill MUST be component materials (R035) with fibre composition for textile covers (R023 pattern); flammability and chemical-content statements that a market requires (Appendix G) MUST be structured records evaluated at preflight, and MUST NOT be projected as claims or facets.

CDS1500-R074 Toppling, stability and anchoring warnings that a market requires on the product, its packaging or in the online description (Appendix G) MUST be structured warning records evaluated per market at preflight and projected to the market's listing where the market requires it; presence MUST be verified under CDS-500 (R061).

CDS1500-R075 Modular and sectional furniture MUST express configurations, matching pieces and replacement parts through relationship records per CDS-1900 §5; a configurator's output MUST be traceable to those records.

14B. Garden and Outdoor Living Extension Profile (*normative; `cds.profile.garden.v0_2`, requires `cds.profile.homewares.v0_2`*)

Garden and outdoor living covers outdoor furniture and shade, planters and garden decor, garden tools and equipment, live plants and seeds, growing media, fertilisers and pest control, and barbecue and outdoor cooking. The profile binds Chapter 16 (sun exposure) and reuses the homewares baseline; powered equipment applies CDS-1800, chemicals and gas products apply CDS-1900 §12, and live plants are an organisation extension.

CDS1500-R076 Outdoor suitability MUST be an explicit governed attribute (indoor, covered outdoor, full outdoor) with the weather-resistance basis recorded; "outdoor" MUST NOT be inferred from category placement.

CDS1500-R077 Plants, seeds and bulbs MUST be identified by governed botanical and common names as separate attributes, with sun exposure bound to the governed `sun_exposure` dictionary where the facet is exposed; growth, hardiness and climate suitability MUST carry the scheme they are stated in (a hardiness-zone or climate-zone system) and are region-specific; the package deliberately ships no plant taxonomy.

CDS1500-R078 Fertilisers, pesticides, herbicides, pool chemicals, fuels and gas products MUST carry dangerous-goods declarations and safety data sheet references per CDS-1900 §12 and per-market registration or restricted-sale records where a market regulates them (Appendix G); they MUST NOT be published to a market whose registration is absent.

CDS1500-R079 Live plants, seeds and growing media MUST carry per-market biosecurity and movement restrictions as governed restricted-sale attributes evaluated at preflight, including sub-national restrictions where they exist (for example interstate plant movement controls).

CDS1500-R080 Barbecues, heaters and gas appliances MUST apply CDS-1800 §12 for gas and electrical compliance records and CDS-1800 §8 for energy and power facts.

15. Homewares Material and Construction (*normative; seed table informative*)

Homewares material data often describes multiple components. The profile supports primary material, component materials and surface finish without flattening them into one tag.

Field	Purpose	Example
MF_material_primary	Dominant structural or customer-relevant material	Oak
MF_material_components	Named component-material relationships	Frame: Oak; Handle: Brass
MF_material_facet	Broad customer filter (governed material facet dictionary, Appendix D seed)	Wood
MF_finish	Surface appearance or treatment	Natural Oil
MF_construction	Manufacturing or textile construction	Hand-woven, Mouth-blown, Tufted
MF_food_contact	Food-contact suitability	Yes / No / Conditional
MF_outdoor_suitability	Declared outdoor use	Indoor only, Covered outdoor, Outdoor

CDS1500-R037 A material-family facet MUST NOT erase the specific canonical material used for product display and channel mapping.

CDS1500-R038 Component materials SHOULD identify the component wherever customer understanding, care, compliance or value depends on it.

Seed material table (*informative — the apparel and homewares material dictionaries are separate dictionaries with distinct `dictionary_keys` (`material_apparel`, `material_homewares`); the same code may therefore carry a different family classification in each without conflict — resolves DICT-5 via DICT-6 scoping*):

Canonical material	Family	Material facet
cotton	textile	cotton
linen	textile	linen

Canonical material	Family	Material facet
wool	textile	wool
polyester	textile_synthetic	synthetic_textile
ceramic	ceramic	ceramic
stoneware	ceramic	ceramic
porcelain	ceramic	porcelain
glass	glass	glass
crystal	glass	glass
stainless_steel	metal	metal
aluminium	metal	metal
brass	metal	metal
iron	metal	metal
oak	wood	wood
pine	wood	wood
teak	wood	wood
rattan	natural_fibre	natural_fibre
bamboo	natural_fibre	natural_fibre
marble	stone	stone
travertine	stone	stone
concrete	mineral	concrete
resin	synthetic	resin
soy_wax	wax	wax
paraffin_wax	wax	wax

16. Homewares Dimensions, Capacity and Units (*normative*)

Measure	Canonical storage	Display guidance
Length / width / height / depth / diameter	Millimetres or another declared canonical SI unit	Localise to cm, mm or inches as appropriate
Weight	Grams or kilograms with typed unit	Distinguish net product weight from shipping weight
Capacity	Millilitres or litres	Never text such as "approximately two cups"
Area	Square metres or declared unit	Rugs, wallpaper, flooring-like goods
Volume	Cubic measure or capacity depending on use	Distinguish container capacity from external volume
Pack count	Integer	Separate pieces from place settings
Burn time	Duration range with declared method	Avoid unsupported precision
Cable length	Length measure	Electrical accessories and lighting

CDS1500-R039 Canonical unit storage MUST be deterministic and conversion-safe.

CDS1500-R040 Display conversions MUST NOT overwrite canonical values.

CDS1500-R041 A range used in filtering or comparison MUST store lower and upper bounds rather than a single text string.

17. Homewares Room, Style, Finish and Shape (*normative; seed tables informative*)

Room, style, finish and shape describe different concepts and require separate dictionaries.

Dictionary	Question answered	Examples
Room	Where is it commonly used?	Living Room, Bedroom, Bathroom
Style	What aesthetic does it support?	Coastal, Minimal, Industrial
Finish	What surface appearance/treatment does it have?	Matte, Brushed, Glazed
Shape	What is its overall geometry?	Round, Square, Organic
Use environment	Where can it safely function?	Indoor, Covered Outdoor, Outdoor

CDS1500-R042 Room MAY be multi-value, but every-room assignment SHOULD be prevented by a declared maximum room count per product.

CDS1500-R043 Style SHOULD be curated from a limited organisational vocabulary and MUST NOT be inferred solely from colour.

Seed room dictionary (*informative*): living_room, bedroom, dining_room, kitchen, bathroom, home_office, entryway, nursery, kids_room, laundry, outdoor, multi_room.

Seed finish dictionary (*informative*): matte, gloss, satin, polished, brushed, hammered, distressed, weathered, painted, powder_coated, natural_oil, lacquered, glazed, unglazed, antique (alias: aged), textured, marbled (*re-homed from patterns per DICT-17*).

Seed shape dictionary (*informative*): round, square, rectangular, oval, cylindrical, tapered, organic (alias: irregular), arched, cone, spherical.

Cross-dictionary id reuse (*DICT-6*): codes such as *tapered* (fit and shape), *casual / formal* (style and occasion) and *outdoor* (room, occasion and use environment) are distinct values in distinct dictionaries. Every dictionary declares a `dictionary_key`; mappings and alias records are always scoped by it (Appendix E).

18. Shared Colour and Visual Appearance Profile (*normative*)

Apparel and homewares share the same top-level colour facet dictionary so customers encounter consistent filter language across the store. Industry profiles MAY use different canonical shade vocabularies beneath those shared facets. (*Preserved — ADR-1500-001.*)

CDS1500-Ro44 A store SHOULD expose one consistent colour-family filter vocabulary across apparel and homewares unless a documented usability finding supports divergence.

CDS1500-Ro45 Transparent, metallic, natural and multi-colour appearances MUST be handled explicitly (as facet values or declared extensions, Appendix C) rather than forced into arbitrary hue families.

CDS1500-Ro46 Wood species, stone type and metal finish MUST remain material or finish values; a canonical material MAY additionally map to a visual colour family where the product appearance warrants it (declared in the dictionary, per-product only via the Ro21 override).

19. Customer Facet Design and Cardinality (*informative — normative facet rules live in CDS-600*)

Facet dictionaries are UX artefacts. Their success is measured by findability, comprehensibility and result quality rather than by maximum data exposure.

Facet	Recommended baseline behaviour	Anti-pattern
Colour	Broad colour families with swatches and text labels	Every supplier shade as a separate checkbox
Material	Customer-recognisable materials or families for the collection	Fibre, fabric, finish and claims mixed in one list
Size	Collection-relevant size systems and sort order	Alphabetical sort of mixed AU/US/EU/alpha sizes
Fit	Controlled product-family fit terms	Ad-hoc phrases from marketing copy
Room	Short curated room list	Every decor product in every room
Style	Governed organisational vocabulary	Hundreds of near-synonymous style tags
Dimensions	Ranges or typed values	Dozens of text-formatted dimensions
Price	Channel-native numeric ranges	Price bands as permanent product attributes

Progressive disclosure for unscannable lists, facet monitoring (cardinality, zero-result rates, selection behaviour under CDS-1400) and **zero-result value handling follow CDS-600's normative rules** (CDS-600 §17–18: zero-result values SHOULD normally be disabled or hidden). The vo.1 MUST NOT here is withdrawn in favour of that single home (*resolves CDS-600-3*).

20. Product Page Attribute Presentation (*informative*)

Product pages present customer-relevant attributes in logical groups rather than internal field order:

Apparel group	Examples
Design and fit	Fit, length, neckline, sleeve, rise, leg shape
Material and construction	Composition, fabric, lining, stretch
Care and origin	Care instructions, country of origin, certifications
Size help	Size chart, garment measurements, model reference

Homewares group	Examples
Dimensions and capacity	Height, width, depth, diameter, capacity, weight
Materials and finish	Primary material, components, finish, construction
Use and care	Room, indoor/outdoor, cleaning, dishwasher or microwave suitability
Contents and compatibility	Pack count, included items, electrical compatibility

Internal workflow, supplier and verification fields are never customer-facing unless intentionally transformed into customer content (rule: CDS-600 §5, CDS-300 QA_ semantics).

21. Channel Projection Guidance (*informative — normative rules: CDS-500 publication/verification, CDS-900 platform mappings*)

Canonical concept	Metafield-style channel (informative)	Feed-style channel (informative)
Product category	Mapped standard category plus internal category	CH_google_product_category plus product_type
Canonical colour shade	Category/custom metafield and display data	color using an accepted customer-readable value
Colour facet	Search/discovery filter metafield	May inform color but never replaces the richer canonical shade

Canonical concept	Metafield-style channel (informative)	Feed-style channel (informative)
Composition	Structured metafield or content block	<code>material</code> , where supported
Size system and value	Variant option plus supporting metafields	<code>size</code> and <code>size_system</code> where required
Room/style/finish	Custom metafields and storefront filters	Channel descriptive fields, or omitted when unsupported
Dimensions	Typed metafields and display content	Channel fields where supported; otherwise descriptive content

Channel projections use the expected representation defined by the mapping layer and are verified under CDS-500; unsupported channel attributes remain canonical in the PIM rather than being deleted or coerced into unrelated fields (CDS-500). Which layer (canonical, display or facet) feeds each channel is declared per attribute in the Attribute Definition (ADR-D4).

22. AI Enrichment and Review (*informative — normative AI rules live in CDS-700*)

Profile-specific application of CDS-700's proposal/evidence/review model:

Task	AI may propose	Deterministic or human control
Colour mapping	Canonical shade and facet family from image and source text	Dictionary validation; review of ambiguous or multi-colour cases; split-shade intake (Ro20) resolved by governed alias, not model preference
Material extraction	Composition and component candidates	Percent totals, canonical dictionary and evidence checks
Category classification	Family and category candidates	Taxonomy rules and human review at low confidence
Style and occasion	Limited curated suggestions	Declared maxima and merchandising-owner approval
Dimensions	Parsed typed measures from supplier documents	Unit validation and plausibility checks
Claims	Potential claim language	No acceptance without evidence and policy review

AI selects from governed dictionaries or returns an unmapped proposal for review; it never manufactures composition percentages, measurements, certifications or care instructions; accepted values retain provenance to evidence and review decision — all per CDS-700 (no restatement here).

23. Governance and Organisational Extensions (*normative*)

CDS1500-Ro47 An organisation MAY add categories, attributes, values and aliases but MUST assign stable identifiers and owners (CDS-800).

CDS1500-Ro48 A new colour shade SHOULD map to an existing colour facet; a new top-level family requires the evidence in Ro49.

CDS1500-Ro49 A proposed new facet value MUST include evidence that it is distinct, understandable and sufficiently populated against declared thresholds in the organisation's facet governance record.

CDS1500-Ro50 Deprecated values MUST retain aliases and replacement mappings for migration and historical interpretation.

CDS1500-Ro51 Industry-profile changes SHOULD be impact-assessed against storefront UX, supplier imports, channel mappings, saved filters, analytics and verification rules before adoption.

24. Conformance Requirements (*normative — claims and levels per CDS-1000*)

CDS1500-Ro52 An implementation claiming the CDS Apparel Profile MUST: maintain product and variant scope correctly for size and sellable colour (Ro07, Ro12–Ro13); separate display shade, canonical shade and colour facet (Ro14); represent fibre composition separately from fabric type and claims (Ro23–Ro25); identify size system and deterministic size ordering (Ro26–Ro27); apply category-specific attribute requirements (Ro09); use governed dictionaries conforming to CDS-400 for material, fit, pattern and other filterable attributes (Ro11); and publish and verify channel representations under CDS-500.

CDS1500-Ro53 An implementation claiming the CDS Homewares Profile MUST: store measurable specifications as typed values with units (Ro34, Ro39–Ro41); separate material, component material and finish (Ro35, Ro37–Ro38); apply category-specific requirements for dimensions, capacity, set contents and compatibility (Ro09); use governed room, style, shape and finish dictionaries where those facets are exposed (Ro11); project broad customer facets without losing canonical specificity (Ro37, Ro46); and publish and verify channel representations under CDS-500.

CDS1500-Ro81 An implementation claiming any industry profile (this chapter or CDS-1600 through CDS-2000) MUST additionally: name profile identifiers from the register (Ro54) and claim the parent of every extension profile claimed (Ro55); maintain a jurisdiction requirement register for every market it publishes to (Ro56); evaluate the register at preflight per market with the declared publication consequences and retain the results (Ro57); keep verification status visible and never present unverified entries as verified (Ro58); map unmapped obligations to declared extensions (Ro59); and project and verify listing elements, warnings and rating labels for each market (Ro61).

CDS1500-Ro82 An implementation claiming the CDS Footwear Profile MUST satisfy the Apparel Profile (Ro52) and Ro33, Ro62–Ro64 (size system and width as variant options with sort order, component materials, governed footwear dictionaries).

CDS1500-Ro83 An implementation claiming the CDS Jewellery and Accessories Profile MUST satisfy the Apparel Profile (Ro52) and Ro65–Ro68 (metal, fineness, plating and stone as separate governed attributes; stone status; size systems; market marking and content obligations evaluated through the register).

CDS1500-Ro84 An implementation claiming the CDS Furniture Profile MUST satisfy the Homewares Profile (Ro53) and Ro69–Ro75 (assembled and per-package dimensions, governed assembly and delivery classes, bedding size systems, governed mattress firmness, component materials with flammability and chemical statements as records, toppling warnings projected and verified, configurations as relationship records).

CDS1500-R085 An implementation claiming the CDS Garden and Outdoor Living Profile MUST satisfy the Homewares Profile (R053) and R076–R080 (explicit outdoor suitability, governed plant naming and sun exposure with region-specific schemes, dangerous-goods and registration records for chemicals and gas products, biosecurity restrictions at preflight, CDS-1800 compliance for gas and electrical appliances).

25. Worked Product Examples *(informative)*

25.1 Apparel — Relaxed Linen Shirt

```
CAT_family = apparel
CAT_product_type = shirt
CAT_internal_category = women > tops > shirts
MF_material_primary = linen
MF_fibre_composition = [{linen: 55}, {cotton: 45}]
MF_fabric_type = woven
MF_fit = relaxed
MF_pattern = plain

Variant (STD_sku = SHIRT-FN-10):
  VAR_colour = colour_french_navy      # canonical value_id at variant scope (ADR-D4)
  VAR_size_system = AU_WOMENS_NUMERIC
  VAR_size_label = 10

Product-scope colour projections (derived from variant canonical values, R013):
  MF_colour_display = French Navy      # display label projection
  MF_colour_facet = blue               # facet projection via governed facet dictionary

CH_google_color = Blue                # channel projection (feeding layer per Attribute Definition)
QA_shopify_colour = MATCH              # verification status per CDS-500 §17
```

25.2 Homewares — Stoneware Table Lamp

```
CAT_family = homewares
CAT_product_type = table_lamp
CAT_internal_category = home > lighting > table_lamps
MF_material_primary = stoneware
MF_finish = glazed
MF_colour_canonical = colour_oatmeal   # product-scope canonical (colour not variant-defining)
MF_colour_display = Oatmeal
MF_colour_facet = beige
MF_room = [living_room, bedroom]
MF_height_mm = 520
MF_width_mm = 280
MF_bulb_base = E27
MF_max_wattage_w = 40
VAR_plug_type = AU                    # VAR_plug_profile retired (CDS-1500-8)
CH_shopify_filter_material = Ceramic
QA_shopify_height = MATCH
```

25.3 Homewares — Printed Cushion

```
CAT_product_type = decorative_cushion
MF_cover_composition = [{cotton: 100}]
MF_fill_material = recycled_polyester
MF_pattern = floral
MF_colour_display = Garden Multi
MF_colour_components = [colour_sage_green, colour_blush_pink, colour_ecru]
MF_colour_dominant = colour_sage_green # canonical layer records component shades (ADR-D4)
MF_colour_facet = multicolour          # optional dominant-shade declaration
MF_shape = square                     # multicolour is a facet/status value, never canonical
MF_width_mm = 500
MF_height_mm = 500
MF_removable_cover = true
VAR_insert_option = cover_only | feather_insert
```

(v0.1 stored MF_colour_canonical = multicolour, destroying shade detail — corrected per CDS-1500-4 and ADR-D4.)

26. Architecture Decision Records *(informative summary — decisions live in the global ADR register per CDS000-R006)*

ADR	Decision	Status
CDS-ADR-1500-001	Shared top-level colour facets across apparel and homewares	Accepted

ADR	Decision	Status
CDS-ADR-1500-002	Canonical shade separate from facet colour	Accepted (now an application of ADR-D4)
CDS-ADR-1500-003	Fibre composition separate from fabric type	Accepted
CDS-ADR-1500-004	Measurements stored in canonical typed units	Accepted
CDS-ADR-1500-005	Style and occasion as governed multi-value attributes with declared maxima	Accepted
CDS-ADR-1500-006	Starter dictionaries published as extensible informative baselines; requirements bind to governed dictionaries (R011)	Accepted (amended v0.2)
CDS-ADR-1500-007	No universal exact size conversion claim	Accepted
CDS-ADR-1500-008	Facet-ambiguous shade names split into distinct canonical shades (charcoal, rust, tan), intake via scoped alias mappings; product-level facet override only as governed record with provenance	Accepted (v0.2; applies ADR-D4/D12; see resolved D12 note at §7)
CDS-ADR-1500-009	One corpus-wide industry profile register (§2.1); extension profiles require their parent; new verticals get their own chapters (CDS-1600 through CDS-2000) selected by global market research (REVIEW-020) rather than being appended to this chapter	Accepted (2026-09-20 revision; open flag IP1, IP8)
CDS-ADR-1500-010	Regulatory obligations are not encoded in requirement levels; each organisation maintains a jurisdiction requirement register (§2.2) evaluated at preflight per market, seeded per chapter with dated, verification-marked entries for AU, NZ, US, EU, GB and CA	Accepted (2026-09-20 revision; open flag IP9)
CDS-ADR-1500-011	The product relationship record, kits and bundles, and the dangerous-goods declaration are homed in CDS-1900 and cited by every other profile; promotion of the relationship record into CDS-200 is deferred to v0.3	Accepted (2026-09-20 revision; open flag IP3)

Appendix A. Apparel Attribute Baseline (*normative — dictionary bindings per CDS1500-R011*)

Field	Scope	Type	Baseline requirement
CAT_product_type	Product	Governed classification reference	R
VAR_colour	Variant	Canonical value_id (colour dictionary)	C — R when colour distinguishes sellable units
MF_colour_canonical	Product	Canonical value_id (colour dictionary)	C — R when colour is not variant-defining
MF_colour_display	Product/Variant	Free text linked to a canonical value (CDS-400 §9)	REC
MF_colour_facet	Product/Variant	Facet value, derived via the governed facet dictionary	R (derived, not independently authored)
VAR_size_label	Variant	Text	C
VAR_size_system	Variant/Product	Governed dictionary reference	C
MF_fibre_composition	Product	Structured percentage list of canonical material ids	R for garments
MF_material_primary	Product	Governed dictionary reference	R
MF_fabric_type	Product	Governed dictionary reference	REC
MF_fit	Product	Governed dictionary reference	REC
MF_pattern	Product	Governed dictionary reference	REC
MF_care_instructions	Product	Structured/text	REC
MF_country_of_origin	Product	Governed country reference	C
MF_size_chart_id	Product	Reference	REC

The v0.1 type "Text from dictionary" for MF_colour_display is corrected: display labels are free text linked to a canonical value, per CDS-400 §9 (resolves CDS-1500-7).

A.1 Footwear Extension Baseline (*normative; adds to Appendix A*)

Field	Scope	Type	Baseline requirement
VAR_size_label / VAR_size_system	Variant	Text; governed dictionary reference (size_system, footwear systems)	R
VAR_width_fitting	Variant	Governed dictionary reference (footwear_width_fitting)	C — R where width is offered
MF_internal_length_mm	Variant	Typed measurement	REC
MF_material_components (upper, lining, sole, insole)	Product	Structured component list with canonical material ids	R
MF_footwear_style	Product	Governed dictionary reference	REC
MF_heel_height_mm / MF_heel_height_band	Product	Typed measurement; derived governed band	C — R for heeled footwear
MF_footwear_closure / MF_toe_shape	Product	Governed dictionary references	REC
MF_care_instructions / MF_country_of_origin	Product	Per Appendix A	Per Appendix A

A.2 Jewellery and Accessories Extension Baseline (*normative; adds to Appendix A*)

Field	Scope	Type	Baseline requirement
MF_jewellery_metal	Product	Governed dictionary reference	R for jewellery and watches

Field	Scope	Type	Baseline requirement
MF_metal_fineness	Product	Typed value with scale (karat or parts per thousand)	C — R where fineness is claimed or a market requires marking
MF_plating	Product	Governed dictionary reference (<code>jewellery_metal</code>) or none	R (explicit none)
MF_jewellery_stone / MF_stone_status	Product	Governed dictionary reference; governed status (natural, laboratory-grown, treated, simulated)	C — R where a stone is present
MF_stone_carat_weight / MF_stone_grading	Product	Typed value; claim record with evidence	C
MF_jewellery_closure / MF_jewellery_style	Product	Governed dictionary references	REC
VAR_size_label / VAR_size_system / measured basis	Variant	Text; governed reference (<code>au_ring</code> , <code>us_ring</code> , <code>eu_ring</code> , <code>head_circumference_cm</code> , chain length); typed measurement	C — R for sized pieces
MF_dimensions / MF_weight_g	Product / Variant	Typed measurements	REC
MF_warning_statements / CMP_compliance_declarations	Product (per market)	Structured records (nickel release, lead and cadmium content, hallmarking)	C — per §2.2

Appendix B. Homewares Attribute Baseline (*normative — dictionary bindings per CDS1500-R011*)

Field	Scope	Type	Baseline requirement
CAT_product_type	Product	Governed classification reference	R
MF_material_primary	Product	Governed dictionary reference	R
MF_material_components	Product	Structured component list	C
MF_finish	Product	Governed dictionary reference	REC
MF_colour_canonical	Product/Variant	Canonical value_id (colour dictionary)	R
MF_colour_display	Product/Variant	Free text linked to a canonical value (CDS-400 §9)	REC
MF_colour_facet	Product/Variant	Facet value, derived via the governed facet dictionary	R (derived)
MF_height_mm / MF_width_mm / MF_depth_mm	Product/Variant	Number (typed unit)	C
MF_capacity_ml	Product/Variant	Number (typed unit)	C
MF_pack_count	Product/Variant	Integer	C
MF_room	Product	Governed dictionary list	REC
MF_shape	Product	Governed dictionary reference	REC
MF_care_instructions	Product	Structured/text	REC

B.1 Furniture Extension Baseline (*normative; adds to Appendix B*)

Field	Scope	Type	Baseline requirement
MF_assembly_type	Product	Governed dictionary reference (<code>assembly_type</code>)	R
MF_dimensions_assembled	Product / Variant	Typed dimensions	R
MF_dimensions_packaged / MF_packaged_weight_kg / MF_package_count	Product / Variant	Typed dimensions and weight per package; integer	R where the organisation ships
MF_weight_kg / MF_load_rating_kg	Product / Variant	Typed measurements	R / C
MF_seat_height_mm / MF_seat_depth_mm / MF_arm_height_mm	Product	Typed measurements	C — R for seating
MF_bed_size_system / VAR_bed_size / mattress dimensions	Variant	Governed reference; text; typed measurements	R for beds and mattresses
MF_mattress_firmness / MF_mattress_depth_mm	Product / Variant	Governed dictionary reference (<code>mattress_firmness</code>); typed measurement	R for mattresses where the facet is exposed
MF_material_components (upholstery, frame, fill)	Product	Structured component list	R
MF_delivery_class	Product / Variant	Governed logistics class (derived)	R where the organisation ships
MF_warning_statements / CMP_compliance_declarations	Product (per market)	Structured records (toppling, flammability, chemical content)	C — per §2.2
Relationship records (configurations, matching pieces, parts)	Product	Records per CDS-1900 §5	C

B.2 Garden and Outdoor Living Extension Baseline (*normative; adds to Appendix B*)

Field	Scope	Type	Baseline requirement
MF_outdoor_suitability	Product	Governed value (indoor, covered outdoor, full outdoor) with basis	R

Field	Scope	Type	Baseline requirement
MF_botanical_name / MF_common_name	Product	Governed organisation dictionary values	R for plants, seeds and bulbs
MF_sun_exposure	Product	Governed dictionary reference (sun_exposure)	REC; R for plants where the facet is exposed
MF_hardiness_or_climate_zone	Product	Typed value with scheme	C
CMP_dangerous_goods / MED_technical_documents (SDS)	Product / Variant	Per CDS-1900 §12	R (explicit value)
CMP_market_registrations / CMP_restricted_sale	Product (per market)	Governed records	R for regulated chemicals, seeds and live plants
Gas and electrical compliance records	Product (per market)	Per CDS-1800 §12	R for appliances
Homewares baseline	—	Per Appendix B	R

Appendix C. Colour Facet Baseline *(informative seed — owned by CDS-1500; CDS-400 and CDS-600 cite this table (D11); becomes governed data on adoption per CDS1500-R011)*

C.1 Baseline families (18)

ID	Label	Definition	Typical canonical shades (deterministic — no per-product judgement)
black	Black	Near-black and black visual colours	black, jet_black, charcoal_black
white	White	White visual colours	pure_white, optic_white
grey	Grey	Neutral grey family	charcoal_grey, slate_grey, dove_grey
silver	Silver	Silver metallic appearance	silver, chrome
brown	Brown	Brown family	chocolate, espresso, tan_brown, rust_brown
beige	Beige	Warm pale brown and neutral family	sand, camel, oatmeal, tan_beige
cream	Cream	Warm off-white family	ivory, ecru, vanilla
yellow	Yellow	Yellow family	lemon, mustard, ochre
orange	Orange	Orange family	rust_orange, terracotta, apricot
red	Red	Red family	scarlet, burgundy, wine
pink	Pink	Pink family	blush_pink, rose, fuchsia
purple	Purple	Purple family	lilac, plum, lavender
blue	Blue	Blue family	navy, french_navy, cobalt, sky_blue, denim_blue
green	Green	Green family	sage_green, olive_green, emerald, teal (primary facet)
gold	Gold	Gold metallic appearance	gold, brass_visual
clear	Clear	Transparent or substantially colourless	clear_glass, transparent_acrylic
multicolour	Multi-colour	No single dominant colour or intentionally mixed palette	<i>(facet/status value only — never a canonical shade; ADR-D4)</i>
natural	Natural	Uncoloured natural appearance where no clearer family applies	raw_rattan, undyed_fibre

The vo.1 appearance-conditional notes ("charcoal may map to Grey unless visually near-black"; "tan may instead map Beige"; "rust may use Orange or Brown based on appearance") are **deleted**; those shades are split into distinct canonicals per CDS1500-R020. Multi-facet shades (teal green|blue, tan_beige-style splits, rose_gold gold|pink) declare an **ordered facet list, primary first** (D28 as resolved in CDS-400).

C.2 Boundary guidance *(informative)*

- **Cream / Beige / Natural / White** — White is visually white; Cream is warm off-white (ivory, ecru); Beige is warm pale brown (sand, oatmeal); Natural is reserved for *uncoloured* material appearance (raw rattan, undyed fibre) and never used where a hue family clearly applies. Organisations running the 12-family core (C.3) fold Cream into White or Beige and Natural into Beige, per shade.
- **Silver / Grey** — Silver requires metallic appearance; a non-metallic pale neutral is Grey. A brushed-metal product may carry material facet Metal with visual family Silver (R046).
- **Gold / Brown** — Gold requires metallic appearance; warm non-metallic browns (caramel, bronze-toned textiles) are Brown or Beige. Brass-*material* products map to Gold only when the visible appearance is gold-metallic.

C.3 Recommended core (~12 families) with category-conditional extensions *(informative)*

For most storefronts a 12-family core filter outperforms the full 18: **black, white, grey, beige, brown, red, orange, yellow, green, blue, purple, pink**, plus **multicolour** as a status value. The remaining families are **category-conditional extensions**, enabled where the assortment warrants them: **silver, gold** (jewellery, hardware, decor with metallic finishes), **clear** (glassware, acrylic), **cream, natural** (textiles and natural-fibre homewares where the distinction demonstrably aids filtering). Extensions follow the same governance as any facet value (R049).

Appendix D. Material Facet Seed *(informative seed — new in vo.2, resolves DICT-4; becomes governed data on adoption per CDS1500-R011)*

dictionary_key: material_facet — the governed value set for MF_material_facet and apparel material facets (previously ungoverned free-text labels in the CSVs).

Facet ID	Label	Scope
cotton	Cotton	shared
linen	Linen	shared
wool	Wool	shared
hemp	Hemp	apparel
cashmere	Cashmere	apparel
silk	Silk	apparel
viscose	Viscose	apparel
lyocell	Lyocell	apparel
modal	Modal	apparel
polyester	Polyester	apparel
nylon	Nylon	apparel
synthetic	Synthetic	apparel
leather	Leather	apparel
faux_leather	Faux Leather	apparel
synthetic_textile	Synthetic Textile	homewares
ceramic	Ceramic	homewares
porcelain	Porcelain	homewares
glass	Glass	homewares
metal	Metal	homewares
wood	Wood	homewares
natural_fibre	Natural Fibre	homewares
stone	Stone	homewares
concrete	Concrete	homewares
resin	Resin	homewares
wax	Wax	homewares

Excluded by design: **Recycled** (claim-driven from MF_recycled_content_percent, R025), **Stretch** (property-driven from MF_stretch, R025), **Other Natural** (forbidden catch-all label; hemp promoted to its own facet).

Appendix E. Reference Dictionary Package (*informative — column contract and audit dispositions*)

The companion CDS-1500 Starter Reference Dictionaries package ships CSVs for colour facets, colour reference values, apparel materials, homewares materials, material facets (new), patterns, fits, styles, occasions, rooms, finishes and shapes, with a manifest and SHA-256 checksums (the SHA256SUMS + manifest pattern is retained — it verified end-to-end in the shipped-CSV audit). The package is versioned per ADR-D5 with folder and manifest versions agreeing (DICT-16). *(Distribution note, 2026-08-05: the companion data now ships as part of the unified CDS Reference Dictionary package (version per its manifest) which merges these profile-bound dictionaries with the E.3 gap-fill vocabularies and the ahead-of-profile industry chapters; content and contract unchanged.)*

The package is **informative starter data** in v0.2. Requirement levels bind to governed dictionaries per CDS1500-R011; an organisation adopting these files governs all subsequent changes under CDS-400/CDS-800 before treating them as production master data. The manifest's no-silent-redefinition note is retained and strengthened: a canonical code, once published, is never silently redefined — changes go through deprecation with replacement mappings (R050).

E.1 Required columns (per DICT-1, DICT-6, DICT-11, DICT-15 and ADR-D4 §5)

Every dictionary file carries, for every row:

Column	Content
dictionary_key	The owning dictionary (e.g. colour, material_apparel, material_homewares, material_facet, fit, pattern, style, occasion, room, finish, shape). Scopes all ids, mappings and aliases; permits cross-dictionary id reuse (DICT-5/6)
row_type	canonical or alias — source and canonical rows are never conflated in one row shape (DICT-11)
value_id / code	Stable identifier (snake_case per CDS-400)
label	Single-name display label — never slash-pair composites (DICT-7): "Viscose / Rayon" becomes label Viscose with alias rayon; likewise Nylon (alias polyamide), Elastane (alias spandex), Plaid (alias tartan), Graphic (alias placement print), Organic (alias irregular), Antique (alias aged)
aliases	Alias column present in all dictionaries, not only colour (9 of 11 v0.1 files had none — DICT-7); alias scope (supplier/locale) optional per ADR-D4
facet_ids	Where applicable: ordered list, primary first; order significance declared in the package documentation (DICT-9/D28)
status	Lifecycle per CDS-400 (proposed/active/deprecated/retired/rejected)
provenance	Origin of the row (seed, import, organisational addition)

Column	Content
version	Row-level version per CDS-400 change rules
locale	Label locale; the seed data declares en-GB (DICT-15)

Package documentation states column semantics, list delimiters (; for aliases, | for ordered facet lists) and case-matching rules (DICT-10). Channel mappings are declared **out of scope** for the starter package in the manifest (DICT-8); organisations add per-value channel mappings under CDS-400 §11 when needed.

E.2 Seed content dispositions from the shipped-CSV audit

- **sky_blue** is confirmed as the canonical code, with `light_blue` recorded as an alias; the CDS-400 draft's `sky → light_blue` example is corrected to cite this baseline (DICT-3 — resolved in the CSV's favour under the no-silent-redefinition rule).
- **teal** declares facet order `green|blue`, ordered primary-first (DICT-9; D28 as resolved in CDS-400 — cited, not restated).
- **charcoal, rust, tan** are split per CDS1500-R020; the old single rows become default aliases to the grey/orange/beige-side canonicals.
- **natural** remains a top-level colour facet here; CDS-400's Appendix B "natural → Beige" exemplar is corrected to cite this appendix (DICT-2, D11).
- **colour_reference_values.csv is retained as the four-layer demonstration** — source ≠ display ≠ canonical ≠ facet, including the hard multi-facet cases (teal, rose_gold, and the now-split tan/rust) — restructured with `row_type` so canonical rows (navy) and alias rows (`french_navy → navy` in the coarse configuration) are distinguishable (DICT-11). (*Preserve item.*)
- **patterns.csv**: `textured` and `marbled` moved to `finishes` (texture, not pattern); `plain` documented as also meaning "no pattern" (DICT-17).
- **material facets** now reference `material_facets.csv` (Appendix D) instead of free-text labels (DICT-4).

E.3 Known-missing referenced dictionaries (DICT-12)

Referenced by this chapter but not shipped as seed files; organisations supply governed content or adopt future package releases: **country of origin** (MF_country_of_origin), **component roles** (MF_material_components), **weather/season relevance, use environment, plant naming** (§14B), **mattress construction** (§14A), **licensed properties and skill levels** (CDS-2000). Since package 0.4.0 the **fabric types, size systems and care instructions** dictionaries listed here in the 2026-08-04 draft are shipped (Chapters 3, 5 and 4); package 0.8.0 adds bedding, ring, bike-frame and head-circumference size systems, mattress firmness (Chapter 15) and the Chapter 17–19 vocabularies bound by CDS-1700 through CDS-2000. Listing here is disclosure, not endorsement of free text: until governed, these fields cannot satisfy R-level requirements (R011).

Appendix F. Source Acknowledgements (*informative*)

The profile applies the product classification, attribute, value, unit and distribution concepts described in Jorij Abraham, *Product Information Management: Theory and Practice* (Springer, 2014), together with the project-documented PIM-first pipeline, tag governance and store-sequencing practices. CDS extends those foundations with explicit facet dictionaries, channel read-back verification, AI governance and human-readable semantic namespaces. Verification statuses, traffic lights and reason codes are defined solely in CDS-500 (ADR-D3); this chapter states none of its own.

Appendix G. Jurisdiction Requirement Register — Seed for Apparel, Footwear, Jewellery, Furniture and Garden Goods (*informative seed of the normative register defined in §2.2*)

Conventions: obligation types are those of §2.2. "Verified" means the primary instrument or regulator page was read on 2026-09-20; "retrieved" means a regulator page was retrieved but the instrument text was not read; "unverified" means the entry is seeded from general knowledge for completeness and MUST be verified before adoption (R060); "not seeded" means the organisation supplies the entry. Jurisdiction codes are those of the `jurisdiction` dictionary (package Chapter 19); scheme codes are those of `regulatory_scheme`. Cosmetics-like goods (fragrance, jewellery cleaners) follow CDS-1600 Appendix D; electrical and battery-powered goods (heated apparel, smart watches, powered furniture) follow CDS-1800 Appendix D; chemicals and gas products follow CDS-1900 Appendix D.

G.1 Australia (AU)

Ref	Instrument / regulator	Applies to	Obligation	Required data elements → CDS field	Trigger / dates	Verification
G-AU-1	Consumer Goods (Care Labelling for Clothing and Textile Products) Information Standard 2023 (AS/NZS 1957 with ISO 3758 symbols permitted) — ACCC (<code>au_care_labelling_2023</code>)	Clothing, household textiles, furnishings, piece goods and yarns, plastic-coated fabrics, suede skins, leathers and furs	<code>labelling_element</code> (permanent care label with care instructions in English, ISO symbols optionally alongside)	Care instructions as structured values → <code>MF_care_instructions</code> (Chapter 4 values with declared symbol set), fibre composition where used to derive care → <code>MF_fibre_composition</code>	Standard commenced 2023 replacing the 2010 standard; transition ended	Verified 2026-09-20 (ACCC mandatory standard page)
G-AU-2	Consumer Goods (Children's Nightwear and Limited Daywear and Paper Patterns for Children's Nightwear) Safety Standard 2017 (AS/NZS 1249) — ACCC (<code>au_childrens_nightwear_2017</code>)	Children's nightwear and limited daywear sizes 00–14	<code>pre_market</code> (fire-hazard category testing); <code>labelling_element</code> (category 1–4 label: "Low fire danger" or "Warning: High fire danger. Keep away from fire")	Fire-hazard category and label text → <code>MF_warning_statements</code> (scheme <code>au_childrens_nightwear_2017</code>), <code>CMP_compliance_declarations</code> (test evidence)	Ongoing	Verified 2026-09-20 (ACCC mandatory standard page)
G-AU-3	Consumer Goods (Toppling Furniture) Information Standard 2024 — ACCC (<code>au_toppling_furniture_2024</code>)	Clothing storage units, bookcases, hall tables, display cabinets, buffets, entertainment	<code>warning_statement</code> (safety warning on the product and packaging); <code>listing_element</code> (warning in the online product	Warning text, anchor supply status, online description warning → <code>MF_warning_statements</code> (scheme <code>au_toppling_furniture_2024</code> , includes online warning text),	Mandatory from 4 May 2025	Verified 2026-09-20 (ACCC toppling furniture guidance)

Ref	Instrument / regulator	Applies to	Obligation	Required data elements → CDS field	Trigger / dates	Verification
		units and similar free-standing furniture ≥ 686 mm high (as defined)	description); labelling_element (anchoring instructions and information at point of sale)	MF_anchor_kit_included, listing projection and verification (R074)		
G-AU-4	Australian Consumer Law s 29 and s 33 (country of origin and safe-harbour rules), Competition and Consumer Act; no mandatory fibre-content labelling standard for clothing (AS/NZS 2622 voluntary) — ACCC	Country of origin claims on apparel and homewares	restricted_content (origin claims must meet the safe-harbour tests)	Claim records with evidence → MF_country_of_origin, claims per R031	Ongoing	Retrieved (not spot-verified)
G-AU-5	Trade Practices (Consumer Product Information Standards) — jewellery: no mandatory hallmarking; lead and cadmium limits for children's jewellery under the Consumer Goods (Lead and Other Elements in Children's Toys) and related standards; Consumer Goods (Products Containing Button/Coin Batteries) standards for jewellery and accessories with batteries	Jewellery and accessories	restricted_content; labelling_element	Content limits and battery warnings → CMP_compliance_declarations, MF_warning_statements (per CDS-1800 D-AU-5)	Ongoing	Unverified — organisation to confirm the applicable children's jewellery instrument
G-AU-6	Agricultural and Veterinary Chemicals Code (APVMA registration of garden pesticides and fertiliser-pesticide products); state biosecurity acts (plant movement and prohibited species, for example interstate quarantine); Consumer Goods (Portable Ladders / Trampolines) standards — APVMA, state departments, ACCC	Garden chemicals, seeds and live plants, outdoor equipment	pre_market (APVMA registration and approved label); restricted_content (biosecurity); labelling_element	APVMA number and label → CMP_market_registrations (scheme au_apvma_agvet_code); movement restrictions → CMP_restricted_sale per state	Ongoing	Retrieved (APVMA); biosecurity entries unverified

G.2 New Zealand (NZ)

Ref	Instrument / regulator	Applies to	Obligation	Required data elements → CDS field	Trigger / dates	Verification
G-NZ-1	Consumer Information Standards (Fibre Content Labelling) Regulations 2000; (Care Labelling) Regulations 2000; (Country of Origin (Clothing and Footwear) Labelling) Regulations 1992 — Commerce Commission (nz_consumer_information_standards_textiles)	Clothing, footwear, textile goods	labelling_element (fibre content, care instructions, country of origin for clothing and footwear)	→ MF_fibre_composition, MF_care_instructions, MF_country_of_origin	Ongoing	Unverified — seeded from general knowledge; confirm current consolidation
G-NZ-2	Product Safety Standards (Children's Nightwear and Limited Daywear Having Reduced Fire Hazard) Regulations 2016 (AS/NZS 1249) — MBIE (nz_childrens_nightwear_2016)	Children's nightwear	pre_market; labelling_element (fire-hazard label)	→ MF_warning_statements (scheme nz_childrens_nightwear_2016)	Ongoing	Unverified — organisation to confirm

G.3 United States (US, with California noted)

Ref	Instrument / regulator	Applies to	Obligation	Required data elements → CDS field	Trigger / dates	Verification
G-US-1	Textile Fiber Products Identification Act and 16 CFR Part 303; Wool Products Labeling Act and 16 CFR Part 300 — FTC (us_ftc_textile_wool_labelling)	Textile and wool apparel and household textiles	labelling_element (generic fibre names and percentages in order of predominance; country of origin; manufacturer or dealer identity by name or Registered Identification Number (RN)); restricted_content (fibre and origin claims in advertising, including online, must be consistent with the label)	→ MF_fibre_composition (generic fibre names per 16 CFR 303.7), MF_country_of_origin, CMP_responsible_person (RN or name), claims per R031	Ongoing	Verified 2026-09-20 (eCFR 16 CFR 303)
G-US-2	Care Labeling Rule, 16 CFR Part 423 — FTC (us_ftc_care_labeling_423)	Textile wearing apparel and certain piece goods	labelling_element (permanent care label with washing or dry-cleaning instructions and warnings; ASTM symbols permitted)	→ MF_care_instructions with declared symbol set	Ongoing	Verified 2026-09-20 (eCFR 16 CFR 423)
G-US-3	Flammable Fabrics Act; 16 CFR 1610 (general wearing apparel), 16 CFR 1615 and 1616 (children's sleepwear sizes 0–6X and 7–14), 16 CFR 1632 and 1633 (mattresses) — CPSC (us_cpsc_flammable_fabrics)	Apparel, children's sleepwear, mattresses and mattress pads	pre_market (testing and certification); labelling_element (sleepwear: flame-resistant or snug-fitting labelling; mattresses: permanent label and manufacturer identification)	→ CMP_compliance_declarations (scheme us_cpsc_flammable_fabrics, test evidence), MF_warning_statements (sleepwear "wear snug-fitting" hang tag where applicable)	Ongoing	Retrieved (not spot-verified)

Ref	Instrument / regulator	Applies to	Obligation	Required data elements → CDS field	Trigger / dates	Verification
G-US-4	STURDY Act; 16 CFR Part 1261 Safety Standard for Clothing Storage Units — CPSC (<code>us_cpsc_sturdy_16_cfr_1261</code>)	Clothing storage units (dressers, chests, armoires) ≥ 27 inches high manufactured after 1 September 2023	<code>pre_market</code> (stability testing; certificate); <code>labelling_element</code> (permanent warning label and product identification; tip-over restraint supplied)	→ <code>CMP_compliance_declarations</code> (scheme <code>us_cpsc_sturdy_16_cfr_1261</code>), <code>MF_warning_statements</code> , <code>MF_anchor_kit_included</code>	Products manufactured after 1 September 2023	Verified 2026-09-20 (CPSC STURDY business guidance)
G-US-5	California Technical Bulletin 117-2013 (upholstered furniture flammability; label stating compliance and whether flame-retardant chemicals were added) and SB 1019 (Business and Professions Code 19094) — Bureau of Household Goods and Services (<code>us_ca_tb117_2013</code>)	Upholstered furniture sold in California (de facto national labelling)	<code>labelling_element</code> (TB 117-2013 compliance label with flame-retardant chemical statement)	→ <code>CMP_compliance_declarations</code> (scheme <code>us_ca_tb117_2013</code>), <code>MF_flame_retardant_added</code> (boolean)	SB 1019 labels from 1 January 2015	Retrieved (not spot-verified)
G-US-6	Proposition 65 (California) — OEHHA (<code>us_ca_prop65_warning</code>)	Apparel, jewellery (lead, cadmium), furniture (flame retardants, formaldehyde), garden chemicals	<code>warning_statement</code> ; <code>listing_element</code>	Per CDS-1800 D-US-3	Effective 1 January 2025	Verified 2026-09-20 (CDS-1800 [E7])
G-US-7	Consumer Product Safety Improvement Act (lead content and lead paint limits for children's products; children's jewellery); California Metal-Containing Jewelry Law (Health and Safety Code 25214.1–25214.4.2); 16 CFR 1500.19 for accessories with small parts — CPSC, DTSC	Children's jewellery and accessories; jewellery sold in California	<code>restricted_content</code> (content limits); <code>pre_market</code> (children's product certificate); documentation	→ <code>CMP_compliance_declarations</code> , <code>MF_metal_fineness</code> and plating facts, CPC reference	Ongoing	Unverified — organisation to confirm current limits
G-US-8	Federal Seed Act (7 CFR 201), state seed laws, state noxious-weed lists; FIFRA (pesticide registration and labelling); state fertiliser registration laws — USDA, EPA, state departments	Seeds, pesticides, fertilisers	<code>pre_market</code> (registration per state and EPA registration number); <code>labelling_element</code> ; <code>restricted_content</code> (state prohibited species; restricted-use pesticides)	→ <code>CMP_market_registrations</code> (EPA Reg. No., state registrations), <code>CMP_restricted_sale</code> per state	Ongoing	Unverified — organisation to confirm

G.4 European Union (EU)

Ref	Instrument / regulator	Applies to	Obligation	Required data elements → CDS field	Trigger / dates	Verification
G-EU-1	Regulation (EU) No 1007/2011 on textile fibre names and related labelling and marking of the fibre composition of textile products (<code>eu_textile_labelling_1007_2011</code>)	Textile products (≥ 80% textile fibres by weight), including apparel, home textiles and upholstery covers	<code>labelling_element</code> (fibre composition using Annex I names in the official language of the member state; non-textile parts of animal origin stated); <code>listing_element</code> (Article 16: the fibre composition must be indicated in a clear, legible and uniform manner and, where products are offered for sale to the consumer, including by electronic means, the information must be visible before purchase)	→ <code>MF_fibre_composition</code> (Annex I fibre names, percentages), <code>MF_animal_material</code> ("Contains non-textile parts of animal origin"), localisation per market, listing projection and verification (R061)	Ongoing	Verified 2026-09-20 (EUR-Lex consolidated text)
G-EU-2	Regulation (EU) 2023/988 General Product Safety Regulation (<code>eu_gpsr_2023_988</code>)	Apparel, footwear, jewellery, furniture, garden goods (non-harmonised)	<code>listing_element</code> ; <code>labelling_element</code> ; <code>pre_market</code> (responsible economic operator in the EU)	Per CDS-1800 D-EU-6 → <code>CMP_responsible_person</code> , <code>CMP_product_identifier</code> , <code>MF_warning_statements</code> , listing projection	Applies from 13 December 2024	Verified 2026-09-20 (CDS-1800 [E13])
G-EU-3	REACH Regulation (EC) No 1907/2006 Annex XVII entries 27 (nickel release from articles in prolonged skin contact), 23 (cadmium in jewellery), 63 (lead in jewellery), 43 (azo dyes in textiles), 72 (CMR substances in textiles and footwear) — ECHA (<code>eu_reach_annex_xvii_articles</code>)	Jewellery, accessories, textiles, footwear	<code>restricted_content</code> (substance limits); documentation (test evidence)	→ <code>CMP_compliance_declarations</code> (scheme <code>eu_reach_annex_xvii_articles</code> , entries and evidence)	Ongoing; entry 72 applies since 1 November 2020	Retrieved (not spot-verified)
G-EU-4	Directive 94/11/EC on the labelling of materials used in the main components of footwear (<code>eu_footwear_labelling_94_11</code>)	Footwear	<code>labelling_element</code> (material of upper, lining and sock, outer sole, by pictogram or text, at least on one shoe of each pair)	→ <code>MF_material_components</code> (upper, lining, sole) with the directive's material categories (leather, coated leather, textile, other)	Ongoing	Unverified — organisation to confirm current status

Ref	Instrument / regulator	Applies to	Obligation	Required data elements → CDS field	Trigger / dates	Verification
G-EU-5	Regulation (EU) 2024/1781 ESPR (textiles and furniture are priority product groups; delegated acts pending; unsold-goods destruction disclosure and ban for apparel and footwear for large enterprises from 19 July 2026) (eu_espr_2024_1781)	Apparel, footwear, furniture (as delegated acts are adopted)	listing_element; documentation; post_market	Digital product passport identifiers and labels per delegated act → CMP_product_identifier, MED_technical_documents	Framework in force 18 July 2024; product-group acts not seeded	Verified 2026-09-20 (CDS-1800 [E15]) for the framework; textile act not seeded
G-EU-6	Regulation (EU) 2016/2031 plant health (plant passports for movement of plants for planting within the EU, including distance sales to final users); Regulation (EU) 2019/1009 (fertilising products; CE marking) and Regulation (EC) 1107/2009 (plant protection products; national authorisation)	Live plants, seeds, fertilisers, pesticides	pre_market (authorisation, plant passport); labelling_element; restricted_content (distance sales of plants require a plant passport)	→ CMP_market_registrations, CMP_restricted_sale, MED_technical_documents (plant passport reference)	Plant health regulation applies since 14 December 2019	Unverified — organisation to confirm

G.5 United Kingdom (GB)

Ref	Instrument / regulator	Applies to	Obligation	Required data elements → CDS field	Trigger / dates	Verification
G-UK-1	Textile Products (Labelling and Fibre Composition) Regulations 2012 (retaining Regulation 1007/2011 as assimilated law) — Trading Standards (uk_textile_labelling_2012)	Textile products	labelling_element; listing_element (fibre composition visible before purchase including online)	As G-EU-1 with English-language fibre names → MF_fibre_composition, listing_projection	Ongoing	Retrieved 2026-09-20 (legislation.gov.uk)
G-UK-2	Furniture and Furnishings (Fire) (Safety) Regulations 1988 (as amended) — OPSS and Trading Standards (uk_furniture_fire_safety_1988)	Upholstered furniture, mattresses, cushions and other filled articles supplied in the UK	pre_market (fillings and covers meet the ignition tests); labelling_element (permanent label with the statutory wording and supplier identity; display label at point of sale — including online product pages by guidance)	→ CMP_compliance_declarations (scheme uk_furniture_fire_safety_1988), MF_warning_statements (statutory label wording), CMP_responsible_person, listing_projection	Ongoing (reform consultations continuing)	Retrieved 2026-09-20 (legislation.gov.uk)
G-UK-3	Hallmarking Act 1973 — assay offices and Trading Standards (uk_hallmarking_act_1973)	Articles described as gold, silver, platinum or palladium above the exemption weights	labelling_element (UK hallmark before sale and description); restricted_content (description of unhallmarked articles)	→ MF_jewellery_metal, MF_metal_fineness, CMP_compliance_declarations (hallmark evidence), dealer's notice display	Ongoing	Unverified — organisation to confirm
G-UK-4	General Product Safety Regulations 2005; UK REACH Annex XVII (nickel, cadmium, lead, azo dyes) — OPSS, HSE	Apparel, footwear, jewellery, furniture, garden goods	restricted_content; labelling_element	As G-EU-2 and G-EU-3 with GB instruments	Ongoing	Unverified — organisation to confirm
G-UK-5	Plant Health (Official Controls and Miscellaneous Provisions) Regulations; UK plant passports; Plant Protection Products Regulations (GB) — Defra, APHA, HSE	Live plants, seeds, garden chemicals	pre_market; restricted_content	→ CMP_market_registrations, CMP_restricted_sale	Ongoing	Unverified — organisation to confirm

G.6 Canada (CA)

Ref	Instrument / regulator	Applies to	Obligation	Required data elements → CDS field	Trigger / dates	Verification
G-CA-1	Textile Labelling Act and Textile Labelling and Advertising Regulations — Competition Bureau (ca_textile_labelling_act)	Consumer textile articles (apparel, household textiles, upholstery covers)	labelling_element (fibre content by generic name for fibres ≥ 5%, in English and French; dealer identity by name and postal address or CA identification number; country of origin for imports under the Customs Tariff marking rules); restricted_content (advertising representations)	→ MF_fibre_composition (generic names, ≥ 5% rule), CMP_responsible_person (CA number), MF_country_of_origin, bilingual localisation per CDS-300 §18	Ongoing	Verified 2026-09-20 (Competition Bureau guide to the Textile Labelling Act)
G-CA-2	Canada Consumer Product Safety Act — Textile Flammability Regulations; Children's Sleepwear Regulations; Children's Jewellery Regulations (lead and cadmium limits); Consumer Chemicals and Containers Regulations for garden chemicals — Health Canada (ca_ccpsa_textiles_jewellery)	Apparel and textiles, children's sleepwear, children's jewellery, consumer chemicals	pre_market (compliance testing); restricted_content (content limits); labelling_element (bilingual warnings)	→ CMP_compliance_declarations, MF_warning_statements (EN and FR), CMP_restricted_sale	Ongoing	Unverified — organisation to confirm

Ref	Instrument / regulator	Applies to	Obligation	Required data elements → CDS field	Trigger / dates	Verification
G-CA-3	Pest Control Products Act (PMRA registration); Fertilizers Act; Seeds Act; Plant Protection Act — Health Canada, CFIA	Garden pesticides, fertilisers, seeds, live plants	pre_market (registration); labelling_element; restricted_content (import and movement conditions)	→ CMP_market_registrations, CMP_restricted_sale	Ongoing	Unverified — organisation to confirm

G.7 Other markets

Jurisdiction	Status	Note
Japan (JP), China (CN), Republic of Korea (KR), India (IN), Brazil (BR) and others	Not seeded	Textile labelling (for example the Japanese Household Goods Quality Labelling Act, China GB 5296.4, India's Legal Metrology packaged-commodity rules), hallmarking (India BIS) and furniture safety regimes differ; organisation supplies entries with local advice.

Commerce Data Standard (CDS)

CDS-1600 — Beauty, Health and Personal Care Industry Profile

Field	Value
Status	vo.2 Review Draft (working source; not an approved standard)
Release	CDS v0.2 (single corpus release per ADR-D5); chapter added in the 2026-09-20 industry-profile expansion
Date	2026-09-20
Supersedes	Nothing — first edition. The Chapter 10 dictionary vocabularies (beauty skin types, hair types, scent families, formulations, makeup finishes) were shipped ahead of this profile and are bound here for the first time.
Normative status	§1, §3, §5–§14, §19 and Appendix A are normative. §2 (pointer), §4, §15–§18, §20, §21 and Appendices B–D are informative; Appendix D seeds the jurisdiction requirement register whose record structure and obligations are normative in CDS-1500 §2.2. Every table is individually marked.
Primary audience	Beauty and health merchandisers, regulatory and compliance owners, data stewards, PIM architects, catalogue managers, UX teams, developers and AI enrichment designers
Depends on	CDS-000 through CDS-1500, especially CDS-200 (entity model), CDS-300 (namespaces, incl. CMP_), CDS-400 (dictionaries, units), CDS-500 (verification, preflight), CDS-600 (facets), CDS-700 (evidence classes, claims), CDS-900 (platform profiles), CDS-1500 (profile model, requirement levels, colour architecture, profile register, jurisdiction requirement register), CDS-1700 (allergen and nutrition rules for ingestible products), CDS-1900 (kits, bundles and dangerous-goods declarations)
Companion package	CDS Reference Dictionary: Chapter 10 (Beauty & Personal Care), Chapter 1 (Colours), Chapter 19 (Jurisdictions & Regulatory Schemes); column contract per CDS-1500 Appendix E
Profile identifier	<code>cds.profile.beauty_health.v0_2</code> (registered in CDS-1500 §2.1)
Research basis	REVIEW-020 (global commerce vertical market research, 2026-09-20): Beauty & Fitness is the third-largest vertical across all Shopify stores worldwide (11.0%) and the second-largest across Shopify Plus stores worldwide (12.5%); Beauty and Health & Wellness show the highest Plus adoption rates of any vertical globally; beauty and personal care is a top-seven worldwide ecommerce revenue segment.

Terminology follows CDS-100. Attribute requirement levels (R, C, REC, O, N/A) are those of CDS-1500 §4 and are not restated here.

1. Purpose and Scope (*normative*)

CDS-1600 defines the industry profile for products applied to, or taken into, the body: cosmetics and skincare, haircare, body and personal care, fragrance, colour cosmetics, sun care, oral care, grooming, and ingestible health products such as vitamins, supplements and complementary medicines. The profile also covers personal-care devices only where their product data is dominated by beauty attributes; devices dominated by technical specifications (hair dryers, electric toothbrushes, massage devices) apply CDS-1800 alongside this profile.

The data shape this profile governs is distinctive: ordered ingredient lists with regulated naming; shade systems layered on the shared colour architecture; net quantities and pack sizes that define sellable units; suitability facts (skin type, hair type) that are frequently confused with benefit claims; ratings under declared regulatory schemes (SPF); and a dense layer of regulated identity, warnings and shelf-life information whose authority is the product's label artwork and product information file, not marketing copy. Every one of those regulated elements differs by jurisdiction, so this profile is the first to be seeded with a per-country requirement register (Appendix D).

CDS1600-R001 An implementation claiming the Beauty and Health Profile MUST represent ingredient lists, active ingredients, net quantities, regulatory identifiers, ratings and warnings as structured, typed attributes conforming to CDS-200 §7 and CDS-400 §18, never solely as free text or as content embedded in a description.

CDS1600-R002 An implementation claiming this profile MUST separate declared product facts (ingredients, quantities, suitability, ratings) from benefit, efficacy and ethical claims, and MUST govern every claim under the evidence model of CDS-700 §7 and the claim rules of §11.

CDS1600-R003 Ingestible health products (supplements, vitamins, complementary and listed medicines) MUST additionally apply the allergen declaration and nutrition-value rules of CDS-1700 §6–§7 as cited in §14.

CDS1600-R004 This profile MUST NOT weaken any rule of CDS-1500 that it reuses (colour architecture, requirement levels, dictionary binding, jurisdiction requirement register); where this chapter is silent, CDS-1500 and the core chapters govern.

2. Profile Model (*informative pointer*)

This profile applies the industry profile model of CDS-1500 §2 (Core → Industry Profile → Product Family → Category Profile), the attribute requirement levels of CDS-1500 §4 and the jurisdiction requirement register of CDS-1500 §2.2 without restatement. Its identifier and dictionary bindings are recorded in the industry profile register, CDS-1500 §2.1; its seeded jurisdiction entries are in Appendix D.

3. Product Model (*normative*)

A beauty product is a formulation or design offered in one or more sellable presentations. The Product is the formulation (the serum, the foundation line, the fragrance); Variants are the sellable presentations that differ by shade, net quantity, pack or dispenser.

```

Product: Radiance Serum                                (informative example)
Product scope:
  CAT_product_type = face_serum
  MF_product_form = serum                               # beauty_formulation dictionary
  MF_ingredient_list = [aqua, glycerin, niacinamide, ...] # ordered INCI list
  MF_skin_type_suitability = [normal, combination, oily]
  MF_period_after_opening_months = 12
Variant scope:
  VAR_net_quantity = {value: 30, unit: millilitre}
  VAR_pack_count = 1
  STD_id / VAR_sku = SER-RAD-030

```

CDS1600-R005 Shade, net quantity, pack count and dispenser or applicator type MUST be modelled at variant scope where they identify distinct sellable units (variant boundary rule: CDS-200 §5); formulation, ingredients, suitability and claims MUST remain at product scope unless the fact genuinely differs between variants.

CDS1600-R006 Where an ingredient list genuinely differs between shades of one product line (for example pigment-dependent colourants), the implementation MUST store the ingredient list at variant scope for the affected variants and MUST NOT publish the product-scope list as if it applied to every shade. *(Informative: several jurisdictions permit a range-level colourant list introduced by "may contain" or "+/-"; where that convention is used the implementation records the convention explicitly, see §5.)*

CDS1600-R007 Gift sets, discovery sets and value packs containing distinct formulations MUST be modelled as kits or bundles with component relationships per CDS-1900 §11, not as a single product whose ingredient list concatenates the components.

CDS1600-R008 Testers, samples and professional-size presentations MUST carry the same formulation identity as the retail product and MUST be distinguished by a governed presentation attribute (MF_presentation_type), never by a different ingredient list.

4. Product Families and Category Profiles *(informative baseline; the inheritance rule of CDS1500-R016 applies)*

Product family / category	Required	Recommended	Typical variant options
Skincare (cleansers, serums, moisturisers, masks)	product form; ingredient list; net quantity; skin type suitability	key active ingredients; period after opening; scent family; texture	size
Sun care (primary sunscreens)	SPF and rating scheme; broad-spectrum status; ingredient list; net quantity; regulatory identity per market	water-resistance duration; product form; skin type suitability	size
Makeup — face	product form; shade (sellable colour); finish; ingredient list; net quantity	coverage; undertone; shade family; skin type suitability	shade, size
Makeup — eyes, lips, nails	product form; shade; finish; ingredient list; net quantity	shade family; applicator type	shade
Haircare (shampoo, conditioner, treatment, styling)	product form; ingredient list; net quantity; hair type suitability	scent family; period after opening	size
Body and personal care (wash, lotion, deodorant, oral care)	product form; ingredient list; net quantity	scent family; skin type suitability; dangerous-goods status for aerosols	size, pack
Fragrance	scent family; fragrance concentration; net quantity; ingredient list	fragrance notes (top/heart/base); dangerous-goods status	size
Grooming (shaving, beard care)	product form; ingredient list; net quantity	skin type suitability; scent family	size, pack
Supplements and listed medicines (§14)	dosage form; active ingredients with quantity per dose; pack quantity; directions; warnings; regulatory identity per market	allergen declarations; nutrition or supplement facts; storage conditions	pack size, flavour
Beauty tools and devices	material; dimensions	power source (CDS-1800)	colour

5. Ingredients *(normative; seed guidance informative)*

Ingredient information is regulated content whose authority is the product's label and product information file. The profile stores it as an ordered, structured list so that it can be validated, searched, projected and compared, and so that free-from statements can be checked mechanically.

Field	Meaning	Type
MF_ingredient_list	Ordered list of ingredient names as declared on the label	Ordered list of ingredient entries (name, optional canonical ingredient reference, optional function, optional percentage)
MF_ingredient_naming_scheme	The naming convention the list follows	Governed value: inci, english_common, aan (Australian Approved Name), chemical_name, mixed_as_labelled
MF_ingredient_order_basis	The ordering rule the label applies	Governed value: descending_by_mass_or_volume, descending_over_one_percent_then_any_order
MF_range_colourant_convention	Whether colourants are listed per shade or for the whole range	Governed value: per_shade, range_may_contain
MF_active_ingredients	Active or key ingredients with quantities	Structured list (canonical ingredient, quantity as typed measurement or percentage, unit basis)
MF_fragrance_disclosure	How fragrance is declared	Governed value: parfum_placeholder, components_listed, fragrance_free

Field	Meaning	Type
CMP_fragrance_allergen_declarations	Declared fragrance allergens under a named jurisdiction list	List of (allergen name, regulatory scheme)
MF_nanomaterial_ingredients	Ingredients present as nanomaterials, where a jurisdiction requires "nano" marking	List of ingredient references

CDS1600-R009 MF_ingredient_list MUST be stored as an ordered list whose order is declared significant in the Attribute Definition (CDS400-R040) and MUST preserve the label order; an implementation MUST NOT re-sort, de-duplicate or "clean" the list without a governed change carrying label evidence.

CDS1600-R010 The ingredient naming scheme and ordering basis MUST be declared per product (or inherited from a category profile) so that a consumer of the data can interpret the list; a list without a declared scheme MUST NOT satisfy an R-level ingredient requirement.

CDS1600-R011 Active ingredients MUST carry a numeric quantity with a declared unit or percentage basis (CDS400-R056) wherever the label declares one; a textual strength such as "high strength" MUST NOT be stored as the quantity.

CDS1600-R012 Ingredient names MAY be linked to a governed ingredient dictionary for search and facet projection, but the label-declared spelling MUST be preserved as the source value with provenance (CDS400-R006, R014).

CDS1600-R013 An ingredient list captured by extraction from label artwork or packaging images (evidence class E2 or E3 per CDS-700 §7) MUST be reviewed by a human before acceptance, regardless of confidence (CDS-700 §19).

Informative — ordering conventions across the seeded jurisdictions. Australia (Consumer Goods (Cosmetics) Information Standard 2020), New Zealand (Cosmetic Products Group Standard 2020), the United States (21 CFR 701.3), the EU (Regulation (EC) 1223/2009 Article 19) and Canada (Cosmetic Regulations) all require descending order by mass, volume or predominance, all permit ingredients under 1% in any order after those above 1%, and all permit colour additives to be listed last. Naming differs: Canada mandates INCI names; Australia and New Zealand accept INCI or English names; the EU uses the common ingredient glossary and requires "nano" in brackets for nanomaterials; the US uses the names established by regulation. The two declared fields above let a PIM record which convention a given label followed rather than guessing; Appendix D records the per-jurisdiction rule with its verification date.

6. Product Form, Finish and Scent (*normative; dictionary bindings informative until adopted*)

Attribute	Dictionary (Chapter 10 unless stated)	Scope
MF_product_form	beauty_formulation (cream, serum, gel, mist, foam, powder, stick, bar, sheet mask, ...)	Product
MF_makeup_finish	beauty_makeup_finish (matte, dewy, satin, shimmer, glossy, natural)	Product
MF_scent_family	beauty_scent_family (floral, citrus, woody, amber, ...)	Product
MF_fragrance_concentration	fragrance_concentration (parfum, eau de parfum, eau de toilette, eau de cologne, eau fraîche, body mist) — new in package 0.8.0	Product
MF_fragrance_notes	Structured top/heart/base note lists (organisation-governed vocabulary)	Product

CDS1600-R014 Product form MUST be a governed dictionary value distinct from the product's classification; "serum" as a form and "face serum" as a product type are related but separately governed facts (classification rules: CDS-200 §6).

CDS1600-R015 Fragrance concentration MUST be a governed value and MUST NOT be inferred from price, size or marketing tier; where the manufacturer declares no concentration the attribute is unknown, not "eau de toilette".

CDS1600-R016 Scent family and fragrance notes are merchandising descriptors and MUST NOT be presented as ingredient disclosure.

7. Suitability: Skin Type, Hair Type and Audience (*normative*)

Suitability values state which skin or hair conditions a product is presented for. They are manufacturer-declared facts about intended use, not efficacy claims, and the package deliberately excludes benefit and age-state terms (anti-ageing, hypoallergenic, mature) from these vocabularies.

CDS1600-R017 MF_skin_type_suitability and MF_hair_type_suitability MUST be multi-value attributes bound to the governed beauty_skin_type and beauty_hair_type dictionaries, with a declared maximum value count per product (the CDS1500-R029 pattern), and MUST derive from manufacturer or brand declarations (evidence class E1 or E2), never from model inference over marketing copy.

CDS1600-R018 A suitability value MUST NOT be projected to a customer facet as a benefit ("treats dry skin"); the facet label MUST express intended use ("for dry skin").

CDS1600-R019 Audience attributes (target gender, age group) MUST be governed values where a channel requires them and MUST NOT be inferred from scent family, colour or packaging.

8. Shades and Colour (*normative*)

Colour cosmetics apply the CDS-1500 §7 colour architecture: the sellable shade is a canonical colour value at variant scope, the shade name is its Display Label, and the customer filter uses the governed colour-family facet. Two beauty-specific additions are required: a deterministic shade order within a range, and an optional undertone dimension.

```
Variant: Foundation, shade "Warm Sand 3.5"      (informative example)
  VAR_colour      = colour_warm_sand_3_5        # canonical shade value_id
  MF_colour_display = Warm Sand 3.5              # Display Label
```

MF_colour_facet	= beige	# governed family facet
VAR_shade_sort_key	= 035	# deterministic order in the range
MF_undertone	= warm	# optional: warm / cool / neutral / olive
MF_shade_range_id	= range_foundation_2026	# the range the shade belongs to

CDS1600-R020 Shades MUST be canonical colour values with stable `value_ids` (CDS400-R009); a brand shade name is a Display Label linked to the canonical value (CDS400-R022) and MUST NOT be the only identity of the shade.

CDS1600-R021 Shades within a range MUST carry a sort key or an equivalent declared ordering mechanism so that storefronts and channels present the range in the brand's intended order rather than alphabetically (the CDS1500-R027 pattern).

CDS1600-R022 Undertone, where used, MUST be a governed single-value attribute separate from the colour family facet; it MUST NOT be encoded in the shade name alone.

CDS1600-R023 Swatch imagery and digital colour values follow CDS400-R061: they MAY be stored but MUST NOT be represented as an exact match of the product on skin.

CDS1600-R024 A shade discontinued from a range MUST follow dictionary deprecation (CDS400-R080) with a replacement pointer where a successor shade exists, so that historical orders, reviews and analytics remain interpretable.

9. Quantity, Pack and Dosage (*normative*)

CDS1600-R025 Net quantity MUST be a typed measurement with a declared unit (millilitre, gram, or count for unit-dose products) per CDS400-R056, stored at variant scope, and MUST NOT be embedded only in the title or variant label.

CDS1600-R026 Pack count (number of identical units in the sellable pack) MUST be an integer attribute separate from net quantity per unit; a "3 × 50 ml" presentation is `VAR_pack_count = 3` and `VAR_net_quantity = 50 ml`, never a net quantity of "150 ml" alone.

CDS1600-R027 Unit-pricing base measures required by a channel or jurisdiction MUST be derived from the typed net quantity and pack count, never authored independently.

CDS1600-R028 Supplements MUST record serving size, servings per container and the dosage form as typed values (§14).

10. Sun Protection and Declared Ratings (*normative*)

Sun protection factor (SPF) is a tested rating under a declared scheme, and in some jurisdictions a primary sunscreen is a regulated therapeutic good or an over-the-counter drug. The profile treats the rating value, the scheme and the regulatory status as three separate governed facts.

Field	Meaning	Type
MF_spf_value	Declared SPF as labelled	Integer (with optional "+" flag for capped labels such as "50+")
MF_spf_rating_scheme	The standard the rating was determined under	Governed value: <code>as_nzs_2604</code> , <code>iso_24444</code> , <code>fda_sunscreen_monograph</code> , <code>other_declared</code>
MF_broad_spectrum	Broad-spectrum (UVA) protection status as labelled	Boolean with scheme reference
MF_water_resistance_minutes	Labelled water-resistance duration	Integer minutes, or null
CMP_regulatory_scheme / CMP_regulatory_id	Regulatory identity per market (§12)	Governed scheme code plus identifier

CDS1600-R029 SPF MUST be stored as a numeric value plus a declared rating scheme; a value without a scheme MUST NOT satisfy an R-level requirement, and the scheme MUST NOT be assumed from the target market.

CDS1600-R030 A sun-protection claim in a title, description or facet MUST be derived from `MF_spf_value` and its evidence; an implementation MUST NOT accept an SPF value whose evidence class is below E2 (CDS-700 §7).

CDS1600-R031 Where a jurisdiction in the organisation's register classifies a sunscreen as a therapeutic good or drug, the regulatory identifier and any mandated label elements for that market (§12, Appendix D) MUST be present before publication to that market; publication preflight (CDS-500 §8) MUST block a therapeutic sunscreen lacking them.

Informative. In Australia primary sunscreens are therapeutic goods regulated by the Therapeutic Goods Administration and carry an AUST L or AUST R number, active-ingredient names and proportions, batch and expiry information and storage conditions; secondary sunscreens may be cosmetics subject to conditions. In New Zealand the Cosmetic Products Group Standard allows a primary sunscreen to be labelled to the Australian Therapeutic Goods Order. In the United States a sunscreen is an over-the-counter drug whose active ingredients are declared in a Drug Facts panel before the cosmetic ingredients. In the EU and UK sunscreens are cosmetics. A retailer selling into several markets therefore records the scheme and identifier per market rather than one global value (Appendix D).

11. Claims, Certifications and Free-From Statements (*normative*)

Benefit claims ("reduces the appearance of wrinkles"), ethical claims ("cruelty-free", "vegan"), sustainability claims ("natural", "organic"), and free-from statements ("paraben-free") are the highest-risk content in this vertical. This section applies the corpus claim pattern established for apparel (CDS1500-R025, R031–R032) with beauty-specific mechanics.

Attribute	Structure	Requirement guidance
MF_claims	List of claim records: claim type (governed), claim text as labelled, evidence reference and class, scope (product or component), jurisdictions in which the claim is published, approving owner, effective/expiry dates	C — required whenever a claim is published
MF_certifications	List of certification records: scheme, certifier, certificate identifier, scope, expiry	C — required when a certification is displayed
MF_free_from_statements	List of governed free-from statement records naming the excluded ingredient class	C
MF_natural_origin_percent / MF_organic_content_percent	Numeric, component-aware	C — required when a quantified natural or organic claim is made

CDS1600-Ro32 Every published claim MUST exist as a claim record with an evidence reference of class E1 or E2 (CDS-700 §7); a claim MUST NOT be created or accepted from supplier or brand marketing copy alone (CDS400-Ro66 pattern).

CDS1600-Ro33 A free-from statement naming an ingredient or ingredient class MUST fail validation when the named ingredient (or a governed member of the named class) appears in MF_ingredient_list or in the variant-scope ingredient list of any variant the statement covers. The validation error MUST be classified as conflicting (CDS200-Ro35) and MUST block publication of the statement.

CDS1600-Ro34 Certification-backed claims (cruelty-free, vegan, organic, halal and comparable schemes) MUST reference the certifying body and certificate scope; expiry of the certificate MUST withdraw the claim projection without altering any product fact (the CDS1500-Ro32 pattern).

CDS1600-Ro35 Benefit and efficacy claims MUST NOT be projected as customer facets unless the organisation declares a claim-driven facet with a documented evidence threshold in its facet governance record (CDS1500-Ro49); such a facet MUST be fed only by accepted claim records.

CDS1600-Ro36 Therapeutic and health claims for ingestible products (§14) MUST be restricted to indications permitted under the applicable regulatory scheme of each market in which they are published and MUST carry the scheme reference in the claim record; a claim record MUST declare the jurisdictions it is published in so that preflight can evaluate it per market.

CDS1600-Ro37 AI MAY extract candidate claim language from packaging or copy, but a claim record MUST NOT be accepted at any autonomy level above suggest-only without human review (CDS-700 §5, §19); absence of evidence MUST NOT become a claim (CDS700-Ro18).

12. Regulatory Identity, Responsible Person, Shelf Life, Batch and Expiry (*normative*)

Regulatory identity is a per-market fact. The profile stores it as governed records keyed by jurisdiction and scheme, drawing scheme codes from the regulatory_scheme dictionary (package Chapter 19) and evaluating obligations through the jurisdiction requirement register (CDS-1500 §2.2; seed in Appendix D).

Field	Meaning	Scope
CMP_market_registrations	List of records: jurisdiction, regulatory scheme (governed), identifier (e.g. AUST L number, product listing reference), status (registered / listed / notified / exempt / out_of_scope), effective dates, evidence	Product (one record per market)
CMP_responsible_person	The responsible person, sponsor or importer of record for the market, with the address that the market requires on the label or listing	Product (per market)
MF_period_after_opening_months	Period after opening as labelled	Product
MF_minimum_durability_policy	Whether the product carries a date of minimum durability or a period-after-opening symbol	Product
MF_shelf_life_days	Shelf life from manufacture, where declared	Product
CMP_country_of_origin	Country of origin (required on the label or listing by several markets)	Product

CDS1600-Ro38 Regulatory scheme and identifier MUST be stored per target market as governed records, never as a single global free-text "registration" field; an implementation MUST be able to answer, for a given market, whether the product is registered, listed, notified, exempt or out of scope, and publication preflight for that market MUST evaluate the answer against the register (CDS1500-Ro57).

CDS1600-Ro39 Batch (lot) numbers and expiry dates are lot-level facts. An implementation MUST NOT store a single batch number or expiry date as a canonical product or variant attribute unless it declares a lot-per-variant model; lot-level dates are observed from the inventory or warehouse authority (CDS-200 §13) and MAY be published as channel fields with their authority declared.

CDS1600-Ro40 Period after opening and minimum durability policy MUST be typed values; a storefront or channel MUST NOT display a computed expiry as a product fact where only a period after opening is known.

CDS1600-Ro41 Withdrawal, recall or regulatory suspension of a product in a market MUST be expressed through lifecycle state and channel withdrawal (CDS-500 §12), never by deleting the product record.

CDS1600-Ro42 The responsible person or sponsor of record MUST be a governed per-market attribute; where a market requires that person's name and address on the listing or packaging (Appendix D), the channel projection for that market MUST include it and verification MUST cover it.

13. Safety, Warnings and Handling (*normative*)

CDS1600-Ro43 Warnings, directions for use and precautionary statements required by a jurisdiction MUST be stored as structured, governed statement records (statement type, text as labelled, jurisdiction, mandating scheme) rather than only within description prose, so that channel projection and verification can confirm their presence per market.

CDS1600-Ro44 Products that are dangerous goods for transport (aerosols, alcohol-based fragrances and sanitisers, certain nail products) MUST carry a dangerous-goods declaration per CDS-1900 §12 before publication to a channel or carrier that requires it; the declaration MUST NOT be inferred from category alone.

CDS1600-Ro45 Age or purchase restrictions applicable in a market MUST be governed attributes evaluated at publication preflight for that market.

14. Ingestible Health Products: Supplements and Listed Medicines (*normative*)

Vitamins, minerals, herbal and nutritional supplements and complementary medicines share the ingredient, claim and regulatory pattern of cosmetics but add dosage, allergen and nutrition-panel obligations. The profile homes them here and cites CDS-1700 for the allergen and nutrition mechanics.

Field	Meaning	Type
MF_dosage_form	Tablet, capsule, softgel, powder, liquid, gummy, effervescent, ...	Governed value (organisation-governed; a starter vocabulary is a candidate for package 0.9)
MF_active_ingredients	Active or medicinal ingredients with quantity per dose unit	Structured list (§5)
MF_serving_size / MF_servings_per_container	Dose definition	Typed
MF_directions	Directions for use as labelled	Structured statement records
CMP_allergen_declarations	Allergen declarations	Per CDS-1700 §6
MF_nutrition_values	Nutrition or supplement-facts values	Per CDS-1700 §7 with scheme <code>supplement_facts</code> or <code>nutrition_information_panel</code>
CMP_market_registrations	Listing, licence or registration per market	§12

CDS1600-Ro46 Active ingredients MUST be stored with quantity per dose unit as typed measurements; "one-a-day" or "high potency" MUST NOT stand in for quantities.

CDS1600-Ro47 Allergen declarations and nutrition values for ingestible products MUST conform to CDS-1700 §6 and §7 respectively, including the distinction between "contains" and precautionary "may contain" declarations and the per-100 versus per-serving basis.

CDS1600-Ro48 Indications and health claims MUST reference the permitted-indication or claim scheme of the target market in the claim record (§11), and publication preflight MUST block an indication not permitted under the declared scheme for that market.

CDS1600-Ro49 Ingestible products MUST carry storage conditions (CDS-1700 §10 dictionary `storage_condition`) and mandatory warning statements (§13) before publication.

15. Customer Facet Design (*informative — normative facet rules live in CDS-600*)

Facet	Recommended baseline behaviour	Anti-pattern
Product form	Governed formulation vocabulary; category-aware (serum is meaningless in fragrance)	Free-text "type" from supplier feeds
Skin / hair type	Suitability vocabulary, multi-select OR	Benefit language as facet labels
Shade family	Colour-family facet from CDS-1500 Appendix C; swatch plus text label (CDS600-Ro36)	Every shade name as a checkbox
Finish	Makeup finish vocabulary in colour-cosmetics contexts only	Finish exposed on skincare
Scent family	Small governed list	Note-level lists of hundreds of ingredients
SPF	Numeric range facet (15+, 30+, 50+) derived from <code>MF_spf_value</code>	Marketing tiers
Size	Net-quantity ranges with unit context	Mixed ml and g in one list
Claims	Only claim-driven facets with declared evidence thresholds (Ro35)	"Clean", "natural", "hypoallergenic" as tags

Fragmentation and coverage thresholds, zero-result handling and accessibility follow CDS-600 §10, §17 and §24.

16. Channel Projection Guidance (*informative — normative rules: CDS-500, CDS-900*)

Canonical concept	Metafield-style channel (informative)	Feed-style channel (informative)
Product form, skin type, scent	Category or custom metafields; storefront filters	Descriptive fields where supported, otherwise <code>product_detail</code>
Shade	Variant option with linked colour entry (display label); family facet as filter metafield	<code>color</code> = landing-page shade name (CDS-900 §5.2 rule); family never published to <code>color</code>
Net quantity, pack count	Typed metafields; variant title composition per declared title policy (CDS-700 §12)	<code>unit_pricing_measure</code> / <code>unit_pricing_base_measure</code> where required; <code>multipack</code> for identical-unit packs
Ingredient list	Rich-text or list metafield generated from the structured list; never hand-edited downstream	Not a standard feed field; carried in description where policy allows
SPF, regulatory identifiers, responsible person	Custom metafields; verified by read-back; per-market listing text generated from the register	Description or <code>product_detail</code> ; regulatory identifiers where the channel requires them
Claims and certifications	Only accepted claim records project; expiry withdraws the projection	Certification attributes where the channel defines them
Warnings mandated for a market	Generated statement block per market, verified by read-back	Description block per market

Which layer feeds each channel is declared per attribute in the Attribute Definition (CDS200-R026). Unit-pricing and multipack attributes are required by some feed channels in specific countries; the dated requirements live in the CDS-900 platform profiles and are verified there, not here.

17. AI Enrichment and Review (*informative — normative AI rules live in CDS-700*)

Task	AI may propose	Deterministic or human control
Ingredient extraction from label images	Ordered candidate list with per-entry confidence	Mandatory human review (R013); order preserved; naming scheme declared
Shade family mapping	Family candidate from swatch and shade name	Dictionary validation; split-shade intake per CDS1500-R020
Suitability values	Candidates only from brand declarations (E1/E2)	Declared maxima; merchandising approval
Claim language	Candidate claim records flagged "evidence required"	Never accepted without evidence and human review (R037)
SPF and ratings	Parsed value and scheme from label	Scheme validation; evidence class check (R030)
Free-from validation	None — deterministic	Mechanical conflict check (R033)
Market obligation gaps	Candidate list of register entries a product may trigger	Register evaluation is deterministic; AI never marks an obligation satisfied

AI never manufactures ingredient lists, quantities, ratings, regulatory identifiers or certifications; abstention is a first-class outcome (CDS-700 §17).

18. Governance and Organisational Extensions (*informative — rules in CDS-1500 §23 and CDS-800*)

Organisations extend the Chapter 10 vocabularies, add ingredient dictionaries and fragrance-note vocabularies, and declare claim-evidence thresholds under the same rules as any dictionary (CDS1500-R047–R051). The jurisdiction requirement register is maintained per market under CDS-1500 §2.2 with the review cadence declared under CDS-800 §30; the Appendix D seed is a starting point, not a substitute for the organisation's own regulatory advice.

19. Conformance Requirements (*normative — claims and levels per CDS-1000*)

CDS1600-R050 An implementation claiming the CDS Beauty and Health Profile MUST: store ingredient lists as ordered structured lists with declared naming scheme and ordering basis (R009–R010); store active-ingredient quantities, net quantities, pack counts and SPF as typed values with declared units or schemes (R011, R025–R026, R029); apply the CDS-1500 colour architecture to shades with deterministic shade ordering (R020–R021); govern suitability values from manufacturer declarations (R017); govern every claim, certification and free-from statement as an evidence-bearing record with the mechanical free-from conflict check (R032–R035); store regulatory identity and responsible person per market and keep lot-level facts out of canonical product data (R038–R039, R042); maintain and evaluate a jurisdiction requirement register for every market published to (CDS-1500 §2.2); apply category-specific requirements per CDS1500-R009; and publish and verify channel representations under CDS-500.

CDS1600-R051 An implementation whose scope includes ingestible health products MUST additionally satisfy R046–R049 and the cited CDS-1700 allergen and nutrition rules; a Beauty and Health Profile claim MUST state whether ingestible products are in scope and which jurisdictions its register covers.

20. Worked Product Examples (*informative*)

20.1 Foundation shade range

```
CAT_family = beauty
CAT_product_type = liquid_foundation
MF_product_form = liquid
MF_makeup_finish = satin
MF_ingredient_naming_scheme = inci
MF_ingredient_order_basis = descending_over_one_percent_then_any_order
MF_range_colourant_convention = range_may_contain
MF_ingredient_list = [aqua, cyclopentasiloxane, glycerin, ..., "+/- ci_77891", "+/- ci_77491"]
MF_skin_type_suitability = [normal, combination]
MF_shade_range_id = range_foundation_2026

Variant (VAR_sku = FND-W535):
  VAR_colour = colour_warm_sand_3_5      # canonical shade
  VAR_shade_sort_key = 035
  MF_undertone = warm
  VAR_net_quantity = {value: 30, unit: millilitre}

Market registrations:
{jurisdiction: EU, scheme: eu_cpnp_notification, id: <reference>, status: notified}
{jurisdiction: GB, scheme: uk_scnp_notification, id: <reference>, status: notified}
{jurisdiction: US, scheme: us_mocra_product_listing, id: <listing>, status: listed}
{jurisdiction: CA, scheme: ca_cosmetic_notification, id: <CNF>, status: notified}
{jurisdiction: AU, scheme: au_cosmetics_information_standard_2020, status: in_scope_no_registration}

Projections:
  MF_colour_display = Warm Sand 3.5
  MF_colour_facet = beige
  CH_google_color = Warm Sand 3.5      # landing-page shade name (CDS-900 §5.2)
  QA_shopify_colour = MATCH
```

20.2 SPF 50+ sunscreen sold in Australia and the United States

```
CAT_product_type = sunscreen_lotion
MF_spf_value = 50 (capped_label_plus = true)
MF_spf_rating_scheme = as_nzs_2604 # AU/NZ testing scheme
MF_broad_spectrum = true
MF_water_resistance_minutes = 240
MF_active_ingredients = [{ingredient: zinc_oxide, quantity: 22, basis: percent_w_w}]
CMP_market_registrations =
  [{jurisdiction: AU, scheme: au_artg_listed, id: "AUST L 000000", status: listed}, # placeholder id
   {jurisdiction: US, scheme: us_fda_otc_drug, status: in_scope, evidence: drug_facts_panel}]
CMP_dangerous_goods = not_regulated # lotion; an aerosol would declare per CDS-1900 §12
VAR_net_quantity = {value: 200, unit: millilitre}
Preflight (AU): ARTG identifier present, TGO 92 label elements present -> publication permitted (R031)
Preflight (US): Drug Facts active-ingredient block present -> publication permitted
```

20.3 Vitamin D supplement

```
CAT_product_type = vitamin_supplement
MF_dosage_form = capsule
MF_active_ingredients = [{ingredient: colecalciferol, quantity: 25, unit: microgram, per: capsule}]
MF_serving_size = 1 capsule; MF_servings_per_container = 90
CMP_allergen_declarations = [{allergen: soy, declaration: contains}] # per CDS-1700 §6
MF_storage_condition = store_below_25c # CDS-1700 §10 dictionary
CMP_market_registrations = [{jurisdiction: AU, scheme: au_artg_listed, id: "AUST L 000001", status: listed},
                             {jurisdiction: CA, scheme: ca_nhp_product_licence, id: "NPN 00000000", status: licensed}]
MF_claims = [{type: health_indication, text: "Supports bone health", scheme: au_permitted_indications,
               jurisdictions: [AU], evidence: E1 sponsor evidence file, owner: regulatory_owner, status: accepted}]
```

21. Reference Validation Cases (*informative — the profile's contribution to the cross-industry validation set, REVIEW-020*)

A test dataset supporting a claim of this profile should include, in addition to the CDS-1000 Appendix B negative cases:

- a product whose ingredient list differs by shade (R006);
- a free-from statement that conflicts with the ingredient list and must be blocked (R033);
- a shade range whose brand order differs from alphabetical order (R021);
- a "3 × 50 ml" pack that must not collapse to "150 ml" (R026);
- an SPF value supplied without a scheme, which must not satisfy an R-level requirement (R029);
- a certification whose expiry must withdraw the claim projection without a product edit (R034);
- a therapeutic sunscreen lacking a market regulatory identifier, which preflight must block for that market while permitting publication to a market where it is a cosmetic (R031, R038);
- a supplier feed carrying a batch number in a product field, which must be quarantined rather than stored canonically (R039);
- a product published to the EU after 31 July 2026 whose fragrance allergen declarations follow the pre-2026 list, which the register must flag (Appendix D, D-EU-2).

Appendix A. Beauty and Health Attribute Baseline (*normative — dictionary bindings per CDS1500-R011*)

Field	Scope	Type	Baseline requirement
CAT_product_type	Product	Governed classification reference	R
MF_product_form	Product	Governed dictionary reference (beauty_formulation)	R for skincare, haircare, body, makeup; N/A for tools
MF_ingredient_list	Product (variant where R006 applies)	Ordered structured list	R for formulated products
MF_ingredient_naming_scheme / MF_ingredient_order_basis	Product	Governed values	R where an ingredient list is R
MF_active_ingredients	Product	Structured list with typed quantities	C — R for sunscreens and ingestibles
VAR_colour (shade)	Variant	Canonical value_id (colour dictionary)	C — R for colour cosmetics
MF_colour_facet	Product/Variant	Derived facet value	R where shade is R (derived)
VAR_shade_sort_key	Variant	Sort key	R where shade is R
MF_undertone	Variant	Governed value	O
MF_makeup_finish	Product	Governed dictionary reference	REC for colour cosmetics
MF_skin_type_suitability / MF_hair_type_suitability	Product	Governed dictionary list with declared maximum	REC
MF_scent_family	Product	Governed dictionary reference	REC for fragrance and scented products

Field	Scope	Type	Baseline requirement
MF_fragrance_concentration	Product	Governed dictionary reference	R for fragrance
VAR_net_quantity	Variant	Typed measurement	R
VAR_pack_count	Variant	Integer	C — R for multi-unit packs
MF_spf_value / MF_spf_rating_scheme / MF_broad_spectrum	Product	Typed value, governed scheme, boolean	R for sun-care products
MF_claims / MF_certifications / MF_free_from_statements	Product	Structured records with evidence	C
CMP_market_registrations	Product (per market)	Governed records (regulatory_scheme dictionary)	C — R where a market regulates the product
CMP_responsible_person	Product (per market)	Governed record	C — R where a market requires it on the label or listing
CMP_country_of_origin	Product	Governed country reference	C
MF_period_after_opening_months / MF_minimum_durability_policy	Product	Integer / governed value	REC; R where a market mandates one of them
MF_warning_statements	Product	Structured statement records	C — R where a jurisdiction mandates warnings
CMP_dangerous_goods	Product/Variant	Declaration record per CDS-1900 §12	C
MF_dosage_form / MF_serving_size / MF_servings_per_container	Product/Variant	Governed value and typed values	R for ingestibles
CMP_allergen_declarations / MF_nutrition_values	Product	Per CDS-1700 §6–§7	R for ingestibles where labelling requires them

Appendix B. Dictionary Bindings (*informative — becomes governed data on adoption per CDS1500-R011*)

Dictionary key	Package chapter	Bound attribute(s)	Status at 0.8.0
beauty_formulation	10	MF_product_form	Bound by this profile
beauty_makeup_finish	10	MF_makeup_finish	Bound
beauty_scent_family	10	MF_scent_family	Bound
beauty_skin_type / beauty_hair_type	10	MF_skin_type_suitability / MF_hair_type_suitability	Bound
fragrance_concentration	10 (new, 0.8.0)	MF_fragrance_concentration	Bound
colour / colour_facet	1	VAR_colour, MF_colour_facet	Bound via CDS-1500 §7
jurisdiction / regulatory_scheme	19 (new, 0.8.0)	CMP_market_registrations, claim and warning records	Bound via CDS-1500 §2.2
allergen / allergen_facet / storage_condition	17 (new, 0.8.0)	CMP_allergen_declarations, MF_storage_condition (ingestibles)	Bound via CDS-1700
dangerous_goods_class	18 (new, 0.8.0)	CMP_dangerous_goods	Bound via CDS-1900 §12

Deliberately not shipped: ingredient dictionaries (organisations adopt an INCI-based source under their own governance), benefit and ethical claim vocabularies (evidence-gated claim types are organisation-declared), dosage forms (candidate for package 0.9 after profile use).

Appendix C. References (*informative; retrieved 2026-09-20 unless stated*)

Ref	Source	Location	Use in this chapter
[B1]	ACCC Product Safety — Cosmetics ingredients labelling mandatory standard (Consumer Goods (Cosmetics) Information Standard 2020)	productsafety.gov.au/business/search-mandatory-standards/cosmetics-ingredients-labelling-mandatory-standard	Ingredient ordering and naming conventions (§5, D-AU-1)
[B2]	Federal Register of Legislation — Consumer Goods (Cosmetics) Information Standard 2020	legislation.gov.au/F2020L01469	Ordering alternatives; flavour and fragrance declaration (§5)
[B3]	AICIS — Personal care, skincare, make-up and other cosmetic products (2025-02-07)	industrialchemicals.gov.au/cosmetics-and-soap	Business registration and ingredient categorisation (D-AU-2)
[B4]	Therapeutic Goods Administration — Australian regulatory guidelines for sunscreens (v1.2, 2019); Labelling medicines to comply with TGO 91 and TGO 92 (March 2026); Understanding labelling and presentation requirements for listed medicines (2025-04-09)	tga.gov.au	Therapeutic sunscreen and listed medicine label elements, AUST L/R (§10, §14, D-AU-3)
[B5]	New Zealand EPA — Cosmetic Products Group Standard 2020 (HSR002552, consolidated)	epa.govt.nz	NZ ingredient listing, batch code, nano marking, alternative compliance (D-NZ-1)
[B6]	U.S. FDA — Cosmetics Labeling Regulations; Summary of Cosmetics Labeling Requirements (2025-11-18); 21 CFR 701.3 (eCFR, current to 2026-08-27)	fda.gov; ecf.gov	US label elements and ingredient declaration (D-US-1)
[B7]	U.S. FDA — Modernization of Cosmetics Regulation Act of 2022 (MoCRA) (2026-08-10); Guidance: Registration and Listing of Cosmetic Product Facilities and Products	fda.gov/cosmetics	Facility registration, product listing, adverse-event contact, enforcement date (D-US-2)

Ref	Source	Location	Use in this chapter
[B8]	U.S. FDA — Dietary Supplement Labeling Guide; 21 CFR 101.4(g) and 101.36	fda.gov	Supplement Facts, ingredient list placement (§14, D-US-4)
[B9]	Regulation (EC) No 1223/2009 on cosmetic products, Article 19 (EUR-Lex; UK retained text on legislation.gov.uk)	eur-lex.europa.eu	EU/UK mandatory label particulars, PAO, batch, CPNP (D-EU-1, D-UK-1)
[B10]	Commission Regulation (EU) 2023/1545 (fragrance allergens); HPRA guidance on application dates	eur-lex.europa.eu/eli/reg/2023/1545; hpra.ie	Expanded fragrance allergen labelling, 31 July 2026 / 31 July 2028 (D-EU-2)
[B11]	GOV.UK — Submit a cosmetic product notification; Making cosmetic products available to consumers in Great Britain; Regulation 1223/2009 and the Cosmetic Products Enforcement Regulations 2013 guidance (2022-04-27)	gov.uk	UK Responsible Person, SCPN, PIF, serious undesirable effects (D-UK-1)
[B12]	Health Canada — Industry Guide for the labelling of cosmetics; Cosmetic Regulations (C.R.C., c. 869) ss. 18, 21.2, 30–31; Cosmetic Notification Form	canada.ca; laws-lois.justice.gc.ca	CNF within 10 days, INCI on outer label, bilingual rule, shade-range colourants (D-CA-1)
[B13]	Natural Health Products Regulations (SOR/2003-196)	laws-lois.justice.gc.ca	Product licence (NPN), medicinal ingredient quantity per dosage unit (D-CA-2)
[B14]	OEHHA — Proposition 65 Clear and Reasonable Warnings (effective 2025-01-01) and FAQs for businesses	oehha.ca.gov; p65warnings.ca.gov	Internet warning before purchase, short-form transition to 2028 (D-US-5)
[B15]	Store Leads — The State of Shopify in 2026; Shopify Plus Statistics 2026 (updated 2026-09-11)	storeleads.app	Market basis (header; REVIEW-020)

Appendix D. Jurisdiction Requirement Register — Seed (*informative seed of the normative register defined in CDS-1500 §2.2; verification column states what was checked on 2026-09-20*)

Obligation types are those of CDS-1500 §2.2 (`pre_market`, `labelling_element`, `listing_element`, `warning_statement`, `rating_label`, `restricted_content`, `transport`, `documentation`, `post_market`). "Verified" means the primary instrument or regulator page was read on the stated date; "retrieved" means a regulator page was retrieved but the instrument text was not read; "not seeded" means the organisation supplies the entry. Jurisdiction codes are those of the `jurisdiction` dictionary (package Chapter 19); scheme codes are those of `regulatory_scheme`.

D.1 Australia (AU)

Ref	Instrument / regulator	Applies to	Obligation	Required data elements → CDS field	Trigger / dates	Verification
D-AU-1	Consumer Goods (Cosmetics) Information Standard 2020 — ACCC (<code>au_cosmetics_information_standard_2020</code>)	Cosmetics made or imported for sale in Australia	<code>labelling_element</code>	Ingredient list on container (or product, or displayed at point of sale where size prevents): descending by mass or volume, or ≥1% descending then <1% any order then colour additives; INCI or English names; flavour/fragrance as such or by components → <code>MF_ingredient_list</code> , <code>MF_ingredient_naming_scheme</code> , <code>MF_ingredient_order_basis</code> , <code>MF_fragrance_disclosure</code> . Hand sanitiser with alcohol as primary active: alcohol % v/v and warning statements → <code>MF_active_ingredients</code> , <code>MF_warning_statements</code>	Products manufactured after the 2020 standard's transition	Verified 2026-09-20 [B1] [B2]
D-AU-2	Industrial Chemicals Act 2019 — AICIS (<code>au_aicis_registration</code>)	Importers and manufacturers of cosmetics containing industrial chemicals	<code>pre_market</code> (business-level)	Business registration before import or manufacture; ingredient categorisation (exempted / reported / assessed); annual declaration by 30 November → organisation-level record; <code>CMP_market_registrations</code> status <code>in_scope_business_registered</code>	Any commercial import; no value threshold	Verified 2026-09-20 [B3]
D-AU-3	Therapeutic Goods Act 1989; TGO 92 (non-prescription medicine labels); Australian regulatory guidelines for sunscreens — TGA (<code>au_artg_listed</code> , <code>au_artg_registered</code>)	Primary sunscreens; listed and registered complementary medicines and supplements	<code>pre_market</code> ; <code>labelling_element</code> ; <code>warning_statement</code>	ARTG listing/registration number (AUST L / AUST R); sponsor name and address; active ingredients by Australian Approved Name with proportions; net quantity; batch number; expiry; storage conditions; permitted indications; mandatory excipient declarations; minimum text sizes → <code>CMP_market_registrations</code> (id, sponsor), <code>MF_active_ingredients</code> , <code>VAR_net_quantity</code> , <code>MF_directions</code> , <code>MF_warning_statements</code> , <code>MF_claims</code> (scheme <code>au_permitted_indications</code>)	Before supply in Australia	Verified 2026-09-20 [B4]
D-AU-4	Poisons Standard (SUSMP) scheduling of ingredients — TGA/states	Cosmetics and health products containing	<code>restricted_content</code>	Scheduling status per ingredient; label statements where scheduled → <code>MF_warning_statements</code>	Ingredient-dependent	Retrieved (not spot-verified)

Ref	Instrument / regulator	Applies to	Obligation	Required data elements → CDS field	Trigger / dates	Verification
		scheduled substances				
D-AU-5	Australian Dangerous Goods Code edition 7.9 – NTC/state regulators (<code>au_adg_code_7_9</code>)	Aerosols, flammable liquids (alcohol-based fragrance, sanitiser), certain nail products	transport	Dangerous-goods declaration per CDS-1900 §12	Mandatory from 1 October 2025	Verified 2026-09-20 (CDS-1900 Appendix C)

D.2 New Zealand (NZ)

Ref	Instrument / regulator	Applies to	Obligation	Required data elements → CDS field	Trigger / dates	Verification
D-NZ-1	Cosmetic Products Group Standard 2020 (HSR002552) – EPA (<code>nz_cosmetic_products_group_standard_2020</code>)	Cosmetic products imported or manufactured in NZ	labelling_element; pre_market (nano notification)	Ingredient list (≥1% descending, <1% any order, colour additives, flavour/fragrance wording) → MF_ingredient_list, MF_ingredient_order_basis; manufacturer source or batch code → lot-level (R039); "nano" marking → MF_nanomaterial_ingredients; nanomaterial notification to the EPA at first import → CMP_market_registrations. Alternative compliance: a label compliant with AU, US, CA, UK or EU cosmetic labelling requirements satisfies the ingredient-list and schedule conditions → CMP_market_registrations status alternative_compliance with the relied-on jurisdiction recorded	Group Standard in force from 30 April 2021; four-year transition to 30 April 2025 for Labelling, SDS and Packaging Notices	Verified 2026-09-20 [B5]
D-NZ-2	Sunscreen (Product Safety Standard) Act 2022 – MBIE (<code>nz_sunscreen_product_safety_standard</code>)	Primary sunscreens	pre_market; rating_label	Compliance with AS/NZS 2604 (SPF, broad-spectrum, water resistance testing) → MF_spf_value, MF_spf_rating_scheme; the Group Standard also permits labelling to the Australian Therapeutic Goods Order for primary sunscreens	From September 2022	Retrieved (not spot-verified)
D-NZ-3	Dietary Supplements Regulations 1985 / natural health products reform – MPI/Medsafe	Supplements	labelling_element	Not seeded – organisation supplies	–	Not seeded

D.3 United States (US, with California noted)

Ref	Instrument / regulator	Applies to	Obligation	Required data elements → CDS field	Trigger / dates	Verification
D-US-1	FD&C Act and Fair Packaging and Labeling Act; 21 CFR parts 701 and 740 – FDA (<code>us_fda_cosmetic_labeling_21cfr701</code>)	Cosmetics sold at retail	labelling_element; warning_statement	Principal display panel: identity statement and net quantity (US customary units, metric optional) → VAR_net_quantity; information panel: name and place of business (with "Manufactured for" / "Distributed by" where applicable) → CMP_responsible_person; ingredient declaration in descending order of predominance, colour additives and ≤1% ingredients in any order, "fragrance"/"flavor" permitted, cosmetic-drug products declare active ingredients first → MF_ingredient_list, MF_ingredient_order_basis, MF_active_ingredients; required warnings (aerosols, feminine deodorant sprays, children's bubble bath, flammables; "safety not determined" where unsubstantiated) → MF_warning_statements; English mandatory	Ongoing	Verified 2026-09-20 [B6]
D-US-2	Modernization of Cosmetics Regulation Act of 2022 (MoCRA), FD&C Act s. 607 – FDA (<code>us_mocra_product_listing</code>)	Cosmetics distributed in the US (small-business exemptions apply, with exclusions)	pre_market (facility registration, product listing); post_market (adverse events); labelling_element	Facility registration (biennial renewal); product listing including ingredients, updated annually; responsible person contact for adverse-event reporting on the label; serious adverse events reported within 15 business days → CMP_market_registrations (listing reference), CMP_responsible_person	Enforcement of registration and listing from 1 July 2024; listing within 120 days of first marketing	Verified 2026-09-20 [B7]
D-US-3	OTC drug labelling (Drug Facts), 21 CFR 201.66 and the sunscreen monograph – FDA (<code>us_fda_otc_drug</code>)	Sunscreens, antiperspirants, anti-dandruff and other cosmetic-drug products	labelling_element	Drug Facts panel with active ingredients and purposes before cosmetic ingredients; SPF and broad-spectrum statements per monograph → MF_active_ingredients, MF_spf_value, MF_spf_rating_scheme= <code>fda_sunscreen_monograph</code> , MF_warning_statements	Ongoing	Verified in part 2026-09-20 [B6] (21 CFR 701.3(d));

Ref	Instrument / regulator	Applies to	Obligation	Required data elements → CDS field	Trigger / dates	Verification
						monograph text not read
D-US-4	Dietary Supplement Health and Education Act; 21 CFR 101.36 — FDA (us_fda_dietary_supplement_labeling)	Dietary supplements	labelling_element	Statement of identity, net quantity, Supplement Facts panel (serving size, servings per container, dietary ingredients and amounts, %DV), ingredient list immediately below the panel, structure/function claim disclaimer, name and place of business → MF_serving_size, MF_servings_per_container, MF_nutrition_values (scheme supplement_facts), MF_ingredient_list, MF_claims, CMP_responsible_person	Ongoing	Verified 2026-09-20 [B8]
D-US-5	Proposition 65 (California), 27 CCR 25602–25603 — OEHHA (us_ca_prop65_warning)	Products causing exposure to listed chemicals, sold to California consumers by businesses with 10+ employees	warning_statement (listing_element for internet sales)	Warning on the product display page, a clearly marked "WARNING" hyperlink, or a warning prominently displayed before purchase; new short-form content names at least one chemical per endpoint; retailers have 60 days to update online warnings after notice → MF_warning_statements (jurisdiction US-CA), projected to listings	Amended regulations effective 1 January 2025; products manufactured and labelled before 1 January 2028 may use the prior short-form	Verified 2026-09-20 [B14]

D.4 European Union (EU)

Ref	Instrument / regulator	Applies to	Obligation	Required data elements → CDS field	Trigger / dates	Verification
D-EU-1	Regulation (EC) No 1223/2009 on cosmetic products, Articles 4, 11, 13, 19 — European Commission and member-state authorities (eu_cosmetics_regulation_1223_2009, eu_cpnp_notification)	Cosmetics made available in the EU	pre_market (Responsible Person, CPNP notification, product information file); labelling_element	Responsible Person name and address (EU) on container and packaging; country of origin for imported products; nominal content by weight or volume (exemptions <5 g/ml, samples, single-application); date of minimum durability (hourglass or "best used before the end of") where durability ≤30 months, otherwise period after opening symbol with months; particular precautions; batch number or product reference; function unless obvious; list of ingredients preceded by "ingredients", descending order by weight at time of addition, <1% any order, colourants may be listed last, "parfum"/"aroma", nanomaterials marked " (nano)", range colourants with "may contain"/"+/-"; product information file kept 10 years; notification of label and packaging photograph → CMP_responsible_person, CMP_country_of_origin, VAR_net_quantity, MF_minimum_durability_policy, MF_period_after_opening_months, MF_warning_statements, MF_ingredient_list, MF_ingredient_order_basis, MF_range_colourant_convention, MF_nanomaterial_ingredients, CMP_market_registrations	Ongoing	Verified 2026-09-20 [B9]
D-EU-2	Commission Regulation (EU) 2023/1545 amending Annex III (fragrance allergens) (eu_fragrance_allergens_2023_1545)	Cosmetics containing listed fragrance allergens above 0.001% (leave-on) or 0.01% (rinse-off)	labelling_element	Individual naming of each listed allergen in the ingredient list in addition to "parfum"/"aroma" → CMP_fragrance_allergen_declarations, MF_ingredient_list	Products placed on the market from 31 July 2026 must comply; products placed earlier may be made available until 31 July 2028	Verified 2026-09-20 [B10]
D-EU-3	Regulation (EU) No 655/2013 (common criteria for claims)	Cosmetic claims	restricted_content	Claims must satisfy legal compliance, truthfulness, evidential support, honesty, fairness and informed decision-making criteria → MF_claims evidence records	Ongoing	Retrieved (not spot-verified)
D-EU-4	Regulation (EU) 2023/988 General Product Safety Regulation (eu_gpsr_2023_988)	Non-harmonised consumer products in this profile (beauty tools,	listing_element; labelling_element	Online offers must show manufacturer name, postal and electronic address, product identifier (type/batch/serial), warnings and safety information and a picture; an EU-established responsible economic operator → CMP_responsible_person, CMP_product_identifier, MF_warning_statements	Applies from 13 December 2024	Verified 2026-09-20 (CDS-1800 Appendix C)

Ref	Instrument / regulator	Applies to	Obligation	Required data elements → CDS field	Trigger / dates	Verification
		accessories) and cross-cutting distance-selling duties				

D.5 United Kingdom (GB)

Ref	Instrument / regulator	Applies to	Obligation	Required data elements → CDS field	Trigger / dates	Verification
D-UK-1	Regulation (EC) 1223/2009 as retained (UK Cosmetics Regulation) and Cosmetic Products Enforcement Regulations 2013 — OPSS (<code>uk_cosmetics_regulation</code> , <code>uk_scpn_notification</code>)	Cosmetics made available in Great Britain	<code>pre_market</code> (UK Responsible Person; SCPN notification before placing on the market; PIF); <code>labelling_element</code> ; <code>post_market</code> (serious undesirable effects)	UK Responsible Person with a UK established address (not a PO box or mail-forwarding address) named on packaging (mandatory for all products from 31 December 2022 for those previously on the EU market); notification through the Submit Cosmetic Product Notification service with category, name, RP, PIF address, urgent contact, nanomaterials, CMR substances, ingredient summary, label image and packaging photograph; label elements as per Article 19 in English → <code>CMP_responsible_person</code> , <code>CMP_market_registrations</code> (scheme <code>uk_scpn_notification</code>), same label fields as D-EU-1	New products since 1 January 2021	Verified 2026-09-20 [B9] [B11]
D-UK-2	Fragrance allergen labelling in GB	Cosmetics with fragrance allergens	<code>labelling_element</code>	GB retained the pre-2023 Annex III allergen list at the end of the transition period; whether GB adopts the expanded list is a register review item → <code>CMP_fragrance_allergen_declarations</code> with scheme <code>uk_fragrance_allergens_retained</code>	Review each release	Not verified — organisation confirms current GB Annex III

D.6 Canada (CA)

Ref	Instrument / regulator	Applies to	Obligation	Required data elements → CDS field	Trigger / dates	Verification
D-CA-1	Food and Drugs Act; Cosmetic Regulations (C.R.C., c. 869) ss. 18, 21.2–21.5, 30–31; Consumer Packaging and Labelling Act — Health Canada (<code>ca_cosmetic_notification</code>)	Cosmetics sold in Canada	<code>pre_market</code> (notification); <code>labelling_element</code>	Cosmetic Notification Form within 10 days after first sale, including product name, function (leave-on / rinse-off), ingredient list with exact concentrations or concentration ranges, form, manufacturer or importer name and address in Canada; revised notification within 10 days of any inaccuracy; ingredient list on the outer label using INCI names (or chemical name where none; Schedule EU technical name / INCI / French equivalent alternatives), botanicals by genus and species, shade-range colourants with "+/-" or "may contain/peut contenir"; all other required label information in English and French; Cosmetic Ingredient Hotlist restrictions → <code>CMP_market_registrations</code> (CNF reference), <code>MF_ingredient_list</code> (naming scheme <code>inci</code>), <code>MF_range_colourant_convention</code> , <code>CMP_responsible_person</code> , localisation dimension per CDS-300 §18	Within 10 days of first sale; sale prohibited after the 10-day period if not notified	Verified 2026-09-20 [B12]
D-CA-2	Natural Health Products Regulations (SOR/2003-196) — Health Canada (<code>ca_nhp_product_licence</code>)	Vitamins, minerals, herbal remedies and other natural health products	<code>pre_market</code> (product licence); <code>labelling_element</code>	Product licence (NPN) before sale; application includes medicinal ingredients with proper and common names, quantity per dosage unit and potency, non-medicinal ingredients with purpose, label text; licence changes notified within 60 days → <code>CMP_market_registrations</code> (id = NPN), <code>MF_active_ingredients</code> , <code>MF_ingredient_list</code>	Before sale	Verified 2026-09-20 [B13]

D.7 Other markets

Jurisdiction	Status	Note
Japan (JP), China (CN), Singapore (SG), Republic of Korea (KR), India (IN), Brazil (BR)	Not seeded	The <code>jurisdiction</code> dictionary carries codes so an organisation can extend the register; China's cosmetics regime (CSAR, NMPA filing and registration) and Japan's Pharmaceutical and Medical Device Act notably differ from the seeded markets and require local regulatory advice before publication.

END OF CDS-1600 v0.2 REVIEW DRAFT

Commerce Data Standard (CDS)

CDS-1700 — Food, Beverage and Grocery Industry Profile

Field	Value
Status	vo.2 Review Draft (working source; not an approved standard)
Release	CDS v0.2 (single corpus release per ADR-D5); chapter added in the 2026-09-20 industry-profile expansion
Date	2026-09-20
Supersedes	Nothing — first edition. Package Chapter 17 (Food & Beverage: allergens, allergen facets, dietary claim types, storage conditions) is shipped with this profile in package o.8.o.
Normative status	§1, §3, §5–§14, §19 and Appendix A are normative. §2 (pointer), §4, §15–§18, §20, §21 and Appendices B–D are informative; Appendix D seeds the jurisdiction requirement register whose record structure and obligations are normative in CDS-1500 §2.2. Every table is individually marked.
Primary audience	Grocery and specialty-food merchandisers, regulatory and food-safety owners, data stewards, PIM architects, catalogue managers, UX teams, developers and AI enrichment designers
Depends on	CDS-000 through CDS-1500, especially CDS-200, CDS-300 (CMP_, PRC_ conditional), CDS-400 (dictionaries, units), CDS-500 (preflight, verification), CDS-600 (facets), CDS-700 (evidence, claims), CDS-900 (platform profiles), CDS-1500 (profile model, requirement levels, profile register, jurisdiction requirement register), CDS-1900 (kits, bundles, dangerous goods), CDS-2000 (pet food cross-reference)
Companion package	CDS Reference Dictionary: Chapter 17 (Food & Beverage), Chapter 19 (Jurisdictions & Regulatory Schemes); column contract per CDS-1500 Appendix E
Profile identifier	<code>cds.profile.food_beverage.v0_2</code> (registered in CDS-1500 §2.1)
Research basis	REVIEW-020 (global commerce vertical market research, 2026-09-20): food and beverages is a top-three worldwide ecommerce revenue segment (estimated at roughly US\$0.9 trillion for 2026, with the fastest growth of the large segments); Food & Drink is the fourth-largest Shopify Plus vertical worldwide (8.3% of Plus stores).

Terminology follows CDS-100. Attribute requirement levels (R, C, REC, O, N/A) are those of CDS-1500 §4 and are not restated here.

1. Purpose and Scope (*normative*)

CDS-1700 defines the industry profile for packaged and fresh food, non-alcoholic and alcoholic beverages, and grocery consumables. It covers ambient, chilled and frozen goods; multipacks, cases and hampers; variable-weight products; and the regulated information that every market attaches to a food label: ingredients, allergens, nutrition, quantity, date marking, storage, origin and alcohol content. Dietary supplements and complementary medicines are homed in CDS-1600 §14 and cite this chapter; pet food is homed in CDS-2000 §12 and cites this chapter's allergen and nutrition mechanics.

The data shape is distinctive because most of the record is regulated content whose authority is the label and the manufacturer's specification, because "contains" and "may contain" have different legal meanings, because nutrition values are numeric tables under jurisdiction-specific schemes, and because the sellable unit hierarchy (each, inner, case) and variable weight break the assumptions of a simple product-variant model. In every seeded market, mandatory food information must be available to the consumer before an online purchase is concluded, which makes the completeness of this data a publication gate, not a merchandising nicety.

CDS1700-R001 An implementation claiming the Food and Beverage Profile **MUST** represent ingredients, allergen declarations, nutrition values, net quantity, date-marking policy, storage conditions, origin and alcohol content as structured, typed attributes conforming to CDS-200 §7 and CDS-400 §18, never solely as free text.

CDS1700-R002 An implementation claiming this profile **MUST** distinguish regulated declarations (allergens, nutrition, origin, alcohol content) from dietary and marketing claims, and **MUST** govern every claim under CDS-700 §7 and §8 of this chapter.

CDS1700-R003 An implementation **MUST** maintain a jurisdiction requirement register (CDS-1500 §2.2) for every market it publishes food to, and publication preflight for a market **MUST** evaluate the mandatory food information required by that market before the offer is made available.

CDS1700-R004 This profile **MUST NOT** weaken any rule of CDS-1500 that it reuses; where this chapter is silent, CDS-1500 and the core chapters govern.

2. Profile Model (*informative pointer*)

This profile applies the industry profile model of CDS-1500 §2, the requirement levels of CDS-1500 §4 and the jurisdiction requirement register of CDS-1500 §2.2 without restatement. Its identifier and dictionary bindings are recorded in CDS-1500 §2.1; its seeded jurisdiction entries are in Appendix D.

3. Product Model (*normative*)

A food product is a recipe or formulation offered in one or more sellable presentations. The Product is the formulation; Variants are the sellable units that differ by net quantity, pack configuration, flavour or vintage. Above the variant sits the sale-unit hierarchy (each, inner pack, case), each level of which may carry its own trade identifier.

Product: Toasted Muesli (informative example)
Product scope:

```

CAT_product_type = breakfast_cereal
MF_ingredient_list = [rolled oats (55%), honey, almonds, ...] # ordered, with characterising %
CMP_allergen_declarations = [{allergen: oats, declaration: contains, gluten: true},
                              {allergen: almond, declaration: contains},
                              {allergen: milk, declaration: may_contain, basis: shared_line}]
MF_nutrition_values = {scheme: nip_au_nz, per_100g: {...}, per_serving: {...}, serving_size_g: 45}
MF_storage_condition = ambient
MF_date_mark_type = best_before
CMP_origin_declaration = {jurisdiction: AU, type: made_in_australia, australian_ingredients_band: gte_90_percent}
Variant scope:
  VAR_net_quantity = {value: 750, unit: gram}
  VAR_gtin = 0930000000001
  VAR_sale_unit_level = each

```

CDS1700-R005 Net quantity, pack configuration (count of identical units), flavour or variety, and vintage MUST be modelled at variant scope where they identify distinct sellable units (variant boundary rule: CDS-200 §5); ingredients, allergens, nutrition per 100 g or ml, origin and claims MUST remain at product scope unless the fact genuinely differs between variants (a flavour variant normally carries its own ingredient list and allergen declarations).

CDS1700-R006 Where a flavour or variety variant has its own ingredient list, allergen declarations or nutrition values, those facts MUST be stored at variant scope for that variant and the product-scope values MUST NOT be published as if they applied to it.

CDS1700-R007 Sale-unit levels (each, inner, case, pallet) MUST be modelled as a declared hierarchy with a level identifier, the quantity of the child unit contained, and the trade identifier of the level (GTIN-13/GTIN-14 or equivalent) where one exists; a case MUST NOT be modelled as an unrelated product.

CDS1700-R008 A multipack of identical units MUST be represented as a variant with a pack count; a bundle or hamper of different products MUST be represented as a kit with component relationships per CDS-1900 §11, with allergen and nutrition information carried by each component and the kit-level declarations derived, never authored.

4. Product Families and Category Profiles (*informative baseline; the inheritance rule of CDS1500-R016 applies*)

Product family / category	Required	Recommended	Typical variant options
Ambient packaged food (pantry, snacks, cereals, confectionery)	ingredient list; allergen declarations; nutrition values; net quantity; date-mark type; storage condition; origin per market	dietary claim records; serving size; characterising ingredient percentages	size, flavour, multipack
Chilled and frozen food	as ambient plus temperature class and shelf-life policy	thawing/handling instructions; minimum remaining shelf life at dispatch	size, flavour
Fresh produce, meat, seafood (variable weight)	net quantity basis (per kg or per unit); origin; storage; catch-weight flag	grade; cut; sustainability certifications	weight band, pack
Bakery	ingredient list; allergens; date-mark type	nutrition (where required); prepacked-for-direct-sale status	size
Non-alcoholic beverages	ingredient list; allergens; nutrition; net volume; storage; caffeine and sweetener warnings where applicable	serving size; sugar content claims	volume, multipack
Alcoholic beverages	alcohol by volume; standard drinks or units where required; allergens; net volume; pregnancy or health warnings per market; age restriction	vintage, varietal, region, closure (wine); style (beer); ingredients and energy where the market requires	volume, vintage, multipack
Infant and special-purpose foods	as ambient plus special labelling regime	—	size
Grocery non-food consumables (cleaning, paper)	net quantity; hazard statements where chemical	—	size, multipack
Meal kits, hampers, gift sets	kit composition; component declarations derived	—	configuration

5. Ingredients (*normative; seed guidance informative*)

Field	Meaning	Type
MF_ingredient_list	Ordered statement of ingredients as declared, with compound ingredients expanded or named per the label	Ordered list of entries (name, optional canonical reference, optional percentage, optional function class and additive code, optional sub-ingredients)
MF_ingredient_order_basis	Ordering rule applied	Governed value: <code>descending_by_ingoing_weight</code> (the rule in every seeded market), with declared exceptions for water, <2% groupings (US) or similar
MF_characterising_ingredient_percentages	Quantitative ingredient declarations (QUID)	Structured list (ingredient, percentage)
MF_additive_declaration_scheme	How additives are named	Governed value: <code>function_class_plus_name_or_number</code> , <code>name_only</code>

CDS1700-R009 MF_ingredient_list MUST be stored as an ordered list whose order is declared significant (CDS400-R040) and MUST preserve the label order; the implementation MUST NOT re-sort or de-duplicate the list without a governed change carrying label or specification evidence.

CDS1700-R010 Compound ingredients MUST be represented so that their sub-ingredients remain identifiable, because allergen declarations (§6) depend on sub-ingredients still present in the finished food.

CDS1700-R011 Characterising ingredient percentages MUST be numeric values linked to the ingredient entry (CDS400-R056), never text within the ingredient name alone.

CDS1700-R012 An ingredient list captured by extraction from label images or supplier documents (evidence class E2 or E3) MUST be reviewed by a human before acceptance regardless of confidence (CDS-700 §19).

6. Allergen Declarations (*normative; dictionary seed informative*)

Allergen information is the highest-consequence data in this vertical. The profile separates three things that are routinely conflated: the regulated declared name of an allergen in a given market, the declaration type (present as an ingredient versus precautionary "may contain"), and the customer-facing facet a storefront chooses to expose.

Field	Meaning	Type
CMP_allergen_declarations	Allergen declaration records	List of records: allergen (allergen dictionary value_id), declaration type (contains, may_contain, free_from_claimed), basis (ingredient, processing aid, cross-contact, shared line), gluten flag where the allergen is a gluten-containing cereal, market naming override where a market requires a different required name
CMP_allergen_scheme	The market allergen scheme the declarations are evaluated against	Governed value from regulatory_scheme (e.g. au_fsanx_1_2_3_allergens, eu_fic_1169_2011_annex_ii, us_falcpa_faster, ca_fdr_priority_allergens)

CDS1700-R013 Allergen declarations MUST be stored as structured records bound to the governed allergen dictionary, one record per allergen, and MUST NOT be stored only as prose inside the ingredient list or description.

CDS1700-R014 The declaration type MUST distinguish an allergen present as an ingredient or processing aid (contains) from a precautionary cross-contact statement (may_contain); the two MUST NOT be merged into one list, and a channel projection MUST label them differently.

CDS1700-R015 The declared name projected to a market MUST be the required name of that market's scheme (for example the individual tree-nut name required in Australia, or the "nuts" group emphasis in the EU); the canonical dictionary value MUST resolve to the required name through the market mapping, never through free text authored per product.

CDS1700-R016 Allergen declarations MUST NOT be derived automatically from the ingredient list without human review; AI or rules MAY propose a declaration from an ingredient, but acceptance MUST require review regardless of confidence, and the absence of an allergen in the ingredient list MUST NOT generate a "free from" declaration (CDS700-R018).

CDS1700-R017 A "free from" or "allergen-free" statement is a claim (§8) and MUST fail validation when the named allergen appears in a contains declaration for the same product or variant (the CDS1600-R033 pattern); a may_contain declaration for the same allergen MUST trigger review of the claim.

CDS1700-R018 Allergen declarations MUST be evaluated at publication preflight for each market against the market's scheme: an allergen required by the market that is present in the ingredient list without a declaration MUST block publication to that market (CDS400-R033 pattern).

Informative — seeded schemes. The package allergen dictionary carries the regulated required names of the seeded jurisdictions as canonical values with an ordered allergen_facet mapping to the fourteen EU groups (cereals containing gluten, crustaceans, eggs, fish, peanuts, soybeans, milk, nuts, celery, mustard, sesame, sulphites, lupin, molluscs). Australia's Plain English Allergen Labelling (Standard 1.2.3 and Schedule 9, mandatory from 25 February 2024) requires individual tree nuts, individual gluten cereals (barley, oats, rye when containing gluten), wheat, fish, crustacean, mollusc, egg, milk, lupin, peanut, soy, sesame and sulphites at 10 mg/kg or more, declared in bold in the ingredient list and in a "Contains" summary statement. The United States requires nine major allergens (milk, eggs, fish, crustacean shellfish, tree nuts, peanuts, wheat, soybeans, sesame) with species or type named. Canada's priority allergens add mustard and sulphites. Appendix D records the per-market rules with verification dates.

7. Nutrition Values (*normative*)

Field	Meaning	Type
MF_nutrition_values	Nutrition declaration	Structured object: scheme (governed: nip_au_nz, eu_nutrition_declaration, us_nutrition_facts, ca_nutrition_facts_table, supplement_facts, typical_analysis_pet_food), basis columns (per 100 g / 100 ml and/or per serving as the scheme requires), serving size as typed measurement, servings per package, nutrient rows (nutrient identifier, value, unit, rounding rule applied, %DV or %RI where the scheme uses it)
MF_energy_kj / MF_energy_kcal	Energy values per basis	Typed numbers (derived from the nutrition object)
MF_front_of_pack_rating	Voluntary or mandatory front-of-pack scheme result	Structured (scheme, value, calculation version)

CDS1700-R019 Nutrition values MUST be numeric, unit-bearing values under a declared scheme (CDS400-R056–R057); a scheme MUST be declared because nutrient sets, mandatory bases, rounding and %DV reference amounts differ by market, and one value set MUST NOT be presented as satisfying a scheme it was not prepared for.

CDS1700-R020 Where a market requires both a per-100 basis and a per-serving basis (Australia and New Zealand), both MUST be present; where a market requires per-serving with a defined serving size (United States, Canada), the serving size MUST be a typed measurement with its household-measure label where the scheme requires it.

CDS1700-R021 Nutrition values MUST NOT be generated, estimated or "filled in" by AI; extraction from a label image or specification is permitted only as a proposal with mandatory human review (CDS-700 §19), and an accepted value MUST retain provenance to the specification or label it came from (E1/E2).

CDS1700-R022 A front-of-pack rating (for example a health star rating, a nutrition symbol or a nutrient-profile score) MUST be stored with its scheme and calculation version and MUST NOT be recomputed by the PIM unless the organisation declares itself the calculating authority under CDS-200 §13.

8. Dietary Suitability, Claims and Certifications (*normative; dictionary seed informative*)

Dietary suitability is the primary purchase driver in this vertical and the most abused data. The profile treats every dietary statement as a claim record whose type comes from the governed `dietary_claim_type` dictionary and whose acceptance depends on evidence appropriate to the type: a regulated definition (gluten free), a certification (organic, halal, kosher), or a manufacturer declaration where no regulated definition exists (vegan, vegetarian).

Attribute	Structure	Requirement guidance
MF_dietary_claims	List of claim records: claim type (governed), evidence class and reference, certifier and certificate where applicable, jurisdictions in which the claim is published, owner, effective/expiry	C — required when any dietary statement is published
MF_nutrition_content_claims	Nutrition content claims ("low fat", "source of fibre") with the scheme and condition satisfied	C
MF_health_claims	Health claims with the scheme, the permitted relationship and any required dietary-context statement	C
MF_certifications	Certification records (scheme, certifier, identifier, scope, expiry)	C

CDS1700-R023 A dietary claim MUST exist as a claim record with a governed claim type and an evidence reference; a claim type whose definition is regulated in a market (for example gluten free) MUST carry the regulated threshold or definition satisfied for that market in the claim record.

CDS1700-R024 Dietary claims MUST NOT be inferred from the ingredient list by rules or AI without human review; a product with no animal ingredients is not thereby "vegan" in the canonical data (CDS400-R066, CDS700-R018).

CDS1700-R025 Nutrition content claims and health claims MUST reference the scheme under which they are permitted in each market they are published to; publication preflight MUST block a health claim not permitted under the declared scheme for that market or lacking a required accompanying statement.

CDS1700-R026 A customer facet for dietary suitability MUST be fed only by accepted claim records (a claim-driven facet, the CDS1500-R025 pattern) and MUST NOT be assigned per product from marketing copy or supplier tags.

CDS1700-R027 Certification expiry MUST withdraw the dependent claim projection without editing any product fact (CDS1500-R032).

9. Quantity, Pack Hierarchy, Unit Pricing and Variable Weight (*normative*)

CDS1700-R028 Net quantity MUST be a typed measurement with a declared unit (gram, kilogram, millilitre, litre, or count) at variant scope; drained weight, where declared, MUST be a separate typed value.

CDS1700-R029 Pack count MUST be an integer separate from unit net quantity (a "6 × 375 ml" multipack is pack count 6 and net quantity 375 ml), and the total net quantity MUST be derived, never authored.

CDS1700-R030 Where a market or channel requires unit pricing, the unit-price base measure MUST be derived from the typed net quantity and pack count and the price authority (CDS-200 §13; `PRC_` fields only where the PIM is the declared authority), never keyed by hand.

CDS1700-R031 Variable-weight (catch-weight) products MUST carry a catch-weight flag, a pricing basis (per kilogram or per unit), and a nominal or average weight as a typed value; they MUST NOT be presented as fixed-weight products, and the sellable unit's actual weight is an order-level observation, not canonical product data.

CDS1700-R032 Trade identifiers at each sale-unit level MUST be stored with their scheme (GTIN-8/12/13/14 or an organisation-declared alternative) and MUST validate per scheme (CDS-1100 GTIN rules).

10. Storage, Temperature, Shelf Life and Date Marking (*normative; dictionary seed informative*)

Field	Meaning	Type
MF_storage_condition	Governed storage condition (<code>storage_condition</code> dictionary: <code>ambient</code> , <code>store_below_25c</code> , <code>refrigerate</code> , <code>refrigerate_after_opening</code> , <code>frozen</code> , <code>keep_frozen_do_not_refreeze</code> , <code>store_in_cool_dry_place</code>)	Product
MF_temperature_class	Logistics temperature class (<code>ambient</code> / <code>chilled</code> / <code>frozen</code>)	Governed value
MF_date_mark_type	The date-mark regime the product carries (<code>best_before</code> , <code>use_by</code> , <code>baked_on</code> , <code>packed_on</code> , <code>none_exempt</code>)	Governed value
MF_shelf_life_days	Shelf life from production under declared storage	Integer
MF_minimum_remaining_shelf_life_days	Policy: minimum remaining life at dispatch	Integer
MF_storage_instructions	Storage and after-opening instructions as labelled	Structured statement records

CDS1700-R033 Storage condition and temperature class MUST be governed values; free text such as "keep cool" MUST NOT satisfy the requirement.

CDS1700-R034 The date-mark type MUST be a governed value that distinguishes a safety-based "use by" from a quality-based "best before"; the actual date on a unit is lot-level and MUST NOT be stored as a canonical product attribute (the CDS1600-R039 pattern), but the shelf-life policy MAY be.

CDS1700-R035 A product whose date-mark type is `use_by` MUST carry a minimum-remaining-shelf-life policy before publication to a channel that ships to consumers, so that fulfilment can be verified against it.

11. Origin and Provenance (*normative*)

Field	Meaning	Type
CMP_country_of_origin	Country of origin under the applicable determination rule	Governed country reference
CMP_origin_declarations	Per-market origin declaration records	List: jurisdiction, scheme, declaration type (e.g. <code>grown_in</code> , <code>produced_in</code> , <code>made_in</code> , <code>packed_in</code> , <code>product_of</code> , <code>origin_statement</code>), percentage-of-local-ingredients band where the scheme uses one, mark or logo eligibility
MF_region_of_origin	Sub-national origin (wine region, PDO/PGI name)	Governed value where a scheme protects the name

CDS1700-R036 Origin MUST be stored as a governed country reference plus per-market declaration records; a single free-text origin string MUST NOT be projected to markets whose schemes require a specific declaration form.

CDS1700-R037 Protected geographical names (designations of origin and geographical indications) MUST be governed values with evidence and MUST NOT be assigned from supplier descriptions without verification.

12. Alcoholic Beverages (*normative*)

Field	Meaning	Type
MF_alcohol_by_volume	Alcoholic strength by volume at 20 °C	Decimal percentage (typed)
MF_standard_drinks / MF_alcohol_units	Standard drinks (or units) per container under the market's definition	Decimal with scheme (e.g. <code>au_nz_standard_drink_10g</code> , <code>uk_unit_8g</code>)
CMP_alcohol_warnings	Mandatory warning statements per market (pregnancy warning, government warning)	Structured statement records with scheme, pictogram version and size class
MF_beverage_style	Wine varietal and vintage; beer style; spirit category	Governed values
CMP_age_restriction	Minimum purchase age per market	Per-market records

CDS1700-R038 Alcohol by volume MUST be a typed decimal percentage; where a market requires a different expression below a threshold (for example a "contains not more than X% alcohol by volume" statement), the projection MUST derive it from the typed value.

CDS1700-R039 Standard drinks or units MUST be stored with the definition scheme they were calculated under and MUST NOT be recalculated across schemes without a declared conversion rule (CDS400-R057).

CDS1700-R040 Mandatory alcohol warnings MUST be structured statement records with the scheme and the pictogram or wording version, evaluated per market at preflight; a product above a market's threshold lacking the market's warning MUST NOT be published to that market.

CDS1700-R041 Age restriction MUST be a governed per-market attribute evaluated at preflight and projected to channels that support age gating.

13. Warnings, Advisory Statements and Restricted Sale (*normative*)

CDS1700-R042 Warning and advisory statements required by a market (caffeine, phenylalanine, sweeteners, quinine, kava, royal jelly, choking hazards, "contains a source of..." statements) MUST be structured statement records with the mandating scheme, evaluated per market at preflight; the trigger conditions (ingredient present, quantity threshold) MUST be expressed as rules (CDS200-R034).

CDS1700-R043 Products subject to sale restrictions in a market (alcohol, restricted supplements, tobacco where in scope) MUST carry a governed restriction attribute per market and MUST NOT be published to a channel or market whose capability declaration cannot enforce the restriction.

14. Safety, Recalls and Dangerous Goods (*normative*)

CDS1700-R044 A recall or withdrawal MUST be expressed through lifecycle state and channel withdrawal (CDS-500 §12) with the affected lot range recorded as an observation, never by deleting the product; lot-level recall scope is outside canonical product data but MUST be linkable to the product.

CDS1700-R045 Grocery consumables that are hazardous chemicals or dangerous goods for transport (aerosols, high-strength spirits, cleaning chemicals) MUST carry a dangerous-goods declaration per CDS-1900 §12 and, where a market requires it, a safety data sheet reference (CDS-1900 §12).

15. Customer Facet Design (*informative — normative facet rules live in CDS-600*)

Facet	Recommended baseline behaviour	Anti-pattern
Dietary suitability	Claim-driven facet fed by accepted claim records (R026); labels express the claim type (Gluten free, Vegan)	Facet populated from supplier tags or inferred from ingredients
Allergen-free browsing	Exclusion filters driven by <code>contains</code> declarations only, with a visible note that precautionary statements are not filtered unless the organisation declares otherwise	Treating "may contain" as absence
Category and product form	Classification-driven	Cuisine and occasion mixed with product type

Facet	Recommended baseline behaviour	Anti-pattern
Size / pack	Net-quantity ranges and pack counts	Mixed grams and millilitres in one list
Origin	Country and region facets from governed values	Free-text origin
Storage	Ambient / chilled / frozen for delivery planning	—
Alcohol	Style, varietal, region, ABV range	Health claims

Zero-result handling, coverage thresholds and accessibility follow CDS-600 §10, §17 and §24.

16. Channel Projection Guidance (*informative — normative rules: CDS-500, CDS-900*)

Canonical concept	Metafield-style channel (informative)	Feed-style channel (informative)
Ingredient list, allergens, nutrition	Structured metafields generated from canonical records (Shopify's standard taxonomy exposes allergen, dietary and flavour attributes for food categories); read back and verified per market	Description blocks per market; no standard nutrition fields in the seeded feed channels
Net quantity, pack count, unit pricing	Typed metafields; variant title composition per declared policy	unit_pricing_measure / unit_pricing_base_measure where required by country; multipack; is_bundle for kits
Dietary claims	Only accepted claim records project; certification expiry withdraws	Attributes where the channel defines them
Alcohol	Age-gating and warning blocks per market	Channel alcohol policies restrict listings; the dated rules live in CDS-900
Storage and temperature class	Fulfilment metafields	Not published
Mandatory food information before purchase	Generated per-market information block on the product page (Appendix D)	—

Which layer feeds each channel is declared per attribute (CDS200-R026); feed-channel requiredness by country is a dated CDS-900 platform fact.

17. AI Enrichment and Review (*informative — normative AI rules live in CDS-700*)

Task	AI may propose	Deterministic or human control
Ingredient extraction from labels or specifications	Ordered candidate list	Mandatory review (R012); order preserved
Allergen declaration proposals	Candidate contains records from ingredients	Mandatory review regardless of confidence (R016); never auto-accepted
Nutrition value extraction	Parsed table with scheme candidate	Mandatory review (R021); unit and rounding validation
Dietary claim candidates	Candidate claim records flagged "evidence required"	Never accepted without evidence (R024)
Origin and protected names	Candidates from specification documents	Verification of protected names (R037)
Market obligation gaps	Candidate list of register entries triggered	Register evaluation deterministic; AI never marks obligations satisfied

18. Governance and Organisational Extensions (*informative — rules in CDS-1500 §23 and CDS-800*)

Organisations extend the allergen, storage and claim-type dictionaries under CDS-400 governance; regulated allergen lists change (Australia's 2024 required names, the US sesame addition) and are handled as governed dictionary migrations, never silent edits. The jurisdiction requirement register is reviewed on the cadence declared under CDS-800 §30, with a mandatory review whenever a market changes a mandatory particular.

19. Conformance Requirements (*normative — claims and levels per CDS-1000*)

CDS1700-R046 An implementation claiming the CDS Food and Beverage Profile MUST: store ingredients as ordered structured lists with compound ingredients identifiable (R009–R011); store allergen declarations as structured records distinguishing "contains" from "may contain", bound to the governed allergen dictionary, with market required-name mapping and preflight evaluation (R013–R018); store nutrition values as numeric, unit-bearing values under a declared scheme with the bases the scheme requires (R019–R022); govern every dietary, nutrition-content and health claim as an evidence-bearing record with claim-driven facets (R023–R027); store net quantity, pack count, sale-unit hierarchy and variable-weight facts as typed values (R028–R032); store storage, date-mark type and shelf-life policy as governed values with lot dates kept out of canonical data (R033–R035); store origin and alcohol facts per market (R036–R041); maintain and evaluate a jurisdiction requirement register for every market published to (R003, CDS-1500 §2.2); apply category-specific requirements per CDS1500-R009; and publish and verify channel representations under CDS-500.

CDS1700-R047 A Food and Beverage Profile claim MUST state which markets its register covers and whether alcoholic beverages and variable-weight products are in scope.

20. Worked Product Examples (*informative*)

20.1 Granola with tree nuts and a precautionary milk statement (Australia and EU)

```
CAT_product_type = breakfast_cereal
MF_ingredient_list = [rolled oats (55%), honey (12%), almonds (8%), sunflower oil, coconut, pepitas, cinnamon]
CMP_allergen_declarations =
  [{allergen: oats, declaration: contains, gluten: true, basis: ingredient},
   {allergen: almond, declaration: contains, basis: ingredient},
   {allergen: milk, declaration: may_contain, basis: shared_line}]
Market projections:
  AU (scheme au_fsanz_1_2_3_allergens): ingredient list with "oats", "almonds" in bold;
    summary statement "Contains: gluten, oats, almond"; precautionary line "May contain milk"
  EU (scheme eu_fic_1169_2011_annex_ii): "oats" and "almonds" emphasised; allergen group facets
    gluten_containing_cereals, tree_nuts; precautionary statement kept separate
MF_nutrition_values = {scheme: nip_au_nz, serving_size: 45 g, servings_per_package: 16.7,
  per_100g: {energy_kj: 1850, protein_g: 9.1, fat_g: 18.2, saturated_g: 4.0,
    carbohydrate_g: 55.0, sugars_g: 14.2, sodium_mg: 12},
  per_serving: {...}}
MF_dietary_claims = [] # "gluten free" would fail R017 (oats contain gluten)
VAR_net_quantity = {value: 750, unit: gram}
CMP_origin_declarations = [{jurisdiction: AU, scheme: au_cool_2016, type: made_in_australia,
  australian_ingredients_band: gte_90_percent, standard_mark: true}]
Preflight AU: allergens declared for every required allergen present -> pass
```

20.2 Craft gin (Australia, UK)

```
CAT_product_type = gin
MF_alcohol_by_volume = 42.0
MF_standard_drinks = {value: 23.3, scheme: au_nz_standard_drink_10g} # 700 ml container
MF_alcohol_units = {value: 29.4, scheme: uk_unit_8g}
CMP_alcohol_warnings = [{jurisdiction: AU, scheme: au_fsanz_2_7_1_alcohol, statement: pregnancy_warning_mark,
  size_class: "over 200 ml to 800 ml"}]
CMP_age_restriction = [{jurisdiction: AU, minimum_age: 18}, {jurisdiction: GB, minimum_age: 18}]
CMP_allergen_declarations = [] # none; ingredient list not required >1.2% ABV in GB/EU
VAR_net_quantity = {value: 700, unit: millilitre}
CMP_dangerous_goods = {regulated: true, un_number: "UN3065", class: "3", packing_group: "III",
  limited_quantity_eligible: true} # per CDS-1900 §12
Preflight AU: pregnancy warning present, age restriction enforceable on channel -> pass
```

20.3 Chilled catch-weight product

```
CAT_product_type = whole_chicken
MF_catch_weight = true
MF_pricing_basis = per_kilogram
MF_nominal_weight = {value: 1.6, unit: kilogram, tolerance: "±0.3"}
MF_temperature_class = chilled
MF_storage_condition = refrigerate
MF_date_mark_type = use_by
MF_minimum_remaining_shelf_life_days = 4
CMP_origin_declarations = [{jurisdiction: AU, scheme: au_cool_2016, type: grown_in_australia}]
Order-level observation: actual weight 1.52 kg -> price computed by the pricing authority, not the PIM
```

21. Reference Validation Cases (*informative — the profile's contribution to the cross-industry validation set, REVIEW-020*)

- a product whose ingredient list contains a required allergen with no declaration, which preflight must block per market (R018);
- a "gluten free" claim on a product with a contains declaration for a gluten cereal, which must fail (R017);
- a "may contain" statement that a channel projection must keep distinct from "contains" (R014);
- an allergen whose required name differs between Australia (individual tree nut) and the EU (nuts group), resolved by dictionary mapping (R015);
- a nutrition panel prepared for one scheme offered to a market with another scheme, which must not satisfy the requirement (R019);
- a flavour variant with its own ingredient list (R006);
- a "6 × 375 ml" multipack that must not collapse to "2.25 l" (R029);
- a catch-weight product presented as fixed weight (R031);
- a use-by product without a minimum-remaining-shelf-life policy (R035);
- an alcoholic beverage above threshold lacking the market's warning (R040);
- a case-level GTIN-14 that must validate and link to its each-level GTIN (R032);
- a hamper whose kit-level allergen summary must derive from components (R008).

Appendix A. Food and Beverage Attribute Baseline (*normative — dictionary bindings per CDS1500-R011*)

Field	Scope	Type	Baseline requirement
CAT_product_type	Product	Governed classification reference	R

Field	Scope	Type	Baseline requirement
MF_ingredient_list / MF_ingredient_order_basis	Product (variant per Ro06)	Ordered structured list; governed value	R for multi-ingredient foods; N/A for single-ingredient exempt foods
MF_characterising_ingredient_percentages	Product	Structured list	C — R where a market requires quantitative declaration
CMP_allergen_declarations / CMP_allergen_scheme	Product (variant per Ro06)	Structured records (allergen dictionary)	R
MF_nutrition_values	Product (variant per Ro06)	Structured object with declared scheme	R unless exempt in every target market
MF_front_of_pack_rating	Product	Structured (scheme, value)	C
MF_dietary_claims / MF_nutrition_content_claims / MF_health_claims / MF_certifications	Product	Structured records with evidence	C
VAR_net_quantity	Variant	Typed measurement	R
VAR_pack_count	Variant	Integer	C — R for multipacks
VAR_sale_unit_level / VAR_gtin	Variant and sale-unit levels	Governed level; identifier with scheme	R where trade identifiers exist
MF_catch_weight / MF_pricing_basis / MF_nominal_weight	Product/Variant	Boolean; governed value; typed measurement	R for variable-weight products
MF_storage_condition / MF_temperature_class	Product	Governed dictionary references	R
MF_date_mark_type / MF_shelf_life_days / MF_minimum_remaining_shelf_life_days	Product	Governed value; integers	R / REC / C (R for use-by products sold online)
CMP_country_of_origin / CMP_origin_declarations	Product	Governed country; per-market records	R where a market requires origin
MF_alcohol_by_volume / MF_standard_drinks / CMP_alcohol_warnings / CMP_age_restriction	Product/Variant	Typed values; statement records; per-market records	R for alcoholic beverages
MF_warning_statements	Product	Structured statement records	C — R where a market mandates a statement
CMP_responsible_person	Product (per market)	Governed record	R where a market requires the food business operator's name and address
CMP_dangerous_goods	Product/Variant	Declaration record per CDS-1900 §12	C
CMP_market_registrations	Product (per market)	Governed records	C

Appendix B. Dictionary Bindings (*informative — becomes governed data on adoption per CDS1500-R011*)

Dictionary key	Package chapter	Bound attribute(s)	Status at 0.8.0
allergen	17 (new)	CMP_allergen_declarations	Bound by this profile
allergen_facet	17 (new)	Customer-facing allergen grouping; EU group mapping	Bound
dietary_claim_type	17 (new)	MF_dietary_claims.type	Bound (claim-type registry; evidence-gated)
storage_condition	17 (new)	MF_storage_condition	Bound
jurisdiction / regulatory_scheme	19 (new)	CMP_* market records, claim and warning schemes	Bound via CDS-1500 §2.2
dangerous_goods_class	18 (new)	CMP_dangerous_goods	Bound via CDS-1900 §12

Deliberately not shipped: nutrient identifier lists per scheme (organisations adopt their regulator's nutrient set), country lists (ISO 3166 at adoption), protected geographical name registers, flavour vocabularies (open-ended), wine varietal and region lists (organisation- or registry-sourced).

Appendix C. References (*informative; retrieved 2026-09-20 unless stated*)

Ref	Source	Location	Use in this chapter
[F1]	FSANZ — Allergen labelling for food businesses (Standard 1.2.3, Schedule 9; PEAL from 25 February 2024); DAFF IFN 10-25	foodstandards.gov.au/business/labelling/allergen-labelling; agriculture.gov.au	Required allergen names, bold and summary statement (§6, D-AU-1)
[F2]	FSANZ — Standard 1.2.8 Nutrition information requirements (compilation 2024-10-29); Food Standards Code compilation (March 2026)	legislation.gov.au/F2015L00395	NIP content, per-serving and per-unit-quantity bases (§7, D-AU-2)
[F3]	FSANZ — Standard 1.2.7 Nutrition, health and related claims (compilation 2025-08-13); Schedule 4	legislation.gov.au/F2015L00394	General and high-level health claims, NPSC, dietary context statements (§8, D-AU-3)
[F4]	FSANZ — Standard 2.7.1 Labelling of alcoholic beverages (compilation 2025-08-13); Labelling of alcoholic beverages (2025-09-10)	legislation.gov.au/F2015L00469; foodstandards.gov.au	ABV statement, standard drinks, pregnancy warning label sizes (§12, D-AU-4)
[F5]	Country of Origin Food Labelling Information Standard 2016; business.gov.au guidance	legislation.gov.au/F2016L00528; business.gov.au	Standard mark, bar chart, kangaroo logo, origin statement (§11, D-AU-5)

Ref	Source	Location	Use in this chapter
[F6]	FSANZ — Call for information: Health Star Rating and NIP (Nov 2024)	consultations.foodstandards.gov.au	HSR voluntary status and 70% uptake target by 14 November 2025 (D-AU-6)
[F7]	Consumer Information Standards (Origin of Food) Regulations 2021 (NZ)	legislation.govt.nz/regulation/public/2021/0097	NZ origin disclosure for regulated fresh and single-ingredient foods from 12 February 2022 (D-NZ-1)
[F8]	U.S. FDA — Food Labeling Guide; 21 CFR 101.3, 101.4, 101.5, 101.7, 101.9 (eCFR)	fda.gov; ecfr.gov	Identity, net quantity, ingredient order, Nutrition Facts, name and place of business (§5, §7, §9, D-US-1)
[F9]	U.S. FDA — Food Allergies; FALCPA; FASTER Act (sesame from 1 January 2023); Guidance Edition 5 (2025)	fda.gov/food/nutrition-food-labeling-and-critical-foods/food-allergies	Nine major allergens, species and type naming, "Contains" statement (§6, D-US-2)
[F10]	EFSA — Food allergens (EU 14) (2025-04-22); Regulation (EU) No 1169/2011 Articles 9, 12, 14, 18, 21 (EUR-Lex; UK retained text)	efsa.europa.eu; eur-lex.europa.eu	EU mandatory particulars, distance-selling availability before purchase (§1, §6, D-EU-1)
[F11]	Regulation (EU) 2021/2117 (wine ingredient and nutrition labelling)	eur-lex.europa.eu	Wine sector labelling from 8 December 2023 (D-EU-2)
[F12]	Food Standards Agency (UK) — PPDS allergen labelling (Natasha's Law, 1 October 2021); Food allergen labelling technical guidance; Food Information Regulations 2014 (as at 2021-10-01)	food.gov.uk; legislation.gov.uk/ukxi/2014/1855	UK allergen rules, gluten-free 20 mg/kg, distance selling (D-UK-1)
[F13]	The Food (Promotion and Placement) (England) Regulations 2021; Food (Promotion and Presentation) (Wales) Regulations 2025	legislation.gov.uk/ukxi/2021/1368; legislation.gov.uk/wsi/2025/395	HFSS online placement and volume price promotion restrictions (D-UK-2)
[F14]	Canadian Food Inspection Agency — Food labelling requirements checklist; Front-of-package nutrition symbol (compliance 1 January 2026); Health Canada FOP guide v3	inspection.canada.ca; canada.ca	Bilingual labelling, NFI, priority allergens, FOP symbol (D-CA-1)
[F15]	Store Leads — Shopify Plus Statistics 2026 (updated 2026-09-11); global ecommerce revenue by segment (Statista Market Insights, March 2026, via secondary reporting)	storeleads.app; statista.com	Market basis (REVIEW-020)

Appendix D. Jurisdiction Requirement Register — Seed (*informative seed of the normative register defined in CDS-1500 §2.2*)

Conventions as in CDS-1600 Appendix D.

D.1 Australia (AU)

Ref	Instrument / regulator	Applies to	Obligation	Required data elements → CDS field	Trigger / dates	Verification
D-AU-1	Australia New Zealand Food Standards Code, Standard 1.2.3 and Schedule 9 (allergen declarations) — FSANZ; enforced by state and territory agencies and DAFF for imports (au_fsan_z_1_2_3_allergens)	Food for sale requiring a label (and unlabelled food via other means)	labelling_element	Required names: wheat, fish, crustacean, mollusc, egg, milk, lupin, peanut, soy, sesame, almond, Brazil nut, cashew, hazelnut, macadamia, pecan, pine nut, pistachio, walnut, barley/oats/rye when containing gluten, sulphites ≥10 mg/kg; declared in bold in the statement of ingredients and in a separate bold "Contains" summary statement in the same field of view; "gluten" listed with wheat and for gluten-containing cereals → CMP_allergen_declarations (scheme au_fsan_z_1_2_3_allergens), projection rules	Food manufactured or produced from 25 February 2024	Verified 2026-09-20 [F1]
D-AU-2	Standard 1.2.8 Nutrition information requirements; Schedule 12 format (au_fsan_z_1_2_8_nip)	Packaged food unless exempt (e.g. standardised alcoholic beverages, herbs, tea, single-ingredient produce, small packages)	labelling_element	Nutrition information panel: servings per package, serving size (g/ml), unit quantity (100 g/ml), energy (kJ, optionally kcal), protein, fat, saturated fat, carbohydrate, sugars (g), sodium (mg), plus any claimed nutrient; per serving and per 100 g/ml; ≤3 significant figures → MF_nutrition_values (scheme nip_au_nz)	Ongoing	Verified 2026-09-20 [F2]
D-AU-3	Standard 1.2.7 Nutrition, health and related claims; Schedule 4 (permitted claims); Schedule 5 (NPSC) (au_fsan_z_1_2_7_claims)	Foods bearing nutrition content or health claims	restricted_content	Health claims only where the food meets the nutrient profiling scoring criterion and the food–health relationship is in Schedule 4 (high level) or Schedule 4 / self-substantiated and notified (general level); dietary context statement; claims prohibited on kava, infant formula and most alcohol >1.15% → MF_health_claims, MF_nutrition_content_claims (scheme au_fsan_z_1_2_7_claims)	Ongoing	Verified 2026-09-20 [F3]

Ref	Instrument / regulator	Applies to	Obligation	Required data elements → CDS field	Trigger / dates	Verification
D-AU-4	Standard 2.7.1 Labelling of alcoholic beverages (au_fsanx_2_7_1_alcohol)	Beverages ≥0.5% ABV; food containing alcohol	labelling_element; warning_statement	Alcohol content as % ABV (or mL/100 mL) for >1.15%; "contains not more than X% alcohol by volume" for 0.5–1.15%; statement of approximate number of standard drinks (10 g ethanol) for >0.5%; pregnancy warning pictogram or mark for >1.15% with size classes by container volume (≤200 ml pictogram ≥8 mm; >200–800 ml mark ≥6 mm; >800 ml ≥9 mm; outer packs ≥11 mm); energy statement for prescribed beverages → MF_alcohol_by_volume, MF_standard_drinks (scheme au_nz_standard_drink_10g), CMP_alcohol_warnings	Pregnancy warning mandatory since 1 August 2023	Verified 2026-09-20 [F4]
D-AU-5	Country of Origin Food Labelling Information Standard 2016 – ACCC (au_cool_2016)	Food sold in Australia (priority and non-priority foods; imported foods)	labelling_element	Standard mark (kangaroo logo where grown, produced or made in Australia; bar chart with percentage of Australian ingredients; explanatory text; box) for priority foods grown, produced, made or packed in Australia; country of origin statement ("Made in X", "Product of X") for non-priority and imported foods, boxed for packaged priority foods → CMP_origin_declarations (scheme au_cool_2016, declaration type, ingredient band, mark eligibility)	Ongoing	Verified 2026-09-20 [F5]
D-AU-6	Health Star Rating system (voluntary front-of-pack) – Food ministers/FSANZ (au_health_star_rating)	Packaged foods	rating_label (voluntary)	HSR value and calculator version where displayed; ministers set a 70% uptake target by 14 November 2025 with mandating under consideration → MF_front_of_pack_rating (scheme au_health_star_rating)	Voluntary; review 2026	Verified 2026-09-20 [F6]
D-AU-7	Standards 1.2.4 (statement of ingredients), 1.2.5 (date marking), 1.2.6 (directions for use and storage), 1.2.10 (characterising ingredients)	Food requiring a label	labelling_element	Ingredients in descending ingoing weight; best-before / use-by regime; storage directions; characterising ingredient percentages → MF_ingredient_list, MF_date_mark_type, MF_storage_instructions, MF_characterising_ingredient_percentages	Ongoing	Retrieved 2026-09-20 (Code compilation index; standard texts not read in full)
D-AU-8	Unit Pricing Code (Competition and Consumer (Industry Codes—Unit Pricing) Regulations) – ACCC (au_unit_pricing_code)	Grocery retailers above the floor-area threshold and online grocery retailers	listing_element	Unit price per prescribed unit of measure displayed with the selling price online → derived per Ro30	Ongoing	Retrieved (not spot-verified)

D.2 New Zealand (NZ)

Ref	Instrument / regulator	Applies to	Obligation	Required data elements → CDS field	Trigger / dates	Verification
D-NZ-1	Australia New Zealand Food Standards Code (joint standards 1.2.3, 1.2.7, 1.2.8, 2.7.1 apply in NZ) – MPI	Food for sale in NZ	as D-AU-1 to D-AU-4	Same data elements as the joint standards; Standard 1.2.10 (characterising ingredients) is Australia-only; country of origin under the Code does not apply in NZ → CMP_allergen_scheme, MF_nutrition_values, CMP_alcohol_warnings as for AU	Ongoing	Verified 2026-09-20 [F2] (Code compilation notes application)
D-NZ-2	Consumer Information Standards (Origin of Food) Regulations 2021 – MBIE/Commerce Commission (nz_origin_of_food_2021)	Regulated single-ingredient fresh or thawed fruit, vegetables, meat, fish and seafood (and certain frozen items)	listing_element; labelling_element	Origin information disclosed when the item is supplied, offered or advertised for sale, including online; replacement statements where origin changes often → CMP_origin_declarations (scheme nz_origin_of_food_2021)	From 12 February 2022 (frozen items later per the regulations)	Verified 2026-09-20 [F7]

D.3 United States (US)

Ref	Instrument / regulator	Applies to	Obligation	Required data elements → CDS field	Trigger / dates	Verification
D-US-1	FD&C Act; Fair Packaging and Labeling Act; 21 CFR part 101 – FDA (us_fda_food_labeling_21cfr101)	Packaged food under FDA jurisdiction (USDA regulates meat, poultry and egg products)	labelling_element	Statement of identity and net quantity on the principal display panel (weight, measure or count; fluid measure for liquids); ingredient statement by common or usual name in descending order of predominance by weight (≤2% grouping permitted); name and place of business ("Manufactured for"/"Distributed by" qualifiers); Nutrition Facts per 21 CFR 101.9 (serving size in household measure plus metric, servings per container, calories, mandatory nutrients incl. added sugars, vitamin D, calcium, iron, potassium, %DV, dual-column rules) → VAR_net_quantity, MF_ingredient_list, CMP_responsible_person, MF_nutrition_values (scheme us_nutrition_facts)	Nutrition Facts compliance since 2020/2021	Verified 2026-09-20 [F8]

Ref	Instrument / regulator	Applies to	Obligation	Required data elements → CDS field	Trigger / dates	Verification
D-US-2	FALCPA 2004 and FASTER Act 2021 (FD&C Act s. 403(w)) – FDA (us_falcpa_faster)	FDA-regulated packaged foods and dietary supplements	labelling_element	Nine major allergens (milk, egg, fish, crustacean shellfish, tree nuts, peanuts, wheat, soybeans, sesame) declared by food source in the ingredient list or in a "Contains" statement immediately after it; type of tree nut, species of fish and crustacean named → CMP_allergen_declarations (scheme us_falcpa_faster), projection rules	Sesame effective 1 January 2023	Verified 2026-09-20 [F9]
D-US-3	FDA gluten-free labeling rule (21 CFR 101.91) (us_fda_gluten_free)	Foods labelled "gluten-free"	restricted_content	Claim permitted only where gluten is below 20 ppm and no gluten-containing grain ingredient is used unless processed to remove gluten → MF_dietary_claims (type gluten_free, threshold 20 ppm)	Ongoing	Retrieved (not spot-verified)
D-US-4	USDA National Organic Program (7 CFR part 205) (us_usda_nop_organic)	Foods labelled organic	restricted_content; documentation	"100% organic", "organic" (≥95%), "made with organic" categories; certifier named on the information panel → MF_dietary_claims (type organic), MF_certifications	Ongoing	Retrieved (not spot-verified)
D-US-5	Federal Alcohol Administration Act; 27 CFR parts 4, 5, 7 and 16 – TTB (us_ttb_alcohol_labeling)	Wine, distilled spirits, malt beverages	labelling_element; warning_statement; pre_market (COLA)	Brand name, class or type, alcohol content, net contents, name and address, country of origin for imports, sulfite declaration ≥10 ppm, Government Warning statement; certificate of label approval before sale in interstate commerce → MF_alcohol_by_volume, VAR_net_quantity, CMP_alcohol_warnings (scheme us_ttb_alcohol_labeling), CMP_market_registrations (COLA)	Ongoing	Retrieved (not spot-verified)
D-US-6	Proposition 65 (California) – OEHHHA (us_ca_prop65_warning)	Foods causing exposure to listed chemicals sold to California consumers	warning_statement; listing_element	Food-specific short-form or full warning (no triangle symbol required for food; food web address); internet warning before purchase → MF_warning_statements (jurisdiction US-CA)	Amended regulations effective 1 January 2025	Verified 2026-09-20 (CDS-1600 [B14])

D.4 European Union (EU)

Ref	Instrument / regulator	Applies to	Obligation	Required data elements → CDS field	Trigger / dates	Verification
D-EU-1	Regulation (EU) No 1169/2011 on food information to consumers (FIC), Articles 9, 12, 14, 16, 18, 21, 30 and Annexes II–V – European Commission and member states (eu_fic_1169_2011)	Prepacked food; distance selling of prepacked and non-prepacked food	labelling_element; listing_element	Mandatory particulars: name of the food; list of ingredients (descending by weight as recorded at use); allergens (Annex II: 14 groups) emphasised in the ingredient list; quantity of certain ingredients (QUID); net quantity; date of minimum durability or use-by date; special storage conditions or conditions of use; name or business name and address of the food business operator; country of origin or place of provenance where required; instructions for use where needed; actual alcoholic strength for beverages >1.2% ABV; nutrition declaration (energy, fat, saturates, carbohydrate, sugars, protein, salt per 100 g/ml; exemptions in Annex V; not mandatory for >1.2% ABV). Distance selling (Article 14): all mandatory information except the date mark must be available before the purchase is concluded on the material supporting the sale, free of charge; all particulars at delivery → MF_ingredient_list, CMP_allergen_declarations (scheme eu_fic_1169_2011_annex_ii), MF_characterising_ingredient_percentages, VAR_net_quantity, MF_date_mark_type, MF_storage_instructions, CMP_responsible_person, CMP_origin_declarations, MF_alcohol_by_volume, MF_nutrition_values (scheme eu_nutrition_declaration); preflight rule Roo3	Ongoing	Verified 2026-09-20 [F10]
D-EU-2	Regulation (EU) 2021/2117 amending Regulation (EU) No 1308/2013 (wine) and (EU) No 251/2014 (aromatised wine) (eu_wine_labelling_2021_2117)	Wine and aromatised wine products	labelling_element	Nutrition declaration and list of ingredients compulsory; energy value may be on-label with full declaration and ingredients provided electronically (no tracking or marketing); allergens remain on the label; de-alcoholised products <10% ABV carry a date of minimum durability → MF_nutrition_values, MF_ingredient_list, CMP_allergen_declarations, electronic-label link record	Applies to wine produced from 8 December 2023 (existing stocks may be sold through)	Verified 2026-09-20 [F11]
D-EU-3	Regulation (EC) No 1924/2006 nutrition and health claims (eu_claims_1924_2006); Commission Implementing Regulation (EU) No 828/2014 (gluten) (eu_gluten_828_2014)	Foods bearing claims	restricted_content	Only authorised health claims and defined nutrition claims; "gluten-free" ≤20 mg/kg, "very low gluten" ≤100 mg/kg → MF_health_claims, MF_nutrition_content_claims, MF_dietary_claims	Ongoing	Verified in part 2026-09-20 [F12] (gluten thresholds via FSA); claims regulation retrieved
D-EU-	Regulation (EU) 2018/848 organic production and labelling	Foods labelled	restricted_content; documentation	EU organic logo conditions, code number of the control body, origin statement → MF_dietary_claims (type organic),	Ongoing	Retrieved (not spot-verified)

Ref	Instrument / regulator	Applies to	Obligation	Required data elements → CDS field	Trigger / dates	Verification
4	(eu_organic_2018_848)	organic		MF_certifications		

D.5 United Kingdom (GB)

Ref	Instrument / regulator	Applies to	Obligation	Required data elements → CDS field	Trigger / dates	Verification
D-UK-1	Assimilated Regulation (EU) No 1169/2011 and the Food Information Regulations 2014 (England; Wales and NI equivalents; Scotland separate) — FSA/FSS and local authorities (uk_fir_2014)	Prepacked, non-prepacked and prepacked-for-direct-sale (PPDS) food; distance selling	labelling_element; listing_element	Same mandatory particulars as D-EU-1 (assimilated); 14 allergens emphasised; PPDS food must carry the name and a full ingredients list with allergens emphasised (from 1 October 2021); distance selling: mandatory allergen information available before purchase and at delivery; gluten-free ≤20 mg/kg; alcohol >1.2% ABV exempt from ingredient list but allergens must be declared → as D-EU-1 with scheme uk_fir_2014	PPDS from 1 October 2021	Verified 2026-09-20 [F12]
D-UK-2	The Food (Promotion and Placement) (England) Regulations 2021; Food (Promotion and Presentation) (Wales) Regulations 2025 — DHSC/Welsh Government (uk_hfss_promotion_placement)	Qualifying businesses (50+ employees) offering specified less-healthy (HFSS) prepacked food online	listing_element (placement); restricted_content (promotions)	Specified food must not be offered on the online home page, in non-related browsing or search results (with exceptions), on pop-ups, favourites (except previously bought), or checkout pages; volume price promotions (multibuy, extra free) prohibited; nutrient profiling score determines "specified food" → MF_hfss_classification (jurisdiction GB), channel placement rules	Placement restrictions since October 2022; England volume-price promotion restrictions from 1 October 2025; Wales regulations 2025	Verified 2026-09-20 [F13]
D-UK-3	Alcohol labelling (assimilated FIC Article 16; Weights and Measures; voluntary UK unit guidance) (uk_alcohol_labelling)	Alcoholic beverages	labelling_element	ABV >1.2%; units (8 g ethanol) and pregnancy logo are voluntary industry practice → MF_alcohol_by_volume, MF_alcohol_units (scheme uk_unit_8g)	Ongoing	Retrieved (not spot-verified)

D.6 Canada (CA)

Ref	Instrument / regulator	Applies to	Obligation	Required data elements → CDS field	Trigger / dates	Verification
D-CA-1	Food and Drugs Act and Regulations; Safe Food for Canadians Act and Regulations; Consumer Packaging and Labelling — CFIA and Health Canada (ca_fdr_food_labelling)	Prepackaged food sold in Canada	labelling_element	Common name; net quantity in metric; list of ingredients with priority allergens, gluten sources and added sulphites declared by prescribed source names in the list or a "Contains" statement, with cross-contamination statements permitted; Nutrition Facts table (serving size in household measure and metric, energy and 12 core nutrients, %DV, format families, small-package exemptions); bilingual English and French for mandatory information (dealer name and address may be in one language); country of origin declaration "Product of [country]" where required; date marking and storage → MF_ingredient_list, CMP_allergen_declarations (scheme ca_fdr_priority_allergens), MF_nutrition_values (scheme ca_nutrition_facts_table), VAR_net_quantity, CMP_origin_declarations, localisation dimension per CDS-300 §18	Ongoing	Verified 2026-09-20 [F14]
D-CA-2	Food and Drug Regulations B.01.350–B.01.354 (front-of-package nutrition symbol) — Health Canada/CFIA (ca_fop_nutrition_symbol)	Prepackaged foods at or above thresholds of saturated fat, sugars or sodium (≥15% DV default; ≥10% for reference amounts ≤30 g/ml; ≥30% for main dishes), with exemptions and prohibitions	rating_label	Nutrition symbol on the principal display panel in prescribed bilingual format; multipack and assortment rules → MF_front_of_pack_rating (scheme ca_fop_nutrition_symbol)	Compliance from 1 January 2026, no enforcement discretion	Verified 2026-09-20 [F14]

D.7 Other markets

Jurisdiction	Status	Note
Japan (JP), China (CN), Singapore (SG), Republic of Korea (KR), India (IN), Brazil (BR) and others	Not seeded	China's GB 7718 labelling standard and Japan's Food Labeling Act differ materially (allergen lists, nutrition bases, language); the organisation extends the register with local advice before publishing food to those markets.

Commerce Data Standard (CDS)

CDS-1800 — Consumer Electronics, Appliances and Technical Goods Industry Profile

Field	Value
Status	vo.2 Review Draft (working source; not an approved standard)
Release	CDS vo.2 (single corpus release per ADR-D5); chapter added in the 2026-09-20 industry-profile expansion
Date	2026-09-20
Supersedes	Nothing — first edition. The Chapter 11 dictionary vocabularies (connectivity, screen technologies, power sources, installation types, energy rating systems) were shipped ahead of this profile and are bound here for the first time; battery chemistries and product conditions are added in package 0.8.0.
Normative status	§1, §3, §5–§12, §17 and Appendix A are normative. §2 (pointer), §4, §13–§16, §18, §19 and Appendices B–D are informative; Appendix D seeds the jurisdiction requirement register whose record structure and obligations are normative in CDS-1500 §2.2. Every table is individually marked.
Primary audience	Electronics and appliance merchandisers, compliance and product-safety owners, data stewards, PIM architects, catalogue managers, UX teams, developers and AI enrichment designers
Depends on	CDS-000 through CDS-1500, especially CDS-200 (entity model, relationships), CDS-300 (CMP_, MED_, SUP_), CDS-400 (dictionaries, units), CDS-500 (preflight, verification), CDS-600 (facets), CDS-700 (evidence), CDS-900 (platform profiles), CDS-1100 (identifier and measurement contracts), CDS-1500 (profile model, requirement levels, profile register, jurisdiction requirement register), CDS-1900 (product relationship record, kits and bundles, dangerous-goods declarations)
Companion package	CDS Reference Dictionary: Chapter 11 (Electronics & Appliances), Chapter 18 (Parts, Automotive & Industrial: product conditions, dangerous-goods classes), Chapter 19 (Jurisdictions & Regulatory Schemes)
Profile identifier	<code>cds.profile.electronics.v0_2</code> (registered in CDS-1500 §2.1)
Research basis	REVIEW-020 (global commerce vertical market research, 2026-09-20): consumer electronics is the largest worldwide ecommerce category by revenue (estimated at roughly US\$1 trillion for 2026) and the largest category of Amazon's global gross merchandise value (38% in 2025); electronics stores carry among the largest catalogues on Shopify (average above 2,400 products per store), so the vertical's PIM intensity far exceeds its store count.

Terminology follows CDS-100. Attribute requirement levels (R, C, REC, O, N/A) are those of CDS-1500 §4 and are not restated here. "Technical goods" in this chapter means products whose purchase decision is dominated by measured specifications and compatibility rather than by appearance.

1. Purpose and Scope (*normative*)

CDS-1800 defines the industry profile for consumer electronics (phones, computers, audio, video, cameras, gaming, wearables, smart home), small and large domestic appliances, and technical accessories (cables, chargers, mounts, storage media). It also governs the technical-specification layer of products homed in other profiles (a beauty device, a power tool, a bicycle computer) where those chapters cite it.

The data shape is distinctive for four reasons: identity is manufacturer-anchored (brand plus manufacturer part number, usually with a GTIN) and variants differ by capacity, colour, region or plug rather than by size; the record is dominated by dozens to hundreds of measured, unit-bearing specifications whose set depends on the product type; compatibility ("works with", "fits", "requires") is a relationship between products or standards, not an attribute; and a large regulated layer (electrical safety marks, radio approvals, energy labels, battery and transport rules, repairability and take-back obligations) differs by jurisdiction and, in several markets, must be shown online next to the price.

CDS1800-R001 An implementation claiming the Electronics Profile MUST represent technical specifications as typed, unit-bearing attribute values governed by Attribute Definitions (CDS-200 §7, CDS-400 §18), never solely as specification text, tables in descriptions or images of specification sheets.

CDS1800-R002 An implementation claiming this profile MUST represent compatibility, inclusion and accessory relationships as product relationship records per CDS-1900 §5, never as tags, free text or category membership.

CDS1800-R003 An implementation MUST maintain a jurisdiction requirement register (CDS-1500 §2.2) for every market it publishes electrical or electronic goods to, and publication preflight for a market MUST evaluate the market's pre-market, marking, energy-label, battery, transport and listing obligations before the offer is made available.

CDS1800-R004 This profile MUST NOT weaken any rule of CDS-1500 that it reuses; where this chapter is silent, CDS-1500 and the core chapters govern.

2. Profile Model (*informative pointer*)

This profile applies the industry profile model of CDS-1500 §2, the requirement levels of CDS-1500 §4 and the jurisdiction requirement register of CDS-1500 §2.2 without restatement. Its identifier and dictionary bindings are in CDS-1500 §2.1; its seeded jurisdiction entries are in Appendix D.

3. Product Model (*normative*)

A technical product is a manufacturer model offered in one or more sellable configurations. The Product is the model (the television model, the laptop model line as configured by the manufacturer); Variants are the sellable configurations that differ by capacity, memory, screen size within a model line, colour, region, plug type or bundled accessories. Model families (a generation, a series) are relationships, not product attributes.

```
Product: 65-inch 4K OLED television, model X65Q      (informative example)
Product scope:
  STD_brand = Example Electronics
  SUP_manufacturer_part_number = X65Q-AU
  CAT_product_type = television
  MF_screen_size_in = 65
  MF_screen_technology = oled                      # screen_technology dictionary
  MF_resolution = {horizontal: 3840, vertical: 2160}
  MF_connectivity = [hdmi, wifi, bluetooth, ethernet]
  MF_hdmi_port_count = 4
  MF_power_consumption_w = 180
  CMP_energy_ratings = [{jurisdiction: AU, scheme: au_energy_rating, value: 4.5, registration_id: ...},
                        {jurisdiction: EU, scheme: eu_energy_label, class: F, eprel_id: ...}]

Variant scope:
  VAR_region = AU
  VAR_plug_type = AU
  VAR_gtin = 09300000000010
```

CDS1800-R005 Capacity, memory, storage, screen size within a model line, colour, region and plug type MUST be modelled at variant scope where they identify distinct sellable units (variant boundary rule: CDS-200 §5); specifications shared by every configuration MUST remain at product scope.

CDS1800-R006 Where a specification differs between configurations (battery capacity, weight, energy rating), the implementation MUST store the differing value at variant scope for the affected variants and MUST NOT publish a product-scope value as if it applied to every configuration.

CDS1800-R007 Model families, generations and successor relationships MUST be product relationship records (CDS-1900 §5) with typed relationship types, never encoded only in titles or in a free-text "series" field.

CDS1800-R008 Where a channel's variant-option limit is smaller than the model's configuration dimensions, the implementation MUST apply a documented decomposition strategy (CDS900-R024) and MUST preserve the canonical configuration dimensions in the PIM.

4. Product Families and Category Profiles *(informative baseline; the inheritance rule of CDS1500-R016 applies)*

Product family / category	Required	Recommended	Typical variant options
Mobile phones and tablets	brand; MPN; GTIN; screen size and technology; storage; memory; battery capacity; connectivity; operating system; region; charging port type and power; battery chemistry and transport class	ingress-protection rating; energy label per market; repairability class where a market issues one; included charger status	storage, colour, region
Computers	brand; MPN; processor; memory; storage; screen size; operating system; ports; battery capacity (portables)	dimensions and weight; energy label per market	configuration, colour
Audio (headphones, speakers)	brand; MPN; connectivity; power source; battery capacity where rechargeable	driver size; codec support; ingress rating; case battery	colour
Television and video	brand; MPN; screen size; screen technology; resolution; connectivity and port counts; energy rating per market; installation type	HDR formats; refresh rate; smart platform	size
Cameras and optics	brand; MPN; sensor format; resolution; lens mount; battery	video capabilities; weather sealing	kit configuration
Small appliances	brand; MPN; power source; wattage; capacity; dimensions; electrical safety compliance per market	energy or water rating where regulated; installation type	colour
Large and white goods	brand; MPN; capacity; dimensions; energy and water ratings per market; installation type; electrical compliance	noise level; programme features; delivery class	colour, finish
Smart home and networking	brand; MPN; connectivity standards and versions; power source; compatibility relationships	hub or ecosystem requirements as relationships	colour
Accessories (cables, chargers, mounts, cases)	brand; MPN; connector or mount standard; compatibility relationships; power delivery where applicable	length; material	length, colour
Batteries and power banks	chemistry; capacity (Wh and mAh); voltage; cell count; transport classification	charging ports; power delivery	capacity

5. Identity and Identifiers *(normative)*

Field	Meaning	Type
STD_brand	Brand as marketed	Governed brand dictionary value
SUP_manufacturer	Legal manufacturer where different from brand	Governed value
SUP_manufacturer_part_number	Manufacturer part or model number (MPN)	Identifier
MF_model_name	Marketed model name	Text
VAR_gtin	Trade item identifier with scheme	Identifier (GTIN-8/12/13/14 or declared alternative)
MF_identifier_exists	Whether a manufacturer-assigned trade identifier exists	Boolean
MF_series / MF_generation	Family and generation	Relationship records (R007) plus optional governed labels
MF_release_date / lifecycle.end_of_sale_date	Market release and end-of-sale dates	Dates

CDS1800-R009 Brand and manufacturer part number MUST be governed identity attributes; a GTIN MUST be stored with its scheme and MUST validate per scheme (CDS-1100), and the absence of a manufacturer-assigned identifier MUST be an explicit declaration (`MF_identifier_exists = false`), never an empty field.

CDS1800-R010 A retailer SKU MUST NOT be substituted for the manufacturer part number in the MPN attribute; the two are distinct identifiers with distinct authority.

CDS1800-R011 Model name and model number MUST be separate attributes; a regional suffix on a part number (for example a plug-region code) MUST be represented at variant scope, not by forking the product.

CDS1800-R012 Where a market requires a product identification element (type, batch or serial number) and the manufacturer's identity and address on the product or in the online offer (Appendix D), the product identifier MUST be a governed attribute and the responsible-person record (CDS-1600 §12 pattern; `CMP_responsible_person`) MUST be maintained per market.

CDS1800-R013 Release date and end-of-sale date MUST be typed dates in the product lifecycle group (CDS-1100 §8); "new" and "discontinued" storefront states MUST derive from them under governed rules (CDS600-R086).

6. Technical Specifications (*normative*)

Specifications are the heart of this profile. Every specification is an Attribute Definition with a declared data type, unit, cardinality, validation range and comparison strategy; the set of specifications required for a product type is declared in the category profile, not improvised per product.

Rule area	Mechanism
Specification sets	Each category profile declares its specification set with requirement levels (R/C/REC/O) — for example a television declares screen size, technology, resolution, refresh rate, port counts, power consumption, dimensions with and without stand, weight
Units	Every numeric specification carries a governed unit; storage capacity declares its base (decimal gigabyte or binary gibibyte) per organisation policy
Standards and versions	Interface and protocol standards (Wi-Fi 6E, Bluetooth 5.3, HDMI 2.1, USB4, DisplayPort 2.1, Thunderbolt 4) are governed dictionary values with version as part of the value, not free text
Ranges and "up to"	Manufacturer maximums ("up to 20 hours") are stored as typed values with a basis qualifier (<code>manufacturer_rated</code> , <code>measured_under_scheme</code>), never as bare numbers implying measurement
Provenance	Each accepted specification value carries provenance to the manufacturer specification sheet or regulatory database (evidence class E1) or to an official product page (E2)

CDS1800-R014 Every specification exposed for search, filtering, comparison or channel publication MUST be an Attribute Definition with a declared data type, unit (where numeric), validation range and comparison strategy (CDS200-R023, CDS400-R056); a specification that exists only inside description text MUST NOT be treated as populated.

CDS1800-R015 The specification set for a product type MUST be declared in the category profile with requirement levels; requiredness MUST NOT be evaluated against a flat universal specification list (CDS1500-R009).

CDS1800-R016 Interface, protocol and standard versions MUST be governed dictionary values that include the version where the version is commercially meaningful (the `connectivity` dictionary pattern), and MUST NOT be free text.

CDS1800-R017 A manufacturer-rated maximum or typical value MUST carry a basis qualifier distinguishing it from a value measured under a declared scheme; the two MUST NOT be compared as equivalent.

CDS1800-R018 Specification values MUST carry provenance to an eligible source (CDS-700 §7 E1 or E2); an AI-extracted specification MUST be a proposal with evidence and MUST be validated against the Attribute Definition's unit and range before acceptance, and an implementation MUST NOT accept a specification value with no eligible evidence (CDS700-R018).

CDS1800-R019 Where two eligible sources disagree on a specification, the conflict MUST be surfaced for review (CDS700-R019); the implementation MUST NOT silently prefer the more favourable value.

7. Compatibility and Standards Compliance (*normative*)

Compatibility is the single most-asked question in this vertical and the most-often faked. The profile represents it as relationships and standards, never as inference.

CDS1800-R020 Compatibility between products (an accessory and the devices it fits, a component and the systems it works in) MUST be expressed as product relationship records of a governed type (`compatible_with`, `accessory_for`, `requires`, `fits`) per CDS-1900 §5, with the target expressed as a product, a model family, a governed standard or an application record; it MUST NOT be expressed as tags, category placement or free text alone.

CDS1800-R021 "Universal" compatibility MUST be an explicit declaration bound to a governed standard (for example a connector or mount standard) rather than an absence of relationship records.

CDS1800-R022 Compatibility MUST NOT be inferred from category similarity, title similarity or shared brand by rules or AI; a proposed relationship requires eligible evidence and human review before publication.

CDS1800-R023 Standards-compliance and certification claims (certified to a wireless, connector, safety or performance standard) MUST be governed claim or certification records with the certifying scheme and evidence (CDS-1600 §11 pattern; `MF_certifications`), and MUST NOT be projected as facets unless fed by accepted records.

CDS1800-R024 Port and slot counts MUST be typed integer specifications per governed interface value (for example MF_hdmi_port_count), never a comma-separated list.

8. Power, Batteries and Energy Ratings (*normative; dictionary bindings informative*)

Field	Meaning	Type
MF_power_source	Governed value (power_source dictionary: mains, rechargeable battery, replaceable battery, USB powered, ...)	Product
MF_input_voltage_range_v / MF_input_frequency_hz	Electrical input	Typed measurements or ranges
VAR_plug_type	Plug or socket standard supplied	Governed value (variant scope; CDS-1500 §13)
MF_rated_power_w / MF_power_consumption_w	Rated power and consumption	Typed measurements
MF_battery_chemistry	Governed value (battery_chemistry dictionary: lithium_ion, lithium_polymer, lithium_metal, nickel_metal_hydride, alkaline, lead_acid, sodium_ion, ...)	Product/Variant
MF_battery_capacity_wh / MF_battery_capacity_mah / MF_battery_voltage_v	Battery capacity and voltage	Typed measurements (watt-hours mandatory for lithium transport classification)
MF_battery_removable / MF_battery_included / MF_battery_count	Battery facts	Booleans and integer
CMP_dangerous_goods	Transport declaration	Record per CDS-1900 §12 (UN 3480/3481 lithium-ion, 3090/3091 lithium metal, 3551/3552 sodium-ion, and others)
MF_charging_port / MF_charging_power_min_w / MF_charging_power_max_w / MF_usb_pd_supported / MF_charger_included	Charging capabilities as labelled	Governed value; typed watts; booleans
CMP_energy_ratings	Per-market energy or water rating records	List: jurisdiction, scheme (energy_rating_system dictionary), rating value or class, class range in force, registration identifier, label artefact reference, product information sheet reference, verification date

CDS1800-R025 Power source, input electrical characteristics and plug type MUST be governed or typed values; plug type MUST be variant-scoped where it defines the sellable unit.

CDS1800-R026 Battery chemistry MUST be a governed value and battery capacity MUST be a typed measurement in watt-hours (with milliampere-hours and voltage where labelled) for every product containing or supplied with a battery; these values MUST be present before a dangerous-goods declaration is made.

CDS1800-R027 Every product containing or packed with a lithium or sodium-ion battery MUST carry a dangerous-goods declaration per CDS-1900 §12 before publication to a channel or carrier that ships it; the declaration MUST distinguish batteries contained in equipment, packed with equipment and shipped alone, and MUST NOT be inferred from category.

CDS1800-R028 Energy and water ratings MUST be stored as per-market records with the scheme, the rating value or class, the class range in force for the scheme (where a scheme uses a range), the registration identifier where the scheme registers models, and a reference to the label artefact and product information sheet; a rating MUST NOT be stored as a single global number or as a marketing claim.

CDS1800-R029 Where a market requires the energy label, class arrow, product information sheet or rating icon to be shown online close to the price (Appendix D), the channel projection for that market MUST include the required artefact or class-and-range display, generated from the rating record, and verification MUST cover it.

CDS1800-R030 Charging capabilities required to be labelled in a market (charging port standard, minimum and maximum charging power, protocol support, whether a charger is included) MUST be typed attributes and, where the market requires the pictogram or label online close to the price, MUST be projected and verified for that market.

CDS1800-R031 A regulated energy or water rating MUST derive from the scheme's registration or supplier documentation (E1); it MUST NOT be estimated or copied from a similar model.

CDS1800-R032 Efficiency programme membership that is voluntary in a market (for example a certification programme) is a certification claim under R023, not a rating; the energy_rating_system dictionary marks such schemes as evidence-gated.

9. Physical Attributes, Installation and Packaging (*normative*)

CDS1800-R033 Product dimensions and weight MUST be typed measurements distinguishing the product (with and without stand or accessories where relevant) from the packaged unit; packaged dimensions and weight MUST be present for products published to channels that compute shipping.

CDS1800-R034 Installation and mounting types MUST be governed values (installation_type dictionary; mounting standards such as display-mount patterns as governed specifications).

CDS1800-R035 Delivery class (parcel, oversized, two-person, installation required) MUST be a governed logistics attribute where the organisation ships the product, derived from packaged measurements and category rules, not authored per product.

10. Contents, Kits and Accessories (*normative*)

CDS1800-Ro36 Items included in the box MUST be represented as `includes` relationship records with quantities (CDS-1900 §11), not only as a descriptive list, so that channel content, warranty and returns processes can rely on them.

CDS1800-Ro37 Bundles combining a device with accessories or services MUST be modelled as kits with component relationships per CDS-1900 §11, with each component retaining its own identity, specifications and compliance records.

11. Condition, Warranty, Lifecycle and Repair (*normative*)

Field	Meaning	Type
MF_condition	Governed value (<code>product_condition</code> dictionary: new, refurbished, remanufactured, used)	Variant
MF_refurbished_grade	Organisation-governed grade where condition is refurbished	Governed value
MF_warranty_months / MF_warranty_type / MF_warranty_provider	Warranty term, type (manufacturer, retailer, extended) and provider, per market	Per-market records
MF_software_support_until	Declared operating-system or security update period	Date
MF_spare_parts_availability_years / MF_repairability_class	Repair information where a market requires or scores it	Typed values with scheme

CDS1800-Ro38 Condition MUST be a governed value at variant scope; a refurbished or used product MUST NOT share a variant with a new one, and channels that require a condition attribute MUST receive it from this value.

CDS1800-Ro39 Warranty MUST be stored per market as typed records; a warranty statement in a description MUST derive from the record.

CDS1800-Ro40 Where a market requires repair, spare-part availability or software-support information to be made available (Appendix D), the information MUST be a typed attribute with the scheme and MUST be projected for that market.

CDS1800-Ro41 Product recalls and safety withdrawals MUST be expressed through lifecycle state and channel withdrawal (CDS-500 §12), with the affected serial or batch range recorded as an observation, never by deleting the product.

CDS1800-Ro42 Region locking, regional software or regional service restrictions MUST be governed variant attributes evaluated at preflight for the target market.

12. Compliance, Safety Marks and Technical Documentation (*normative*)

Field	Meaning	Type
CMP_compliance_declarations	Per-market compliance records	List: jurisdiction, scheme (<code>regulatory_scheme</code> dictionary), obligation type, mark or marking applied, registration or certificate identifier, responsible supplier or economic operator, declaration of conformity reference, evidence, status, dates
CMP_responsible_person	Responsible supplier, importer, authorised representative or economic operator per market	Per-market record
CMP_product_identifier	Type, batch or serial identification element required by a market	Governed attribute
MED_technical_documents	Typed document set	Records: type (user manual, quick start, specification sheet, declaration of conformity, energy label, product information sheet, safety data sheet, test summary), language, version, URL or asset reference
MF_warning_statements	Mandatory warnings per market (for example button-battery warnings)	Structured statement records with scheme

CDS1800-Ro43 Compliance with a market's electrical-safety, electromagnetic-compatibility, radio, energy, water, battery, take-back or general-safety regime MUST be a governed compliance record per market referencing the scheme and its evidence; a conformity mark in an image or a phrase in a description MUST NOT be treated as the compliance fact.

CDS1800-Ro44 Compliance declarations and conformity marks MUST NOT be projected as customer facets or claims unless fed by accepted records; they MAY be displayed as information where the market requires or permits it.

CDS1800-Ro45 Technical documents MUST be typed media records (`MED_`) with language and version, linked to the product or variant they describe; documents a market requires to be made available online (Appendix D) MUST be projected for that market and their presence verified.

CDS1800-Ro46 Mandatory warnings applying to a product type in a market (for example products containing button or coin batteries) MUST be structured statement records evaluated per market at preflight and projected to the market's listing where the market requires or recommends an online warning.

CDS1800-Ro47 Where a market requires a battery test summary, a declaration of conformity or a product information sheet to be available to purchasers on request or online, the document reference MUST be present in `MED_technical_documents` before publication to that market.

13. Customer Facet Design (*informative — normative facet rules live in CDS-600*)

Facet	Recommended baseline behaviour	Anti-pattern
Brand	Governed brand dictionary	Supplier spellings
Specification ranges (screen size, capacity, wattage)	Numeric range facets with units (CDS600-R043)	Text buckets typed per product
Connectivity and standards	Governed dictionary values, multi-select OR	Free-text feature lists
Compatibility ("works with")	Fed by relationship records; target-model facets only where the catalogue supports them	Tags such as "iphone-compatible"
Energy rating	Per-market facet from rating records; display class and range as the market requires	A global star facet applied to every market
Condition	Governed condition values	"Like new" free text
Installation type	Governed values	—

14. Channel Projection Guidance (*informative — normative rules: CDS-500, CDS-900*)

Canonical concept	Metafield-style channel (informative)	Feed-style channel (informative)
Specifications	Category metafields where the taxonomy defines them (connectivity, display technology, battery, compatibility attributes exist for many electronics categories), custom metafields otherwise; verified by read-back	<code>product_detail</code> name/value pairs; identifiers <code>brand</code> , <code>gtin</code> , <code>mpn</code> , <code>identifier_exists</code>
Compatibility	Relationship projections (complementary-product and compatibility metafields where supported)	Not a standard feed field in the seeded channels; title and <code>product_detail</code>
Energy rating	Per-market metafield plus label artefact; nested class arrow for markets that require it	<code>energy_efficiency_class</code> and related attributes are accepted only for specific countries in the Google profile (verified 2026-09-20: Switzerland, Norway, UK); <code>certification</code> attribute for EU energy labels — dated facts in CDS-900
Condition	Condition metafield or variant option	<code>condition</code> (new / refurbished / used)
Battery and transport	Fulfilment metafields; carrier declarations from the DG record	Not published to consumers; carrier integration
Warnings and compliance information	Per-market generated blocks; verified	Description blocks where policy allows

15. AI Enrichment and Review (*informative — normative AI rules live in CDS-700*)

Task	AI may propose	Deterministic or human control
Specification extraction from specification sheets (E1)	Typed candidates with units	Unit and range validation; provenance required (R018); conflicts surfaced (R019)
Standards and version normalisation	Dictionary candidates	Governed dictionary; unknown versions quarantined
Compatibility proposals	Candidate relationships with evidence	Never accepted without evidence and review (R022)
Energy and battery data	None — regulatory database or supplier document only	R031, R026
Market obligation gaps	Candidate register entries triggered by category and attributes	Register evaluation deterministic

16. Governance and Organisational Extensions (*informative — rules in CDS-1500 §23 and CDS-800*)

Specification sets per category are organisational schema decisions governed under CDS-800 §23 with impact assessment on facets, channels and comparison. External specification models (ETIM classes and features, GS1 GPC brick attributes, marketplace category attributes) may be adopted as the source of a category's specification set through Taxonomy Mappings and attribute crosswalks (CDS-1900 §8). The jurisdiction requirement register for electrical goods changes frequently (energy-label rescaling, battery regulation phase-ins, charger rules) and is reviewed on the declared cadence with a mandatory review on each market change.

17. Conformance Requirements (*normative — claims and levels per CDS-1000*)

CDS1800-R048 An implementation claiming the CDS Electronics Profile MUST: govern brand, manufacturer part number and trade identifiers with declared schemes and explicit identifier-absence (R009–R011); store every exposed specification as a typed, unit-bearing Attribute Definition value with a declared category specification set, basis qualifiers and provenance (R014–R019); express compatibility, inclusion, family and successor relationships as governed relationship records (R007, R020–R022, R036–R037); store power, battery and per-market energy or water ratings as typed and governed records with dangerous-goods declarations for battery products (R025–R032); store condition, warranty and lifecycle as governed values (R038–R042); maintain per-market compliance records, responsible-person records, typed technical documents and mandatory warnings (R043–R047); maintain and evaluate a jurisdiction requirement register for every market published to (R003); apply category-specific requirements per CDS1500-R009; and publish and verify channel representations under CDS-500, including market-mandated online label and information displays (R029, R030, R045).

CDS1800-R049 An Electronics Profile claim MUST state which markets its register covers and whether battery-containing products and large appliances are in scope.

18. Worked Product Examples (informative)

18.1 Television sold in Australia and the EU

```
STD_brand = Example Electronics; SUP_manufacturer_part_number = X65Q
CAT_product_type = television
MF_screen_size_in = 65; MF_screen_technology = oled; MF_resolution = {3840, 2160}; MF_refresh_rate_hz = 120
MF_connectivity = [hdmi, wifi, bluetooth, ethernet]; MF_hdmi_port_count = 4; MF_hdmi_version = hdmi_2_1
MF_power_consumption_w = 180; MF_installation_type = freestanding (wall_mounted supported; mount pattern 300x300)
CMP_energy_ratings =
  [{jurisdiction: AU, scheme: au_energy_rating, value: 4.5, registration_id: <GEMS registration>, label_ref: MED-..., verified: 2026-09-20},
   {jurisdiction: EU, scheme: eu_energy_label, class: F, class_range: "A-G", eprel_id: <EPREL model id>, label_ref: MED-..., pis_ref: MED-...}]
CMP_compliance_declarations =
  [{jurisdiction: AU, scheme: au_eess_rcm, level: 2, registration: <EESS>, responsible_supplier: ...},
   {jurisdiction: AU, scheme: au_acma_rcm, ...},
   {jurisdiction: EU, scheme: eu_ce_marking_electrical, directives: [LVD, EMC, RED, RoHS], doc_ref: MED-...},
   {jurisdiction: EU, scheme: eu_weee_2012_19, marking: crossed_out_bin, producer_registration: ...}]
Variant AU: VAR_plugin_type = AU; VAR_gtin = 09300000000010
Variant EU: VAR_plugin_type = EU; VAR_gtin = 04000000000017
EU projection: class arrow "F" with range A-G adjacent to price; link labelled "Product Information Sheet" -> verified
AU projection: Energy Rating Icon generated from the registration data -> verified
```

18.2 Wireless earbuds with charging case

```
CAT_product_type = wireless_earbuds
MF_connectivity = [bluetooth]; MF_bluetooth_version = bluetooth_5_3
MF_battery_chemistry = lithium_polymer
MF_battery_capacity_wh = 0.2 (each earbud); MF_case_battery_capacity_wh = 1.9
MF_battery_removable = false; MF_charging_port = usb_c; MF_charger_included = false
CMP_dangerous_goods = {regulated: true, un_number: "UN3481", proper_shipping_name: "Lithium ion batteries packed with equipment",
  class: "9", watt_hours_per_battery: 1.9, transport_modes: [road, air], test_summary_ref: MED-...}
CMP_compliance_declarations = [{jurisdiction: EU, scheme: eu_common_charger_2022_2380, pictogram: charger_not_included,
  charging_label: {min_w: 3, max_w: 5, usb_pd: false}}, ...]
EU projection: charger pictogram and charging label shown close to the price -> verified
```

18.3 Dishwasher with energy and water ratings

```
CAT_product_type = dishwasher
MF_installation_type = freestanding; MF_capacity_place_settings = 14
MF_dimensions_mm = {h: 850, w: 600, d: 600}; MF_packaged_weight_kg = 52; MF_delivery_class = two_person
CMP_energy_ratings =
  [{jurisdiction: AU, scheme: au_energy_rating, value: 4.0, registration_id: ...},
   {jurisdiction: AU, scheme: wels_water_rating, value: 5.5, registration_id: ...},
   {jurisdiction: CA, scheme: ca_energyguide, annual_kwh: 240, verification_mark: <certification body>},
   {jurisdiction: US, scheme: us_ftc_energyguide, label_ref: MED-..., annual_cost_usd: 30}]
US projection: EnergyGuide label image or icon hyperlink close to the price -> verified
```

19. Reference Validation Cases (informative — the profile's contribution to the cross-industry validation set, REVIEW-020)

- a specification present only in description text, which must count as unpopulated (R014);
- two eligible sources disagreeing on battery capacity, which must surface a conflict (R019);
- an accessory whose compatibility is asserted only by a tag, which must fail (R020);
- a lithium-battery product lacking watt-hours or a dangerous-goods declaration, which preflight must block for shipping channels (R026–R027);
- a single global "energy rating" number offered to two markets with different schemes, which must not satisfy either (R028);
- an EU listing without the class arrow and product information sheet link, which verification must report (R029);
- a refurbished unit sharing a variant with a new unit (R038);
- a product containing a button battery published to Australia without the mandated warning text in the listing (R046, D-AU-5);
- a phone published to the EU after 20 June 2025 without its energy label and reparability class (D-EU-3);
- a manufacturer part number overwritten by a retailer SKU (R010).

Appendix A. Electronics Attribute Baseline (normative — dictionary bindings per CDS1500-R011)

Field	Scope	Type	Baseline requirement
STD_brand / SUP_manufacturer_part_number / VAR_gtin / MF_identifier_exists	Product / Variant	Governed identity values	R
CAT_product_type	Product	Governed classification reference	R
Category specification set	Product / Variant	Typed Attribute Definitions per category profile	Per category profile (R/C/REC/O)
MF_connectivity	Product	Governed dictionary list	R where the product has interfaces

Field	Scope	Type	Baseline requirement
MF_screen_technology / MF_screen_size_in / MF_resolution	Product	Governed value; typed measurements	R for display products
MF_power_source / MF_input_voltage_range_v / VAR_plug_type / MF_rated_power_w	Product / Variant	Governed and typed values	R for mains-powered products
MF_battery_chemistry / MF_battery_capacity_wh / MF_battery_removable / MF_battery_included	Product / Variant	Governed value; typed measurement; booleans	R for battery products
CMP_dangerous_goods	Product / Variant	Declaration per CDS-1900 §12	R for battery and other regulated products
MF_charging_port / MF_charging_power_min_w / MF_charging_power_max_w / MF_charger_included	Product	Governed and typed values	C — R where a market mandates charging labelling
CMP_energy_ratings	Product / Variant (per market)	Rating records (energy_rating_system dictionary)	R where a market regulates the product type
MF_installation_type	Product	Governed dictionary reference	REC; R for appliances
MF_dimensions / MF_weight / packaged equivalents	Product / Variant	Typed measurements	R
Relationship records (compatible_with, includes, accessory_for, requires, successor_of)	Product	Records per CDS-1900 §5	C — R where compatibility or contents are published
MF_condition	Variant	Governed dictionary reference (product_condition)	R
MF_warranty_months / MF_warranty_type / MF_warranty_provider	Product (per market)	Typed records	REC; R where a market requires warranty disclosure
CMP_compliance_declarations / CMP_responsible_person / CMP_product_identifier	Product (per market)	Governed records	R where a market regulates the product
MED_technical_documents	Product / Variant	Typed media records	REC; R where a market requires online availability
MF_warning_statements	Product	Structured statement records	C — R where a market mandates a warning
MF_spare_parts_availability_years / MF_repairability_class / MF_software_support_until	Product	Typed values with scheme	C

Appendix B. Dictionary Bindings (*informative — becomes governed data on adoption per CDS1500-R011*)

Dictionary key	Package chapter	Bound attribute(s)	Status at 0.8.0
connectivity	11	MF_connectivity and interface versions	Bound by this profile
screen_technology	11	MF_screen_technology	Bound
power_source	11	MF_power_source	Bound
installation_type	11	MF_installation_type	Bound
energy_rating_system	11	CMP_energy_ratings.scheme	Bound
battery_chemistry	11 (new, 0.8.0)	MF_battery_chemistry	Bound
product_condition	18 (new, 0.8.0)	MF_condition	Bound (shared with CDS-1900)
relationship_type	18 (new, 0.8.0)	Relationship records	Bound via CDS-1900 §5
dangerous_goods_class	18 (new, 0.8.0)	CMP_dangerous_goods	Bound via CDS-1900 §12
jurisdiction / regulatory_scheme	19 (new, 0.8.0)	CMP_* records	Bound via CDS-1500 §2.2

Deliberately not shipped: per-category specification sets (adopt ETIM, GS1 GPC or marketplace attribute sets under CDS-1900 §8), brand dictionaries, mount-pattern and connector registries beyond the connectivity vocabulary, vendor ecosystem lists (excluded by the package rules).

Appendix C. References (*informative; retrieved 2026-09-20 unless stated*)

Ref	Source	Location	Use in this chapter
[E1]	EESS — Registration of in-scope electrical equipment; Marking of electrical equipment; In-scope equipment definitions and risk levels v4.3 (2024); ACMA requirements — use of RCM	eess.gov.au	AU Levels 1–3 registration, RCM marking, responsible supplier (D-AU-1, D-AU-2)
[E2]	Energy Rating (energyrating.gov.au) — Displaying the label and icon; Register a product	energyrating.gov.au	AU/NZ GEMS registration, mandatory label products, online icon (D-AU-3)
[E3]	ACCC Product Safety — Button and coin batteries mandatory standards; Products containing button/coin batteries information standard (2026-02-19); Consumer Goods (Products Containing Button/Coin Batteries) Safety and Information Standards 2020	productsafety.gov.au; legislation.gov.au	Secure compartments, warnings on packaging, instructions and online listings (D-AU-5)
[E4]	WorkSafe New Zealand — Supplier declaration of conformity; High and medium risk products; Electrical safety compliance; Electricity (Safety) Regulations 2010 reg 83 (as at 2026-01-15); RSM Step 4 (SDoC)	worksafe.govt.nz; legislation.govt.nz; rsm.govt.nz	NZ declared medium/high risk articles, SDoC, RCM, radio SDoC (D-NZ-1, D-NZ-2)

Ref	Source	Location	Use in this chapter
[E5]	FCC — Equipment Authorization: RF Device; KDB 784748 Do1 (labelling) and Do2 (e-labelling)	fcc.gov; apps.fcc.gov	SDoC and certification labelling, FCC ID, e-labels (D-US-1)
[E6]	FTC — 16 CFR 305.27 Paper catalogs and websites; EnergyGuide Labeling FAQs	ecfr.gov; ftc.gov	EnergyGuide label or icon online near price (D-US-2)
[E7]	OEHHA — Proposition 65 warnings for internet purchases (regulations effective 2025-01-01)	oehha.ca.gov; p65warnings.ca.gov	Online warning before purchase (D-US-3)
[E8]	California SB 244 Right to Repair Act (Public Resources Code 42488–42488.3), operative 2024-07-01	leginfo.legislature.ca.gov	Parts, tools and documentation availability 3 or 7 years (D-US-4)
[E9]	European Commission — EPREL: Dealers; Suppliers; Commission Implementing Regulation (EU) 2024/994; Dealers webinar on smartphones and slate tablets (June 2025)	energy-efficient-products.ec.europa.eu	Online label and product information sheet display, nested arrow, QR (D-EU-2, D-EU-3)
[E10]	Commission Delegated Regulation (EU) 2023/1669 (energy labelling of smartphones and slate tablets); Commission Regulation (EU) 2023/1670 (ecodesign)	eur-lex.europa.eu	Label content, reparability class, spare parts and information requirements from 20 June 2025 (D-EU-3)
[E11]	Regulation (EU) 2023/1542 concerning batteries (consolidated 2024-07-18)	eur-lex.europa.eu	Separate collection symbol from 18 August 2025, general label from 18 August 2026, QR code from 18 February 2027 (D-EU-4)
[E12]	Directive (EU) 2024/884 amending the WEEE Directive 2012/19/EU (marking per EN 50419:2022)	eur-lex.europa.eu	Crossed-out wheeled bin marking (D-EU-5)
[E13]	Regulation (EU) 2023/988 General Product Safety Regulation (applies from 13 December 2024)	eur-lex.europa.eu	Traceability, responsible economic operator, online offer information (D-EU-6)
[E14]	Directive (EU) 2022/2380 (common charger)	eur-lex.europa.eu	USB-C receptacle, charging label and pictogram close to the price in distance selling from 28 December 2024 (laptops from 28 April 2026) (D-EU-7)
[E15]	Regulation (EU) 2024/1781 Ecodesign for Sustainable Products (ESPR)	eur-lex.europa.eu	Digital product passport framework, dealer obligations in distance selling (D-EU-8)
[E16]	GOV.UK — Placing UKCA or CE marked products on the market in Great Britain (2026-03-31); Product regulations by sector (2026-03-31); Electrical Equipment (Safety) Regulations 2016 guidance; Product Safety and Metrology etc. (Amendment) Regulations 2024 explanatory memorandum	gov.uk; legislation.gov.uk	Indefinite CE recognition from 1 October 2024, UKCA labelling flexibilities to 31 December 2027, manufacturer and importer labelling (D-UK-1)
[E17]	The Waste Electrical and Electronic Equipment Regulations 2013 (as amended to 2025-02-27)	legislation.gov.uk	Distance-seller producer obligations, crossed-out bin and date mark (D-UK-2)
[E18]	Natural Resources Canada — Introduction to the Regulations (2026-06-19); EnerGuide label for appliances; Energy Efficiency Regulations, 2016 (SOR/2016-311) ss. 4, 5, 13; Energy Efficiency Act s. 4	natural-resources.canada.ca; laws-lois.justice.gc.ca	Verification mark, energy efficiency report, EnerGuide label products (D-CA-1)
[E19]	CSA Group — US/Canada (FCC/ISED) certification for household appliances (2019)	csagroup.org	ISED SDoC and IC certification pattern (D-CA-2; retrieved)
[E20]	IATA — Lithium Battery Guidance Document (2026 edition)	iata.org	UN numbers 3480/3481/3090/3091/3551/3552, 30% state of charge, test summary availability (\$8)
[E21]	ECDB — Amazon product categories by revenue 2025; secondary reporting of Statista Market Insights ecommerce revenue by segment (2026)	ecdb.com; statista.com	Market basis (REVIEW-020)

Appendix D. Jurisdiction Requirement Register — Seed (*informative seed of the normative register defined in CDS-1500 §2.2*)

Conventions as in CDS-1600 Appendix D.

D.1 Australia (AU)

Ref	Instrument / regulator	Applies to	Obligation	Required data elements → CDS field	Trigger / dates	Verification
D-AU-1	Electrical Equipment Safety System (EESS) under state and territory electrical safety laws; AS/NZS 4417 — ERAC/state regulators (au_eess_rcm)	In-scope electrical equipment (household-type, ≤ certain ratings) sold in participating jurisdictions	pre_market (responsible supplier registration; Level 2 and 3 equipment registration in the national database; Level 3 certificate of conformity); labelling_element (RCM marking)	Responsible supplier registration; equipment level (1/2/3); equipment registration and, for Level 3, certificate of conformity; compliance folder or evidence of compliance with the relevant standard; RCM marked per AS/NZS 4417.1 near the model identification → CMP_compliance_declarations (scheme au_eess_rcm, level, registration, certificate), CMP_responsible_person, marking flag	Ongoing; Level 3 registrations for 1, 2 or 5 years	Verified 2026-09-20 [E1]

Ref	Instrument / regulator	Applies to	Obligation	Required data elements → CDS field	Trigger / dates	Verification
D-AU-2	Radiocommunications (Compliance Labelling) notices incl. EMC Notice 2017 – ACMA (au_acma_rcm)	Radiocommunications, EMC, telecommunications and EME-regulated devices	pre_market (responsible supplier registration on the national database); labelling_element (RCM); documentation (compliance records)	Supplier registration (ACMA-only registration has no fee); compliance level 1/2/3 (ACMA levels do not correlate with EESS levels); RCM applied; compliance records held → CMP_compliance_declarations (scheme au_acma_rcm, level), CMP_responsible_person	Ongoing	Verified 2026-09-20 [E1]
D-AU-3	Greenhouse and Energy Minimum Standards (GEMS) Act 2012 and product determinations – Energy Rating (au_gems_energy_rating)	Regulated products (clothes dryers, washers, dishwashers, refrigerators and freezers, televisions, computer monitors, air conditioners up to 30 kW, pool pumps and other MEPS products)	pre_market (registration before supply); rating_label (label in stores; icon for online sales)	Registration in the Energy Rating Product Registration System (brand, model number, energy performance); Energy Rating Label displayed when supplying in Australian retail stores; the Energy Rating Icon may be used online and in advertising and can be generated from the regulated product data extract → CMP_energy_ratings (scheme au_energy_rating, star rating, annual kWh, registration id, icon artefact)	Ongoing; cancelled registrations on determination updates	Verified 2026-09-20 [E2]
D-AU-4	Water Efficiency Labelling and Standards (WELS) Act 2005 (au_wels_water_rating)	Washing machines, dishwashers, taps, showers, toilets, urinals, flow controllers	pre_market (registration); rating_label	WELS registration; star rating and water consumption on the label; display of the label or rating in advertising and online → CMP_energy_ratings (scheme wels_water_rating)	Ongoing	Retrieved (not spot-verified)
D-AU-5	Consumer Goods (Products Containing Button/Coin Batteries) Safety Standard 2020 and Information Standard 2020; Consumer Goods (Button/Coin Batteries) Safety and Information Standards 2020 – ACCC (au_button_battery_standards_2020)	Consumer goods containing or supplied with button or coin batteries; button and coin batteries	pre_market (compliance testing of representative samples); labelling_element; warning_statement; listing_element (recommended)	Secure battery compartment (tool or two independent movements) for consumer-replaceable batteries; batteries must not release under foreseeable use or misuse; captive fasteners; warnings in instructions, on packaging (safety alert symbol on the front panel) or attached to unpackaged goods covering: alert word, symbol, hazard and keep-away statement, lithium 2-hour severe-injury statement or non-lithium serious-injury statement, seek-immediate-medical-attention advice; child-resistant packaging for batteries; online listings should include the warning in the product description → MF_battery_chemistry (button/coin form), MF_warning_statements (scheme au_button_battery_standards_2020), CMP_compliance_declarations (test evidence), listing projection	Mandatory from 22 June 2022	Verified 2026-09-20 [E3]
D-AU-6	Australian Dangerous Goods Code 7.9 (lithium batteries UN 3480/3481 etc.) (au_adg_code_7_9)	Battery products for road and rail transport	transport	Dangerous-goods declaration per CDS-1900 §12	Mandatory from 1 October 2025	Verified 2026-09-20 (CDS-1900 Appendix C)

D.2 New Zealand (NZ)

Ref	Instrument / regulator	Applies to	Obligation	Required data elements → CDS field	Trigger / dates	Verification
D-NZ-1	Electricity (Safety) Regulations 2010 regs 81–86A – WorkSafe Energy Safety (nz_electricity_safety_regulations_2010)	Declared medium-risk articles (SDoC required; list includes lithium-ion/polymer batteries) and declared high-risk articles (approval or recognised certification, or EESS registration by an NZ-registered responsible or affiliated supplier deemed approval)	pre_market; documentation (SDoC available to purchasers on request within 10 days)	Supplier Declaration of Conformity (description, standard cited from Schedule 4 or AS/NZS 3820, ISO/IEC 17050-1 form) with supporting test report; approval number or RCM on high-risk articles; SDoC available to consumers and to WorkSafe within 10 days → CMP_compliance_declarations (scheme nz_electricity_safety_regulations_2010, article class, SDoC reference), MED_technical_documents (SDoC, test report)	Ongoing	Verified 2026-09-20 [E4]

Ref	Instrument / regulator	Applies to	Obligation	Required data elements → CDS field	Trigger / dates	Verification
D-NZ-2	Radiocommunications Act 1989 supplier compliance — RSM (nz_rsm_sdoc)	Radio and EMC products	pre_market (SDoC per product or significant variation); labelling_element (R-NZ mark or RCM)	SDoC for Level of Conformity 1 radio products and all Level 2, 3, A1–A3 products; Australian SDoC recognised under the mutual recognition notice → CMP_compliance_declarations (scheme nz_rsm_sdoc)	Ongoing	Verified 2026-09-20 [E4]
D-NZ-3	Energy Efficiency (Energy Using Products) Regulations 2002 — EECA (nz_eeca_meps)	MEPS and MEPL regulated products	pre_market (registration in NZ or in Australia under trans-Tasman mutual recognition); rating_label	Registration and Energy Rating Label per the joint scheme; icon rules differ from Australia → CMP_energy_ratings (scheme au_energy_rating, jurisdiction NZ)	Ongoing	Verified in part 2026-09-20 [E2][E4]

D.3 United States (US, with California noted)

Ref	Instrument / regulator	Applies to	Obligation	Required data elements → CDS field	Trigger / dates	Verification
D-US-1	47 CFR Parts 2, 15 and 18 equipment authorization — FCC (us_fcc_equipment_authorization)	RF devices: unintentional radiators (most digital devices) via Supplier's Declaration of Conformity; intentional radiators (transmitters) via certification	pre_market (SDoC or certification before marketing or import); labelling_element (FCC ID for certified devices; unique identification and compliance information statement for SDoC; e-labelling permitted with a temporary physical label); documentation (compliance statement, Part 15 user information)	FCC ID or SDoC compliance information (responsible party in the US, identification, statement), Part 15.19/15.105 statements in manuals or packaging, e-label access in ≤3 steps → CMP_compliance_declarations (scheme us_fcc_equipment_authorization, procedure, FCC ID, responsible party), MED_technical_documents	Ongoing	Verified 2026-09-20 [E5]
D-US-2	Energy Labeling Rule, 16 CFR Part 305 (§305.27 websites and catalogs) — FTC (us_ftc_energyguide)	Covered products (refrigerators, freezers, room and portable air conditioners, clothes washers, dishwashers, ceiling fans, water heaters, pool heaters, central air conditioners, heat pumps, furnaces, televisions, lamps; plumbing products by text disclosure)	rating_label (EnergyGuide/Lighting Facts label); listing_element (websites)	Website pages with price and detailed description must display a recognisable, legible label image close to the price, or hyperlink via the FTC EnergyGuide icon without requiring download; manufacturers post labels online and report data via DOE CCMS → CMP_energy_ratings (scheme us_ftc_energyguide, annual cost or efficiency, label artefact), listing_projection	Website display since 15 January 2014	Verified 2026-09-20 [E6]
D-US-3	Proposition 65 (California), 27 CCR 25602–25603 — OEHHA (us_ca_prop65_warning)	Products exposing California consumers to listed chemicals (common for cables, cords, batteries, plastics)	warning_statement; listing_element	Warning on the product display page, "WARNING" hyperlink or prominent pre-purchase warning; new short-form names a chemical per endpoint; 60-day online update window → MF_warning_statements (jurisdiction US-CA)	Effective 1 January 2025; transition to 1 January 2028	Verified 2026-09-20 [E7]
D-US-4	Right to Repair Act (California SB 244, PRC 42488) and comparable state laws (us_right_to_repair_state)	Electronic and appliance products first sold in California on or after 1 July 2021 with wholesale price ≥ US\$50	post_market (parts, tools and documentation for 3 years at US\$50–99.99 wholesale, 7 years at ≥ US\$100)	Availability of documentation, functional parts and tools on fair and reasonable terms; independent repairer notices → MF_spare_parts_availability_years (scheme us_ca_sb244), MED_technical_documents	Operative 1 July 2024; other states (for example New York, Minnesota, Oregon, Colorado) have their own effective dates — not seeded	Verified 2026-09-20 [E8] (California); other states retrieved (not spot-verified)
D-US-5	DOE conservation standards and certification (10 CFR 429/430); UL/NRTL safety listing (voluntary but retailer-required); 49 CFR hazardous materials (batteries)	Appliances; electrical products; battery shipments	pre_market; documentation; transport	DOE certification in CCMS; NRTL listing marks; hazmat declarations → CMP_compliance_declarations, CMP_dangerous_goods	Ongoing	Retrieved (not spot-verified)

D.4 European Union (EU)

Ref	Instrument / regulator	Applies to	Obligation	Required data elements → CDS field	Trigger / dates	Verification
D-EU-1	Low Voltage Directive 2014/35/EU, EMC Directive 2014/30/EU, Radio Equipment Directive 2014/53/EU, RoHS Directive 2011/65/EU — CE marking (eu_ce_marking_electrical)	Electrical, electronic and radio equipment	pre_market (conformity assessment, EU declaration of conformity, technical file); labelling_element (CE marking; manufacturer and importer name, trade mark, postal address; type, batch or serial number)	CE marking; manufacturer and importer identification; product identification element; DoC available; instructions and safety information in the required language → CMP_compliance_declarations (scheme eu_ce_marking_electrical, directives, DoC ref), CMP_responsible_person, CMP_product_identifier, MED_technical_documents	Ongoing	Retrieved (not spot-verified this pass; directive texts not read)
D-EU-2	Energy Labelling Regulation (EU) 2017/1369; product-specific delegated regulations; Commission Implementing Regulation (EU) 2024/994 (EPREL) (eu_energy_label_2017_1369)	Energy-labelled product groups (dishwashers, washing machines, washer-dryers, refrigerators, electronic displays, light sources, tyres, smartphones and tablets, and others)	pre_market (supplier registers the model in EPREL before placing on the market); rating_label; listing_element (dealers, including online stores)	Online stores must show the energy label and product information sheet close to the price (nested display permitted: class arrow pointing left, letter at least the size of the price, full label on click or hover; link named "Product Information Sheet" in the local language; also on list pages and basket pages); QR code readable; suppliers communicate the EPREL registration number to dealers; advertisements show class and range → CMP_energy_ratings (scheme eu_energy_label, class, range, EPREL id, label and PIS artefacts), listing projection and verification	Ongoing; EPREL implementing act 2024	Verified 2026-09-20 [E9]
D-EU-3	Commission Delegated Regulation (EU) 2023/1669 (energy labelling of smartphones and slate tablets) and Commission Regulation (EU) 2023/1670 (ecodesign) (eu_smartphone_energy_label_2023_1669)	Smartphones and slate tablets (labelling); also mobile and cordless phones (ecodesign)	rating_label; listing_element; post_market (spare parts, OS updates, repair information)	Label: energy efficiency class A–G and range, battery endurance per cycle, repeated free-fall reliability class, repairability class, battery endurance in cycles, ingress protection rating; product information sheet in EPREL; dealers show label and PIS online near the price; suppliers publish ingress rating, minimum battery endurance in cycles, recycled content, critical raw material ranges, and make spare parts available to professional repairers for 7 years after end of placement → CMP_energy_ratings (scheme eu_smartphone_energy_label_2023_1669, class, endurance, repairability class, IP rating), MF_spare_parts_availability_years, MF_repairability_class, MED_technical_documents	Units placed on the market from 20 June 2025	Verified 2026-09-20 [E9][E10]
D-EU-4	Regulation (EU) 2023/1542 concerning batteries and waste batteries (eu_batteries_2023_1542)	Batteries, including batteries incorporated in appliances, sold by distance contracts	labelling_element; documentation; post_market (producer responsibility; distance sellers established elsewhere are producers)	Separate collection symbol on all batteries from 18 August 2025; general information label from 18 August 2026; QR code linking to information (and, for certain batteries, the battery passport) from 18 February 2027; carbon footprint declaration accompanying industrial and EV batteries until accessible via QR; producer registration → CMP_compliance_declarations (scheme eu_batteries_2023_1542, markings applied, producer registration), MF_battery_chemistry, MF_battery_capacity_wh	Phased 2025–2027	Verified 2026-09-20 [E11]
D-EU-5	WEEE Directive 2012/19/EU as amended by Directive (EU) 2024/884 (eu_weee_2012_19)	Electrical and electronic equipment	labelling_element (crossed-out wheeled bin per EN 50419:2022, date mark); post_market (producer registration, distance sellers via authorised representative)	Crossed-out bin marking (on packaging, instructions and warranty only in exceptional cases); producer registration number per member state → CMP_compliance_declarations (scheme eu_weee_2012_19)	Ongoing	Verified 2026-09-20 [E12]
D-EU-6	Regulation (EU) 2023/988 General Product Safety Regulation (eu_gpsr_2023_988)	All consumer products (cross-cutting distance-selling duties also for harmonised products)	listing_element; labelling_element; pre_market (responsible economic operator established in the EU)	Product bears type, batch or serial number and manufacturer name, trade mark, postal and electronic address (importer details where applicable); online offers show the manufacturer's (or responsible operator's) name, address and contact, product identification information, warnings and safety information, and a picture; warnings and instructions in a language easily understood; traceability records 6 years → CMP_responsible_person, CMP_product_identifier, MF_warning_statements, listing projection	Applies from 13 December 2024	Verified 2026-09-20 [E13]
D-EU-	Directive (EU) 2022/2380 amending the Radio Equipment Directive (common charger)	Handheld mobile phones,	labelling_element; listing_element	USB Type-C receptacle; label with charging specifications (minimum and maximum power in	28 December 2024; laptops	Verified 2026-09-20

Ref	Instrument / regulator	Applies to	Obligation	Required data elements → CDS field	Trigger / dates	Verification
7	(eu_common_charger_2022_2380)	tablets, cameras, headphones, headsets, handheld videogame consoles, portable speakers, e-readers, keyboards, mice, navigation systems, earbuds (from 28 December 2024); laptops (from 28 April 2026) rechargeable by wired charging		watts, USB PD support) printed in the instructions and on packaging; pictogram indicating whether a charging device is included; in distance selling the pictogram and label displayed close to the price → MF_charging_port, MF_charging_power_min_w, MF_charging_power_max_w, MF_usb_pd_supported, MF_charger_included, listing projection	28 April 2026	[E14]
D-EU-8	Regulation (EU) 2024/1781 Ecodesign for Sustainable Products (ESPR) and delegated acts (eu_espr_2024_1781)	Product groups as delegated acts are adopted (digital product passport, labels, unsold-goods rules)	listing_element; documentation	Dealers and online marketplaces make the data carrier or unique product identifier accessible to potential customers in distance selling; labels displayed as delegated acts specify → CMP_product_identifier (DPP link), MED_technical_documents	Framework in force 18 July 2024; obligations per delegated act	Verified 2026-09-20 [E15] (framework); product-group acts not seeded

D.5 United Kingdom (GB)

Ref	Instrument / regulator	Applies to	Obligation	Required data elements → CDS field	Trigger / dates	Verification
D-UK-1	Electrical Equipment (Safety) Regulations 2016; Electromagnetic Compatibility Regulations 2016; Radio Equipment Regulations 2017; RoHS Regulations 2012; Product Safety and Metrology etc. (Amendment) Regulations 2024 — OPSS (uk_electrical_equipment_regulations_ukca_ce)	Electrical, electronic and radio equipment placed on the GB market	pre_market (UK or EU declaration of conformity; UKCA or CE marking — CE recognition continued indefinitely from 1 October 2024; "Fast-Track UKCA" on EU conformity assessment); labelling_element (manufacturer name, registered trade name or mark, postal address, type/batch/serial; importer details, which may be on packaging or documents until 31 December 2027; UKCA on a label or accompanying document permitted until 31 December 2027)	Conformity marking choice, DoC reference, manufacturer and importer identification, product identifier → CMP_compliance_declarations (scheme uk_electrical_equipment_regulations_ukca_ce), CMP_responsible_person, CMP_product_identifier	CE recognition indefinite from 1 October 2024; labelling flexibilities to 31 December 2027	Verified 2026-09-20 [E16]
D-UK-2	The Waste Electrical and Electronic Equipment Regulations 2013 (as amended, incl. online marketplace provisions 2025) — Environment Agency (uk_weee_2013)	EEE placed on the UK market, including by distance sellers established outside the UK and via	labelling_element (crossed-out wheeled bin; date mark); post_market (producer scheme membership; distributor take-back)	Crossed-out bin symbol and post-13 August 2005 date mark; producer registration; distributor take-back or scheme membership → CMP_compliance_declarations (scheme uk_weee_2013)	Ongoing; marketplace provisions 2025	Verified 2026-09-20 [E17]

Ref	Instrument / regulator	Applies to	Obligation	Required data elements → CDS field	Trigger / dates	Verification
		online marketplaces				
D-UK-3	Energy Information Regulations 2011 (GB energy label) and Ecodesign for Energy-Related Products Regulations 2010 (uk_energy_information_2011)	Energy-labelled product groups	rating_label; listing_element	GB energy label and product information sheet displayed online close to the price in the same manner as the EU scheme (UK label, no EPREL) → CMP_energy_ratings (scheme uk_energy_label, existing energy_rating_system value)	Ongoing; CE recognition for ecodesign continued from 1 October 2024	Retrieved 2026-09-20 [E16] (sector table); regulation text not read
D-UK-4	Waste Batteries and Accumulators Regulations 2009; Batteries and Accumulators (Placing on the Market) Regulations 2008 — Defra (uk_batteries_2009)	Batteries and battery-containing products	labelling_element (crossed-out bin); post_market (producer registration; distributor take-back)	Markings and producer registration; no conformity marking required in GB (EU battery CE marking applies only in Northern Ireland) → CMP_compliance_declarations (scheme uk_batteries_2009)	Ongoing; UK considering its approach to the EU regulation	Retrieved 2026-09-20 [E16] (sector table)

D.6 Canada (CA)

Ref	Instrument / regulator	Applies to	Obligation	Required data elements → CDS field	Trigger / dates	Verification
D-CA-1	Energy Efficiency Act (S.C. 1992, c. 36) and Energy Efficiency Regulations, 2016 (SOR/2016-311) — Natural Resources Canada (ca_energy_efficiency_regulations_2016)	Regulated energy-using products imported or shipped between provinces for sale or lease (appliances, water heaters, HVAC, lighting, electronics, commercial equipment)	pre_market (energy efficiency report to NRCan before import or interprovincial shipment; product meets the standard; energy efficiency verification mark from an SCC-accredited certification body or equivalent province); rating_label (bilingual EnerGuide label on clothes washers, dryers, dishwashers, refrigerators, freezers, ranges, room air conditioners and others; lighting labels; walk-in nameplates)	Verification mark (readily visible; on packaging for chargers, power supplies and lamps); energy efficiency report (product name, brand, model number, manufacturer, certification body); EnerGuide label with annual kWh (or CEER) attached before first retail sale; NRCan does not currently require labels to be posted online → CMP_energy_ratings (scheme ca_energiguide, annual kWh, verification body), CMP_compliance_declarations (report reference)	Ongoing	Verified 2026-09-20 [E18]
D-CA-2	Radio Standards Specifications and ICES standards — ISSED (ca_ised_radio_equipment); electrical product safety certification required by provincial and territorial electrical safety authorities (ca_electrical_safety_certification)	Radio equipment (IC certification or SDoC per RSS/ICES); electrical products (certification mark from an accredited body before sale in each province)	pre_market; labelling_element	IC certification number or SDoC statement; recognised certification mark (for example cCSAus, cULus) → CMP_compliance_declarations	Ongoing	Retrieved 2026-09-20 [E19] (not spot-verified against ISSED or provincial authorities)
D-CA-3	Consumer Packaging and Labelling Act and Regulations; Canada Consumer Product Safety Act — CFIA/Health Canada	Prepackaged consumer products	labelling_element	Bilingual product identity, net quantity and dealer identity; mandatory safety information → localisation dimension per CDS-300 §18, CMP_responsible_person	Ongoing	Retrieved (not spot-verified)

D.7 Other markets

Jurisdiction	Status	Note
Singapore (SG), Japan (JP), China (CN)	Energy-label schemes only	The energy_rating_system dictionary already carries sg_energy_label, jp_energy_label and cn_energy_label; safety, radio and transport regimes for those markets are not seeded.
Republic of Korea (KR), India (IN), Brazil (BR) and others	Not seeded	Organisation supplies entries with local advice (for example KC marking, BIS registration, INMETRO certification).

Commerce Data Standard (CDS)

CDS-1900 — Parts, Automotive, Industrial and Fitment Industry Profile

Field	Value
Status	vo.2 Review Draft (working source; not an approved standard)
Release	CDS vo.2 (single corpus release per ADR-D5); chapter added in the 2026-09-20 industry-profile expansion
Date	2026-09-20
Supersedes	Nothing — first edition. Package Chapter 18 (Parts, Automotive & Industrial: relationship types, part origin types, product conditions, dangerous-goods classes) is shipped with this profile in package 0.8.0.
Normative status	§1, §3, §5–§14, §19 and Appendix A are normative. §2 (pointer), §4, §15–§18, §20, §21 and Appendices B–D are informative; Appendix D seeds the jurisdiction requirement register whose record structure and obligations are normative in CDS-1500 §2.2. Every table is individually marked. This chapter is the single normative home (CDS000-R005) for three cross-industry mechanisms: the Product Relationship record (§5), kits, bundles and assemblies (§11) and the dangerous-goods declaration (§12). Other industry chapters cite these sections and do not restate them.
Primary audience	Parts and industrial merchandisers, catalogue and fitment data managers, compliance and dangerous-goods owners, B2B ecommerce operators, data stewards, PIM architects, developers and AI enrichment designers
Depends on	CDS-000 through CDS-1500, especially CDS-200 (entity model, taxonomy mappings, relationships), CDS-300 (CMP_, SUP_, PRC_ and INV_ conditional), CDS-400 (dictionaries, units, related values), CDS-500 (preflight, verification, observation coverage), CDS-600 (facets), CDS-700 (evidence), CDS-900 (platform profiles, supplier import profile), CDS-1100 (entity references, identifiers), CDS-1500 (profile model, requirement levels, profile register, jurisdiction requirement register), CDS-1800 (technical specification rules, cited)
Companion package	CDS Reference Dictionary: Chapter 18 (Parts, Automotive & Industrial), Chapter 19 (Jurisdictions & Regulatory Schemes)
Profile identifier	<code>cds.profile.parts_fitment.v0_2</code> (registered in CDS-1500 §2.1)
Research basis	REVIEW-020 (global commerce vertical market research, 2026-09-20): automotive and hardware-and-tools stores carry the largest catalogues of any Shopify vertical (averages above 5,700 and 4,200 products per store respectively), DIY and hardware is a top-five worldwide ecommerce revenue segment (roughly US\$0.4 trillion for 2026), and industrial and scientific categories are among the fastest-growing on Amazon; store counts are modest, PIM intensity per store is the highest of any archetype.

Terminology follows CDS-100. Attribute requirement levels (R, C, REC, O, N/A) are those of CDS-1500 §4 and are not restated here. "Fitment" means the declared applicability of a part to a vehicle, machine, appliance or other host; "interchange" means equivalence between part numbers.

1. Purpose and Scope (*normative*)

CDS-1900 defines the industry profile for automotive and powersports parts and accessories, industrial and maintenance supplies, tools and hardware, electrical and plumbing components, hobby components (radio-control, model, craft hardware) and spare parts for appliances and equipment. It also defines, for the whole corpus, how products relate to one another (compatibility, fitment, inclusion, interchange, supersession), how kits and bundles are composed, and how dangerous goods are declared for transport.

The data shape is distinctive because the purchase decision is "will this fit?" rather than "do I like this?": the record is dominated by relationships to hosts and to other parts, by part-number equivalences and supersession chains, by typed technical attributes in the language of an external classification standard, by units of measure and pack quantities that determine what a sale actually delivers, and by hazardous-materials data that determines whether the item can be shipped at all.

CDS1900-R001 An implementation claiming the Parts and Fitment Profile **MUST** represent fitment, compatibility, inclusion, interchange and supersession as product relationship records (§5), never as tags, free text, title conventions or category placement.

CDS1900-R002 An implementation claiming this profile **MUST** represent units of measure, pack quantities, sale-unit hierarchies and variable-length or variable-quantity sale forms as typed attributes (§10) so that the delivered quantity of every order line is deterministic.

CDS1900-R003 An implementation **MUST** declare a dangerous-goods status for every product (§12) and **MUST** maintain a jurisdiction requirement register (CDS-1500 §2.2) for every market it publishes parts, chemicals or equipment to.

CDS1900-R004 This profile **MUST NOT** weaken any rule of CDS-1500 that it reuses; where this chapter is silent, CDS-1500, CDS-1800 (for technical specifications) and the core chapters govern.

2. Profile Model (*informative pointer*)

This profile applies the industry profile model of CDS-1500 §2, the requirement levels of CDS-1500 §4 and the jurisdiction requirement register of CDS-1500 §2.2 without restatement. Its identifier and dictionary bindings are in CDS-1500 §2.1; its seeded jurisdiction entries are in Appendix D.

3. Product Model (*normative*)

A part is identified by its manufacturer part number under a brand. The Product is the part; Variants are rare and arise only from sellable-unit differences (pack quantity, finish, length cut) — never from the hosts the part fits.

```

Product: Front brake pad set, brand ExampleBrake, part number EB-1234      (informative example)
Product scope:
  STD_brand = ExampleBrake
  SUP_manufacturer_part_number = EB-1234
  CAT_product_type = brake_pad_set
  MF_part_origin_type = aftermarket          # part_origin_type dictionary
  MF_position = front
  MF_quantity_per_application = 1           # one set per vehicle
  MF_unit_of_sale = set; MF_units_per_sale = 4    # four pads per set
relationships:
  - {type: fits, target: application {make: Example, model: Sedan, years: 2018–2023, engine: 2.0L}, position: front}
  - {type: alternative_to, target: external_part {brand: OEM, number: 04465-00000}}
  - {type: superseded_by, target: product EB-1234A, effective: 2026-03-01}
Variant scope:
  VAR_gtin = 09300000000027

```

CDS1900-R005 Fitment MUST NOT be modelled as variants; a part that fits many hosts is one product with many `fits` relationship records (§6).

CDS1900-R006 Pack quantity, finish, colour, length cut and hand (left/right) MAY be variant options where they identify distinct sellable units; position on the host (front, rear, left, right) is a fitment qualifier (§6) unless a distinct part number exists per position, in which case it is a separate product.

CDS1900-R007 Brand and manufacturer part number MUST be governed identity attributes (CDS-1800 §5 rules R009–R011 apply); the same part number under two brands is two products related by `cross_reference`, never one product with two brands.

4. Product Families and Category Profiles *(informative baseline; the inheritance rule of CDS1500-R016 applies)*

Product family / category	Required	Recommended	Typical variant options
Automotive replacement parts	brand; MPN; part origin type; fitment records with declared vocabulary source; position and quantity per application; condition; interchange records	supersession; technical attributes per category; core charge status; warranty; compliance marks per market (E-mark, DOT, CARB EO)	pack quantity
Automotive accessories and consumables (oils, fluids, care)	brand; MPN; fitment or universal declaration; dangerous-goods declaration; SDS reference where hazardous; net quantity	specification standards (viscosity grade, approvals as certifications)	size, multipack
Tyres and wheels	brand; MPN; size designation; load index; speed rating; regulated tyre label per market; DOT/E-mark compliance	season; run-flat; noise class	—
Tools and hardware	brand; MPN; technical attributes (size, drive, material, rating); unit of sale and pack quantity; hazardous status for chemicals	compatibility with systems (battery platforms as relationships); external classification code	size, pack
Electrical and plumbing components	brand; MPN; class attributes per an adopted classification model (e.g. ETIM class features); certification records per market; unit of sale (each / metre / box)	—	length, pack
Industrial MRO supplies	brand; MPN; classification code (UNSPSC/ETIM/GPC); unit of sale; MOQ and order multiple where B2B	SDS; shelf life for chemicals	pack
Hobby components (RC, model, craft)	brand; MPN; compatibility records; scale or standard; battery and DG where applicable	included items (kits)	colour, pack
Appliance and equipment spare parts	brand; MPN; fitment to appliance models (fits records); original vs compatible origin type	exploded-view diagram reference	—

5. Product Relationship Record *(normative — single home for the corpus)*

CDS-200's entity model gives products a `relationships` collection of entity references; this section defines the record that collection carries. A product relationship is a governed, directed, typed link from a product (or variant) to a target, with the evidence and lifecycle of any other governed fact.

Property	Meaning	Requirement
<code>relationship_id</code>	Stable identifier	Required
<code>source</code>	Product or variant reference	Required
<code>relationship_type</code>	Governed value from the <code>relationship_type</code> dictionary (package Chapter 18): <code>fits</code> , <code>compatible_with</code> , <code>accessory_for</code> , <code>requires</code> , <code>includes</code> , <code>part_of_set</code> , <code>replaces</code> , <code>superseded_by</code> , <code>alternative_to</code> , <code>cross_reference</code> , <code>successor_of</code> , <code>variant_family_of</code>	Required
<code>target_kind</code>	<code>product</code> , <code>variant</code> , <code>model_family</code> , <code>standard</code> , <code>application</code> , <code>external_part</code> , <code>external_product</code>	Required
<code>target</code>	Reference or embedded record appropriate to the kind (a product reference; a governed standard value; an application record per §6; an external part record with brand and number)	Required
<code>quantity</code>	Quantity of the target per source (for <code>includes</code> , <code>requires</code>) or of the source per application (for <code>fits</code>)	Conditional
<code>qualifiers</code>	Governed qualifiers (position, side, drive, engine code, option code, notes from a governed qualifier vocabulary)	Optional
<code>direction</code>	<code>directed</code> or <code>symmetric</code> (declared per relationship type in the dictionary)	Required
<code>status</code>	Lifecycle per CDS-400 §17.1 (proposed / active / deprecated / retired / rejected)	Required

Property	Meaning	Requirement
effective_from / effective_to	Validity window (supersession dates, model years)	Optional
provenance	Source, evidence class, actor, timestamps (CDS-400 §23; CDS-700 for AI proposals)	Required
version	Record version	Required

CDS1900-R008 Every product relationship MUST be a record conforming to this section, carried in the product's `relationships` collection (CDS-1100 §8) or an equivalent governed structure, with a governed `relationship_type`, a typed target and provenance.

CDS1900-R009 Relationship types MUST be governed dictionary values whose direction (directed or symmetric) is declared in the dictionary; an implementation MUST NOT invent relationship semantics per product.

CDS1900-R010 Relationships MUST NOT be stored as tags, in free-text fields, in titles or in category membership as the system of record; such projections MAY be generated from relationship records.

CDS1900-R011 A relationship whose target is another product in the catalogue MUST reference that product's stable identifier; a relationship whose target is outside the catalogue (an OEM number, a competitor part, a vehicle) MUST use the external target kinds with the identifying scheme declared.

CDS1900-R012 Relationships MUST carry provenance and evidence class (CDS-700 §7); an AI-proposed relationship MUST be a proposal with evidence and MUST NOT be published without human review (CDS-700 §19), because a wrong fitment or compatibility is a safety and returns event.

CDS1900-R013 Relationships are versioned governed facts: changes follow CDS-400 §24 (versioning, deprecation with replacement, breaking-change migration) and a superseded relationship MUST be retained with its validity window, never deleted.

CDS1900-R014 Where a channel accepts relationship data (compatibility lists, complementary products, fitment tables), the projection MUST be generated from relationship records, declared in the Field Mapping with its comparison strategy, and verified under CDS-500; where the channel exposes no read-back for relationships, the verification status MUST be UNOBSERVABLE, not MATCH.

CDS1900-R015 Symmetric relationships (`compatible_with`, `alternative_to`, `part_of_set`) MUST be stored once and evaluated in both directions; directed relationships MUST NOT be assumed to hold in reverse.

CDS1900-R016 Cycles in `superseded_by` and `replaces` chains MUST be rejected at validation (the CDS400-R036 pattern).

6. Fitment and Application Data (*normative*)

Fitment is a `fits` relationship whose target is an application record: a description of the host (vehicle, machine, appliance model) drawn from a declared vocabulary. The corpus ships no vehicle or equipment tables — the leading reference databases are licensed (for example the Auto Care Association's vehicle configuration database used by the ACES standard, or manufacturer model lists) — so the organisation declares which vocabulary and release each application record uses.

Application record property	Meaning
vocabulary_source	Governed value naming the reference vocabulary and release (for example <code>aces_vcdb_<release></code> , <code>manufacturer_model_list_<brand>_<version></code> , <code>organisation_vehicle_dictionary_v3</code>)
host_type	vehicle, powersport, marine, equipment, appliance, other
make / model / submodel / body / series	Governed values from the vocabulary
year_from / year_to (or model year list)	Typed integers
engine / transmission / drive / fuel / region qualifiers	Governed qualifier values from the vocabulary
equipment or appliance model identifiers	Governed values where the host is not a vehicle
position	Governed value (front, rear, left, right, upper, lower, inner, outer, ...)
quantity_per_application	Integer
notes	Governed qualifier statements only (from a qualifier vocabulary), never free text

CDS1900-R017 Every fitment MUST be a `fits` relationship with an application record whose vocabulary source and release are declared; application values MUST be governed values of that vocabulary, never free text.

CDS1900-R018 Universal fitment MUST be an explicit declaration (`MF_fitment_scope` = `universal` with the governed scope it is universal within), never the absence of fitment records; a part with neither fitment records nor a universal declaration MUST NOT satisfy an R-level fitment requirement.

CDS1900-R019 Position and quantity per application MUST be typed qualifiers on the fitment record where the part's applicability depends on them.

CDS1900-R020 Fitment MUST NOT be derived from titles, descriptions, supplier category names or image text by rules or AI without human review; a proposed fitment carries evidence (manufacturer application catalogue, E1; certified data feed, E2) and review state.

CDS1900-R021 Model years MUST be typed integers or ranges; a fitment record MUST NOT encode years in free text.

CDS1900-R022 When the declared vocabulary releases a new version, the organisation MUST record the migration of affected application records as a governed dictionary migration (CDS400-R082 pattern); records bound to a retired vocabulary release MUST be flagged for review, not silently re-pointed.

CDS1900-R023 Fitment published to a channel (a vehicle compatibility table, a fitment metafield, a "fits your vehicle" lookup) MUST be generated from fitment records and verified under CDS-500 where the channel exposes read-back; otherwise reported UNOBSERVABLE (R014).

CDS1900-R024 A storefront "shop by vehicle" or "shop by model" selector is an application filter over fitment records, distinct from customer facets (CDS-600); it MUST use the same governed vocabulary as the fitment records and MUST NOT be populated from free-text vehicle tags.

7. Interchange, Cross-Reference and Supersession (*normative*)

CDS1900-R025 Original-equipment part numbers, competitor and aftermarket equivalents MUST be `alternative_to` or `cross_reference` relationships with external part targets that carry the issuing brand and its identifier scheme; the organisation's own SKU MUST NOT be stored in a manufacturer- or OEM-number attribute (CDS-1800 R010).

CDS1900-R026 Supersession MUST be a directed `superseded_by` relationship with an effective date; a superseded part's lifecycle state MUST reflect its status and its record MUST be retained (R013) so that searches for the old number resolve to the replacement.

CDS1900-R027 Equivalence ("alternative_to") MUST declare its basis (form-fit-function equivalent, functional equivalent, manufacturer-declared) as a governed qualifier; an equivalence asserted without a basis MUST be treated as a proposal.

CDS1900-R028 Where a channel or marketplace uses interchange data for search, the projection MUST be generated from relationship records (R014).

CDS1900-R029 Cross-reference numbers MUST be searchable through governed search synonyms or identifier indexing (CDS600-R075) without being exposed as facets.

8. Technical Attributes and External Classification (*normative*)

Technical attributes follow the specification rules of CDS-1800 §6 (typed, unit-bearing, category specification sets, basis qualifiers, provenance). This section adds the mechanism for adopting an external classification and attribute model as the source of a category's specification set.

External model (informative)	What it provides
ETIM (electro-technical, HVAC, building, tools, shipbuilding sectors)	Classes with typed features (alphanumeric from value lists, numeric with units, logical, range); one class per product; released in versions
GS1 Global Product Classification (GPC)	Segment / family / class / brick with brick attributes and values; 8-digit codes; updated twice yearly
UNSPSC	Procurement commodity codes (segment / family / class / commodity); no attributes
Auto Care Association PCdb and PAdb (used by ACES/PIES)	Part terminology and part attributes for the automotive aftermarket; licensed
Marketplace category attribute sets	Channel-specific required attributes

CDS1900-R030 An organisation MAY adopt an external classification model as the source of a category's specification set; where it does, each external class or brick MUST be mapped to an internal category through a Taxonomy Mapping (CDS-200 §6) with the model's version, and each external feature MUST be mapped to an Attribute Definition with a declared crosswalk (name, unit, value list mapping).

CDS1900-R031 External classification codes MUST be stored as Taxonomy Mappings or governed identifiers with their model version, never as free text, and a product MUST be assigned to exactly one class in any one model.

CDS1900-R032 Values drawn from an external value list MUST be governed dictionary values with alias mappings for the external codes (CDS-400 §7), so that a change of model version is a governed migration.

CDS1900-R033 Dimensional, threaded and tolerance attributes MUST be structured (nominal value, unit, tolerance, thread standard) rather than a single text token such as "M8x1.25".

CDS1900-R034 Weight and dimensions used for logistics MUST be typed measurements of the sale unit (§10), distinguished from the technical dimensions of the part.

9. Origin Type, Brand, Manufacturer and Condition (*normative; dictionary bindings informative*)

Field	Meaning	Dictionary
MF_part_origin_type	Genuine (original equipment as sold by the vehicle or equipment maker), original equipment supplier (OES), aftermarket	<code>part_origin_type</code> (Chapter 18)
MF_condition	new, refurbished, remanufactured, used	<code>product_condition</code> (Chapter 18)
STD_brand / SUP_manufacturer / MF_remanufacturer	Brand as sold; legal manufacturer; remanufacturer where condition is remanufactured	Governed values
PRC_core_charge	Core charge where the PIM is the pricing authority	Conditional (CDS-300 §14)

CDS1900-R035 Part origin type MUST be a governed value and MUST NOT be inferred from brand alone; a "genuine" claim MUST carry evidence (E1/E2) because it is a claim about the maker's supply chain.

CDS1900-R036 Condition MUST be a governed variant-scope value; remanufactured MUST be distinguished from refurbished and from used, and a channel that supports fewer condition values MUST receive a declared mapping (for example remanufactured → refurbished) recorded on the Attribute Definition.

CDS1900-R037 Brand, manufacturer and remanufacturer MUST be separate attributes where they differ.

CDS1900-R038 Core-charge and exchange-part terms MUST be typed attributes under the pricing authority and MUST NOT be encoded in titles.

10. Units of Measure, Pack Quantity and B2B Terms (*normative*)

Field	Meaning	Type
MF_unit_of_sale	The unit one order line delivers (each, pair, set, box, roll, metre, litre, kilogram)	Governed value
MF_units_per_sale	Number of base units in the sale unit (pads per set, pieces per box)	Integer
MF_sold_by_measure	Whether the item is sold by cut length, weight or volume rather than by count	Boolean
MF_measure_increment / MF_measure_minimum / MF_measure_maximum	Increments and bounds for cut-length or measured sales	Typed measurements
MF_minimum_order_quantity / MF_order_multiple	B2B ordering constraints	Integers (conditional on authority)
VAR_sale_unit_level	each / inner / case / pallet with contained quantities and identifiers	Per CDS-1700 R007 pattern

CDS1900-R039 Every product MUST declare its unit of sale and the number of base units per sale unit; a "set of four" and a single unit MUST be distinguishable without reading the title.

CDS1900-R040 Products sold by cut length, weight or volume MUST declare the measure, its increment and bounds as typed values; the delivered quantity is an order-level fact and MUST NOT be a product variant per length unless the organisation sells fixed pre-cut lengths.

CDS1900-R041 Sale-unit levels (each, inner, case) MUST follow the hierarchy rules of CDS1700-R007 with trade identifiers per level where they exist.

CDS1900-R042 Minimum order quantities, order multiples and B2B price breaks MUST be typed attributes under the declared pricing or commercial authority (CDS-200 §13; `PRC_` / `INV_` only where the PIM is the authority) and MUST be evaluated at preflight for channels that enforce them.

CDS1900-R043 Logistics measurements (packaged dimensions and weight of the sale unit) MUST be typed and distinct from technical dimensions (R034); oversized and hazardous handling classes MUST be derived under governed rules.

11. Kits, Bundles and Assemblies (*normative — single home for the corpus*)

A kit is a product whose sellable content is composed of other products. The corpus distinguishes a **stocked kit** (a manufacturer or retailer-assembled item with its own identifier and inventory), a **virtual bundle** (assembled at order time from component inventory) and an **assembly** (a product whose components are also sold separately as spare parts).

CDS1900-R044 A kit, bundle or assembly MUST be a product whose composition is expressed as `includes` relationship records with quantities (§5); component identity, specifications, compliance records and declarations remain on the components.

CDS1900-R045 The kit kind (stocked kit, virtual bundle, assembly) MUST be a governed attribute (`MF_kit_kind`), because inventory authority, trade identifiers and channel projection differ: a stocked kit MAY carry its own GTIN; a virtual bundle MUST NOT claim a trade identifier it does not have.

CDS1900-R046 Kit-level declarations that depend on components (allergens and nutrition for food hampers per CDS-1700; ingredients for beauty sets per CDS-1600; dangerous-goods status per §12; battery facts per CDS-1800) MUST be derived from the component records under declared rules and MUST NOT be independently authored.

CDS1900-R047 Where a channel supports bundle semantics (a bundle flag, component listing, bundle pricing), the projection MUST be generated from the relationship records and declared in the Field Mapping; availability of a virtual bundle MUST be derived from component availability under the inventory authority and MUST be verifiable (CDS-500).

CDS1900-R048 Removing or replacing a component MUST be a governed relationship change (R013) with the kit's revision advanced (CDS-200 §14), so that a published kit state can be traced to its composition.

12. Hazardous Materials and Dangerous Goods (*normative — single home for the corpus*)

Whether a product can be shipped, by which mode, in what packaging and with which documents is determined by its classification under the transport-of-dangerous-goods regimes, and whether it must be accompanied by a safety data sheet is determined by workplace and consumer chemical regimes. The declaration is a product fact; the carrier's documentation is a projection of it.

Property	Meaning
CMP_dangerous_goods.regulated	Boolean: whether the product (or any component of a kit) is regulated for transport in any declared mode; an explicit false is a declaration, not a default
un_number	UN number (e.g. UN3480)
proper_shipping_name	Proper shipping name

Property	Meaning
class_or_division	Governed value from the <code>dangerous_goods_class</code> dictionary (Chapter 18)
subsidiary_hazards	Governed values
packing_group	I / II / III or none
quantity_per_package / net_quantity_basis	Typed measurements used for limited and excepted quantity determination
limited_quantity_eligible / excepted_quantity_eligible	Booleans with the regime they were evaluated under
transport_modes	Modes and regimes evaluated (road/rail under the national code, sea under IMDG, air under ICAO/IATA)
battery_specifics	Watt-hours or lithium content, cells or batteries per package, contained-in/packed-with/alone, state of charge constraints, test summary reference (CDS-1800 §8)
sds_reference	Safety data sheet document (MED_) with format (GHS revision) and date
hazard_communication	GHS pictograms, signal word, hazard and precautionary statements as labelled
declared_by / declared_at / evidence	Provenance

CDS1900-R049 Every product MUST carry a dangerous-goods declaration record whose `regulated` value is an explicit declaration; a product with no record MUST NOT be published to a channel or carrier integration that requires shipping data, and the absence of a record MUST be reported as a missing required value (CDS200-R035), never treated as "not regulated".

CDS1900-R050 A regulated product's declaration MUST carry the UN number, proper shipping name, class or division and packing group (where applicable) from the governing regime, and the quantity facts needed to evaluate limited or excepted quantity provisions.

CDS1900-R051 Dangerous-goods classification MUST NOT be inferred from category, title or description by rules or AI; it MUST derive from the safety data sheet, the manufacturer's declaration or a competent classification (E1) and MUST be reviewed by a person accountable for dangerous-goods compliance before acceptance.

CDS1900-R052 Where a product is a hazardous chemical in a market that requires a safety data sheet to be prepared or supplied (Appendix D), the SDS MUST be a typed document record with its GHS revision and date, linked to the product, and its presence MUST be evaluated at preflight for that market; a consumer-market exemption from SDS supply MUST be recorded as the reason where relied on.

CDS1900-R053 GHS hazard communication elements as labelled (pictograms, signal word, hazard and precautionary statements) MUST be structured statement records where a market requires them, and MUST NOT be projected as customer facets.

CDS1900-R054 Kits inherit the most restrictive dangerous-goods status of their components under declared rules (R046); a kit containing a regulated component MUST be declared regulated.

CDS1900-R055 Carrier and channel shipping declarations MUST be projections of the record, declared in the Field Mapping with write semantics and verified under CDS-500 where read-back exists; a carrier rejection MUST be imported as an observed quality event, not written into the canonical record.

CDS1900-R056 Products whose transport is prohibited in a mode (for example damaged, defective or recalled lithium batteries by air) MUST carry the prohibition as a governed attribute evaluated at preflight for channels shipping by that mode.

13. Compliance, Certifications and Vehicle Standards (*normative*)

CDS1900-R057 Regulatory compliance marks and approvals for parts (vehicle-component type approvals, emissions executive orders, design-rule compliance, electrical and machinery conformity marks, tyre labels) MUST be governed compliance records per market (the CDS-1800 §12 pattern: scheme, identifier, evidence, responsible party, status, dates) and MUST NOT be treated as populated because a mark appears in an image or a phrase in copy.

CDS1900-R058 Where a market requires a label or class information to be shown online close to the price (tyre labels in the EU and GB, Appendix D), the projection MUST be generated from the compliance record and verified.

CDS1900-R059 Performance and quality standards a part is claimed to meet (viscosity grades, industry specifications, manufacturer approvals) MUST be certification or claim records with evidence (CDS-1600 §11 pattern); they MAY feed claim-driven facets only under CDS1500-R049 governance.

CDS1900-R060 Country of origin and tariff classification, where required for cross-border sale, MUST be governed attributes (`CMP_country_of_origin`, `CMP_tariff_code` with scheme and version) with provenance.

14. Media and Technical Documentation (*normative*)

CDS1900-R061 Exploded-view diagrams, installation instructions, technical data sheets, safety data sheets and certificates MUST be typed media records (MED_) with language, version and the product or relationship they document; a diagram callout that identifies a part MUST link to the part's product record.

CDS1900-R062 Documents a market requires to be available to purchasers (Appendix D) MUST be present before publication to that market and their presence verified.

15. Customer Facet and Navigation Design (*informative — normative facet rules live in CDS-600*)

Construct	Recommended baseline behaviour	Anti-pattern
Shop by vehicle / model	Application filter over fitment records using the governed vocabulary (R024)	Free-text vehicle tags; category per vehicle

Construct	Recommended baseline behaviour	Anti-pattern
Brand	Governed brand dictionary	Supplier spellings
Position / side	Governed qualifier values	Free text
Origin type	Genuine / OES / aftermarket from the governed value	Unevidenced "genuine"
Condition	Governed condition values	—
Technical attributes	Typed specification facets with units and ranges per category	Text buckets
Universal fit	Explicit declaration facet	Absence of fitment treated as universal
Hazardous handling	Never a customer facet; a fulfilment attribute	—

16. Channel Projection Guidance (*informative — normative rules: CDS-500, CDS-900*)

Canonical concept	Metafield-style channel (informative)	Feed-style channel (informative)
Fitment	Fitment metafields or app-managed compatibility tables generated from records; read back where the app exposes it, otherwise UNOBSERVABLE	The seeded feed channels (Google Merchant Center, Meta) define no fitment attributes (verified 2026-09-20); carry vehicle applicability in <code>product_detail</code> and titles under a declared policy. Marketplaces with fitment models (for example automotive parts marketplaces) are candidates for future CDS-900 platform profiles
Interchange and supersession	Search synonyms and redirects generated from records	Not published
Unit of sale and pack	Typed metafields; title composition policy	<code>multipack</code> , <code>is_bundle</code> , <code>unit_pricing_measure</code> where relevant
Condition	Condition metafield or option	<code>condition</code> ; remanufactured mapped per Ro36
Dangerous goods	Fulfilment metafields; carrier declarations	Not consumer-facing
Compliance labels required online (tyre label)	Generated per-market block near price	Description where policy allows

17. AI Enrichment and Review (*informative — normative AI rules live in CDS-700*)

Task	AI may propose	Deterministic or human control
Fitment extraction from application catalogues	Candidate <code>fits</code> records bound to the declared vocabulary	Mandatory review (Ro20); vocabulary validation
Cross-reference proposals	Candidate <code>alternative_to</code> / <code>cross_reference</code> records with evidence	Mandatory review; equivalence basis required (Ro27)
Technical attribute extraction	Typed candidates (CDS-1800 §15)	Unit and range validation; provenance
External class assignment	Candidate ETIM/GPC class	Taxonomy mapping governance; one class per model (Ro31)
Dangerous-goods classification	None — SDS or manufacturer declaration only	Ro51
Kit composition	Candidate component lists from documents	Relationship governance (Ro44)

18. Governance and Organisational Extensions (*informative — rules in CDS-1500 §23 and CDS-800*)

The relationship-type, origin-type and condition dictionaries are extended under CDS-400; the declared application vocabulary is a governed external dictionary with release-based migration (Ro22). Dangerous-goods classification changes (a new UN entry, a regime edition such as the Australian Code 7.9 replacing 7.8) are handled as governed dictionary and record migrations with an accountable dangerous-goods owner (CDS-800 §5). The jurisdiction requirement register for parts and chemicals is reviewed on the declared cadence and whenever a market changes a transport, chemical or vehicle-equipment rule.

19. Conformance Requirements (*normative — claims and levels per CDS-1000*)

CDS1900-Ro63 An implementation claiming the CDS Parts and Fitment Profile MUST: represent every fitment, compatibility, inclusion, interchange and supersession fact as a governed product relationship record with provenance, versioning and verification (Ro01, Ro08–Ro16); bind fitment to a declared application vocabulary with typed years, positions and quantities and explicit universal-fit declarations (Ro17–Ro24); keep OEM and cross-reference numbers as external targets and supersession as dated directed relationships (Ro25–Ro29); adopt external classification models only through governed Taxonomy Mappings and attribute crosswalks (Ro30–Ro34); govern origin type, condition, brand and manufacturer (Ro35–Ro38); declare unit of sale, pack quantity, measured sale forms and B2B constraints as typed values (Ro39–Ro43); model kits as `includes` relationships with derived kit-level declarations (Ro44–Ro48); declare dangerous-goods status explicitly for every product with SDS and hazard-communication records where required (Ro49–Ro56); maintain per-market compliance records and required documents (Ro57–Ro62); maintain and evaluate a jurisdiction requirement register for every market published to (Ro03); apply category-specific requirements per CDS1500-Ro09; and publish and verify channel representations under CDS-500.

CDS1900-Ro64 A Parts and Fitment Profile claim MUST name the application vocabulary sources and releases in use, state whether hazardous chemicals are in scope, and state which markets its register covers. Implementations claiming other industry profiles that cite §5, §11 or §12 of this chapter MUST satisfy the cited sections without claiming this profile.

20. Worked Product Examples (*informative*)

20.1 Brake pad set with fitment, cross-reference and supersession

```
STD_brand = ExampleBrake; SUP_manufacturer_part_number = EB-1234; CAT_product_type = brake_pad_set
MF_part_origin_type = aftermarket; MF_condition = new
MF_unit_of_sale = set; MF_units_per_sale = 4; MF_quantity_per_application = 1
relationships:
  {type: fits, target: application {vocabulary_source: aces_vcdb_2026_08, host_type: vehicle,
    make: Example, model: Sedan, submodel: GT, year_from: 2018, year_to: 2023, engine: "2.0L L4 turbo"},
    qualifiers: {position: front}, provenance: {source: manufacturer_application_catalogue, class: E1}}
  {type: fits, target: application {..., model: Wagon, year_from: 2019, year_to: 2023}, qualifiers: {position: front}}
  {type: cross_reference, target: external_part {brand: ExampleMotors, scheme: oem_part_number, number: "04465-00000"},
    qualifiers: {basis: form_fit_function}}
  {type: superseded_by, target: product EB-1234A, effective_from: 2026-03-01}
CMP_dangerous_goods = {regulated: false, declared_by: dg_owner, declared_at: 2026-09-01}
CMP_compliance_declarations = [{jurisdiction: EU, scheme: eu_unece_type_approval, mark: "E-mark R90", certificate: ...}]
Channel: fitment table generated from `fits` records; feed channel carries applicability in product_detail;
        verification: metafield MATCH; feed UNOBSERVABLE (no read-back of fitment)
```

20.2 Hydraulic hose sold by the metre

```
CAT_product_type = hydraulic_hose
MF_sold_by_measure = true; MF_unit_of_sale = metre; MF_measure_increment = {0.5, metre}
MF_measure_minimum = {1, metre}; MF_measure_maximum = {50, metre}
MF_inside_diameter = {nominal: 12.7, unit: millimetre}; MF_working_pressure = {value: 210, unit: bar}
Taxonomy mapping: ETIM class EC0xxxxx (release 10.0) -> internal category Hydraulics > Hoses; feature crosswalk declared
Order line: 7.5 m -> delivered quantity computed at order time, not a variant
```

20.3 Engine oil (dangerous goods and SDS)

```
CAT_product_type = engine_oil
VAR_net_quantity = {value: 5, unit: litre}
MF_viscosity_grade = sae_5w_30 (governed); MF_certifications = [{scheme: api_service_category, value: "SP", evidence: E1}]
CMP_dangerous_goods = {regulated: false, transport_modes: [road, air], basis: "not classified under ADG 7.9 / IATA",
  sds_reference: MED-sds-2026-01 (GHS rev 7, 2026-01-10), declared_by: dg_owner}
CMP_country_of_origin = AU; CMP_tariff_code = {scheme: hs_2022, code: "2710.19"}
AU preflight: SDS present (hazardous chemical for workplace supply; consumer retail exemption not relied on) -> pass
```

20.4 Radio-control servo with compatibility

```
CAT_product_type = rc_servo
relationships:
  {type: compatible_with, target: standard {standard: "servo_connector_jr_3_pin"}, direction: symmetric}
  {type: accessory_for, target: model_family {brand: ExampleRC, family: "Crawler 1/10"}}
MF_voltage_range_v = {min: 4.8, max: 8.4}; MF_torque_kg_cm = 25
MF_kit_kind = none; MF_unit_of_sale = each
```

21. Reference Validation Cases (*informative — the profile's contribution to the cross-industry validation set, REVIEW-020*)

- a fitment held only in a tag or title, which must fail (R001, R010);
- a part with neither fitment records nor a universal declaration offered as "fits all", which must not satisfy an R-level requirement (R018);
- a fitment with free-text years ("2018–current") (R021);
- a vocabulary release change that must produce a governed migration rather than silent re-pointing (R022);
- a superseded_by cycle that validation must reject (R016);
- an OEM number stored in the organisation's own SKU attribute (R025);
- a "set of four" whose unit of sale is not declared, making the delivered quantity ambiguous (R039);
- a product with no dangerous-goods record published to a shipping channel, which must be reported as missing (R049);
- a kit containing a lithium-battery component declared "not regulated" at kit level (R054);
- a dangerous-goods classification proposed by AI from a category, which must be refused (R051);
- a tyre listed in the EU without the tyre label near the price (R058, D-EU-3);
- an aftermarket emissions part sold into California without an Executive Order reference (D-US-3);
- a "genuine" origin claim without evidence (R035).

Appendix A. Parts and Fitment Attribute Baseline (*normative — dictionary bindings per CDS1500-R011*)

Field	Scope	Type	Baseline requirement
STD_brand / SUP_manufacturer_part_number / VAR_gtin	Product / Variant	Governed identity values	R
CAT_product_type	Product	Governed classification reference	R

Field	Scope	Type	Baseline requirement
MF_part_origin_type	Product	Governed dictionary reference (part_origin_type)	R
MF_condition	Variant	Governed dictionary reference (product_condition)	R
fits relationship records or MF_fitment_scope = universal	Product	Relationship records / declaration	R for application-specific parts
alternative_to / cross_reference / superseded_by / replaces records	Product	Relationship records	C — R where interchange or supersession is published
includes records / MF_kit_kind	Product	Relationship records / governed value	R for kits and bundles
Category technical attributes	Product	Typed Attribute Definitions (CDS-1800 §6; external crosswalk per §8)	Per category profile
CMP_external_classifications	Product	Taxonomy Mappings with model and version	REC; R where a trading partner requires the code
MF_unit_of_sale / MF_units_per_sale	Product	Governed value; integer	R
MF_sold_by_measure / MF_measure_increment / MF_measure_minimum / MF_measure_maximum	Product	Boolean; typed measurements	R for measured sale forms
VAR_sale_unit_level	Variant / levels	Governed level with quantities and identifiers	C
MF_minimum_order_quantity / MF_order_multiple	Product	Integers (authority-conditional)	C
Logistics dimensions and weight of the sale unit	Product / Variant	Typed measurements	R where the organisation ships
CMP_dangerous_goods	Product / Variant	Declaration record (§12)	R (explicit regulated value for every product)
MED_technical_documents (SDS, diagrams, certificates, instructions)	Product	Typed media records	C — R where a market requires availability
CMP_compliance_declarations	Product (per market)	Governed records	C — R where a market regulates the part
CMP_country_of_origin / CMP_tariff_code	Product	Governed values	C — R for cross-border sale
MF_warranty_months / MF_warranty_type	Product (per market)	Typed records	REC

Appendix B. Dictionary Bindings (*informative — becomes governed data on adoption per CDS1500-R011*)

Dictionary key	Package chapter	Bound attribute(s)	Status at o.8.o
relationship_type	18 (new)	Product relationship records (§5)	Bound (corpus-wide)
part_origin_type	18 (new)	MF_part_origin_type	Bound
product_condition	18 (new)	MF_condition	Bound (shared with CDS-1800)
dangerous_goods_class	18 (new)	CMP_dangerous_goods.class_or_division	Bound (corpus-wide)
jurisdiction / regulatory_scheme	19 (new)	CMP_* records	Bound via CDS-1500 §2.2
connectivity, power_source, battery_chemistry	11	Technical attributes where relevant	Bound via CDS-1800

Deliberately not shipped: vehicle and equipment application vocabularies (licensed or manufacturer-sourced; declared per organisation), part terminology and attribute databases (licensed), the full UN dangerous-goods list (regime-published; only the class and division vocabulary is seeded), units of measure (CDS-400 §18 reference data), qualifier statement vocabularies (organisation- or standard-sourced).

Appendix C. References (*informative; retrieved 2026-09-20 unless stated*)

Ref	Source	Location	Use in this chapter
[P1]	Auto Care Association — Data Standards; ACES 5.0 and PIES 8.0 release (2026-04-02); supporting databases (VCdb, Qdb, PCdb, PAdb, Brand Table)	autocare.org	Fitment and product information exchange model, licensed reference databases (§6, §8)
[P2]	ETIM International — Classification Guidelines (v3, 2020)	etim-international.com	Class/feature/value model, one class per product, IXF release format (§8)
[P3]	GS1 — Global Product Classification Development & Implementation Guide (2023); How GPC works	gs1.org	Segment/family/class/brick with attributes; 8-digit coding (§8)
[P4]	National Transport Commission — Australian Dangerous Goods Code edition 7.9 (usable from 1 October 2024, mandatory from 1 October 2025)	ntc.gov.au	Classes and divisions, limited and excepted quantities, lithium battery mark (§12, D-AU-1)
[P5]	Safe Work Australia — Adoption of GHS 7 (mandatory from 1 January 2023); model Code of Practice: Preparation of safety data sheets (2023); Suppliers and users information sheet (June 2023)	safeworkaustralia.gov.au	SDS obligations for hazardous chemicals incl. consumer products supplied to workplaces (D-AU-2)
[P6]	IATA — Lithium Battery Guidance Document (2026)	iata.org	UN 3480/3481/3090/3091/3551/3552, 30% state of charge, test summary availability (§12)
[P7]	Regulation (EU) 2020/740 on the labelling of tyres (applies from 1 May 2021); UK retained text	eur-lex.europa.eu; legislation.gov.uk	Tyre label content and display close to the price in distance selling; product information sheet (D-EU-3, D-UK-2)

Ref	Source	Location	Use in this chapter
[P8]	Regulation (EU) 2023/988 General Product Safety Regulation (applies from 13 December 2024)	eur-lex.europa.eu	Traceability and online offer information (D-EU-4)
[P9]	ICC Compliance Center — Consumer chemical products and GHS SDS requirements (Canada CCCR 2001 / HPR; OSHA HazCom consumer-use exemption; REACH Article 31(4)) (2022)	thecompliancecenter.com	Consumer-product SDS exemptions by jurisdiction (D-CA-1, D-US-2, D-EU-2; retrieved, secondary)
[P10]	OEHHA — Proposition 65 amendments effective 2025-01-01 including tailored warnings for passenger or off-highway motor vehicle parts and recreational marine vessel parts	oehha.ca.gov	Vehicle-parts warning content and internet warnings (D-US-4)
[P11]	StoreInspect — Best Shopify categories to target in 2026 (updated 2026-09-19); ECDB Amazon marketplace report (2026)	storeinspect.com; ecdb.com	Catalogue-size and growth basis (REVIEW-020)

Appendix D. Jurisdiction Requirement Register — Seed (*informative seed of the normative register defined in CDS-1500 §2.2*)

Conventions as in CDS-1600 Appendix D. Vehicle-equipment type-approval and emissions regimes are seeded at summary level; most were retrieved but not spot-verified this pass and are marked accordingly.

D.1 Australia (AU)

Ref	Instrument / regulator	Applies to	Obligation	Required data elements → CDS field	Trigger / dates	Verification
D-AU-1	Australian Code for the Transport of Dangerous Goods by Road and Rail, edition 7.9 (aligned to UN Model Regulations 23rd revised edition), given force by state and territory dangerous goods transport laws — NTC and state competent authorities (au_adg_code_7_9)	Dangerous goods of Classes 2–6, 8 and 9 consigned by road or rail (Classes 1 and 7 under other laws)	transport; documentation	Classification (UN number, proper shipping name, class/division, packing group), packaging and marking incl. limited-quantity and lithium battery marks, transport documentation and emergency information; very small consignment and limited quantity provisions → CMP_dangerous_goods (all properties), carrier projection	Usable from 1 October 2024; mandatory from 1 October 2025	Verified 2026-09-20 [P4]
D-AU-2	Model WHS Regulations (as adopted by each jurisdiction) — GHS Revision 7 classification, labelling and safety data sheets — Safe Work Australia and state regulators (au_ghs7_sds)	Hazardous chemicals manufactured or imported for use, handling or storage at workplaces, including hazardous chemicals intended as consumer products (exemptions include potable liquids that are consumer products at retail premises)	documentation (SDS prepared before first supply to a workplace, reviewed at least every 5 years, Australian manufacturer or importer contact details); labelling_element (GHS 7 labels)	SDS in the Australian 16-section format with GHS 7 classification, signal word, pictograms, hazard and precautionary statements, transport information; supplier must provide the SDS with supply to a workplace → MED_technical_documents (SDS, GHS rev 7, date), CMP_dangerous_goods.hazard_communication	GHS 7 only from 1 January 2023	Verified 2026-09-20 [P5]
D-AU-3	Road Vehicle Standards Act 2018 and Australian Design Rules; state vehicle standards for modifications — Department of Infrastructure and state regulators (au_adr_vehicle_standards)	Replacement components and accessories subject to design rules (lighting, tyres, child restraints via mandatory standards, seatbelts) and modifications	pre_market (component certification where an ADR applies); labelling_element (markings)	ADR compliance evidence and markings per component; tyre markings → CMP_compliance_declarations (scheme au_adr_vehicle_standards)	Ongoing	Retrieved (not spot-verified)
D-AU-4	Proposition-style warnings do not apply; Australian Consumer Law mandatory safety and information standards apply to specific products (for example quad bikes, portable ladders, vehicle jacks) — ACCC	Specific consumer products	labelling_element; warning_statement	Product-specific mandatory standard elements → MF_warning_statements, CMP_compliance_declarations	Product-specific	Not seeded (organisation supplies per product type)

D.2 New Zealand (NZ)

Ref	Instrument / regulator	Applies to	Obligation	Required data elements → CDS field	Trigger / dates	Verification
D-NZ-1	Land Transport Rule: Dangerous Goods 2005 (aligned to the UN Model Regulations / ADG) — Waka Kotahi NZ Transport Agency (nz_dangerous_goods_rule_2005)	Dangerous goods transported by land	transport; documentation	Classification and documentation as for AU with NZ variations → CMP_dangerous_goods	Ongoing	Retrieved (not spot-verified)

Ref	Instrument / regulator	Applies to	Obligation	Required data elements → CDS field	Trigger / dates	Verification
D-NZ-2	Hazardous Substances and New Organisms Act 1996; Hazardous Substances (Labelling) Notice 2017 and (Safety Data Sheets) Notice 2017 — EPA and WorkSafe (nz_hsno_labelling_sds)	Hazardous substances, including consumer chemical products	labelling_element; documentation (SDS)	GHS-aligned labels; SDS supplied with substances for workplaces → MED_technical_documents (SDS), CMP_dangerous_goods.hazard_communication	Ongoing (Group Standards provide alternative compliance routes)	Retrieved (not spot-verified)

D.3 United States (US, with California noted)

Ref	Instrument / regulator	Applies to	Obligation	Required data elements → CDS field	Trigger / dates	Verification
D-US-1	Hazardous Materials Regulations, 49 CFR parts 171–180 — PHMSA/DOT (us_49cfr_hazmat)	Hazardous materials offered for transport in commerce (road, rail, air, water; air also under ICAO/IATA)	transport; documentation	UN number, proper shipping name, hazard class, packing group, limited quantity and ORM-D-style consumer commodity provisions (as amended), lithium battery mark and test summary → CMP_dangerous_goods	Ongoing	Retrieved (not spot-verified)
D-US-2	Hazard Communication Standard, 29 CFR 1910.1200 (updated 2024 to GHS Revision 7) — OSHA (us_oshahazcom)	Hazardous chemicals in workplaces; consumer products exempt where workplace use mirrors consumer use in duration and frequency	documentation (SDS); labelling_element	16-section SDS and GHS labels for workplace-supplied hazardous chemicals → MED_technical_documents (SDS), exemption reason where relied on (R052)	2024 update with phased compliance	Retrieved 2026-09-20 [P9] (secondary; not spot-verified)
D-US-3	California Air Resources Board aftermarket parts Executive Orders (Vehicle Code 27156) (us_carb_executive_order)	Aftermarket emissions-related parts sold or installed in California (and states adopting California standards)	pre_market (Executive Order exemption); labelling_element (EO number)	CARB Executive Order number for the part and applications → CMP_compliance_declarations (scheme us_carb_executive_order, EO number), fitment scope	Ongoing	Retrieved (not spot-verified)
D-US-4	Proposition 65 (California), 27 CCR 25607.50–25607.53 tailored warnings for passenger or off-highway motor vehicle parts and recreational marine vessel parts; internet warning rules — OEHHHA (us_ca_prop65_warning)	Vehicle and vessel parts causing exposure to listed chemicals sold to California consumers	warning_statement; listing_element	Tailored safe-harbour warning content; warning on the product display page, "WARNING" hyperlink or prominent pre-purchase display → MF_warning_statements (jurisdiction US-CA, scheme with the vehicle-parts variant), listing projection	Effective 1 January 2025	Verified 2026-09-20 [P10]
D-US-5	Federal Motor Vehicle Safety Standards and equipment marking (49 CFR part 571; DOT tire identification number 49 CFR 574) — NHTSA (us_fmvss_equipment)	Regulated motor vehicle equipment (lighting, brake hoses, tyres, glazing, child restraints)	pre_market (self-certification); labelling_element (DOT symbol, tire identification number)	DOT marking and TIN → CMP_compliance_declarations	Ongoing	Retrieved (not spot-verified)

D.4 European Union (EU)

Ref	Instrument / regulator	Applies to	Obligation	Required data elements → CDS field	Trigger / dates	Verification
D-EU-1	European Agreement concerning the International Carriage of Dangerous Goods by Road (ADR) as applied by Directive 2008/68/EC; IMDG and ICAO/IATA for sea and air (eu_adr_transport)	Dangerous goods in transport	transport; documentation	Classification, packaging, marking, documentation → CMP_dangerous_goods	Biennial ADR editions	Retrieved (not spot-verified)
D-EU-2	REACH Regulation (EC) No 1907/2006 Article 31 (safety data sheets); CLP Regulation (EC) No 1272/2008 (classification, labelling, packaging) — ECHA and member states (eu_reach_sds_article_31, eu_clp_1272_2008)	Hazardous substances and mixtures	documentation (SDS to professional users; for products sold to the general public an SDS need not be supplied where sufficient information is provided, but must be supplied on request by a downstream user or distributor); labelling_element (CLP pictograms, signal word, hazard and precautionary statements)	SDS document, CLP hazard communication elements → MED_technical_documents (SDS), CMP_dangerous_goods.hazard_communication	Ongoing	Verified in part 2026-09-20 [P9] (Article 31(4) via secondary source; regulation text not read)

Ref	Instrument / regulator	Applies to	Obligation	Required data elements → CDS field	Trigger / dates	Verification
D-EU-3	Regulation (EU) 2020/740 on the labelling of tyres (eu_tyre_label_2020_740)	C1, C2 and C3 tyres (passenger, light and heavy commercial), retreads once a test method exists	rating_label; listing_element; documentation (product information sheet); pre_market (supplier enters values in the product database, EPREL)	Tyre label with fuel efficiency class (A–E), wet grip class (A–E), external rolling noise class and dB value, severe-snow and ice-grip pictograms; product information sheet (supplier, tyre type identifier, size designation, load index, speed symbol, classes, production start and end dates); label displayed close to the price in distance selling and online (nested display permitted), PIS accessible and printable on request; hosting service providers enable the display → CMP_compliance_declarations (scheme eu_tyre_label_2020_740, classes, EPREL id), MF_tyre_size_designation, MF_load_index, MF_speed_rating, MED_technical_documents (label, PIS), listing projection	Applies from 1 May 2021	Verified 2026-09-20 [P7]
D-EU-4	Regulation (EU) 2023/988 General Product Safety Regulation (eu_gpsr_2023_988)	Non-harmonised consumer products (many tools, accessories, hobby components) and cross-cutting online duties	listing_element; labelling_element; pre_market (responsible economic operator)	Manufacturer name, address and electronic contact; product identifier; warnings and safety information; picture in online offers → CMP_responsible_person, CMP_product_identifier, MF_warning_statements	Applies from 13 December 2024	Verified 2026-09-20 [P8]
D-EU-5	UNECE type-approval regulations for vehicle components (E-mark), Regulation (EU) 2018/858 and (EU) 2019/2144 (general safety); Machinery Regulation (EU) 2023/1230 (CE marking for machinery and tools, applies from 20 January 2027) (eu_unece_type_approval, eu_machinery_regulation_2023_1230)	Replacement components subject to type approval (lighting, braking components, tyres, mirrors, glazing); machinery and power tools	pre_market (type approval; conformity assessment); labelling_element (E-mark with approval number; CE marking)	Approval number and marking → CMP_compliance_declarations	Ongoing; machinery regulation from 20 January 2027	Retrieved (not spot-verified)

D.5 United Kingdom (GB)

Ref	Instrument / regulator	Applies to	Obligation	Required data elements → CDS field	Trigger / dates	Verification
D-UK-1	Carriage of Dangerous Goods and Use of Transportable Pressure Equipment Regulations 2009 (ADR as applied in GB); UK REACH and GB CLP — HSE (uk_carriage_dangerous_goods_2009, uk_reach_clp)	Dangerous goods in transport; hazardous substances and mixtures	transport; documentation (SDS); labelling_element	As for D-EU-1 and D-EU-2 with GB-specific formats → CMP_dangerous_goods, MED_technical_documents (SDS)	Ongoing	Retrieved (not spot-verified)
D-UK-2	Regulation (EU) 2020/740 as retained in GB (tyre labelling) — DfT/OPSS (uk_tyre_label_retained_2020_740)	C1, C2 and C3 tyres	rating_label; listing_element	Tyre label and product information sheet displayed close to the price online, as in the EU scheme (no EPREL) → CMP_compliance_declarations (scheme uk_tyre_label_retained_2020_740), listing projection	Ongoing	Verified in part 2026-09-20 [P7] (retained text on legislation.gov.uk read; post-2021 GB amendments not checked)
D-UK-3	Supply of Machinery (Safety) Regulations 2008; UKCA or CE marking with indefinite CE recognition from 1 October 2024 — OPSS (uk_machinery_regulations_2008)	Machinery and power tools	pre_market; labelling_element	Conformity marking choice, DoC, manufacturer and importer identification → CMP_compliance_declarations	CE recognition indefinite from 1 October 2024	Verified in part 2026-09-20 (CDS-1800 [E16] sector table lists machinery among the 2024 regulations)

D.6 Canada (CA)

Ref	Instrument / regulator	Applies to	Obligation	Required data elements → CDS field	Trigger / dates	Verification
D-CA-1	Transportation of Dangerous Goods Act 1992 and Regulations — Transport Canada (<code>ca_tdg_regulations</code>); Hazardous Products Act and Hazardous Products Regulations (WHMIS 2015, GHS-aligned) for workplace products; Consumer Chemicals and Containers Regulations 2001 (CCCR) under the Canada Consumer Product Safety Act for consumer chemical products (no SDS requirement, consumer labelling instead) — Health Canada (<code>ca_whmis_2015</code> , <code>ca_cccr_2001</code>)	Dangerous goods in transport; workplace hazardous products; consumer chemical products sold at retail	transport; documentation (SDS for workplace products only); labelling_element (CCCR hazard symbols and bilingual statements for consumer products)	TDG classification → <code>CMP_dangerous_goods</code> ; SDS for products not sold at retail in the same container size → <code>MED_technical_documents</code> with exemption reason (<code>Ro52</code>); CCCR consumer labelling → <code>MF_warning_statements</code> (bilingual)	Ongoing	Verified in part 2026-09-20 [P9] (secondary source on the HPA consumer-product exclusion; regulations not read)
D-CA-2	Motor Vehicle Safety Act; Canada Motor Vehicle Safety Standards (CMVSS) — Transport Canada (<code>ca_cmvs_equipment</code>)	Regulated motor vehicle equipment (tyres, lighting, brake hoses, child restraints)	<code>pre_market</code> ; labelling_element (national safety mark or DOT equivalents as recognised)	Compliance marking → <code>CMP_compliance_declarations</code>	Ongoing	Retrieved (not spot-verified)

D.7 Other markets

Jurisdiction	Status	Note
Japan (JP), China (CN), Republic of Korea (KR), India (IN), Brazil (BR) and others	Not seeded	Vehicle-equipment approval (for example JIS marks, CCC certification, KC marks, BIS, INMETRO) and chemical regimes differ; organisation supplies entries with local advice. All UN Model Regulations-aligned transport regimes share the UN number, class and packing-group vocabulary seeded in the <code>dangerous_goods_class</code> dictionary.

END OF CDS-1900 v0.2 REVIEW DRAFT

Commerce Data Standard (CDS)

CDS-2000 — Sports and Outdoor, Toys and Games, and Pet Supplies Industry Profiles

Field	Value
Status	vo.2 Review Draft (working source; not an approved standard)
Release	CDS vo.2 (single corpus release per ADR-D5); chapter added in the 2026-09-20 industry-profile expansion
Date	2026-09-20
Supersedes	Nothing — first edition. The Chapter 12 (Sports & Outdoor), Chapter 13 (Toys & Games) and Chapter 14 (Pet Supplies) dictionary vocabularies were shipped ahead of these profiles and are bound here for the first time; pet food forms are added in package 0.8.0.
Normative status	§1, §3, §5–§8, §13 and Appendix A are normative. §2 (pointer), §4, §9–§12, §14, §15 and Appendices B–D are informative; Appendix D seeds the jurisdiction requirement register whose record structure and obligations are normative in CDS-1500 §2.2. Every table is individually marked.
Primary audience	Sports, outdoor, toy and pet merchandisers, product-safety and compliance owners, data stewards, PIM architects, catalogue managers, UX teams, developers and AI enrichment designers
Depends on	CDS-000 through CDS-1500, especially CDS-200, CDS-300, CDS-400, CDS-500, CDS-600, CDS-700, CDS-900, CDS-1100, CDS-1500 (profile model, requirement levels, profile register, jurisdiction requirement register, apparel size and colour rules), CDS-1600 (claims, ingredients and restricted-sale patterns), CDS-1700 (ingredient, nutrition and net-quantity patterns), CDS-1800 (technical specifications, batteries), CDS-1900 (relationships, kits, dangerous goods)
Companion package	CDS Reference Dictionary: Chapters 12, 13, 14, 5 (size systems), 19 (Jurisdictions & Regulatory Schemes)
Profile identifiers	<code>cds.profile.sports_outdoor.v0_2</code> , <code>cds.profile.toys_games.v0_2</code> , <code>cds.profile.pets.v0_2</code> (registered in CDS-1500 §2.1)
Research basis	REVIEW-020 (global commerce vertical market research, 2026-09-20): sports, toys and hobbies, and pet supplies are each mid-sized worldwide store populations (roughly 2–6% of Shopify Plus stores each) with above-average catalogue depth (sports stores average more than 1,200 products) and, for toys, the densest product-safety regulation of any consumer vertical. They are grouped in one chapter because they share three mechanisms — activity or audience targeting, safety-driven age and warning data, and technical attributes inherited from other profiles — rather than because they share a market.

Terminology follows CDS-100. Attribute requirement levels (R, C, REC, O, N/A) are those of CDS-1500 §4 and are not restated here. Each of the three profiles in this chapter is claimed independently (§13).

1. Purpose and Scope (*normative*)

CDS-2000 defines three industry profiles: **Sports and Outdoor** (equipment, activewear and footwear where sold by activity, camping and outdoor gear, cycling, water sports, fitness), **Toys and Games** (toys, games, puzzles, hobby kits, ride-ons, children's play equipment) and **Pet Supplies** (pet food and treats, accessories, habitats, grooming, health products). Apparel and footwear sold in these verticals apply CDS-1500 for size, colour and composition; electronic and battery-powered goods apply CDS-1800; kits, compatibility and dangerous goods apply CDS-1900; ingredients and nutrition apply CDS-1700 patterns; claims apply CDS-1600 §11.

What distinguishes these verticals is who or what the product is for and what it is for: a sport or activity, a child's age and developmental stage, an animal's species, life stage and size. Those targeting attributes drive navigation ("shop by sport", "shop by age", "shop by pet") and, for toys and pet food, they are also regulated facts (age warnings, nutritional adequacy) that must be kept separate from the merchandising attributes that resemble them.

CDS2000-R001 An implementation claiming any profile in this chapter **MUST** represent the targeting attributes of that profile (activity, age band, pet type, life stage, pet size) as governed dictionary values, never as tags or free text, and **MUST** keep merchandising targeting distinct from regulated safety or adequacy statements (§6, §7).

CDS2000-R002 An implementation claiming any profile in this chapter **MUST** apply CDS-1500, CDS-1800 and CDS-1900 for the apparel, technical and relationship layers of its products where those layers are present, without claiming those profiles.

CDS2000-R003 An implementation **MUST** maintain a jurisdiction requirement register (CDS-1500 §2.2) for every market it publishes toys, children's products, protective equipment, pet food or pet health products to, and publication preflight for a market **MUST** evaluate the market's obligations before the offer is made available.

CDS2000-R004 These profiles **MUST NOT** weaken any rule of CDS-1500 that they reuse; where this chapter is silent, CDS-1500 and the core chapters govern.

2. Profile Model (*informative pointer*)

The three profiles apply the industry profile model of CDS-1500 §2, the requirement levels of CDS-1500 §4 and the jurisdiction requirement register of CDS-1500 §2.2 without restatement. Their identifiers and dictionary bindings are in CDS-1500 §2.1; their seeded jurisdiction entries are in Appendix D.

3. Shared Product Model (*normative*)

Products in these verticals follow the standard product and variant model. Variant options are typically size (apparel and footwear sizes under CDS-1500 §9; equipment sizes under a declared size system), colour, pack or bag size (pet food), and configuration (piece count, player count edition). Targeting attributes are product-scope descriptive attributes, never variant options.

```

Product: Adult cycling helmet, model Ridge          (informative example)
Product scope:
  CAT_product_type = bicycle_helmet
  MF_sport_activity = [cycling, mountain_biking]    # sport_activities dictionary
  MF_certifications = [{scheme: as_nzs_2063, evidence: E1}, {scheme: cpsc_16_cfr_1203, evidence: E1}]
  MF_size_system = head_circumference_cm           # size_systems dictionary (0.8.0)
  MF_weight_g = 310
Variant scope:
  VAR_size_label = "M (55-59 cm)"; VAR_size_min = 55; VAR_size_max = 59
  VAR_colour = matte_black

```

CDS2000-R005 Sport or activity, age band, play type, pet type, life stage and pet size MUST be product-scope governed attributes; where a product genuinely serves several values (a multi-sport shoe, a toy for two age bands, a food for dogs and cats) the attribute MUST be multi-valued with a declared maximum (the CDS-1500 §10 pattern), not duplicated products.

CDS2000-R006 Equipment sizes (helmets, frames, boards, wetsuits, harnesses, pet collars and coats) MUST identify their size system (the CDS1500-R026 pattern) and MUST store measured bounds (head circumference, frame size, chest or neck girth, pet weight range) as typed measurements where the size is defined by a measurement, so that size guides and facets derive from data.

CDS2000-R007 Kits, sets and bundles (a tent with footprint, a game with expansion, a starter aquarium) MUST follow CDS-1900 §11; batteries and rechargeable products MUST follow CDS-1800 §8 and CDS-1900 §12.

CDS2000-R008 Included and required-but-not-included items (batteries, balls, mounting hardware, food bowls) MUST be `includes` and `requires` relationship records (CDS-1900 §5) so that "batteries not included" and similar statements derive from data.

4. Product Families and Category Profiles (*informative baseline; the inheritance rule of CDS1500-R016 applies*)

Profile	Product family / category	Required	Recommended	Typical variant options
Sports & Outdoor	Activewear and sports footwear	CDS-1500 apparel or footwear baseline; sport activity	performance attributes with basis qualifiers; weather suitability	size, colour
Sports & Outdoor	Cycling (bikes, components)	brand; MPN; frame size system and size; wheel size; compatibility records (CDS-1900 §5); weight	groupset, gearing, geometry as typed values; e-bike battery and motor per CDS-1800	frame size, colour
Sports & Outdoor	Water sports (surf, swim, paddle)	sport activity; wetsuit style, thickness, zip entry and seam (wetsuits); board dimensions and volume	fin system as relationship; buoyancy certification (PFDs)	size, colour
Sports & Outdoor	Camping and outdoor gear	capacity (persons, litres); packed and assembled dimensions; weight; season or temperature rating with basis	materials; waterproof rating with scheme; fuel type (stoves) with dangerous goods per CDS-1900 §12	size, colour
Sports & Outdoor	Fitness equipment	dimensions; weight; user weight limit; power source (if powered)	assembly type; connectivity per CDS-1800	weight/resistance level
Sports & Outdoor	Protective equipment (helmets, guards, eyewear)	size system and size; certification records per market; sport activity	replacement date guidance; lens category (sunglasses)	size, colour
Toys & Games	Toys (0–12 years)	retail age band; play type; safety age warning where applicable; piece count; battery facts; compliance records per market	material; dimensions; educational themes as governed values	colour
Toys & Games	Games and puzzles	player count (min, max); recommended age; play time; piece count; language	complexity or difficulty as governed value	edition, language
Toys & Games	Hobby kits and models	scale; skill level as governed value; included and required items; paint or adhesive dangerous goods	assembled dimensions	—
Toys & Games	Ride-ons and outdoor play	age band and weight limit; dimensions; assembly type; compliance records; battery facts for powered ride-ons	—	colour
Pet Supplies	Pet food and treats	pet type; life stage; pet size where formulated by size; food form; ingredient list; typical or guaranteed analysis; nutritional adequacy statement and scheme; net quantity; feeding guide reference	breed size; dietary claims with evidence; storage condition	bag size, flavour
Pet Supplies	Pet health and supplements	pet type; active ingredients; registration per market; restricted sale status; dosage form	—	pack size
Pet Supplies	Accessories (collars, leads, beds, apparel)	pet type; size system and measured bounds; material	colour facet; washable status	size, colour
Pet Supplies	Habitats and equipment (aquariums, cages, feeders)	pet type; dimensions and capacity; power source where powered (CDS-1800)	compatibility records for consumables (filters, cartridges)	size

5. Sports and Outdoor Profile (*normative; dictionary bindings informative*)

Field	Meaning	Dictionary / type
MF_sport_activity	Activities the product is designed for	<code>sport_activity</code> (Chapter 12), multi-valued with declared maximum

Field	Meaning	Dictionary / type
MF_wetsuit_style / MF_wetsuit_thickness / MF_wetsuit_zip / MF_wetsuit_seam	Wetsuit construction	Chapter 12 dictionaries
MF_size_system + VAR_size_label + measured bounds	Equipment sizing	size_system (Chapter 5, incl. bike_frame_cm, head_circumference_cm at 0.8.o)
MF_temperature_rating / MF_season_rating / MF_waterproof_rating	Performance ratings	Typed values with scheme and basis qualifier (CDS-1800 Ro17 pattern)
MF_user_weight_limit_kg / MF_capacity_persons / MF_capacity_l	Limits and capacities	Typed measurements
MF_certifications	Safety and performance certifications (helmet, PFD, eyewear, climbing standards)	Certification records with scheme and evidence (CDS-1600 §11 pattern)
Relationship records	Compatibility (components, fin systems, mounts), included items	CDS-1900 §5

CDS2000-Ro09 Sport or activity **MUST** be a governed multi-valued attribute bound to the `sport_activity` dictionary (or its governed extension) and **MUST NOT** be derived from category placement alone; the "shop by sport" facet is a projection of this attribute (CDS-600).

CDS2000-Ro10 Performance ratings (temperature or season rating, waterproof or breathability rating, buoyancy, load rating) **MUST** be typed values carrying the rating scheme and a basis qualifier (manufacturer-rated or measured under a named standard); ratings with different schemes **MUST NOT** be compared or filtered as equivalent.

CDS2000-Ro11 Protective equipment (helmets, personal flotation devices, eye protection, climbing and fall-protection equipment) **MUST** carry certification records naming the standard and the evidence for each market it is sold to; a certification **MUST NOT** be projected as a facet or claim unless fed by an accepted record, and a market that mandates the certification (Appendix D) **MUST** block publication when the record is absent.

CDS2000-Ro12 Wetsuit style, thickness, zip entry and seam construction **MUST** be governed values bound to the Chapter 12 dictionaries where those facets are exposed; thickness **MUST** be stored as the governed thickness value with its millimetre components, not as free text.

CDS2000-Ro13 Cycling and other component-based equipment **MUST** express component compatibility (frame, wheel size, drivetrain, mount standards) through relationship records or governed standard values per CDS-1900 §5 and §7; "fits most bikes" **MUST** be an explicit universal-fit declaration within a governed scope (CDS1900-Ro18).

CDS2000-Ro14 Powered sports equipment (e-bikes, e-scooters, electric surfboards, fitness machines) **MUST** apply CDS-1800 §8 and §12 for battery, energy and compliance facts, and CDS-1900 §12 for transport declarations.

CDS2000-Ro15 Skill level, intended use environment and performance tier, where exposed, **MUST** be governed organisation dictionaries (the package deliberately ships none because they are not comparable across brands); they **MUST NOT** be projected as claims.

CDS2000-Ro16 Fuel, gas cartridges, bear spray, marine flares and similar outdoor consumables **MUST** carry dangerous-goods declarations per CDS-1900 §12 and restricted-sale flags where a market restricts them.

CDS2000-Ro17 Apparel and footwear sold under this profile **MUST** satisfy the CDS-1500 apparel or footwear baseline; this profile adds activity and performance attributes and does not replace them.

6. Toys and Games Profile (*normative; dictionary bindings informative*)

The profile separates three age facts that are routinely conflated: the **retail age band** (a merchandising recommendation), the **manufacturer's recommended age** (a product fact from the maker), and the **safety age warning** (a regulated statement such as "Not suitable for children under 36 months" with its hazard, required by law in most markets and, in several, required to be visible online before purchase).

Field	Meaning	Dictionary / type
MF_age_band	Retail age band for navigation	toy_age_range (Chapter 13), multi-valued with declared maximum
MF_recommended_age_min_months / MF_recommended_age_max_months	Manufacturer's recommended age	Integers (months)
MF_safety_age_warning	Regulated age warning with hazard statement and scheme	Structured statement record (MF_warning_statements)
MF_play_type	Play type	play_type (Chapter 13)
MF_piece_count	Number of pieces or parts	Integer
MF_player_count_min / MF_player_count_max / MF_play_time_min / MF_play_time_max	Game facts	Integers
MF_language	Language content of games and books	Governed language codes (CDS-300 §18)
MF_small_parts / MF_contains_magnets / MF_contains_button_batteries / MF_projectile / MF_cord_length_mm	Hazard-relevant facts driving warnings	Booleans and typed measurements
Battery fields	Per CDS-1800 §8	—
CMP_compliance_declarations / CMP_responsible_person / CMP_product_identifier	Per-market compliance (toy safety marking, children's product certificate, tracking label, importer identity)	Governed records (CDS-1800 §12 pattern)
MF_weight_limit_kg	User weight limit for ride-ons and play equipment	Typed measurement

CDS2000-R018 Retail age band, manufacturer's recommended age and safety age warning MUST be three separate attributes; a safety age warning MUST NOT be derived from the retail age band or the manufacturer's recommendation, and the retail age band MUST NOT contradict a safety age warning (a product warned as unsuitable under 36 months MUST NOT carry the 0–2 years band).

CDS2000-R019 Safety age warnings and hazard warnings (small parts, magnets, cords, projectiles, balloons, small balls, button batteries) MUST be structured statement records with the applicable scheme per market, derived from hazard-relevant facts under governed rules with human review, and MUST be projected to a market's listing where the market requires the warning to be visible before purchase (Appendix D); their presence MUST be verified under CDS-500.

CDS2000-R020 Hazard-relevant facts (small parts present, magnets, button or coin batteries, projectiles, cords, sharp points or edges) MUST be explicit boolean or typed declarations with provenance, never inferred from category by rules or AI; an unknown value MUST be quarantined, not defaulted to false.

CDS2000-R021 Play type and age band MUST be governed values bound to the Chapter 13 dictionaries (or governed extensions); "educational" and developmental-benefit statements are claims under CDS-1600 §11 pattern rules and MUST NOT be facets unless fed by accepted records.

CDS2000-R022 Game facts (player count, play time, recommended age, language, edition) MUST be typed attributes; player count MUST be a bounded range with integer minimum and maximum (CDS1500-R041 pattern).

CDS2000-R023 Per-market toy-safety compliance (conformity marking, declaration or certificate of conformity, third-party test reports, tracking or batch labels, importer or responsible-person identity) MUST be governed compliance records (CDS-1800 §12 pattern) and MUST be evaluated at preflight for each market; a conformity mark visible in an image MUST NOT be treated as the compliance fact.

CDS2000-R024 Products with batteries MUST satisfy CDS-1800 §8 and CDS-1900 §12; products containing button or coin batteries MUST additionally carry the market-mandated warnings (R019) and secure-compartment compliance evidence where required.

CDS2000-R025 Piece count MUST be an integer; where a product's piece count varies by variant (edition, size), it MUST be variant-scoped.

CDS2000-R026 Licensed character and franchise properties MUST be governed values in an organisation dictionary distinct from brand, so that licence-based facets and expiry are governable; they MUST NOT be stored as tags.

CDS2000-R027 Toys that are also apparel (dress-ups), electronics (children's tablets) or sports equipment (junior bikes) MUST satisfy the applicable layers of CDS-1500, CDS-1800 or §5 in addition to this profile.

7. Pet Supplies Profile (*normative; dictionary bindings informative*)

Field	Meaning	Dictionary / type
MF_pet_type	Species the product is for	pet_type (Chapter 14), multi-valued with declared maximum
MF_pet_life_stage	Life stage the product is formulated or designed for	pet_life_stage (Chapter 14)
MF_pet_size	Size class where formulated or designed by size	pet_size (Chapter 14) with measured bounds (weight range)
MF_breed_suitability	Breed or breed-size targeting	Governed organisation dictionary (the package ships none)
MF_pet_food_form	Food form (dry kibble, wet, raw, freeze-dried, treats, ...)	pet_food_form (Chapter 14, new at 0.8.0)
MF_ingredient_list / MF_ingredient_naming_scheme	Ordered ingredients	CDS-1700 §5 pattern
MF_nutrition_values	Typical or guaranteed analysis	CDS-1700 R019 with scheme typical_analysis_pet_food or guaranteed_analysis_pet_food
MF_nutritional_adequacy	Adequacy statement, scheme and life stage (complete and balanced / complementary / treat)	Structured record with scheme (AAFCO, FEDIAF, AS 5812 or market scheme) and evidence
MF_feeding_guide	Feeding directions	Structured table or document reference
VAR_net_quantity	Net weight or volume	CDS-1700 §9 pattern
MF_storage_condition / MF_date_mark_type	Storage and date marking	CDS-1700 §10 pattern
CMP_market_registrations / CMP_restricted_sale	Registration and restricted-sale status for veterinary medicines, parasitocides and supplements	CDS-1600 §12 and §14 patterns
MF_active_ingredients / MF_dosage_form / MF_dosage_by_weight	Health-product facts	CDS-1600 §14 pattern

CDS2000-R028 Pet type MUST be a governed multi-valued attribute bound to the pet_type dictionary (or its governed extension) and MUST be present for every pet product; life stage and pet size MUST be governed values where the product is formulated or designed by them.

CDS2000-R029 Pet size classes MUST carry measured bounds (the weight range each class covers, as declared by the organisation or the manufacturer) so that "medium dog" is comparable across brands only where bounds match; a size class without bounds MUST NOT feed a numeric filter.

CDS2000-R030 Pet food ingredients MUST be stored as ordered structured lists per CDS-1700 §5 (naming scheme declared; composite ingredients expanded where the market requires); "grain-free", "single-protein" and similar dietary statements are claims requiring evidence under CDS-1700 §8 pattern rules.

CDS2000-R031 Typical or guaranteed analysis MUST be stored as typed nutrition values with the analysis scheme and basis (as-fed or dry-matter) per CDS-1700 R019, never as text; a channel that displays the analysis MUST receive it from the record.

CDS2000-R032 Nutritional adequacy MUST be a structured record naming the scheme (a feeding-trial or nutrient-profile standard), the life stage it applies to and whether the product is complete and balanced, complementary or a treat; the statement MUST NOT be inferred from ingredients or from the merchandising life stage, and a market requiring the statement (Appendix D) MUST block publication when it is absent.

CDS2000-R033 Food form MUST be a governed value bound to the `pet_food_form` dictionary; net quantity, unit pricing and pack counts follow CDS-1700 §9.

CDS2000-R034 Pet health products (veterinary medicines, parasiticides, supplements, prescription diets) MUST carry per-market registration and restricted-sale records under the CDS-1600 §12 and §14 patterns; a product restricted to veterinary prescription or authorised sellers in a market MUST be blocked from channels in that market that cannot enforce the restriction.

CDS2000-R035 Accessories designed by size (collars, harnesses, coats, crates, beds) MUST declare the size system and measured bounds (neck or chest girth, pet weight, crate internal dimensions) per R006.

CDS2000-R036 Live animals, live feeder insects and live plants for aquaria MUST NOT be modelled under this profile without an organisation extension declaring the welfare, biosecurity and carriage rules that apply; the package ships no live-animal vocabulary.

CDS2000-R037 Consumable compatibility (filter cartridges for a filter model, litter for a system, replacement parts for feeders) MUST be `compatible_with`, `accessory_for` or `fits` relationship records per CDS-1900 §5.

8. Warnings, Restricted Sale and Recalls (normative)

CDS2000-R038 All mandatory warnings under the three profiles MUST be structured statement records (`MF_warning_statements`) with scheme, market and hazard, evaluated per market at preflight and projected to listings where a market requires or recommends online display (Appendix D); free-text warnings in descriptions MUST derive from the records.

CDS2000-R039 Restricted-sale status (age-restricted sporting goods such as knives and air guns, prescription pet medicines, products banned in a market) MUST be governed per-market attributes evaluated at preflight (the CDS-1700 §12 pattern).

CDS2000-R040 Recalls and safety withdrawals MUST be handled through lifecycle state and channel withdrawal (CDS-500 §12) with affected batch or date ranges recorded as observations; deletion of the product record is prohibited (CDS-200 §14).

9. Customer Facet Design (informative – normative facet rules live in CDS-600)

Facet	Recommended baseline behaviour	Anti-pattern
Shop by sport	Governed <code>sport_activity</code> values, multi-select OR, top-level navigation entry	Category per sport duplicating products
Shop by age	Governed <code>toy_age_range</code> bands; never the safety warning	Bands typed per product; "3+" as text
Play type	Governed values	Marketing theme tags
Players / play time	Numeric range facets	Text buckets
Shop by pet	Governed <code>pet_type</code> values; life stage and size as secondary facets	Species from category name
Food form / dietary	Governed <code>pet_food_form</code> ; dietary claims only from accepted claim records	"Grain-free" tag without evidence
Size (equipment)	Size-system-qualified facets with measured bounds	Unqualified "M" across systems
Certification	Never a facet unless fed by accepted records	"CE certified" tag

10. Channel Projection Guidance (informative – normative rules: CDS-500, CDS-900)

Canonical concept	Metafield-style channel (informative)	Feed-style channel (informative)
Sport activity, age band, pet type	Category metafields where the taxonomy defines them (activity, age group, pet type attributes exist in the seeded channel taxonomy), custom metafields otherwise	Title and <code>product_detail</code> ; Google <code>age_group</code> is an apparel audience attribute (newborn, infant, toddler, kids, adult), not a toy age band — do not map toy age bands to it
Safety age and hazard warnings	Per-market generated blocks near the price or in the description, verified	Description; where a market requires visibility before purchase, the listing must carry it
Nutritional adequacy and analysis	Structured metafields; generated description block	<code>product_detail</code>
Certification and compliance	Information blocks generated from records	Not published as attributes
Restricted sale	Channel exclusion rules	Channel exclusion; <code>adult</code> flag where the channel uses it

11. AI Enrichment and Review (informative – normative AI rules live in CDS-700)

Task	AI may propose	Deterministic or human control
Activity, play type, pet type from copy and images	Dictionary candidates with evidence	Governed dictionaries; review for multi-value maxima
Age band suggestion	Retail band candidates	Never the safety warning (R018–R020)
Hazard facts	None from category; extraction from packaging or test reports (E1) as proposals	Human review; unknown quarantined (R020)
Ingredient and analysis extraction	Structured candidates from labels (E1)	Scheme and unit validation (CDS-1700 §5–§6)
Nutritional adequacy	None — manufacturer statement only	R032
Compatibility (components, consumables)	Candidate relationships with evidence	CDS-1900 §5 review rules

12. Governance and Organisational Extensions (*informative — rules in CDS-1500 §23 and CDS-800*)

The Chapter 12–14 vocabularies are extended under CDS-400. Organisations add skill-level, breed, franchise, use-environment and theme dictionaries as governed extensions (§5 Ro15, §6 Ro26, §7 Ro28). Toy-safety and pet-food regimes change on multi-year cycles with long transition periods (for example the EU toy regulation's 2030 application date); the register review cadence should align to those transitions and to any market's list changes for mandatory standards.

13. Conformance Requirements (*normative — claims and levels per CDS-1000*)

CDS2000-Ro41 An implementation claiming the CDS Sports and Outdoor Profile MUST: govern sport or activity as a multi-valued dictionary attribute (Ro05, Ro09); identify equipment size systems with measured bounds (Ro06); store performance ratings with scheme and basis (Ro10); maintain certification records for protective equipment with per-market preflight (Ro11); bind wetsuit attributes to governed dictionaries where exposed (Ro12); express compatibility and inclusion through relationship records (Ro08, Ro13); apply CDS-1800 and CDS-1900 for powered equipment and consumables (Ro14, Ro16); satisfy the CDS-1500 apparel and footwear baselines for apparel sold under the profile (Ro17); maintain warnings, restricted-sale and recall handling (Ro38–Ro40); maintain a jurisdiction requirement register for every market published to (Ro03); and publish and verify channel representations under CDS-500.

CDS2000-Ro42 An implementation claiming the CDS Toys and Games Profile MUST: keep retail age band, manufacturer's recommended age and safety age warning as separate attributes (Ro18); maintain structured hazard facts and market-specific warnings projected and verified where required online (Ro19–Ro20, Ro24, Ro38); bind age band and play type to governed dictionaries (Ro21); store game facts as typed values (Ro22, Ro25); maintain per-market compliance records evaluated at preflight (Ro23); govern licensed properties (Ro26); apply other profiles' layers where present (Ro27); handle restricted sale and recalls (Ro39–Ro40); maintain a jurisdiction requirement register for every market published to (Ro03); and publish and verify channel representations under CDS-500.

CDS2000-Ro43 An implementation claiming the CDS Pet Supplies Profile MUST: govern pet type, life stage and size with measured bounds (Ro28–Ro29, Ro35); store ingredients, analysis and nutritional adequacy as structured records with schemes and evidence (Ro30–Ro32); bind food form to the governed dictionary (Ro33); maintain per-market registration and restricted-sale records for health products (Ro34); exclude live animals absent a declared extension (Ro36); express consumable compatibility through relationship records (Ro37); handle warnings, restricted sale and recalls (Ro38–Ro40); maintain a jurisdiction requirement register for every market published to (Ro03); and publish and verify channel representations under CDS-500.

14. Worked Product Examples (*informative*)

14.1 Toy — Magnetic building set

```
CAT_product_type = construction_toy; MF_play_type = [construction, stem]
MF_age_band = [age_3_5, age_6_8]; MF_recommended_age_min_months = 36
MF_contains_magnets = true; MF_small_parts = true; MF_piece_count = 100
MF_safety_age_warning = {scheme: eu_toy_safety_2009_48, statement: "Warning. Not suitable for children under 36 months. Small parts. Choking hazard.", hazard: [small_parts]}
MF_warning_statements += {scheme: eu_toy_safety_2009_48, statement: "Warning. This toy contains magnets or magnetic components..."},
                        {scheme: us_cpsc_16_cfr_1500_19, statement: "WARNING: CHOKING HAZARD – Small parts. Not for children under 3 yrs."},
                        {scheme: au_toys_containing_magnets_2020, statement: <mandatory warning text>}
CMP_compliance_declarations = [{jurisdiction: EU, scheme: eu_toy_safety_2009_48, mark: CE, doc_ref: MED-..},
                                {jurisdiction: US, scheme: us_cpsc_16_cfr_1500_19, mark: MED-.., tracking_label: true},
                                {jurisdiction: AU, scheme: au_toys_containing_magnets_2020, test_report: MED-..}]
EU listing: warning visible before purchase -> verified; US listing: cautionary statement displayed -> verified
```

14.2 Pet food — Dry dog food, large breed adult

```
CAT_product_type = dog_food_dry; MF_pet_type = [dog]; MF_pet_life_stage = adult; MF_pet_size = large {weight_min_kg: 25, weight_max_kg: 45}
MF_pet_food_form = dry_kibble
MF_ingredient_list = [chicken_meal, brown_rice, ...] (scheme: manufacturer_declared, order: descending_by_weight)
MF_nutrition_values = {scheme: guaranteed_analysis_pet_food, basis: as_fed, values: [{crude_protein, min, 26, %}, {crude_fat, min, 12, %},
{crude_fibre, max, 4, %}, {moisture, max, 10, %}]}
MF_nutritional_adequacy = {scheme: aafco_dog_food_nutrient_profiles, life_stage: adult_maintenance, status: complete_and_balanced, evidence: E1}
Variants: 3 kg (VAR_net_quantity = {3, kilogram}), 12 kg, 20 kg; unit price per kilogram derived
AU: complies with AS 5812 (voluntary) recorded as certification; US: state feed registration recorded per state where required
```

14.3 Sports — Wetsuit

```
CAT_product_type = wetsuit; MF_sport_activity = [surfing]
MF_wetsuit_style = full_suit; MF_wetsuit_thickness = t_3_2 (3/2 mm); MF_wetsuit_zip = chest_zip; MF_wetsuit_seam = glued_blind_stitched
VAR_size_system = <brand_wetsuit_size_system>; VAR_size_label = "MT"; measured bounds: height 178–185 cm, chest 96–101 cm
MF_temperature_rating = {value_min: 12, value_max: 17, unit: celsius, basis: manufacturer_rated}
```

15. Reference Validation Cases (*informative — the profiles' contribution to the cross-industry validation set, REVIEW-020*)

- a safety age warning derived from the retail age band (must fail Ro18);
- a toy with MF_contains_magnets unknown defaulted to false (must quarantine, Ro20);
- an EU toy listing without the age warning visible before purchase (verification must report, Ro19, D-EU-1);

- a US internet listing of a small-parts toy without the cautionary statement (R019, D-US-1);
- a helmet sold into Australia without a bicycle-helmet certification record (must block, R011, D-AU-2);
- a sport activity encoded as a category rather than an attribute (R009);
- a temperature rating compared across two different schemes (R010);
- a pet food with adequacy "inferred" from ingredients (R032);
- a "medium dog" size class with no weight bounds feeding a weight filter (R029);
- a prescription pet medicine published to a marketplace that cannot enforce the restriction (R034);
- a wetsuit thickness stored as free text "3/2mm" (R012);
- a "batteries not included" statement with no `requires` record (R008).

Appendix A. Attribute Baselines (*normative — dictionary bindings per CDS1500-R011*)

A.1 Sports and Outdoor

Field	Scope	Type	Baseline requirement
CAT_product_type	Product	Governed classification reference	R
MF_sport_activity	Product	Governed dictionary list (<code>sport_activity</code>)	R
VAR_size_system / VAR_size_label / measured bounds	Variant	Governed system; text; typed measurements	R for sized equipment and apparel
MF_wetsuit_style / MF_wetsuit_thickness / MF_wetsuit_zip / MF_wetsuit_seam	Product	Governed dictionary references	R for wetsuits where exposed; REC otherwise
MF_temperature_rating / MF_waterproof_rating / MF_season_rating	Product	Typed values with scheme and basis	C
MF_user_weight_limit_kg / MF_capacity_persons / MF_capacity_l	Product	Typed measurements	C — R where a limit exists
MF_certifications	Product (per market)	Certification records	R for protective equipment
Relationship records (<code>compatible_with</code> , <code>includes</code> , <code>requires</code> , <code>fits</code>)	Product	Records per CDS-1900 §5	C
Battery and power fields	Product / Variant	Per CDS-1800 §8	R for powered products
CMP_dangerous_goods	Product / Variant	Declaration per CDS-1900 §12	R (explicit value)
MF_warning_statements / CMP_restricted_sale	Product (per market)	Structured records	C
Apparel and footwear baseline	—	Per CDS-1500 Appendices A and A.1	R where applicable

A.2 Toys and Games

Field	Scope	Type	Baseline requirement
CAT_product_type	Product	Governed classification reference	R
MF_age_band	Product	Governed dictionary list (<code>toy_age_range</code>)	R
MF_recommended_age_min_months / MF_recommended_age_max_months	Product	Integers	REC; R where a market requires the manufacturer's age
MF_safety_age_warning	Product	Structured statement record	C — R where a market mandates it
MF_play_type	Product	Governed dictionary list (<code>play_type</code>)	REC
MF_piece_count	Product / Variant	Integer	C
MF_player_count_min / MF_player_count_max / MF_play_time_min / MF_play_time_max	Product	Integers	R for games
MF_language	Product / Variant	Governed language codes	C
MF_small_parts / MF_contains_magnets / MF_contains_button_batteries / MF_projectile / MF_cord_length_mm	Product	Booleans; typed measurement	R (explicit values; unknown quarantined)
MF_warning_statements	Product (per market)	Structured records	C — R where mandated
CMP_compliance_declarations / CMP_responsible_person / CMP_product_identifier	Product (per market)	Governed records	R where a market regulates toys
Battery and power fields / CMP_dangerous_goods	Product / Variant	Per CDS-1800 §8, CDS-1900 §12	R for battery products
MF_licensed_property	Product	Governed organisation dictionary	C
MF_weight_limit_kg	Product	Typed measurement	R for ride-ons and play equipment

A.3 Pet Supplies

Field	Scope	Type	Baseline requirement
CAT_product_type	Product	Governed classification reference	R
MF_pet_type	Product	Governed dictionary list (pet_type)	R
MF_pet_life_stage	Product	Governed dictionary reference (pet_life_stage)	C — R for food formulated by life stage
MF_pet_size	Product	Governed dictionary reference (pet_size) with bounds	C
MF_pet_food_form	Product	Governed dictionary reference (pet_food_form)	R for food and treats
MF_ingredient_list / MF_ingredient_naming_scheme	Product	Ordered structured list	R for food, treats and supplements
MF_nutrition_values	Product	Typed values with scheme and basis	R for food where a market requires analysis; REC otherwise
MF_nutritional_adequacy	Product	Structured record with scheme	R for complete foods; C otherwise
MF_feeding_guide	Product	Structured table or document	REC; R where a market requires feeding directions
VAR_net_quantity	Variant	Typed measurement	R
MF_storage_condition / MF_date_mark_type	Product	Governed values	C
CMP_market_registrations / CMP_restricted_sale / MF_active_ingredients / MF_dosage_form	Product (per market)	Governed records	R for health products
Relationship records (compatible_with , accessory_for , fits)	Product	Records per CDS-1900 §5	C
MF_warning_statements	Product (per market)	Structured records	C

Appendix B. Dictionary Bindings (*informative — becomes governed data on adoption per CDS1500-R011*)

Dictionary key	Package chapter	Bound attribute(s)	Status at 0.8.0
sport_activity	12	MF_sport_activity	Bound by this chapter
wetsuit_style / wetsuit_thickness / wetsuit_zip / wetsuit_seam	12	MF_wetsuit_*	Bound
play_type / toy_age_range	13	MF_play_type / MF_age_band	Bound
pet_type / pet_life_stage / pet_size	14	MF_pet_type / MF_pet_life_stage / MF_pet_size	Bound
pet_food_form	14 (new, 0.8.0)	MF_pet_food_form	Bound
size_system (bike_frame_cm, head_circumference_cm added at 0.8.0)	5	VAR_size_system	Bound
storage_condition, allergen, dietary_claim_type	17	Per CDS-1700 patterns where used	Bound via CDS-1700
battery_chemistry, product_condition, relationship_type, dangerous_goods_class	11, 18	Per CDS-1800 and CDS-1900	Bound via those chapters
jurisdiction / regulatory_scheme	19	CMP_* records, warning schemes	Bound via CDS-1500 §2.2

Deliberately not shipped: skill levels, performance tiers, breed lists, licensed-property registries, theme and character vocabularies, live-animal vocabularies, pet nutrient profiles (scheme-published).

Appendix C. References (*informative; retrieved 2026-09-20 unless stated*)

Ref	Source	Location	Use in this chapter
[S1]	ACCC Product Safety — mandatory standards: Toys for children up to and including 36 months (Consumer Goods (Toys for Children up to and including 36 months of age) Safety Standard 2023, AS/NZS ISO 8124.1:2023); Toys containing magnets (2020); Projectile toys (2020); Bicycle helmets (2024); Sunglasses and fashion spectacles (2017); Products containing button/coin batteries (2020)	productsafety.gov.au; legislation.gov.au	D-AU-1 to D-AU-4 (retrieved; instrument texts not read this pass except button batteries, see CDS-1800 [E3])
[S2]	Standards Australia — AS 5812:2023 Manufacturing and marketing of pet food (voluntary; referenced by the Pet Food Industry Association of Australia); APVMA — Agvet Code registration of veterinary chemical products	standards.org.au; apvma.gov.au	D-AU-5, D-AU-6 (retrieved)
[S3]	New Zealand — Product Safety Standards (Children's Toys) Regulations 2005; ACVM Act 1997 (pet food exemptions and standards)	legislation.govt.nz; mpi.govt.nz	D-NZ-1, D-NZ-2 (retrieved)
[S4]	US CPSC — Consumer Product Safety Improvement Act 2008 (children's product certificate, third-party testing, tracking labels); 16 CFR 1500.19 (small parts and other cautionary labelling), 16 CFR 1500.20 (labelling in internet and catalogue advertising), 16 CFR 1501 (small parts test), 16 CFR 1250 (ASTM F963 toy standard), 16 CFR 1203 (bicycle helmets)	cpsc.gov; ecfr.gov	D-US-1 to D-US-3 (retrieved; not spot-verified this pass)
[S5]	US FDA — Pet food labelling (21 CFR 501); AAFCO — Model Regulations for pet food and specialty pet food (product name, guaranteed analysis, ingredient statement, nutritional adequacy statement, feeding directions, quantity, manufacturer); state feed control laws	fda.gov; aaeco.org	D-US-4 (retrieved)
[S6]	Directive 2009/48/EC on the safety of toys (warnings determining the decision to purchase visible before purchase including online); Regulation (EU) 2025/2509 on the safety of toys (in force 1 January 2026; applies from 1 August 2030; digital product passport; online offers)	eur-lex.europa.eu	D-EU-1 (regulation verified 2026-09-20; directive detail retrieved)
[S7]	Regulation (EU) 2016/425 on personal protective equipment; Regulation (EU) 2023/988 (GPSR)	eur-lex.europa.eu	D-EU-2, D-EU-3

Ref	Source	Location	Use in this chapter
[S8]	Regulation (EC) No 767/2009 on the placing on the market and use of feed (labelling of pet food); Regulation (EC) No 1831/2003 (feed additives); FEDIAF Code of Good Labelling Practice for Pet Food	eur-lex.europa.eu; europeanpetfood.org	D-EU-4 (retrieved)
[S9]	UK — Toys (Safety) Regulations 2011; Personal Protective Equipment (Enforcement) Regulations 2018 with retained 2016/425; General Product Safety Regulations 2005; retained Regulation 767/2009 — OPSS, Defra	legislation.gov.uk; gov.uk	D-UK-1 to D-UK-3 (retrieved)
[S10]	Canada — Canada Consumer Product Safety Act; Toys Regulations (SOR/2011-17); Consumer Packaging and Labelling Act (bilingual); CFIA guidance on pet food (imports and labelling)	laws-lois.justice.gc.ca; inspection.canada.ca	D-CA-1, D-CA-2 (retrieved)
[S11]	StoreLeads (Shopify Plus category distribution, 2026-09-11); StoreInspect (average products per store by category, 2026-09-19)	storeleads.app; storeinspect.com	Market basis (REVIEW-020)

Appendix D. Jurisdiction Requirement Register — Seed (*informative seed of the normative register defined in CDS-1500 §2.2*)

Conventions as in CDS-1600 Appendix D. Most entries in this chapter were retrieved from regulator pages without reading the instrument text in this pass and are marked accordingly; organisations verify before relying on them.

D.1 Australia (AU)

Ref	Instrument / regulator	Applies to	Obligation	Required data elements → CDS field	Trigger / dates	Verification
D-AU-1	Consumer Goods (Toys for Children up to and including 36 months of age) Safety Standard 2023 (AS/NZS ISO 8124.1:2023 provisions on small parts, sharp points and edges, cords); Consumer Goods (Toys Containing Magnets) Safety Standard 2020; Consumer Goods (Projectile Toys) Safety Standard 2020 — ACCC and state regulators (au_toys_under_36_months_2023, au_toys_containing_magnets_2020, au_projectile_toys_2020)	Toys for children up to 36 months; toys containing magnets; projectile toys	pre_market (compliance with the referenced standard provisions); warning_statement (magnet toys: mandatory warning on packaging or the toy); labelling_element	Small-parts, magnet and projectile compliance evidence; magnet warning text → MF_small_parts, MF_contains_magnets, MF_projectile, MF_warning_statements, CMP_compliance_declarations (test evidence)	Ongoing (2023 standard replaced the 2020 standard)	Retrieved 2026-09-20 [S1] (not spot-verified)
D-AU-2	Consumer Goods (Bicycle Helmets) Safety Standard 2024 (AS/NZS 2063 and recognised international standards) — ACCC (au_bicycle_helmets_2024)	Bicycle helmets supplied in Australia	pre_market (certification to an accepted standard); labelling_element (certification marking)	Standard and certification evidence → MF_certifications (scheme au_bicycle_helmets_2024), CMP_compliance_declarations	Ongoing	Retrieved 2026-09-20 [S1]
D-AU-3	Consumer Goods (Sunglasses and Fashion Spectacles) Safety Standard 2017 (AS/NZS 1067.1:2016 lens categories) — ACCC (au_sunglasses_2017)	Sunglasses and fashion spectacles	labelling_element (lens category and required markings)	Lens category 0–4 and marking → MF_lens_category, MF_warning_statements	Ongoing	Retrieved 2026-09-20 [S1]
D-AU-4	Button and coin battery safety and information standards 2020 — ACCC (au_button_battery_standards_2020)	Toys and products containing button or coin batteries	As CDS-1800 D-AU-5	Per CDS-1800 D-AU-5 → MF_contains_button_batteries, MF_warning_statements, listing projection	Mandatory from 22 June 2022	Verified 2026-09-20 (CDS-1800 [E3])
D-AU-5	AS 5812:2023 Manufacturing and marketing of pet food (voluntary industry standard; PFIAA membership requirement) (au_as_5812_pet_food)	Pet food	documentation (voluntary labelling: product name, species, life stage, complete or complementary statement, ingredient list, typical analysis, feeding guide, net quantity, batch and best-before, manufacturer details)	Elements → MF_pet_type, MF_pet_life_stage, MF_nutritional_adequacy (scheme au_as_5812_pet_food), MF_ingredient_list, MF_nutrition_values, MF_feeding_guide, VAR_net_quantity	Voluntary (no mandatory national pet-food standard as at 2026-09-20; state stock-food laws apply in some states)	Retrieved 2026-09-20 [S2]
D-AU-6	Agricultural and Veterinary Chemicals Code Act 1994 — APVMA (au_apvma_agvet_code)	Veterinary medicines, parasiticides and treatments for animals	pre_market (registration and approved label); restricted_content (scheduled medicines; prescription-only products); labelling_element	APVMA registration number, approved label, schedule and supply restrictions → CMP_market_registrations (scheme au_apvma_agvet_code), CMP_restricted_sale, MF_active_ingredients	Ongoing	Retrieved 2026-09-20 [S2]

D.2 New Zealand (NZ)

Ref	Instrument / regulator	Applies to	Obligation	Required data elements → CDS field	Trigger / dates	Verification
D-NZ-1	Product Safety Standards (Children's Toys) Regulations 2005 (AS/NZS ISO 8124.1 provisions for toys for children under 36 months) — MBIE and Commerce Commission (nz_childrens_toys_regulations_2005)	Toys for children under 36 months	pre_market (compliance with the referenced provisions)	Small parts, sharp points, cords compliance → MF_small_parts, CMP_compliance_declarations	Ongoing	Retrieved 2026-09-20 [S3]
D-NZ-2	Agricultural Compounds and Veterinary Medicines Act 1997 — MPI (nz_acvm_act_1997)	Pet food (exempt from registration subject to conditions) and	pre_market (veterinary medicines); documentation (pet	Registration or exemption status → CMP_market_registrations (scheme nz_acvm_act_1997)	Ongoing	Retrieved 2026-09-20 [S3]

Ref	Instrument / regulator	Applies to	Obligation	Required data elements → CDS field	Trigger / dates	Verification
		veterinary medicines (registration)	food exemption conditions)			

D.3 United States (US)

Ref	Instrument / regulator	Applies to	Obligation	Required data elements → CDS field	Trigger / dates	Verification
D-US-1	Federal Hazardous Substances Act; 16 CFR 1500.19 (cautionary labelling for small parts, small balls, marbles, balloons) and 16 CFR 1500.20 (cautionary statements in internet and catalogue advertising) — CPSC (us_cpssc_16_cfr_1500_19)	Toys and games intended for children 3 to under 6 years with small parts; balloons; small balls; marbles	warning_statement; listing_element (the same cautionary statement must appear in internet advertising that offers direct purchase)	Statutory cautionary statement text and format → MF_safety_age_warning / MF_warning_statements (scheme us_cpssc_16_cfr_1500_19), listing projection and verification	Ongoing	Retrieved 2026-09-20 [S4]
D-US-2	Consumer Product Safety Improvement Act 2008; 16 CFR 1250 (ASTM F963 toy safety standard); 16 CFR 1107 (third-party testing); 16 CFR 1130 (tracking labels) — CPSC (us_cpssia_children_product)	Children's products (designed or intended primarily for children 12 and under)	pre_market (Children's Product Certificate based on accredited third-party testing); labelling_element (tracking label with manufacturer, location and date of production, batch); documentation	CPC reference, test reports, tracking label data → CMP_compliance_declarations (scheme us_cpssia_children_product, CPC ref), CMP_product_identifier, MED_technical_documents	Ongoing	Retrieved 2026-09-20 [S4]
D-US-3	16 CFR 1203 Safety Standard for Bicycle Helmets — CPSC (us_cpssc_16_cfr_1203)	Bicycle helmets	pre_market (certification); labelling_element	Certification evidence and label → MF_certifications (scheme us_cpssc_16_cfr_1203)	Ongoing	Retrieved 2026-09-20 [S4]
D-US-4	Federal Food, Drug, and Cosmetic Act and 21 CFR 501 (animal food labelling) — FDA; state feed laws adopting the AAFCO Model Regulations (product name, net quantity, guaranteed analysis, ingredient statement, nutritional adequacy statement, feeding directions, manufacturer or distributor name and address; state registration and licensing) (us_fda_21_cfr_501_pet_food, us_aafco_state_feed_law)	Pet food and treats sold in the US	labelling_element; documentation; pre_market (state product registration where required)	Elements → MF_nutrition_values (scheme guaranteed_analysis_pet_food), MF_nutritional_adequacy (scheme aafco_*_nutrient_profiles or feeding trial), MF_ingredient_list, MF_feeding_guide, VAR_net_quantity, CMP_responsible_person, CMP_market_registrations per state	Ongoing	Retrieved 2026-09-20 [S5]
D-US-5	Proposition 65 (California) — OEHHA (us_ca_prop65_warning)	Toys, sporting goods and pet products causing listed-chemical exposure	warning_statement; listing_element	Per CDS-1800 D-US-3	Effective 1 January 2025	Verified 2026-09-20 (CDS-1800 [E7])

D.4 European Union (EU)

Ref	Instrument / regulator	Applies to	Obligation	Required data elements → CDS field	Trigger / dates	Verification
D-EU-1	Directive 2009/48/EC on the safety of toys (CE marking, EC declaration of conformity, manufacturer and importer identification, warnings including age warnings and specific warnings for toys in Annex V; warnings that determine the decision to purchase must be clearly visible before purchase including online) — to be replaced by Regulation (EU) 2025/2509 (applies from 1 August 2030; digital product passport; online offers to display warnings and safety information) (eu_toy_safety_2009_48, eu_toy_safety_regulation_2025_2509)	Toys (products designed or intended for use in play by children under 14)	pre_market (conformity assessment; DoC; technical file); labelling_element (CE marking; manufacturer and importer name and address; type/batch/serial); warning_statement; listing_element	Warning texts (preceded by "Warning"), age warning with hazard where applicable, CE marking, identification, DoC reference → MF_safety_age_warning, MF_warning_statements (scheme eu_toy_safety_2009_48), CMP_compliance_declarations, CMP_responsible_person, CMP_product_identifier, listing projection	Directive in force until 1 August 2030; regulation in force 1 January 2026	Regulation verified 2026-09-20 [S6]; directive online-visibility detail retrieved (not spot-verified)
D-EU-2	Regulation (EU) 2016/425 on personal protective equipment (eu_ppe_2016_425)	Helmets, eye protection, flotation aids, protective sports equipment classified as PPE (Category I–III)	pre_market (conformity assessment by category; DoC; notified body for II/III); labelling_element (CE marking; identification; instructions)	Category, standard, notified body number where applicable, DoC → MF_certifications, CMP_compliance_declarations (scheme eu_ppe_2016_425), MED_technical_documents	Ongoing	Retrieved 2026-09-20 [S7] (not spot-verified)
D-EU-	Regulation (EU) 2023/988 General Product Safety Regulation (eu_gpsr_2023_988)	Non-harmonised	listing_element; labelling_element;	Per CDS-1800 D-EU-6	Applies from 13	Verified 2026-09-20 (CDS-

Ref	Instrument / regulator	Applies to	Obligation	Required data elements → CDS field	Trigger / dates	Verification
3		sports equipment, pet accessories and other consumer products; cross-cutting online-offer duties	pre_market (responsible economic operator)		December 2024	1800 [E13]]
D-EU-4	Regulation (EC) No 767/2009 on the placing on the market and use of feed (Chapter 4 labelling: type of feed, feed business operator, net quantity, composition, analytical constituents, additives, moisture, best-before, batch, instructions for proper use, species and category of animal, complete or complementary designation); Regulation (EC) No 1831/2003 (additives); FEDIAF labelling code (eu_feed_labelling_767_2009)	Pet food (feed for pet animals)	labelling_element; documentation	Elements → MF_pet_type, MF_pet_life_stage, MF_nutritional_adequacy (complete/complementary; scheme fediaf_nutritional_guidelines where claimed), MF_ingredient_list (composition by category or ingredient), MF_nutrition_values (scheme eu_analytical_constituents), MF_feeding_guide, VAR_net_quantity, MF_date_mark_type, CMP_responsible_person	Ongoing	Retrieved 2026-09-20 [S8] (not spot-verified)
D-EU-5	Regulation (EU) 2019/6 on veterinary medicinal products (eu_veterinary_medicines_2019_6)	Veterinary medicines and medicated products	pre_market (marketing authorisation); restricted_content (prescription status; online retail rules per member state)	Authorisation and prescription status → CMP_market_registrations, CMP_restricted_sale	Applies from 28 January 2022	Retrieved (not spot-verified)

D.5 United Kingdom (GB)

Ref	Instrument / regulator	Applies to	Obligation	Required data elements → CDS field	Trigger / dates	Verification
D-UK-1	Toys (Safety) Regulations 2011 — OPSS (uk_toys_safety_regulations_2011)	Toys placed on the GB market	pre_market (UKCA or CE marking with indefinite CE recognition from 1 October 2024; DoC; technical file); labelling_element; warning_statement (warnings visible before purchase including online)	As D-EU-1 with GB marking and importer identification → MF_safety_age_warning, MF_warning_statements (scheme uk_toys_safety_regulations_2011), CMP_compliance_declarations, CMP_responsible_person	Ongoing; CE recognition indefinite from 1 October 2024	Retrieved 2026-09-20 [S9]
D-UK-2	Regulation (EU) 2016/425 as retained (PPE) with the Personal Protective Equipment (Enforcement) Regulations 2018; General Product Safety Regulations 2005 — OPSS (uk_ppe_retained_2016_425, uk_gpsr_2005)	PPE (helmets, eyewear, flotation); non-harmonised consumer products	pre_market; labelling_element	Certification and identification → MF_certifications, CMP_compliance_declarations	Ongoing	Retrieved 2026-09-20 [S9]
D-UK-3	Regulation (EC) 767/2009 as retained (animal feed labelling); Veterinary Medicines Regulations 2013 (as amended 2024) — Defra, VMD (uk_feed_labelling_retained_767_2009, uk_veterinary_medicines_regulations)	Pet food; veterinary medicines	labelling_element; pre_market; restricted_content (distribution categories POM-V, POM-VPS, NFA-VPS, AVM-GSL)	As D-EU-4; authorisation and distribution category → CMP_market_registrations, CMP_restricted_sale	Ongoing	Retrieved 2026-09-20 [S9]

D.6 Canada (CA)

Ref	Instrument / regulator	Applies to	Obligation	Required data elements → CDS field	Trigger / dates	Verification
D-CA-1	Canada Consumer Product Safety Act; Toys Regulations (SOR/2011-17) (small parts, magnets, cords, noise, toxic substances, age-related requirements); Consumer Packaging and Labelling Act (bilingual identity, net quantity, dealer) — Health Canada (ca_toys_regulations_2011)	Toys and children's products	pre_market (compliance; no certificate regime but records on request); warning_statement (bilingual); labelling_element	Bilingual warnings and identity → MF_warning_statements (scheme ca_toys_regulations_2011, EN and FR), CMP_responsible_person, localisation per CDS-300 §18	Ongoing	Retrieved 2026-09-20 [S10]
D-CA-2	Pet food: Consumer Packaging and Labelling Act and Competition Act (labelling and representations); Feeds Act (largely excludes pet food); Health of Animals Act	Pet food sold in Canada	labelling_element (bilingual identity, net quantity, dealer identity); documentation (import conditions)	Elements → MF_pet_type, VAR_net_quantity, CMP_responsible_person; voluntary	Ongoing	Retrieved 2026-09-20 [S10]

Ref	Instrument / regulator	Applies to	Obligation	Required data elements → CDS field	Trigger / dates	Verification
	import conditions — CFIA and Competition Bureau (ca_pet_food_labelling)			AAFCO-style adequacy statements → MF_nutritional_adequacy		

D.7 Other markets

Jurisdiction	Status	Note
Japan (JP), China (CN), Republic of Korea (KR), India (IN), Brazil (BR) and others	Not seeded	Toy safety marks (ST mark, CCC, KC), PPE and pet-food regimes differ; organisation supplies entries with local advice.

END OF CDS-2000 v0.2 REVIEW DRAFT

Appendix A. Architecture Decision Register

ADR	Status
ADR-D1 — Single Conformance Ladder; Governance as Capstone; AI as Overlay Profile	Accepted (owner, 2026-08-03)
ADR-D2 — MF_ Semantics and the STD_/MF_ Boundary by Enumeration	Accepted (owner, 2026-08-03)
ADR-D24 — DF_ → CH_ Transition (Consolidated ADR)	Accepted (consolidates and supersedes: CDS-200 ADR-003 "Provisional", CDS-300 ADR-009 "Accepted for new design", CDS-ADR-900-002/003, CDS-ADR-1300-005)
ADR-D3 — One Verification Status Enum, Reason Codes, and the Traffic-Light Mapping	Accepted (owner, 2026-08-03)
ADR-D4 — Value-Layer Terminology: Source → Alias Mapping → Canonical → Projections	Accepted (owner, 2026-08-03)
ADR-D5 — Corpus Version Policy and Schema Identifier Scheme	Accepted (owner, 2026-08-03)

All vo.1 chapter-local ADRs are superseded by or consolidated into this register and their chapters' vo.2 sections.

Appendix B. Requirements Index (summary)

1,320 requirements: 765 MUST · 334 MUST NOT · 161 SHOULD · 32 MAY · 11 SHOULD NOT · 17 constitutive/indicative (counted by each requirement's primary keyword). Full machine-readable index: `requirements-index.csv`.

Appendix C. Bibliography and Evidence Base

- An anonymised legacy implementation study covering product-data flow, tag governance and implementation sequencing.
- Platform documentation (priority 3, verified 2026-08-03/04, full register REVIEW-006/006A): [help.shopify.com](#) (product category; variants; Search & Discovery filters; store permissions; activity logs); [shopify.dev](#) (productUpdate; ProductUpdateInput; storefront filtering; webhooks; metafields); [support.google.com/merchants](#) (7052112; 6324436; 6324487); [facebook.com/business/help](#) (365831587397584; 125074381480892).
- Literature (priority 4): Jorij Abraham, *Product Information Management — Theory and Practice*, Springer 2014 (cross-check REVIEW-002B).
- Licence practice research: REVIEW-006A (OpenAPI, JSON Schema, AsyncAPI, schema.org, W3C, CloudEvents; accessed 2026-08-04).
- Industry-profile expansion research (REVIEW-020, retrieved 2026-09-20): global store-population data (StoreLeads worldwide and Shopify Plus category distributions and catalogue-size bands; StoreInspect average products per category), revenue segmentation (Statista Market Insights as reported by ECDB; ECDB Amazon category GMV), regional cross-checks (Australia Post Inside Australian Online Shopping 2026; StoreLeads Australia), channel taxonomies (Google product taxonomy 2021-09-21; Shopify Standard Product Taxonomy 2026-11-unstable; Google Merchant Center attribute specifications) and industry data standards (Auto Care Association ACES 5.0/PIES 8.0; ETIM; GS1 GPC; IATA lithium battery guidance 2026).
- Regulatory instruments and regulator publications cited per jurisdiction in the Appendix D tables of CDS-1600 through CDS-2000 and Appendix G of CDS-1500 (ACCC, TGA, FSANZ, EESS/ERAC, ACMA, Energy Rating, Safe Work Australia, NTC, APVMA; WorkSafe NZ, RSM, MPI; FDA, FTC, CPSC, FCC, OEHHHA, CARB, USDA, TTB; EUR-Lex, ECHA, EPREL; GOV.UK and [legislation.gov.uk](#), OPSS, HSE, VMD; Health Canada, CFIA, NRCan, ISED, Competition Bureau), each entry marked verified, retrieved or unverified as at 2026-09-20.